

SURGERY 1 – GENERAL SURGERY

SURGERY BANK

Surgery 1 – General Surgery

MCQs and EMQs with short explanations and step-by-step management flowcharts.

3RD EDITION / 2026

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118 TEAM

3RD EDITION / 2026

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COLOPHON

THIS EDITION

The 3rd Edition of the Surgery Bank for Surgery 1 (general surgery and urology). 2,454 questions — 1461 MCQs and 993 EMQ items in 39 chapters — with answers and explanations.

WHAT CHANGED IN THE 3RD EDITION

Every key was checked against current guidance and standard texts (Bailey & Love 28th edition, NICE, ATLS 10th edition, EAU, ESMO, WSES); 777 questions were corrected, and every explanation was rewritten and then cut to two or three lines that say why the answer is right. 28 questions whose idea had already been asked were removed: the question kept is the one from the earliest source in the chapter, and a highlighted note under it says how many times the idea was asked and in which exams or books. Numbering now restarts for each source: the local exams of a chapter (39th batch block, end-induction and final papers, ministerial, formative and previous-years exams) share one sequence, and each international book has its own; EMQs are numbered by theme. The 39th batch papers were added to the chapters they belong to; their recalled questions were completed and keyed, and their short-answer tasks recast as single-best-answer questions. All changes are logged in the corrections file that accompanies this book.

MANAGEMENT FLOWCHARTS

The old summary tables were removed. Each chapter now opens with step-by-step flowcharts that combine the summaries of the source books with current guidelines, so that the next step in management can be read at a glance.

ARTWORK

Part and cover artwork was drawn for this edition; each illustration shows the subject of its part.

CONTRIBUTORS

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IMPORTANT

This book is a revision aid. Guidelines change: always check current local and national guidance before applying anything to patient care.

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CONTENTS

NECK LUMPS & ENDOCRINE

01	Neck Lumps, Thyroid & Parathyroid Salivary, thyroid, parathyroid and other neck swellings: triangles of the neck, FNA, thyroid cancer and hyperparathyroidism.	95 Q	13
02	Endocrine Surgery Adrenal, pituitary and pancreatic endocrine tumours, MEN syndromes and the surgical endocrine work-up.	20 Q	31
03	Neck Lumps & Endocrine EMQs Extended matching questions on neck swellings, dysphagia, thyroid disease and skin lumps.	43 Q	39

BREAST

04	Benign Breast Disorders Breast assessment and triple assessment, fibroadenoma, cysts, mastalgia, nipple discharge, infection and gynecomastia.	47 Q	47
05	Breast Cancer Screening, risk and BRCA, DCIS and invasive cancer, staging, surgery, sentinel node biopsy and adjuvant therapy.	62 Q	59
06	Breast EMQs Extended matching questions on breast lumps, nipple discharge, investigation and treatment of breast cancer.	75 Q	75

GASTROINTESTINAL TRACT

07	Salivary Glands Parotid and submandibular swellings: sialolithiasis, sialadenitis, pleomorphic adenoma, Warthin tumour and the facial nerve.	14 Q	87
08	Oesophagus Dysphagia, achalasia, reflux and hiatus hernia, Barrett oesophagus, perforation and oesophageal cancer.	40 Q	93
09	Stomach & Duodenum Peptic ulcer and its complications, gastric outlet obstruction, gastric cancer, GIST and bariatric anatomy.	40 Q	105
10	Small & Large Intestine Intestinal obstruction, appendicitis, inflammatory bowel disease, diverticular disease, volvulus, polyps and colorectal cancer.	236 Q	117
11	Anorectal Disorders Haemorrhoids, fissure, abscess and fistula, pilonidal disease, prolapse and anal cancer.	44 Q	155
12	Gastrointestinal Bleeding Upper and lower GI bleeding: resuscitation, risk scores, endoscopy, varices, angiodysplasia and diverticular bleeding.	49 Q	166
13	Gastrointestinal Tumours Oesophageal, gastric, small-bowel, colorectal and anal tumours: polyps, polyposis syndromes, staging and treatment.	82 Q	179
14	Gastrointestinal EMQs		

Extended matching questions on the acute abdomen, dysphagia, GI bleeding, hernias, anorectal and hepatobiliary disease.	511 Q	195
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PAEDIATRIC SURGERY

15	Paediatric Surgery Pyloric stenosis, intussusception, malrotation, Hirschsprung disease, atresias, NEC, hernias and the neonatal abdomen.	65 Q	249
16	Paediatric Surgery EMQs Extended matching questions on vomiting infants, neonatal obstruction and the paediatric acute abdomen.	31 Q	263

ABDOMEN

17	Acute Abdomen & Abdominal Trauma Assessment of the acute abdomen, peritonitis and perforation, and blunt and penetrating abdominal trauma.	29 Q	269
18	Liver Liver abscess, hydatid cyst, benign and malignant liver tumours, portal hypertension and liver trauma.	56 Q	279
19	Gallbladder & Biliary Tree Gallstones, cholecystitis, cholangitis, obstructive jaundice, bile-duct injury and cholangiocarcinoma.	81 Q	293
20	Pancreas Acute and chronic pancreatitis, pseudocyst, pancreatic cancer and periampullary tumours.	53 Q	311
21	Spleen Splenic trauma, splenomegaly, indications for splenectomy and post-splenectomy sepsis.	12 Q	324
22	Adrenal Gland Pheochromocytoma, Conn and Cushing syndromes, incidentaloma and adrenal surgery.	16 Q	330
23	Hernias & Abdominal Wall Inguinal, femoral, umbilical, incisional and rare hernias: anatomy, complications and repair.	51 Q	336
24	Abdomen EMQs Extended matching questions on abdominal pain, masses, splenomegaly, pancreatic and hepatobiliary disease.	79 Q	349

UROLOGY

25	Urinary Tract Infection Cystitis, pyelonephritis, prostatitis, renal abscess and urinary tuberculosis.	19 Q	361
26	Urinary Stones Renal colic, imaging, medical expulsive therapy, ESWL, ureteroscopy and PCNL.	32 Q	369
27	Renal Tumours Renal cell carcinoma, Wilms tumour, angiomyolipoma and the renal mass.	13 Q	378
28	Urological Investigation & Haematuria Haematuria work-up, urinalysis, imaging and endoscopy of the urinary tract.	22 Q	385
29	Urological Emergencies & Trauma Urinary retention, testicular torsion, priapism, Fournier gangrene and renal, bladder and urethral injury.	41 Q	393
30	Prostate		

	Benign prostatic hyperplasia, LUTS, PSA and prostate cancer.	36 Q	404
31	Congenital Urology Undescended testis, hypospadias, PUJ obstruction, vesicoureteric reflux and posterior urethral valves.	22 Q	415
32	Scrotum & Testis Hydrocele, varicocele, epididymo-orchitis, scrotal swellings and testicular tumours.	27 Q	423
33	Bladder Tumours Haematuria, transitional cell carcinoma, TURBT, cystectomy and urinary diversion.	22 Q	432
34	Urology EMQs Extended matching questions on haematuria, LUTS, scrotal swellings, urological investigation and emergencies.	149 Q	439

TRAUMA & CRITICAL CARE

35	Trauma ATLS primary survey, airway, chest, head and limb injuries, and burns.	79 Q	459
36	Metabolic Response to Trauma The stress response, fluids, electrolytes and nutrition in the surgical patient.	15 Q	476
37	Shock Classes of haemorrhage, types of shock, resuscitation and transfusion.	31 Q	483
38	Obesity & Bariatric Surgery Indications for bariatric surgery, operations and their complications.	10 Q	493
39	Trauma & Perioperative EMQs Extended matching questions on shock, trauma, burns, perioperative care and postoperative complications.	105 Q	497

BACK MATTER

—	Index		511
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HOW TO USE THIS BOOK

FLOWCHARTS FIRST

Each chapter opens with management flowcharts updated to current guidelines: follow the arrows from the presenting problem, answer the ? decision boxes, and read the next step.

MCQS AND EMQS

MCQs are numbered 1, 2, 3 ... within their source. EMQs are numbered **Theme 1, Theme 2 ...**; each theme lists its options A, B, C ... and its scenarios are numbered 1, 2, 3

REPEATED QUESTIONS

When an idea was asked more than once, one question is kept — the one from the earliest source — and a highlighted note under it says how many times and where else it was asked. Repeats are a strong signal of a high-yield topic.

REPEATED 3* · ALSO ASKED IN
Previous-years exam · Bailey & Love

UPDATED ANSWERS AND INDEX

Answers marked **UPDATED** were corrected in this edition. The index lists diseases, drugs, tests and signs with the **page numbers** of the questions that test them.

LOCAL EXAMS FIRST, THEN THE BOOKS

Questions come grouped by source. **Local Exams** — the 39th-batch block, end-induction and final papers, ministerial, formative and previous-years exams — come first and share one numbering. The essay questions converted to MCQs and each international book (Bailey & Love, Lange, SBA, Oxford, PreTest, Get Ahead, Crash Course, Irfan) follow, each with its own numbering from 1.

ANSWERS AFTER EVERY SET

Answers follow each set of up to 25 questions, each with a two- or three-line explanation. Tap a question number to jump to its answer and the answer number to jump back.

TRUE/FALSE QUESTIONS

Questions that ask *which statements are true* may have several correct options; the answer chip lists all of them (for example **A C E**) and the explanation names the false ones.

AT A GLANCE

CHAPTER	MCQS	EMQ SCENARIOS	REPEATED IDEAS
01 Neck Lumps, Thyroid & Parathyroid	95	0	0
02 Endocrine Surgery	20	0	0
03 Neck Lumps & Endocrine EMQs	0	43	0
04 Benign Breast Disorders	47	0	1
05 Breast Cancer	62	0	0
06 Breast EMQs	0	75	0
07 Salivary Glands	14	0	0
08 Oesophagus	40	0	0
09 Stomach & Duodenum	40	0	0
10 Small & Large Intestine	236	0	1
11 Anorectal Disorders	44	0	0
12 Gastrointestinal Bleeding	49	0	0
13 Gastrointestinal Tumours	82	0	0
14 Gastrointestinal EMQs	0	511	4
15 Paediatric Surgery	65	0	10
16 Paediatric Surgery EMQs	0	31	0
17 Acute Abdomen & Abdominal Trauma	29	0	0
18 Liver	56	0	3
19 Gallbladder & Biliary Tree	81	0	0
20 Pancreas	53	0	0
21 Spleen	12	0	0
22 Adrenal Gland	16	0	0
23 Hernias & Abdominal Wall	51	0	1
24 Abdomen EMQs	0	79	0
25 Urinary Tract Infection	19	0	0
26 Urinary Stones	32	0	0
27 Renal Tumours	13	0	0
28 Urological Investigation & Haematuria	22	0	1
29 Urological Emergencies & Trauma	41	0	2
30 Prostate	36	0	0
31 Congenital Urology	22	0	0
32 Scrotum & Testis	27	0	1
33 Bladder Tumours	22	0	0
34 Urology EMQs	0	149	1
Total	1461	993	28

CHAPTER	MCQS	EMQ SCENARIOS	REPEATED IDEAS
■ 35 Trauma	79	0	1
■ 36 Metabolic Response to Trauma	15	0	2
■ 37 Shock	31	0	0
■ 38 Obesity & Bariatric Surgery	10	0	0
■ 39 Trauma & Perioperative EMQs	0	105	0
Total	1461	993	28

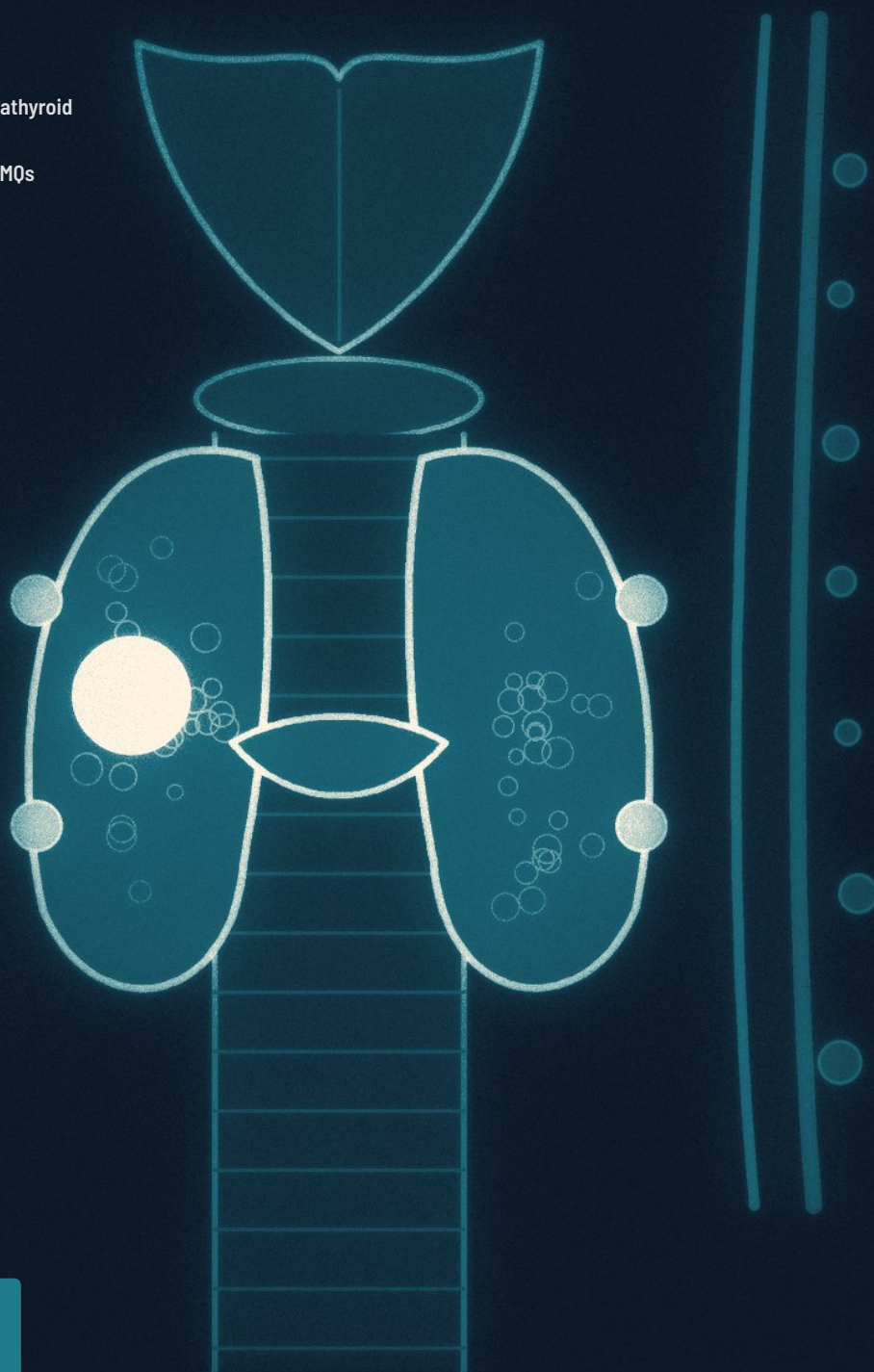
QUESTION SOURCES

In every chapter the banks appear in this order. “Merged” counts questions removed because their idea was already asked in an earlier or better question; each is credited in the repeat caption of the question that was kept.

SOURCE	QUESTIONS	MERGED
39th Batch Exams Block, end-induction and final papers of the 39th batch, recalled	174	4
Ministerial Exam Ministerial evaluation examinations	51	—
Formative College formative examinations	19	2
Previous Years Previous-years college examinations	137	1
Essay to MCQ College essay questions converted to MCQs	96	3
Bailey & Love After Bailey & Love's Short Practice of Surgery MCQs & EMQs	534	1
Lange After Lange Review of Surgery	145	3
SBA Single best answers in surgery	182	3
Oxford After the Oxford Specialty Training / Oxford surgery questions	166	2
PreTest After PreTest Surgery	197	2
Get Ahead After Get ahead! Surgery MCQs & EMQs	384	4
Crash Course After Crash Course Surgery SBAs & EMQs	219	3
Irfan Irfan surgery EMQ collection	150	—
Total	2454	28

PART 01 / 3 CHAPTERS

- 01 Neck Lumps, Thyroid & Parathyroid
- 02 Endocrine Surgery
- 03 Neck Lumps & Endocrine EMQs



01

NECK LUMPS & ENDOCRINE

Neck Lumps, Thyroid & Parathyroid

Salivary, thyroid, parathyroid and other neck swellings: triangles of the neck, FNA, thyroid cancer and hyperparathyroidism.

95

MCQs

0

EMQ SCENARIOS

7

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

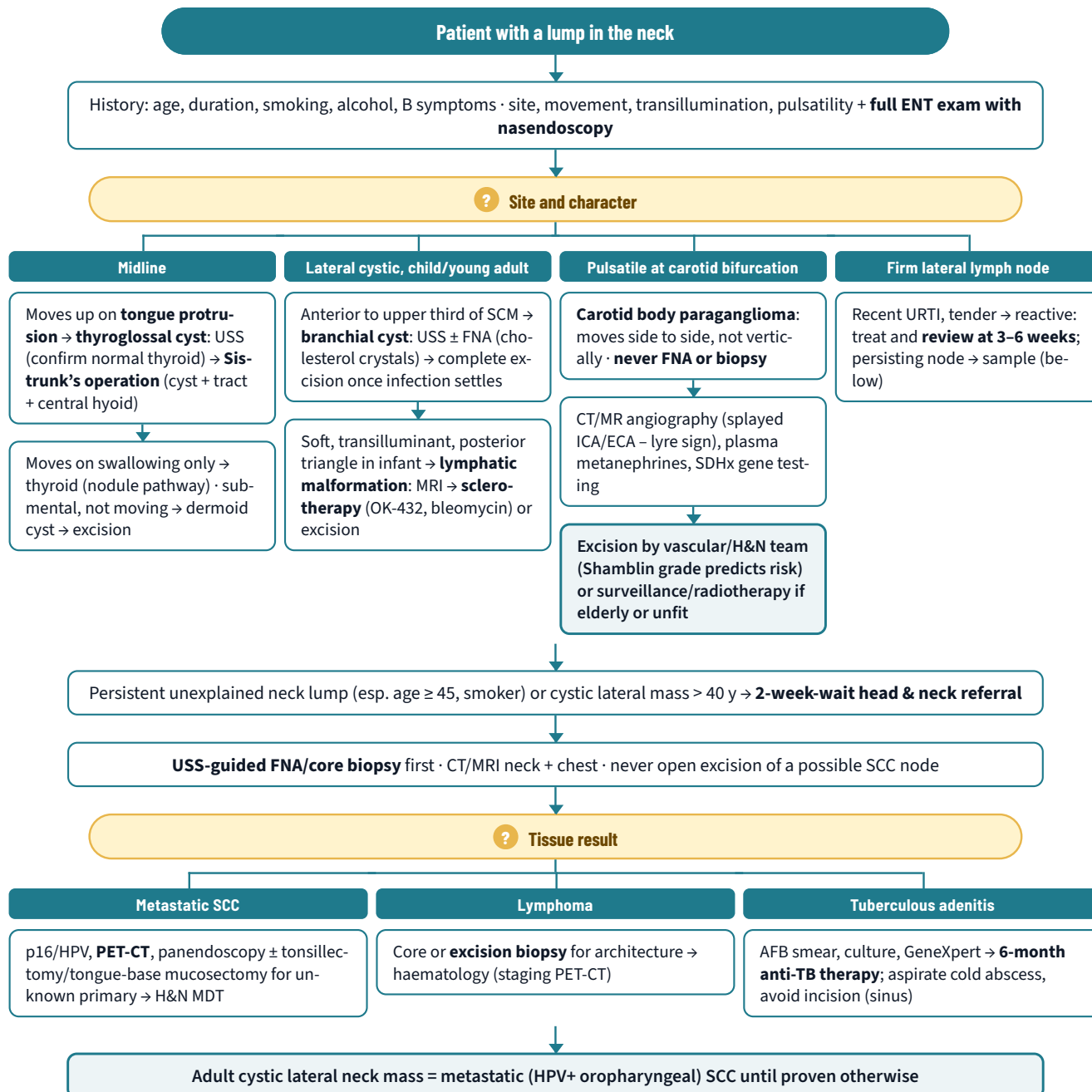
01	Management flowcharts	3 charts	page 14
02	Local Exams MCQs	4 Q	page 17
03	Essay to MCQ MCQs	12 Q	page 18
04	Bailey & Love MCQs	25 Q	page 20
05	Lange MCQs	11 Q	page 23
06	SBA MCQs	9 Q	page 24
07	PreTest MCQs	19 Q	page 26
08	Get Ahead MCQs	15 Q	page 29

01

Step by step

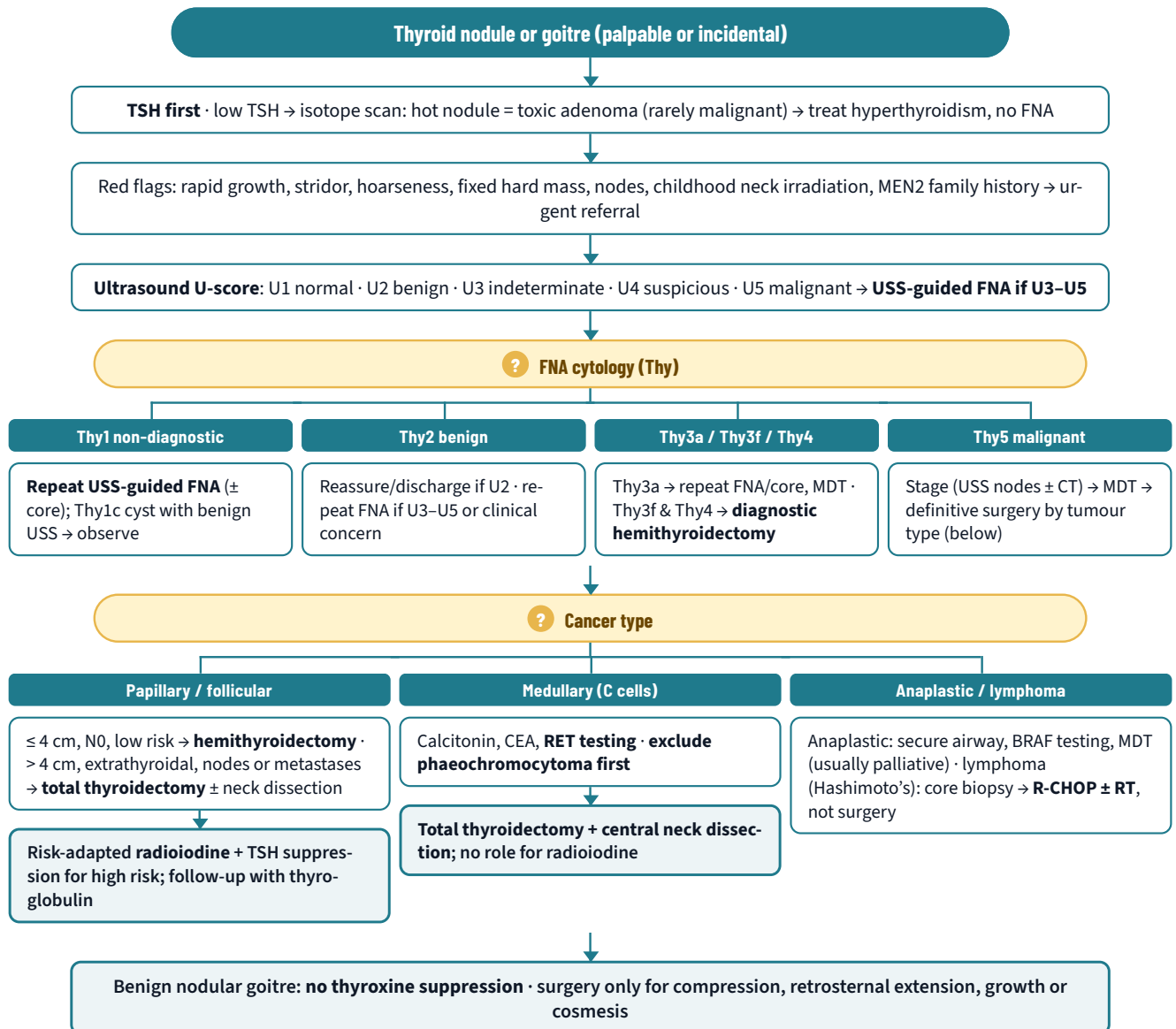
Neck lump: from site to diagnosis

NICE NG12 & NG36 • UK HEAD & NECK MDT 2016 • BAILEY & LOVE 28E



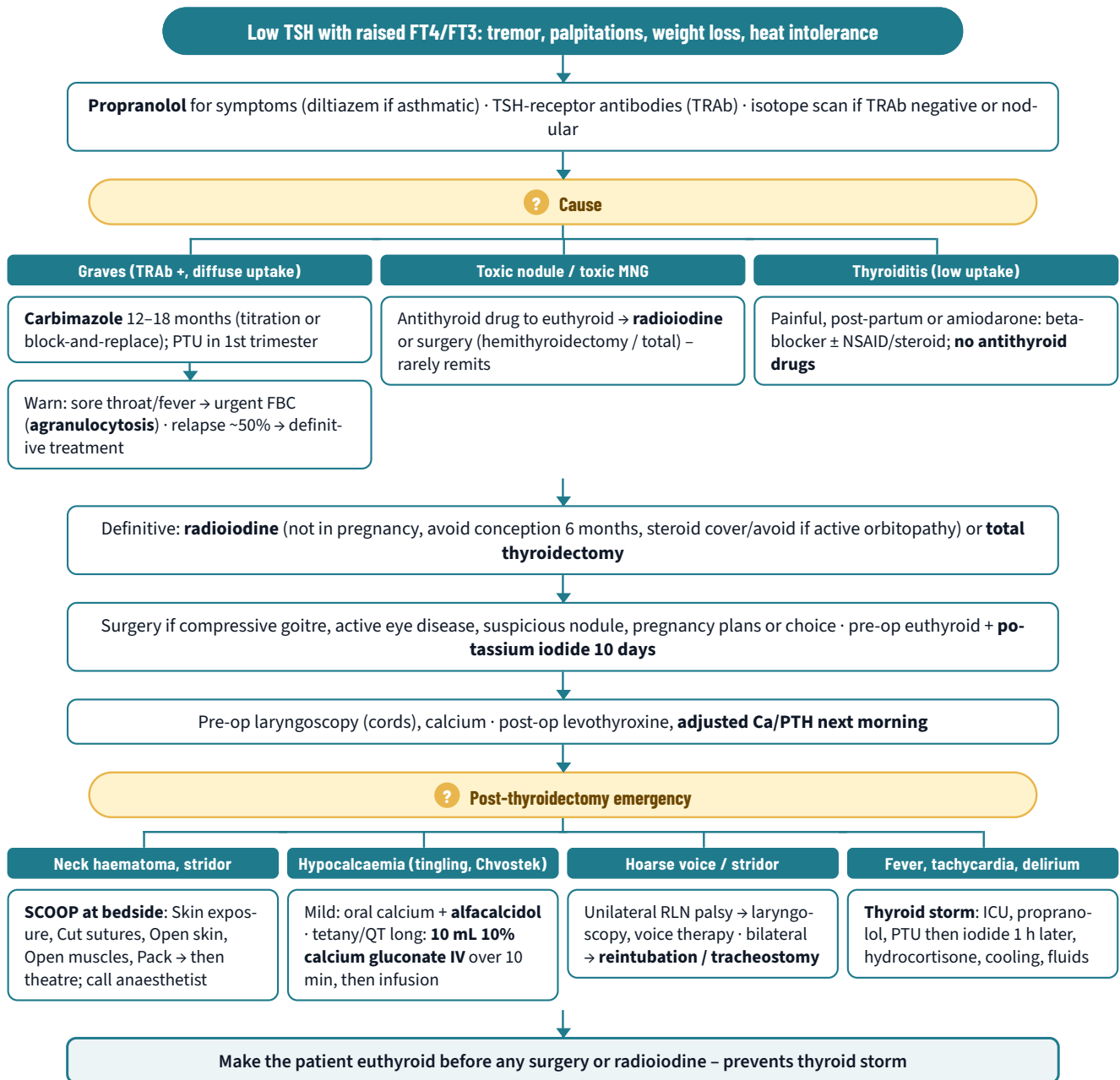
Thyroid nodule and goitre

BTA 2014 THYROID CANCER · NICE NG145 (2019) & NG230 (2022)



Thyrotoxicosis and safe thyroidectomy

NICE NG145 (2019) · ETA 2018 GRAVES · DAS/BAETS 2022 HAEMATOMA



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 20-year-old male presented with a tender and fixed swelling in the submandibular region associated with fever and malaise. Three days prior, a dental abscess was drained due to dental caries. What is the most likely diagnosis?
 - A Submandibular salivary gland tumor.
 - B Salivary gland retention cyst.
 - C Reactionary cervical lymphadenopathy.
 - D Branchial cyst infection.
 - E Infected thyroglossal cyst.
- 2 A 33-year-old pregnant woman presents with weight loss, palpitations, and increased anxiety over the past 3 months. Physical examination reveals a diffusely enlarged vascular thyroid gland, and mild exophthalmos. Thyroid function tests indicate elevated free T3 and T4 levels, with suppressed TSH. Thyroid scan: diffuse increased uptake of both lobes. Which of the following is the most appropriate initial treatment?
 - A Immediate surgical intervention with total thyroidectomy.
 - B Radioactive iodine (I^{131}) therapy.
 - C Propylthiouracil therapy.
 - D Carbimazole therapy.
 - E High-dose iodine supplementation.
- 3 A 35-year-old female presented with loss of weight and palpitation. Examination revealed fine tremor in the hands and bulging eyes. Which of the following antibodies is diagnostic of this condition?
 - A Thyroid-stimulating receptor antibodies
 - B Thyroglobulin antibodies
 - C Thyroid peroxidase antibodies
 - D Anticardiolipin antibodies
 - E Antimicrosomal antibodies
- 4 A 46-year-old professional singer had a left thyroid lobectomy for a large thyroid nodule with compression symptoms. Postoperatively, she notices that she is having trouble hitting some notes and projecting her voice. The ligation of which vessel most likely had resulted in this nerve injury?
 - A Inferior thyroid artery.
 - B Superior thyroid artery.
 - C Inferior thyroid vein.
 - D Middle thyroid vein.
 - E Thyrocervical trunk.

Answers 1-4

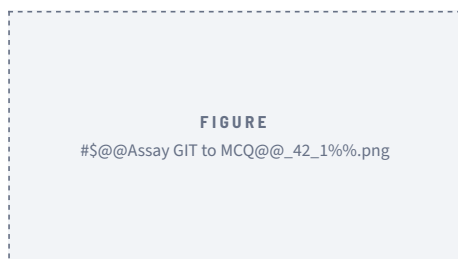
LOCAL EXAMS MCQs

- 1 **C Reactionary cervical lymphadenopathy.** — History of dental abscess drainage with consequent reactive lymphadenopathy.
- 2 **C Propylthiouracil therapy.** — Propylthiouracil is preferred in pregnancy for hyperthyroidism.
- 3 **A Thyroid-stimulating receptor antibodies** — TSH receptor antibodies confirm Graves' disease.
- 4 **B Superior thyroid artery.** — Ligation of the superior thyroid artery may injure the external laryngeal nerve, affecting voice.

12 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQs

Essay to MCQ MCQs

- 1** A 54-year-old female presents with a right preauricular lump slowly growing over 5 years. Workup is shown below. What are these studies?



- A** CT scan and ultrasound
B T1 and T2 MRI
C Plain X-ray and CT scan
D Ultrasound and Doppler study
E PET scan and MRCP
- 2** What are the findings?
- A** A cystic lesion in the parotid gland
B Enlargement of the right parotid gland
C A mass in the submandibular gland
D Normal parotid gland
E Lymphadenopathy in the neck
- 3** What is the next step to reach the diagnosis?
- A** Core needle biopsy
B Fine needle aspiration biopsy (FNAB) of the parotid gland
C Open surgical biopsy
D CT guided biopsy
E Excisional biopsy
- 4** The previous test results were inconclusive, and her surgeon advised her for surgery. What is the best surgical option?
- A** Total parotidectomy
B Radical parotidectomy
C Superficial parotidectomy
D Excisional biopsy
E Simple enucleation
- 5** Postoperatively the patient has difficulty in closing her right eye. Explain.
- A** This is due to damage to the optic nerve
B This is due to injury to the facial nerve during parotidectomy
C This is a complication of general anesthesia
D This is due to local infection
E This is due to a stroke

- 6** If the final histopathology was malignant, what is the next step in managing this lady?

A Observation
B Radiation therapy
C Radical parotidectomy since it is indicated in patients with clear histological evidence of malignancy
D Chemotherapy alone
E Repeat FNAB

- 7** Eight years later, the patient presents with a right submandibular mass. What is this lump and what is the cause?

A Benign lymphadenopathy due to infection
B Salivary gland stone
C Enlargement of the submandibular gland due to the development of a new tumor or spread of cancer cells from the previous parotid gland malignancy
D Cystic hygroma
E Thyroid nodule

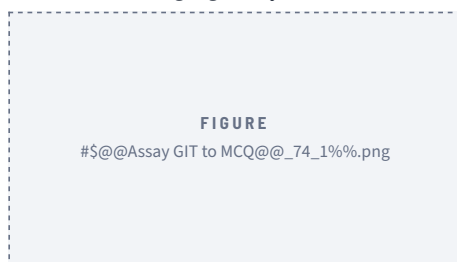
- 8** How can you prove the diagnosis?

A CT scan of the neck
B Ultrasound of the neck
C FNAB
D MRI
E PET scan

- 9** What is the treatment now?

A Radiation therapy only
B Chemotherapy only
C Surgical excision & chemotherapy to prevent recurrence
D Observation and follow-up
E Palliative care

- 10** A 31-year-old male is complaining of a painful swelling on the right side of his neck below the jaw that started a few days ago. He has experienced similar episodes in the past, which resolved spontaneously but were short in duration. The pain worsens when eating sour food. What is the imaging study and what are the findings?



- A CT scan showing a lymph node enlargement
- B Ultrasound showing a cystic lesion
- C Submandibular sialography reveals a filling defect in the submandibular gland, indicating a potential obstruction or stone in the duct
- D MRI of the neck showing a tumor
- E Plain X-ray of the neck

- 11** What is the treatment?
- A Immediate surgical removal
 - B High-dose steroids
 - C Conservative treatment if the sialolith is small, including removal through the duct orifice with bimanual palpation, heat application, and antibiotics; or sialendoscopy/surgical excision if necessary
 - D Chemotherapy
 - E Radiation therapy
- 12** After a while, the patient came with a weird smile on one side of his face, what is the diagnosis?
- A Parotid gland tumor
 - B Stroke
 - C Injury to the marginal mandibular branch of the facial nerve during submandibular gland excision
 - D Bell's palsy
 - E Temporomandibular joint disorder

Answers 1-12

ESSAY TO MCQ MCQS

- 1** **B T1 and T2 MRI** — The studies are T1 and T2 MRI sequences.
- 2** **B Enlargement of the right parotid gland** — Imaging shows enlargement of the parotid gland.
- 3** **B Fine needle aspiration biopsy (FNAB) of the parotid gland** — FNAB is the appropriate next step for tissue diagnosis.
- 4** **C Superficial parotidectomy** — Superficial parotidectomy is preferred when tests are inconclusive.
- 5** **B This is due to injury to the facial nerve during parotidectomy** — Facial nerve injury is a known risk during parotid surgery.
- 6** **C Radical parotidectomy since it is indicated in patients with clear histological evidence of malignancy** — Radical parotidectomy is indicated for malignant lesions.
- 7** **C Enlargement of the submandibular gland due to the development of a new tumor or spread of cancer cells from the previous parotid gland malignancy** — The mass suggests a new tumor or metastasis from prior malignancy.
- 8** **C FNAB** — FNAB is the investigation of choice to confirm the diagnosis.
- 9** **C Surgical excision & chemotherapy to prevent recurrence** — **UPDATED** Recurrent malignant salivary tumour: surgical excision with neck dissection, then adjuvant (chemo)radiotherapy; chemotherapy alone has a limited role.
- 10** **C Submandibular sialography reveals a filling defect in the submandibular gland, indicating a potential obstruction or stone in the duct** — Sialography demonstrates a filling defect consistent with a sialolith.
- 11** **C Conservative treatment if the sialolith is small, including removal through the duct orifice with bimanual palpation, heat application, and antibiotics; or sialendoscopy/surgical excision if necessary** — Treatment is based on stone size; small stones can be managed conservatively.
- 12** **C Injury to the marginal mandibular branch of the facial nerve during submandibular gland excision** — **UPDATED** An asymmetric smile after submandibular gland surgery is weakness of depressor anguli oris from injury to the marginal mandibular branch of the facial nerve.

Bailey & Love MCQs

- 1 Which lymph nodes represent the primary echelon of drainage for the thyroid gland?
 - A Retropharyngeal nodes.
 - B Jugulodigastric nodes.
 - C Central compartment nodes.
 - D Parapharyngeal nodes.
 - E Mediastinal nodes.
- 2 Which of the following conditions is usually present with hyperthyroidism?
 - A Graves' disease.
 - B Toxic nodule.
 - C Thyroid malignancy.
 - D De Quervain's thyroiditis.
 - E Toxic multinodular goitre.
- 3 What are the manifestations of thyrotoxicosis?
 - A Irritability.
 - B Hair loss.
 - C Muscle weakness and wasting.
 - D Hyperkinesias.
 - E Heart failure.
- 4 Which of the following are associated with Graves' disease?
 - A Pretibial myxoedema.
 - B Exposure keratitis.
 - C Optic neuropathy.
 - D Chemosis.
 - E Lymphoid hyperplasia.
- 5 Which of the following are true regarding medical therapy for thyrotoxicosis?
 - A Propranolol and nadolol reduce free T3 (fT3) and free T4 (fT4) levels.
 - B Antithyroid drugs most often cure thyrotoxicosis due to a toxic nodule.
 - C Carbimazole can be safely given in pregnancy and lactation.
 - D Agranulocytosis is an uncommon problem with antithyroid drugs.
 - E Patients with ophthalmopathy respond best to medical management.
- 6 Surgery is the preferred option of treatment in which of the following cases of thyrotoxicosis?
 - A A diffuse toxic goitre.
 - B Severe manifestations of Graves' ophthalmopathy.
 - C Pregnant mothers not adequately controlled with medications.
 - D Relapse of Graves' disease.
 - E Presence of local compressive symptoms.
- 7 Which of the following are indications for radioiodine treatment?
 - A Relapsed Graves' disease.
 - B Thyrotoxicosis in young children.
 - C Multinodular goitre.
 - D Severe ophthalmopathy.
 - E Pregnancy and lactation.
- 8 Which of the following are true in neonatal hypothyroidism?
 - A Endemic cretinism is due to iodine deficiency.
 - B Macroglossia is a clinical feature.
 - C Biochemical screening for hypothyroidism is carried out on all neonates in the United Kingdom.
 - D Radioactive iodine treatment is safe after the first trimester of pregnancy.
 - E Women on antithyroid drugs may give birth to a hypothyroid baby.
- 9 Myxoedema can present as the following:
 - A Complete heart block.
 - B Ventricular tachycardia.
 - C Mania.
 - D Shortness of breath.
 - E Sepsis.
- 10 Biochemically, hypothyroidism can be associated with the following:
 - A Hyponatraemia.
 - B Hyperlipidaemia.
 - C Thyroid peroxidase (TPO) autoantibodies.
 - D Raised TSH.
 - E Low fT3 and fT4.
- 11 Hypothyroidism could be a presenting feature in which of the following conditions?
 - A Sarcoidosis.
 - B Pituitary lesions.
 - C Benzodiazepine treatment.
 - D Thyroiditis.
 - E Iodine deficiency.
- 12 Which of the following statements regarding thyroid neoplasms are true?
 - A Papillary carcinoma is the most common.
 - B Men and women are equally affected.
 - C Thyroid cancer is most common after the age of 70 years.
 - D Medullary carcinoma originates from the C-cells.
 - E Anaplastic carcinoma is the least common.

- 13** Which of the following statements regarding thyroglobulin are true?
- A It is part of normal thyroid function tests.
 - B It is secreted only by cancerous cells.
 - C Autoantibodies may interfere with its levels.
 - D It is a sensitive marker of recurrence of differentiated thyroid cancer.
 - E Radioiodine treatment affects its concentration.
- 14** Regarding papillary and follicular cancer, which of the following statements are true?
- A Lymph node metastasis is more common in follicular cancer than in papillary cancer.
 - B In papillary cancer, central lymph node metastases are associated with decreased survival.
 - C Distant metastasis is more common in the follicular variety.
 - D Hurthle cell carcinoma is a type of follicular cancer.
 - E Papillary carcinoma is usually slow-growing.
- 15** Which of the following statements are true when a preoperative diagnosis of a differentiated thyroid cancer is made?
- A Whole-body MRI or CT imaging is essential.
 - B Lobectomy is recommended for tumours less than 1 cm.
 - C Elective lateral nodal dissection should be performed.
 - D Routine central compartment node clearance is not essential for small tumours.
 - E The absence of a rising thyroglobulin excludes recurrence.
- 16** Which of the following statements regarding medullary thyroid cancers are true?
- A It can present as multiple endocrine neoplasia.
 - B It is not TSH-dependent.
 - C It is associated with a poor prognosis.
 - D Diarrhoea is a common feature.
 - E High levels of calcitonin and carcinoembryonic antigen (CEA) are typical.
- 17** Regarding primary hyperparathyroidism, which of the following statements are true?
- A Primary hyperparathyroidism is usually sporadic.
 - B Hypercalcaemia triggers the release of parathyroid hormone (PTH).
 - C Familial hyperparathyroidism commonly presents as an adenoma.
 - D Familial hyperparathyroidism is mostly sporadic.
 - E Hyperparathyroidism can be associated with a pituitary adenoma.
- 18** Which of the following suggest a biochemical diagnosis of primary hyperparathyroidism?
- A Raised ionised calcium and suppressed PTH levels.
 - B Raised total calcium and elevated PTH levels.
 - C Low serum phosphate levels.
 - D Low urine calcium level.
 - E Raised ionised calcium and elevated PTH levels.
- 19** Which of the following are associated with hypercalcaemia?
- A Thyrotoxicosis.
 - B Chronic renal failure (CRF).
 - C Familial hypocalciuric hypercalcaemia (FHH).
 - D Sarcoidosis.
 - E Milk alkali syndrome.
- 20** Which of the following are operative indications for primary hyperparathyroidism?
- A Renal stones.
 - B Low bone density.
 - C Renal impairment.
 - D Serum calcium of 2.75 mmol/L.
 - E Young age.
- 21** Regarding operation for primary hyperparathyroidism, which of the following statements are correct?
- A Permanent hypoparathyroidism is likely.
 - B PTH can be measured intraoperatively.
 - C Recurrent laryngeal nerve damage can occur in about 5%–7% of cases.
 - D Endoscopic technique is the most favoured operative mode.
 - E Gamma probes can be used for exploratory purposes.
- 22** Which of the following statements are true regarding imaging of the parathyroid gland?
- A High-frequency ultrasound can identify nearly 75% of enlarged glands.
 - B Technetium-99m-labelled sestamibi scan identifies around 90% of enlarged glands.
 - C Single photon emission computed tomography (SPECT) scan can influence surgical approach.
 - D CT and MRI must be undertaken prior to first-time neck exploration.
 - E Adenomas weighing less than 500 mg show reduced concordance with imaging.
- 23** Which of the following statements are true regarding operative strategies for the parathyroid?
- A In a conventional approach, a transverse collar incision is made.
 - B In a targeted approach, a 2–3-cm incision is made over the site of the adenoma.
 - C In patients with four-gland disease, transcervical thymectomy is not recommended.
 - D In multiple endocrine neoplasia type 1 (MEN-1), total parathyroidectomy reduces the risk of recurrence.
 - E Preoperative imaging can identify nearly 10% of patients with mediastinal adenoma.

- 24** Which of the following statements are correct regarding hypoparathyroidism?
- A** Serum calcium must be checked within 24 hours of total thyroidectomy.
 - B** Serum calcium levels below 1.9 mmol/L can pose a medical emergency.
 - C** Chvostek's sign is suggestive of hypercalcaemia.
 - D** DiGeorge syndrome is a medical cause of hypoparathyroidism.
 - E** Cardiac arrhythmias can occur during an episode of hypocalcaemia.

- 25** Which of the following statements are true regarding the medical management of primary hypercalcaemia?
- A** Use of diuretics could reduce serum calcium levels by increasing their excretion via the kidneys.
 - B** Cinacalcet is a bisphosphonate used to reduce calcium levels.
 - C** Palmidronate is a bisphosphonate used to reduce calcium levels.
 - D** Intravenous saline is the first line of management in hypercalcaemia.
 - E** Serum calcium over 3.5 mmol/L is a medical emergency.

Answers 1-25

BAILEY & LOVE MCQS

- 1** **C** **Central compartment nodes.** — Primary drainage is to central compartment lymph nodes.
- 2** **A** **Graves' disease.** — Graves' disease is the most common cause of primary hyperthyroidism.
- 3** **ABCDE** **True: A, B, C, D, E** — Excess thyroid hormone causes neuropsychiatric symptoms, hair loss, muscle weakness and may lead to heart failure.
- 4** **ABCDE** **True: A, B, C, D, E** — Graves' disease involves eye signs including myxoedema, keratitis, optic neuropathy, chemosis and lymphoid changes.
- 5** **D** **Agranulocytosis is an uncommon problem with antithyroid drugs.** — Only agranulocytosis is true; the beta-blockers do not lower thyroid hormones, antithyroid drugs rarely cure toxic nodules and carbimazole is contraindicated in pregnancy.
- 6** **ABCDE** **True: A, B, C, D, E** — Surgery is indicated for diffuse goitre, severe ophthalmopathy, inadequate control in pregnancy, relapse and compressive symptoms.
- 7** **AC** **True: A, C** — Radioiodine treatment is indicated for relapsed Graves' disease and multinodular goitre only.
- 8** **ABCE** **True: A, B, C, E** — Screening is routine; iodine deficiency, macroglossia and maternal antithyroid drugs are implicated, while radioactive iodine is contraindicated.
- 9** **ABCDE** **True: A, B, C, D, E** — Severe hypothyroidism (myxoedema) may cause conduction abnormalities, arrhythmias, neuropsychiatric issues, dyspnea and sepsis.
- 10** **ABCDE** **True: A, B, C, D, E** — All of these biochemical features are seen in hypothyroidism.
- 11** **ABDE** **True: A, B, D, E** — It presents in sarcoidosis, pituitary lesions, thyroiditis and iodine deficiency; benzodiazepine treatment is not a cause.
- 12** **ADE** **True: A, D, E** — Papillary carcinoma is most common, medullary carcinoma arises from C-cells and anaplastic carcinoma is rare.
- 13** **CDE** **True: C, D, E** — Autoantibodies can interfere with its measurement, and thyroglobulin is used as a marker for recurrence and is affected by radioiodine therapy.
- 14** **CDE** **True: C, D, E** — **UPDATED** Follicular cancer more often has distant metastases, Hurthle cell carcinoma is a variant and papillary carcinoma is generally indolent.
- 15** **BD** **True: B, D** — For small differentiated thyroid cancers, a lobectomy with limited central node clearance is recommended.
- 16** **ABDE** **True: A, B, D, E** — Medullary carcinoma is part of MEN syndromes, is TSH-independent, may cause diarrhoea and is characterized by high calcitonin/CEA.
- 17** **AE** **True: A, E** — **UPDATED** It is usually sporadic and may be associated with pituitary adenomas; normally hypercalcaemia suppresses PTH.
- 18** **BCE** **True: B, C, E** — High total and ionised calcium with elevated PTH and low phosphate indicate primary hyperparathyroidism.
- 19** **ABCDE** **True: A, B, C, D, E** — **UPDATED** All are causes of hypercalcaemia; in chronic renal failure it arises with tertiary hyperparathyroidism or calcium/vitamin D therapy.
- 20** **ABCE** **True: A, B, C, E** — Operative treatment is indicated for renal stones, low bone density, renal impairment and young age; a mildly elevated calcium is not sufficient.
- 21** **ABE** **True: A, B, E** — There is a risk of permanent hypoparathyroidism, intraoperative PTH measurement is used and gamma probes may assist; nerve injury rates and endoscopic use are lower.
- 22** **ACE** **True: A, C, E** — Ultrasound has ~75% sensitivity, SPECT can guide surgery and small adenomas are harder to detect.
- 23** **ABD** **True: A, B, D** — Conventional and targeted incisions are correct and in MEN-1 total parathyroidectomy is advised; the other statements are false.
- 24** **ABDE** **True: A, B, D, E** — Calcium must be monitored post-thyroidectomy, levels below 1.9 mmol/L are dangerous, DiGeorge syndrome causes hypoparathyroidism and hypocalcaemia may lead to arrhythmias.
- 25** **CDE** **True: C, D, E** — Palmidronate, IV saline and calcium >3.5 mmol/L being an emergency are true; diuretics and cinacalcet are not used as stated.

Lange MCQs

- 1** A 19-year-old man presents with a penetrating wound to the right neck (zone II), with hematoma, odynophagia, and violation of the platysma. Vascular and aerodigestive injury is suspected. Which of the following is the best next step?

 - A** intubation and observation in the ICU
 - B** admission to the ICU for close observation without intubation
 - C** observation in the ICU only if carotid angiogram is normal
 - D** observation in the ICU only if carotid angiogram, contrast esophagram, and bronchoscopy are normal
 - E** neck exploration
- 2** A 39-year-old woman presents with a palpable 2.5-cm nodule in the left thyroid lobe. She has a history of low-dose chest irradiation in infancy. Which part of her history increases the risk for thyroid cancer?

 - A** age group of 20–40 years
 - B** female gender
 - C** low-dose irradiation during infancy
 - D** chronicity of the nodule
 - E** past history of Hashimoto's disease
- 3** For the same thyroid nodule, what is the most appropriate next step in management?

 - A** ultrasound of the neck
 - B** thyroid scintiscan
 - C** MRI of the neck
 - D** CT scan of the neck and chest
 - E** FNA of the nodule
- 4** A previously healthy 45-year-old man presents with a 9-month history of a slow-growing, painless right neck mass over the angle of the mandible. Which is the most likely diagnosis?

 - A** metastatic carcinoma
 - B** infectious parotitis
 - C** pleomorphic adenoma of the parotid
 - D** Hodgkin's disease
 - E** reactive cervical lymphatic hyperplasia
- 5** What is the best next step in management for the neck mass described above?

 - A** antibiotics
 - B** excisional biopsy
 - C** observation with re-evaluation in 2–4 weeks
 - D** superficial parotidectomy
 - E** chest x-ray
- 6** A 52-year-old woman presents with a calcium level of 11.5 mg/dL found on routine labs. What is the most appropriate next step?

 - A** Repeat her calcium level next year at her annual physical.
 - B** Reassure patient that a calcium level of 11.5 mg/dL is normal.
 - C** Repeat her calcium level and obtain a parathyroid hormone (PTH) level.
 - D** Obtain a CT scan of her neck.
 - E** Obtain positron emission tomography (PET) scan to look for bony metastases.
- 7** A 22-year-old man has a 2-cm thyroid nodule. His father, now deceased, had thyroid carcinoma and an abdominal tumour. Calcitonin is raised and fine-needle aspiration shows spindle cells with amyloid stroma. Which type of thyroid carcinoma is this?

 - A** Papillary carcinoma
 - B** Follicular carcinoma
 - C** Anaplastic carcinoma
 - D** Medullary carcinoma
 - E** Thyroid lymphoma
- 8** A 22-year-old man has a 2-cm thyroid nodule. His family history includes thyroid carcinoma and an unknown abdominal tumour in his father, now deceased; his BP is 145/90 mmHg. Fine-needle aspiration shows medullary thyroid carcinoma. Screening for which of the following must be undertaken before treatment of the thyroid carcinoma?

 - A** an adrenocorticotrophic hormone (ACTH)-producing tumor
 - B** Zollinger-Ellison syndrome
 - C** pituitary adenoma
 - D** pheochromocytoma
 - E** ganglioneuroma
- 9** Which of the following is the minimal treatment for the patient's thyroid carcinoma?

 - A** ipsilateral subtotal lobectomy
 - B** ipsilateral lobectomy and isthmectomy
 - C** subtotal thyroidectomy and isthmectomy
 - D** total thyroidectomy
 - E** total thyroidectomy and parathyroidectomy
- 10** Screening for abnormalities in which protooncogene should be offered to his siblings?

 - A** myc
 - B** ret
 - C** ras
 - D** myb
 - E** erbB

- 11** A 65-year-old woman evaluated for depression with a history of a recent kidney stone has hypercalcemia and elevated PTH. What is the most likely etiology?
- A parathyroid adenoma
 - B parathyroid hyperplasia
 - C multiple endocrine neoplasia type I (MEN I)
 - D multiple endocrine neoplasia type IIa (MEN IIa)
 - E parathyroid carcinoma

Answers 1-11

LANGE MCQs

- 1** **E neck exploration** — Zone II penetrating neck trauma with clinical signs mandates surgical exploration.
- 2** **C low-dose irradiation during infancy** — Past low-dose irradiation is a known risk factor for thyroid cancer.
- 3** **E FNA of the nodule** — FNA is the most accurate diagnostic test for thyroid nodules.
- 4** **C pleomorphic adenoma of the parotid** — A discrete, slow-growing neck mass in this location is most consistent with a pleomorphic adenoma.
- 5** **D superficial parotidectomy** — Definitive treatment of pleomorphic adenoma is superficial parotidectomy.
- 6** **C Repeat her calcium level and obtain a parathyroid hormone (PTH) level.** — Workup for hyperparathyroidism includes repeat calcium and PTH assay.
- 7** **D Medullary carcinoma** — Raised calcitonin with amyloid stroma is medullary carcinoma (parafollicular C cells); a familial pattern suggests MEN2 (RET).
- 8** **D pheochromocytoma** — **UPDATED** Familial medullary carcinoma suggests MEN2: exclude pheochromocytoma (plasma metanephrines) first, as an unrecognised tumour can cause a fatal crisis at thyroidectomy.
- 9** **D total thyroidectomy** — Total thyroidectomy is the treatment for medullary carcinoma.
- 10** **B ret** — RET gene mutations are associated with MEN II.
- 11** **A parathyroid adenoma** — A parathyroid adenoma is the common cause in older women.

9 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** An 18-year-old school student of East African origin presents with a 2-week history of lethargy, myalgia, rigours with hot flushes and intermittent pyrexia. Neither she nor any close contacts have been travelling recently. Examination reveals enlarged anterior and posterior chain lymph nodes in the neck. A blood film reveals the presence of Downey bodies, thrombocytosis and an increase in the lymphocyte count to 50% of total leucocytes. This girl is likely to have
- A Tuberculosis
 - B Toxoplasmosis
 - C Human immunodeficiency virus (HIV) infection
 - D Lymphoma
 - E Infectious mononucleosis
- 2** A 16-year-old girl presents with a smooth, round and painless lump in the midline of the neck, of which she feels very self-conscious. On examination, it is firm, transilluminates and is painless to touch. In addition, it moves up when the patient is asked to take a sip of water. This is likely to be a
- A Papillary carcinoma of the thyroid
 - B Goitre
 - C Thyroglossal cyst
 - D Lingual thyroid
 - E Sebaceous cyst
- 3** A 45-year-old female business executive presents with a swelling in the midline of the neck and features of hyperthyroidism secondary to Graves' disease. Following a course of antithyroid medication, a subtotal thyroidectomy is performed. Raised titres of which immunoglobulin would be expected in this individual?
- A Anti-thyroglobulin antibody
 - B Rheumatoid factor
 - C Antinuclear antibody
 - D Antineutrophil cytoplasmic antibody
 - E Anti-thyroid-stimulating hormone receptor antibody

- 4** You are asked to assess a 25-year-old patient who returned from theatre 2 hours ago following a thyroidectomy for a large, hyperplastic goitre and is now complaining of difficulty swallowing sips of water. On examination you note that she is very short of breath with a respiratory rate of 30 breaths/min, using her accessory muscles of respiration and only able to answer your questions in two or three words. In addition, there appears to be a fluctuant mass in the midline of the neck underlying the surgical clips. Immediate management of this patient would be
- A High-flow oxygen via Hudson mask
 - B Removal of surgical clips at the bedside
 - C Intravenous access with two large-bore cannulae and fluid resuscitation
 - D Removal of surgical clips in theatre under general anaesthesia
 - E Call your senior and wait for him/her to remove the clips
- 5** A 28-year-old secretary presents with a lump in the midline of the neck that has grown progressively larger over the past several months. On examination, there is a palpable lymph node in the left submandibular region. An aspirate is taken, confirming a malignant cancer of the thyroid gland. The origin of this is most likely to be
- A Follicular
 - B Anaplastic
 - C Medullary
 - D Lymphoma
 - E Papillary
- 6** A 37-year-old patient is diagnosed with medullary carcinoma of the thyroid gland. The concentration of which electrolyte could be reduced in this patient?
- A Sodium
 - B Potassium
 - C Chloride
 - D Calcium
 - E Magnesium
- 7** A 68-year-old man presents with a 10-week history of a rapidly growing swelling in the midline of the neck. Prior to this problem developing he has been in excellent health. Over the past 10 weeks, he has begun to feel more worn out and is unable to complete his daily 3-mile walk without getting short of breath, which he was previously able to manage comfortably. His wife also reports that his voice has become increasingly hoarse in the last 2 weeks. The presentation is likely to be consistent with
- A Recurrent laryngeal nerve damage
 - B Tracheal trauma
 - C Medullary carcinoma of the thyroid
 - D Anaplastic carcinoma of the thyroid
 - E Papillary carcinoma of the thyroid
- 8** A 59-year-old fisherman presents with longstanding weight loss, anorexia and lethargy. Physical examination reveals a palpable swelling in the left supraclavicular fossa. A disseminated adenocarcinoma of the stomach is diagnosed following a gastroscopy and computed tomography scan of the thorax and abdomen. This patient has presented with
- A Virchow's node
 - B Battle's sign
 - C Cloquet's node
 - D Troisier's sign
 - E Trousseau's sign
- 9** Which of the following structures does not form a border of the anterior triangle of the neck?
- A Midline of the neck
 - B Anterior border of sternocleidomastoid muscle
 - C Lower border of the mandible
 - D Investing fascia
 - E Middle third of the clavicle

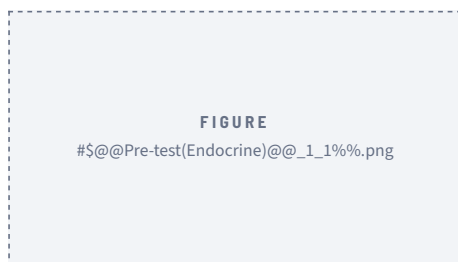
Answers 1-9

- 1** **E Infectious mononucleosis** — Infectious mononucleosis is caused by Epstein-Barr virus, resulting in reactive lymphocytosis with Downey bodies; diagnosis is usually confirmed by a positive Monospot test.
- 2** **C Thyroglossal cyst** — Thyroglossal cysts are remnants of the thyroglossal duct that move with swallowing and are managed by surgical excision.
- 3** **E Anti-thyroid-stimulating hormone receptor antibody** — Graves' disease is associated with increased anti-TSH receptor antibodies.
- 4** **B Removal of surgical clips at the bedside** — Airway compromise from a tension hematoma post-thyroidectomy necessitates immediate bedside clip removal.
- 5** **E Papillary** — Papillary carcinoma is the most common thyroid cancer, often multifocal and typically spreads via lymphatics.
- 6** **D Calcium** — Medullary carcinoma produces calcitonin from C cells, which lowers serum calcium levels.
- 7** **D Anaplastic carcinoma of the thyroid** — Anaplastic thyroid carcinoma is aggressive, with rapid growth and compressive symptoms such as hoarseness and breathlessness.
- 8** **D Troisier's sign** — An enlarged left supraclavicular node (Troisier's sign) suggests metastatic spread from an abdominal malignancy such as gastric adenocarcinoma.
- 9** **E Middle third of the clavicle** — The middle third of the clavicle is not included as a border of the anterior triangle of the neck.

19 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1** A 45-year-old woman complains to her primary care physician of nervousness, sweating, tremulousness, and weight loss. The thyroid scan shown here exhibits a pattern that is most consistent with which of the following disorders?



- A** Hypersecreting thyroid nodule
B Graves disease
C Papillary carcinoma of the thyroid
D Parathyroid adenoma
E Hashimoto's thyroiditis
- 2** A 35-year-old woman with a history of previous right thyroidectomy for a benign thyroid nodule now undergoes completion thyroidectomy for a suspicious thyroid mass. Several hours postoperatively, she develops progressive swelling under the incision, stridor, and difficulty breathing. Orotracheal intubation is successful. Which of the following is the most appropriate next step?
- A** Fiberoptic laryngoscopy to rule out bilateral vocal cord paralysis
B Administration of intravenous calcium
C Administration of broad-spectrum antibiotics and debridement of the wound
D Wound exploration
E Administration of high-dose steroids and antihistamines
- 3** A 52-year-old woman sees her physician with complaints of fatigue, headache, flank pain, hematuria, and abdominal pain. She undergoes a sestamibi scan that demonstrates persistent uptake in the right superior parathyroid gland at 2 hours. Which of the following laboratory values is most suggestive of her diagnosis?
- A** Elevated serum phosphate levels
B Serum alkaline phosphatase above 120 IU/L
C Serum calcium above 11 mg/dL
D Elevated vitamin D levels
E Parathyroid hormone levels below 5 pmol/L

- 4** A 55-year-old woman presents with a slow-growing painless mass in her neck. A complete neck ultrasound demonstrates a 1-cm nodule in the right thyroid without masses in the contralateral lobe or lymph node metastasis in the central and lateral neck compartments. A fine-needle aspiration of the nodule shows a well-differentiated papillary carcinoma. With regards to this patient, which of the following is associated with a poor prognosis?

- A** Age
B Sex
C Grade of tumor
D Size of tumor
E Lymph node status

- 5** A 55-year-old woman presents with a 6-cm right thyroid mass and palpable cervical lymphadenopathy. Fine-needle aspiration (FNA) of one of the lymph nodes demonstrates the presence of thyroid tissue. Which of the following best describes the management of this thyroid disorder?

- A** Screening for pancreatic endocrine neoplasms and hypercalcemia
B Total thyroidectomy with modified radical neck dissection
C Total thyroidectomy with frozen section intraoperatively, with modified radical neck dissection reserved for patients with extra-capsular invasion
D Right thyroid lobectomy followed by iodine 131 (131I) therapy
E Right thyroid lobectomy

- 6** A 53-year-old woman presents with complaints of weakness, anorexia, malaise, constipation, and back pain. While being evaluated, she becomes somewhat lethargic. Laboratory studies include a normal chest x-ray, serum albumin 3.2 mg/dL, serum calcium 14 mg/dL, serum phosphorus 2.6 mg/dL, serum chloride 108 mg/dL, blood urea nitrogen (BUN) 32 mg/dL, and creatinine 2.0 mg/dL. Which of the following is the most appropriate initial management?

- A** Intravenous normal saline infusion
B Administration of thiazide diuretics
C Administration of intravenous phosphorus
D Use of mithramycin
E Neck exploration and parathyroidectomy

- 7** Which of the following patients with primary hyperparathyroidism should undergo parathyroidectomy?
- A A62-year-old asymptomatic woman
 - B A54-year-old woman with fatigue and depression
 - C A42-year-old woman with a history of kidney stones
 - D A59-year-old woman with mildly elevated 24-hour urinary calcium excretion
 - E A60-year-old woman with mildly decreased bone mineral density measured at the hip of less than 2 standard deviations below peak bone density
- 8** A29-year-old woman with a history of difficulty becoming pregnant presents to her primary care physician and is diagnosed with Graves disease on iodine uptake scan; her thyrotropin (TSH) level is markedly suppressed and her free thyroxine (T4) level is elevated. She desires to conceive as soon as possible and elects to undergo thyroidectomy. After she is rendered euthyroid with medications preoperatively, which of the following management strategies should also be employed to reduce the risk of developing thyroid storm in the operating room?
- A Drops of Lugol iodine solution daily beginning 10 days preoperatively
 - B Preoperative treatment with phenoxybenzamine for 3 weeks
 - C Preoperative treatment with propranolol for 1 week
 - D Twenty-four hours of corticosteroids preoperatively
 - E No other preoperative medication is required
- 9** A30-year-old woman presents with hypertension, weakness, bone pain, and a serum calcium level of 15.2 mg/dL. Plain films of bilateral hands show osteitis fibrosa cystica. Which of the following is the most likely cause of these findings?
- A Sarcoidosis
 - B Vitamin D intoxication
 - C Paget disease
 - D Metastatic carcinoma
 - E Primary hyperparathyroidism
- 10** A35-year-old woman presents with a serum calcium level of 15.2 mg/dL and an elevated parathyroid hormone level. Following correction of the patient's hypercalcemia with hydration and furosemide, which of the following is the best therapeutic approach?
- A Administration of steroids
 - B Radiation treatment to the neck
 - C Neck exploration and resection of all 4 parathyroid glands
 - D Neck exploration and resection of a parathyroid adenoma
 - E Avoidance of sunlight, vitamin D, and calcium-containing dairy products
- 11** A58-year-old man presents with tachycardia, fever, confusion, and vomiting. Workup reveals markedly elevated (triiodothyronine) T3 and (thyroxine) T4 levels. He is diagnosed as having a thyroid storm. Which of the following is the most appropriate next step in the management of this patient?
- A Emergent subtotal thyroidectomy
 - B Emergent total thyroidectomy
 - C Emergent hemodialysis
 - D Administration of fluid, antithyroid drugs, β -blockers, iodine solution, and steroids
 - E Emergent radiation therapy to the neck
- 12** A36-year-old woman, 20 weeks pregnant, presents with a 1.5-cm right thyroid mass. FNA is consistent with a papillary neoplasm. The mass is cold on scan and solid on ultrasound. Which of the following methods of treatment is contraindicated in this patient?
- A Right thyroid lobectomy
 - B Subtotal thyroidectomy
 - C Total thyroidectomy
 - D Total thyroidectomy with lymph node dissection
 - E ^{131}I radioactive ablation of the thyroid gland
- 13** A63-year-old woman notices lumps on both sides of her neck. A fine-needle aspirate is nondiagnostic, and she undergoes total thyroidectomy. Final pathology reveals a 4-cm Hürthle cell carcinoma. Which of the following is the most appropriate postsurgical management of this patient?
- A No further therapy is indicated
 - B Chemotherapy
 - C External beam radiotherapy
 - D Radioiodine ablation
 - E Chemotherapy, external beam radiotherapy, and radioiodine ablation
- 14** A51-year-old man presents with a 2-cm left thyroid nodule. Thyroid scan shows a cold lesion. FNA cytology demonstrates follicular cells. Which of the following is the most appropriate initial treatment of this patient?
- A External beam radiation to the neck
 - B Multidrug chemotherapy
 - C TSH suppression by thyroid hormone
 - D Prophylactic neck dissection is indicated along with a total thyroidectomy
 - E Thyroid lobectomy

- 15** A 52-year-old woman presents with a history of abdominal pain and constipation. As part of her workup, she was noted to have a calcium level of 12 mg/dL. For each of the following test results, choose the most appropriate etiology of hypercalcemia. Each lettered option may be used once, more than once, or not at all. Genetic testing showing mutation in a tumor suppressor gene on chromosome 11q12-13
- A Parathyroid adenoma
 - B Parathyroid hyperplasia
 - C Familial hypocalciuric hypercalcemia
 - D Parathyroid carcinoma
 - E Paraneoplastic syndrome
- 16** Low 24-hour urinary calcium level (<100 mg/d)
- A Parathyroid adenoma
 - B Parathyroid hyperplasia
 - C Familial hypocalciuric hypercalcemia
 - D Parathyroid carcinoma
 - E Paraneoplastic syndrome
- 17** Serum calcium level 15 mg/dL and parathyroid hormone level 5 times normal
- A Parathyroid adenoma
 - B Parathyroid hyperplasia
 - C Familial hypocalciuric hypercalcemia
 - D Parathyroid carcinoma
 - E Paraneoplastic syndrome
- 18** Very low level of intact PTH
- A Parathyroid adenoma
 - B Parathyroid hyperplasia
 - C Familial hypocalciuric hypercalcemia
 - D Parathyroid carcinoma
 - E Paraneoplastic syndrome
- 19** What is the necessary drop in intraoperative PTH during parathyroidectomy?
- A 30%
 - B 10%
 - C At least 50%
 - D 90%
 - E 20%

Answers 1-19

PRETEST MCQs

- 1** **A Hypersecreting thyroid nodule** — The thyroid scan shows a single focus of increased isotope uptake, consistent with hypersecreting thyroid nodule.
- 2** **D Wound exploration** — The clinical presentation is consistent with a wound hematoma and necessitates exploration of the wound, drainage.
- 3** **C Serum calcium above 11 mg/dL** — Elevated parathyroid hormone (PTH) levels in conjunction with elevated calcium levels are diagnostic for hyperparathyroidism.
- 4** **A Age** — **UPDATED** In differentiated thyroid cancer, age is the strongest prognostic factor (AJCC 8th: ≥55 years is high risk); a 1-cm node-negative tumour otherwise has an excellent outlook.
- 5** **B Total thyroidectomy with modified radical neck dissection** — Treatment of high-risk papillary carcinomas consists of total thyroidectomy with modified radical neck dissection.
- 6** **A Intravenous normal saline infusion** — The initial management of hypercalcemia involves intravenous normal saline to restore intravascular volume and promote renal calcium excretion.
- 7** **C A42-year-old woman with a history of kidney stones** — Patients with symptomatic primary hyperparathyroidism, such as those with kidney stones, should undergo parathyroidectomy.
- 8** **A Drops of Lugol iodine solution daily beginning 10 days preoperatively** — Drops of Lugol iodine solution daily beginning 10 days preoperatively reduce thyroid hormone synthesis and vascularity, lowering the risk of thyroid storm.
- 9** **E Primary hyperparathyroidism** — Osteitis fibrosa cystica is a condition associated with primary hyperparathyroidism, characterized by hypercalcemia and high PTH levels.
- 10** **D Neck exploration and resection of a parathyroid adenoma** — Primary hyperparathyroidism is typically treated surgically with neck exploration and resection of the affected parathyroid gland (adenoma).
- 11** **D Administration of fluid, antithyroid drugs, β-blockers, iodine solution, and steroids** — Thyroid storm requires immediate management with fluid replacement, antithyroid drugs, β-blockers, iodine solutions, and steroids.
- 12** **E 131I radioactive ablation of the thyroid gland** — Radioactive iodine (131I) is contraindicated in pregnancy as it can cross the placenta and harm the fetal thyroid.
- 13** **D Radioiodine ablation** — The treatment of Hürthle cell carcinoma involves surgical resection followed by radioiodine ablation to reduce recurrence risk.
- 14** **E Thyroid lobectomy** — For follicular neoplasms, thyroid lobectomy is the initial treatment to assess capsular or vascular invasion on histology, which determines malignancy.
- 15** **B Parathyroid hyperplasia** — Mutation in tumor suppressor gene on 11q12-13 is associated with parathyroid hyperplasia.
- 16** **C Familial hypocalciuric hypercalcemia** — Low 24-hour urinary calcium is indicative of familial hypocalciuric hypercalcemia (FHH).
- 17** **D Parathyroid carcinoma** — High calcium and markedly elevated PTH suggest parathyroid carcinoma.
- 18** **E Paraneoplastic syndrome** — Low PTH in hypercalcemia suggests a paraneoplastic syndrome, likely PTHrP production.
- 19** **C At least 50%** — **UPDATED** A fall in PTH of at least 50% from the highest pre-excision value 10 minutes after gland removal (Miami criterion) predicts cure.

15 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 76-year-old man presents with a rapidly growing lump in his neck causing difficulty swallowing. Examination reveals a 4 cm hard mass in the front of the neck fixed to the overlying skin. What is the most likely diagnosis?
 - A Anaplastic carcinoma
 - B Follicular carcinoma
 - C Graves disease
 - D Papillary carcinoma
 - E Toxic multinodular goitre
- 2 A 12-year-old girl notices a painless swelling on the left side of her neck after a recent cold. On examination, it is smooth, fluctuant, non-tender, anterior to the sternocleidomastoid. What is the most likely diagnosis?
 - A Branchial cyst
 - B Cervical rib
 - C Cystic hygroma
 - D Sternocleidomastoid tumour
 - E Thyroglossal cyst
- 3 A 41-year-old woman presents with a 1-week history of tiredness, muscle aches, neck pain, and restlessness. On examination, she has a smooth enlarged thyroid and a heart rate of 120/min. What is the most likely diagnosis?
 - A De Quervain thyroiditis
 - B Hashimoto thyroiditis
 - C Pharyngitis
 - D Riedel thyroiditis
 - E Thyroglossal cyst
- 4 A 17-year-old boy has paroxysmal sweating, anxiety, palpitations, and a history of a thyroid tumour. He is tall and thin with long fingers. What is the most likely diagnosis?
 - A Carcinoid syndrome
 - B Multiple endocrine neoplasia type I
 - C Multiple endocrine neoplasia type IIa
 - D Multiple endocrine neoplasia type IIb
 - E Nelson syndrome
- 5 A 12-month-old girl has a large mass in the right side of her neck noticed for months. It is fleshy, compressible, and lies posterior to the sternocleidomastoid. What is the most likely diagnosis?
 - A Branchial cyst
 - B Cystic hygroma
 - C Pleomorphic adenoma
 - D Sternocleidomastoid tumour
 - E Thyroglossal cyst
- 6 A 21-year-old man collapses after dizziness and right arm pain while sanding a table. A bony lump is palpable in the right side of his neck. Neurological and vascular exams are normal now. Most likely diagnosis?
 - A Cerebrovascular attack
 - B Deep venous thrombosis
 - C Embolus
 - D Subclavian steal syndrome
 - E Transient ischaemic attack
- 7 A 72-year-old man presents with difficulty swallowing and regurgitation of food. He has also had several chest infections over the last few months. On examination, you see a soft swelling in the anterior triangle of the neck. On palpation, the swelling makes a gurgling sound. What is the most likely cause of his dysphagia?
 - A Carotid artery aneurysm
 - B Enlarged lymph node
 - C Laryngocoele
 - D Pharyngeal pouch
 - E Thyroglossal cyst
- 8 A 42-year-old man presents to the GP with a slowly enlarging mass in the left side of his neck which has been present for 3 months. On examination, the mass is 2 cm in diameter, non-tender, pulsatile and can be moved from side to side, but not up or down. What is the most likely diagnosis?
 - A Branchial cyst
 - B Cervical rib
 - C Chemodectoma
 - D Pleomorphic adenoma
 - E Virchow node
- 9 A 36-year-old woman presents to the GP with a 2-month history of tremors. This is interfering with her work as a typist and the worry has caused her to lose over a stone in weight over the last few weeks. On examination, the thyroid appears normal although a bruit can be heard overlying it. What is the most likely diagnosis?
 - A De Quervain thyroiditis
 - B Graves disease
 - C Thyroid adenoma
 - D Thyroid storm
 - E Toxic multinodular goitre

- 10** A 58-year-old woman is brought to the emergency department having been found unresponsive by her husband at home. On examination, you note that she is overweight with coarse features. Observations include heart rate 42/min, blood pressure 100/68 mmHg and temperature 34.6°C. Which of the following would be the most appropriate next step in your management?
- A Intravenous thyroxine
 - B Intravenous propranolol
 - C Oral propylthiouracil
 - D Oral thyroxine
 - E Thyroidectomy
- 11** A 32-year-old man presents to the GP with multiple lesions on his trunk which have appeared over the last few months. Each lesion is raised, firm and rubbery, and causes a tingling sensation when touched. Which of the following is the most likely diagnosis?
- A Ganglion
 - B Granuloma annulare
 - C Kaposi sarcoma
 - D Neurofibroma
 - E Sebaceous cyst
- 12** A 42-year-old woman presents to the emergency department with swelling and redness in her left cheek and malaise which has been present for 1 day. On examination, the area of erythema is well demarcated and is hot to touch. Her observations include: heart rate 106/min, blood pressure 108/68 mmHg and temperature 38.6°C. Which of the following is the best course of management?
- A 5-Fluorouracil
 - B Analgesia
 - C Intravenous antibiotics
 - D Oral antibiotics
 - E Surgical debridement
- 13** A 46-year-old woman attends her general practice complaining of low mood and tiredness. She has had this for a few months and has had to take time off work. On further questioning she admits to being constipated and to having gained 2 stone in weight over the previous month. On examination, no abnormality is apparent in the neck and no lymphadenopathy is palpable. What is the most likely diagnosis?
- A Haemorrhage into a cyst
 - B Hashimoto thyroiditis
 - C Medullary carcinoma
 - D Myxoedema coma
 - E Primary myxoedema
- 14** You are called to see a 52-year-old woman who has just undergone a thyroidectomy. She complains of a 20-minute history of neck pain and shortness of breath. A rapidly expanding mass is noted around the operation site. Which of the following would you do next?
- A Anaesthetic review
 - B Bedside opening of incision
 - C Book next available theatre slot
 - D Intravenous antibiotics
 - E Laryngoscopy
- 15** A 6-year-old girl is brought to the GP by her father with a smooth lump in the midline of her neck. She is otherwise well. On examination, there is a 1 cm painless, fluctuant swelling which moves upwards on swallowing and on tongue protrusion. What is the most likely diagnosis?
- A Branchial cyst
 - B Chemodectoma
 - C Cystic hygroma
 - D Thyroid adenoma
 - E Thyroglossal cyst

Answers 1-15

GET AHEAD MCQS

- 1** **A Anaplastic carcinoma** — A rapidly enlarging, fixed thyroid mass in an elderly patient suggests anaplastic carcinoma.
- 2** **A Branchial cyst** — A painless, fluctuant swelling in the anterior triangle consistent with a branchial cyst.
- 3** **A De Quervain thyroiditis** — A painful subacute thyroiditis with transient hyperthyroid features, often viral in origin.
- 4** **D Multiple endocrine neoplasia type IIb** — MEN IIb: medullary thyroid carcinoma, pheochromocytoma, marfanoid habitus, mucosal neuromas.
- 5** **B Cystic hygroma** — A congenital lymphatic lesion in the posterior triangle of the neck, often transilluminates.
- 6** **D Subclavian steal syndrome** — Cervical rib compresses subclavian artery causing vertebral artery flow reversal under exertion.
- 7** **D Pharyngeal pouch** — A pharyngeal (Zenker) pouch can cause dysphagia, regurgitation, halitosis, and a gurgling neck swelling.
- 8** **C Chemodectoma** — Carotid body tumors (chemodectomas) are pulsatile, move side to side, and lie at the bifurcation.
- 9** **B Graves disease** — An autoimmune hyperthyroidism with a diffuse goitre and bruit is classic for Graves disease.
- 10** **A Intravenous thyroxine** — Myxoedema coma requires urgent IV thyroxine and supportive care.
- 11** **D Neurofibroma** — Benign nerve sheath tumours that can present as firm, rubbery skin lesions.
- 12** **C Intravenous antibiotics** — Erysipelas is a superficial streptococcal infection often managed with IV antibiotics if severe.
- 13** **E Primary myxoedema** — Spontaneous atrophic hypothyroidism with typical features of weight gain, constipation, low mood.
- 14** **B Bedside opening of incision** — Expanding haematoma threatening the airway requires immediate release of sutures.
- 15** **E Thyroglossal cyst** — Midline cystic swelling moves on tongue protrusion, consistent with thyroglossal duct remnant.

Endocrine Surgery

Adrenal, pituitary and pancreatic endocrine tumours, MEN syndromes and the surgical endocrine work-up.

20

MCQs

0

EMQ SCENARIOS

4

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

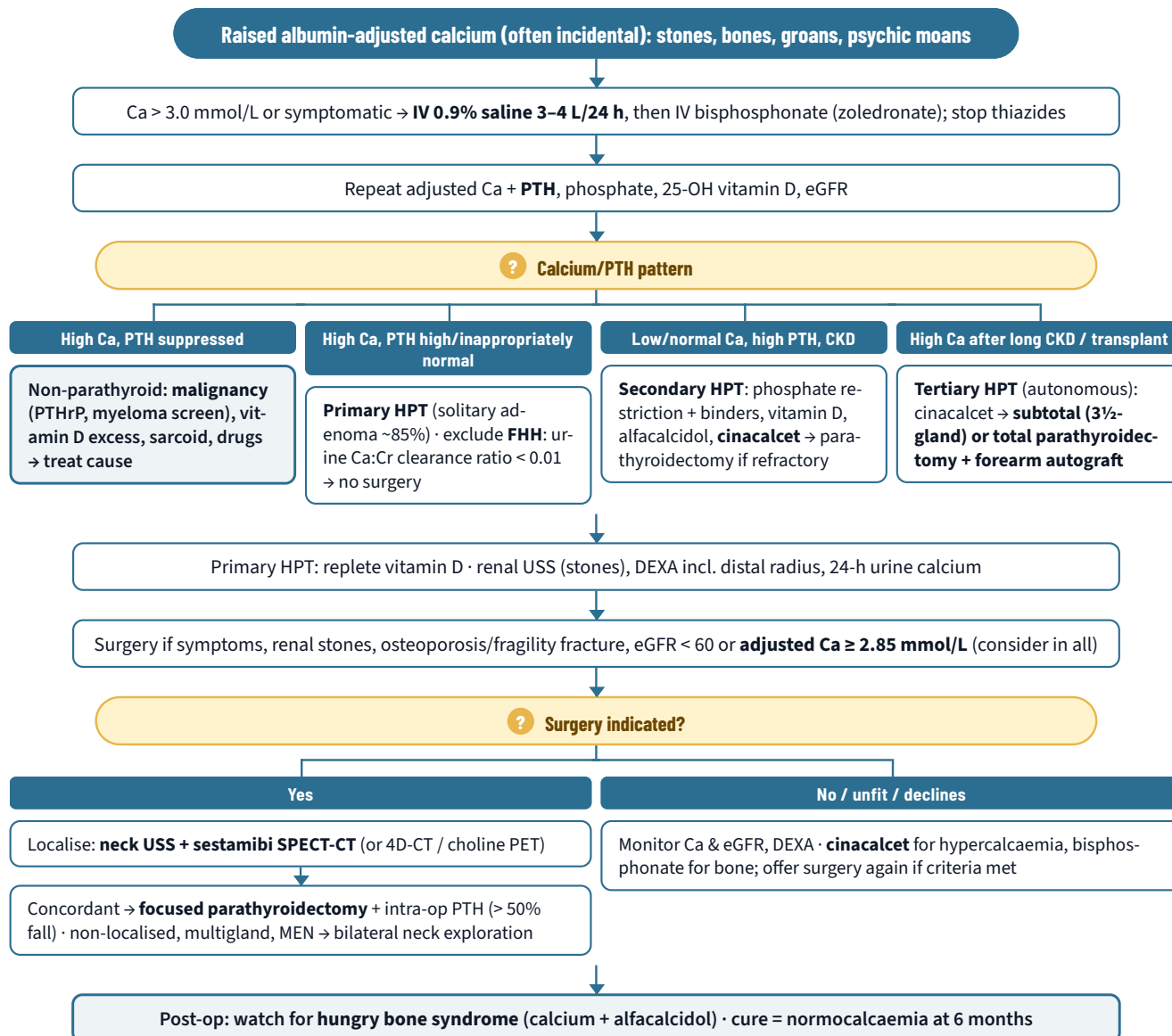
01	Management flowcharts	3 charts	page 32
02	Bailey & Love MCQs	5 Q	page 35
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02

Step by step

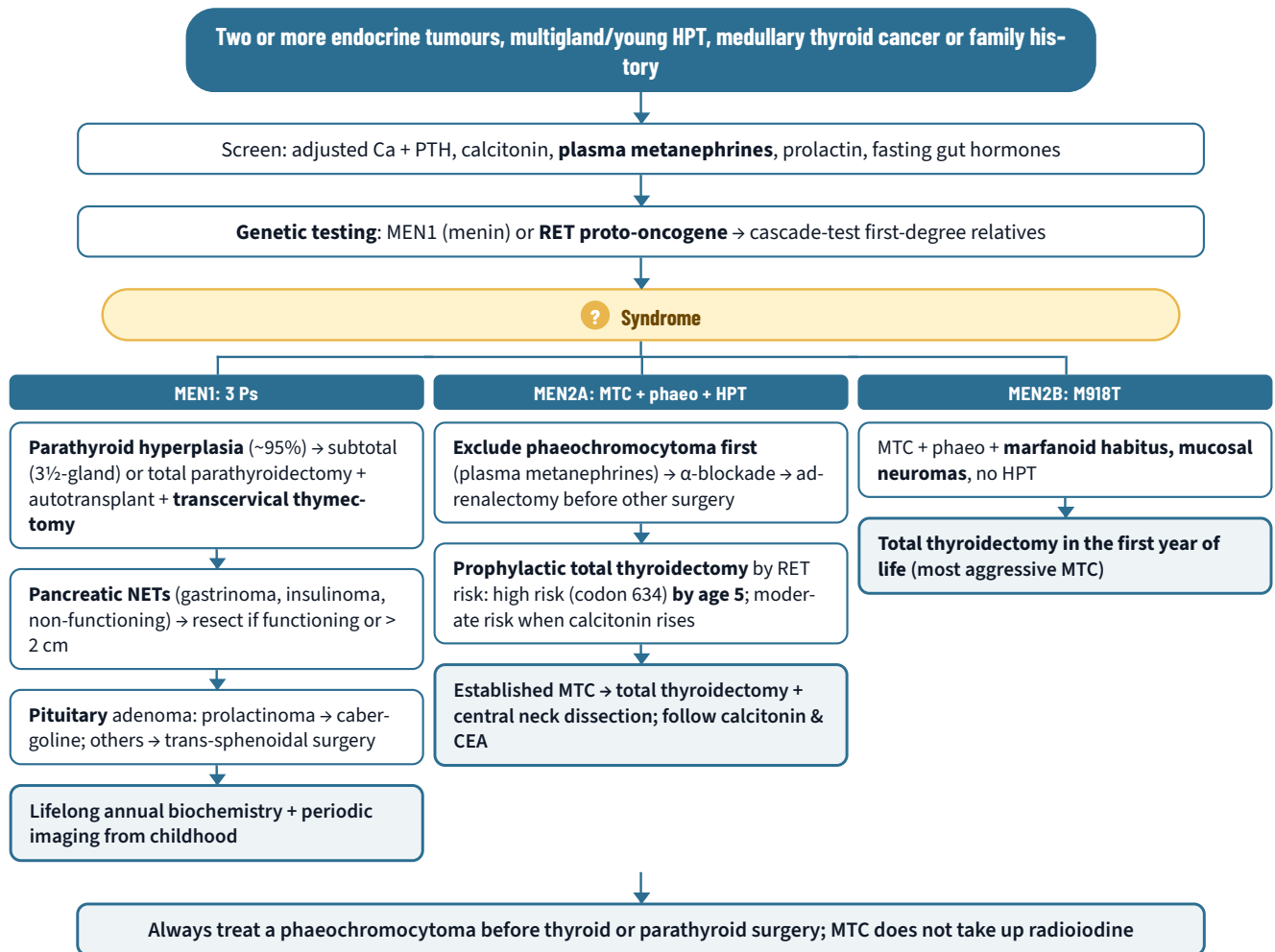
Hypercalcaemia and hyperparathyroidism

NICE NG132 (2019) • 5TH INTERNATIONAL WORKSHOP PHPT 2022 • KDIGO 2017 CKD-MBD



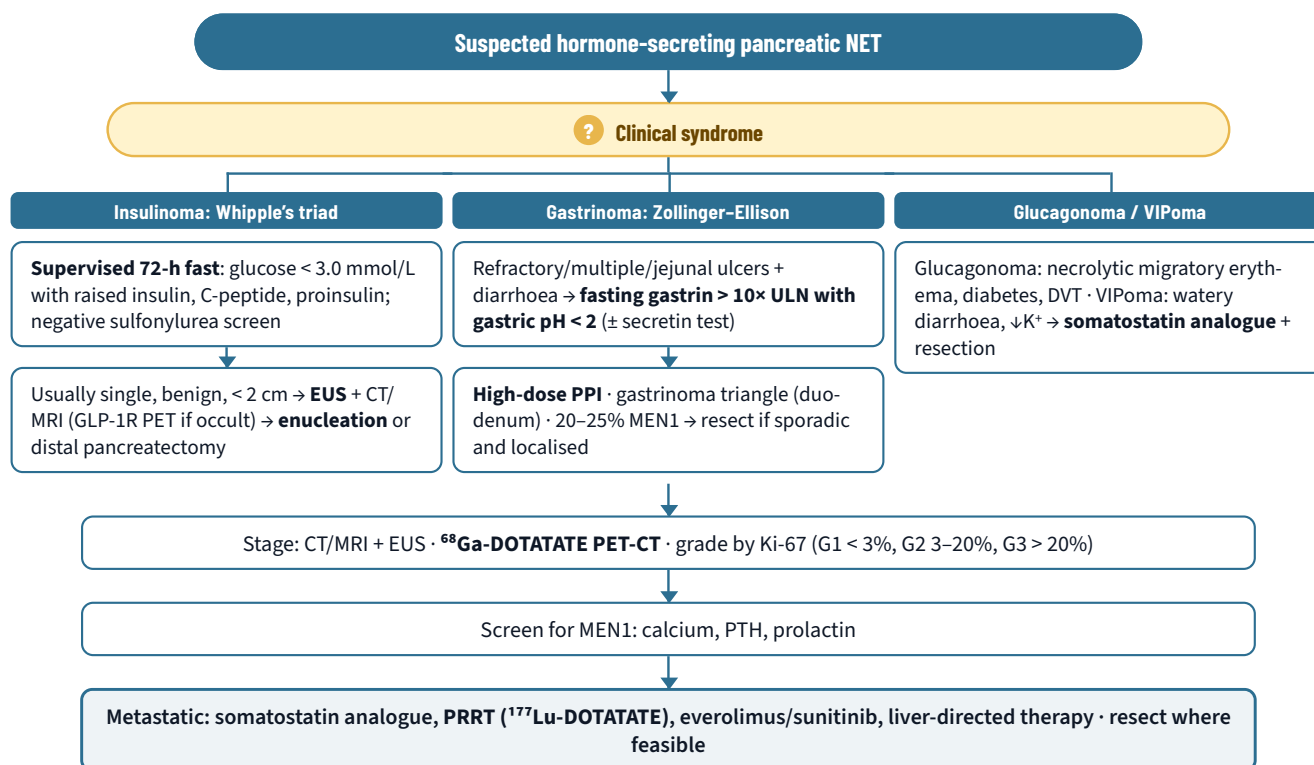
Multiple endocrine neoplasia (MEN)

ATA 2015 MEDULLARY THYROID CANCER · THAKKER 2012 MEN1 · BAILEY & LOVE 28E



Functioning pancreatic neuroendocrine tumour

ENETS 2023 FUNCTIONING PNET · ESMO 2020 GEP-NEN



Bailey & Love MCQs

- 1 A 60-year-old retired nurse presents with weight loss, fainting episodes, and tiredness. A blood glucose is 2.5 mmol/L and an insulinoma is suspected. Which of the following tests are appropriate?
 - A A short synacthen test.
 - B Insulin levels.
 - C C-peptide levels.
 - D A 24-hour fast.
 - E All of the above
- 2 Which of the following statements are true? (MEN Epidemiology and Genetics)
 - A MEN1 is characterised by tumours in the pituitary, pancreas and parathyroids.
 - B Medullary thyroid carcinoma (MTC) plays a key role in MEN2 subtypes.
 - C The mode of inheritance is autosomal dominant in both MEN1 and 2.
 - D The penetrance of the mutation is low.
 - E Screening of family members of the affected individual is essential.
- 3 The following statements are true except: (MEN1 Clinical Presentation)
 - A Nephrolithiasis is a common clinical presentation.
 - B Pancreatic endocrine tumours (PET) are a common cause of death.
 - C Insulinomas are the most common pancreatic tumours in MEN1.
 - D All pituitary tumours must be surgically resected.
 - E Adrenocortical carcinomas frequently develop.
- 4 Which of the following statements are true? (Surgical Treatment in MEN1)
 - A Selective resection of the parathyroid is a preferred treatment option.
 - B MEN1 gastrinomas and insulinomas must be operated on.
 - C All nonfunctional adrenal tumours can be left alone.
 - D All nonfunctional PETs can be left alone.
 - E Most procedures related to the pituitary can be performed through the trans-sphenoidal approach.
- 5 Which of the following statements are true? (MEN2)
 - A MTC is a late manifestation of MEN2a.
 - B Primary hyperparathyroidism (pHPT) is an early manifestation of MEN2.
 - C Diagnostic work-up for MEN2 must include scintigraphy.
 - D Mutation carriers of MEN2 must be offered a thyroidectomy.
 - E Unilateral or bilateral sub-total resection of a pheochromocytoma in MEN2 is appropriate.

Answers 1-5

- 1 **BCD** True: B, C, D — **UPDATED** High insulin and C-peptide during a 24-hour fast confirm an insulinoma; synacthen test is not indicated.
- 2 **ABCE** True: A, B, C, E — **UPDATED** MEN1 and MEN2 are autosomal dominant with high penetrance; screening is essential and tumours involve the pituitary, pancreas, parathyroids and MTC in MEN2.
- 3 **CDE** True: C, D, E — **UPDATED** False: Insulinomas are not most common, not all pituitary tumours require surgery and adrenocortical carcinoma is rare in MEN1.
- 4 **BE** True: B, E — **UPDATED** Functional pancreatic tumours require surgery and pituitary procedures are mainly trans-sphenoidal.
- 5 **CDE** True: C, D, E — **UPDATED** In MEN2, scintigraphy is used, prophylactic thyroidectomy is advised and subtotal resection of pheochromocytoma is acceptable; the other statements are false.

2 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 28-year-old woman, who was hospitalized 2 months ago following a head injury, attends the outpatient clinic with a 6-week history of polyuria and polydipsia and no other symptoms. Her blood pressure is 117/83 mmHg and her heart rate is 68 beats/min. From the list below, select the most appropriate management option.
- A Carbimazole
 - B Desmopressin
 - C Spironolactone
 - D Thyroxine
 - E Octreotide
- 2** During a ward round, you are asked by your surgical registrar about the management of a pheochromocytoma. Select from the list below the most appropriate management plan for a pheochromocytoma.
- A Surgical resection, followed by β blockade, followed by α blockade
 - B Lifelong β and α blockade
 - C Surgical resection
 - D β blockade, followed by α blockade, followed by surgical resection
 - E α blockade, followed by β blockade followed by surgical resection

Answers 1-2

SBA MCQs

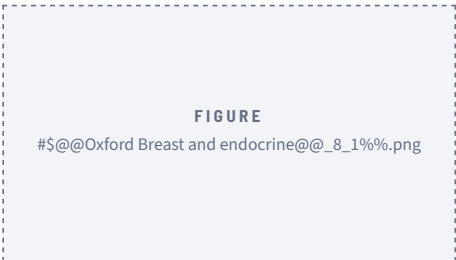
- 1** **B Desmopressin** — Central diabetes insipidus post-head injury is treated with desmopressin.
- 2** **E α blockade, followed by β blockade followed by surgical resection** — Pheochromocytoma management requires α blockade, then β blockade, before surgical resection.

9 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** A 35-year-old woman has been recently diagnosed by her GP with Graves' disease and has been referred to the outpatient department for further management. She has no other past medical history. She is a housewife and looks after her three young children aged 1 to 6. She has a large goitre with signs of severe exophthalmos (see Figure 5.1). Which is the single most appropriate long term treatment option for this lady?
- FIGURE

#\$@@@Oxford Breast and endocrine@@_2_1%%.png
- A Carbimazole 20mg twice a day
 - B Propylthiouracil 200mg twice a day
 - C Radioactive iodine
 - D Subtotal thyroidectomy
 - E Thyroid lobectomy
- 2** A 50-year-old woman has had 3 weeks of pain in front of her neck. She also complains of general malaise, episodes of feeling feverish, and palpitations. She has no past medical history and is not on any regular medications. She has a diffusely tender, slightly enlarged thyroid gland. Which is the single most likely diagnosis?
- A De Quervain's thyroiditis
 - B Graves' disease
 - C Hashimoto's thyroiditis
 - D Physiological thyroid enlargement
 - E Thyroid papillary carcinoma
- 3** A 90-year-old woman has a swelling in her neck which has increased in size over the last 2 weeks. She reports difficulty in swallowing, shortness of breath, coughing, and a hoarse voice. She has a large fixed mass in the left anterior triangle of her neck with palpable cervical lymphadenopathy. Which is the single most likely diagnosis?
- A Anaplastic cancer
 - B Follicular cancer
 - C Lymphoma
 - D Medullary cancer
 - E Papillary cancer

- 4 A 35-year-old woman has had a painful lump in her neck for 1 month. She has no other related symptoms. She has a solitary thyroid nodule that is slightly tender. A fine-needle aspiration biopsy (FNAB) gives a result of Thy2. Which is the single most likely diagnosis?
- Benign colloid nodule
 - Follicular lesion
 - Graves' disease
 - Medullary carcinoma of the thyroid
 - Normal thyroid
- 5 A 40-year-old woman has undergone a total thyroidectomy for a multinodular goitre. She is a member of her local choir and has noticed that she is no longer able to reach high notes as she used to prior to her surgery and that her voice fatigues. Which is the single most likely nerve that has been injured at surgery?
- External laryngeal nerve
 - Glossopharyngeal nerve
 - Internal laryngeal nerve
 - Recurrent laryngeal nerve
 - Ansa cervicalis
- 6 Figure 5.2 is a graphic illustration of the formation of various thyroid hormones by thyroid epithelial cells. I- is ionic iodine, deI indicates the process of de-iodinization, and AA circulating amino acids. Four structurally different hormones are indicated by the letters W, X, Y, and Z. Which is the single correct name of the hormone X?
- 
- FIGURE
#\$\$\$@Oxford Breast and endocrine@@_8_1%%.png
- Di-iodothyronine
 - Di-iodotyrosine
 - Reverse tri-iodothyronine
 - Thyroxine
 - Tri-iodothyronine
- 7 A 40-year-old man has had 1 month of spontaneous episodes of palpitations, headaches, sweating, and chest pain lasting for up to 15min. A diagnosis of pheochromocytoma is considered. Which is the single most appropriate initial investigation that will help to confirm the diagnosis?
- Aldosterone suppression test
 - Clonidine suppression test
 - Long synacthen test
 - Short synacthen test
 - 24-h urine vanillylmandelic acid and noradrenaline
- 8 A 65-year-old woman is referred by her GP following a routine blood test which showed a raised serum calcium. Over the last few months she has become constipated with intermittent bouts of abdominal pain. She is lethargic and her mood has been very low. Clinical examination is unremarkable. Biochemical results show a corrected serum calcium of 2.95mmol/L and a serum parathyroid hormone of 24pmol/L. Which is the single most likely diagnosis?
- Addison's disease
 - Lymphoma
 - C 'Milk-alkali' syndrome
 - Multiple myeloma
 - Primary hyperparathyroidism
- 9 A 65-year-old man has recently gained a significant amount of weight, his hair has become very dry and brittle, and his mood is low. He has a round face with a prominent fat pad along his back. Which is the single most appropriate subsequent investigation?
- Abdominal CT scan
 - Abdominal MRI scan
 - Dexamethasone suppression test
 - Early morning cortisol level
 - Pituitary MRI scan

Answers 1-9

- 1 **D Subtotal thyroidectomy** — **UPDATED** Graves disease with severe ophthalmopathy is best treated surgically (now usually total thyroidectomy); radioiodine can worsen the eye disease and drugs alone often relapse.
- 2 **A De Quervain's thyroiditis** — Painful tender swelling of the thyroid is suggestive of de Quervain's thyroiditis.
- 3 **A Anaplastic cancer** — Anaplastic carcinoma is most likely due to large fixed mass and lymphadenopathy.
- 4 **A Benign colloid nodule** — FNAB result Thy2 indicates a benign colloid nodule.
- 5 **A External laryngeal nerve** — Injury to the external laryngeal nerve impairs high-pitched voice production.
- 6 **E Tri-iodothyronine** — Hormone X corresponds to Tri-iodothyronine (T3).
- 7 **E 24-h urine vanillylmandelic acid and noradrenaline** — **UPDATED** Catecholamine excess is confirmed biochemically first (plasma free or 24-h urinary metanephrines; here 24-h urinary VMA/noradrenaline), then the tumour is localised by CT/MRI.
- 8 **E Primary hyperparathyroidism** — Primary hyperparathyroidism is indicated by raised calcium and PTH levels.

- 9 **C Dexamethasone suppression test** — Clinical signs indicate Cushing's syndrome; dexamethasone suppression test is appropriate.

4 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 15-year-old otherwise healthy female high school student begins to notice galactorrhea. A pregnancy test is negative. Which of the following is a frequently associated physical finding?
 - A Gonadal atrophy
 - B Bitemporal hemianopia
 - C Exophthalmos and lid lag
 - D Episodic hypertension
 - E Buffalo hump
- 2 A 53-year-old woman presents with weight loss and a persistent rash to her lower abdomen and perineum. She is diagnosed with necrolytic migrating erythema and additional workup demonstrates a hemoglobin A1c of 7.5%, hematocrit of 26%, and a large mass in the tail of the pancreas. Which of the following is the most likely diagnosis?
 - A Verner-Morrison syndrome (VIPoma)
 - B Glucagonoma
 - C Somatostatinoma
 - D Insulinoma
 - E Gastrinoma
- 3 A 50-year-old man presents with intractable peptic ulcer disease, severe esophagitis, and abdominal pain. Which of the following is most consistent with the diagnosis of Zollinger-Ellison syndrome?
 - A Hypercalcemia
 - B Fasting gastrin level of 10 pg/mL
 - C Fasting gastrin level of 100 pg/mL
 - D Increase in gastrin level (>200 pg/mL) after administration of secretin
 - E Decrease in gastrin level (>200 pg/mL) after administration of secretin
- 4 A 34-year-old woman has recurrent fainting spells induced by fasting. She also reports palpitations, trembling, diaphoresis, and confusion prior to the syncopal episodes. She has relief of symptoms with the administration of glucose. Which of the following findings is most consistent with the diagnosis of an insulinoma?
 - A Serum glucose level >50 mg/dL, elevated serum insulin levels, elevated C-peptide levels
 - B Serum glucose level >50 mg/dL, elevated serum insulin levels, decreased C-peptide levels
 - C Serum glucose level <50 mg/dL, elevated serum insulin levels, elevated C-peptide levels
 - D Serum glucose level <50 mg/dL, elevated serum insulin levels, decreased C-peptide levels
 - E Serum glucose level <50 mg/dL, decreased serum insulin levels, decreased C-peptide levels

Answers 1-4

PRETEST MCQS

- 1 **B Bitemporal hemianopia** — **UPDATED** Prolactin-secreting tumors in the pituitary gland may cause bitemporal hemianopia because of compression of the optic chiasm.
- 2 **B Glucagonoma** — **UPDATED** A large pancreatic tail mass with necrolytic migrating erythema is consistent with glucagonoma.
- 3 **D Increase in gastrin level (>200 pg/mL) after administration of secretin** — **UPDATED** A paradoxical increase in serum gastrin after administration of secretin is diagnostic of Zollinger-Ellison syndrome.
- 4 **C Serum glucose level <50 mg/dL, elevated serum insulin levels, elevated C-peptide levels** — **UPDATED** In insulinoma, endogenous insulin production leads to low glucose, high insulin, and high C-peptide levels.

Neck Lumps & Endocrine EMQs

Extended matching questions on neck swellings, dysphagia, thyroid disease and skin lumps.

0

MCQs

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EMQ SCENARIOS

3

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

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03

Oxford EMQs

THEME 1 Neck lumps

- A Branchial cyst
- B Cervical rib
- C Chemodectoma
- D Cystic hygroma
- E Lipoma
- F Lymphoma
- G Thyroid tumour
- H Sebaceous cyst
- I Thyroglossal cyst

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 30-year-old man has had a lump on the right side of his neck for years which intermittently discharges. There is a superficial, mobile lump in the posterior triangle with a visible punctum.

- 2 A 20-year-old man has an asymptomatic swelling just to the right of midline in the anterior neck. It feels firm and moves on tongue protrusion.
- 3 A 15-year-old boy has a smooth, non-tender, fluctuant mass along the lower anteromedial border of the sternocleidomastoid. It occasionally increases in size, causing discomfort.
- 4 A 50-year-old man has a soft, mobile, non-tender lump in the left anterior triangle and several smaller similar lumps on the right.
- 5 A 35-year-old woman has an enlarging left-sided neck swelling, 9 kg weight loss, lethargy, and a diffuse, rubbery, firm mass in the left anterior triangle with cervical and axillary lymphadenopathy.

Answers • Theme 1

OXFORD EMQS

Theme 1 • Neck lumps

- 1 **H Sebaceous cyst** — Sebaceous cysts are common, superficial, mobile, and have a central punctum.
- 2 **I Thyroglossal cyst** — Thyroglossal cysts move with tongue protrusion and lie near midline.

- 3 **A Branchial cyst** — Branchial cysts occur along the anterior border of SCM and are fluctuant.
- 4 **E Lipoma** — Lipomas are soft, mobile, painless fatty tumours; can be multiple.
- 5 **F Lymphoma** — Lymphoma presents with rubbery lymphadenopathy and systemic symptoms.

Get Ahead EMQs

THEME 1 Neck lumps

- A Acute bacterial sialolithiasis
- B Branchial cyst
- C Chemodectoma
- D Cystic hygroma
- E Pleomorphic adenoma
- F Salivary duct carcinoma
- G Salivary gland stone
- H Sternocleidomastoid tumour
- I Thyroglossal cyst
- J Virchow's node

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 1-year-old boy is brought to general practice by his mother with a large mass in his neck that has been present for a while. On examination, the mass is located posterior to the left sternocleidomastoid muscle. It is soft, compressible, and transilluminates.

- 2 A 64-year-old woman presents with a rapidly growing mass in her neck associated with drooping of the right side of her face. On examination, there is a 4 cm hard mass in the neck that is attached to the overlying skin.
- 3 A 13-year-old boy presents with a painless neck lump on the left side. On examination, there is a 2 cm lump at the anterior border of the sternocleidomastoid that is smooth and fluctuant but does not transilluminate.
- 4 An 8-year-old girl presents with a smooth lump in the midline of the neck. On examination, there is a 1 cm painless, fluctuant swelling in the midline that moves upward on swallowing and with tongue protrusion.
- 5 A 56-year-old man presents to the emergency department with pain and swelling in his neck that occurs every time he eats and lasts around an hour. He has experienced this for over a fortnight. On examination, there is no obvious swelling or erythema.

THEME 2 Thyroid disease

- A Anaplastic carcinoma
- B de Quervain thyroiditis
- C Follicular carcinoma
- D Graves disease
- E Hashimoto thyroiditis
- F Haemorrhage into a cyst
- G Medullary carcinoma
- H Myxoedema coma
- I Primary myxoedema
- J Thyroid storm
- K Toxic multinodular goitre

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 50-year-old woman attends her general practice complaining of low mood and lethargy. She has had this for a few months and has missed much time at work. On further questioning, she admits to being constipated and to having gained weight. On examination, no abnormality is apparent in the neck and no lymphadenopathy is palpable.
- 2 A 28-year-old woman goes to her general practitioner following a sudden, sharp pain in her neck earlier in the day. Since then, she has found it difficult to breathe. On examination, you find a swelling on the right side of her neck. The patient tells you that this swelling has been there for some time and has never previously caused her any trouble.
- 3 A 32-year-old woman presents with a 5-week history of resting tremor and diarrhoea. She mentions that just before these symptoms occurred, she had some time off work with a cough and cold. On examination, her pulse is 96/min and regular. The thyroid gland is generally enlarged and slightly tender to touch. No cervical lymph nodes are palpable.
- 4 A 22-year-old anxious woman has a month's history of worsening diarrhoea and has lost 10 pounds despite a good appetite. She has a slight tremor, an irregular pulse of 110/min and a bruit over the thyroid, with no obvious deformity or palpable nodes.
- 5 A 62-year-old woman is brought by her husband to the emergency department with reduced consciousness. On examination, she has a temperature of 30°C and her pulse is 48/min. Her husband tells you that she has a history of depression and has recently been refusing to take her regular medications.

THEME 3 Thyroid disease

- A Anaplastic carcinoma
- B de Quervain thyroiditis
- C Follicular carcinoma
- D Graves disease
- E Hashimoto thyroiditis
- F Haemorrhage into a cyst
- G Medullary carcinoma
- H Papillary carcinoma
- I Riedel thyroiditis
- J Struma ovarii
- K Toxic multinodular goitre

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 25-year-old woman presents to the clinic with a slowly enlarging lump on the left side of her neck. She denies any other symptoms. On examination, there is a 2 cm smooth, regular, firm lump that moves up with swallowing but not with tongue protrusion. Lymph nodes are palpable on the left side of the neck.
- 2 A 54-year-old man presents to clinic with a midline neck mass that has been increasing in size over a few months. On examination, the thyroid gland is enlarged, firm, and irregular, although the patient does not complain of any pain. No cervical nodes are palpable. A core biopsy is taken and the histology report denies the presence of malignancy.
- 3 A 42-year-old woman presents to her general practitioner with a long-standing goitre. More recently, she has been suffering with constipation, lethargy, and weight gain. On examination, her pulse is 64/min and regular. She has a diffusely enlarged, smooth, firm goitre. No lymph nodes are palpable.
- 4 A 46-year-old woman presents to the emergency department with palpitations. She is clearly nervous and describes having multiple paroxysms of fast palpitations over the preceding fortnight. On examination, you find an enlarged, lumpy thyroid gland that is not fixed to the overlying skin.
- 5 An 84-year-old woman presents with a lump in her neck that was first noticed last month and has since been growing rapidly. On further questioning, she admits to problems swallowing her food. On examination, there is a 5 cm irregular, hard mass on the left side that is fixed to the overlying skin.

Answers · Themes 1-3

GET AHEAD EMQS

Theme 1 · Neck lumps

- 1 **D Cystic hygroma** — Cystic hygroma is a congenital lymphatic malformation in the posterior triangle that transilluminates.
- 2 **F Salivary duct carcinoma** — Rapid growth and skin fixation suggest salivary duct carcinoma.

- 3 **B Branchial cyst** — Branchial cyst is a fluctuant mass along the anterior border of SCM.
- 4 **I Thyroglossal cyst** — Movement with swallowing and tongue protrusion confirms thyroglossal cyst.

- 5 **G Salivary gland stone** — Pain on eating indicates a salivary gland stone obstructing the duct.

Theme 2 · Thyroid disease

- 1 **I Primary myxoedema** — Hypothyroid symptoms of lethargy, constipation, and weight gain indicate primary myxoedema.
- 2 **F Haemorrhage into a cyst** — Sudden pain in a pre-existing thyroid nodule suggests haemorrhage into a cyst.
- 3 **B de Quervain thyroiditis** — Tender enlarged thyroid after viral illness indicates de Quervain thyroiditis.
- 4 **D Graves disease** — Weight loss, tachycardia, tremor, and thyroid bruit are characteristic of Graves disease.
- 5 **H Myxoedema coma** — Severe hypothermia, bradycardia, and altered consciousness in untreated hypothyroidism indicate myxoedema coma.

Theme 3 · Thyroid disease

- 1 **H Papillary carcinoma** — Papillary carcinoma presents as a firm painless nodule in young women.
- 2 **I Riedel thyroiditis** — Riedel thyroiditis causes a fibrotic goitre with benign histology.
- 3 **E Hashimoto thyroiditis** — Hashimoto thyroiditis presents with a firm goitre and hypothyroid symptoms.
- 4 **K Toxic multinodular goitre** — Toxic multinodular goitre causes hyperthyroid symptoms with nodular enlargement.
- 5 **A Anaplastic carcinoma** — Anaplastic carcinoma is an aggressive thyroid cancer in the elderly.

5 THEMES · 23 SCENARIOS · IRFAN SURGERY EMQ COLLECTION

Irfan EMQs

THEME 1 Swelling in the neck

- A Branchial cyst
- B Thyroid nodule
- C Cervical rib
- D Lymphoma
- E Chemodectoma
- F Sternomastoid tumour
- G Dermoid cyst
- H Pharyngeal pouch
- I Thyroglossal cyst
- J Cystic hygroma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 43-year-old man presents to his GP with a 6-month history of a painless pulsatile mass at the angle of the jaw.
- 2 A 23-year-old girl complains of intermittent numbness and paraesthesiae in her right hand for the past 2 months. On examination there is a fixed, hard, 1 cm x 2 cm swelling in the right supraclavicular fossa.
- 3 A 3-year-old boy is seen by his GP with an enlarging midline swelling that has been present for the past year. It is smooth and rounded, located just below the hyoid bone, measuring 2 cm x 2 cm, and rises on protrusion of the tongue.
- 4 A 32-year-old woman presents to her GP with a neck lump enlarging for the last 3 years. It measures 1 cm x 1.5 cm and is located behind the junction of the upper and middle thirds of the left sternocleidomastoid muscle. In the past this lump has become infected, resolving with oral antibiotics.

- 5 A 23-year-old man presents to his GP with a 2 cm x 3 cm painless lump at the angle of the jaw; it has been there for 2 months. He also complains of weight loss, night sweats and fever, over the same period. Hepatosplenomegaly is detected on examination of the abdomen.

THEME 2 Dysphagia

- A Foreign body
- B Pharyngeal pouch
- C Oesophageal varices
- D Hypopharyngeal carcinoma
- E Mallory-Weiss syndrome
- F Globus pharyngeus
- G Thyroid goitre
- H Motor neuron disease
- I Oesophageal atresia
- J Diffuse oesophageal spasm

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 70-year-old man presents with dysphagia and regurgitation of undigested food after about 5 minutes.
- 2 A 75-year-old woman presents with a 3-month history of increasing dysphagia to solids, pain on swallowing and weight loss.
- 3 A 35-year-old woman presents with a 3-month history of a sensation of a lump in the throat. There is no dysphagia and the sensation improves when she swallows.
- 4 A 50-year-old woman presents with a gradually increasing swelling in the lower part of the neck. Occasionally food can 'stick' when swallowed. The swelling was found to move on swallowing.

- 5** A 75-year-old man wakes up in the middle of the night with sudden onset of pain in the neck and difficulty swallowing his saliva. He has no symptoms during the day.

THEME 3 Presentation with a lump

- A histiocytoma
- B myosarcoma
- C ganglion
- D abscess
- E lipoma
- F carbuncle
- G furuncle
- H neurofibroma
- I sebaceous cyst
- J keloid
- K Marjolin's ulcer
- L keratoacanthoma
- M osteoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 22-year-old man presents with a lump on the scalp. Examination reveals a smooth, spherical tense lump. A small punctum can be seen on the surface.
- 2** A 33-year-old man presents with a swelling on the upper arm which has been growing slowly for a number of years. Examination reveals a soft, compressible, non-tender lobulated mass.
- 3** A 28-year-old man presents with a painless swelling on the dorsum of the right hand. Examination reveals a smooth, spherical, tense, 1.5-cm swelling. The overlying skin can be drawn over it.
- 4** A 29-year-old man presents with two mobile fusiform-shaped lumps on the forearm. Each swelling feels like firm rubber and causes tingling in the hand upon pressure.
- 5** A 65-year-old woman presents with a rapidly growing lump just below the eye. Examination reveals a 2-cm, smooth, round, skin-coloured lump with a black central core. The lump is freely mobile over subcutaneous tissues.

THEME 4 Thyroid conditions (i)

- A follicular carcinoma
- B Hashimoto's disease
- C anaplastic carcinoma
- D papillary carcinoma
- E lymphoma
- F Riedel's thyroiditis
- G Graves' disease
- H follicular adenoma
- I medullary carcinoma
- J de Quervain's thyroiditis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 75-year-old woman presents with acute airway obstruction and a thyroid mass after several weeks of worsening shortness of breath. Postmortem reveals a thyroid mass that has invaded the trachea and surrounding structures. Histology shows giant cells containing pleomorphic hyperchromatic nuclei.
- 2** A 38-year-old patient presents with a 2.5-cm right-sided thyroid mass. On examination there is mild enlargement of the cervical lymph nodes. A thyroidectomy is performed and histology reveals a non-encapsulated infiltrative mass. Psammoma bodies and epithelial cells with large clear areas within the nuclei are noted.
- 3** A 24-year-old woman presents with a diffusely enlarged thyroid gland. On examination she has a fine tremor and exophthalmos.
- 4** A 30-year-old woman presents with a tender thyroid swelling associated with fever. Investigations reveal no evidence of infection.

THEME 5 Thyroid conditions (ii)

- A follicular carcinoma
- B Hashimoto's disease
- C anaplastic carcinoma
- D endemic goitre
- E lymphoma
- F Riedel's thyroiditis
- G Graves' disease
- H follicular adenoma
- I medullary carcinoma
- J De Quervain's thyroiditis
- K papillary carcinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 36-year-old woman presents with a history of increasing weight gain, tiredness and neck swelling. On examination the patient has a firm rubbery goitre. Fine-needle aspiration shows a diffuse lymphocytic and plasma cell infiltrate. The report also mentions the presence of lymphoid follicles and parenchymal atrophy.
- 2** A 40-year-old woman presents with a history of a slowly growing right-sided lump at the base of the neck. On examination there is a thyroid nodule with associated lymphadenopathy. She has a recent history of adrenalectomy.
- 3** A 45-year-old woman presents with a painless, firm, left-sided neck swelling. Fine-needle aspiration reveals the presence of multiple compact follicles. Histology shows evidence of capsular and blood vessel invasion.

- 4 A 55-year-old African man is referred to the ENT clinic with a large, smooth goitre that has been increasing in size for at least 20 years.

Answers · Themes 1-5

IRFAN EMQS

Theme 1 · Swelling in the neck

- 1 **E Chemodectoma** — A pulsatile mass at the carotid bifurcation that moves side to side but not up and down is a chemodectoma (carotid body tumour); duplex/CT angiography, then excision.
- 2 **C Cervical rib** — A cervical rib arises from the 7th cervical vertebra and may compress neurovascular structures causing upper limb symptoms such as paraesthesia and weakness. Appears as a hard posterior neck mass.
- 3 **I Thyroglossal cyst** — A thyroglossal cyst is a midline neck mass that moves with tongue protrusion. It arises from persistent thyroglossal duct remnants. Common in children. Treated with Sistrunk's operation.
- 4 **A Branchial cyst** — A branchial cyst is a congenital neck swelling along the anterior border of the sternocleidomastoid. It can become infected and usually presents in young adults. Excision is the treatment.
- 5 **D Lymphoma** — Lymphoma presents as painless lymphadenopathy with systemic symptoms such as weight loss and night sweats. Common in young adults. Diagnosis confirmed by biopsy and imaging.

Theme 2 · Dysphagia

- 1 **B Pharyngeal pouch** — Pharyngeal pouch typically presents with dysphagia and regurgitation of undigested food several minutes after eating, often in elderly men.
- 2 **D Hypopharyngeal carcinoma** — Hypopharyngeal carcinoma can cause progressive dysphagia, odynophagia, and weight loss in elderly patients.
- 3 **F Globus pharyngeus** — Globus pharyngeus is a sensation of a lump in the throat without true dysphagia, often improves with swallowing and is linked to anxiety or reflux.
- 4 **G Thyroid goitre** — Thyroid goitre can compress the oesophagus causing intermittent dysphagia; moves with swallowing.
- 5 **A Foreign body** — Foreign body impaction may cause sudden nocturnal onset of neck pain and difficulty swallowing, especially with saliva, indicating acute obstruction.

Theme 3 · Presentation with a lump

- 1 **I sebaceous cyst** — The small punctum seen on the scalp lump indicates a blocked sebaceous gland, which is typical of a sebaceous cyst; the lesion's smooth, spherical, and tense nature further supports this diagnosis.
- 2 **E lipoma** — The slow-growing, soft, compressible, and lobulated mass in the arm is consistent with a lipoma – a benign tu-

mor composed of fat cells that typically presents as a painless and mobile swelling.

- 3 **C ganglion** — The painless, mobile, and tense swelling on the dorsum of the hand, with skin that can be drawn over it, is characteristic of a ganglion, which commonly arises near joints or tendon sheaths.
- 4 **H neurofibroma** — Firm, rubbery fusiform lumps that tingle on pressure lie along a nerve: neurofibromas (benign nerve sheath tumours).
- 5 **L keratoacanthoma** — The rapidly growing, smooth, round lesion just below the eye with a central black core is indicative of a keratoacanthoma, a skin lesion that grows quickly and often features a central keratin plug.

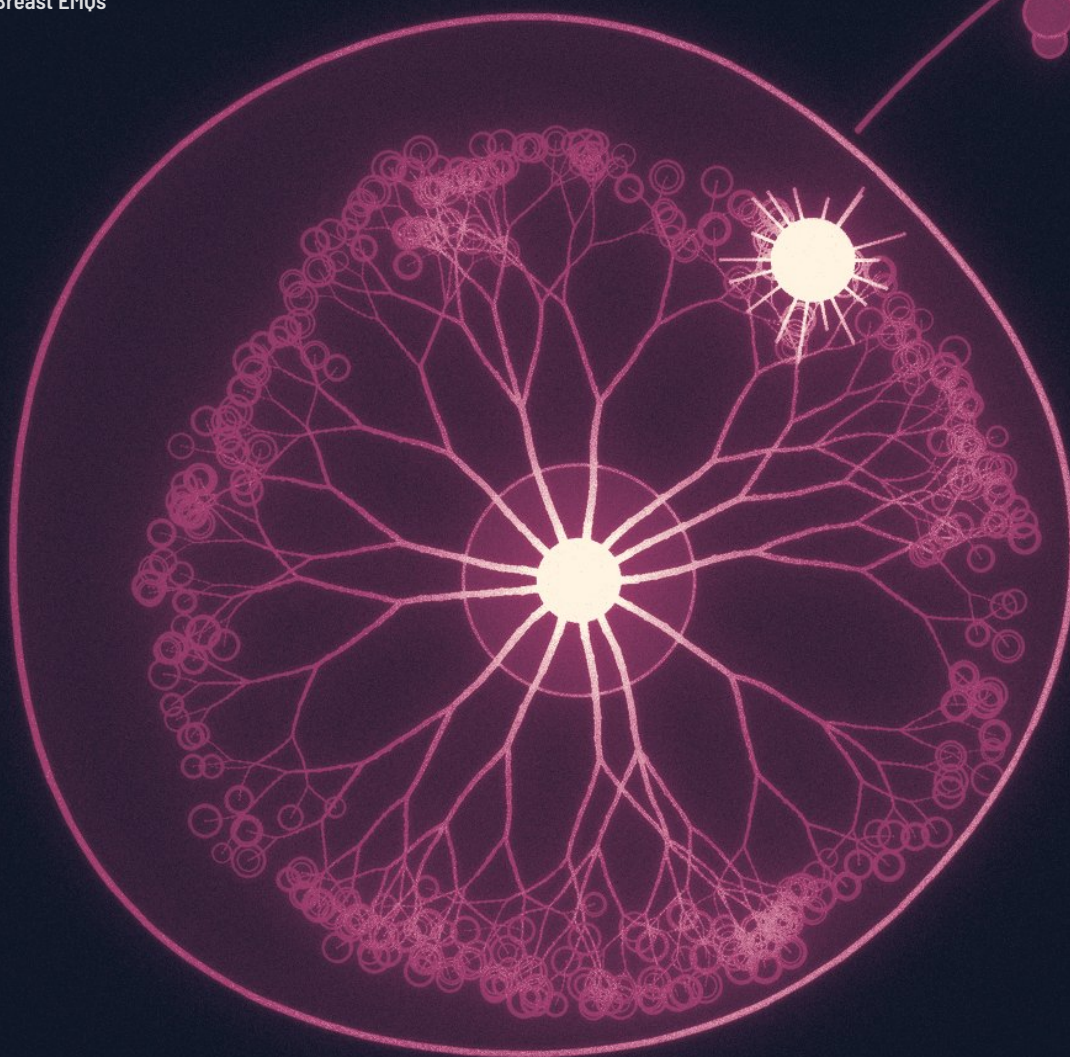
Theme 4 · Thyroid conditions (i)

- 1 **C anaplastic carcinoma** — An elderly patient with rapid airway obstruction, invasive thyroid mass and giant pleomorphic cells on histology indicates anaplastic carcinoma, which is aggressive and locally invasive.
- 2 **D papillary carcinoma** — A middle-aged patient presenting with a thyroid mass exhibiting psammoma bodies and clear nuclear features with cervical lymphadenopathy is classic for papillary carcinoma.
- 3 **G Graves' disease** — A young woman with diffuse thyroid enlargement accompanied by tremor and exophthalmos points to Graves' disease, an autoimmune hyperthyroid condition affecting both the thyroid and orbital tissues.
- 4 **J de Quervain's thyroiditis** — A patient with a tender thyroid swelling and fever, without signs of infection elsewhere, suggests de Quervain's thyroiditis, a subacute granulomatous inflammation usually following a viral illness.

Theme 5 · Thyroid conditions (ii)

- 1 **B Hashimoto's disease** — Hypothyroid features with a firm rubbery goitre and lymphocytic infiltrate with lymphoid follicles on FNA is Hashimoto thyroiditis.
- 2 **I medullary carcinoma** — Medullary carcinoma: A 40-year-old woman with a thyroid nodule, lymphadenopathy and a history of adrenalectomy (suggesting MEN2 syndrome) points to medullary carcinoma.
- 3 **A follicular carcinoma** — Capsular and vascular invasion on histology (not cytology) is what distinguishes follicular carcinoma from follicular adenoma.
- 4 **D endemic goitre** — Endemic goitre: A 55-year-old man with a long-standing (20 years) large, smooth goitre is consistent with an iodine deficiency-induced endemic goitre.

- 04 Benign Breast Disorders
- 05 Breast Cancer
- 06 Breast EMQs



02

BREAST

Benign Breast Disorders

Breast assessment and triple assessment, fibroadenoma, cysts, mastalgia, nipple discharge, infection and gynaecomastia.

47

MCQS

0

EMQ SCENARIOS

8

SOURCES

1

REPEATED IDEAS

IN THIS CHAPTER

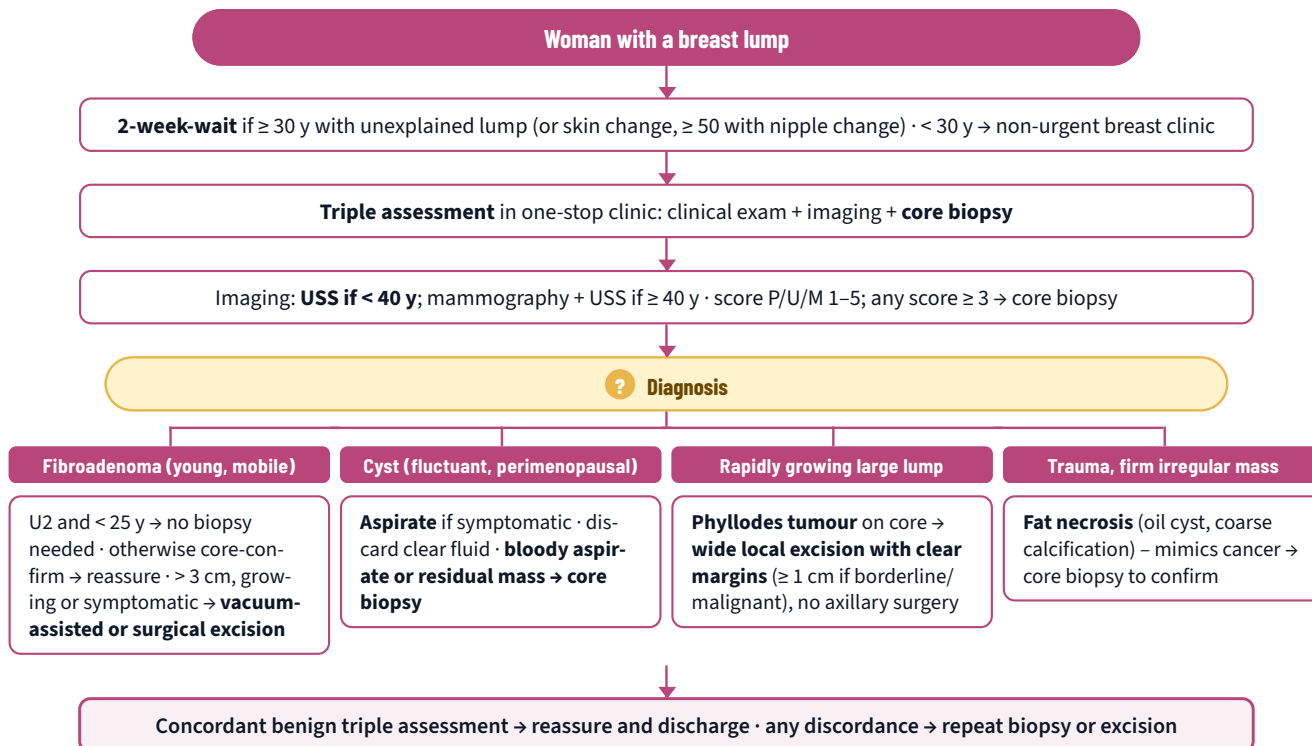
01	Management flowcharts	3 charts	page 48
02	Local Exams MCQs	6 Q	page 50
03	Bailey & Love MCQs	1 Q	page 51
04	Lange MCQs	6 Q	page 51
05	SBA MCQs	14 Q	page 52
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07	PreTest MCQs	2 Q	page 55
08	Get Ahead MCQs	6 Q	page 56
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04

Step by step

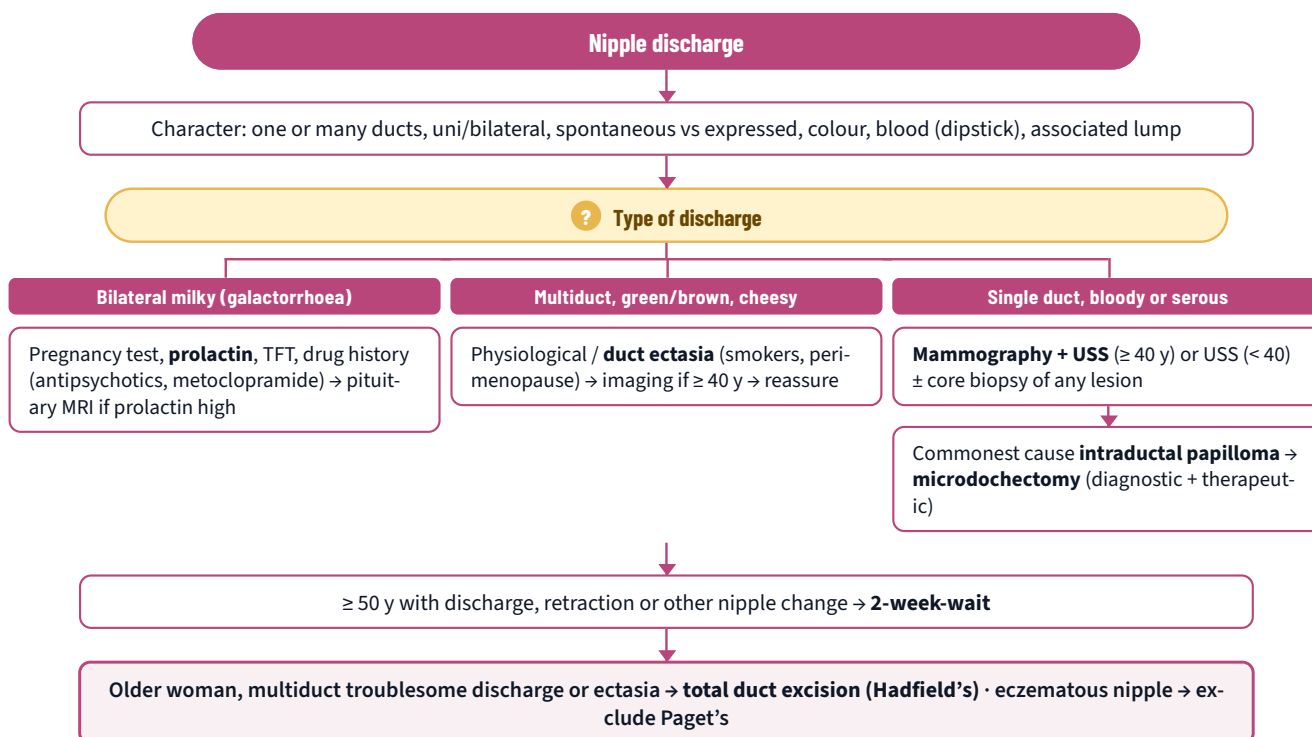
Breast lump: triple assessment

NICE NG12 • ABS 2019 BEST-PRACTICE DIAGNOSTIC GUIDELINES • BAILEY & LOVE 28E



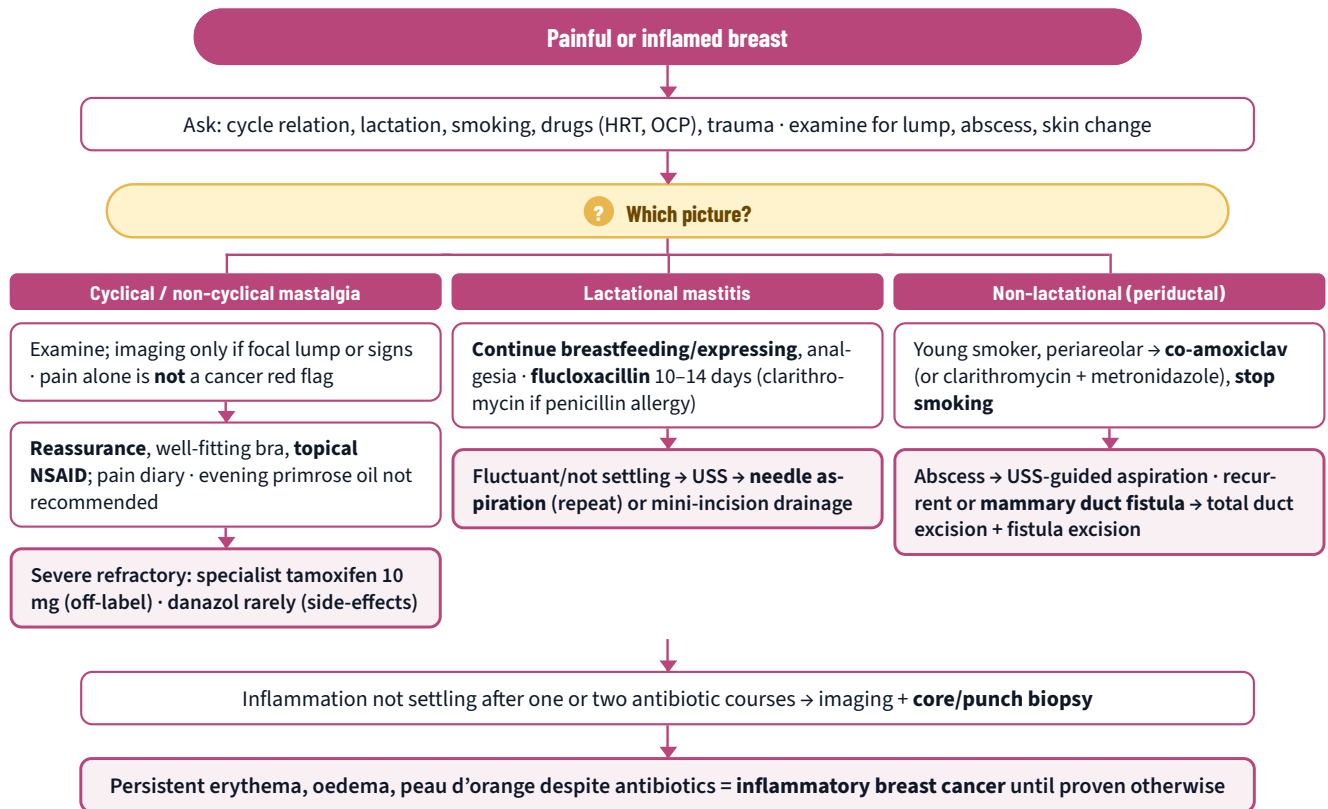
Nipple discharge

NICE NG12 • ABS 2019 • BAILEY & LOVE 28E



Breast pain and breast infection

NICE CKS 2023 MASTITIS & MASTALGIA · ABS 2019 · BAILEY & LOVE 28E



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-3

A 28-year-old woman, a smoker with five children (all breast-fed), has had bloody nipple discharge from the right breast for 6 months. At times the area swells with a foul-smelling discharge; antibiotics did not help.

- 1 Tests show mildly dilated multiple lactiferous ducts. What is the management?
 - A Lumpectomy
 - B Simple mastectomy
 - C Modified radical mastectomy
 - D Reassurance, smoking cessation and follow-up
 - E Drainage and antibiotics
- 2 Which of the following is NOT useful in investigating her condition?
 - A Bilateral breast examination
 - B Mammography
 - C Breast MRI
 - D Breast ultrasound
 - E Cytology of the discharge
- 3 What is the diagnosis?
 - A Duct ectasia
 - B Ductal carcinoma
 - C Lobular carcinoma
 - D Breast abscess
 - E Fibroadenoma

MINISTERIAL EXAM

Ministerial evaluation examinations

- 4 A 36-year-old multiparous woman attends your clinic with a 7-day history of left breast pain after being involved in a car accident. On examining her breast, you notice a 3-cm hard, irregular, immobile, and tender lump. You also notice some skin tethering and overlying bruising at the region of the lump. Ultrasound features suggest a benign pathology. The most likely diagnosis at this point is:
 - A Fat necrosis.
 - B Breast cyst.
 - C Mammary duct ectasia.
 - D Breast abscess.
 - E Fibroadenosis.
- REPEATED 2x • ALSO ASKED IN SBA**
- 5 A 32-year-old woman came to the breast clinic with five weeks history of redness and swelling around the right nipple. There is no past medical history but she is a heavy smoker. On examination, there is peri-areolar skin inflammation with a small palpable tender mass just behind the nipple. The most likely diagnosis is:
 - A Lactation mastitis
 - B Peri-ductal mastitis
 - C Fat necrosis
 - D Fibrocystic disease
 - E Mondor's disease
 - 6 A 36-year-old woman complains of a 3-month history of bloody discharge from the nipple. On examination, a small nodule is found deep to the areola. Careful palpation of the nipple-areola complex results in blood appearing at one duct. Mammogram findings are normal. What is the most likely diagnosis?
 - A Intraductal papilloma
 - B Breast cyst
 - C Intraductal carcinoma
 - D Carcinoma in situ
 - E Fat necrosis

Answers 1-6

LOCAL EXAMS MCQs

- 1 **D Reassurance, smoking cessation and follow-up** — Duct ectasia/periductal mastitis is benign: reassure, stop smoking and follow up; total duct excision only for persistent discharge or recurrent sepsis.
- 2 **C Breast MRI** — Triple assessment (examination, ultrasound/mammography, cytology) suffices; MRI adds nothing in benign duct ectasia.

- 3 **A Duct ectasia** — Smoking, multiparity and multiple dilated ducts with discharge and recurrent periareolar inflammation are typical of duct ectasia/periductal mastitis.
- 4 **A Fat necrosis.** — **UPDATED** A hard, tethered, tender lump with bruising after trauma is fat necrosis; it mimics carcinoma clinically, so triple assessment (imaging ± core biopsy) is still required.

- 5 **B** **Peri-ductal mastitis** — Presentation and smoking history indicate peri-ductal mastitis.

- 6 **A** **Intraductal papilloma** — Bloody discharge with a nodule suggests intraductal papilloma.

1 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Which of the following statements are true? (Benign Breast Disease)
- A Nipple retraction always indicates an underlying malignancy.
 - B Staphylococcus aureus is the most common cause for lactational mastitis.
 - C Fibroadenomas should always be offered excision, as they are associated with a high risk of malignancy.
 - D The histology of phyllodes tumour may be benign, intermediate or malignant.
 - E Blood-stained nipple discharge is usually due to a breast carcinoma.

Answer 1

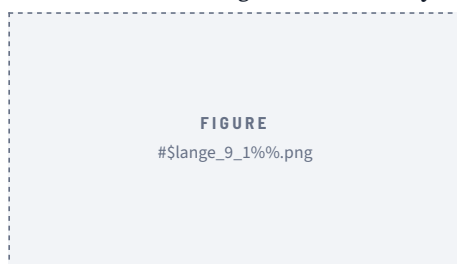
BAILEY & LOVE MCQS

- 1 **BD** **True: B, D** — Staphylococcus aureus is the usual cause of lactational mastitis and phyllodes tumours may be benign, intermediate or malignant.

6 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 51-year-old woman presents with a 2-month history of right breast blood-tinged nipple discharge from a single duct; mammogram is negative, cytology is non-diagnostic, ductogram shows a solitary filling defect. Which of the following is the most likely diagnosis?



- A invasive carcinoma
- B intraductal carcinoma
- C intraductal papilloma
- D fibrocystic disease
- E duct ectasia

- 2 Which of the following is the most appropriate next step in management for the intraductal papilloma?
- A collection of discharge for repeat cytologic analysis
 - B observation, with repeat examination and imaging in 3–6 months
 - C modified radical mastectomy
 - D central lumpectomy (including removal of the nipple/areolar complex)
 - E terminal duct excision (microdochectomy)
- 3 A 25-year-old woman presents with right breast pain localized to the nipple areolar complex with a subareolar fluid collection. Which of the following is the most likely diagnosis?

- A inflammatory carcinoma of the breast
- B periductal mastitis with abscess
- C Paget's disease of the nipple
- D Mondor's disease (thrombophlebitis)
- E intraductal papilloma

- 4** Which of the following is the most appropriate next step in management of the breast mass?
- A** aspiration of the fluid and antibiotics
 - B** central lumpectomy, axillary node dissection, and radiation therapy
 - C** incision and drainage
 - D** antibiotics alone with re-evaluation in 1 week
- 5** A 33-year-old woman with a 2-month left breast mass that is firm, mobile, and well-defined is most consistent with?
- A** invasive ductal carcinoma
 - B** ductal carcinoma in situ
 - C** fibrocystic changes
 - D** fibroadenoma
 - E** intraductal papilloma
- 6** What management is recommended for the above breast mass?
- A** discontinuation of birth control pills
 - B** re-examination after the next menstrual cycle
 - C** needle biopsy
 - D** excisional biopsy
 - E** repeat ultrasound in 6 months

Answers 1-6

LANGE MCQs

- 1** **C intraductal papilloma** — **UPDATED** Intraductal papilloma. Solitary papilloma is the most common cause of pathologic nipple discharge in one duct.
- 2** **E terminal duct excision (microdochectomy)** — **UPDATED** Terminal duct excision (microdochectomy). Definitive diagnosis and treatment by excision of the involved duct.
- 3** **B periductal mastitis with abscess** — Presentation is consistent with periductal mastitis.
- 4** **A aspiration of the fluid and antibiotics** — Management of periductal mastitis with abscess involves aspiration and antibiotics.
- 5** **D fibroadenoma** — **UPDATED** The features most closely match a fibroadenoma.
- 6** **C needle biopsy** — Needle biopsy is advised to confirm the diagnosis.

14 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 55-year-old man with known atrial fibrillation presents to his general practitioner with a 3-month history of gynaecomastia. Which of his following medications is not associated with gynaecomastia?
- A** Digoxin
 - B** Cimetidine
 - C** Spironolactone
 - D** Furosemide
 - E** Metronidazole
- 2** A worried 23-year-old woman, who started taking the combined contraceptive pill 3 months ago, presents with a 1-day history of discovering a painless lump in the right breast. The patient states that the lump was not there a month ago. On examination, a slightly mobile, discrete, well-defined, non-tender, firm 1 cm diameter lump is found. There is no lymphadenopathy. The most likely diagnosis here is
- A** Breast cyst
 - B** Lipoma
 - C** Fibroadenoma
 - D** Sebaceous cyst
 - E** Carcinoma of the breast
- 3** A 33-year-old, non-smoking, breastfeeding woman is 10 days postpartum. She has a 4-day history of a slight crack on the surface of her left nipple. She presents with a 2-day history of severe continuous pain in the left breast, spiking pyrexia up to 38.8 with rigours which has prevented her from sleeping. On examination, you find the outer quadrants of the left breast to be red, warm and tender with a hard 3 cm lump at the edge of the left nipple. The most likely diagnosis is
- A** Acute mastitis
 - B** Breast cyst
 - C** Fat necrosis
 - D** Breast abscess
 - E** Periductal mastitis

- 4** A 43-year-old woman presents to your clinic with a 2-month history of localized dull pain in the right breast. The pain intensifies just before her period. On examination, you find a discrete 2.5 cm mobile, tense, tender, fluctuant lump in the lower inner quadrant of the right breast. The most likely diagnosis here is
- A Fibroadenosis
 - B Periductal mastitis
 - C Breast cyst
 - D Fat necrosis
 - E Fibroadenoma
- 5** A 47-year-old perimenopausal woman presents with a 3-week history of green discharge from the right nipple. On examination, the right nipple is non-tender, has a 'slit-like' appearance and is retracted. The most likely diagnosis is
- A Galactorrhoea
 - B Duct papilloma
 - C Breast carcinoma
 - D Mammary duct ectasia
 - E Fibroadenoma
- 6** A 25-year-old woman presents to your clinic after discovering, for the first time, two lumps in the inner lower quadrant of her left breast. On examination you find these lumps to be 2 cm in size, solid, discrete, mobile and non-tender. The right breast is normal and there is no lymphadenopathy. The most appropriate course of management is
- A Request a mammogram
 - B Reassure the patient and discharge her
 - C Request an ultrasound of the left breast
 - D Request fine needle aspiration
 - E Request a core biopsy
- 7** A 21-year-old nulliparous woman presents to your clinic with a 1-month history of bilateral breast pain. The pain, which is dull and achy in nature, is poorly localized and widespread across both breasts. The pain gradually increases in severity and is worse just before her menses. The pain usually starts to get better once her menses start. On examination, both breasts are tender. There are no lumps, skin changes or obvious swellings. The most likely diagnosis here is
- A Non-cyclical mastalgia
 - B Tietze's syndrome
 - C Cyclical mastalgia
 - D Acute bacterial mastitis
 - E Traumatic fat necrosis
- 8** A 25-year-old woman is diagnosed with a 1.5 cm palpable area of fat necrosis of the left breast by core biopsy following a traumatic injury 14 days earlier. She has slight bruising of the lower outer quadrant of the left breast with moderate tenderness. What would be the most appropriate course of management?
- A Reassurance and discharge
 - B Follow-up appointment for ultrasound in 3 months
 - C Wide local excision
 - D Left mastectomy
 - E Follow-up appointment for ultrasound in 6 months
- 9** A 38-year-old woman, and mother of two healthy children, is diagnosed with a fluid-filled simple cyst after triple assessment. On ultrasound the inner surface of the cyst looks entirely smooth. The woman does not have any significant family history of carcinoma and the cyst is located in the outer-lower quadrant of the right breast. What would be the most appropriate course of action?
- A Wide local excision
 - B Follow-up appointment in 3 months
 - C Annual follow-up
 - D Reassure and discharge
 - E Core biopsy
- 10** A 30-year-old woman who is 12 days postpartum and breastfeeding is diagnosed with acute mastitis of the left breast. Four days earlier, she discovered a painful crack in the region of the left nipple and noticed that the surrounding skin was tender, warm and red in colour. The patient is not allergic to penicillin and you decide to prescribe a course of antibiotics. What would be the most appropriate antibiotic for treating this condition?
- A Erythromycin
 - B Amoxicillin
 - C Ciprofloxacin
 - D Flucloxacillin
 - E Cephalixin
- 11** After a triple assessment, including core biopsy, a 28-year-old woman is diagnosed with a fibroadenoma of the left breast. The patient has a significant family history of breast carcinoma. The non-tender lump is situated in the inner lower quadrant of the left breast. The lump is approximately 1.5 cm × 1.5 cm. What is the most appropriate course of management?
- A Excision of the lump
 - B Reassure and follow-up after 3 months
 - C The patient should be given the choice of excision or not and if not she could be discharged
 - D Fine needle aspiration
 - E Perform triple assessment again in 6 weeks

- 12** A 45-year-old perimenopausal woman is diagnosed with mammary duct ectasia of the right breast after having had small and infrequent amounts of milky green discharge from multiple ducts of the right nipple for over 2 months. The patient has no significant family history and mammography findings are normal. What is the most appropriate course of management?
- A Reassure and discharge
 - B Surgical resection of the duct system of the right breast (Hadfield's operation)
 - C Commence antibiotic therapy
 - D Perform mammography of the right breast in 3 months
 - E Mastectomy of the right breast
- 13** A 48-year-old perimenopausal woman presents with a 2-month history of a painful lump in her right breast. On examination you find a 2.5 cm tense, fluctuant, mobile lump in the outer lower quadrant of the right breast. The most appropriate next course of action is
- A Computed tomography scan
 - B Mammography and ultrasound
 - C Fine needle aspiration
 - D Core biopsy
 - E Mammography and core biopsy
- 14** A 34-year-old premenopausal woman presents to your clinic with a lump in her right breast. On examination you find a 2.5 cm fluctuant, mobile, tender lump in the inner lower quadrant of the right breast. The ultrasound report suggests a benign fluid-filled cyst. The most appropriate course of action is
- A Breast magnetic resonance imaging
 - B Ultrasound guided fine needle aspiration
 - C Ultrasound guided core biopsy
 - D Reassure and discharge
 - E Mammography

Answers 1-14

SBA MCQS

- 1** **D Furosemide** — Furosemide is not typically associated with gynaecomastia, unlike the other medications listed.
- 2** **C Fibroadenoma** — Fibroadenomas typically occur in young women presenting as painless, mobile, firm lumps, differentiating them from cysts, lipomas, and carcinomas.
- 3** **D Breast abscess** — In a lactating woman with a nipple crack and severe localized pain, the findings favor a breast abscess over other conditions.
- 4** **C Breast cyst** — A mobile, fluctuant lump in a woman over 40 suggests a benign breast cyst.
- 5** **D Mammary duct ectasia** — The slit-like nipple with green discharge in a perimenopausal woman points to mammary duct ectasia.
- 6** **C Request an ultrasound of the left breast** — In a 25-year-old with new breast lumps, ultrasound is the most appropriate next step in triple assessment.
- 7** **C Cyclical mastalgia** — Bilateral breast pain that worsens premenstrually is characteristic of cyclical mastalgia.
- 8** **A Reassurance and discharge** — Traumatic fat necrosis usually resolves without intervention, thus reassurance and discharge is appropriate.
- 9** **D Reassure and discharge** — **UPDATED** A simple cyst (smooth wall, anechoic) after triple assessment needs no follow-up: aspirate if symptomatic and discharge; complex or recurrent cysts need biopsy.
- 10** **D Flucloxacillin** — In a breastfeeding woman with acute mastitis, flucloxacillin is the treatment of choice.
- 11** **C The patient should be given the choice of excision or not and if not she could be discharged** — **UPDATED** A core-biopsy-proven fibroadenoma under 3 cm can be left; excision is offered by patient choice or if large, growing or symptomatic. Family history alone is not an indication.
- 12** **A Reassurance and discharge** — Normal findings in mammary duct ectasia warrant reassurance and discharge.
- 13** **B Mammography and ultrasound** — In a woman over 35, combined mammography and ultrasound is the appropriate next step.
- 14** **B Ultrasound guided fine needle aspiration** — For a benign cystic lesion on ultrasound, fine needle aspiration is the next appropriate step.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 A 40-year-old woman has had 3 weeks of unilateral breast pain. The pain has been constant with no relation to her menstrual cycle. She has a past medical history of asthma with a recent episode of exacerbation. There is no evidence of any discrete lumps or axillary lymphadenopathy. There is localized tenderness on palpation of the medial half of the left breast and over the 4th and 5th costochondral junctions. Which is the single most likely diagnosis?
 - A Atypical angina
 - B Breast abscess
 - C Cyclical mastalgia
 - D Mastitis
 - E Tietze's syndrome
- 2 A 50-year-old woman has had intermittent nipple discharge. She has noticed that the discharge seems to be coming from one point on the left nipple. The discharge is slightly blood-stained. There are no discrete lumps and no axillary lymphadenopathy. On gentle squeezing of the left nipple there is a single point of discharge. A mammogram is normal. Which is the single most likely diagnosis?
 - A Acute suppurative mastitis
 - B Duct papilloma
 - C Duct ectasia
 - D Fibrocystic disease
 - E Periductal mastitis

Answers 1-2

OXFORD MCQS

- 1 **E Tietze's syndrome** — Tietze's syndrome is the most likely diagnosis.
- 2 **B Duct papilloma** — Duct papilloma occurs within large, central mammary lactiferous ducts and presents with nipple discharge and small-volume bleeding.

2 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 49-year-old woman undergoes biopsy of a 5.0-cm left breast mass; she has no palpable axillary lymph nodes. Biopsy of the mass shows cystosarcoma phyllodes. What is the most appropriate management for this patient?
 - A Radical mastectomy
 - B Wide local excision
 - C Lumpectomy with adjuvant radiation therapy
 - D Modified radical mastectomy
 - E Simple mastectomy with sentinel lymph node biopsy
- 2 A 51-year-old (premenopausal) woman undergoes needle localization biopsy for microcalcifications. Pathology reveals sclerosing adenosis. What is the most appropriate next step in management?
 - A Lumpectomy with adjuvant radiation therapy
 - B Modified radical mastectomy
 - C Simple mastectomy with sentinel lymph node biopsy
 - D Lumpectomy with axillary lymph node dissection and radiation therapy
 - E No further surgical intervention

Answers 1-2

PRETEST MCQS

- 1 **B Wide local excision** — Cystosarcoma phyllodes tumors are treated with wide local excision with at least 1-cm margins.
- 2 **E No further surgical intervention** — Sclerosing adenosis is a benign lesion and does not require further treatment.

6 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 30-year-old woman presents with multiple bilateral breast swellings which cause discomfort, particularly just before her periods. They have been present for several years but appear to be getting worse. What is the most likely diagnosis?
 - A Duct ectasia
 - B Fat necrosis
 - C Fibroadenoma
 - D Fibrocystic disease
 - E Peau d'orange
- 2 A 38-year-old woman presents with subareolar pain in the left breast and occasional blood-stained nipple discharge. Examination is unremarkable. What is the most likely diagnosis?
 - A Duct ectasia
 - B Galactocoele
 - C Intraductal papilloma
 - D Paget disease
 - E Prolactinoma
- 3 A 27-year-old woman presents with a single lump in the left breast that has been present for 6 months. On examination, there is a 3 cm smooth lump in the upper outer quadrant, mobile and not attached to the skin. What is the most likely diagnosis?
 - A Breast abscess
 - B Breast cyst
 - C Fibroadenoma
 - D Gynaecomastia
 - E Phyllodes tumour
- 4 A 34-year-old woman presents with 3 weeks of left nipple discharge, described as thick and greenish. No mass is palpable. Beta-hCG is negative. Most likely diagnosis?
 - A Breast cancer
 - B Duct ectasia
 - C Intraductal papilloma
 - D Lactating breast
 - E Prolactinoma
- 5 A 28-year-old woman comes to see you at the GP practice having noticed a lump in the right breast. She is concerned that it may be cancer. On examination, you find a smooth, mobile, non-tender mass around 2 cm in diameter in the upper outer quadrant of the breast. There are no overlying skin changes. Which would be the most appropriate action to take?
 - A Advise the patient to return if the lump gets bigger or starts becoming painful
 - B Prescribe antibiotics
 - C Reassure the patient that there is no evidence the lump is cancerous and discharge her
 - D Refer the patient for urgent mammography
 - E Refer the patient for urgent ultrasound scan
- 6 A 22-year-old woman presents to the GP complaining of a 2-week history of milky discharge from both her breasts. She is otherwise well. Clinical examination of the breasts is unremarkable and a pregnancy test is negative. Which of the following investigations would be indicated in the first instance?
 - A Dopamine levels
 - B Fine needle aspiration
 - C Mammogram
 - D MRI of the breast
 - E Prolactin levels

Answers 1-6

GET AHEAD MCQS

- 1 **D Fibrocystic disease** — Multiple painful nodular changes related to menstrual cycle are characteristic of fibrocystic disease.
- 2 **C Intraductal papilloma** — A small tumour in the lactiferous duct often presents with bloody discharge and no palpable lump.
- 3 **C Fibroadenoma** — A painless, mobile 'breast mouse' that is common in younger women.
- 4 **B Duct ectasia** — Greenish discharge from dilated subareolar ducts is typical of duct ectasia.
- 5 **E Refer the patient for urgent ultrasound scan** — Younger women have denser breasts, so ultrasound is more sensitive than mammography.
- 6 **E Prolactin levels** — Bilateral galactorrhoea suggests hyperprolactinaemia; serum prolactin is first-line test.

10 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQs

Crash Course MCQs

- 1 You are a house officer in breast surgery and are clerking in a 45-year-old woman for excision of a benign breast lump. She has no known past medical or surgical history. Which investigation is most highly recommended prior to her surgery?
 - A Full blood count
 - B Electrocardiogram (ECG)
 - C Chest X-ray
 - D Urea and electrolytes
 - E None of the above
- 2 A 45-year-old woman presents with a unilateral erythematous, itchy rash around her nipple and inversion of the nipple. Most likely diagnosis?
 - A Ductal carcinoma in-situ
 - B Infective mastitis
 - C Lobular carcinoma in-situ
 - D Medullary carcinoma
 - E Nipple eczema (dermatitis)
- 3 A 35-year-old woman has a lump in the upper outer quadrant of her right breast that becomes tender pre-menses. Most likely diagnosis?
 - A Localised fibroadenosis
 - B Fibroadenoma
 - C Cyst
 - D Fat necrosis
 - E Carcinoma
- 4 A 31-year-old woman has a nontender, rubbery, mobile lump in her left breast. Most likely diagnosis?
 - A Localised fibroadenosis
 - B Cyst
 - C Fat necrosis
 - D Fibroadenoma
 - E Carcinoma
- 5 A 67-year-old woman 2 weeks post breast-conserving surgery notices a painless lump beneath the incision, nontender, firm, well-circumscribed. Most likely diagnosis?
 - A Abscess
 - B Cystic disease
 - C Fat necrosis
 - D Fibroadenoma
 - E Recurrent cancer
- 6 A 35-year-old woman has a tense, mobile breast lump that yields straw-coloured fluid on fine-needle aspiration. Most likely diagnosis?
 - A Localised fibroadenosis
 - B Fibroadenoma
 - C Fat necrosis
 - D Carcinoma
 - E Cyst
- 7 A 34-year-old female smoker has a green discharge from both nipples, with nipple retraction and lumpiness. Most likely diagnosis?
 - A Mammary duct ectasia
 - B Duct papilloma
 - C Galactorrhoea
 - D Duct carcinoma
 - E Paget disease
- 8 A middle-aged woman has a serous discharge from one nipple on deep palpation. The breast otherwise appears normal. Most likely diagnosis?
 - A Mammary duct ectasia
 - B Duct papilloma
 - C Galactorrhoea
 - D Duct carcinoma
 - E Paget disease
- 9 A 28-year-old woman has a discrete tense, mobile breast lump suspicious of a cyst. Which investigation is most appropriate?
 - A Mammography
 - B CT chest
 - C MRI chest
 - D Ultrasound
 - E Chest X-ray
- 10 A 52-year-old woman has a large painless breast lump, fleshy with a tear-drop shape. Most likely diagnosis?
 - A Fibroadenoma
 - B Duct papilloma
 - C Phyllodes tumour
 - D Duct carcinoma
 - E Paget disease

Answers 1-10

CRASH COURSE MCQs

- 1 **E None of the above** — NICE guidance suggests no routine tests if no comorbidities/low-risk surgery.
- 2 **A Ductal carcinoma in-situ** — (Paget disease of nipple) Unilateral eczematous rash of nipple often indicates Paget (DCIS).

- 3 **A Localised fibroadenosis** — Cyclic breast pain/lumpiness, worse pre-menses.
- 4 **D Fibroadenoma** — 'Breast mouse': rubbery, mobile, painless.
- 5 **C Fat necrosis** — Benign post-trauma/surgery; can mimic recurrence but typically non-tender.
- 6 **E Cyst** — Discrete tense lump with straw-coloured fluid on aspiration.
- 7 **A Mammary duct ectasia** — Greenish discharge + retraction in smokers => duct ectasia.
- 8 **B Duct papilloma** — Serous or bloodstained discharge from a single duct.
- 9 **D Ultrasound** — Best to distinguish solid vs cystic, especially <35 yrs (dense breasts).
- 10 **C Phyllodes tumour** — Leaf-like tumour, can be large/fleshy, tear-drop shape.

Breast Cancer

Screening, risk and BRCA, DCIS and invasive cancer, staging, surgery, sentinel node biopsy and adjuvant therapy.

62

MCQS

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

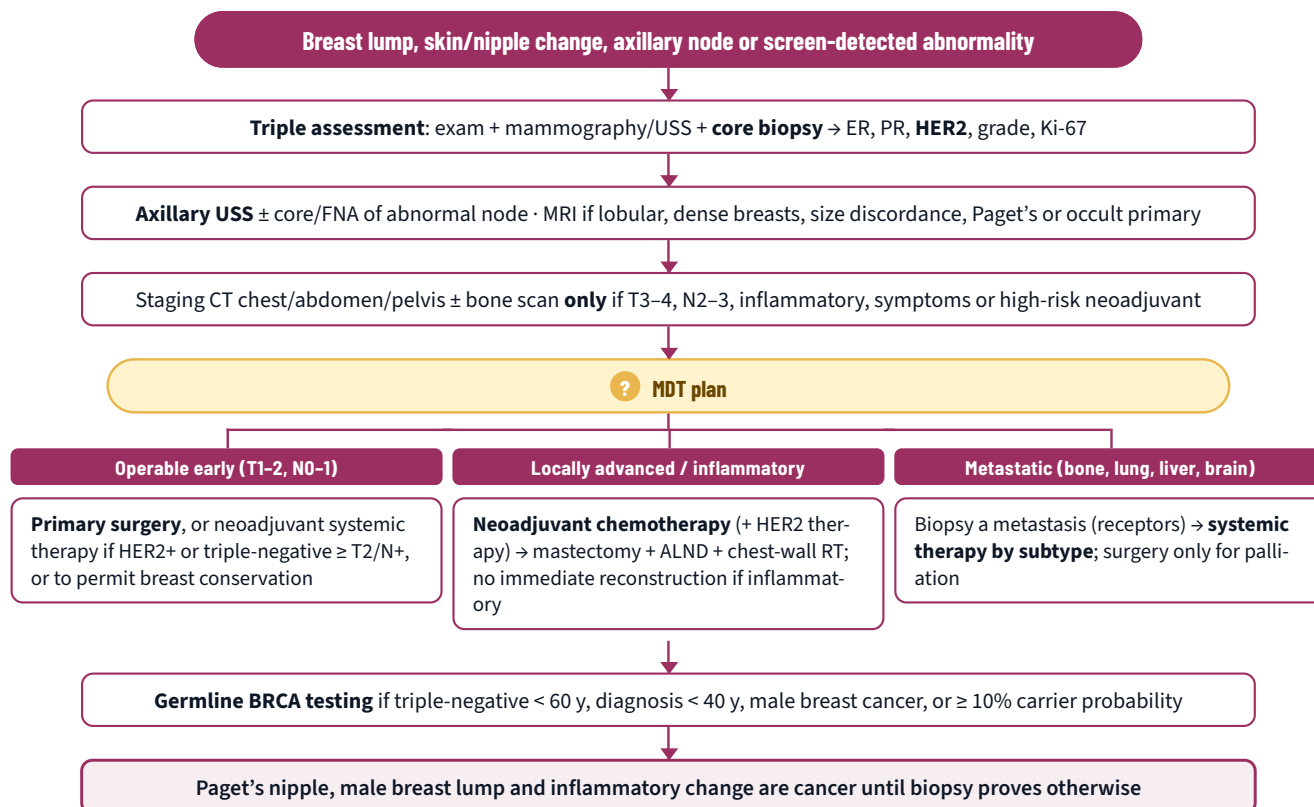
01	Management flowcharts	3 charts	page 60
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05

Step by step

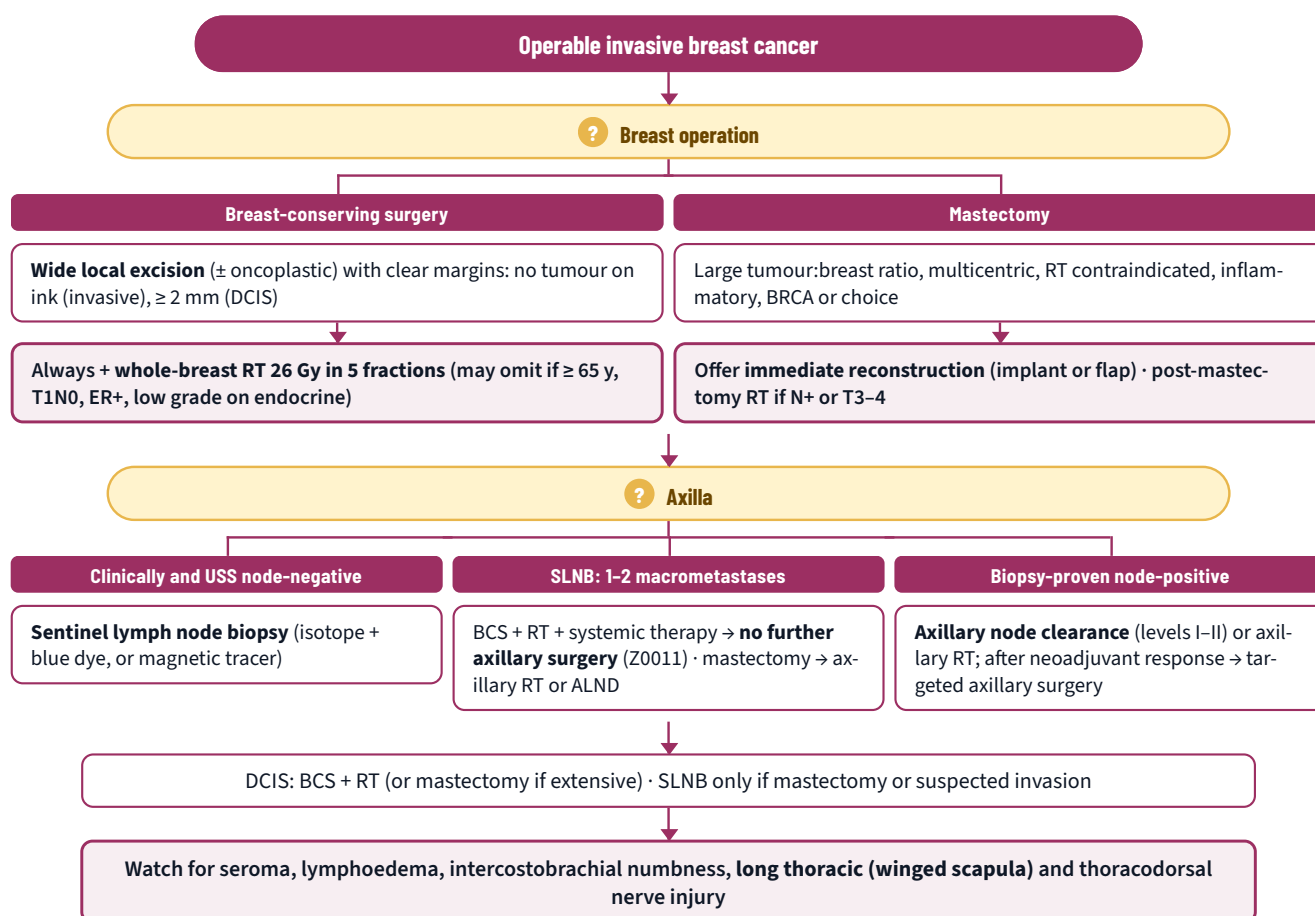
Suspected breast cancer: diagnosis and staging

NICE NG101 (2018, UPDATED) • NICE NG12 • ESMO 2024 EARLY BREAST CANCER



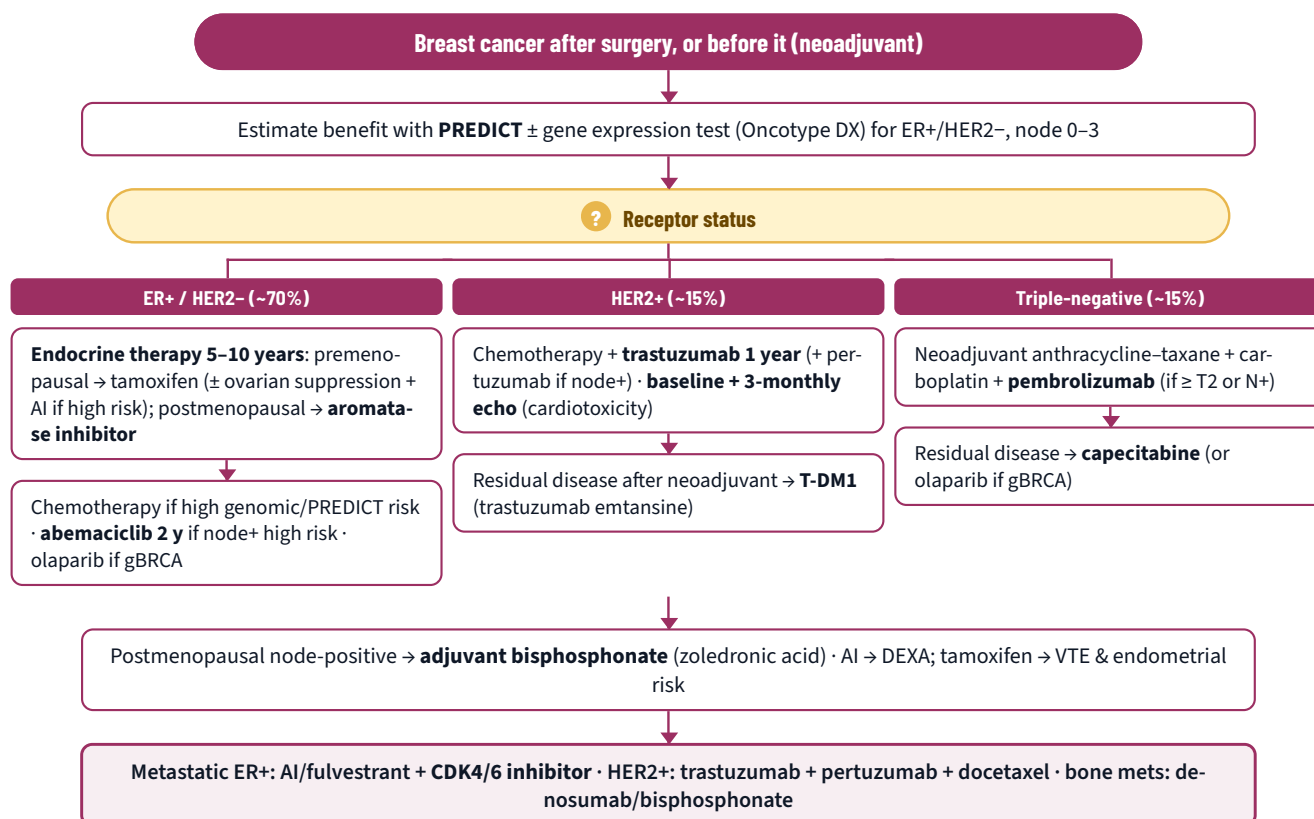
Breast and axillary surgery

NICE NG101 · ABS 2023 · Z0011 / AMAROS / FAST-FORWARD TRIALS



Adjuvant systemic therapy by receptor status

NICE NG101 · ESMO 2024 EARLY BREAST CANCER · NICE TA810/TA851/TA886



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1** A 55-year-old woman has a 3-month painless lump in the upper outer quadrant of the left breast and nipple retraction for 2 weeks. Core biopsy of a palpable axillary node shows invasive carcinoma, hormone-receptor negative; PET shows no metastasis. What is the best next step in management?
- A** Wide local excision
 - B** Simple mastectomy
 - C** Modified radical mastectomy
 - D** Adjuvant chemotherapy
 - E** Radiotherapy

- 2** 8 months after her surgery she has difficulty raising her arm and her medial scapula is prominent. What is the cause?
- A** Frozen shoulder
 - B** Serratus anterior denervation
 - C** Tear of the long head of biceps
 - D** Supraspinatus tendon injury
 - E** Thoracodorsal nerve injury

MINISTERIAL EXAM

Ministerial evaluation examinations

- 3** A 42-year-old female underwent radical mastectomy with axillary clearance. At her first follow-up, the inferior angle of her ipsilateral scapula translocates medially with any attempt at overhead activity. Injury to which of the following structures would most likely cause this abnormality?
- A** Spinal accessory nerve.
 - B** C8 and T1 nerve roots.
 - C** Upper and lower subscapular nerves.
 - D** Thoraco-dorsal nerve.
 - E** Long thoracic nerve.

Answers 1-3

- 1** **C Modified radical mastectomy** — Node-positive cancer with nipple involvement needs mastectomy with axillary clearance; neoadjuvant chemotherapy is often given first for ER-negative disease.
- 2** **B Serratus anterior denervation** — Winging of the scapula follows injury to the long thoracic nerve (serratus anterior) during axillary dissection; thoracodorsal injury weakens latissimus dorsi.
- 3** **E Long thoracic nerve.** — Radical mastectomy can injure the long thoracic nerve, causing scapular winging.

Bailey & Love MCQs

- 1** Which of the following statements are false? (Carcinoma of the breast)
- A** Hormone replacement therapy (HRT) increases the risk of breast cancer.
 - B** Inflammatory breast cancer is caused by infection in the breast.
 - C** Medullary carcinoma in the breast is associated with a poor prognosis.
 - D** Van Nuy's scoring system is used to decide further management after local excision in patients with ductal carcinoma in situ (DCIS).
 - E** Oestrogen receptor negative tumours are treated with tamoxifen or aromatase inhibitors.
- 2** Which of the following statements are true? (Carcinoma of the breast)
- A** Paget's disease of the breast is usually associated with an underlying invasive carcinoma.
 - B** The survival outcome following breast-conserving treatment and mastectomy are equal.
 - C** Positive resected margins are a significant predictor of recurrence after breast-conserving surgery.
 - D** Sentinel lymph node biopsy should not be offered in clinically node-negative axilla.
 - E** The pectoralis minor and major muscles are removed in Patey's mastectomy.

- 3** Which of the following statements are false? (Carcinoma of the Breast)
- A** Adjuvant chemotherapy is not given to post-menopausal women with ER-positive breast cancer.
 - B** Tamoxifen reduces the risk of contralateral breast cancer.
 - C** Cyclophosphamide chemotherapy is as effective as anthracycline- and taxane-based chemotherapy.
 - D** Neoadjuvant chemotherapy improves breast conservation rates in women with slightly larger primary tumour.

- 4** Which of the following statements are true? (Carcinoma of the Breast)
- A** Lymphoedema in the arm after treatment usually indicates a recurrence.
 - B** Radiotherapy after implant reconstruction of the breast gives poor cosmetic results.
 - C** Chemotherapy is safe in second and third trimester in women with pregnancy-associated breast cancer.
 - D** Male breast cancer carries a poor prognosis when compared with female breast cancer.

Answers 1-4

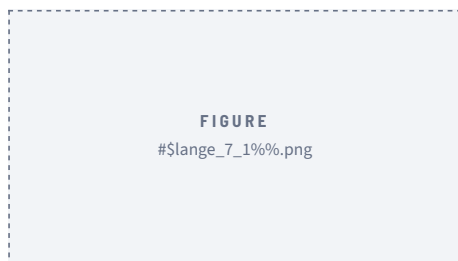
BAILEY & LOVE MCQS

- 1** **BCE** True: B, C, E — **UPDATED** Inflammatory breast cancer is not infective, medullary carcinoma has a better prognosis and ER-negative tumours do not respond to hormone therapy.
- 2** **ABC** True: A, B, C — **UPDATED** Paget's disease is usually linked to underlying invasive carcinoma, survival is equivalent between breast-conserving therapy and mastectomy, and positive margins increase recurrence risk.
- 3** **AC** True: A, C — **UPDATED** False: post-menopausal ER-positive women may still need chemotherapy, and anthracycline/taxane regimens beat cyclophosphamide alone. Neoadjuvant chemotherapy does increase breast conservation.
- 4** **BCD** True: B, C, D — Radiotherapy can worsen cosmetic outcomes post-implant, chemotherapy is safe in later pregnancy and male breast cancer often has a poorer prognosis.

13 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 55-year-old woman has suspicious calcifications on a screening mammogram but denies any breast masses or symptoms. Which of the following is the most likely diagnosis (based on Figure 6-3)?



- A** milk of calcium
 - B** lobular carcinoma in situ (LCIS) with or without an invasive component
 - C** ductal carcinoma in situ (DCIS) with or without an invasive component
 - D** involuting fibroadenoma
 - E** phyllodes tumor
- 2** Which of the following is the most appropriate next step in management (same mammogram scenario)?
- A** observation, with repeat mammogram in 6–12 months
 - B** ultrasound
 - C** biopsy
 - D** lumpectomy, radiation therapy, and sentinel lymph node (SLN) biopsy
 - E** total mastectomy

- 3** A 56-year-old woman presents to the physician's office with complaints of a new left breast mass. She denies any pain, nipple discharge, or skin dimpling. She has a prior history of breast cysts 5 years ago, treated by aspiration at that time. Her last mammogram was at age 53. Past history is pertinent for a 30 pack-year smoking history, prior total abdominal hysterectomy-bilateral salpingo-oophorectomy (TAH-BSO) at age 54 for leiomyomas, and current use of hormone replacement therapy (HRT). Family history is negative for breast disease. Examination reveals a firm, well-defined, mobile, 1.5-cm nodule in the upper outer quadrant of the left breast without any regional lymphadenopathy. Which of the following is the most appropriate next step in management?

- A** fine-needle aspiration (FNA) biopsy
- B** discontinuation of HRT and reexamination in 4–6 weeks
- C** breast imaging
- D** open surgical biopsy
- E** core needle biopsy

4 A 56-year-old woman presents to the clinic for routine health screening. She is concerned about the development of breast cancer. Her past history includes fibrocystic changes with atypical ductal hyperplasia and a fibroadenoma, and her family history is positive only for breast cancer in her mother diagnosed at age 85. Mammograms show dense breasts without suspicious findings. Which of the following is the most common risk factor in evaluating women for breast cancer?

- A fibrocystic changes with atypical ductal hyperplasia
- B alcohol consumption
- C positive family history
- D HRT
- E age

5 A 42-year-old woman returns to the clinic following an uneventful biopsy for a well-defined, mobile mass. The pathology report describes the mass as a fibroadenoma, but lobular carcinoma in situ (LCIS) is identified in the adjacent breast parenchyma extending to the resection margin. With no other risk factors and negative family history, what is the most appropriate management?

- A re-excision of the biopsy cavity to gain negative margins of resection
- B ipsilateral mastectomy
- C contralateral breast biopsy
- D observation including examinations and mammography
- E bilateral total mastectomies

6 A 35-year-old woman with a strong family history of breast cancer (including a sister with a deleterious BRCA1 mutation) is found to carry the same mutation. Which range represents the lifetime risk of breast cancer that should be quoted to her?

- A 0–30%
- B 10–40%
- C 20–50%
- D 50–80%
- E 70–100%

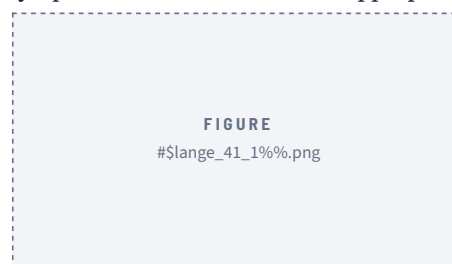
7 For the same patient with a BRCA1 mutation, which of the following management strategies is an accepted option for her high-risk status?

- A Yearly self-breast examination
- B Six-monthly mammography
- C Bilateral breast irradiation
- D Prophylactic unilateral mastectomy
- E Chemoprevention with tamoxifen

8 A 65-year-old woman with a 2.5-cm mass in the upper outer quadrant of the left breast and a palpable axillary node undergoes evaluation. Imaging shows no other lesions. Which of the following is the most appropriate management option for locoregional control yielding results equally effective as mastectomy?

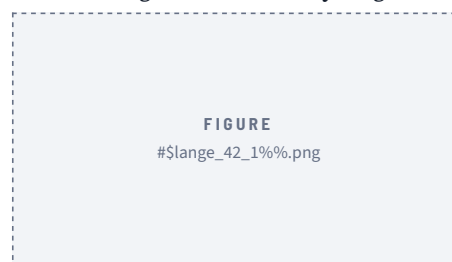
- A radical mastectomy
- B lumpectomy, irradiation, and axillary node dissection
- C lumpectomy and axillary node dissection
- D irradiation of the breast and axilla
- E quadrantectomy, irradiation, and axillary node dissection

9 The same patient asks if she is a candidate for SLN biopsy instead of complete axillary dissection to avoid lymphedema. What is the most appropriate answer?



- A Yes, and if the SLN is positive, then a complete axillary dissection should be performed.
- B Yes, and if the SLN is negative, then an axillary dissection can be avoided.
- C No, because the success of SLN biopsy in patients over age 60 is decreased.
- D No, because SLN biopsy is contraindicated when a palpable axillary node is suspicious for metastatic disease.
- E No, because SLN biopsy is contraindicated for tumors greater than 2 cm.

10 A 65-year-old woman with an abnormal screening mammogram but no symptoms undergoes imaging. The MLO and magnification CC views of the right breast reveal clustered microcalcifications. Which of the following is the most likely diagnosis?



- A milk of calcium
- B LCIS with or without an invasive component
- C DCIS with or without an invasive component
- D involuting fibroadenoma
- E phyllodes tumor

- 11** Following the findings in the previous question, what is the most appropriate next step in management?
- A** observation, with repeat mammogram in 6 months
 - B** observation, with repeat mammogram on an annual basis
 - C** biopsy
 - D** lumpectomy, radiation therapy, and SLN biopsy
 - E** total mastectomy
- 12** An 83-year-old woman undergoing screening mammography is found to have a mass in the lateral right breast. She is asymptomatic. Which of the following is the most likely diagnosis?
- FIGURE**

#\$lange_44_1%%.png
- A** papilloma
 - B** invasive carcinoma
 - C** cystosarcoma phyllodes
 - D** DCIS
 - E** fat necrosis
- 13** What is the most appropriate next step in management for the mass detected in the previous question?
- A** incisional biopsy
 - B** needle biopsy
 - C** lumpectomy, axillary dissection, and irradiation
 - D** total mastectomy
 - E** modified radical mastectomy

Answers 1-13

LANGE MCQs

- 1** **D involuting fibroadenoma** — Coarse popcorn-like calcifications are characteristic of a degenerating fibroadenoma.
- 2** **A observation, with repeat mammogram in 6–12 months** — Characteristic benign calcifications require only follow-up imaging to confirm stability.
- 3** **C breast imaging** — Imaging (diagnostic mammography ± sonography) is required before biopsy.
- 4** **E age** — Age is the most common risk factor even though many risk factors exist.
- 5** **D observation including examinations and mammography** — LCIS is a marker for bilateral risk; close observation is appropriate.
- 6** **D 50–80%** — **UPDATED** BRCA1 carriers have a lifetime breast cancer risk of about 60–70% (quoted as 50–80%), with a high risk of ovarian cancer too.
- 7** **E Chemoprevention with tamoxifen** — **UPDATED** Options for BRCA carriers are annual MRI (± mammography) surveillance, chemoprevention (tamoxifen) and bilateral risk-reducing mastectomy ± salpingo-oophorectomy; unilateral mastectomy is not enough.
- 8** **B lumpectomy, irradiation, and axillary node dissection** — Breast-conserving therapy with lumpectomy plus radiation is equivalent to mastectomy in early-stage disease.
- 9** **D No, because SLN biopsy is contraindicated when a palpable axillary node is suspicious for metastatic disease.** — A palpable suspicious node contraindicates SLN biopsy.
- 10** **C DCIS with or without an invasive component** — The mammographic features (clustered, irregular calcifications) are most consistent with DCIS.
- 11** **C biopsy** — Biopsy is needed to establish a histologic diagnosis.
- 12** **B invasive carcinoma** — The mammographic features of an irregular, stellate mass in an older woman are most consistent with invasive carcinoma.
- 13** **B needle biopsy** — A needle (FNA or core) biopsy is indicated to obtain a tissue diagnosis.

9 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 65-year-old nulliparous woman presents to your clinic with a lump in her left breast, which was discovered 7 months ago. On examination you find a hard, ill-defined, non-tender, 3.5 cm lump behind the left nipple. The patient has also had bloody, non-purulent discharge from a single duct on the left nipple for over 3 months. The most likely diagnosis here is

 - A Mammary duct ectasia
 - B Breast carcinoma
 - C Duct papilloma
 - D Periductal mastitis
 - E Acute mastitis
- 2** A 67-year-old woman, with a 25-year smoking history, on hormone replacement therapy, presents to clinic expressing concerns regarding an increase in the size of her right breast over the last 4 months. On examination, you find a non-tender, mobile, lobulated 10 cm mass with relatively smooth edges in the right breast. The right breast is significantly larger than the left and has a 'teardrop' appearance and the skin looks normal. The most likely diagnosis here is

 - A Paget's disease of the nipple
 - B Inflammatory breast carcinoma
 - C Breast abscess
 - D Malignant phylloides tumour
 - E Fibroadenoma
- 3** A worried 59-year-old city worker arrives at your clinic with a 1-month history of having noticed a non-itchy, persistent, burning rash in the region of her right breast. On examination, you find the right nipple and the skin overlying the areola to be red and eczematous. The most likely cause is

 - A Breast abscess
 - B Malignant phyllodes tumour
 - C Paget's disease of the nipple
 - D Basal cell carcinoma
 - E Mastitis
- 4** You are attending a breast multidisciplinary team (MDT) meeting where the core biopsy histology results of a suspicious breast lesion are being discussed in a 55-year-old woman presenting with a right sided breast lump. The histopathologist states that the breast lesion possesses 'B5b' histology features. What is the most likely diagnosis?

 - A Fibroadenoma
 - B Benign breast cyst
 - C Ductal carcinoma in-situ
 - D Invasive breast carcinoma
 - E No breast abnormality
- 5** A 60-year-old woman was found to have one focal area of microcalcification (approximately 20 mm in diameter) in the left breast. A stereotactic core biopsy of this area was taken for histological assessment, which revealed low-grade ductal carcinoma in situ. In light of this, what would be the most appropriate treatment modality for this patient?

 - A Mastectomy
 - B Mastectomy + postoperative radiotherapy
 - C Wide local excision + postoperative radiotherapy
 - D Mastectomy + axillary clearance + postoperative radiotherapy
 - E Wide local excision + axillary clearance + postoperative radiotherapy
- 6** A 47-year-old woman is diagnosed with a unifocal 2.5-cm grade 3 invasive ductal carcinoma of the right breast with a clinically and sonographically normal axilla. Following MDT discussion, which of the following is the most appropriate management plan?

 - A Wide local excision
 - B Wide local excision and axillary clearance
 - C Mastectomy and sentinel node biopsy
 - D Wide local excision and sentinel node biopsy
 - E Mastectomy and axillary clearance
- 7** A 46-year-old man is diagnosed with an oestrogen receptor positive invasive ductal carcinoma of the right breast after having discovered a lump 3 months before. The patient is found to have multiple involved axillary lymph nodes and the tumour is of an aggressive phenotype. The most appropriate treatment option for this patient is

 - A Cytotoxic chemotherapy and Tamoxifen but no surgery
 - B Mastectomy + axillary clearance + systemic chemotherapy + radiotherapy and tamoxifen
 - C Wide local excision and Tamoxifen only
 - D Mastectomy + postoperative radiotherapy only
 - E Palliative care programme
- 8** In which one of the following scenarios is the complication of lymphoedema of the arm more likely to occur after resection of a breast tumour and axillary clearance?

 - A Mastectomy + axillary clearance + postoperative radiotherapy to the chest wall
 - B Mastectomy + axillary clearance + systemic chemotherapy
 - C Mastectomy + axillary clearance + postoperative radiotherapy to the axilla
 - D Mastectomy + axillary clearance
 - E Mastectomy + postoperative radiotherapy

- 9** A 58-year-old postmenopausal woman has been seen in clinic following discovery of a 3 cm, non-tender, irregular, firm lump in the upper outer quadrant of the left breast. Mammography and ultrasound imaging respectively reveal that the lump has areas of calcification and is a solid mass. The most appropriate course of action is
- A** Repeat mammography and ultrasound scans in 6 months
 - B** Reassure and discharge
 - C** Repeat mammography and ultrasound scans in 3 months
 - D** Fine needle aspiration to ensure that the lump is not really fluid filled
 - E** Core biopsy

Answers 1-9

SBA MCQS

- 1** **B Breast carcinoma** — A postmenopausal woman with a hard lump and bloody nipple discharge is highly suspicious for breast carcinoma.
- 2** **D Malignant phylloides tumour** — Rapid breast enlargement with a large, mobile mass suggests a malignant phylloides tumour.
- 3** **C Paget's disease of the nipple** — Eczematous nipple changes with redness and lymphadenopathy favor Paget's disease of the nipple.
- 4** **D Invasive breast carcinoma** — A B5b histology result confirms invasive breast carcinoma.
- 5** **C Wide local excision + postoperative radiotherapy** — For DCIS, wide local excision with radiotherapy is the preferred treatment for small, low-grade lesions.
- 6** **D Wide local excision and sentinel node biopsy** — UPDATED
A unifocal tumour suitable for breast conservation with a negative axilla: wide local excision with sentinel node biopsy, then radiotherapy and systemic therapy by histology.
- 7** **B Mastectomy + axillary clearance + systemic chemotherapy + radiotherapy and tamoxifen** — In male breast cancer with nodal involvement, mastectomy with adjuvant therapy is indicated.
- 8** **C Mastectomy + axillary clearance + postoperative radiotherapy to the axilla** — Axillary clearance with radiotherapy to the axilla carries the highest risk of lymphoedema.
- 9** **E Core biopsy** — A core biopsy is indicated for a suspicious solid mass with calcifications.

4 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** A 57-year-old woman has noticed a lump in the right breast for 4 weeks. She is otherwise fit and well and is not on any regular medications. On triple assessment she is found to have two foci of invasive ductal carcinoma with widespread ductal carcinoma in situ and FNAB of enlarged axillary lymph nodes confirms tumour involvement. Which is the single most appropriate surgical treatment for this patient?
 - A** Mastectomy and axillary node clearance
 - B** Mastectomy and radiotherapy
 - C** Mastectomy and sentinel node biopsy
 - D** Wide local excision and axillary node clearance
 - E** Wide local excision and sentinel node biopsy
- 2** A 40-year-old premenopausal woman has recently been diagnosed with breast cancer. She has had a wide local excision and sentinel node biopsy. Histology of the tumour reveals a 1.8cm grade 2 invasive ductal carcinoma which is oestrogen receptor positive. The sentinel node did not contain tumour. Which is the single most appropriate hormonal treatment for this patient?
 - A** Anastrozole
 - B** Letrozole
 - C** None
 - D** Tamoxifen
 - E** Goserelin

- 3** A 70-year-old woman has had a lump in her left breast for 12 months. This lump has gradually increased in size and she has recently noticed that the shape of her breast has changed. She denies any other symptoms. She has evidence of skin dimpling in the outer aspect of the left breast and has a hard fixed mass with palpable axillary lymph nodes. Which is the single most likely diagnosis?
- A Breast cancer
 - B Fat necrosis
 - C Fibrocystic disease
 - D Mastitis
 - E Surgical scar
- 4** A 52-year-old woman has noticed 10 days of diffuse pain, redness, and swelling of the right breast. A course of antibiotics for the last 5 days has had no effect. Her nipple has changed shape with some associated discharge. A mammogram last year was normal but there is skin and subcutaneous oedema seen on ultrasound scan which is uncomfortable. Her breast is swollen, tender, and firm, as shown in Figure 5.3. Which is the single most likely diagnosis?
- FIGURE**
#5@@@Oxford Breast and endocrine@@_12_1%%.png
- A Breast abscess
 - B Graves' disease
 - C Inflammatory breast cancer
 - D Insect bite
 - E Mastitis

Answers 1-4

OXFORD MCQS

- 1** **A Mastectomy and axillary node clearance** — Mastectomy with axillary node clearance is indicated due to multifocal widespread disease and lymph node involvement.
- 2** **D Tamoxifen** — Tamoxifen is the most appropriate hormonal treatment for a premenopausal woman with ER-positive breast cancer.
- 3** **A Breast cancer** — Skin dimpling and fixed mass with lymphadenopathy strongly suggest breast cancer.
- 4** **C Inflammatory breast cancer** — Diffuse pain, redness, and lack of response to antibiotics suggest inflammatory breast cancer.

20 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1** A 62-year-old woman presents with invasive ductal carcinoma of the right breast. Which of the following findings would still allow her to receive breast-conservation surgery (partial mastectomy)?
- A Diffuse suspicious microcalcifications throughout the breast
 - B Multifocal disease
 - C Previous treatment of a breast cancer with lumpectomy and radiation
 - D Large tumor relative to breast size
 - E Persistently positive margins after multiple reexcisions of the breast cancer
- 2** A 40-year-old woman presents with a 6-month history of erythema and edema of the right breast with palpable axillary lymphadenopathy. A punch biopsy of the skin reveals neoplastic cells in the dermal lymphatics. She has a full staging workup, including bilateral mammogram, ultrasound of the bilateral breasts and axillae, and CT scan of the chest, abdomen, and pelvis. She has no other sites of disease. Which of the following is the best sequence of steps in her management?
- A A course of nafcillin to treat the overlying cellulitis and then neoadjuvant chemotherapy for breast cancer
 - B Modified radical mastectomy followed by adjuvant chemotherapy
 - C Modified radical mastectomy followed by hormonal therapy
 - D Combined modality chemotherapy and radiation therapy to the right breast with surgery reserved for residual disease
 - E Combined modality therapy with chemotherapy, surgery, and radiation

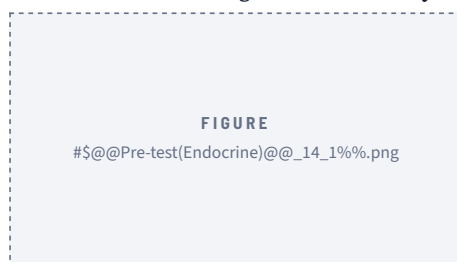
3 A 35-year-old woman presents with a lump in the left breast. Her family history is negative for breast cancer. On examination the mass is rubbery, mobile, and nontender to palpation. There are no overlying skin changes and the axilla is negative for lymphadenopathy. An ultrasound demonstrates a simple 1-cm cyst in the area of the palpable mass in the left breast. Which of the following represents the most appropriate management of this patient?

- A Reassurance and reexamination
- B Immediate excisional biopsy
- C Aspiration of the cyst with cytologic analysis
- D Needle localized excisional biopsy
- E Mammography

4 A 45-year-old woman is found to have suspicious appearing calcifications in the right breast on a screening mammogram. Stereotactic biopsy of the calcifications shows lobular carcinoma in situ (LCIS). On examination, both breasts are dense without palpable masses. The cervical, supraclavicular, and bilateral axilla are negative for lymphadenopathy. Which of the following is the most appropriate management of this patient?

- A Frequent self-breast examinations and yearly screening mammograms
- B Chemotherapy
- C Radiation
- D Right total mastectomy with sentinel lymph node biopsy
- E Bilateral modified radical mastectomy

5 A 14-year-old African American girl has her right breast removed because of a large mass. The tumor weighs 1400 g and has a bulging, very firm, lobulated surface with a whorl-like pattern, as illustrated here. Which of the following is the most likely diagnosis?



- A Cystosarcoma phyllodes
- B Intraductal carcinoma
- C Malignant lymphoma
- D Fibroadenoma
- E Juvenile hypertrophy

6 A 33-year-old pregnant woman notices a persistent, painless lump in the left breast. On examination the left breast has a single mobile mass without evidence of skin changes or lymphadenopathy in the neck or axilla. An ultrasound demonstrates a solid, 1-cm mass in the upper outer quadrant of the breast. A core-needle biopsy shows invasive ductal carcinoma. The patient is in her first trimester of pregnancy. Which of the following is the most appropriate management of this patient?

- A Termination of the pregnancy followed by total mastectomy
- B Immediate administration of chemotherapy followed by modified radical mastectomy after delivery of the baby
- C Segmental mastectomy, sentinel lymph node biopsy, and adjuvant radiation therapy shortly after surgery
- D Total mastectomy with sentinel lymph node biopsy
- E Modified radical mastectomy

7 A 40-year-old woman presents with a rash involving the nipple-areola complex for the last month with associated itching. On physical examination there is crusting and ulceration of the nipple with surrounding erythema involving the areola and surrounding skin, no palpable breast masses, and no cervical or axillary lymphadenopathy. Which of the following is the most appropriate next step in the management of this patient?

- A Reexamine the patient in 1 month
- B Corticosteroid cream to the affected area
- C Administration of oral antibiotics
- D Mammogram and biopsy of the affected area
- E Modified radical mastectomy

8 A 35-year-old female has a 6-month history of a 2-cm breast mass which was biopsied and consistent with a fibroadenoma. She now comes into the office reporting that the mass has rapidly enlarged and is now 10 cm. Which of the following is the most appropriate initial management strategy for this lesion?

- A Wide local excision with a rim of normal tissue
- B Lumpectomy and axillary lymphadenectomy
- C Modified radical mastectomy
- D Excision and postoperative radiotherapy
- E Excision, postoperative radiotherapy, and systemic chemotherapy

9 A 41-year-old woman has noted unilateral nipple discharge from her right breast. No associated mass is identified on ultrasound or mammogram. Which of the following is the next appropriate step in her management?

- A No further intervention
- B Administration of antibiotics
- C Administration of tamoxifen
- D Major duct excision
- E Nipple areolar excision with central lumpectomy

- 10** A 36-year-old woman whose mother has just undergone treatment for breast cancer is asking about how this affects her and what can be done to lessen her chances of having the disease. Which of the following is the least significant risk factor for breast cancer?
- A Dietary fat intake
 - B Paternal relative with breast cancer BRCA1 mutation
 - C Excessive estrogen exposure—early menarche, late menopause, nulliparity
 - D Previous biopsy with atypical hyperplasia
 - E Exposure to ionizing radiation
- 11** Tumor not palpable, clinically positive ipsilateral axillary lymph nodes fixed to one another; no evidence of metastases.
- A Stage 0
 - B Stage I
 - C Stage II
 - D Stage III
 - E Stage IV
- 12** Tumor 4.0 cm; clinically positive, movable axillary ipsilateral lymph nodes; no evidence of metastases.
- A Stage 0
 - B Stage I
 - C Stage II
 - D Stage III
 - E Stage IV
- 13** Tumor 2.1 cm, clinically negative lymph nodes; no evidence of metastases. Final pathology shows only ductal carcinoma in situ.
- A Stage 0
 - B Stage I
 - C Stage II
 - D Stage III
 - E Stage IV
- 14** Tumor not palpable, but breast diffusely enlarged and erythematous, clinically positive supraclavicular nodes; no evidence of metastases.
- A Stage 0
 - B Stage I
 - C Stage II
 - D Stage III
 - E Stage IV
- 15** Tumor 0.5 cm, clinically negative lymph nodes, pathological lumbar fracture.
- A Stage 0
 - B Stage I
 - C Stage II
 - D Stage III
 - E Stage IV
- 16** A 42-year-old woman has a mammogram that demonstrates diffuse suspicious mammographic calcifications suggestive of multicentric disease. Biopsy of one of the lesions reveals ductal carcinoma in situ (DCIS). What is the most appropriate surgical management?
- A Lumpectomy with axillary lymph node dissection and radiation therapy
 - B Wide local excision
 - C Lumpectomy with adjuvant radiation therapy
 - D Modified radical mastectomy
 - E Simple mastectomy with sentinel lymph node biopsy
- 17** A 49-year-old woman has a 6-cm palpable mass that is biopsy-proven ductal adenocarcinoma. She undergoes neoadjuvant chemotherapy which reduces the tumor to 3 cm in size. However, she has palpable axillary lymph nodes; FNA demonstrates adenocarcinoma. She desires breast-conservation therapy if possible. What is the most appropriate surgical option?
- A Modified radical mastectomy
 - B Radical mastectomy
 - C Wide local excision
 - D Lumpectomy with axillary lymph node dissection and radiation therapy
 - E Simple mastectomy with sentinel lymph node biopsy
- 18** A neglected 82-year-old woman presents with a locally advanced breast cancer that is invading the pectoralis major muscle over a broad base. She is otherwise in good health. What is the most appropriate treatment?
- A Lumpectomy with adjuvant radiation therapy
 - B Wide local excision
 - C Lumpectomy with axillary lymph node dissection and radiation therapy
 - D Simple mastectomy with sentinel lymph node biopsy
 - E Radical mastectomy
- 19** What is the recommended chemotherapy regimen for breast cancer?
- A Methotrexate/vinblastine/Herceptin
 - B Cisplatin/cyclophosphamide/doxorubicin
 - C Adriamycin/cyclophosphamide/Taxol (AC followed by T) or Taxol/cyclophosphamide/Herceptin/Pertuzumab (TCHP)
 - D Tamoxifen and radiation therapy
 - E Letrozole and trastuzumab
- 20** What are indications for post-mastectomy radiation therapy?
- A Positive estrogen receptor status
 - B Tumor <2 cm with clear margins
 - C ≥ 4 positive lymph nodes or tumor >5 cm in size
 - D Ductal carcinoma in situ (DCIS) with no lymphovascular invasion
 - E Triple negative breast cancer without lymph node involvement

Answers 1-20

PRETEST MCQs

- 1 **B Multifocal disease** — Multifocal disease can still allow breast-conservation surgery if lesions are small relative to breast size.
- 2 **E Combined modality therapy with chemotherapy, surgery, and radiation** — **UPDATED** Inflammatory breast cancer (dermal lymphatic invasion) is treated with trimodality therapy: neoadjuvant chemotherapy, modified radical mastectomy, then radiotherapy.
- 3 **A Reassurance and reexamination** — Most cysts in the breast are benign; reassurance and reexamination are appropriate when there are no suspicious features.
- 4 **A Frequent self-breast examinations and yearly screening mammograms** — Lobular carcinoma in situ (LCIS) is considered a risk factor for invasive breast carcinoma and is managed with frequent self-exams and yearly mammograms.
- 5 **D Fibroadenoma** — Fibroadenomas are well-demarcated and firm and can be large in young African American girls.
- 6 **D Total mastectomy with sentinel lymph node biopsy** — Immediate surgery, such as total mastectomy with sentinel lymph node biopsy, is recommended during the first trimester.
- 7 **D Mammogram and biopsy of the affected area** — This patient has Paget disease of the breast and needs a mammogram and biopsy to confirm and manage accordingly.
- 8 **A Wide local excision with a rim of normal tissue** — Large or rapidly enlarging fibroadenomas may represent phyllodes tumors, which require wide excision with a rim of normal tissue to reduce recurrence risk.
- 9 **D Major duct excision** — Pathologic nipple discharge should be further investigated with duct excision if no mass is identified.
- 10 **A Dietary fat intake** — Dietary fat intake has not been strongly linked to breast cancer risk compared to other factors.
- 11 **D Stage III** — Fixed or matted nodes are associated with poor prognosis, corresponding to Stage III.
- 12 **C Stage II** — Tumor >2cm but ≤5cm with positive lymph nodes corresponds to Stage II.
- 13 **A Stage 0** — Ductal carcinoma in situ is stage 0, as it is in situ.
- 14 **D Stage III** — Inflammatory breast cancer with fixed supraclavicular nodes is Stage III.
- 15 **E Stage IV** — Pathological lumbar fracture indicates distant metastasis, corresponding to Stage IV.
- 16 **E Simple mastectomy with sentinel lymph node biopsy** — Multicentric DCIS is best treated with simple mastectomy and sentinel lymph node biopsy.
- 17 **D Lumpectomy with axillary lymph node dissection and radiation therapy** — Lumpectomy with axillary dissection and radiation is appropriate in patients desiring breast conservation with confirmed nodal metastasis.
- 18 **E Radical mastectomy** — Radical mastectomy may be indicated in cases of extensive invasion of the pectoralis major muscle.
- 19 **C Adriamycin/cyclophosphamide/Taxol (AC followed by T) or Taxol/cyclophosphamide/Herceptin/Pertuzumab (TCHP)** — **UPDATED** Standard regimens are anthracycline–cyclophosphamide followed by a taxane, with trastuzumab/pertuzumab added for HER2-positive disease (e.g. TCHP).
- 20 **C ≥4 positive lymph nodes or tumor >5 cm in size** — Post-mastectomy radiation therapy is indicated when the risk of locoregional recurrence is high, such as in patients with 4 or more positive lymph nodes or large primary tumors >5 cm.

2 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1 A 58-year-old woman presents with itchy, dry skin around the left areola. On examination, the skin is dry and cracked, and there appears to be a 1 cm nodule underlying the affected area. What is the most likely diagnosis?
 - A Atopic dermatitis
 - B Fibrocystic disease
 - C Mondor disease
 - D Paget disease
 - E Peau d'orange
- 2 A 46-year-old pre-menopausal woman has a 5 cm lump over her right breast which has been confirmed to be an invasive tumour. She is otherwise fit and well. Which is the next most appropriate course of action?
 - A Mastectomy
 - B Radiotherapy
 - C Tamoxifen
 - D Trastuzumab
 - E Wide local excision

Answers 1-2

GET AHEAD MCQs

- 1 **D Paget disease** — Eczematous changes of the nipple plus a subareolar lump suggest Paget disease of the breast.
- 2 **A Mastectomy** — Large tumours (>4 cm) typically require mastectomy rather than breast-conserving surgery.

7 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQs

Crash Course MCQs

- 1 A 57-year-old woman with strong family history of breast cancer is told certain chromosomes are linked. Which chromosome is most associated?
- A Chromosome 14
B Chromosome 15
C Chromosome 16
D Chromosome 17
E Chromosome 18
- 2 A 45-year-old woman asks about screening mammograms in the UK. At what ages do women undergo screening as part of the National Programme?
- A 50-70
B 45-59
C 55-69
D 46-60
E 56-70
- 3 A 54-year-old woman with a 5 cm right breast lump and fixed axillary nodes, no supraclavicular nodes, moderately differentiated adenocarcinoma, no distant mets. What is the TNM stage?
- A Stage 1
B Stage 2
C Stage 3
D Stage 4
E Stage 5
- 4 A 52-year-old woman with a 4 cm breast lump, no palpable nodes. Wide excision and clearance show well-differentiated tumour, no lymph involvement. Approx 5-year survival?
- A 84%
B 71%
C 59%
D 48%
E 18%
- 5 A 64-year-old woman with a painless breast lump, skin dimpling, nipple retraction. Biopsy: adenocarcinoma with fibrosis. Most common type?
- A Ductal
B Lobular
C Medullary
D Colloid
E Sarcoma
- 6 A 39-year-old woman has left-sided breast cancer, ER-, well-differentiated, ~6.5 cm, ipsilateral nodes, no distant mets. Next step?
- A Neoadjuvant chemo then surgery + axillary clearance
B Primary surgery without axillary dissection
C Radiotherapy
D Tamoxifen
E Herceptin
- 7 A middle-aged woman has a scaly red rash on her nipple/areola, sore, bleeds occasionally. Most likely diagnosis?
- A Mammary duct ectasia
B Duct papilloma
C Galactorrhoea
D Duct carcinoma
E Paget disease

Answers 1-7

CRASH COURSE MCQs

- 1 **D Chromosome 17** — BRCA1 on 17q, BRCA2 on 13q linked with breast cancer.
- 2 **A 50-70** — **UPDATED** The NHS Breast Screening Programme invites women aged 50 to 70 (up to their 71st birthday) for mammography every 3 years.
- 3 **C Stage 3** — Large tumour + fixed ipsilateral axillary nodes, no metastasis => Stage 3.
- 4 **B 71%** — **UPDATED** A 4-cm node-negative well-differentiated tumour is T2N0 (stage IIA); about 70% 5-year survival in the source (higher with modern adjuvant therapy).

- 5** **A Ductal** — Most common histological type is invasive ductal carcinoma.
- 6** **A Neoadjuvant chemo then surgery + axillary clearance** — Stage III ER- tumour => neoadjuvant chemo + definitive surgery.
- 7** **E Paget disease** — Scaly erythematous nipple lesion is classic Paget's of breast (DCIS).

Breast EMQs

Extended matching questions on breast lumps, nipple discharge, investigation and treatment of breast cancer.

0

MCQS

75

EMQ SCENARIOS

6

SOURCES

0

REPEATED IDEAS

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06

Local Exams EMQs

FORMATIVE

College formative examinations

THEME 1 Breast Condition

- A Mastitis
- B Ductal carcinoma
- C Fibroadenoma
- D Phyllodes tumor
- E Ductal ectasia
- F Breast abscess

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A lactating woman presents with a tender, erythematous, and warm area in the breast. There is no fluctuation. Bilateral axillary lymph nodes are tender.
- 2 A 46-year-old woman presents with a painless, hard lump in the upper outer quadrant of the breast, along with palpable ipsilateral axillary lymphadenopathy.
- 3 A woman in her early 20s presents with a 2 cm, firm, freely mobile lump in the breast.

Answers • Theme 1

LOCAL EXAMS EMQS

Theme 1 • Breast Condition

- 1 **A Mastitis** — Common during lactation. Pain, redness, and tenderness with systemic signs are key features.

- 2 **B Ductal carcinoma** — Hard, painless mass with lymph node involvement in a middle-aged woman is suggestive of malignancy.
- 3 **C Fibroadenoma** — Young women often present with mobile, well-circumscribed benign lumps like fibroadenoma.

Bailey & Love EMQs

THEME 1 Breast Conditions – Diagnosis and Investigations

- A ANDI (Aberration of normal development and involution)
- B BRCA 1 and 2 tests
- C Breast abscess
- D Breast cyst
- E Core needle biopsy
- F Duct ectasia
- G Fibroadenoma
- H Galactocele
- I Gynaecomastia
- J Male breast cancer
- K MRI scan of the breasts
- L Phyllodes tumour
- M Vacuum-assisted biopsy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 26-year-old breast-feeding mother presents as an emergency with pain, swelling in the right breast and fever for 2 days.
- 2 A 58-year-old woman underwent a screening mammogram and was recalled for further assessment. She was found to have fine linear branching micro-calcifications.
- 3 A 49-year-old woman woke up to find a lump in the right breast with pain. She is very anxious.
- 4 A 35-year-old woman presents with breast pain and lumpiness in both breasts. It is particularly worse before her periods.
- 5 A 32-year-old woman attends the breast clinic with a strong family history of breast cancer. Two of her sisters, her mother and her maternal aunt's daughter had breast cancer at an early age. She wants to know her risk of getting breast cancer.
- 6 A 42-year-old woman presented with a painless mass in the right breast, which has been rapidly increasing in size. On examination she has a large mass in the right breast with skin stretched and shiny over it.
- 7 A 48-year-old woman feels a hard lump that is painless and gradually increasing in size over the past 4 months. Her mammogram shows a spiculated mass in the right breast.
- 8 A 26-year-old young woman presents with a painless mass in the left breast. She stopped breastfeeding 1 month ago, following which she noticed this lump.

- 9 A 14-year-old boy presents with enlarging breasts which are embarrassing him.
- 10 A 45-year-old woman has been diagnosed with a lobular carcinoma in the right breast.
- 11 A 52-year-old woman presented with a greenish black nipple discharge on both sides.
- 12 A 28-year-old woman presents with a discrete lump in the breast. The lump is painless and moves freely within the breast.
- 13 A 49-year-old man presents with a painless mass just under the nipple and areola on the right side of 3 months duration.

Answers • Theme 1

BAILEY & LOVE EMQS

Theme 1 • Breast Conditions – Diagnosis and Investigations

- 1 **C Breast abscess** — Fever, pain and swelling in a breast-feeding mother is lactational infection progressing to abscess (*S. aureus*); ultrasound-guided aspiration plus flucloxacillin.
- 2 **M Vacuum-assisted biopsy** — Fine linear branching microcalcification suggests DCIS; vacuum-assisted (stereotactic) biopsy samples calcifications more reliably than standard core biopsy.
- 3 **D Breast cyst** — A sudden painful discrete lump in a perimenopausal woman is a cyst; ultrasound confirms it and aspiration relieves it.
- 4 **A ANDI (Aberration of normal development and involution)** — Bilateral cyclical pain and lumpiness worse before periods is ANDI (fibroadenosis/cyclical mastalgia): reassurance, a supportive bra and simple analgesia.
- 5 **B BRCA 1 and 2 tests** — Several young relatives with breast cancer means genetic counselling and BRCA1/2 testing before planning surveillance or risk-reducing surgery.
- 6 **L Phyllodes tumour** — A large, rapidly growing mass with stretched shiny skin in a woman over 40 is a phyllodes tumour; treat by wide excision with clear margins.
- 7 **E Core needle biopsy** — A hard enlarging lump with a spiculated mammographic mass is cancer until proven otherwise: core needle biopsy gives histology and receptor status.
- 8 **H Galactocoele** — A painless lump appearing after stopping breast-feeding is a galactocoele (milk-retention cyst); aspiration of milky fluid is diagnostic and curative.
- 9 **I Gynaecomastia** — Breast enlargement in a pubertal boy is physiological gynaecomastia, which usually settles within 2 years; exclude drugs (e.g. cannabis) and hormonal causes.
- 10 **K MRI scan of the breasts** — Invasive lobular carcinoma is often multifocal and bilateral and underestimated on mammography, so MRI is used to define its extent before surgery.
- 11 **F Duct ectasia** — Bilateral greenish-black multiduct discharge in a postmenopausal woman is duct ectasia; reassure if imaging is normal.
- 12 **G Fibroadenoma** — A painless, discrete, very mobile lump in a young woman is a fibroadenoma ('breast mouse'); confirm with ultrasound and core biopsy.
- 13 **J Male breast cancer** — A painless unilateral subareolar lump in a middle-aged man is male breast cancer until proven otherwise; gynaecomastia is usually bilateral, soft and tender.

2 THEMES • 10 SCENARIOS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford EMQs

THEME 1 Breast problems

- A Breast carcinoma
- B Breast cyst
- C Duct ectasia
- D Fat necrosis
- E Fibroadenoma
- F Galactocoele
- G Lactational mastitis
- H Lipoma
- I Periductal mastitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 20-year-old woman is seen in the breast clinic with a lump in her right breast for 1 month. She has a 1cm discrete, mobile lump in the lower outer quadrant of the right breast which is non-tender.
- 2 A 50-year-old woman has suffered a painful lump in the right breast for 3 weeks. Over the last few months she has had irregular periods and hot flushes. There is a 2cm discrete, moderately tender lump in the lower inner quadrant of her right breast with normal-looking breast skin. Ultrasound scan shows a well-defined lesion and needle aspiration yielded 12 mL of a green-yellow fluid.
- 3 A 40-year-old woman has found an asymptomatic lump in the central portion of her right breast. Past medical history includes diabetes mellitus and a fractured collar bone in a road traffic collision 3 months ago. There is a semimobile irregular lump in the central portion of the right breast with overlying skin haemosiderin deposition. There is no evidence of axillary lymphadenopathy.

- 4** A 32-year-old woman has had 5 weeks of redness and swelling around the right nipple. There is no past medical history but she is a heavy smoker. There is periareolar skin inflammation with a small palpable tender mass just behind the nipple.
- 5** A 60-year-old woman has noted that in the last few months both nipples have gradually retracted. She is also complaining of significant nipple discharge that is thick and yellow/green in colour. Both nipples are retracted and slit-like with no palpable lumps or lymphadenopathy.

THEME 2 Breast lumps

- A** Anastrozole
- B** Excision biopsy
- C** External beam radiotherapy
- D** Neoadjuvant chemotherapy
- E** Mastectomy and sentinel node biopsy
- F** Mastectomy and axillary node clearance
- G** Tamoxifen
- H** Trastuzumab
- I** Wire-guided wide local excision
- J** Wide local excision and sentinel node biopsy
- K** Wide local excision and axillary node clearance

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** An 84-year-old woman presents with a 2-month history of a lump in her left breast. She has ischaemic heart disease and previous deep vein thrombosis, with shortness of breath on minimal exertion. Triple assessment reveals a grade 2 invasive ductal carcinoma which is oestrogen receptor positive.
- 2** A 50-year-old woman is diagnosed with a small area of microcalcification in the left breast measuring approximately 12 mm under the national breast cancer screening programme. Stereotactic core biopsies confirm low-grade ductal carcinoma in situ.
- 3** A 60-year-old woman is diagnosed with a palpable invasive ductal carcinoma. She undergoes a wide local excision and sentinel node biopsy. Histology confirms a 10 mm grade 1 invasive ductal carcinoma completely excised with no lymph node involvement. The tumour is oestrogen receptor negative.
- 4** An 82-year-old woman is 3 years post mastectomy and axillary node clearance for a lymph node positive, grade 3 breast cancer. She has developed spinal metastases causing back pain without cord compression.
- 5** A 32-year-old woman presents with a 3-month history of a large lump in the right breast. There is an 8 cm mass and palpable axillary lymphadenopathy. Core biopsy confirms invasive ductal carcinoma and FNAB of nodes is positive.

Answers • Themes 1-2

OXFORD EMQS

Theme 1 • Breast problems

- 1** **E Fibroadenoma** — Fibroadenoma is a benign breast lump seen in young women under 30; discrete, mobile, painless.
- 2** **B Breast cyst** — Breast cysts are common in perimenopausal women; aspiration yields green-yellow fluid.
- 3** **D Fat necrosis** — Fat necrosis may follow trauma; haemosiderin deposition indicates tissue injury.
- 4** **I Periductal mastitis** — Periductal mastitis occurs in smokers; presents with periareolar inflammation and tender subareolar mass.
- 5** **C Duct ectasia** — Duct ectasia involves duct dilatation in involution; causes cheesy discharge and slit-like nipple retraction.

Theme 2 • Breast lumps

- 1** **A Anastrozole** — Anastrozole is first-line hormonal therapy for an elderly patient unfit for surgery.
- 2** **I Wire-guided wide local excision** — Wire-guided wide local excision is required to remove a small, impalpable area of DCIS.
- 3** **C External beam radiotherapy** — External beam radiotherapy reduces recurrence after breast-conserving surgery.
- 4** **C External beam radiotherapy** — External beam radiotherapy provides rapid pain relief for bone metastases.
- 5** **D Neoadjuvant chemotherapy** — Neoadjuvant chemotherapy downstages large tumours and treats micrometastases.

6 THEMES • 30 SCENARIOS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead EMQs

SET 1 OF 2

THEME 1 Diagnosis of breast disease

- A Breast cancer
- B Breast cyst
- C Duct ectasia
- D Fat necrosis
- E Fibroadenoma
- F Fibrocystic disease
- G Galactocele
- H Gynaecomastia
- I Lactating breast
- J Paget disease
- K Peau d'orange
- L Phyllodes tumour
- M Prolactinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 17-year-old female presents with multiple bilateral cystic swellings, which she admits to having for years. These are causing her considerable discomfort, especially premenstrually.
- 2 A 25-year-old woman presents to her general practitioner with a single 4 cm firm, smooth lump in the upper outer quadrant of the breast. On examination, it is noted that the lump is not adherent to the overlying skin and is freely mobile within the breast.
- 3 A 42-year-old woman attends the breast clinic with a rapidly growing irregular mass in her right breast. On examination, it is 6 cm in diameter. Core biopsy later reveals mixed epithelial and connective tissue elements.
- 4 A 62-year-old woman complains of dry, cracked skin around the right areola. On examination, there is a discrete firm nodule underlying the affected area.
- 5 A 72-year-old woman went to her general practitioner complaining of a swelling in the left breast. On examination, you notice dimpling of the skin in the left breast and a firm 3 cm lump in the lower outer quadrant.

THEME 2 Diagnosis of breast disease

- A Breast abscess
- B Breast cancer
- C Breast cyst
- D Fat necrosis
- E Fibroadenoma
- F Galactocele
- G Galactorrhoea
- H Gynaecomastia
- I Lactating breast
- J Mondor disease
- K Paget disease
- L Peau d'orange
- M Simple obesity

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 43-year-old woman attends the breast clinic complaining of a sudden pain and swelling in her right breast. On examination, a single, fluctuant swelling is palpable. Fine-needle aspiration reveals clear fluid.
- 2 A 24-year-old woman presents 5 weeks after the birth of her second child with a painful left breast. On examination, there is a 4 cm area of inflammation and erythema lateral to the nipple with an associated underlying lump.
- 3 A 52-year-old obese woman comes to the breast clinic with an irregular 2 cm firm lump in her left breast. On questioning, she admits to recent trauma to the area. Core biopsy of the lesion is reported as B2.
- 4 A 24-year-old man complains of a soft swelling around his left nipple that has increased in size over the last year. He has no significant past medical history but admits to being a frequent user of cannabis. His body mass index is 21. On examination, there is no discharge, skin changes, or axillary lymphadenopathy.
- 5 A 54-year-old woman presents with a 1-week history of pain in her left breast. On examination, you feel an indurated, tender, cord-like structure running down the anterior surface of the breast.

THEME 3 INVESTIGATION OF BREAST DISEASE

- A Aspirate culture
- B Aspirate cytology
- C Core biopsy
- D CT scan
- E Fine-needle aspiration cytology
- F Magnetic resonance imaging
- G Mammography
- H No further investigation
- I Ultrasound scan

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 34-year-old woman has lumpiness and tenderness in both breasts that is worse in the days before menstruation. An ultrasound scan shows multiple small cysts bilaterally, and a tissue sample shows fibrosis and epithelial hyperplasia.
- 2 A 28-year-old woman has a 1 cm soft, painless lump in her right breast that appeared suddenly last week; there is no other abnormality on examination.
- 3 A 42-year-old woman has a hard 2 cm lump in the upper outer quadrant of the left breast with axillary lymphadenopathy. She has had bilateral cosmetic breast augmentation, and the mammogram is too hazy to delineate any lesion.
- 4 A 52-year-old woman has a 4 cm lump in her left breast. Fine-needle aspiration cytology reports dysplastic cells.
- 5 A 45-year-old woman has a long-standing lump in her right breast. Ultrasound shows a cyst, which is aspirated under ultrasound guidance; the aspirate is blood-stained.

THEME 4 NIPPLE DISCHARGE

- A Breast cancer
- B Breast cyst
- C Duct ectasia
- D Fat necrosis
- E Fibroadenoma
- F Fibrocystic disease
- G Galactocele
- H Gynaecomastia
- I Intraductal papilloma
- J Lactating breast
- K Peau d'orange
- L Phyllodes tumour
- M Prolactinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 21-year-old woman presents with a 4-month history of unilateral, painless, blood-stained discharge from the left nipple. On examination, no lumps are palpable on either side and there is no axillary lymphadenopathy. Cytology of the discharge reveals no malignant cells.

- 2 A 32-year-old woman has had multiple visits to her general practitioner over the last 6 months complaining of early morning headaches unresponsive to analgesia. On this occasion, she also admits to having a milky discharge from both nipples. A β -hCG urine test is negative and she has not had any children before.
- 3 A 45-year-old smoker presents to her general practitioner with a 6-month history of pain and a cheesy discharge from her left nipple associated with some inflammation of the areola. A mass can be felt under the areola, but no other lumps were palpable.
- 4 A 28-year-old woman is referred to you by her general practitioner with a 7-month history of amenorrhoea. She also complains of a whitish discharge from her nipples bilaterally. Her body mass index is 42 and she has no neurological symptoms. Breast examination is otherwise normal. She is worried as her mother has recently been diagnosed with metastatic breast cancer.
- 5 A 50-year-old woman presents with a 4-month history of a firm 3 cm lump in her right breast, with associated nipple inversion, a bloody discharge and axillary lymphadenopathy. Fine-needle aspiration cytology was reported as C5.

THEME 5 MANAGEMENT OF BREAST CANCER

- A Adjuvant chemotherapy
- B Aromatase inhibitor
- C Modified radical mastectomy
- D Neoadjuvant chemotherapy
- E Radiotherapy
- F Tamoxifen
- G Trastuzumab
- H Wide local excision and axillary clearance
- I Wide local excision and sentinel node biopsy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 32-year-old woman is found to have a 5 cm invasive ductal carcinoma in the upper outer quadrant. You have explained that she may need a mastectomy, but she is keen to preserve the breast.
- 2 A 65-year-old woman has a 6.5 cm central invasive breast tumour. She is otherwise fit and well.
- 3 A 42-year-old woman has recently had a wide local excision and axillary clearance for a 2 cm breast carcinoma. Histology showed clear excision margins, a grade I oestrogen receptor-negative tumour and no positive axillary lymph nodes. She returns to the outpatient clinic and wants to know what further treatment she will be having.
- 4 A 35-year-old woman who has recently had a 3 cm invasive breast tumour resection has returned to clinic. The histology report states that the tumour is oestrogen receptor-negative but is overexpressing HER2.

- 5 A 64-year-old woman has just had a radical mastectomy and radiotherapy for a large invasive breast tumour. Histology showed an oestrogen receptor-positive tumour. Apart from suffering a left deep vein thrombosis during the postoperative recovery period, she is well.

Answers · Themes 1-5

GET AHEAD EMQS

Theme 1 · Diagnosis of breast disease

- 1 **F Fibrocystic disease** — Fibrocystic disease causes bilateral cyclical cystic breast changes.
- 2 **E Fibroadenoma** — Fibroadenoma is a firm, mobile, well-circumscribed benign breast lump.
- 3 **L Phyllodes tumour** — Phyllodes tumour is a rapidly enlarging biphasic breast mass.
- 4 **J Paget disease** — Paget disease presents with eczematous nipple changes and underlying carcinoma.
- 5 **A Breast cancer** — A firm lump with skin dimpling (tethering) in a 72-year-old is breast cancer; peau d'orange is skin oedema from lymphatic blockage, not a diagnosis.

Theme 2 · Diagnosis of breast disease

- 1 **C Breast cyst** — Fluctuant swelling with clear fluid on aspiration indicates a breast cyst.
- 2 **A Breast abscess** — Painful, inflamed lump postpartum is a breast abscess.
- 3 **D Fat necrosis** — History of trauma and benign histology suggests fat necrosis.
- 4 **H Gynaecomastia** — Soft swelling under the nipple in a male indicates gynaecomastia.
- 5 **J Mondor disease** — Tender cord-like structure is Mondor disease (superficial thrombophlebitis).

Theme 3 · INVESTIGATION OF BREAST DISEASE

- 1 **H No further investigation** — Fibrocystic changes are benign; no further tests.
- 2 **I Ultrasound scan** — Ultrasound is first-line for a young woman's breast lump.

- 3 **F Magnetic resonance imaging** — MRI clarifies lesions when mammography is inconclusive in augmented breasts.
- 4 **C Core biopsy** — Core biopsy provides definitive histology for dysplasia.
- 5 **B Aspirate cytology** — Blood-stained cyst fluid requires cytological analysis.

Theme 4 · NIPPLE DISCHARGE

- 1 **I Intraductal papilloma** — Intraductal papilloma is the most common benign cause of bloody nipple discharge.
- 2 **M Prolactinoma** — Prolactinoma causes galactorrhea and headaches from pituitary enlargement.
- 3 **C Duct ectasia** — Duct ectasia presents with cheesy discharge and subareolar inflammation.
- 4 **J Lactating breast** — Amenorrhoea with bilateral milky discharge and no neurological features: exclude pregnancy first (β -hCG) — a lactating breast — before checking prolactin.
- 5 **A Breast cancer** — Features of lump, inversion, bloody discharge, and positive cytology indicate breast cancer.

Theme 5 · MANAGEMENT OF BREAST CANCER

- 1 **D Neoadjuvant chemotherapy** — Neoadjuvant chemotherapy downstages tumour for conservation.
- 2 **C Modified radical mastectomy** — Modified radical mastectomy for large central tumours.
- 3 **E Radiotherapy** — Radiotherapy reduces local recurrence after conservation.
- 4 **G Trastuzumab** — Trastuzumab targets HER2-positive disease.
- 5 **B Aromatase inhibitor** — Aromatase inhibitors are standard in postmenopausal ER-positive cancer.

SET 2 OF 2

THEME 6 TUMOUR MARKERS

- A α -fetoprotein
- B β -hCG
- C CA 125
- D CA 15-3
- E CA 19-9
- F Calcitonin
- G Carcinoembryonic antigen
- H Paraproteins
- I Prostate-specific antigen
- J S-100

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 47-year-old woman presents with a 3 cm firm lump in the left breast that is fixed to the overlying skin.
- 2 A 76-year-old man complains of new constipation. He has a history of passing fresh blood mixed with his stools and has recently lost weight.
- 3 A 68-year-old man presents with difficulty passing urine and dribbling at the end of the stream.

4 A 34-year-old woman presents with a 1 cm flat, brown lesion on her left leg. On examination, the lesion is irregularly pigmented and occasionally itches and bleeds.

5 A 62-year-old woman presents with jaundice that has been worsening over the previous weeks. She denies having any pain. On examination, the gallbladder is palpable.

Answers • Theme 6

GET AHEAD EMQS

Theme 6 • TUMOUR MARKERS

- 1** **D** **CA 15-3** — CA 15-3 is the breast cancer marker, used to monitor treatment and recurrence rather than for diagnosis.
- 2** **G** **Carcinoembryonic antigen** — Carcinoembryonic antigen (CEA) is a marker for colorectal cancer, fitting his presentation of blood in the stool, constipation and weight loss.

- 3** **I** **Prostate-specific antigen** — Prostate-specific antigen (PSA) is used in the detection and management of prostate cancer, which is consistent with his urinary symptoms.
- 4** **J** **S-100** — S-100 is frequently elevated in malignant melanoma, corresponding to the irregular, pigmented lesion that bleeds occasionally.
- 5** **E** **CA 19-9** — CA 19-9 is associated with pancreatic head adenocarcinoma; its elevation accounts for the painless jaundice and palpable gallbladder (Courvoisier's sign).

2 THEMES • 10 SCENARIOS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course EMQs

THEME 1 MANAGEMENT OF BREAST CANCER

- A** Cyclophosphamide
- B** Trastuzumab (Herceptin)
- C** Tamoxifen
- D** Anastrozole
- E** Doxorubicin
- F** Raloxifene
- G** Vincristine
- H** Dexamethasone
- I** Paclitaxel
- J** Faslodex

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A therapeutic agent highly recommended in the treatment of lymph-node-positive HER2-positive breast cancer.
- 2** A selective oestrogen-receptor modulator drug useful in the treatment of breast cancer but associated with risks of endometrial cancer and pulmonary embolism.
- 3** An aromatase inhibitor useful in the treatment of early breast cancer in postmenopausal women.
- 4** A therapeutic agent useful in the treatment of breast cancer but known to cause left ventricular failure and pulmonary toxicity.
- 5** A therapeutic agent useful in the treatment of breast cancer but known to cause bladder cancer.

THEME 2 BREAST LUMPS

- A** Breast abscess
- B** Cyst
- C** Duct ectasia
- D** Fat necrosis
- E** Fibroadenoma
- F** Fibrocystic change
- G** Galactorrhoea
- H** Lipoma
- I** Mastitis
- J** Papilloma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 19-year-old woman presents with a highly mobile lump in her breast. Fine-needle aspiration shows no malignancy.
- 2** A 28-year-old woman, four weeks postpartum, presents to her GP with a tender lump in the upper outer quadrant of her left breast. On examination it is ill-defined, erythematous and hot to touch.
- 3** A 38-year-old very anxious woman presents with a hard, irregular lump in her breast. On examination it is craggy and well-defined. Fine-needle aspiration shows no evidence of malignancy. On further questioning, she reveals that she was involved in a car crash 6 months previously.
- 4** A 42-year-old woman presents with a yellow-brown discharge from her nipples. On examination her discharge appears to be coming from multiple ducts. Examination is otherwise normal and mammography is unremarkable.

- 5 A 32-year-old woman presents with a lump in her breast which is painless. On examination it is well demarcated, fluctuant and transilluminable.

Answers · Themes 1-2

CRASH COURSE EMQS

Theme 1 · MANAGEMENT OF BREAST CANCER

- 1 **B Trastuzumab (Herceptin)** — Trastuzumab (anti-HER2) is given for HER2-positive breast cancer, especially node-positive disease, with chemotherapy.
- 2 **C Tamoxifen** — Tamoxifen is a SERM: antagonist in the breast but agonist in the endometrium and clotting system (endometrial cancer, VTE).
- 3 **D Anastrozole** — Anastrozole, an aromatase inhibitor, blocks peripheral oestrogen synthesis and is used in postmenopausal ER-positive disease.
- 4 **B Trastuzumab (Herceptin)** — Trastuzumab causes (usually reversible) left ventricular dysfunction and can cause pulmonary toxicity, so LVEF is monitored.
- 5 **A Cyclophosphamide** — Cyclophosphamide's metabolite acrolein causes haemorrhagic cystitis and bladder cancer (mesna is protective).

Theme 2 · BREAST LUMPS

- 1 **E Fibroadenoma** — A highly mobile lump in a young woman with benign cytology is a fibroadenoma.
- 2 **I Mastitis** — An ill-defined, hot, red, tender area in a breast-feeding woman is lactational mastitis; a fluctuant mass would suggest abscess.
- 3 **D Fat necrosis** — A hard, craggy lump after trauma with benign cytology is fat necrosis; it mimics carcinoma, so triple assessment is still needed.
- 4 **C Duct ectasia** — Yellow-brown discharge from multiple ducts with a normal mammogram is duct ectasia.
- 5 **B Cyst** — A well-demarcated, fluctuant, transilluminable lump is a cyst; ultrasound and aspiration confirm it.

2 THEMES · 9 SCENARIOS · IRFAN SURGERY EMQ COLLECTION

Irfan EMQs

THEME 1 Breast conditions

- A Paget's disease
- B phyllodes tumour
- C mammary haemangioma
- D galactocoele
- E chronic breast abscess
- F carcinoma of breast
- G benign eczema of nipple
- H fibroadenoma
- I postpartum fat necrosis
- J gynaecomastia
- K mastitis
- L dermatitis herpetiformis
- M duct ectasia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 75-year-old woman presents with a breast lump in the upper outer quadrant. On examination the lump is hard and irregular. There is axillary lymphadenopathy.
- 2 A 53-year-old woman presents with nipple retraction and a greenish-yellow discharge. Ultrasound shows dilated breast ducts.

- 3 A 21-year-old woman presents with a smooth, non-tender, highly mobile mass on the upper outer quadrant of the right breast. Fine needle aspiration shows benign breast epithelial cells (C2).
- 4 A 70-year-old woman presents with a worsening eczema-like rash overlying the areola and nipple. The rash does not itch. On examination a palpable mass can be felt under the rash.
- 5 A 26-year-old woman presents a fortnight postpartum with a painful, enlarged left breast. On examination she is pyrexial and her breast is tender and inflamed. There are no palpable masses.

THEME 2 Treatment of breast cancer

- A Tamoxifen
- B Cytotoxic chemotherapy
- C Mastectomy and immediate reconstruction
- D Quadrantectomy
- E Immunotherapy
- F Radical mastectomy
- G Radical mastectomy plus radiotherapy
- H Wide local excision, sentinel node biopsy plus tamoxifen
- I Wide local excision, axillary node clearance and chemotherapy
- J No further intervention

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 92-year-old woman presents with oestrogen receptor positive stage III invasive ductal carcinoma.
- 2 A 35-year-old woman is diagnosed with ductal carcinoma-in-situ involving all quadrants of the breast.
- 3 A 60-year-old patient presents with a 2-cm invasive ductal carcinoma of the outer quadrant of the right breast. The tumour is oestrogen receptor positive.
- 4 A 32-year-old patient presents with a 2.5-cm invasive ductal carcinoma. She has a palpable axillary lymph node which reveals cancer cells on cytology. The tumour is oestrogen receptor negative.

Answers · Themes 1-2

IRFAN EMQS

Theme 1 · Breast conditions

- 1 **F carcinoma of breast** — A hard, irregular lump with axillary nodes in an elderly woman is breast carcinoma; confirm by triple assessment.
- 2 **M duct ectasia** — Nipple retraction with green-yellow discharge and dilated ducts on ultrasound is duct ectasia.
- 3 **H fibroadenoma** — A smooth, highly mobile lump in a 21-year-old with benign cytology (C2) is a fibroadenoma.
- 4 **A Paget's disease** — A non-itchy eczema-like rash of the nipple and areola with an underlying mass is Paget disease of the nipple, usually with underlying carcinoma.
- 5 **K mastitis** — A tender, inflamed breast with fever 2 weeks postpartum and no mass is lactational mastitis; continue breast-feeding and give flucloxacillin.

Theme 2 · Treatment of breast cancer

- 1 **A Tamoxifen** — A 92-year-old with ER-positive disease is treated with primary endocrine therapy (tamoxifen or an aromatase inhibitor) rather than surgery.
- 2 **C Mastectomy and immediate reconstruction** — DCIS throughout all quadrants cannot be excised with conservation: mastectomy, with immediate reconstruction in a young woman.
- 3 **H Wide local excision, sentinel node biopsy plus tamoxifen** — A 2-cm ER-positive cancer with a clinically negative axilla: wide local excision and sentinel node biopsy, then radiotherapy and endocrine therapy.
- 4 **I Wide local excision, axillary node clearance and chemotherapy** — Cytology-proven axillary nodes need axillary node clearance, and an ER-negative tumour in a young woman needs chemotherapy.

PART 03 / 8 CHAPTERS

- 07 Salivary Glands
- 08 Oesophagus
- 09 Stomach & Duodenum
- 10 Small & Large Intestine
- 11 Anorectal Disorders
- 12 Gastrointestinal Bleeding
- 13 Gastrointestinal Tumours
- 14 Gastrointestinal EMQs

03

GASTROINTESTINAL TRACT



Salivary Glands

Parotid and submandibular swellings: sialolithiasis, sialadenitis, pleomorphic adenoma, Warthin tumour and the facial nerve.

14

MCQS

0

EMQ SCENARIOS

4

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

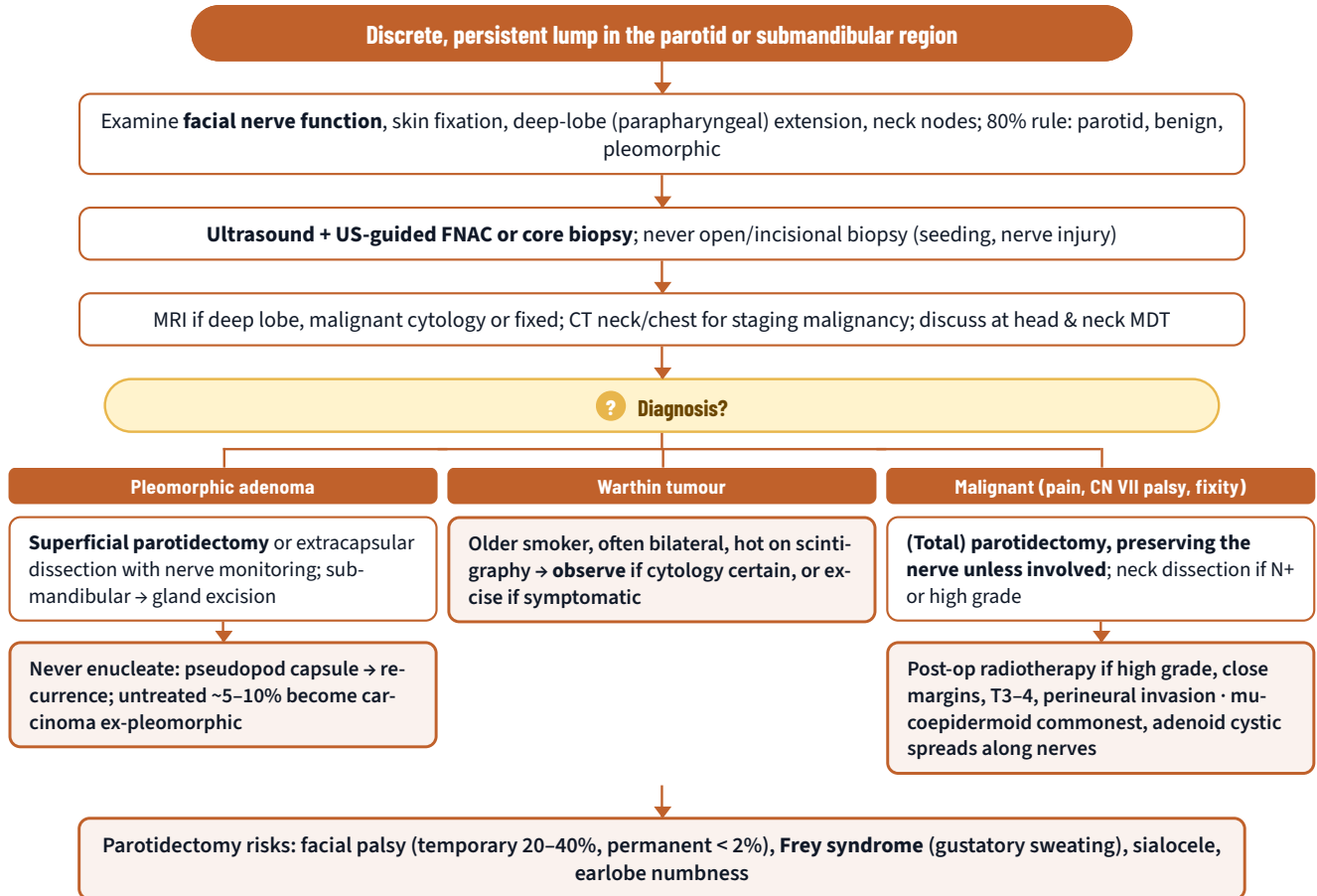
01	Management flowcharts	3 charts	page 88
02	Local Exams MCQs	1 Q	page 90
03	Bailey & Love MCQs	7 Q	page 90
04	SBA MCQs	4 Q	page 91
05	Get Ahead MCQs	2 Q	page 92

07

Step by step

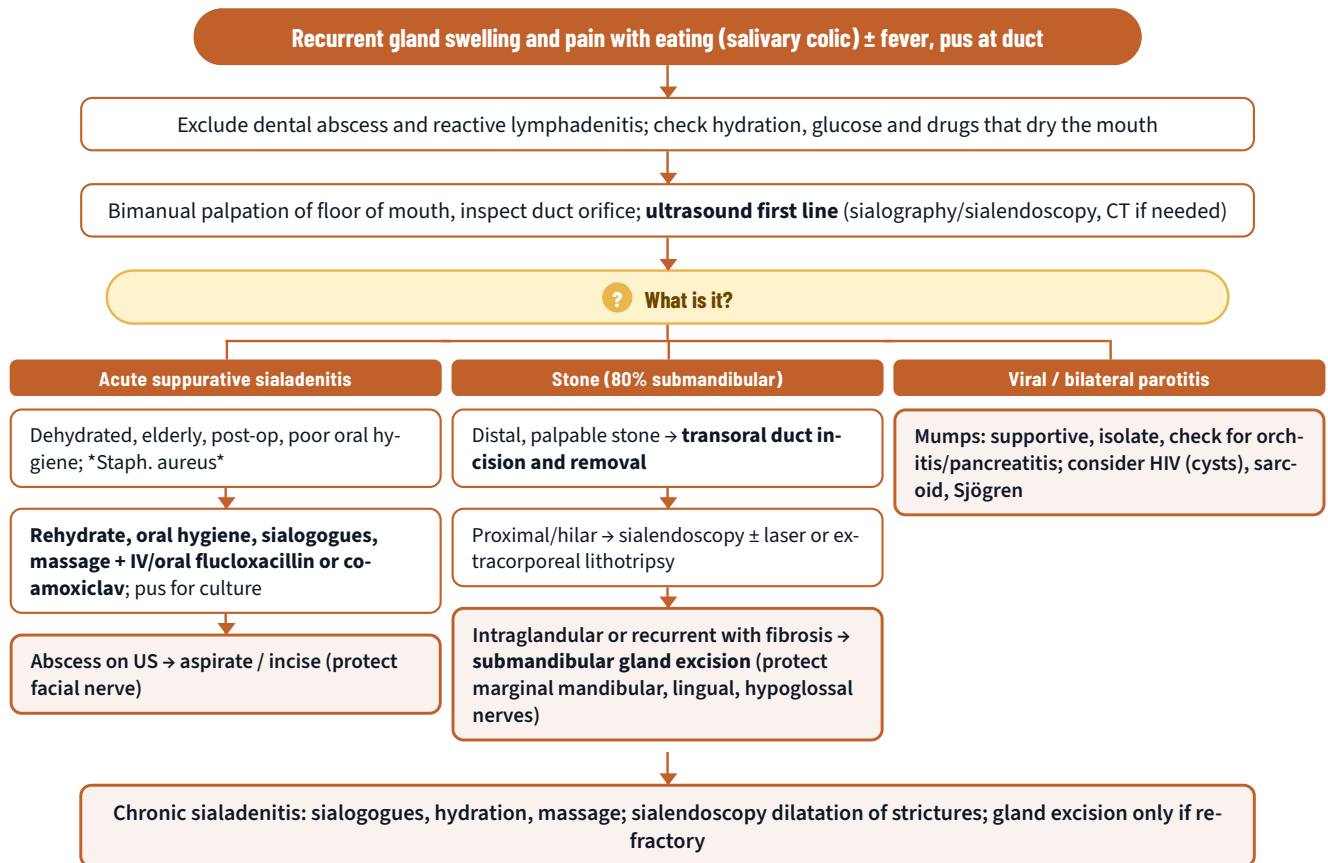
Salivary gland lump (parotid / submandibular)

BAHNO/ENT UK 2020 · NICE NG36 · BAILEY & LOVE 28E



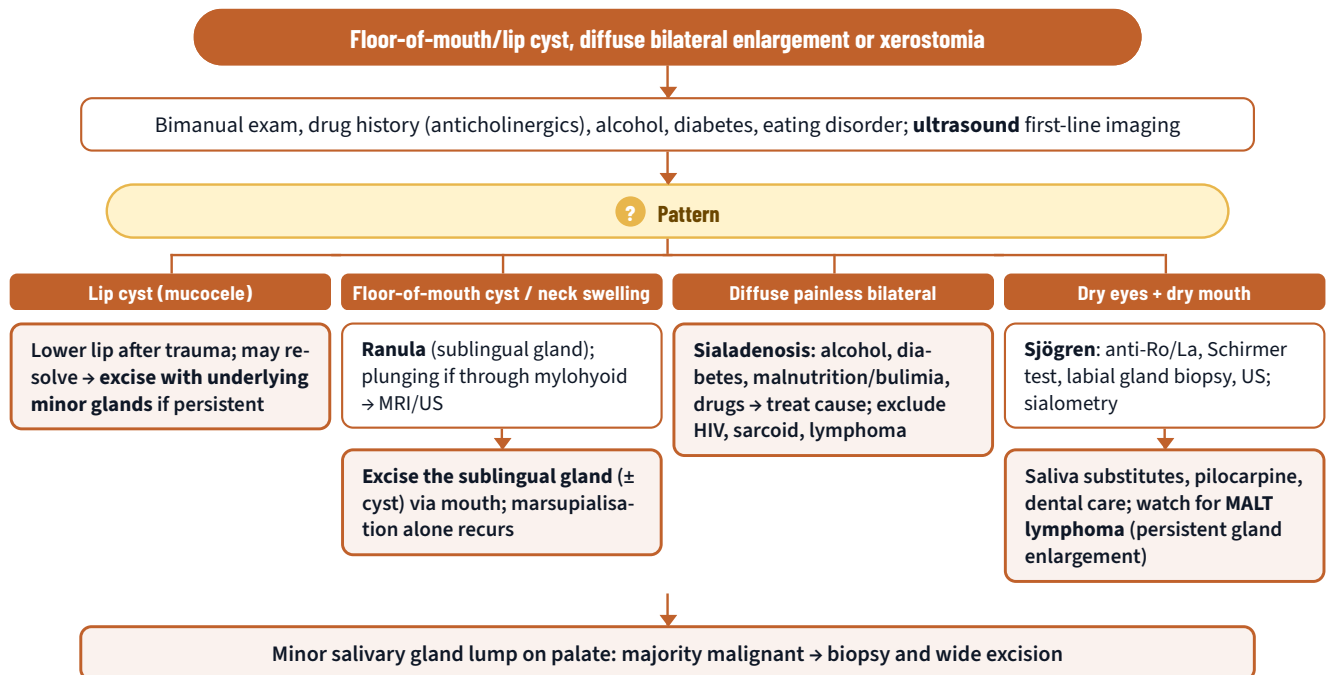
Painful swelling with meals: stones and sialadenitis

ENT UK · BAILEY & LOVE 28E



Other swellings and dry mouth

BSR 2017 SJÖGREN · ACR/EULAR 2016 CRITERIA · BAILEY & LOVE 28E



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 40-year-old female presents with an acute onset of fever, pain, and swelling at her right cheek. On examination, there is tenderness and swelling at the area of the right parotid gland. What is the most likely diagnosis?
- A Parotid gland tumor.
 - B Frey's syndrome.
 - C Parotid gland sialadenitis.
 - D Parotid gland abscess.
 - E Sjögren's syndrome.

Answer 1

LOCAL EXAMS MCQS

- 1 **C** Parotid gland sialadenitis. — Acute fever and parotid tenderness indicate sialadenitis.

7 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Which of the following statements are true?
- A There are three pairs of minor salivary glands.
 - B Minor salivary glands contribute 10% of overall saliva flow.
 - C Most minor salivary gland tumours are benign.
 - D Presumed benign tumours of the minor salivary glands >1 cm should be managed with excisional biopsy.
 - E Malignant minor salivary gland tumours are managed with primary surgical resection.
- 2 Which of the following statements are true?
- A The sublingual glands occupy the anterior floor of mouth.
 - B The sublingual glands drain both directly into the mouth and into the submandibular duct.
 - C Tumours of the sublingual glands are rarely malignant.
 - D The term ranula is applied to mucous retention cysts of the sublingual gland.
 - E A plunging ranula is the term used to describe a ranula that extends beyond the floor of mouth and into the neck.
- 3 Which of the following statements are true?
- A The two lobes of the submandibular gland communicate around the posterior belly of the digastric muscle.
 - B 80% of salivary calculi occur in the submandibular duct.
 - C Calculi identified posterior to the point where the duct crosses the lingual nerve should be removed via a transoral approach.
 - D The marginal mandibular nerve is at risk during removal of the gland.
 - E The lingual nerve is divided during removal of the gland to facilitate dissection of the deep lobe of the gland.
- 4 Which of the following statements are true?
- A 80% of submandibular gland tumours are benign.
 - B Submandibular gland masses can be differentiated from lymph nodes by bimanual palpation.
 - C Facial nerve weakness is a sign suggestive of malignancy.
 - D Open surgical biopsy is indicated to confirm malignant disease prior to surgery.
 - E Radiation is the primary modality of treatment for most submandibular gland malignancies.

- 5** Which of the following statements are true?
- A** 80% of the parotid gland lies deep to the facial nerve.
 - B** During parotidectomy, the greater auricular nerve should be included in the skin flap to preserve function.
 - C** The facial nerve is identified using landmarks including the tragal pointer and the posterior belly of digastric.
 - D** If a branch of the facial nerve must be sacrificed during dissection, immediate grafting should be performed.
 - E** Frey's syndrome results from inappropriate innervation of sweat glands with parasympathetic fibres within the parotid bed.
- 6** Which of the following statements is true?
- A** Viral infection of the parotid glands affects both glands symmetrically.
 - B** Dehydrated patients are at risk from bacterial parotitis.
 - C** Recurrent parotitis of childhood tends to result in abscess formation.
 - D** HIV might present with multiple parotid cysts.
 - E** Chronic trauma to the parotid duct may result in obstruction.
- 7** Which of the following statements are true?
- A** The most common tumour of the parotid gland is a pleomorphic adenoma.
 - B** Tumour excision for benign disease involves enucleation of the lesion.
 - C** Malignant tumours should be excised with a cuff of normal tissue including the facial nerve.
 - D** Symmetrical swelling of the parotid glands may be due to excess alcohol intake.
 - E** Sjogren's syndrome is associated with B cell lymphoma.

Answers 1-7

BAILEY & LOVE MCQs

- 1** **BE** True: B, E — B and E are correct because minor salivary glands produce about 10% of saliva and malignant tumours require primary surgical resection; the other options are incorrect.
- 2** **ABE** True: A, B, E — A, B and E are correct based on gland location, drainage pattern and the definition of a plunging ranula; options C and D are incorrect.
- 3** **BD** True: B, D — B and D are correct; A, C and E are false due to anatomical inaccuracies.
- 4** **BC** True: B, C — B and C are correct; option A is false since only about 50% are benign and D and E are not standard practice.
- 5** **CE** True: C, E — C and E are correct; options A, B and D are incorrect based on surgical anatomy.
- 6** **BDE** True: B, D, E — B, D and E are correct; A and C are not consistently true.
- 7** **ADE** True: A, D, E — A, D and E are correct; B and C are incorrect regarding surgical management.

4 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 45-year-old woman is diagnosed with a submandibular calculus, having presented with a tender lump below the jaw on eating. Which structure is likely to be obstructed?
- A** Warthin's duct
 - B** Stensen's duct
 - C** Biliary duct
 - D** Lingual nerve
 - E** Facial nerve
- 2** A 40-year-old man presents with a month-long history of intermittent, left-sided pain and swelling in the anterior neck under the left jaw. His symptoms are associated with meal times and regress shortly after completing each meal. In recent days however, the pain appears to have become more intense and the swelling more firm. On examination a firm lump is palpable in the left submandibular region, bimanual palpation of which causes the patient to complain of a foul taste in his mouth. The most appropriate diagnostic investigation will be
- A** Plain radiographs of the mouth
 - B** Blood calcium level
 - C** Biopsy of submandibular tissue
 - D** Sialogram
 - E** None of the above

- 3** A 54-year-old builder attends your outpatient clinic with his wife, reporting an 8-month history of a lump on the left side of his face, in front of his ear. It is painless and does not trouble him. He has attended on the insistence of his wife, who is worried that it may be growing larger. On examination, the swelling appears to lie anterior to the angle of the jaw on the left side. It is non-tender and approximately 5 cm in diameter with a clear edge. The skin is easily moved on top of it and the lump itself feels rubbery to touch. There is no cervical lymphadenopathy or facial droop evident on examination. An aspirate is taken which shows cells of many different types. A decision is made at a follow-up appointment to excise the lump under anaesthesia. This is likely to be
- A Pleomorphic adenoma
 - B Adenolymphoma (Warthin's tumour)
 - C Mikulicz's syndrome
 - D Sjögren's syndrome
 - E Carcinoma of the parotid gland
- 4** A 6-year-old boy presents with his mother to the general practice at which you are based, with a 3-week history of bilateral cheek swelling. The family is newly registered and medical records have not yet been transferred to the practice. His mother informs you that the young boy has not received any of his childhood vaccinations, as she has been worried about their possible detrimental effects. The clinical suspicion is of mumps parotitis. Which class of infectious agent is responsible for mumps?
- A RNA viruses
 - B DNA viruses
 - C Gram-negative bacteria
 - D Gram-positive bacteria
 - E Fungi

Answers 1-4

SBA MCQS

- 1** **A Warthin's duct** — Submandibular calculi typically obstruct Warthin's duct, leading to pain and swelling during meals.
- 2** **D Sialogram** — A sialogram is used to evaluate salivary duct obstruction.
- 3** **A Pleomorphic adenoma** — Pleomorphic adenoma is the most common benign parotid tumor, presenting as a rubbery and mobile mass.
- 4** **A RNA viruses** — Mumps is caused by an RNA virus from the paramyxovirus family.

2 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1** A 56-year-old woman presents with a lump on the left side of her neck which has been slowly enlarging over the last few months. On examination, there is a 2 cm firm, painless, mobile lump near the angle of the left jaw. She is still able to move her facial muscles. What is the most likely diagnosis?
- A Chemodectoma
 - B Pleomorphic adenoma
 - C Salivary duct carcinoma
 - D Salivary duct stone
 - E Sternocleidomastoid tumour
- 2** A 71-year-old man presents to the GP with a painful, enlarging lump just in front of his right ear, which has been present for 2 months. He woke this morning to find that the right side of his face was drooping. What is the most likely cause of his symptoms?
- A Acute bacterial sialolithiasis
 - B Parotitis
 - C Pleomorphic adenoma
 - D Salivary gland calculi
 - E Salivary gland carcinoma

Answers 1-2

GET AHEAD MCQS

- 1** **B Pleomorphic adenoma** — Most common benign parotid tumour, usually painless, no facial nerve involvement.
- 2** **E Salivary gland carcinoma** — Malignant parotid tumour often presents with rapid growth and facial nerve involvement.

Oesophagus

Dysphagia, achalasia, reflux and hiatus hernia, Barrett oesophagus, perforation and oesophageal cancer.

40

MCQs

0

EMQ SCENARIOS

9

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

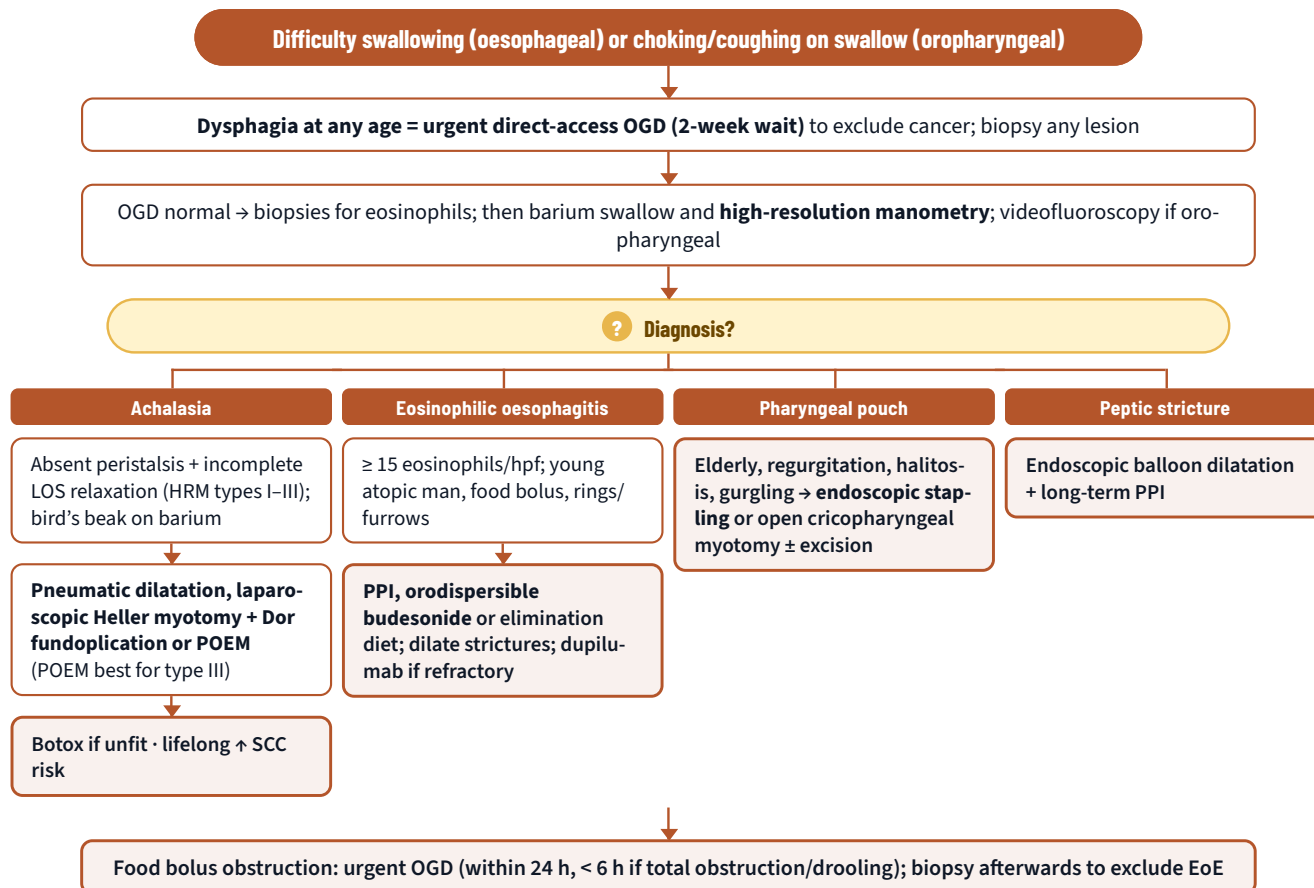
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08

Step by step

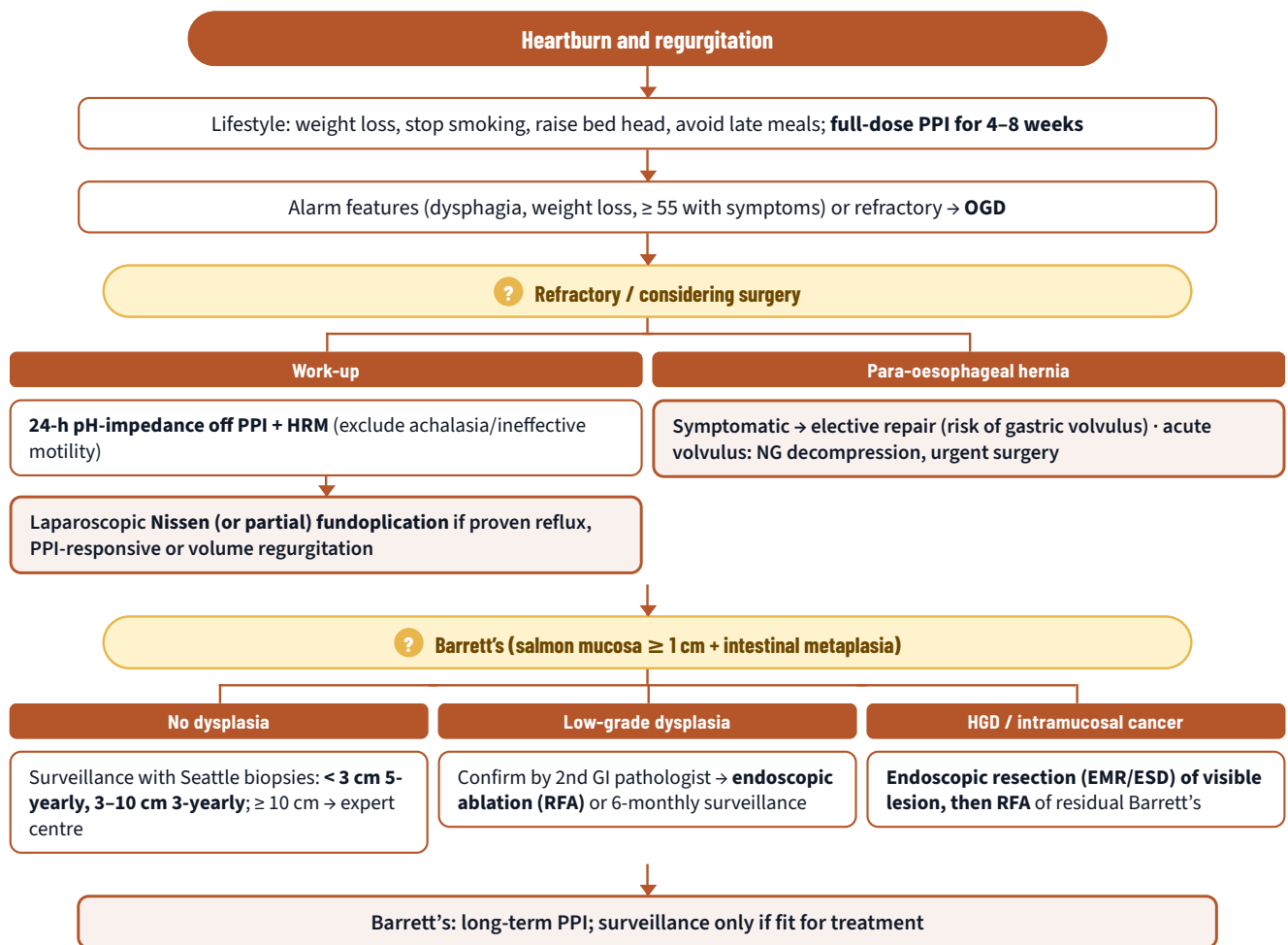
Dysphagia

NICE NG12 • CHICAGO CLASSIFICATION 4.0 • ESNM/UEG 2020 ACHALASIA



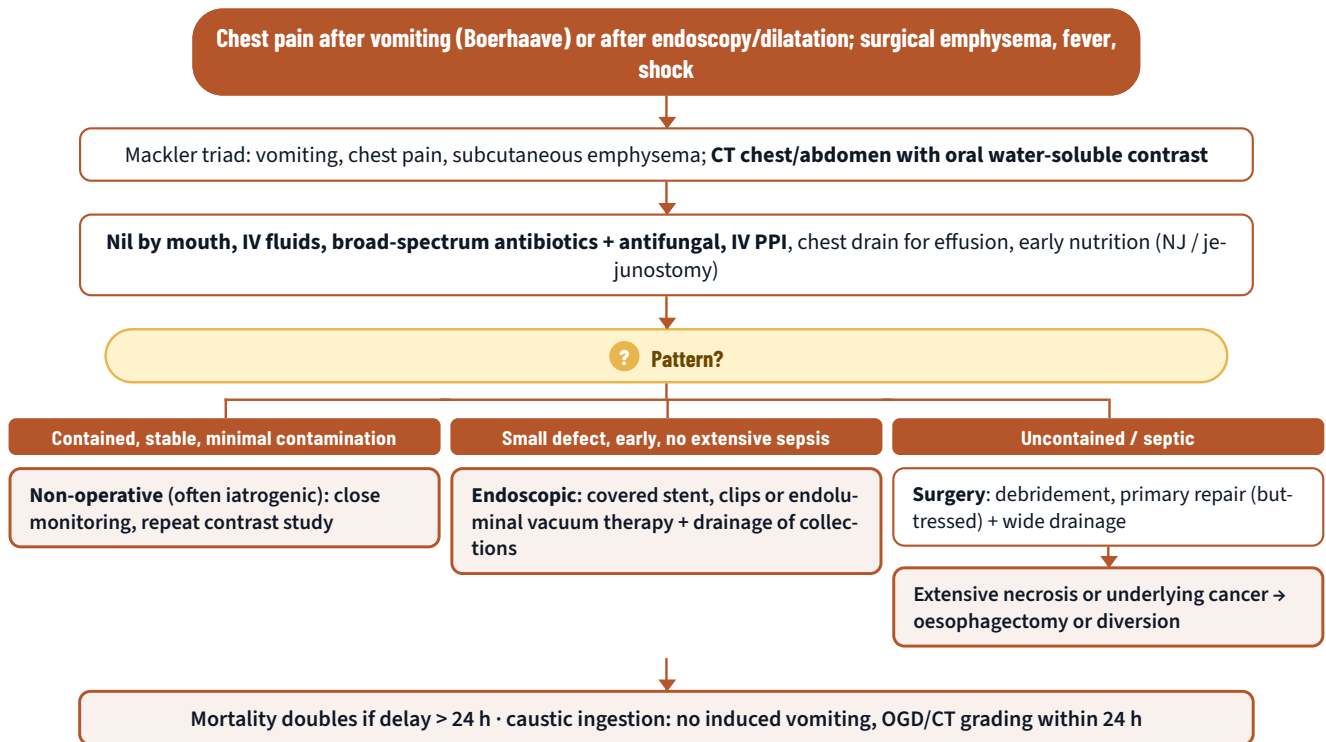
GORD, hiatus hernia and Barrett's oesophagus

NICE CG184 · BSG 2014 BARRETT'S · ESGE 2023 BARRETT'S



Oesophageal perforation

WSES 2019 OESOPHAGEAL PERFORATION · BAILEY & LOVE 28E

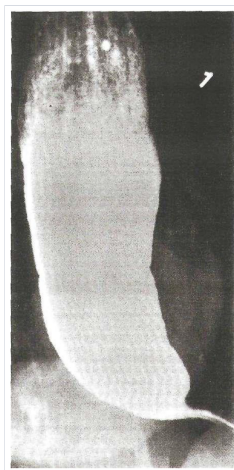


Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1 A barium swallow is shown. What is the most likely diagnosis on this image?



- A Achalasia
B Oesophageal carcinoma
C Diffuse oesophageal spasm
D Pharyngeal pouch
E Hiatus hernia
- 2 A barium swallow of a patient with progressive dysphagia and weight loss is shown. What is the most likely diagnosis on this image?



- A Oesophageal carcinoma
B Achalasia
C Benign peptic stricture
D Oesophageal web
E Diffuse oesophageal spasm

CASE • QUESTIONS 3-5

A 55-year-old man has had intermittent dysphagia for solids and liquids with central chest pain for 4 months. ECG and troponin are normal; sublingual nitrates relieve the pain.

- 3 What is the diagnosis?
A Achalasia
B Diffuse oesophageal spasm
C Oesophageal carcinoma
D Plummer-Vinson syndrome
E Gastro-oesophageal reflux disease
- 4 Which investigation confirms the diagnosis?
A Oesophageal manometry
B Upper GI endoscopy
C CT chest
D Ultrasound
E Laparotomy
- 5 What is the best management now?
A Calcium channel blocker
B Peroral endoscopic myotomy (POEM)
C Heller myotomy
D Oesophagectomy
E Proton-pump inhibitor alone

MINISTERIAL EXAM

Ministerial evaluation examinations

- 6 A 54-year-old man with recurrent problems with acid reflux despite changes to his diet and reduction of his alcohol intake. An endoscopy is performed which reveals Barrett's oesophagus. Which of the following is the optimum management for this patient in regards to symptom control and disease monitoring?
A Endoscopy only every year
B Long term PPI and endoscopy every year
C Long term PPI and endoscopy every five years
D Trial of PPI and endoscopy every five years
E Trial of PPI only
- 7 Shortly after an esophagoscopy, a 30-year-old patient develops shortness of breath, chest pain, and fever. A contrast study shows extravasation of contrast into the left chest cavity. The most appropriate course of action is:
A Perform a cervical esophagostomy, gastrostomy, and insert a chest tube.
B Perform Immediate left thoracotomy and repair the esophageal tear.
C Insert a nasogastric tube and begin high dose antibiotic therapy
D Deploy an endoscopic intra-esophageal stent to plug the hole.
E Stop ingestion of food, insert a chest tube and begin high dose antibiotic therapy.

Answers 1-7

LOCAL EXAMS MCQS

- 1 **A Achalasia** — A grossly dilated oesophagus tapering smoothly at the cardia (bird beak) is achalasia.
- 2 **A Oesophageal carcinoma** — An irregular stricture with shouldering and proximal dilatation is oesophageal carcinoma.
- 3 **B Diffuse oesophageal spasm** — Intermittent dysphagia for solids and liquids with chest pain relieved by nitrates is diffuse oesophageal spasm (corkscrew oesophagus on barium).
- 4 **A Oesophageal manometry** — High-resolution manometry shows premature simultaneous contractions; endoscopy is done first to exclude obstruction.
- 5 **A Calcium channel blocker** — Oesophageal spasm is treated medically first (calcium channel blockers or nitrates); POEM is reserved for refractory cases.
- 6 **C Long term PPI and endoscopy every five years** — Barrett's oesophagus requires acid suppression and periodic surveillance with endoscopy.
- 7 **B Perform Immediate left thoracotomy and repair the oesophageal tear.** — **UPDATED** Iatrogenic thoracic perforation recognised early, with free leak into the pleura, is repaired primarily via a left thoracotomy with drainage; stenting or conservative care suits contained leaks or unfit patients.

1 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

- 1 After undergoing total gastrectomy, the patient presents with chest pain and sore throat. What could be the cause, and how should it be managed?
 - A Myocardial infarction; manage with thrombolytics
 - B Esophageal stricture; manage with dilation
 - C GERD; manage with lifestyle modifications and proton pump inhibitors
 - D Pulmonary embolism; manage with anticoagulants
 - E Infection; manage with antibiotics

Answer 1

ESSAY TO MCQ MCQS

- 1 **C GERD; manage with lifestyle modifications and proton pump inhibitors** — Post-gastrectomy, GERD is common and is managed with lifestyle changes and PPIs.

6 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Which of the following statements are true?
 - A The oesophagus is 30 cm long.
 - B The oesophagus has five natural constrictions.
 - C The gastro-oesophageal junction is 40 cm from the incisors.
 - D The oesophagus is lined by stratified squamous epithelium.
 - E The oesophageal opening in the diaphragm is at the level of T10 vertebra.
- 2 The following statements are true except:
 - A The lower oesophageal sphincter (LOS) is an area of high pressure.
 - B The normal length of the LOS is 3 to 4 cm.
 - C The three phases of swallowing, oral, pharyngeal and oesophageal, are involuntary.
 - D Pharyngeal diverticulum occurs because of the incoordination of the pharyngeal phase of deglutition.
 - E The pressure in the LOS is 10–25 mmHg.

- 3** Which of the following statements are false?
- A** Dysphagia is a cardinal symptom of oesophageal carcinoma.
 - B** Dysphagia associated with cough is a symptom of impending tracheo-oesophageal fistula.
 - C** Dysphagia in the oral and pharyngeal phases of swallowing indicates a neurological rather than a mechanical problem.
 - D** Odynophagia is always of cardiac origin.
 - E** Regurgitation and reflux are the same and caused by dysphagia.
- 4** The following statements are true except:
- A** Anginal pain can be confused with heartburn.
 - B** Heartburn with weight loss is a 'red flag' symptom.
 - C** Barium swallow is the ideal investigation in a patient with heartburn.
 - D** Dysphagia to liquids more than solids is typical of achalasia of cardia.
 - E** Oesophagogastroduodenoscopy (OGD) is the first line investigation in suspected spontaneous rupture of oesophagus.
- 5** Which of the following statements are true?
- A** A barium meal should be the first line of investigation in a patient presenting with heartburn.
 - B** 24-hour pH monitoring is the most accurate method to assess gastro-oesophageal reflux disease (GORD).
 - C** Laparoscopy should be undertaken prior to oesophagectomy.
 - D** Manometry is essential in the diagnosis of motility disorders.
 - E** Raman spectroscopy detects early molecular changes in Barrett's mucosal neoplasia.
- 6** Which of the following statements are false?
- A** Injection of botulinum toxin is a modality of treatment in achalasia.
 - B** The operation for resection of carcinoma of middle one-third of oesophagus is Heller's procedure.
 - C** Endoscopic mucosal resection is a surgical option in very early mucosal neoplasia in Barrett's oesophagus.
 - D** All patients with oesophageal perforation should be treated with surgical repair.
 - E** The ideal surgical treatment for a Zenker's diverticulum is excision through an incision on the left side of the neck followed by cricopharyngeal myotomy.

Answers 1-6

BAILEY & LOVE MCQs

- 1** **CDE** True: C, D, E — **UPDATED** True: the GOJ lies about 40 cm from the incisors, the lining is stratified squamous and the hiatus is at T10. The oesophagus is ~25 cm long with three constrictions.
- 2** **C** The three phases of swallowing, oral, pharyngeal and oesophageal, are involuntary. — Because the oral phase is voluntary while the pharyngeal and oesophageal phases are involuntary; LOS measurements are correct.
- 3** **DE** True: D, E — Odynophagia is typically due to oesophageal causes and regurgitation and reflux are distinct phenomena.
- 4** **CE** True: C, E — OGD is preferred for heartburn evaluation and in suspected spontaneous oesophageal rupture other imaging is indicated.
- 5** **BCDE** True: B, C, D, E — **UPDATED** 24-h pH monitoring is the most accurate test for GORD; staging laparoscopy, manometry (motility disorders) and Raman spectroscopy all have roles. OGD, not barium, is first-line for heartburn.
- 6** **BDE** True: B, D, E — Heller's myotomy is used for achalasia—not cancer—and not all perforations mandate surgery while endoscopic stapling is preferred for Zenker's diverticulum.

6 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A patient has a symptomatic pharyngeal pouch (Zenker diverticulum). What is the most important aspect of treatment?
- A** resection of the diverticulum
 - B** cricopharyngeal muscle myotomy
 - C** H₂ blockers
 - D** elevation of the head of the bed
 - E** diverticulopexy
- 2** A 55-year-old man with 10 years of heartburn, chronic cough, and regurgitation—whose endoscopy shows Barrett's esophagus with high-grade dysplasia—would most likely have been prevented by which early treatment?
- A** H₂ blockers
 - B** diet modification
 - C** Nissen fundoplication
 - D** proton-pump inhibitor
 - E** regular schedule of antacids

- 3** For the same patient with Barrett's esophagus and high-grade dysplasia, what is recommended?
- A** double dose proton-pump inhibitor and repeat endoscopy in 3 months
 - B** proton-pump inhibitor and H₂ blocker, followed by repeat endoscopy in 3 months
 - C** antibiotic treatment for *Helicobacter pylori* with proton-pump inhibitor
 - D** Nissen fundoplication
 - E** Endoscopic resection and ablation (oesophagectomy if endoscopic therapy is not possible)
- 4** A 55-year-old man with acute substernal chest and epigastric pain after prolonged vomiting, subcutaneous emphysema, and a chest X-ray showing pneumomediastinum has which diagnosis?
- A** bronchopleural fistula
 - B** esophageal perforation
 - C** Mallory-Weiss tear
 - D** spontaneous pneumothorax
 - E** gastric perforation
- 5** Which diagnostic test is most appropriate for suspected spontaneous oesophageal perforation (Boerhaave syndrome)?
- A** computed tomography (CT) scan
 - B** esophagogram
 - C** analysis of the pleural fluid
 - D** esophagoscopy
 - E** tracheobronchogram
- 6** Which factor most influences outcome in esophageal perforation?
- A** choice of antibiotics
 - B** time to definitive treatment
 - C** intensive care management
 - D** early intubation and mechanical ventilation
 - E** history of smoking

Answers 1-6

LANGE MCQS

- 1** **B cricopharyngeal muscle myotomy** — **UPDATED** Cricopharyngeal muscle myotomy Myotomy is key to relieve the underlying neuromotor dysfunction.
- 2** **C Nissen fundoplication** — Early antireflux surgery may prevent progression to Barrett's esophagus.
- 3** **E Endoscopic resection and ablation (oesophagectomy if endoscopic therapy is not possible)** — **UPDATED** Barrett oesophagus with high-grade dysplasia is now treated by endoscopic eradication (EMR of visible lesions plus radiofrequency ablation); oesophagectomy is reserved for failures or invasive disease.
- 4** **B esophageal perforation** — **UPDATED** The findings are indicative of Boerhaave's syndrome.
- 5** **B esophagogram** — **UPDATED** A water-soluble contrast oesophagogram is the classic test to confirm and localise the leak; CT with oral contrast is used if it is equivocal or the patient is unwell.
- 6** **B time to definitive treatment** — **UPDATED** Prompt treatment is critical for survival.

8 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** The blood supply to the oesophagus is derived from which three vessels?
- A** Inferior thyroid artery, descending aorta, left gastric artery
 - B** Internal carotid, descending aorta, oesophageal artery
 - C** Lateral thoracic artery, phrenic artery, right gastric artery
 - D** Pharyngeal artery, long thoracic artery, phrenic artery
 - E** Ascending aorta, common hepatic artery, left gastric artery
- 2** A 58-year-old patient presents with a 6-week history of increasing difficulty swallowing. He first noticed problems when eating meat which became stuck 'behind his heart', but this gradually began to include other foods. He is now struggling with thick fluids and has noticed some unintentional weight loss. What is the most appropriate investigation?
- A** Staging computed tomography
 - B** Barium meal
 - C** Upper gastrointestinal endoscopy
 - D** Barium swallow
 - E** Electrocardiography

- 3** A 27-year-old patient presents with a 3-month history of increasing difficulty in swallowing – beginning with fluids and later with solids – accompanied by regurgitation and unintentional weight loss; a chest radiograph shows a widened mediastinum. What is the most likely diagnosis?
- A Thoracic aortic aneurysm
 - B Oesophageal malignancy
 - C Plummer-Vinson syndrome
 - D Achalasia
 - E Oesophageal spasm
- 4** CREST syndrome is an autoimmune condition associated with atrophy and fibrosis of the oesophageal musculature resulting in dysphagia and reflux-type symptoms. Which of the following is not a feature of CREST syndrome?
- A Raynaud's phenomenon
 - B Erythematous malar rash
 - C Sclerodactyly
 - D Soft tissue calcification
 - E Telangiectasia
- 5** The gastro-oesophageal sphincter does not rely on a ring of contractile muscle but on several mechanisms to prevent reflux. Which of the following does not contribute to its function?
- A Left crus of diaphragm
 - B Smooth muscle of the lower oesophageal sphincter
 - C Angle of His
 - D Right crus of diaphragm
 - E Intra-abdominal pressure
- 6** A 58-year-old builder with a long history of retrosternal chest pain and a hiatus hernia is found on endoscopy to have grade 2 oesophagitis. What is the most appropriate management?
- A Triple eradication therapy
 - B Proton pump inhibitor
 - C Nissen's fundoplication
 - D Yearly endoscopic surveillance and biopsy
 - E Supportive gusset
- 7** Which of the following is not a lifestyle change that might reduce reflux-type symptoms?
- A Smoking cessation
 - B Abdominal exercise
 - C Avoid tight clothing
 - D Eat smaller portions more often
 - E Raise the head of the bed at night
- 8** A 65-year-old patient with symptoms of oesophagitis and endoscopic findings of Barrett's oesophagus with low grade dysplasia is best managed by
- A Referral for oesophagectomy
 - B Staging CT chest abdomen and pelvis
 - C Oesophageal ablation
 - D High dose PPI and repeat endoscopy
 - E Triple eradication therapy

Answers 1–8

- 1** **A Inferior thyroid artery, descending aorta, left gastric artery** — **UPDATED** The cervical oesophagus is supplied by the inferior thyroid artery, the thoracic part by aortic branches and the abdominal part by the left gastric artery.
- 2** **C Upper gastrointestinal endoscopy** — Endoscopy is the gold standard investigation to assess suspected oesophageal cancer.
- 3** **D Achalasia** — Achalasia, due to failure of the lower oesophageal sphincter to relax, is the most likely cause.
- 4** **B Erythematous malar rash** — An erythematous malar rash is not a characteristic feature of CREST syndrome.
- 5** **A Left crus of diaphragm** — The left crus does not contribute to the maintenance of sphincter tone.
- 6** **B Proton pump inhibitor** — Acid suppression with a PPI is the first-line treatment for oesophagitis associated with reflux.
- 7** **B Abdominal exercise** — Abdominal exercise may worsen reflux rather than improve it.
- 8** **D High dose PPI and repeat endoscopy** — **UPDATED** Low-grade dysplasia in Barrett oesophagus: high-dose PPI and repeat endoscopy at 6 months; confirmed persistent LGD is then offered radiofrequency ablation.

Oxford MCQs

- 1 28-year-old woman has dysphagia with low retrosternal 'sticking' and an occasional sensation of choking. A video barium swallow shows this appearance in the oesophagus (see Figure 6.1). Which is the single most likely diagnosis?



- A Achalasia
- B Carcinoma of the oesophagus
- C Diffuse oesophageal spasm
- D Foreign body ingestion
- E Reflux oesophagitis

- 2 72-year-old man has had a sensation of retrosternal burning that is worse at night and after meals for the last 4 months. It partially responds to 'over the counter' antacid treatment. Which is the single most appropriate initial diagnostic investigation?

- A Barium meal
- B Barium swallow
- C CT scan thorax
- D OGD
- E 24h pH studies

Answers 1-2

OXFORD MCQS

- 1 **A Achalasia** — Achalasia (failure of relaxation of the distal oesophageal musculature associated with degeneration of the myenteric plexus) is among the most common causes of dysphagia in the younger population.

- 2 **D OGD** — **UPDATED** New reflux symptoms in a 72-year-old need OGD first, to see the mucosa, take biopsies and exclude malignancy; pH studies come later if the diagnosis is unclear.

PreTest MCQs

- 1 A 41-year-old man complains of regurgitation of saliva and of undigested food. An esophagram reveals a dilated esophagus and a bird's-beak deformity. Manometry shows a hypertensive lower esophageal sphincter with failure to relax with deglutition. Which of the following is the safest and most effective treatment of this condition?

- A Medical treatment with sublingual nitroglycerin, nitrates, or calcium-channel blockers
- B Repeated bougie dilations
- C Injections of botulinum toxin directly into the lower esophageal sphincter
- D Dilation with a Gruntzig-type (volume-limited, pressure-control) balloon
- E Surgical esophagomyotomy

- 2 A 45-year-old executive experiences increasingly painful retrosternal heartburn, especially at night. He has been chewing antacid tablets. EKG and chest x-ray are normal. An esophagram shows a hiatal hernia. In determining the proper treatment for a sliding hiatal hernia, which of the following is the most useful modality?

- A Barium swallow with cinefluoroscopy during Valsalva maneuver
- B Flexible endoscopy
- C Twenty-four-hour monitoring of esophageal pH
- D Measurement of the size of the hernia on upper GI
- E Assessment of the patient's smoking and drinking history

- 3** A 65-year-old man has been noticing progressive difficulty swallowing, first solid food and now liquids as well. A barium study shows a narrowing just below the carina. Endoscopic biopsy confirms squamous cell carcinoma. Which of the following provides the most accurate information regarding the T stage of an esophageal carcinoma?
- A Computed tomography
 - B Positron emission tomography
 - C Magnetic resonance imaging
 - D Endoscopic ultrasound
 - E Bronchoscopy
- 4** A 56-year-old woman has nonspecific complaints that include an abnormal sensation when swallowing. An esophagram and EGD are performed. Which of the following is most likely to require surgical correction?
- A Large sliding esophageal hiatal hernia
 - B Paraesophageal hiatal hernia
 - C Traction diverticulum of esophagus
 - D Schatzki ring of distal esophagus
 - E Esophageal web

Answers 1-4

PRETEST MCQS

- 1** **E Surgical esophagomyotomy** — Surgical esophagomyotomy is the most effective treatment for achalasia.
- 2** **B Flexible endoscopy** — Endoscopy is useful in evaluating esophageal changes in hiatal hernia.
- 3** **D Endoscopic ultrasound** — Endoscopic ultrasound is the most accurate modality for T staging in esophageal carcinoma.
- 4** **B Paraesophageal hiatal hernia** — Paraesophageal hernias pose a risk of strangulation and typically require surgical repair.

4 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1** A 40-year-old man presents with increasing difficulty swallowing over the last few months. He has been working in Mexico for the last two years. He is otherwise well and denies any other symptoms. Which of the following is the most likely cause of his symptoms?
- A Chagas disease
 - B Gastro-oesophageal reflux disease
 - C Myasthenia gravis
 - D Plummer-Vinson syndrome
 - E Zenker diverticulum
- 2** A 56-year-old woman complains of episodes of severe retrosternal chest pain postprandial, with a bitter taste in her mouth, but no difficulty swallowing. Examination is unremarkable. Which investigation is most useful?
- A 24-hour lower oesophageal pH
 - B Barium swallow
 - C Erect chest X-ray
 - D Manometry
 - E Upper gastrointestinal endoscopy
- 3** A 33-year-old woman presents to the surgical outpatient clinic with increasing dysphagia for solids and liquids, retrosternal chest pain on eating, regurgitation of undigested food and weight loss. Oesophageal and gastric biopsy at endoscopy was normal. Oesophageal manometry showed impaired relaxation of the lower oesophageal sphincter and barium swallow revealed a dilated proximal oesophagus with distal narrowing. Which of the following surgical procedures would be indicated?
- A Billroth I
 - B Billroth II
 - C Heller myotomy
 - D Ramstedt procedure
 - E Whipple procedure
- 4** A 68-year-old man presents with acute lower chest and upper abdominal pain following an episode of vomiting. On examination, he has marked tenderness in the epigastrium and you note the presence of subcutaneous emphysema. ECG shows a sinus tachycardia with no ST segment changes, and a chest X-ray demonstrates mediastinal air. Which of the following is the most likely diagnosis?
- A Boerhaave syndrome
 - B Chest infection
 - C Mallory-Weiss tear
 - D Myocardial infarction
 - E Pulmonary embolus

Answers 1-4

GET AHEAD MCQS

- 1 **A Chagas disease** — Trypanosoma cruzi infection can cause megaesophagus presenting with progressive dysphagia.
- 2 **A 24-hour lower oesophageal pH** — GORD is best diagnosed by prolonged oesophageal pH monitoring.
- 3 **C Heller myotomy** — Achalasia is treated surgically by incising the LES (Heller myotomy).
- 4 **A Boerhaave syndrome** — Forceful vomiting caused transmural oesophageal rupture leading to mediastinal air.

2 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 45-year-old man has difficulty swallowing liquids/solids. Barium swallow shows narrowing in lower oesophagus; endoscopy excludes organic lesion. Which long-term therapy?
 - A Antispasmodics
 - B Balloon dilatation
 - C Heller myotomy
 - D Botulinum toxin
 - E Proton pump inhibitors
- 2 A 10-year-old boy ingests household bleach and has severe mouth/stomach pain. He is febrile, tachycardic. Next most appropriate initial investigation?
 - A Endoscopy
 - B Chest CT scan
 - C Chest X-ray
 - D Barium swallow
 - E Oesophageal manometry

Answers 1-2

CRASH COURSE MCQS

- 1 **C Heller myotomy** — Achalasia definitive relief from surgical myotomy.
- 2 **C Chest X-ray** — UPDATED Fever, tachycardia and severe pain after caustic ingestion suggest perforation: erect chest X-ray (then CT) first; endoscopy is contraindicated if perforation is suspected.

Stomach & Duodenum

Peptic ulcer and its complications, gastric outlet obstruction, gastric cancer, GIST and bariatric anatomy.

40

MCQS

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

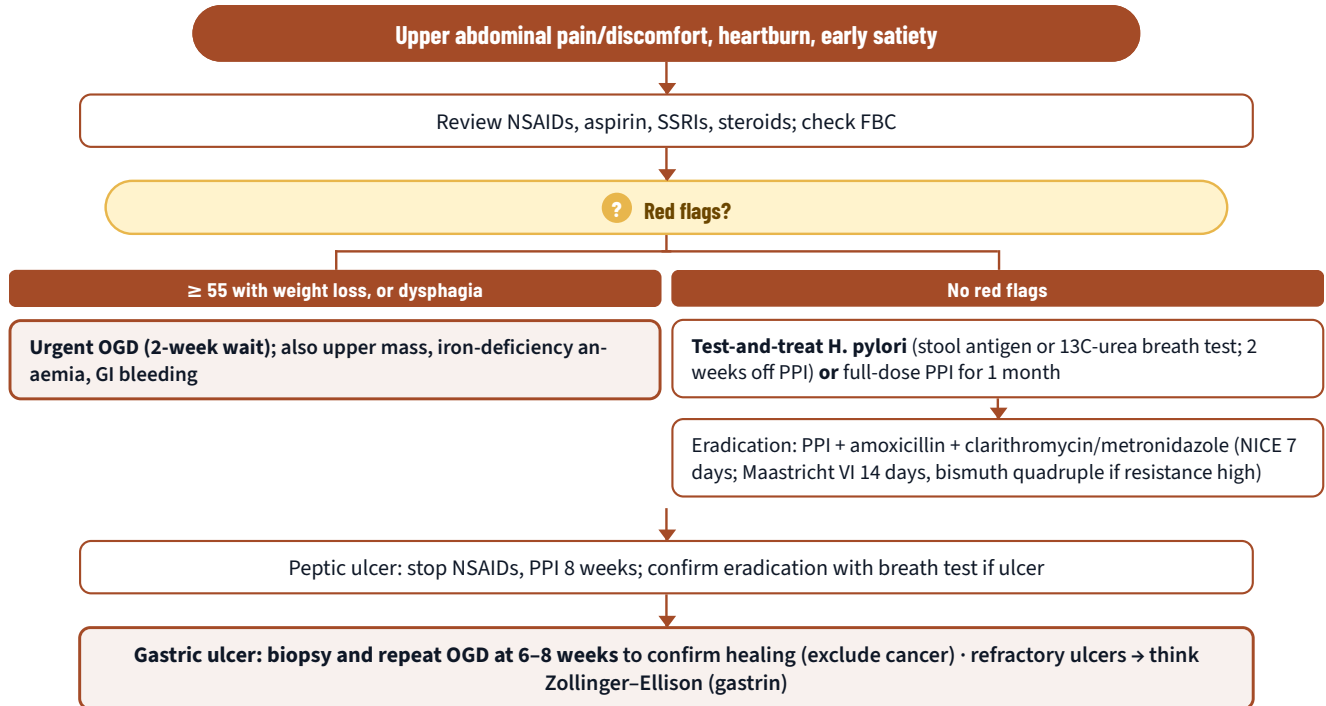
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09

Step by step

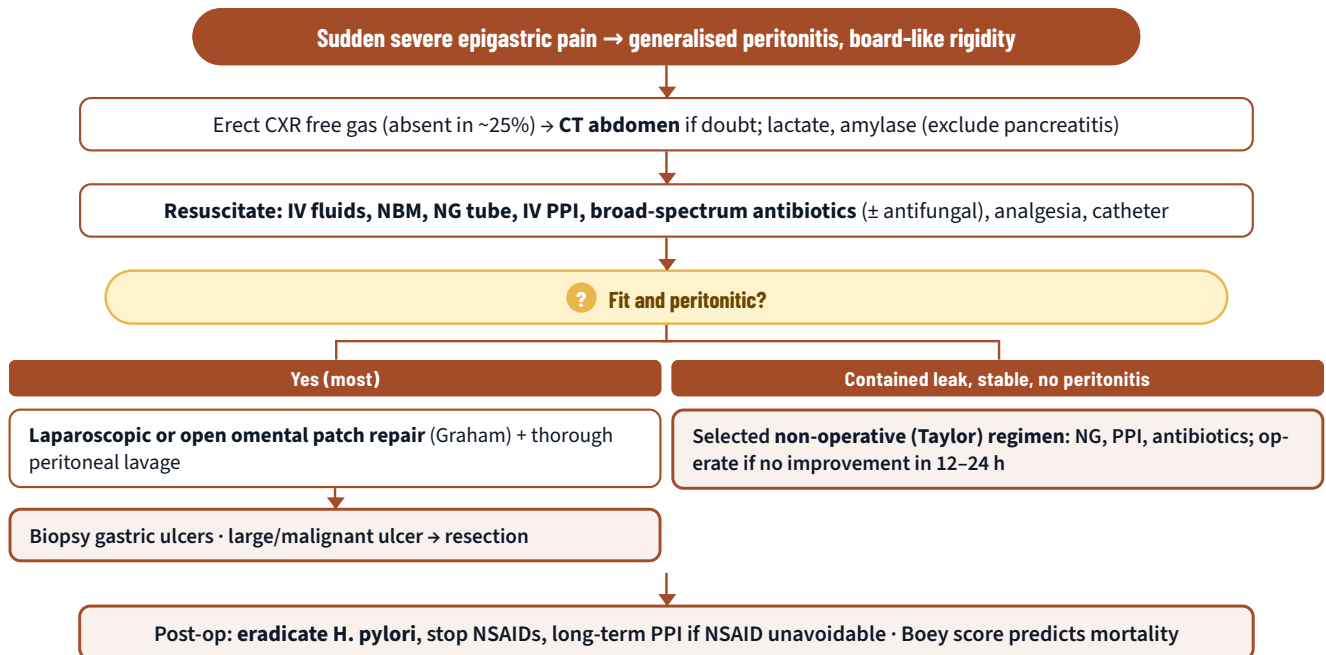
Dyspepsia, H. pylori and peptic ulcer

NICE CG184 • NICE NG12 • MAASTRICHT VI 2022



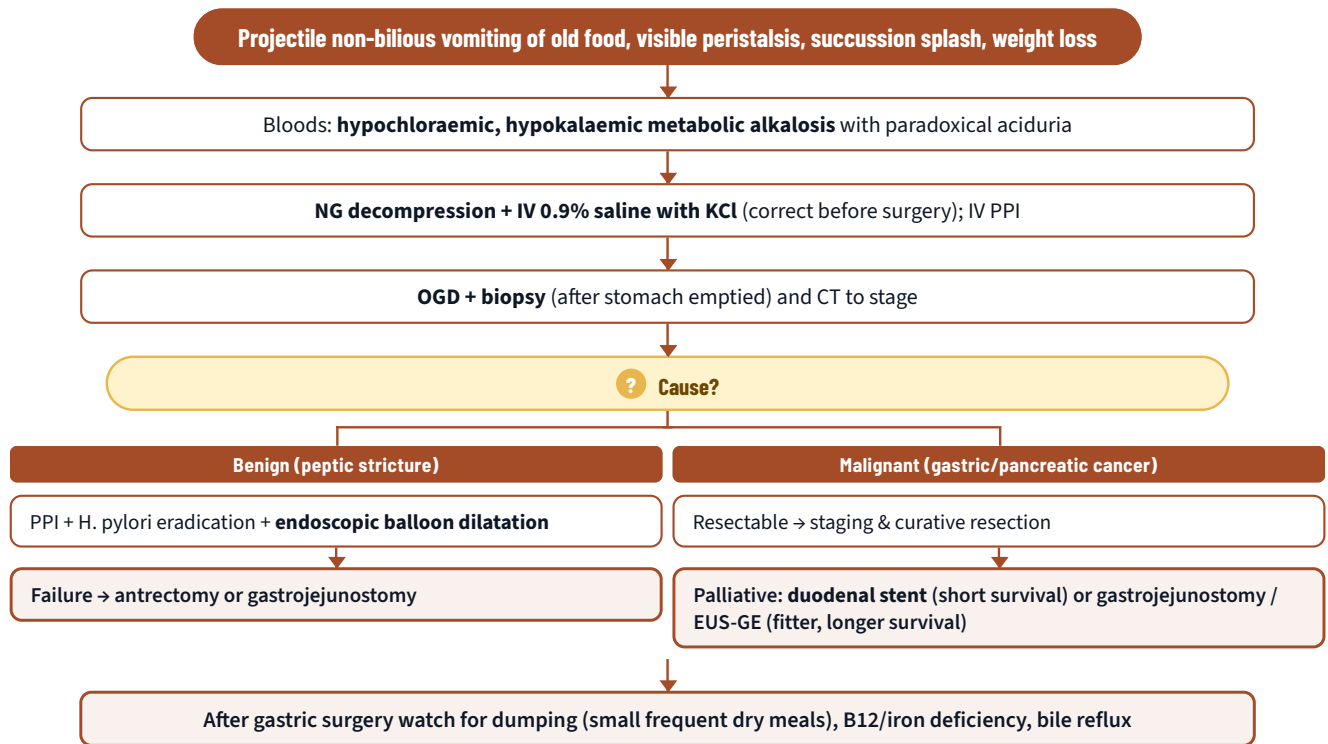
Perforated peptic ulcer

WSER 2020 PERFORATED & BLEEDING PEPTIC ULCER



Gastric outlet obstruction

BAILEY & LOVE 28E · ASGE 2021 G00



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 66-year-old woman with a history of peptic ulcer disease is diagnosed with osteoarthritis. She had prescribed ibuprofen for several days. Which of the following is the most appropriate precaution to take in order to reduce the risk of recurrence of peptic ulcer?
 - A Add omeprazole 20 mg
 - B Commence H. pylori eradication therapy
 - C Send for gastroscopy
 - D Switch to celecoxib
 - E Urea breath test followed by H. pylori eradication if positive
- 2 A 45-year-old woman presents to her primary care physician with complaints of recurrent epigastric pain that occurs shortly after meals. She reports a history of chronic stress due to work and family responsibilities. She denies any history of alcohol consumption or smoking. On further questioning, she mentions frequent use of medications for headaches relieve. Physical examination is unremarkable. What is the most likely diagnosis?
 - A Acute pancreatitis
 - B Cholecystitis
 - C Peptic ulcer
 - D Irritable bowel disease
 - E Chronic mesenteric ischemia
- 3 A 35-year-old female presents to her primary care physician with complaints of recurrent epigastric pain, bloating, and occasional nausea. She reports that the symptoms worsen when she skips meals or consumes spicy foods. There is no significant past medical history. Which of the following is the most appropriate initial investigation?
 - A Serologic testing for H. pylori antibodies
 - B Urea breath test (UBT)
 - C Upper gastrointestinal endoscopy with culture of gastric biopsy
 - D Upper gastrointestinal endoscopy with histopathological examination of gastric biopsy
 - E Rapid urease test (RUT) on gastric biopsy sample
- 4 A 57-year-old man who is a heavy smoker presents to his family physician with epigastric pain, occasional vomiting, tiredness and easy fatigability. Clinical examination reveals signs of anemia and epigastric tenderness but no masses or organomegaly. Routine blood tests confirm mild iron deficiency anemia but no other abnormalities. Which action by the family physician is the most appropriate?
 - A She should refer him for an urgent upper GI endoscopy (OGD)
 - B She should check his H. pylori serology and start him on eradication therapy if positive
 - C She should start him immediately on H. pylori eradication therapy
 - D She should start him on a course of PPIs and review him in 2 months for repeat blood tests
 - E She should arrange for him to have an urgent barium meal

Answers 1-4

LOCAL EXAMS MCQS

- 1 **E Urea breath test followed by H. pylori eradication if positive** — **UPDATED** A patient with previous ulcer starting an NSAID should be tested and treated for H. pylori and given PPI gastroprotection.
- 2 **C Peptic ulcer** — The clinical features suggest peptic ulcer disease likely related to analgesic use.
- 3 **B Urea breath test (UBT)** — Urea breath test is a reliable non-invasive method for detecting H. pylori infection.
- 4 **A She should refer him for an urgent upper GI endoscopy (OGD)** — Given his age and iron deficiency anemia, an urgent upper GI endoscopy is warranted to rule out malignancy.

9 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

- 1 Multiparous woman with vague upper abdominal pain relieved after defecation, flatulence, and an above-umbilicus mass. What is the cause of upper abdominal pain?
 - A Peptic ulcer disease
 - B Irritable bowel syndrome due to pain relieved after defecation, distention, and psychological factors
 - C Gallstones causing biliary colic
 - D Acute cholecystitis
 - E Pancreatitis
- 2 An 18-year-old female from Al-Haydaria presented with upper abdominal discomfort, and her GP decided to prescribe eradication therapy for *H. pylori*. (Note: This case is missing some information) Q1/ Was the doctor right about giving the empirical therapy initially?
 - A No, the therapy was inappropriate
 - B Yes, because *H. pylori* is the commonest cause of upper abdominal discomfort
 - C No, further testing was required before therapy
 - D Yes, but only if the patient had a confirmed peptic ulcer
 - E No, empirical therapy should never be initiated
- 3 A middle-aged male, approximately 41 years old, with hypertension, presents with new-onset epigastric pain, dyspepsia, weight loss, and tarry stool. What is the most important investigation needed to confirm the diagnosis? Why?
 - A Abdominal ultrasound
 - B Oesophago-gastro duodenoscopy (OGD) with biopsy
 - C CT scan
 - D MRI
 - E Barium swallow
- 4 OGD biopsy in the same patient shows high-grade lymphoid tissue suggestive of gastric MALT lymphoma. Besides chemoradiation, what is the appropriate medical treatment?
 - A Immediate surgery
 - B Observation
 - C Eradication of *H. pylori* with an anti-*H. pylori* regimen (PPI + amoxicillin + clarithromycin) as high-grade lymphoid tissue suggests MALToma
 - D Only chemotherapy
 - E Only radiotherapy
- 5 What non-invasive investigation would you send to assess the response to treatment?
 - A CT scan
 - B MRI
 - C Urea breath test
 - D Endoscopy
 - E Ultrasound
- 6 A 51-year-old heavy smoker male presents with increasing upper abdominal pain (especially with meals) and episodes of black stool. He had partial response to *H. pylori* eradication. What is the next logical step to establish the diagnosis?
 - A Colonoscopy
 - B CT scan
 - C Esophagogastrosocopy (EGD)
 - D Abdominal ultrasound
 - E Barium swallow
- 7 Previous testing shows positive *H. pylori* resistant to metronidazole & clarithromycin. Suggest the best treatment option now.
 - A PPI + metronidazole
 - B PPI + clarithromycin
 - C PPI + amoxicillin + rifampicin
 - D PPI + Amoxicillin + Levofloxacin
 - E PPI + bismuth
- 8 A 59-year-old male smoker presents with vague upper abdominal pain over the past three weeks, with mild nausea and fullness in the epigastric area. With a BMI of 41 and dyslipidemia, a specialist surgeon decided to start with anti-*H. pylori* treatment. Was the surgeon's decision right or not? Explain.
 - A No, because the patient does not have any evidence of *H. pylori* infection
 - B Yes, because upper abdominal pain with partial response to antacids suggests *H. pylori* infection with a peptic ulcer
 - C No, because the patient should have been managed for his dyslipidemia first
 - D Yes, but only after a positive urea breath test
 - E No, the treatment should have been directed at his prostate issues
- 9 What kind of non-pharmacological measures would you suggest to prevent the recurrence of his abdominal complaint?
 - A Increased protein intake
 - B Dietary modifications (avoid large fatty or spicy meals), weight loss, smoking cessation, and minimizing NSAIDs
 - C Bed rest and hydration
 - D Increased physical activity only
 - E Vitamin supplementation

Answers 1-9

ESSAY TO MCQ MCQS

- 1 **B Irritable bowel syndrome due to pain relieved after defecation, distention, and psychological factors** — Pain relief with defecation and associated psychological factors suggest IBS.
- 2 **C No, further testing was required before therapy** — **UPDATED** Young dyspeptic patients without alarm features are managed by 'test and treat': confirm *H. pylori* (urea breath or stool antigen test) before giving eradication therapy.
- 3 **B Oesophago-gastro duodenoscopy (OGD) with biopsy** — OGD with biopsy is crucial to evaluate for gastritis, ulceration, or malignancy.
- 4 **C Eradication of *H. pylori* with an anti-*H. pylori* regimen (PPI + amoxicillin + clarithromycin) as high-grade lymphoid tissue suggests MALToma** — **UPDATED** Pylori eradication is appropriate when high-grade lymphoid tissue is found, suggesting MALToma.
- 5 **C Urea breath test** — A urea breath test is a non-invasive method to assess *H. pylori* eradication.
- 6 **C Esophagogastroscope (EGD)** — EGD is indicated to further evaluate upper GI pathology in the context of melena.
- 7 **D PPI + Amoxicillin + Levofloxacin** — A regimen including levofloxacin is recommended for resistant *H. pylori* strains.
- 8 **D Yes, but only after a positive urea breath test** — **UPDATED** Eradication therapy should follow a positive test (urea breath or stool antigen); a 59-year-old with persistent symptoms also needs OGD if they do not settle.
- 9 **B Dietary modifications (avoid large fatty or spicy meals), weight loss, smoking cessation, and minimizing NSAIDs** — Lifestyle changes are key to preventing recurrence of peptic ulcer symptoms.

11 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Which of the following statements are true?
 - A It is easy to get views beyond the ligament of Treitz during an oesophagogastroduodenoscopy (OGD).
 - B Current endoscope is fibreoptic endoscope.
 - C It is not necessary to stop clopidogrel before colonoscopic polypectomy.
 - D Verbal consent is an accepted practice.
 - E Perforation and haemorrhage are uncommon but significant complications of the procedure.
- 2 Gross Anatomy – Blood Supply of the Stomach and Duodenum. Which of the following statements are true regarding blood supply?
 - A The stomach has an arterial supply on both lesser and greater curves.
 - B The right gastric artery arises from the common hepatic artery.
 - C The gastroduodenal artery passes behind the second part of duodenum.
 - D The short gastric arteries are branches of the left gastric artery.
 - E The right gastroepiploic artery runs along the greater curvature of the stomach.
- 3 Gross Anatomy – Nerve Supply of the Stomach and Duodenum. Which of the following statements are true regarding the nerve supply?
 - A The intrinsic nerves exist principally in two plexuses, the myenteric plexus of Auerbach and the submucosal plexus of Meissner.
 - B There are fewer ganglia in the submucosal plexus of the antrum of the stomach.
 - C The extrinsic supply is mainly derived from the vagus nerves.
 - D The vagal fibres are only efferent.
 - E The sympathetic fibres are derived from the thoracic ganglia.
- 4 Microscopic Anatomy of the Stomach and Duodenum. The following statements are true except:
 - A The parietal cells are in the body of the stomach.
 - B The chief cells produce pepsinogen.
 - C The endocrine cells are in the antrum and produce gastrin.
 - D The endocrine cells in the duodenum produce secretin and cholecystokinin.
 - E Histologically the chief cells are found at the mucosal surface.

- 5** Physiology of the Stomach and Duodenum – Gastric Acid Secretion. Which of the following statements are true regarding gastric acid secretion?
- A** Hydrogen ions are produced in the parietal cells by the proton pump.
 - B** The most important factor acting on the chief cell is histamine.
 - C** Vagus nerve is responsible for the gastric phase of acid secretion.
 - D** Secretin stimulates acid secretion.
 - E** Somatostatin inhibits acid secretion.
- 6** Physiology of the Stomach – Functions of the Stomach. Which of the following are functions of the stomach?
- A** Reservoir for ingested food.
 - B** Mechanically breaks down ingested food.
 - C** Starts the process of digestion.
 - D** Produces mucus.
 - E** Proximal stomach exhibits peristaltic activity.
- 7** Investigations of the Stomach and Duodenum. Which of the following statements are true?
- A** Flexible endoscopy is the primary investigation of choice.
 - B** Endoscopic ultrasound is useful in determining the 'T' status of tumours.
 - C** CT scan is useful in the staging of gastric cancer.
 - D** PET/CT scan can help avoid radical surgery in approximately 10% of patients with gastric cancer.
 - E** Laparoscopy is helpful in evaluating posterior extension of tumours.
- 8** Type A and B Gastritis. The following statements are true except:
- A** Type A gastritis is an autoimmune condition.
 - B** Type A gastritis produces hypochlorhydria and achlorhydria.
 - C** Type B gastritis is associated with *Helicobacter pylori*.
 - D** Type B gastritis most often affects the body and fundus of the stomach.
 - E** Both type A and B gastritis predispose to gastric cancer.
- 9** Erosive Gastritis. Which of the following statements are true?
- A** It is caused by increased production of acid.
 - B** It is associated with the inhibition of COX-1 receptor enzyme.
 - C** The production of cytoprotective prostaglandins is reduced in this condition.
 - D** Long-term use of selective COX-2 inhibitors may cause cardio-vascular side effects.
 - E** The beneficial anti-inflammatory effects of NSAIDs are mediated via COX-2 receptors.
- 10** Menetrier's Disease. The following statements are true except:
- A** Menetrier's disease is characterised by gross hypertrophy of the gastric mucosal folds.
 - B** Hyperchlorhydria is an associated finding of Menetrier's disease.
 - C** Menetrier's disease is caused by the reduced expression of transforming growth factor alpha (TGF- α).
 - D** Menetrier's disease might present with hypoproteinaemia and anaemia.
 - E** The treatment for Menetrier's disease is gastrectomy.
- 11** Stress Gastritis. Which of the following statements are true?
- A** Usually it occurs in patients who are in the intensive care unit (ICU).
 - B** It is due to avascularity of the gastric mucosa.
 - C** The usual presentation is perforation of an ulcer.
 - D** Prevention is the keystone in management.
 - E** *H. pylori* has no role in its aetiology.

Answers 1-11

- 1** **E** Perforation and haemorrhage are uncommon but significant complications of the procedure. — Modern endoscopes are video-based; OGD rarely visualizes beyond the ligament of Treitz, clopidogrel is usually stopped, and written consent is standard.
- 2** **ABE** True: A, B, E — The stomach receives blood along both curvatures; note that the gastroduodenal artery passes behind the first part of the duodenum and short gastrics arise from the splenic artery.
- 3** **AC** True: A, C — The stomach's innervation includes intrinsic enteric plexuses and a dominant vagal (both afferent and efferent) parasympathetic supply; sympathetic fibres come via the coeliac ganglia.
- 4** **E** Histologically the chief cells are found at the mucosal surface. — Chief cells are actually located deep within the gastric glands rather than at the mucosal surface.
- 5** **AE** True: A, E — **UPDATED** Parietal cells secrete H^+ via the H^+/K^+ ATPase and somatostatin inhibits acid. Histamine acts on parietal cells, the vagus mediates the cephalic phase and secretin inhibits acid.
- 6** **ABCD** True: A, B, C, D — The stomach stores food, mechanically and chemically digests it, and secretes mucus for protection; the proximal part mainly relaxes rather than showing strong peristalsis.
- 7** **ABCD** True: A, B, C, D — Endoscopy is first-line; EUS, CT and PET/CT all have important roles in staging, while laparoscopy is less useful for assessing posterior extension.
- 8** **D** Type B gastritis most often affects the body and fundus of the stomach. — Type B gastritis predominantly affects the antrum, not the body and fundus.

- 9 **BCDE** True: B, C, D, E — Erosive gastritis results from compromised mucosal protection due to reduced prostaglandin synthesis; acid overproduction is not the primary cause.
- 10 **BC** True: B, C — **UPDATED** False: Ménétrier disease causes hypochlorhydria and results from increased (not reduced) TGF- α

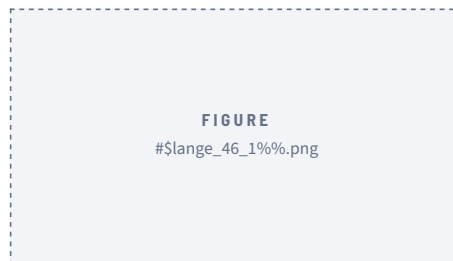
signalling. It gives giant folds, protein loss and anaemia; gastrectomy is a last resort.

- 11 **ABDE** True: A, B, D, E — Stress gastritis is seen in critically ill patients due to mucosal ischemia and is best managed by prophylaxis; perforation is uncommon.

4 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 30-year-old man with a 6-month history of chronic epigastric pain that is relieved by antacids now presents with sudden onset severe epigastric pain, vomiting, and a rigid abdomen. Plain radiographs reveal pneumoperitoneum. Which of the following is the most likely diagnosis?
- A small-bowel obstruction
 - B dead bowel
 - C perforated colon carcinoma
 - D perforated duodenal ulcer
 - E perforated gastric ulcer
- 2 A 55-year-old man presents with a 2-month history of gnawing upper abdominal pain exacerbated by food, nausea, and 20-lb weight loss. Which of the following tests is the most reliable method for diagnosing a gastric ulcer?
- A UGI barium x-rays
 - B fiberoptic upper endoscopy
 - C CT scan
 - D endoscopic ultrasound
 - E MRI
- 3 In a patient with a benign gastric ulcer that persists after 3 months of proton-pump inhibitor and triple antibiotic therapy for *H. pylori*, what is the most appropriate next step in management?
- A a second trial of proton-pump inhibitors with triple antibiotics and re-evaluation in 2 months
 - B a trial of H2 blockers with triple antibiotics and re-evaluation in 2 months
 - C a trial of sucralfate and re-evaluation in 2 months
 - D surgical management
 - E a trial of prostaglandins and re-evaluation in 2 months
- 4 A 65-year-old woman presents with 6 months of epigastric discomfort, poor appetite and weight loss. Abdominal ultrasound reveals cholelithiasis, and an upper GI contrast series is shown. What is the most likely diagnosis?



- A Cholecystoenteric fistula
- B Duodenal ulcer
- C Gastric ulcer
- D Gastric diverticulum
- E Duodenal diverticulum

Answers 1-4

LANGE MCQS

- 1 **D perforated duodenal ulcer** — **UPDATED** Perforated duodenal ulcer The sudden onset and radiographic pneumoperitoneum in a patient with a history of peptic pain point to a perforated duodenal ulcer.
- 2 **B fiberoptic upper endoscopy** — **UPDATED** Fiberoptic upper endoscopy Endoscopy is the most reliable diagnostic tool for gastric ulcers.
- 3 **D surgical management** — **UPDATED** Surgical management Persistent ulceration despite adequate medical therapy is an indication for surgical intervention.
- 4 **C Gastric ulcer** — **UPDATED** The barium study shows a gastric ulcer; with weight loss it needs endoscopic biopsy to exclude malignancy. The gallstones are incidental.

4 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** Haematemesis A 55-year-old man, with a 2-year history of dyspepsia, is brought to the emergency department following a sudden onset of severe epigastric pain. The pain is made worse on movement and the patient has also experienced one episode of haematemesis. On examination, the patient is cold, sweating profusely and taking shallow breaths. The abdomen is rigid and bowel sounds are absent. A plain film chest radiograph reveals free air under the diaphragm. The most likely diagnosis is

 - A Perforated appendicitis
 - B Acute cholecystitis
 - C Acute pancreatitis
 - D Myocardial infarction
 - E Perforated peptic ulcer
- 2** Vomiting and left upper quadrant pain A 55-year-old woman presents with severe epigastric and left upper quadrant pain. Since admission, the patient has vomited. On examination you notice the patient is retching (which is non-productive), tachycardic and hypotensive. There is marked tenderness in the upper abdomen and bowel sounds are slightly raised. There is failure to pass a nasogastric tube. A chest radiograph reveals a dilated stomach and large fluid level behind the heart. Which is the most likely diagnosis?

 - A Small bowel obstruction
 - B Perforated peptic ulcer
 - C Gastro-oesophageal obstruction secondary to a gastric volvulus
 - D Gastro-oesophageal obstruction secondary to an adenocarcinoma of the stomach
 - E Sigmoid volvulus
- 3** A 45-year-old patient in shock with sudden-onset generalized upper abdominal pain radiating to the right iliac fossa and tip of the right shoulder, with a rigid abdomen and absent bowel sounds, is most likely suffering from

 - A Caecal volvulus
 - B Pancreatitis
 - C Perforated duodenal ulcer
 - D Ascending cholangitis
 - E Appendicitis
- 4** Which syndrome is associated with multiple duodenal ulcers that are refractory to conventional therapy?

 - A Hereditary non-polyposis colon cancer (HNPCC)
 - B Plummer-Vinson syndrome
 - C Gardiner's syndrome
 - D Zollinger–Ellison syndrome
 - E CREST syndrome

Answers 1–4

SBA MCQs

- 1** **E Perforated peptic ulcer** — Free air under the diaphragm suggests perforation of a hollow viscus.
- 2** **C Gastro-oesophageal obstruction secondary to a gastric volvulus** — Failed NG tube and radiograph findings suggest a gastric volvulus.
- 3** **C Perforated duodenal ulcer** — The clinical picture is most consistent with a perforated duodenal ulcer.
- 4** **D Zollinger–Ellison syndrome** — High gastrin levels in Zollinger–Ellison syndrome lead to refractory duodenal ulcers.

1 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 47-year-old woman has had intermittent epigastric discomfort and dyspepsia after food for 10 months. OGD reveals features of moderate inflammation in the body and antrum of the stomach without gastric or duodenal ulceration. A rapid urease test from a biopsy is positive. Which is the single most appropriate treatment prescription for her?
- A Gaviscon 10mL PO as needed + omeprazole 20mg once daily
 - B Omeprazole 20mg PO twice daily + amoxicillin 250mg PO three times daily
 - C Omeprazole 20mg PO twice daily + metronidazole 400mg PO twice daily + clarithromycin 250mg PO twice daily
 - D Omeprazole 20mg PO twice daily + ranitidine 150mg PO twice daily
 - E Ranitidine 150mg PO twice daily + metronidazole 400mg PO twice daily + amoxicillin 250mg PO three times daily

Answer 1

OXFORD MCQS

- 1 **C** Omeprazole 20mg PO twice daily + metronidazole 400mg PO twice daily + clarithromycin 250mg PO twice daily — Combination therapy for *H. pylori* eradication.

3 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 45-year-old man with a history of chronic peptic ulcer disease undergoes a truncal vagotomy and antrectomy with a Billroth II reconstruction (gastrojejunostomy) for gastric outlet obstruction. Six weeks after surgery, he returns, complaining of postprandial weakness, sweating, lightheadedness, crampy abdominal pain, and diarrhea. Which of the following would be the best initial management strategy?
- A Treatment with a long-acting somatostatin analog
 - B Dietary advice and counseling that symptoms will probably abate within 3 months of surgery
 - C Dietary advice and counseling that symptoms will probably not abate but are not dangerous
 - D Workup for neuroendocrine tumor (eg, carcinoid)
 - E Preparation for revision to Roux-en-Y gastrojejunostomy
- 2 A 55-year-old man complains of chronic intermittent epigastric pain. A gastroscopy demonstrates a 2-cm prepyloric ulcer. Biopsy of the ulcer yields no malignant tissue. After a 6-week trial of medical therapy, the ulcer is unchanged. Which of the following is the best next step in his management?
- A Repeat trial of medical therapy
 - B Local excision of the ulcer
 - C Highly selective vagotomy
 - D Treatment for *Helicobacter pylori*
 - E Vagotomy and pyloroplasty

- 3** A 42-year-old man with no history of use of NSAIDs presents with recurrent gastritis. The patient was diagnosed and treated for *H. pylori* 6 months ago. EGD at that time was normal. Which of the following tests provides the least invasive method to document eradication of the infection?
- A Serology testing for *H. pylori*
 - B Carbon-labeled urea breath test
 - C Rapid urease assay
 - D Histologic evaluation of gastric mucosa
 - E Culturing of gastric mucosa

Answers 1-3

PRETEST MCQs

- 1** **B Dietary advice and counseling that symptoms will probably abate within 3 months of surgery** — Dumping syndrome is managed initially with dietary modifications and reassurance.
- 2** **D Treatment for *Helicobacter pylori*** — Persistent gastric ulcer requires *H. pylori* treatment and possible surgical resection to rule out malignancy.
- 3** **B Carbon-labeled urea breath test** — The urea breath test is the least invasive method to document *H. pylori* eradication.

4 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1** A 33-year-old man presents with a 2-hour history of upper abdominal pain radiating to the back and shoulders. On examination, he is sweaty, with a rigid abdomen. HR 106/min, BP 98/58 mmHg. What is the most likely cause?
- A Epigastric hernia incarceration
 - B Gastric carcinoma
 - C Gastritis
 - D Hiatus hernia
 - E Perforated peptic ulcer
- 2** A 54-year-old woman has epigastric pain with food. Endoscopy shows a gastric ulcer, and a rapid urease test is positive for *H. pylori*. Which treatment is correct?
- A Augmentin and metronidazole
 - B Omeprazole, clarithromycin and metronidazole
 - C Omeprazole, prednisolone and augmentin
 - D Omeprazole and ibuprofen
 - E Ranitidine
- 3** A 45-year-old man presents with episodic upper abdominal pain, severe heartburn, episodes of dark vomitus and diarrhoea. He has been taking over-the-counter indigestion medication with no relief. On endoscopy, he has multiple ulcers throughout the duodenum, some of which are large. Which of the following is most likely to have caused this?
- A Alcohol
 - B Aspirin use
 - C Gastrinoma
 - D *Helicobacter pylori* infection
 - E Smoking
- 4** A 27-year-old man presents to the emergency department with severe epigastric pain which started soon after eating his lunch. On examination, the upper abdomen is rigid. The patient says he has recently been suffering from indigestion which started after medical treatment for a meniscal injury to his knee. Which of the following medications could have contributed to his condition?
- A Aspirin
 - B Codeine
 - C Gaviscon
 - D Omeprazole
 - E Paracetamol

Answers 1-4

GET AHEAD MCQS

- 1 **E Perforated peptic ulcer** — Acute onset epigastric pain radiating to the back, rigid abdomen indicates perforation with peritonitis.
- 2 **B Omeprazole, clarithromycin and metronidazole** — Triple therapy to eradicate *H. pylori* and heal peptic ulcer.
- 3 **C Gastrinoma** — Excessive gastrin secretion (Zollinger-Ellison) causes multiple refractory peptic ulcers.
- 4 **A Aspirin** — NSAID use can predispose to peptic ulcer and perforation.

Small & Large Intestine

Intestinal obstruction, appendicitis, inflammatory bowel disease, diverticular disease, volvulus, polyps and colorectal cancer.

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MCQS

0

EMQ SCENARIOS

9

SOURCES

1

REPEATED IDEAS

IN THIS CHAPTER

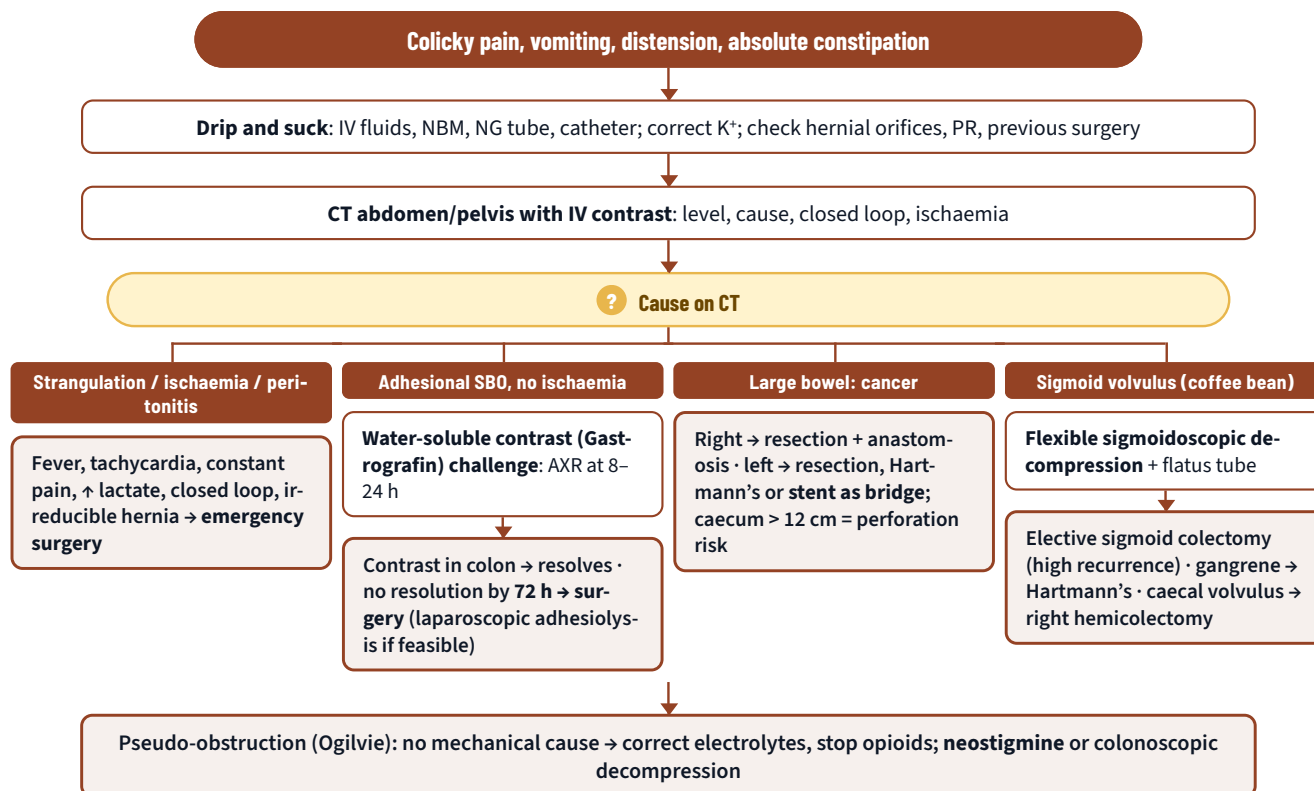
01	Management flowcharts	3 charts	page 118
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08	PreTest MCQs	18 Q	page 146
09	Get Ahead MCQs	16 Q	page 150
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10

Step by step

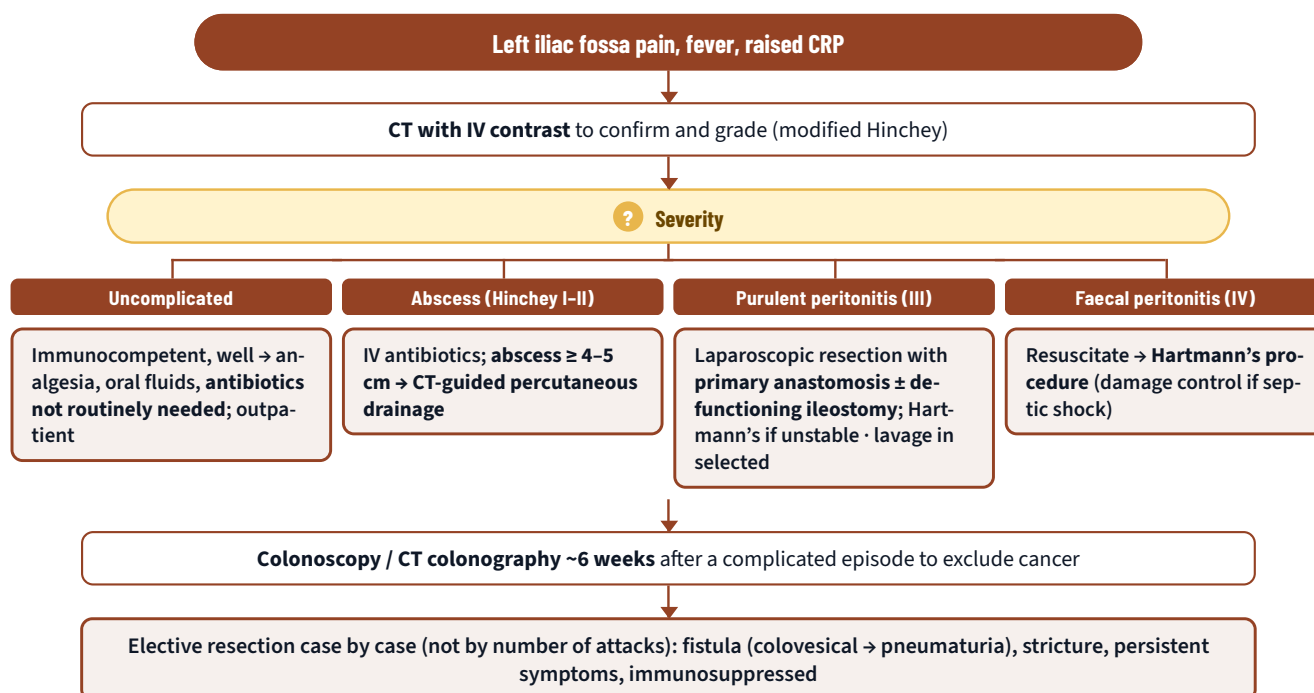
Intestinal obstruction

WSES 2017 ASBO (BOLOGNA) • ESCP/WSES 2021 LBO • BAILEY & LOVE 28E



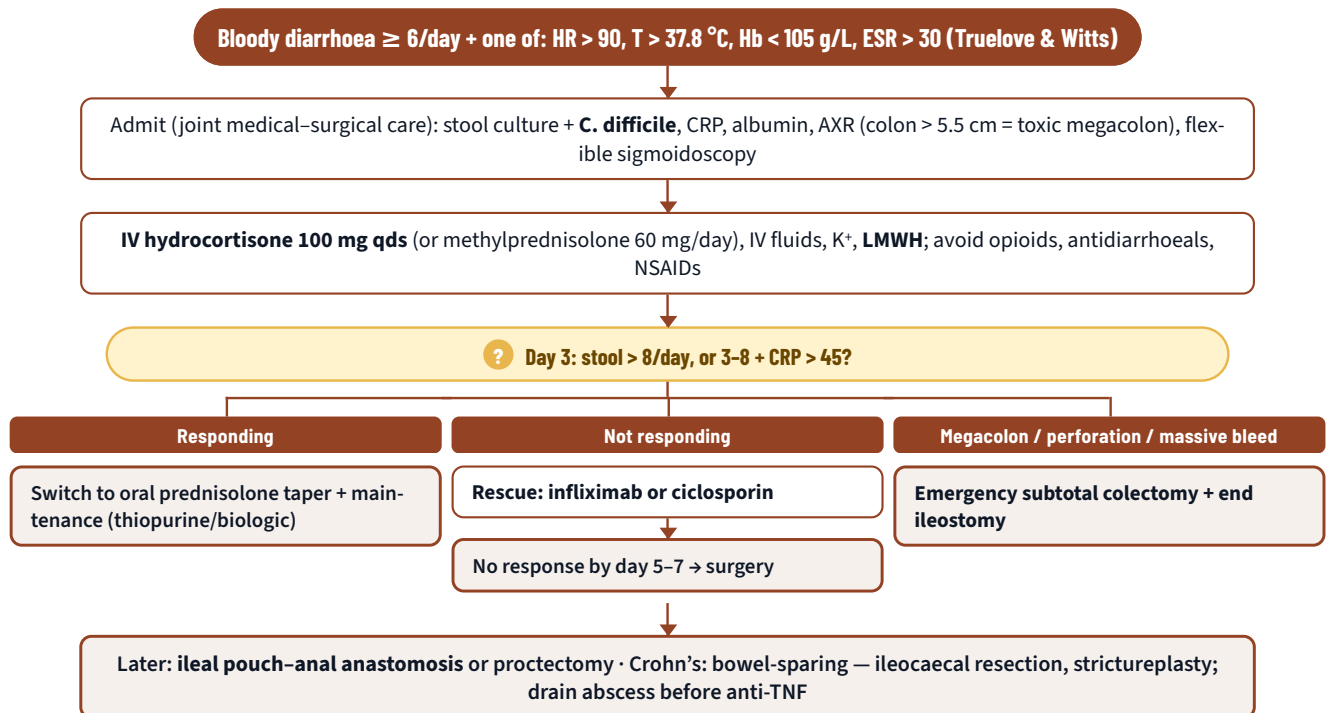
Acute diverticulitis

NICE NG147 • WSES 2020 DIVERTICULITIS • ESCP 2020



Acute severe ulcerative colitis (and surgery in IBD)

BSG 2019 IBD · ECCO 2022 UC



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1–3

An elderly man with previous bloody stools and lower abdominal pain has a sudden attack of left iliac fossa pain with fever. Ultrasound shows thickening of the sigmoid colon.

- 1 What is the most likely diagnosis?
 - A Diverticulitis
 - B Ischaemic colitis
 - C Sigmoid carcinoma
 - D Crohn disease
 - E Appendicitis
- 2 What is the best investigation now?
 - A Colonoscopy
 - B Barium enema
 - C Contrast-enhanced CT
 - D Plain abdominal X-ray
 - E MRI
- 3 What is the treatment?
 - A Colostomy
 - B Sigmoid colectomy
 - C Bowel rest and antibiotics
 - D Percutaneous drainage
 - E Laparoscopic lavage

CASE • QUESTIONS 4–6

An elderly woman in a care home who takes antipsychotics and has chronic constipation presents with a distended, painless, non-tender abdomen. X-ray shows a hugely dilated sigmoid loop reaching the upper abdomen.

- 4 What is the diagnosis?
 - A Sigmoid volvulus
 - B Diverticular disease
 - C Obstructing sigmoid tumour
 - D Acute colonic pseudo-obstruction
 - E Toxic megacolon
- 5 Which procedure is both diagnostic and therapeutic?
 - A Endoscopic (flexible sigmoidoscopic) decompression
 - B Barium enema
 - C CT abdomen
 - D Laparotomy
 - E Rectal tube without endoscopy

- 6 At surgery the descending and sigmoid colon are resected. Which artery is ligated?
 - A Ileocolic
 - B Superior mesenteric
 - C Inferior mesenteric
 - D Middle colic
 - E Coeliac trunk

CASE • QUESTIONS 7–8

A 65-year-old man with distension, persistent vomiting and absent bowel sounds has intestinal obstruction; an NG tube is ordered for decompression.

- 7 How is NG tube placement confirmed?
 - A Aspirate pH ≤ 5.5 , or chest X-ray if no aspirate or pH is high
 - B Auscultation of injected air (whoosh test)
 - C Bubbling when the end is placed in water
 - D Litmus paper colour change to blue
 - E Patient able to talk
- 8 Which is a contraindication to NG tube insertion?
 - A Suspected basal skull fracture
 - B Intestinal obstruction
 - C Postoperative ileus
 - D Upper GI bleeding
 - E Gastric outlet obstruction
- 9 A plain abdominal X-ray of a distended elderly patient is shown. What is the most likely diagnosis on this image?



- A Sigmoid volvulus
- B Small bowel obstruction
- C Caecal volvulus
- D Toxic megacolon
- E Gallstone ileus

- 10** An erect abdominal X-ray is shown. What is the most likely diagnosis on this image?



- A Small bowel obstruction
- B Pneumoperitoneum
- C Sigmoid volvulus
- D Normal film
- E Paralytic ileus of the colon

MINISTERIAL EXAM

Ministerial evaluation examinations

- 11** A young woman visits the endocrinology clinic due to short stature and osteoporosis. She complains of chronic diarrhea and iron deficiency anemia. She has a positive anti-tTG IgA Ab. Which of the following statements about celiac disease is TRUE?
- A Duodenal mucosal villous atrophy is not specific to celiac disease
 - B Genetic study of HLA DQ2 DQ8 is diagnostic in equivocal cases
 - C Usually the duodenum is infected with *H pylori* bacteria
 - D Response to the standard treatment of steroid is usually good
 - E Endoscopy is the best initial investigation in suspected cases
- 12** A 29-year-old man presents with anaemia, bleeding tendency, diarrhoea and abdominal pain. Examination reveals a palpable mass in the right lower quadrant and anal skin tags. What is the most likely underlying condition?
- A Chronic pancreatitis
 - B Coeliac disease
 - C Crohn's disease
 - D Intestinal lymphoma
 - E Ulcerative colitis

- 13** A 50-year-old man underwent a laparoscopic closure of a perforated duodenal ulcer. His postoperative period during the first 4 to 5 days was uneventful. Thereafter, he complained of hiccups with pain in the right subcostal area and at the tip of the right shoulder. He had swinging pyrexia and looked toxic, respiratory rate: 25/minute. Abdominal examination revealed tenderness in the right upper quadrant with oedema of the overlying skin. The most likely diagnosis is:
- A Idiopathic retroperitoneal fibrosis
 - B Mesenteric cyst.
 - C Retroperitoneal liposarcoma
 - D Right sub-hepatic abscess.
 - E Small bowel obstruction from adhesions.

- 14** A 23-year old male patient presents with recurrent episodes of abdominal pain and intermittent rectal bleeding. The patient is noticed to have multiple dark pigmented spots on his lips. On endoscopic examination, multiple dark pigmented polyps are noted in the mucosa of his intestine. What is the most likely diagnosis?
- A Crohn's disease
 - B Gastro-intestinal stromal tumor (GIST)
 - C Ulcerative colitis
 - D Bowel lymphoma
 - E None of the above

- 15** A 22-year-old lady, who was diagnosed as a case of typhoid fever two weeks ago and was kept on medical treatment, presents to the ER with sudden onset of severe abdominal pain. On examination, her abdomen is rigid with tachycardia and abdominal x-ray shows free gas under right diaphragm. The most probable diagnosis is:
- A Acute mesenteric ischemia
 - B Perforation of the Ileum
 - C Small bowel volvulus
 - D Actinomycosis
 - E Campylobacteriosis

- 16** A 65-year-old woman complains of left iliac fossa pain associated with passage of dark-red blood per rectum. She is habitually constipated and takes regular laxatives for a satisfactory bowel action. She is overweight. On examination, her abdomen is tender at the left iliac fossa. What is the most likely diagnosis?
- A Colorectal carcinoma
 - B Crohn's disease
 - C Diverticular disease
 - D Ischemic colitis
 - E Ulcerative colitis

- 17** A 9-year-old child presents with abdominal pain near the umbilical region, no vomiting, and normal bowel motion. On examination: there is tenderness with rebound tenderness in the right iliac fossa. Which one of the following conditions is most commonly confused with an acute appendicitis in the child?
- A Basal Pneumonia
 - B Pelvic inflammatory diseases
 - C Acute pancreatitis
 - D Diverticulitis
 - E None of the above
- 18** A 35-year-old female known to have ulcerative colitis that is controlled by mesalazine for the last 5 years came to the consultation clinic. She is complaining of relapsing diarrhoea of around 6 times daily and mild abdominal pain for the last three days. On examination: temperature 38.2°C, pulse rate 110/min, hypotensive, dehydrated, with mild abdominal tenderness. General stool exam shows RBCs and mucus. Abdominal plain x-ray shows dilated transverse colon up to 4-5 cm in diameter. After resuscitation, the next most appropriate step in treatment should be:
- A Prednisolone 40 mg daily
 - B Hydrocortisone 400 mg daily
 - C Surgical intervention
 - D Metronidazole 800 mg three times daily
 - E Change to sulfasalazine
- 19** A 60-year-old male patient was admitted to the hospital with anemia and a positive stool for occult blood test. He stated that his appetite was decreased in the past 6 weeks but his bowel habit was not changed. Examination of his abdomen showed fullness on lower right side of abdomen. What is the most likely diagnosis?
- A Caecal tumor
 - B Cecal tuberculosis
 - C Appendicular mass
 - D Intussusception
 - E Crohn's disease of the bowel
- 20** A 32-year-old male with third degree internal hemorrhoids underwent an operative hemorrhoidectomy in the surgical center and was discharged home. Which is one of the most common early postoperative complications after hemorrhoidectomy?
- A Urinary retention
 - B Fistula in ano
 - C Fecal Incontinence
 - D Perirectal abscess
 - E Anal fissure
- 21** A 59-year-old woman presented with right lower quadrant pain, nausea, and vomiting. She underwent an uncomplicated laparoscopic appendectomy. Postoperatively, the pathology revealed a 2.5 cm mucinous adenocarcinoma with lymphatic invasion. Staging workup (including colonoscopy, chest x-ray, and computed tomography scan of the abdomen and pelvis) was negative. Which of the following is the most appropriate next step in her management?
- A No further intervention at this time, follow up every 6 months for 2 years
 - B Chemotherapy
 - C Neoadjuvant chemotherapy followed by right hemicolectomy
 - D Right hemicolectomy
 - E Pelvic irradiation therapy
- REPEATED 2x · ALSO ASKED IN PreTest**
- FORMATIVE**
College formative examinations
- 22** A 31-year-old female presents with nausea, vomiting, loss of appetite, and pain that started in the right upper quadrant and then shifted to the right lower quadrant. On examination, she is non-tender, has a low-grade fever, GUE shows pus ++, and her WBC count is 8000. Suspecting acute appendicitis, what is her Alvarado score?
- A 4
 - B 5
 - C 6
 - D 8
 - E 7
- 23** She was treated with broad-spectrum antibiotics and returned after two weeks. CT scan shows fecalith obstructing the appendix. What is the most likely location of the appendix in this case?
- A Retrocecal
 - B Pelvic
 - C Post-iliac
 - D Pre-ileal
 - E Subhepatic
- 24** A patient develops abdominal pain and vomiting after previous laparoscopic surgery. Examination shows periumbilical tenderness and swelling. What is the likely diagnosis?
- A Adhesive intestinal obstruction
 - B Acute appendicitis
 - C Port-site hernia (incisional hernia)
 - D Paralytic ileus
 - E Gastroenteritis

Answers 1-24

LOCAL EXAMS MCQS

- 1 **A Diverticulitis** — LIF pain, fever and sigmoid wall thickening in an older patient is acute diverticulitis.
- 2 **C Contrast-enhanced CT** — Contrast CT confirms diverticulitis and shows abscess or perforation; colonoscopy and barium enema are avoided acutely (perforation risk).
- 3 **C Bowel rest and antibiotics** — Uncomplicated diverticulitis is treated conservatively (bowel rest, fluids, antibiotics if systemically unwell); drainage is for abscess > 4 cm.
- 4 **A Sigmoid volvulus** — A huge sigmoid loop rising from the pelvis (coffee-bean sign) in an elderly, constipated, institutionalised patient is sigmoid volvulus.
- 5 **A Endoscopic (flexible sigmoidoscopic) decompression** — Flexible sigmoidoscopy confirms the twist, checks viability and decompresses with a flatus tube; elective sigmoid colectomy follows to prevent recurrence.
- 6 **C Inferior mesenteric** — The descending and sigmoid colon are supplied by the inferior mesenteric artery (left colic and sigmoid branches).
- 7 **A Aspirate pH ≤ 5.5, or chest X-ray if no aspirate or pH is high** — Use gastric aspirate pH ≤ 5.5 first-line and X-ray if that fails; the whoosh and bubble tests are unreliable and unsafe.
- 8 **A Suspected basal skull fracture** — Basal skull or severe mid-face fractures risk intracranial passage, so use the orogastric route; recent oesophageal surgery or atresia are others.
- 9 **A Sigmoid volvulus** — A huge inverted-U (coffee-bean) loop arising from the pelvis is sigmoid volvulus.
- 10 **A Small bowel obstruction** — Multiple central air–fluid levels in dilated small-bowel loops indicate small bowel obstruction.
- 11 **A Duodenal mucosal villous atrophy is not specific to celiac disease** — Villous atrophy can be seen in conditions other than celiac disease.
- 12 **C Crohn's disease** — The presence of anal skin tags and right lower quadrant mass is suggestive of Crohn's disease.
- 13 **D Right sub-hepatic abscess.** — **UPDATED** Swinging fever, hiccups, right shoulder-tip pain and RUQ tenderness days after repair of a perforated ulcer is a right subhepatic/subphrenic abscess: CT, then percutaneous drainage.
- 14 **E None of the above** — **UPDATED** Mucocutaneous pigmentation of the lips with hamartomatous GI polyps is Peutz–Jeghers syndrome, which is not listed; polyps cause bleeding and intussusception.
- 15 **B Perforation of the ileum** — **UPDATED** Sudden peritonitis with free gas in the 2nd–3rd week of typhoid fever is perforation of a Peyer patch ulcer in the terminal ileum; resuscitate and operate.
- 16 **C Diverticular disease** — **UPDATED** Left iliac fossa pain with dark-red rectal bleeding in a constipated, overweight older woman suggests diverticular disease; colorectal cancer must still be excluded by colonoscopy.
- 17 **A Basal Pneumonia** — **UPDATED** In children, right lower lobe (basal) pneumonia commonly mimics appendicitis through referred pain from the T10–T12 nerves; examine the chest and get a CXR.
- 18 **B Hydrocortisone 400 mg daily** — **UPDATED** Acute severe UC with fever, tachycardia and colonic dilatation needs admission, IV hydrocortisone and daily review; no response in 3–5 days means rescue therapy or colectomy.
- 19 **A Caecal tumor** — **UPDATED** Iron-deficiency anaemia, occult blood and a right iliac fossa mass in a 60-year-old is caecal carcinoma until proven otherwise: colonoscopy and CT.
- 20 **A Urinary retention** — **UPDATED** Urinary retention is the commonest early complication of haemorrhoidectomy (pain, spinal anaesthesia, excess fluid); bleeding is the other early concern.
- 21 **D Right hemicolectomy** — **UPDATED** Appendiceal adenocarcinoma with lymphatic invasion needs a right hemicolectomy for nodal clearance; adjuvant chemotherapy depends on nodal stage. Neoadjuvant chemotherapy is not standard.
- 22 **A 4** — **UPDATED** Alvarado: migratory pain 1, anorexia 1, nausea/vomiting 1, fever 1; no RIF tenderness, rebound, leucocytosis or left shift = 4 (low probability; pyuria suggests UTI).
- 23 **A Retrocecal** — Retrocecal appendix is the most common position and often presents with minimal tenderness.
- 24 **C Port-site hernia (incisional hernia)** — Port-site hernia is a known complication of laparoscopic surgery, leading to acute intestinal obstruction.

34 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQs

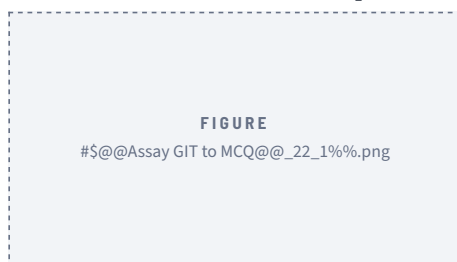
Essay to MCQ MCQs

SET 1 OF 2

- 1** A 37-year-old diabetic woman has had severe loose motions for 24 hours with vague left-sided abdominal pain and tenesmus; the last motion contained blood. There is mild lower abdominal tenderness; pulse 103/min, BP 121/82 mmHg, temperature 37.3 °C; WBC 12 000/mm³, HbA1c 7.8%. Ultrasound shows small asymptomatic gallstones. What is the most likely diagnosis and the first investigation?
- A Amebic or bacillary dysentery. General stool examination
 - B Inflammatory bowel disease. Colonoscopy with biopsy
 - C Viral gastroenteritis. Stool culture
 - D Ischemic colitis. CT scan of the abdomen
 - E Acute appendicitis. Ultrasound abdomen
- 2** Advocate the best treatment option
- A Intravenous fluids and supportive care
 - B High fiber diet and oral rehydration solutions
 - C Oral rehydration with antiemetics
 - D Empirical treatment with metronidazole (tinidazole) with anti Gram-ve AB as aminoglycoside, 2nd or 3rd generation cephalosporins or quinolones
 - E Immediate surgical intervention
- 3** Two year later, she is admitted to the ER with increasing abdominal pain with repeated vomiting over the past two days. The vomitus is bitter & bluish-green. The pain is episodic and colicky. She is dehydrated. Examination revealed periumbilical tenderness and swelling Explain the cause of this presentation.
- A Acute intestinal obstruction due to port-site hernia (incisional hernia)
 - B Acute gastroenteritis due to bacterial infection
 - C Small bowel obstruction due to adhesions
 - D Acute pancreatitis secondary to gallstones
 - E Paralytic ileus due to electrolyte imbalance
- 4** A 70-year-old man has been sent to the emergency department from a nursing home, complaining of intermittent sharp abdominal pain. He has not opened his bowels for days. He suffered a major stroke in the past and requires constant nursing care. He has a history of chronic constipation. Previous medical history includes chronic obstructive airways disease for which he is on regular inhalers. He is allergic to penicillin and is an ex-smoker. On examination, his blood pressure is 110/70 mmHg, pulse rate is 112 bpm, temperature is 37.8°C, and his abdomen is grossly distended and tympanic. Digital rectal examination reveals an empty rectum. Scattered rhonchi and a soft systolic murmur were found upon chest auscultation. His past surgical history is positive for a right hip replacement. Q1/ Is Erect CXR necessary for this patient's initial work up? Explain.
- FIGURE**

#\$\$@@Assay GIT to MCQ@@@_16_1%%.png
- A No, it is not necessary as the patient only shows abdominal symptoms
 - B Yes, it is essential to detect air under the diaphragm and to evaluate the chest due to COPD
 - C No, a CT scan would suffice
 - D Yes, but only to assess cardiac function
 - E No, the digital rectal exam is sufficient
- 5** What is the study shown in the scenario?
- A Abdominal ultrasound
 - B CT scan of the abdomen
 - C Plain Abdominal X-ray
 - D MRI of the abdomen
 - E Barium enema

- 6** A 59-year-old male, heavy smoker, alcoholic, and diabetic, presents with severe left-sided abdominal pain and moderate-grade fever over several days. Workup is shown below. What is this workup?



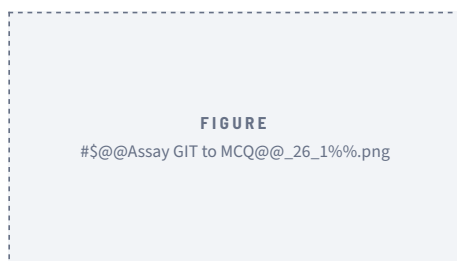
- A Ultrasound
- B Plain abdominal X-ray
- C CT scan
- D MRI
- E Barium enema

- 7** What are the findings?
- A Normal colonic anatomy
 - B Diffuse colonic thickening
 - C Outpouching of the colonic mucosa (Diverticulosis)
 - D A mass lesion in the colon
 - E Intussusception

- 8** What is the most likely diagnosis?
- A Colonic cancer
 - B Irritable bowel syndrome
 - C Diverticulitis
 - D Inflammatory bowel disease
 - E Ischemic colitis

- 9** Suggest the treatment
- A Immediate surgery
 - B Conservative management with dietary restriction, resuscitation, analgesia, and antibiotics
 - C Chemotherapy
 - D Endoscopic stenting
 - E High fiber diet alone

- 10** In spite of maximal treatment, the patient did not improve with high grade fever and rigors. Vitals: PR-125, BP-110. Workup is shown below. What is the diagnosis?



- A Simple diverticulosis
- B Acute appendicitis
- C Complicated diverticulitis with abscess
- D Perforated peptic ulcer
- E Colonic carcinoma

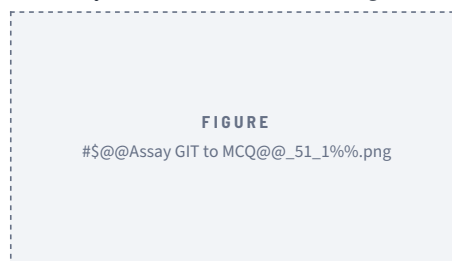
- 11** What is the best treatment option now?
- A Emergency laparotomy
 - B Conservative management with IV fluids
 - C Percutaneous drainage of abscess
 - D High-dose steroids
 - E Repeat colonoscopy

- 12** Four weeks later, the patient is almost asymptomatic except for mild left lower abdominal pain. Abdominal exam is normal. What is the gold standard test to reach the cause of his disease?
- A CT scan
 - B MRI
 - C Colonoscopy
 - D Barium enema
 - E Ultrasound

- 13** What is the best surgical option?
- A Right hemicolectomy
 - B Total colectomy
 - C Sigmoidectomy with primary anastomosis
 - D Appendectomy
 - E Laparoscopic cholecystectomy

- 14** What vital structure is at risk of such surgery?
- A Inferior vena cava
 - B Ureters and inferior mesenteric artery
 - C Portal vein
 - D Hepatic artery
 - E Splenic flexure

- 15** A male patient presented with constipation and left lower quadrant pain; workup is shown below. What is the study? and what are the findings?



- A CT scan showing colonic wall thickening
- B Plain abdominal X-ray showing air-fluid levels
- C Barium enema showing barium filled multiple outpouchings of the sigmoid colon (Diverticulosis)
- D Ultrasound showing bowel dilation
- E Colonoscopy showing ulcerations

- 16** What is the best modality for investigation?
- A MRI
 - B Ultrasound
 - C Endoscopy
 - D CT scan with IV contrast
 - E X-ray

- 17** Later, the patient developed tenderness, generalized rigidity and decreased bowel sounds. What is the cause?
- A Acute appendicitis
 - B Intussusception
 - C Diverticular perforation leading to peritonitis
 - D Bowel obstruction due to adhesions
 - E Gastroenteritis
- 18** What is the surgical management now?
- A Laparoscopic cholecystectomy
 - B Appendectomy
 - C Sigmoidectomy with primary anastomosis
 - D Total colectomy
 - E Hartmann's procedure
- 19** A young man presents with central abdominal pain for 3 hours referring to the right iliac fossa, with vomiting and a 12-hour fasting history. Later, he develops wound erythema, pain, swelling, and a hot right leg. What is your professional diagnosis?
- A Acute cholecystitis
 - B Acute pancreatitis
 - C Acute appendicitis
 - D Small bowel obstruction
 - E Diverticulitis
- 20** How should the patient be prepared before surgery?
- A Immediate surgery without any preparation
 - B Only IV fluids
 - C Only preoperative imaging
 - D IV fluids and antibiotics (Cephalosporins + metronidazole)
 - E Oral medications and discharge
- 21** What test should be done to confirm the diagnosis?
- A Endoscopy
 - B CT scan
 - C Ultrasound
 - D MRI
 - E X-ray
- 22** What is the most reliable radiological test?
- A X-ray
 - B Ultrasound
 - C MRI
 - D CT scan
 - E PET scan
- 23** What is the surgery of choice for such a patient?
- A Cholecystectomy
 - B Colectomy
 - C Appendectomy
 - D Hernia repair
 - E Laparotomy
- 24** Name the radiological sign seen in this study.
- A String sign
 - B Double bubble sign
 - C Coffee bean appearance of sigmoid colon
 - D Target sign
 - E Air-fluid level
- 25** What's the most likely diagnosis?
- A Acute appendicitis
 - B Small bowel obstruction
 - C Sigmoid volvulus
 - D Intestinal perforation
 - E Mesenteric ischemia

Answers 1-25

ESSAY TO MCQ MCQS

- 1** **A Amebic or bacillary dysentery.** General stool examination — **UPDATED** Presentation consistent with dysentery; stool exam is the investigation of choice.
- 2** **D Empirical treatment with metronidazole (tinidazole) with anti Gram-ve AB as aminoglycoside, 2nd or 3rd generation cephalosporins or quinolones** — Empirical antibiotic therapy is indicated for dysentery.
- 3** **A Acute intestinal obstruction due to port-site hernia (incisional hernia)** — Presentation suggests obstruction due to incisional hernia.
- 4** **B Yes, it is essential to detect air under the diaphragm and to evaluate the chest due to COPD** — Erect CXR is indicated to check for free air and evaluate chest in COPD.
- 5** **C Plain Abdominal X-ray** — Plain abdominal X-ray is the study shown.
- 6** **C CT scan** — The workup described is a CT scan.
- 7** **C Outpouching of the colonic mucosa (Diverticulosis)** — CT scan shows diverticulosis.
- 8** **C Diverticulitis** — Clinical and CT findings support diverticulitis.
- 9** **B Conservative management with dietary restriction, resuscitation, analgesia, and antibiotics** — Uncomplicated diverticulitis is managed conservatively.
- 10** **C Complicated diverticulitis with abscess** — Persistent fever and rigors indicate an abscess formation.
- 11** **C Percutaneous drainage of abscess** — Percutaneous drainage is indicated for an abscess.
- 12** **C Colonoscopy** — Colonoscopy is the gold standard for colonic evaluation post-diverticulitis.
- 13** **C Sigmoidectomy with primary anastomosis** — Sigmoidectomy with primary anastomosis is recommended for complicated diverticulitis.
- 14** **B Ureters and inferior mesenteric artery** — Ureters and the inferior mesenteric artery are at risk during sigmoidectomy.
- 15** **C Barium enema showing barium filled multiple outpouchings of the sigmoid colon (Diverticulosis)** — Barium enema reveals diverticulosis.

- 16 **D CT scan with IV contrast** — CT scan with IV contrast is the optimal investigation.
- 17 **C Diverticular perforation leading to peritonitis** — Perforation of a diverticulum causing peritonitis is indicated.
- 18 **C Sigmoidectomy with primary anastomosis** — UPDATED
Generalised peritonitis from perforated diverticulitis needs sigmoid resection: primary anastomosis ($\pm$ defunctioning ileostomy) if stable, Hartmann procedure if unstable or faecal contamination.
- 19 **C Acute appendicitis** — Presentation is most suggestive of acute appendicitis.
- 20 **D IV fluids and antibiotics (Cephalosporins + metronidazole)** — Preparation includes IV fluids and antibiotics.
- 21 **C Ultrasound** — Ultrasound is used to help confirm appendicitis.
- 22 **D CT scan** — CT scan is the most reliable for diagnosing appendicitis.
- 23 **C Appendectomy** — Appendectomy is the treatment of choice for appendicitis.
- 24 **C Coffee bean appearance of sigmoid colon** — The coffee bean sign is characteristic of sigmoid volvulus.
- 25 **C Sigmoid volvulus** — Findings point to sigmoid volvulus.

SET 2 OF 2

- 26 Suggest the best minimally invasive procedure to treat him.
- A Laparoscopic appendectomy
 - B Open surgical decompression
 - C Endoscopic decompression - detorsion
 - D Conservative management with nasogastric tube
 - E Laparotomy
- 27 What acid-base disturbance is expected in this patient?
- A Metabolic alkalosis
 - B Respiratory alkalosis
 - C Metabolic acidosis due to hyperglycemia and lactic acidosis
 - D Respiratory acidosis
 - E No acid-base disturbance
- 28 A 63-year-old male has been experiencing fresh bleeding per rectum over the past 2 months with left-sided abdominal discomfort. DRE reveals small second-degree piles. What is the gold standard test for this patient?
- A Abdominal CT scan
 - B Barium enema
 - C Colonoscopy
 - D Digital rectal examination
 - E Sigmoidoscopy
- 29 What kind of preparations are needed for such a test?
- A Fasting for 24 hours only
 - B No preparation is required
 - C Bowel preparation including a clear liquid diet and laxatives such as polyethylene glycol given in split dosing
 - D Intravenous hydration only
 - E Sedation with midazolam
- 30 During the planned procedure, the patient experiences sudden undue abdominal pain and bloating, and the test is postponed. Hours later, his condition deteriorates with PR=115 and temperature=37.9. Examination reveals diffuse lower abdominal tenderness with rigidity and sluggish bowel sounds. Bowel perforation is suspected. What are two radiological tests that can be used to confirm bowel perforation?
- A MRI of the abdomen and ultrasound
 - B Abdominal X-ray and ultrasound
 - C Chest X-ray to reveal air under the diaphragm and CT scan to localize the perforation
 - D Endoscopy and PET scan
 - E Barium swallow and CT scan
- 31 The patient was put on conservative treatment and improved, but he still complains of mucus diarrhea. What is the appropriate treatment now?
- A Increase the dose of laxatives
 - B Initiate chemotherapy
 - C Oral rehydration solutions and empirical metronidazole treatment while awaiting stool examination results
 - D Immediate surgical intervention
 - E High fiber diet and probiotics
- 32 After a full recovery, the patient's workup revealed sigmoidal diverticulosis. What is the best surgical treatment for this condition?
- A Appendectomy
 - B Laparoscopic cholecystectomy
 - C Sigmoid resection with primary anastomosis
 - D Total colectomy
 - E Herniorrhaphy
- 33 From which main artery did the bleeding originate?
- A Superior mesenteric artery
 - B Renal artery
 - C Celiac trunk
 - D Inferior mesenteric artery (specifically the sigmoid arteries)
 - E Aorta

34 What vital structure is at risk during such a surgery?

- A The liver
- B The spleen
- C The ureters and the inferior mesenteric artery
- D The stomach
- E The pancreas

Answers 26–34

ESSAY TO MCQ MCQS

26 **C Endoscopic decompression - detorsion** — Endoscopic detorsion is preferred for sigmoid volvulus.

27 **C Metabolic acidosis due to hyperglycemia and lactic acidosis** — Hyperglycemia and tissue hypoxia lead to metabolic acidosis.

28 **C Colonoscopy** — Colonoscopy is the gold standard for evaluating rectal bleeding and colonic pathology.

29 **C Bowel preparation including a clear liquid diet and laxatives such as polyethylene glycol given in split dosing** — Proper bowel preparation is essential for an effective colonoscopy.

30 **C Chest X-ray to reveal air under the diaphragm and CT scan to localize the perforation** — Combining a chest X-ray and CT scan is effective in confirming bowel perforation.

31 **C Oral rehydration solutions and empirical metronidazole treatment while awaiting stool examination results** — This treatment addresses residual symptoms and potential infection.

32 **C Sigmoid resection with primary anastomosis** — Sigmoid resection with primary anastomosis is the treatment of choice for symptomatic diverticulosis.

33 **D Inferior mesenteric artery (specifically the sigmoid arteries)** — Bleeding in sigmoidal diverticulosis typically originates from the sigmoid arteries.

34 **C The ureters and the inferior mesenteric artery** — The ureters and inferior mesenteric artery are critical structures at risk during sigmoid resection.

76 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

SET 1 OF 4

1 Which of the following manifestations are caused by *Ascaris lumbricoides*?

- A Intestinal symptoms as a larva.
- B Pulmonary symptoms as a larva.
- C Intestinal symptoms as an adult worm.
- D Pulmonary symptoms primarily as an adult worm.
- E Biliary disease as a larva.

2 What percentage of the world's population is affected by *Ascaris lumbricoides* infestation?

- A 2%
- B 10%
- C 25%
- D 50%
- E 75%

3 How is roundworm disease transmitted?

- A Inadvertent ingestion of larvae present in soil.
- B Inadvertent ingestion of fertilised eggs present in soil.
- C Through penetration of unbroken skin by larvae.
- D Inadvertent ingestion of fertilised eggs in contaminated meat.
- E The larva is transmitted to the lung as a droplet infection.

4 Which of the following are true of the life cycle of *Ascaris lumbricoides*?

- A Eggs release larvae in the lumen of the intestine, which then develop directly into adult worms.
- B Eggs release larvae into the portal blood and then into the liver before reaching the lungs.
- C Eggs penetrate the intestinal wall, travel to the liver where they develop into adult worms, and are later released into the intestinal lumen via the biliary tract.
- D The larvae are released in the lungs; after developing there, they are swallowed in sputum to complete maturation in the intestine.
- E Eggs are released into the environment in stool as resilient forms.

5 How is Loeffler's syndrome characterized?

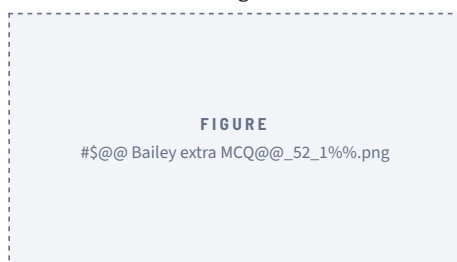
- A Chest pain with productive cough, friction rub, and a chest x-ray showing pleural effusion.
- B Retrosternal burning with discomfort on ingestion of food, fever, and mediastinal widening on chest x-ray.
- C Epigastric pain radiating to the right with a tender, palpable liver and an elevated right diaphragm.
- D Chest pain, dry cough, dyspnoea, and fever with fluffy exudates on chest x-ray.
- E Haemoptysis.

- 6** Which of the following complications can be caused by roundworm infestation?
- A** Malnutrition and failure to thrive.
 - B** Mechanical intestinal obstruction.
 - C** Intestinal perforation.
 - D** Ascending cholangitis, obstructive jaundice, and acute pancreatitis.
 - E** Chest symptoms.

- 7** Which of the following may be found in roundworm infestations?
- A** A high eosinophil count.
 - B** Charcot–Leyden crystals in stool.
 - C** Larvae detected in sputum or on bronchoscopic lavage.
 - D** Fluffy exudates visible on chest x-ray.
 - E** A barium meal showing a worm lying freely in the intestinal lumen.

- 8** Which of the following are recognized treatments for uncomplicated roundworm intestinal obstruction?
- A** Urgent laparotomy.
 - B** Nasogastric suction, IV saline, and hypertonic saline enemas.
 - C** Kneading the worm bolus into the large intestine at laparotomy followed by hypertonic saline enemas.
 - D** Gastrojejunostomy.
 - E** Exteriorisation of the bowel in cases of long-standing small intestinal perforation with heavy worm load.

- 9** A patient presents with sudden severe upper abdominal pain with chills and rigors. He has icterus and is diffusely tender over the right upper abdomen. His blood tests show an elevated white cell count (WCC) with high polymorphs and obstructive jaundice. The ultrasound scan (Figure 6.2) shows stones in the gall bladder and linear shadow in a dilated common bile duct (CBD). The magnetic resonance cholangiopancreatogram (MRCP) shows a linear shadow in the CBD, which in real time is found to change its position. Which of the following statements are true?

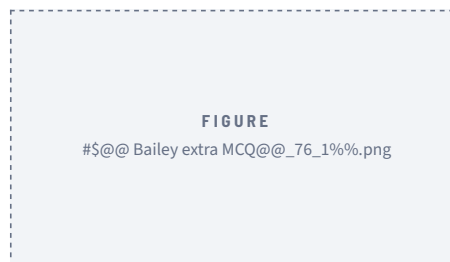


- A** The most likely diagnosis is a line of stones in the CBD.
- B** The condition is likely due to a live worm in the CBD.
- C** The condition should be treated with pyrantel pamoate.
- D** The condition should be treated with albendazole.
- E** Following an anthelmintic, endoscopic removal of the worm followed by cholecystectomy is indicated.

- 10** Which of the following types of intestinal lesions may be caused by *Mycobacterium tuberculosis*?
- A** Transverse ulcers with undermined edges in the ileum.
 - B** Tubercles on the serosal surface of the intestine.
 - C** Apple-core lesions of the colon.
 - D** Hyperplasia and thickening of the terminal ileum.
 - E** Transmural inflammation with a propensity for fistula formation.

- 11** What presenting features may be seen in tuberculosis of the small intestine?
- A** Weight loss, vague abdominal pain, and fever with an evening rise.
 - B** A “doughy” feel of the abdomen from localized ascites.
 - C** A palpable mass in the right iliac fossa.
 - D** A water-can perineum (perianal fistulas with watery discharge).
 - E** Histology showing caseating granulomas.

- 12** A 24-year-old woman with recurrent abdominal pain and distension has a barium study showing terminal ileum narrowing with the caecum pulled subhepatically. Which features suggest tuberculosis?



- A** She is from a developing country with a poor socioeconomic background.
- B** There is a history or evidence of pulmonary tuberculosis.
- C** There are skip lesions of the intestine.
- D** The barium study shows narrowing of the terminal ileum with a normally located caecum.
- E** The caecum is displaced to a subhepatic position.

- 13** Which of the following surgical procedures may be performed for ileal and ileocaecal tuberculosis?

- A** Strictureplasty.
- B** Limited ileocolic resection.
- C** Right hemicolectomy.
- D** Ileo-transverse anastomosis.
- E** All of the above.

- 14** Which of the following are true concerning tuberculous perforation?

- A** Free gas under the diaphragm is always present.
- B** Management includes resuscitation followed by resection of the affected segment.
- C** It may be managed by resuscitation and strictureplasty through the perforation.
- D** Resection with exteriorisation as a first step followed by delayed restoration of bowel continuity after antituberculous therapy is an accepted approach.
- E** It should always be managed conservatively.

- 15** Which of the following are true of typhoid fever?
- A** The bacteria enter the body via infected blood or contaminated needles.
 - B** The organism colonizes the Peyer's patches in the terminal ileum, leading to hyperplasia followed by necrosis and ulceration.
 - C** If untreated, the ulcers may bleed or perforate.
 - D** A typical presentation includes high fever for 2–3 weeks after travel to an endemic area, with abdominal distension due to paralytic ileus.
 - E** Patients may present as emergencies with melaena, hypovolaemia, peritonitis, and shock.
- 16** Which of the following statements regarding the diagnosis of typhoid fever are true?
- A** It is diagnosed by positive blood and stool cultures.
 - B** The Widal test (though now largely obsolete) detects bacteria within red blood cells.
 - C** In cases where cultures are negative due to prior treatment, assays such as the Multi-Test Dip-S-Ticks, Tubex, or TyphiDot may be used.
 - D** In the second or third week, deterioration with abdominal pain should raise concern for perforation until proven otherwise.
 - E** Abdominal distension in the later weeks should be treated by high bowel washout to eliminate toxins.
- 17** Which of the following statements are true regarding the treatment of typhoid perforation?
- A** Management requires vigorous resuscitation with IV fluids and broad-spectrum antibiotics.
 - B** Chloramphenicol is the drug of choice.
 - C** Multiple surgical options exist depending on the patient's condition, number and location of perforations, and peritoneal soiling.
 - D** Postoperative management may include leaving the skin/subcutaneous tissue open for delayed closure due to high rates of wound infection.
 - E** Perforations occurring in the first week tend to have a better prognosis due to better nutritional/immunological status.
- 18** Capsule endoscopy It is not used in which of the following situations?
- A** Diagnosis of obscure gastrointestinal bleeding.
 - B** Diagnosis of small bowel Crohn's disease.
 - C** Assessment of coeliac disease.
 - D** Screening and surveillance of polyps in familial polyposis syndrome.
 - E** Assessment of small bowel obstruction.
- 19** In which of the following conditions is colonoscopy indicated?
- A** Assessment of rectal bleeding with loose stools.
 - B** Management of iron deficiency anaemia.
 - C** Assessment of chronic diarrhoea (> 6 weeks).
 - D** Follow up of colorectal cancer and adenomatous polyp.
 - E** Diagnosis of fulminant colitis.
- 20** Which of the following statements is not true?
- A** Abdominal wall and parietal peritoneum are supplied by the somatic nervous system.
 - B** Abdominal organs and the visceral peritoneum are innervated by the autonomic nervous system.
 - C** Pain arising from pathology in the midgut is localised to the epigastrium.
 - D** Colicky abdominal pain indicates underlying obstructive pathology.
 - E** Pain away from the site of abdominal pathology denotes referred pain.
- 21** Which of the following statements are true?
- A** The proximal 40% is the jejunum.
 - B** The arterial supply in the form of arcades distinguishes jejunum from ileum.
 - C** The fat in the ileal mesentery is thicker.
 - D** The small bowel has autonomic innervation from the splanchnic nerves.
 - E** The jejunum has rich lymphoid aggregates in the form of Peyer's patches.
- 22** The following statements are true except:
- A** The taenia coli are responsible for the colonic haustrations.
 - B** The colon on both sides are closely related to the ureters.
 - C** The sigmoid colon has no mesentery at the back.
 - D** The marginal artery of Drummond is the anastomosis between the superior and inferior mesenteric arteries.
 - E** The superior mesenteric artery supplies the large bowel until the proximal two thirds of the transverse colon.
- 23** Which of the following statements are true?
- A** The principal function of the small intestine is the digestion of food and absorption of fluid and nutrients.
 - B** Absorption of bile salts and vitamin B12 occur in the jejunum.
 - C** The main function of the colon is to absorb water.
 - D** 1 litre of ileal contents enter the large bowel every 24 hours.
 - E** Faecal residue reaches the caecum 24 hours after a meal.
- 24** Which of the following statements are true in ulcerative colitis (UC)?
- A** In 95% of cases, the disease starts in the rectum and spreads proximally.
 - B** Patients might present as an emergency with fulminating disease in 5% to 10%.
 - C** It is a diffuse disease affecting all the layers of the large bowel.
 - D** Granulomas are a typical microscopic feature.
 - E** The transverse colon is affected in toxic megacolon.
- 25** Which of the following is not a complication of UC?
- A** Carcinoma.
 - B** Primary sclerosing cholangitis.
 - C** Internal fistulae.
 - D** Ankylosing spondylitis.
 - E** Perforation.

Answers 1-25

BAILEY & LOVE MCQS

- 1 **BC** **True: B, C** — Larval migration causes pulmonary symptoms while adult worms cause intestinal issues.
- 2 **C** **25%** — **UPDATED** Approximately 25% of the global population – especially in poor sanitation settings – are affected.
- 3 **B** **Inadvertent ingestion of fertilised eggs present in soil.** — Transmission occurs when a person ingests fertilised eggs from contaminated soil.
- 4 **BDE** **True: B, D, E** — **UPDATED** Eggs passed in stool hatch in the gut; larvae migrate via portal blood and liver to the lungs, are coughed up and swallowed, then mature in the intestine.
- 5 **D** **Chest pain, dry cough, dyspnoea, and fever with fluffy exudates on chest x-ray.** — Loeffler's syndrome presents with a dry cough, chest discomfort and patchy infiltrates.
- 6 **ABCDE** **True: A, B, C, D, E** — Heavy infestation can lead to nutritional deficits, obstruction, perforation, biliary complications and pulmonary signs.
- 7 **ACDE** **True: A, C, D, E** — Eosinophilia and imaging/lab findings help support roundworm infestation; Charcot-Leyden crystals are not typical.
- 8 **BCE** **True: B, C, E** — Conservative management is preferred; surgical options are reserved for complicated cases.
- 9 **BDE** **True: B, D, E** — A mobile filling defect in the CBD is most consistent with a live *Ascaris* worm; treatment is with albendazole and endoscopic removal.
- 10 **ABD** **True: A, B, D** — Intestinal TB characteristically causes transverse ulcers, serosal tubercles and terminal ileum thickening.
- 11 **ABCD** **True: A, B, C, D** — TB may present with constitutional symptoms, a doughy abdomen from ascites, a palpable RIF mass and perianal fistulas.
- 12 **ABE** **True: A, B, E** — A subhepatic caecum, TB history and low socioeconomic background suggest intestinal TB.
- 13 **E** **All of the above.** — A variety of surgical procedures may be utilized based on disease extent in TB.
- 14 **BD** **True: B, D** — Tuberculous perforation requires aggressive management with resection and staged repair; free gas is not reliably present.
- 15 **BCDE** **True: B, C, D, E** — Typhoid fever involves Peyer's patches resulting in ulceration, bleeding and perforation, with a prolonged febrile course.
- 16 **ACD** **True: A, C, D** — Typhoid is confirmed by cultures and, if needed, immunoassays – clinical deterioration later suggests perforation.
- 17 **ACDE** **True: A, C, D, E** — Aggressive resuscitation and tailored surgical management are key in typhoid perforation; chloramphenicol is not used.
- 18 **E** **Assessment of small bowel obstruction.** — Capsule endoscopy is contraindicated in suspected small bowel obstruction due to the risk of capsule retention.
- 19 **E** **Diagnosis of fulminant colitis.** — Colonoscopy is contraindicated in fulminant colitis because of the high risk of perforation.
- 20 **C** **Pain arising from pathology in the midgut is localised to the epigastrium.** — Midgut pathology typically causes periumbilical pain, not epigastric discomfort; the other statements correctly reflect neuroanatomy and pain referral.
- 21 **ABCD** **True: A, B, C, D** — Jejunum constitutes about 40% of the small bowel with long, straight arterial branches and less mesenteric fat; Peyer's patches are found in the ileum.
- 22 **C** **The sigmoid colon has no mesentery at the back.** — Sigmoid colon is suspended by its mesocolon and not fixed directly to the posterior abdominal wall.
- 23 **ACD** **True: A, C, D** — **UPDATED** The small bowel digests and absorbs, the colon mainly absorbs water and ~1 L of ileal content enters the colon daily. B12 and bile salts are absorbed in the terminal ileum.
- 24 **ABE** **True: A, B, E** — **UPDATED** UC begins in the rectum and spreads proximally, is mucosal (not transmural) and lacks granulomas (a Crohn feature); toxic megacolon typically dilates the transverse colon.
- 25 **C** **Internal fistulae.** — Fistula formation is uncommon in UC due to its mucosal and submucosal involvement.

SET 2 OF 4

- 26 Which of the following are barium enema findings in UC?
 - A Loss of haustrations.
 - B Narrow contracted colon.
 - C Increase in the presacral space.
 - D Cobblestone appearance.
 - E Backwash ileitis.
- 27 What is colonoscopy routinely used for in UC?
 - A To assess the extent of the disease.
 - B To distinguish it from Crohn's disease.
 - C To monitor the response to treatment.
 - D To carry out surveillance for the development of cancer.
 - E To assess severity in acute cases.

- 28** Which of the following criteria indicate severe disease in UC?
- A More than six motions per day.
 - B Pyrexia of more than 37.5°C.
 - C Tachycardia >90/minute.
 - D Tachypnoea >20/minute.
 - E Hypoalbuminaemia <30 g/L.
- 29** Which UC patients have an enhanced risk of developing cancer?
- A Those who have had the disease since childhood.
 - B Those who have had the disease for over 10 years.
 - C Those who have had a very severe first attack.
 - D Those who have the entire colon involved.
 - E Those who have another member of the family with the disease.
- 30** Which of the following drugs is not used in the medical treatment of UC?
- A Prednisolone.
 - B 5-ASA compounds.
 - C Infliximab.
 - D Azathioprine.
 - E Isoniazid.
- 31** In the management of a severe attack of UC, which of the following is not true?
- A Every patient needs a proctocolectomy after resuscitation.
 - B Daily plain abdominal x-ray is taken to assess transverse colon dilatation.
 - C Parenteral high-calorie alimentation is instituted.
 - D Intravenous hydrocortisone is given.
 - E Azathioprine or cyclosporin A is given.
- 32** What are the indications for surgery in UC?
- A Severe fulminating disease not responding to vigorous medical treatment.
 - B Severe dysplastic change or cancer on biopsy.
 - C Non-compliance of medical treatment.
 - D Chronic steroid-dependent disease requiring large doses.
 - E Extraintestinal disease.
- 33** The following statements regarding surgery in UC are true except:
- A In the emergency situation, total abdominal colectomy and ileostomy should be the procedure of choice.
 - B Proctocolectomy and ileostomy are associated with the lowest complication rate.
 - C Restorative proctocolectomy with an ileoanal pouch should be considered in all patients.
 - D Colectomy with ileorectal anastomosis is the most favoured procedure.
 - E Ileostomy with a continent intra-abdominal pouch is not often done.
- 34** Which of the following statements are true regarding an ileo-anal pouch procedure?
- A This should be considered in any case of inflammatory bowel disease that requires surgery.
 - B The patient must have a normal anal sphincter.
 - C In 15%, the pouch has to be removed.
 - D Small bowel obstruction as a postoperative complication occurs in 10%–15%.
 - E Pouchitis occurs in 30%.
- 35** Which of the following statements is not true of Crohn's disease (CD)?
- A The ileum is affected in 60% of cases.
 - B It affects the entire thickness of the bowel.
 - C Non-caseating granulomas are found in only 60% of patients.
 - D One in 10 patients have a first-degree relative with the disease.
 - E A patient can be cured of CD once the diseased small or large bowel is removed.
- 36** Which of the following pathological features is not found in CD?
- A Internal fistulae.
 - B Serpiginous and aphthous ulcers.
 - C Chronic inflammation involving all layers of the bowel wall.
 - D Pseudopolyps.
 - E Cobblestone mucosa.
- 37** Which of the following can cause acute presentation of CD?
- A Mimicking acute appendicitis.
 - B Perforation.
 - C Intestinal obstruction.
 - D Toxic megacolon.
 - E All of the above.
- 38** Which of the following are true of large-bowel CD?
- A 50%–75% will have an anal lesion.
 - B Non-caseating giant-cell granulomas are most common in anorectal disease.
 - C In a strictured area, malignancy can occur.
 - D A perianal abscess or a fissure might be the first presenting feature.
 - E Surgery is usually indicated.
- 39** Which of the following is not a clinical presentation of CD?
- A Bloodstained diarrhoea.
 - B Intermittent abdominal pain.
 - C Mass in the right iliac fossa.
 - D Typical evening rise of temperature.
 - E Pneumaturia and urinary tract infections.

- 40** Which of the following statements about imaging in CD are true?
- A** Small-bowel enema is the imaging of choice in small-bowel disease.
 - B** Barium enema and colonoscopy should be done for large-bowel disease.
 - C** MRI is the 'gold standard' for perianal fistulae.
 - D** CT scan is used for suspected intra-abdominal abscess and internal fistulae.
 - E** All of the above.
- 41** Which of the following drugs are used in the treatment of CD?
- A** Steroids.
 - B** 5-ASA compounds.
 - C** Azathioprine.
 - D** Infliximab.
 - E** All of the above.
- 42** The following operations are done in CD depending upon the indications except:
- A** Restorative proctocolectomy with ileoanal pouch.
 - B** Strictureplasty.
 - C** Proctocolectomy and ileostomy.
 - D** Colectomy and ileorectal anastomosis.
 - E** Segmental resections.
- 43** Which of the following statements about inflammatory bowel disease (UC and CD) is not true?
- A** Patients must be managed jointly by the physician and surgeon.
 - B** Surgery, when indicated, must be as radical as possible.
 - C** Patients must be given a good trial of optimum medical treatment prior to surgery.
 - D** There is more chance of a cure after surgery in UC than in CD.
 - E** In emergency presentation, patients must be vigorously resuscitated prior to operation and managed in the ITU postoperatively.
- 44** Which of the following statements are true?
- A** Campylobacter is the most common cause of gastroenteritis in the United Kingdom.
 - B** Yersinia can cause acute appendicitis.
 - C** Actinomycosis might present as an enterocutaneous fistula.
 - D** AIDS patients might develop opportunistic intestinal infections.
 - E** The presence of Clostridium difficile might necessitate an operation.
- 45** Which of the following statements are true?
- A** Duodenal diverticulum might result from a long-standing duodenal ulcer.
 - B** Jejunal diverticula might give rise to malabsorption problems.
 - C** Meckel's diverticulum can cause severe lower gastrointestinal haemorrhage.
 - D** A suspected Meckel's diverticulum is best imaged by a barium meal and follow through.
 - E** Pain originating in a Meckel's diverticulum is located around the umbilicus.
- 46** Which of the following statements are false?
- A** In the Western world, 75% of the population over the age of 70 has diverticular disease.
 - B** A low-fibre diet causes the disease.
 - C** These diverticula consist of mucosa, muscle and serosa.
 - D** Those with perforation have a higher mortality than those with an inflammatory mass.
 - E** Sepsis is the principal cause of morbidity.
- 47** Which of the following is not a complication of diverticular disease of the colon?
- A** Paracolic abscess.
 - B** Fistulae.
 - C** Lower gastrointestinal haemorrhage.
 - D** Carcinoma.
 - E** Stricture.
- 48** Which of the following is not true in complicated diverticular disease?
- A** Urinary symptoms might be the predominant presentation at times.
 - B** Profuse colonic haemorrhage might occur in 17%.
 - C** Fistulae occur in 5% of cases.
 - D** The most common fistula is enterocolic.
 - E** In acute diverticulitis, CT scan is the 'gold standard' for imaging.
- 49** Which of the following is not a cause of colo-vesical fistula?
- A** Carcinoma of rectosigmoid.
 - B** Radiation enteritis.
 - C** Crohn's colitis.
 - D** Diverticular disease.
 - E** Amoebic colitis.
- 50** In the surgical treatment of diverticular disease, which of the following statements are true?
- A** Colonoscopy must be carried out in all elective cases.
 - B** Barium enema is essential prior to elective operation.
 - C** Primary resection and end-to-end anastomosis must be carried out in all cases.
 - D** Hartmann's operation is the procedure of choice in perforated diverticulitis.
 - E** In vesicocolic fistula, a one-stage operation can usually be done.

Answers 26–50

- 26** **ABCE** True: A, B, C, E — UC shows loss of haustral markings, a narrow contracted colon from fibrosis, increased presacral space, and backwash ileitis; cobblestone appearance is more typical of Crohn's disease.
- 27** **ABCD** True: A, B, C, D — Colonoscopy is essential to assess disease extent, differentiate from Crohn's, monitor treatment response and for cancer surveillance, but is avoided in acute settings.
- 28** **ABCE** True: A, B, C, E — Severe UC is marked by >6 stools/d, fever, tachycardia and low albumin; tachypnoea is not a standard criterion.
- 29** **ABCD** True: A, B, C, D — Cancer risk in UC is increased by early onset, long duration, severe initial attacks and pancolitis.
- 30** **E** Isoniazid. — Isoniazid is an antitubercular agent and is not used in UC therapy.
- 31** **A** Every patient needs a proctocolectomy after resuscitation. — Not every patient with severe UC requires immediate surgery; many are managed medically.
- 32** **ABDE** True: A, B, D, E — Indications include refractory disease, dysplasia/cancer, steroid dependence and extraintestinal complications; non-compliance is not an indication.
- 33** **D** Colectomy with ileorectal anastomosis is the most favoured procedure. — Colectomy with ileorectal anastomosis is rarely performed in UC due to rectal involvement.
- 34** **BCDE** True: B, C, D, E — **UPDATED** An ileoanal pouch needs a normal sphincter and is for UC/FAP, not Crohn disease; pouch failure ~15%, obstruction 10–15% and pouchitis ~30%.
- 35** **E** A patient can be cured of CD once the diseased small or large bowel is removed. — Crohn's disease is chronic and relapsing; surgery does not cure the disease.
- 36** **D** Pseudopolyps. — Pseudopolyps are more typical of long-standing UC rather than Crohn's disease.
- 37** **E** All of the above. — CD may acutely present as appendicitis, perforation, obstruction or toxic megacolon.
- 38** **ABCD** True: A, B, C, D — Large-bowel CD often presents with anal lesions, granulomas and fistulas; strictures may harbor malignancy, yet surgery is not routinely indicated.
- 39** **D** Typical evening rise of temperature. — An evening rise in temperature is not a characteristic feature of CD.
- 40** **E** All of the above. — All listed imaging modalities have specific roles in evaluating CD.
- 41** **E** All of the above. — Steroids, 5-ASA, azathioprine and infliximab are all employed in CD management.
- 42** **A** Restorative proctocolectomy with ileoanal pouch. — Restorative proctocolectomy is generally not performed in CD due to recurrence risk.
- 43** **B** Surgery, when indicated, must be as radical as possible. — IBD management is best approached conservatively; radical surgery is not the goal.
- 44** **ACDE** True: A, C, D, E — Campylobacter is common; actinomycosis, opportunistic infections and severe *C. difficile* may require surgery; Yersinia causing appendicitis is not supported.
- 45** **ABCE** True: A, B, C, E — **UPDATED** Meckel diverticulum may bleed (ectopic gastric mucosa) and causes periumbilical pain; it is best shown by a technetium-99m pertechnetate scan, not barium.
- 46** **C** These diverticula consist of mucosa, muscle and serosa. — **UPDATED** False: colonic diverticula are pulsion (false) diverticula of mucosa and submucosa only, herniating where vessels pierce the muscle.
- 47** **D** Carcinoma. — Carcinoma is not considered a direct complication of diverticular disease.
- 48** **D** The most common fistula is enterocolic. — The most common fistula is actually colo-vesical, not enterocolic.
- 49** **E** Amoebic colitis. — Amoebic colitis is not a recognized cause of colo-vesical fistula.
- 50** **ABDE** True: A, B, D, E — Preoperative colonoscopy and barium enema are essential; Hartmann's is preferred in perforation and one-stage resection is feasible for vesicocolic fistula; primary resection is not universal.

SET 3 OF 4

- 51** Which of the following statements are true?
- A** Thrombosis most often affects the origin of the superior mesenteric artery (SMA), while embolism affects the middle colic artery.
 - B** Pain out of proportion to the physical findings is the cardinal clinical feature.
 - C** Angiodysplasia most commonly occurs in the sigmoid colon.
 - D** Ischaemic colitis affects mostly the splenic flexure.
 - E** Ischaemic colitis might result in a colonic stricture.
- 52** The following statements are true except:
- A** A loop transverse colostomy is the ideal method to defunction an anterior resection.
 - B** An end ileostomy is always a permanent procedure.
 - C** A loop ileostomy is always a temporary procedure.
 - D** An end colostomy can be a permanent or a temporary procedure.
 - E** Stoma complications are common.

- 53** Which of the following statements are true?
- A The most common cause is a surgical complication.
 - B In management, attention to nutrition is important.
 - C The 'cut-off point' between a high-output and low-output fistula is 500 mL/day.
 - D At operation, end-to-end anastomosis should be carried out in all cases.
 - E Always look for a cause when a fistula does not heal spontaneously.
- 54** The following statements are true except:
- A Laparoscopic approach is particularly suitable for colonic resection in Crohn's disease.
 - B Radical lymphadenectomy in cancer is as good as in open surgery.
 - C Operative times are longer.
 - D Tattooing the lesion preoperatively is performed as per the unit policy.
 - E The dissection is usually performed in a medial to lateral direction.
- 55** Which of the following statements are true?
- A In dynamic intestinal obstruction 40% are due to adhesions.
 - B Proximal to an obstruction the gas is made up of 90% nitrogen and hydrogen sulphide.
 - C Dehydration in intestinal obstruction is solely due to vomiting.
 - D In dynamic obstruction the bowel distal to the obstruction initially functions normally.
 - E In adynamic obstruction peristalsis is absent or inadequate
- 56** The following statements are true except:
- A Strangulation in intestinal obstruction is life-threatening.
 - B In strangulation venous return is compromised before arterial supply.
 - C If the segment of strangulated bowel is short the systemic effects will not be significant.
 - D Closed-loop obstruction will cause strangulation.
 - E The caecum might rupture in closed-loop obstruction
- 57** Which of the following statements are true?
- A Internal hernia is a rare cause, the diagnosis usually being made at operation.
 - B In the majority of internal hernia, division of the constricting agent should not be undertaken.
 - C Enteric strictures causing obstruction must always be resected.
 - D Bolus obstruction in the small bowel can be caused by stones and worms.
 - E Obstruction from postoperative adhesions almost never involves the large bowel
- 58** Regarding acute intussusception, which of the following statements are true?
- A The condition occurs equally in children and adults.
 - B Intussusception in all age groups is ileocolic.
 - C All cases of intussusception will require an operation.
 - D In children the abdomen is not initially distended.
 - E US of abdomen has a high diagnostic sensitivity in children
- 59** Regarding volvulus, which of the following statements are true?
- A The commonest type in the adult is sigmoid volvulus.
 - B Sigmoid volvulus can be treated by non-surgical means.
 - C The longer the duration of symptoms the worse the prognosis.
 - D Untreated the condition will result in ischaemia.
 - E A secondary volvulus occurs around an acquired adhesion
- 60** Regarding clinical features of intestinal obstruction, which of the following statements are true?
- A Vomiting is early and distension minimal in high small bowel obstruction.
 - B The vomitus is faeculent in distal small bowel obstruction.
 - C The more distal the obstruction in the small bowel, the greater is the abdominal distension.
 - D Absolute constipation is present in all cases.
 - E The administration of enemas can result in misleading findings
- 61** The following statements are true except:
- A Continuous severe pain is suggestive of strangulation.
 - B Tenderness over the right iliac fossa is a sinister sign.
 - C Fever indicates strangulation.
 - D Secondary polycythaemia might be a feature of dehydration.
 - E Normal bowel sounds excludes intestinal obstruction
- 62** Which of the following statements are correct?
- A The mainstay of diagnosis of strangulation is clinical.
 - B Shock suggests ischaemia.
 - C Pain is incessant in established or impending strangulation.
 - D Tenderness, absence of cough impulse and discoloration in an external hernia indicates strangulation.
 - E Generalised tenderness and rigidity requires immediate laparotomy
- 63** Which of the following statements are correct except:
- A Erect abdominal films should be routinely carried out.
 - B In distal small bowel obstruction distended small bowel loops lie transversely and central in position.
 - C The jejunum has typical features of valvulae conniventes and the ileum is featureless.
 - D In colonic obstruction a water-soluble enema should be undertaken.
 - E Fluid levels might be seen in non-obstructive conditions
- 64** Regarding treatment of acute intestinal obstruction, which of the following statements are not true?
- A The patient must have an immediate operation.
 - B Antibiotic therapy is mandatory when surgery is carried out.
 - C In very ill patients at operation exteriorisation should be considered.
 - D After operation for intestinal ischaemia, reperfusion injury might occur.
 - E Dextrose-saline should be the fluid of choice for resuscitation

- 65** Which of the following statements are true?
- A** Obstructed external hernia needs early operation.
 - B** Conservative management should be considered in obstruction due to adhesions.
 - C** On opening the abdomen, once the site of obstruction is determined, the initial step is to remove it.
 - D** A second-look operation might sometimes be necessary.
 - E** In obstruction without ischaemia surgery can be deferred
- 66** Adynamic obstruction : Which of the following statements are true?
- A** In paralytic ileus colicky pain is a prominent feature.
 - B** Metabolic abnormalities can give rise to paralytic ileus.
 - C** CT scan should be carried out in a case that does not settle..
 - D** Pseudo-obstruction usually affects the colon.
 - E** Caecal perforation is a serious complication.
- 67** Acute intestinal obstruction of the newborn: Which of the following statements are true except:
- A** Congenital atresia and stenosis are the most common causes..
 - B** Bilious vomiting is a feature of jejunal atresia.
 - C** Perforation is never a problem
 - D** String of sausages' appearance is due to multiple atresias
 - E** The underlying bowel anomaly is usually an isolated finding.
- 68** The following statements are true except:
- A** The appendicular artery arises from the right colic artery.
 - B** The commonest position of the appendix is retrocaecal.
 - C** The position of the base of the appendix is constant.
 - D** The submucosa is rich in lymphoid follicles.
 - E** Argentaffin cells are found in the base of the crypts.
- 69** Regarding acute appendicitis, which of the following statements are true?
- A** The peak incidence of acute appendicitis is in the teens and early 20s.
 - B** The incidence of acute appendicitis is lowest in those who have a high intake of dietary fibre.
 - C** Obstruction of the appendix lumen by a caecal carcinoma might give rise to acute appendicitis.
 - D** Aerobic and anaerobic organisms are responsible for acute appendicitis.
 - E** A mucocoele of the appendix is a clinical variation of acute appendicitis.
- 70** The following factors increase the chances of peritonitis and systemic sepsis except:
- A** Obesity.
 - B** Extremes of age.
 - C** Immunosuppression.
 - D** Pelvic position of appendix.
 - E** Previous abdominal operation.
- 71** Which of the following are signs in acute appendicitis?
- A** Rovsing's sign.
 - B** Pointing sign.
 - C** Obturator sign.
 - D** Psoas sign.
 - E** Troisier's sign.
- 72** Which of the following statements is not true in acute appendicitis?
- A** It carries a higher mortality in the elderly.
 - B** Acute appendicitis in pregnancy will result in foetal loss in 30%.
 - C** An Alvarado score of 7 or more is suggestive of the diagnosis.
 - D** A decision to operate based on clinical suspicion alone will result in removal of a normal appendix in 15%–30%.
 - E** Cancer of the caecum might masquerade as acute appendicitis.
- 73** In children, which of the following could be confused with acute appendicitis except:
- A** Gastroenteritis.
 - B** Mesenteric adenitis.
 - C** Henoch–Schönlein purpura.
 - D** Urinary tract infection (UTI).
 - E** Lobar pneumonia.
- 74** In the adult male, which of the following conditions are differential diagnoses of acute appendicitis?
- A** Mesenteric cyst.
 - B** Right ureteric colic.
 - C** Perforated peptic ulcer.
 - D** Crohn's disease.
 - E** Right testicular torsion.
- 75** In the adult female, which of the following is not part of the differential diagnosis of acute appendicitis?
- A** Mittelschmerz.
 - B** Pelvic inflammatory disease (PID).
 - C** Acute cholecystitis.
 - D** Ruptured ectopic pregnancy or right tubal abortion.
 - E** Torsion or ruptured ovarian cyst.

Answers 51–75

BAILEY & LOVE MCQS

- 51** **ABDE** True: A, B, D, E — **UPDATED** Thrombosis occurs at the SMA origin; pain out of proportion to signs is the hallmark. Ischaemic colitis favours the splenic flexure and may stricture; angiodysplasia is right-sided.
- 52** **AB** True: A, B — Loop transverse colostomy is no longer ideal and an end ileostomy can be temporary.
- 53** **ABCE** True: A, B, C, E — Surgical complications are the most common cause; nutritional support is key and 500 mL/day is the cutoff; routine anastomosis is not mandatory.
- 54** **A** Laparoscopic approach is particularly suitable for colonic resection in Crohn's disease. — Laparoscopy in Crohn's is often challenging and not considered particularly suitable.

- 55 **ABDE** True: A, B, D, E — Dynamic obstruction is mechanical with intact peristalsis initially, adhesions are common, gas composition and multifactorial dehydration are noted, with absent peristalsis in adynamic cases.
- 56 **C** If the segment of strangulated bowel is short the systemic effects will not be significant. — Even a short strangulated segment can lead to significant systemic effects.
- 57 **ABDE** True: A, B, D, E — Internal hernias are rare and diagnosed intraoperatively; division of constricting bands is avoided; strictures may not always require resection; adhesions typically affect the small bowel.
- 58 **DE** True: D, E — **UPDATED** Intussusception is most common in children with minimal initial distension and ultrasound is highly sensitive.
- 59 **ABDE** True: A, B, D, E — **UPDATED** Sigmoid volvulus is common in adults, may be managed non-surgically, and can lead to ischaemia or secondary volvulus; duration is less prognostic.
- 60 **ABCE** True: A, B, C, E — **UPDATED** High obstruction causes early vomiting, distal obstruction leads to faeculent vomitus and greater distension, while enemas may mislead; constipation is not universal.
- 61 **E** Normal bowel sounds excludes intestinal obstruction — Normal bowel sounds do not rule out intestinal obstruction.
- 62 **ABCDE** True: A, B, C, D, E — All statements are classic signs of strangulation and impending bowel infarction.
- 63 **A** Erect abdominal films should be routinely carried out. — Erect films are not routinely used; supine films are preferred.
- 64 **AE** True: A, E — **UPDATED** Immediate surgery without adequate resuscitation is not advised and dextrose-saline is not appropriate for resuscitation.
- 65 **ABDE** True: A, B, D, E — Surgical management is tailored; immediate removal of the obstruction is not the first step.
- 66 **BCDE** True: B, C, D, E — **UPDATED** Paralytic ileus is painless (no colic); metabolic causes are common, CT is needed if it fails to settle, pseudo-obstruction affects the colon and caecal perforation is the main risk.
- 67 **CE** True: C, E — **UPDATED** False: perforation is a real risk in neonatal obstruction, and associated anomalies (e.g. trisomy 21, cardiac) are common. Atresia/stenosis is the commonest cause.
- 68 **A** The appendicular artery arises from the right colic artery. — The appendicular artery actually arises from the ileocolic artery.
- 69 **ABCD** True: A, B, C, D — **UPDATED** A mucocele is not an acute variation; the other statements are true.
- 70 **A** Obesity. — Obesity may help localize inflammation rather than increasing the risk.
- 71 **ABCD** True: A, B, C, D — Troisier's sign is not associated with appendicitis.
- 72 **B** Acute appendicitis in pregnancy will result in foetal loss in 30%. — **UPDATED** The foetal loss rate in appendicitis during pregnancy is lower than 30%.
- 73 **D** Urinary tract infection (UTI). — UTI is less likely to mimic appendicitis in children.
- 74 **BCDE** True: B, C, D, E — These conditions can mimic appendicitis in adult males, whereas mesenteric cyst is rare.
- 75 **C** Acute cholecystitis. — Acute cholecystitis typically presents with right upper quadrant pain.

SET 4 OF 4

- 76 In the elderly, which of the following is not part of the differential diagnosis of acute appendicitis?
- A Diverticulitis.
 - B Intestinal obstruction.
 - C Mesenteric infarction.
 - D Bladder calculus.
 - E Carcinoma of the caecum.

Answer 76

- 76 **D** Bladder calculus. — Bladder calculus usually presents with urinary symptoms rather than mimicking appendicitis.

20 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 16-year-old girl with a history of ulcerative colitis managed with steroid therapy presents with a 36-hour history of nausea, crampy abdominal pain, and severe bloody diarrhea. On examination, the patient is febrile and pale, with a blood pressure of 90/60 mmHg and heart rate of 130 beats/min. Her abdomen is distended and diffusely tender. A complete blood count (CBC) demonstrates a leukocytosis with a left shift. The patient receives IV fluid resuscitation and nasogastric (NG) tube decompression. Further therapeutic interventions should include which one of the following?

 - A 6-mercaptopurine
 - B azathioprine
 - C opioid antidiarrheals
 - D colonoscopic decompression
 - E high-dose IV steroids and broad-spectrum antibiotics
- 2** After 48 hours, there is no clinical improvement (for the same 16-year-old girl with toxic megacolon). Which of the following is the most appropriate next step in management?

 - A colonoscopic decompression
 - B cyclosporine
 - C abdominal colectomy and ileostomy and Hartmann's procedure
 - D proctocolectomy with ileal pouch-anal anastomosis
 - E abdominal colectomy with ileorectal anastomosis
- 3** A 55-year-old woman presents with 6 months of weight loss, crampy abdominal pain, and intermittent non-bloody diarrhea. CT shows thickening of the terminal ileum and an inflammatory mass in the right lower quadrant. Which of the following is the most likely diagnosis?

 - A ulcerative colitis
 - B appendicitis
 - C Crohn's disease
 - D irritable bowel syndrome
 - E lactose intolerance
- 4** Which of the following is the best diagnostic test to confirm Crohn's disease involving the terminal ileum?

 - A repeat CT scan with delayed imaging
 - B ultrasonography
 - C sigmoidoscopy
 - D colonoscopy
 - E small-bowel radiography
- 5** Initial management of Crohn's disease should include which of the following?

 - A antibiotics and IV fluids
 - B lactose-free diet
 - C antispasmodics
 - D nutritional supplementation and systemic steroids
 - E laparotomy
- 6** A 55-year-old man presents with left lower quadrant abdominal pain of 1 week duration that has worsened over the last 2 days, accompanied by nausea, constipation, and dysuria. CT scan reveals a thickened sigmoid colon with pericolic inflammation. Which is the most likely diagnosis?

 - A colon cancer with contained perforation
 - B ischemic colitis
 - C pseudomembranous colitis
 - D diverticulitis
 - E pyelonephritis
- 7** For the patient with diverticulitis described above, what is the most appropriate management?

 - A NPO, IV fluids, and IV antibiotics for gram-negative and anaerobic coverage
 - B NPO, IV fluid hydration, followed by immediate sigmoid colon resection
 - C NPO, IV fluids, and anticoagulation
 - D NPO, IV fluids, evaluation of stool for *C. difficile* toxin, and either metronidazole or vancomycin
 - E NPO, IV fluids, initiation of bowel preparation for elective sigmoid colon resection
- 8** A 75-year-old woman presents with a 2-day history of abdominal pain, distention, and obstipation. A barium enema demonstrates a classic tapering "bird's beak" appearance. Which of the following is the most likely diagnosis?

FIGURE
#lange_51_1%.png

 - A ischemic colitis with stricture
 - B diverticulitis with obstruction
 - C cecal volvulus
 - D sigmoid volvulus
 - E colon cancer with obstruction

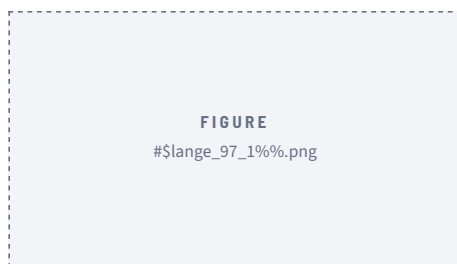
9 After NG tube decompression and resuscitation in a patient with sigmoid volvulus, what is the most appropriate next step in management?

- A urgent sigmoid resection
- B nonoperative reduction by proctoscopy and rectal tube
- C proximal colostomy
- D urgent operative detorsion
- E nonoperative reduction by passage of well-lubricated rectal tube

10 On postoperative day 3 following AAA repair, the patient develops dark diarrhea. Labs show normal electrolytes and renal function; WBC is 15,000. What is the most appropriate next step in management?

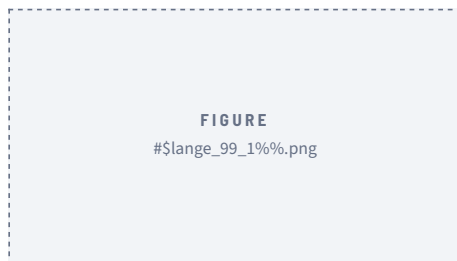
- A stool for *C. difficile* toxin test and institution of metronidazole
- B sigmoidoscopy
- C air contrast barium enema
- D CT scan
- E abdominal x-rays

11 A 10-day-old infant presents with bilious vomiting, paucity of gas on plain radiographs, and duodenal obstruction on UGI contrast study. Which is the most likely diagnosis?



- A congenital hypertrophic pyloric stenosis
- B annular pancreas
- C duodenal atresia
- D midgut volvulus
- E intussusception

12 A 1-day-old infant with Down syndrome presents with feeding intolerance, bilious vomiting, and a double bubble sign on plain radiographs. What is the most likely diagnosis?



- A anorectal malformation
- B esophageal atresia
- C duodenal atresia
- D midgut volvulus
- E intussusception

13 A 55-year-old man with a history of diverticulosis presents with a 2-week history of dysuria, urgency, and pneumaturia. Furthermore, he has a 3-month history of vague, intermittent, left lower quadrant abdominal pain, and irregular bowel habits. On examination, he has fullness on palpation in the left lower quadrant, and urine culture demonstrates a polymicrobial infection. A computed tomography (CT) scan demonstrates an inflammatory mass in the left pelvis. Which of the following investigations is most likely to aid in diagnosis?

- A voiding cystourethrogram
- B CT scan with contrast
- C air contrast barium enema
- D colonoscopy
- E laparoscopy

14 After confirming the diagnosis, the most appropriate initial steps in the management of this patient include which one of the following?

- A broad-spectrum antibiotics and insertion of an indwelling bladder catheter
- B outpatient oral antibiotics and stool softeners
- C clear fluids, magnesium citrate, and oral antibiotics
- D urgent exploratory laparotomy
- E proximal defunctioning colostomy

15 Which of the following is the optimal definitive surgical management for this patient?

- A a proximal defunctioning loop colostomy
- B sigmoid colon resection with primary anastomosis
- C sigmoid colon resection with Hartmann's procedure
- D abdominoperineal resection and permanent colostomy
- E total abdominal colectomy and ileorectal anastomosis

16 An 80-year-old female resident of a nursing home presents with a history of sudden onset of crampy abdominal pain, vomiting, and abdominal distention. Her last bowel movement was 4 days before presentation. Which of the following is the most appropriate next step in the management of this patient?



- A urgent laparotomy
- B admission to the hospital and serial abdominal examinations
- C placement of a rectal decompression tube in the emergency department
- D rigid proctoscopy and directed placement of a colonic decompression tube
- E colonic enemas

- 17** A 68-year-old man with watery diarrhea, crampy abdominal pain, and low-grade fever 3 days after CABG and cephalosporin use presents with an elevated WBC. Which test is most likely to yield a definitive diagnosis?
- A** chest and abdominal x-rays
 - B** stool for ova and parasites
 - C** stool for *Clostridium difficile* toxins
 - D** stool for enterotoxigenic *Escherichia coli*
 - E** CT scan of the abdomen
- 18** Which is the most appropriate next step in management of this patient with suspected *C. difficile* colitis?
- A** IV fluids in the emergency department and discharge on an oral cephalosporin
 - B** IV fluids, admittance to the hospital, and institution of third-generation cephalosporin antibiotics
 - C** IV fluids, admittance to the hospital, discontinuation of cephalosporin, and colonoscopy
 - D** IV fluids, admittance to the hospital, discontinuation of cephalosporin, and institution of metronidazole
 - E** IV fluids in the emergency department and discharge on an oral cephalosporin with an appointment for barium enema
- 19** A 53-year-old man with a history of sigmoid resection presents with crampy abdominal pain, distention, bile-stained vomiting, and no bowel movement for 24 hours. Initial management should include?
- A** IV fluids and nasogastric decompression, followed by plain abdominal radiographs
 - B** IV fluids and analgesics
 - C** plain abdominal radiographs
 - D** a barium enema
 - E** a Fleet's enema and stool softeners
- 20** Over 6 hours without improvement and a radiograph suggestive of complete obstruction, the next step is?
- A** to continue serial examinations and repeat abdominal radiographs in the morning
 - B** insertion of a rectal tube for colonic decompression
 - C** insertion of a Cantor tube for small-bowel decompression
 - D** colonoscopy
 - E** urgent exploratory laparotomy

Answers 1-20

LANGE MCQs

- 1** **E high-dose IV steroids and broad-spectrum antibiotics** — **UPDATED** High-dose IV steroids and broad-spectrum antibiotics Toxic megacolon requires aggressive management with steroids, antibiotics, and supportive care.
- 2** **C abdominal colectomy and ileostomy and Hartmann's procedure** — **UPDATED** Abdominal colectomy and ileostomy and Hartmann's procedure No improvement on medical therapy for toxic megacolon requires urgent surgery; total proctocolectomy is too extensive emergently.
- 3** **C Crohn's disease** — **UPDATED** Crohn's disease Chronic inflammation often involves terminal ileum with mass, strictures, and RLQ pain.
- 4** **D colonoscopy** — **UPDATED** Ileocolonoscopy with terminal ileal biopsy confirms Crohn disease histologically; MR enterography (or small-bowel contrast studies) then defines extent and strictures.
- 5** **D nutritional supplementation and systemic steroids** — **UPDATED** Nutritional supplementation and systemic steroids Acute flare is managed medically with steroids and supportive care.
- 6** **D diverticulitis** — **UPDATED** Diverticulitis The clinical picture and CT findings are most consistent with diverticulitis.
- 7** **A NPO, IV fluids, and IV antibiotics for gram-negative and anaerobic coverage** — **UPDATED** NPO, IV fluids, and IV antibiotics for gram-negative and anaerobic coverage Medical management with bowel rest and IV antibiotics is indicated in uncomplicated diverticulitis.
- 8** **D sigmoid volvulus** — **UPDATED** Sigmoid volvulus The radiographic "bird's beak" appearance is diagnostic for sigmoid volvulus.
- 9** **B nonoperative reduction by proctoscopy and rectal tube** — **UPDATED** Nonoperative reduction by proctoscopy and rectal tube Reduction using proctoscopy and rectal tube is indicated if there are no signs of ischemia.
- 10** **B sigmoidoscopy** — **UPDATED** Sigmoidoscopy The findings raise concern for ischemic colitis; sigmoidoscopy is indicated.
- 11** **D midgut volvulus** — **UPDATED** Midgut volvulus The clinical and radiographic findings are most consistent with midgut volvulus.
- 12** **C duodenal atresia** — **UPDATED** Duodenal atresia The classic double bubble in a Down syndrome infant is diagnostic for duodenal atresia.
- 13** **B CT scan with contrast** — CT scan shows air in the bladder; other studies have low yield.
- 14** **A broad-spectrum antibiotics and insertion of an indwelling bladder catheter** — Initial management involves IV antibiotics and bladder drainage.
- 15** **B sigmoid colon resection with primary anastomosis** — Definitive treatment is elective sigmoid resection with primary anastomosis after inflammation subsides.
- 16** **D rigid proctoscopy and directed placement of a colonic decompression tube** — Sigmoid volvulus is managed with proctoscopic decompression.
- 17** **C stool for *Clostridium difficile* toxins** — Stool toxin assay confirms *C. difficile* colitis.
- 18** **D IV fluids, admittance to the hospital, discontinuation of cephalosporin, and institution of metronidazole** — **UPDATED** Admit, rehydrate and stop the causative antibiotic; oral vancomycin (or fidaxomicin) is now first-line for *C. difficile*, metronidazole only where these are unavailable.
- 19** **A IV fluids and nasogastric decompression, followed by plain abdominal radiographs** — NG decompression and IV fluids before imaging are indicated.

- 20** **E** **urgent exploratory laparotomy** — **UPDATED** Surgical intervention is indicated for complete small-bowel obstruction.

28 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

SET 1 OF 2

- 1** A postoperative patient has been moved to a side room after developing diarrhoea following a course of antibiotics. Faecal samples test positive for *Clostridium difficile* toxin and metronidazole is started. After 10 days the antibiotics are finished and a repeat toxin assay on a formed stool sample remains positive. What is the most appropriate next management step?

 - A No further action required
 - B Continue metronidazole for a further 10 days
 - C Start intravenous vancomycin
 - D Start oral vancomycin
 - E Urgent colonoscopy as this patient is at significant risk of pseudo-membranous colitis
- 2** A 32-year-old female presents with a 6-week history of bloody bowel motions, significant weight loss, increased lethargy and fatigue, and a history of constipation with regular laxative use. What is the most likely diagnosis?

 - A Bowel cancer
 - B Irritable bowel syndrome
 - C Diverticular disease
 - D Inflammatory bowel disease
 - E Anal fissure
- 3** Which one of the following statements regarding stomas is the most correct?

 - A Hartmann's procedure results in a defunctioned rectum and loop colostomy
 - B An abdominoperineal resection may be reversed by formation of an ileal J-pouch
 - C A mucus fistula functions to decompress residual distal bowel
 - D Small bowel obstruction is a common early complication and indicates need for revision
 - E Positioning of the stoma is not important
- 4** An 85-year-old male with a history of chronic constipation presents with acute severe colicky abdominal pain and absolute constipation. A plain abdominal film reveals a grossly dilated sigmoid colon. What is the next step in management?

 - A Laparotomy
 - B Sigmoidoscopy with flatus tube insertion
 - C Sigmoid colectomy with colostomy
 - D Barium swallow
 - E Computed tomography
- 5** Which one of the following does NOT occur as a systemic manifestation of inflammatory bowel disease?

 - A Amyloidosis
 - B Pyoderma gangrenosum
 - C Scleritis
 - D Cardiomyopathy
 - E Sclerosing cholangitis
- 6** Which one of the following statements regarding diverticular disease is the most correct?

 - A Colonic diverticula are true diverticula
 - B Diverticular disease describes the outpouching of colonic mucosa between the muscle layers of the colonic wall
 - C The most common site of diverticula is the rectum
 - D Saint's triad consists of diverticular disease, cholelithiasis and inguinal herniation
 - E 95% of complications from diverticula occur at the sigmoid colon
- 7** A 78-year-old patient is admitted with sudden onset colicky abdominal pain, rectal bleeding, shock, and a history of weight loss. Previous colonoscopy 3 months earlier was normal. What is the diagnosis?

 - A Obstruction secondary to neoplasia
 - B Inflammatory colitis
 - C Ischaemic colitis secondary to embolus
 - D Angiodysplasia
 - E Mesenteric atherosclerosis
- 8** A 27-year-old patient with ulcerative colitis, who poorly tolerates azathioprine and complains of malaise, nausea, and vomiting, is seen in clinic. What is the next treatment option?

 - A Long-term oral steroids
 - B 6-mercaptopurine
 - C Ciclosporin
 - D Infliximab
 - E Methotrexate
- 9** An 18-year-old patient presents with a 12-day history of abdominal pain and fever. On examination, a mass in the right iliac fossa is palpable and CT confirms an appendix mass with an abscess. After 2 days of conservative antibiotic management with no reduction in the mass, what is the next stage of management?

 - A Continue antibiotics for a further 14 days
 - B Proceed to appendicectomy
 - C Percutaneous drainage
 - D Colonoscopy
 - E Laparoscopy

- 10** Which of the following is not encountered when making a Lanz incision during an open appendectomy?
- A External oblique aponeurosis
 - B Rectus abdominis
 - C Peritoneum
 - D Internal oblique muscle
 - E Transversus abdominis
- 11** Which of the following is not a sign associated with acute appendicitis?
- A Murphy's sign
 - B Cope sign
 - C Psoas sign
 - D Rovsing's sign
 - E Pain on rectal examination
- 12** All of the following may cause a right iliac fossa mass that is palpable on abdominal examination, except
- A Mucocele
 - B Ulcerative colitis
 - C Tuberculosis
 - D Appendicitis
 - E Ovarian cancer
- 13** The most common cause of large bowel obstruction in the developing world is
- A Adhesions
 - B Intestinal parasites
 - C Volvulus
 - D Hernia
 - E Neoplasia
- 14** A 45-year-old woman with a history of previous gynaecological surgery presents with central colicky abdominal pain, vomiting, and absolute constipation. Despite resuscitation and NG tube placement, her condition does not improve. What is the next stage of management?
- A Repeat plain abdominal film
 - B Barium meal
 - C Diagnostic laparoscopy
 - D Gastrografin via a nasogastric tube
 - E Radio-opaque contrast enema
- 15** Which of the following is NOT a recognized complication of a Meckel's diverticulum?
- A Volvulus
 - B Adenocarcinoma
 - C Peptic ulceration
 - D Intussusception
 - E Pancreatitis
- 16** Which of the following features is not characteristic of Crohn's disease?
- A Rose thorn abscesses
 - B Cobblestoning
 - C Skip lesions
 - D Lead piping
 - E Serosal involvement
- 17** Management of abdominal pain (1) A 26-year-old woman arrives at the emergency department with unbearable intense right iliac fossa pain. Earlier that day, she was experiencing 'on and off' moderate pain in the umbilical area which gradually moved over to the right iliac fossa. Associated symptoms include anorexia, nausea and vomiting. On examination, the patient is pyrexial and there is rebound tenderness and guarding over the right iliac fossa. A beta-human chorionic gonadotrophin test is negative. What should you do next?
- A Send the patient to the emergency operating theatre for an appendectomy
 - B Alert the obstetrics and gynaecology team, suspecting that she may have a ruptured ectopic pregnancy
 - C Manage the patient medically in the emergency department
 - D Order an ultrasound scan of the abdomen
 - E Send the patient for a plain film radiograph of the abdomen
- 18** Investigation of abdominal pain (1) You see a 55-year-old woman in the emergency department, who was admitted with central colicky abdominal pain and multiple episodes of vomiting. She last opened her bowels 4 hours ago. On examination she appears dehydrated and is in pain. The abdomen is generally tender and slightly distended. Bowel sounds are increased. You suspect a bowel obstruction and decide to order some investigations. What is the most valuable initial investigation that will support your suspected diagnosis?
- A Upper gastrointestinal endoscopy
 - B Colonoscopy
 - C Computed tomography scan of the abdomen
 - D Plain film radiograph of the abdomen
 - E Barium follow-through
- 19** Antibiotic prophylaxis A patient has been sent to theatre for emergency surgery with suspected appendicitis. He is given three doses of intravenous cefuroxime and metronidazole in a timely fashion. When is the best time to administer the first dose of antibiotics?
- A One hour after the first incision is made
 - B One hour before surgery
 - C One hour postoperatively
 - D Just before the surgical incision is made
 - E None of the above

- 20** Diagnosis of left iliac fossa pain An elderly man with chronic constipation experiences acute-onset left iliac fossa pain and tenderness. On examination, the patient has fever and is slightly tachycardic. There is marked tenderness and guarding in the left iliac fossa. Full blood count results reveal a raised white cell count. What is the most likely diagnosis?
- A Diverticular disease
 - B Diverticulitis
 - C Diverticulosis
 - D Perforated diverticulitis
 - E None of the above
- 21** Management of left iliac fossa pain: you are asked by your senior colleague to devise the treatment plan for the elderly man with acute diverticulitis described above. What is the most appropriate treatment plan?
- A Keep nil by mouth and send for emergency laparotomy
 - B Give analgesia and antibiotics
 - C Keep nil by mouth, administer antibiotics and analgesia
 - D Keep nil by mouth, administer intravenous fluids, antibiotics and analgesia
 - E Keep nil by mouth, administer intravenous fluids and analgesia
- 22** Constipation and abdominal pain A 57-year-old man presents with acute colicky pain in the suprapubic area. He has been constipated over the last 2 days and has been feeling bloated. He feels nauseous, but he has not vomited. On examination you notice marked abdominal distension, and increased bowel sounds. What is the most likely diagnosis?
- A Small bowel obstruction
 - B Irritable bowel syndrome
 - C Diverticular disease
 - D Large bowel obstruction
 - E Appendicitis
- 23** Diarrhoea and abdominal pain A 28-year-old man with a 10-year history of ulcerative colitis presents with an acute severe episode of abdominal pain, nausea and vomiting and blood-stained, watery diarrhoea. On examination you notice that the patient has fever and tachycardia and that the abdomen is markedly distended. An abdominal plain film radiograph shows that the transverse colon is dilated at approximately 6.5 cm. What is the most likely diagnosis?
- A Large bowel obstruction
 - B Toxic megacolon
 - C Perforated diverticulitis
 - D Crohn's colitis
 - E None of the above
- 24** Bowel obstruction From the list of options below which one is the most unlikely cause of mechanical intestinal obstruction?
- A Faecal impaction
 - B Caecal volvulus
 - C Paralytic ileus
 - D Congenital intestinal atresia
 - E Crohn's colitis
- 25** Management of abdominal pain (2) You see an 11-year-old boy in the emergency department who is admitted with an acute onset of abdominal pain, nausea and vomiting. There is diffuse pain around the central abdomen and right iliac fossa and it is continuous in nature. On examination, the patient is febrile and there is marked tenderness and rebound in the right iliac fossa. The patient's mother mentions a recent sore throat. You suspect mesenteric adenitis and an ultrasound scan is inconclusive. What is the best next step?
- A Start the patient on analgesia and intravenous antibiotics
 - B Send for a computed tomography scan of the abdomen
 - C Observe patient for the next 2 hours and reassess
 - D Emergency diagnostic laparoscopy ($\pm$ appendicectomy)
 - E None of the above

Answers 1-25

SBA MCQS

- 1** **A No further action required** — A persistently positive toxin assay is expected post-treatment; no further action is needed as the patient is asymptomatic.
- 2** **D Inflammatory bowel disease** — Systemic features point toward inflammatory bowel disease rather than a benign condition.
- 3** **C A mucus fistula functions to decompress residual distal bowel** — A mucus fistula decompresses the distal bowel; this is the correct function in stoma management.
- 4** **B Sigmoidoscopy with flatus tube insertion** — Endoscopic decompression with insertion of a flatus tube is the initial management for sigmoid volvulus.
- 5** **D Cardiomyopathy** — Cardiomyopathy is not a recognized manifestation of IBD.
- 6** **E 95% of complications from diverticula occur at the sigmoid colon** — Complications of diverticular disease predominantly occur in the sigmoid colon.
- 7** **E Mesenteric atherosclerosis** — The presentation suggests intestinal ischemia due to mesenteric atherosclerosis.
- 8** **B 6-mercaptopurine** — 6-mercaptopurine is the appropriate alternative for patients intolerant to azathioprine.
- 9** **C Percutaneous drainage** — Percutaneous drainage is indicated when an appendix mass fails to resolve with conservative management.
- 10** **B Rectus abdominis** — Rectus abdominis is not typically involved in a Lanz incision.
- 11** **A Murphy's sign** — Murphy's sign is indicative of acute cholecystitis, not appendicitis.

- 12 B Ulcerative colitis** — Ulcerative colitis does not typically present with a palpable right iliac fossa mass.
- 13 C Volvulus** — **UPDATED** In Africa, Asia and South America sigmoid volvulus is the commonest cause of large-bowel obstruction; in the West it is colorectal cancer. Hernias mainly obstruct the small bowel.
- 14 D Gastrografin via a nasogastric tube** — Administration of Gastrografin through the NG tube can be both diagnostic and therapeutic in adhesion-related obstruction.
- 15 E Pancreatitis** — Pancreatitis is not a typical complication of Meckel's diverticulum.
- 16 D Lead piping** — Lead piping is characteristic of ulcerative colitis rather than Crohn's disease.
- 17 A Send the patient to the emergency operating theatre for an appendicectomy** — Negative beta-hCG and clinical signs indicate appendicitis requiring surgery.
- 18 D Plain film radiograph of the abdomen** — Plain radiograph is the first investigation for suspected small bowel obstruction.
- 19 B One hour before surgery** — Prophylactic antibiotics are most effective when given one hour before surgery.
- 20 B Diverticulitis** — Clinical signs and raised WBC count point to diverticulitis.
- 21 D Keep nil by mouth, administer intravenous fluids, antibiotics and analgesia** — **UPDATED** Management of diverticulitis includes nil by mouth, IV fluids, antibiotics, and analgesia.
- 22 D Large bowel obstruction** — Marked distension and colicky pain suggest large bowel obstruction.
- 23 B Toxic megacolon** — Severe dilatation of the colon with systemic signs suggests toxic megacolon in ulcerative colitis.
- 24 C Paralytic ileus** — Paralytic ileus is a non-mechanical cause of obstruction.
- 25 D Emergency diagnostic laparoscopy (± appendicectomy)** — **UPDATED** Fever with marked RIF tenderness and rebound means peritonitis; with inconclusive ultrasound, diagnostic laparoscopy is safer than waiting or a CT in a child.

SET 2 OF 2

- 26** Causes of large bowel obstruction You are asked to give your opinion on an abdominal plain film radiograph of a patient with bowel obstruction. The film shows distended loops of large bowel forming a 'U' shape, resembling a big coffee bean. What is the most likely reason for the large bowel obstruction?
- A Faecal impaction
B Sigmoid volvulus
C Obstructing carcinoma
D Foreign body
E None of the above
- 27** Lower gastrointestinal infection A 50-year-old woman, admitted with colicky central abdominal pain and passing blood-stained diarrhoea with mucus per rectum, presents with fever and tachycardia. Abdominal radiography appears normal, and stool analysis reveals *Clostridium difficile* cytotoxins. What is the most likely diagnosis?
- A Ulcerative colitis
B Crohn's colitis
C Ischaemic colitis
D Pseudomembranous colitis
E None of the above
- 28** Investigation of a diverticular abscess A 75-year-old man with a history of diverticular disease is experiencing swinging fevers and left-sided abdominal pain. You suspect a diverticular abscess. Which investigation is most appropriate to confirm your suspicion?
- A Barium enema studies
B Abdominal plain film radiography
C Computed tomography scan of the abdomen
D Colonoscopy
E Flexible sigmoidoscopy

Answers 26-28

SBA MCQS

- 26 B Sigmoid volvulus** — The coffee bean sign is characteristic of sigmoid volvulus.
- 27 D Pseudomembranous colitis** — Stool cytotoxins and clinical picture point to pseudomembranous colitis.
- 28 C Computed tomography scan of the abdomen** — **UPDATED** CT with contrast is the investigation of choice for a suspected diverticular abscess and guides percutaneous drainage.

Oxford MCQs

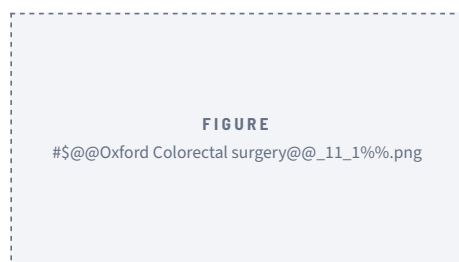
1 A 23-year-old man presents with 6 weeks of increasingly frequent stools with blood-stained mucus mixed with diarrhoea. There is generalized mild abdominal tenderness but no guarding. Temperature is 37.6°C. Which is the single most likely diagnosis?

- A Crohn's disease
- B Diverticular disease
- C Haemorrhoids
- D Rectal cancer
- E Ulcerative colitis

2 A 65-year-old man presents with rapid onset left iliac fossa pain and fever. He is tender in the left iliac fossa but there is no obvious palpable mass. Temperature is 37.7°C, pulse 84bpm, white cell count $14 \times 10^9/L$, and CRP 54 mg/L. Which is the single most likely diagnosis?

- A Acute appendicitis
- B Sigmoid cancer
- C Crohn's disease of the colon
- D Acute sigmoid diverticulitis
- E Ureteric colic

3 A 78-year-old man who is known to have sigmoid diverticular disease has developed left lower quadrant pain over the last 8 days with 48h of constant severe pain, feeling febrile with sweating episodes and shakes. His temperature is fluctuating up to 39.2°C, the pulse is 84bpm, blood pressure 125/74mmHg, and there is localized tenderness in the left lower quadrant but no generalized guarding or tenderness. Intravenous fluids and antibiotics are started. An urgent CT scan of the pelvis is shown in Figure 10.3. Which is the single most appropriate treatment he should be offered?



- A Formation of a defunctioning loop colostomy
- B Colonoscopy and colonic stent insertion
- C Continued IV antibiotics and close monitoring
- D CT guided drainage
- E Sigmoid colectomy and end colostomy (Hartmann's procedure)

4 A 74-year-old man who had a right hemicolectomy 5 days ago is now unwell with shortness of breath, generalized abdominal pain and mild distension. Temperature is 37.8°C, pulse 102bpm and blood pressure 127/67mmHg. There is generalized abdominal tenderness with involuntary guarding on the left side. He is being given supplemental oxygen and has intravenous 0.9% saline running. Which is the single most appropriate first step you should take?

- A Administer IV antibiotics
- B Arrange an abdominopelvic CT scan
- C Arrange an emergency laparotomy
- D Request an erect chest X-ray
- E Request an ultrasound scan and radiological drain

5 48-year-old woman has had progressively worsening colicky abdominal pain for 48h which has become severe and constant for the last 12h. There is a past history of an abdominal hysterectomy. The abdomen is slightly distended but non-tender. Blood investigations are normal except for a white cell count of $13 \times 10^9/L$. Which is the single most likely diagnosis?

- A Acute cholecystitis
- B Acute pancreatitis
- C Ascending cholangitis
- D Ischaemic bowel
- E Perforated peptic ulcer

6 48-year-old woman has recently undergone extensive bowel resection for complications of ischaemia and has been receiving total parenteral nutrition via a conventional central line. There is currently mild abdominal discomfort and a non-productive cough. There is no evidence of anaemia or jaundice. She has developed a swinging pyrexia as high as 39.5°C over the last 12h. Which is the single most likely diagnosis?

- A Acute cholangitis
- B Acute respiratory distress syndrome
- C Central venous line infection
- D Deep vein thrombosis
- E Pneumonia

- 7** The principal differences of macroscopic appearances and functions of the proximal jejunum and distal ileum. Which is the single correct statement comparing the features of each?
- A** Compared to the ileum, the jejunum: absorbs less vitamin B6, is thinner walled, and secretes more fluid during digestion
 - B** Compared to the ileum, the jejunum: absorbs more vitamin B6, secretes less fluid during digestion, and appears redder in colour
 - C** Compared to the ileum, the jejunum: is thicker walled, possesses less prominent lymphoid tissue, and secretes more fluid during digestion
 - D** Compared to the ileum, the jejunum: possesses less prominent lymphoid tissue, paler and appears blue in colour, absorbs less vitamin B6
 - E** Compared to the ileum, the jejunum: possesses more prominent lymphoid tissue, appears paler blue in colour, and secretes less fluid during digestion
- 8** 13-year-old adolescent girl has had 2 days of increasing right iliac fossa and lower abdominal pain with tenderness and guarding. Her temperature is 37.8°C, pulse is 92bpm. A provisional diagnosis of acute appendicitis is made and a laparoscopic appendicectomy planned. Which is the single most appropriate investigation prior to the procedure?
- A** Abdominal X-ray
 - B** Chest X-ray
 - C** CT scan abdomen and pelvis
 - D** Pregnancy test
 - E** Transabdominal pelvic ultrasound scan

Answers 1-8

OXFORD MCQS

- 1** **E Ulcerative colitis** — **UPDATED** Six weeks of bloody mucoid diarrhoea with urgency in a young man is ulcerative colitis until proven otherwise: stool cultures, then colonoscopy with biopsies.
- 2** **D Acute sigmoid diverticulitis** — **UPDATED** Acute LIF pain, fever and raised WCC/CRP in a 65-year-old is acute sigmoid diverticulitis; confirm with CT if severe or uncertain.
- 3** **D CT guided drainage** — **UPDATED** Swinging fever with a pericolic abscess on CT in a stable patient without peritonitis: CT-guided percutaneous drainage plus antibiotics.
- 4** **A Administer IV antibiotics** — **UPDATED** Abdominal pain, fever and tachycardia on day 5 after colectomy is an anastomotic leak until proven otherwise: start IV antibiotics with resuscitation, then CT or re-laparotomy.
- 5** **D Ischaemic bowel** — Disproportion between severe symptoms and minimal physical signs suggest intestinal ischaemia.
- 6** **C Central venous line infection** — Total parenteral nutrition via central line is associated with central venous line infections.
- 7** **C Compared to the ileum, the jejunum: is thicker walled, possesses less prominent lymphoid tissue, and secretes more fluid during digestion** — **UPDATED** The jejunum is thicker walled, redder, with prominent plicae and less lymphoid tissue, and secretes most digestive fluid; the ileum is thinner, paler, with Peyer patches.
- 8** **D Pregnancy test** — **UPDATED** Before appendicectomy in a girl of reproductive age, a pregnancy test is mandatory; other imaging rarely changes a clear clinical decision.

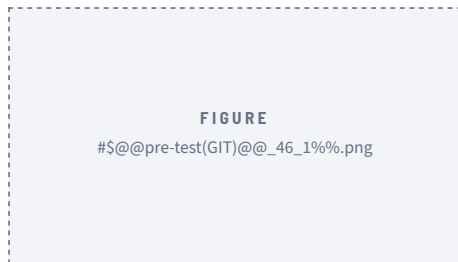
18 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1** A 32-year-old man with a 3-year history of ulcerative colitis (UC) presents for discussion for surgical intervention. The patient is otherwise healthy and does not have evidence of rectal dysplasia. Which of the following is the most appropriate elective operation for this patient?
- A** Total proctocolectomy with end ileostomy
 - B** Total proctocolectomy with ileal pouch-anal anastomosis and diverting ileostomy
 - C** Total proctocolectomy with ileal pouch-anal anastomosis, anal mucosectomy, and diverting ileostomy
 - D** Total abdominal colectomy with ileal-rectal anastomosis
 - E** Total abdominal colectomy with end ileostomy and very low Hartmann
- 2** A 56-year-old woman is referred to you about 3 months after a colostomy subsequent to a sigmoid resection for cancer. She complains that her stoma is not functioning properly. Which of the following is the most common serious complication of an end colostomy?
- A** Bleeding
 - B** Skin breakdown
 - C** Parastomal hernia
 - D** Retraction of the ostomy
 - E** Stomal prolapse

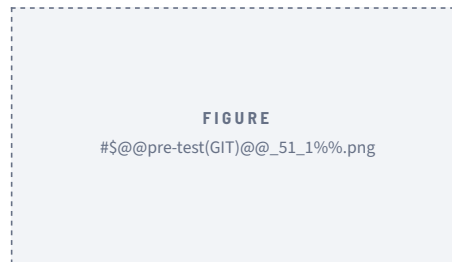
- 3** A 45-year-old woman with history of heavy nonsteroidal antiinflammatory drug ingestion presents with acute abdominal pain which started 30 hours prior to arrival. An abdominal plain film demonstrates pneumoperitoneum. She undergoes exploratory laparotomy and is found to have a perforated duodenal ulcer. Which of the following is the procedure of choice to treat her perforation?
- A Simple closure with omental patch
 - B Washout and placement of drains
 - C Truncal vagotomy and antrectomy
 - D Wedge resection of ulcer
 - E Subtotal gastrectomy
- 4** An 80-year-old man with history of symptomatic cholelithiasis presents with nausea, bilious emesis, and abdominal distention. He has not had a bowel movement for 3 days. Which of the following findings would provide the most help in ascertaining the diagnosis?
- A Coffee-grounds aspirate from the stomach
 - B Pneumobilia
 - C A leukocyte count of 40,000/mL
 - D A pH of 7.5, PCO₂ of 50 kPa, and paradoxically acid urine
 - E A palpable mass in the pelvis
- 5** A 42-year-old man has bouts of intermittent crampy abdominal pain and rectal bleeding. Colonoscopy is performed and demonstrates multiple hamartomatous polyps. The patient is successfully treated by removing as many polyps as possible with the aid of intraoperative endoscopy and polypectomy. Which of the following is the most likely diagnosis?
- A Ulcerative colitis
 - B Villous adenomas
 - C Familial polyposis
 - D Peutz-Jeghers syndrome
 - E Crohn colitis
- 6** A 45-year-old man presents to the emergency room with left lower quadrant pain, fever, and vomiting. CT scan of the abdomen and pelvis reveals a thickened sigmoid colon with inflamed diverticula and a 7-cm by 8-cm rim-enhancing fluid collection in the pelvis. After percutaneous drainage and treatment with antibiotics, the pain and fluid collection resolve. He returns as an outpatient to clinic 1 month later. He undergoes a colonoscopy, which demonstrates only diverticula in the sigmoid colon. Which of the following is the most appropriate next step in this patient's management?
- A Expectant management with sigmoid resection if symptoms recur
 - B Cystoscopy to evaluate for a fistula
 - C Sigmoid resection with end colostomy and rectal pouch (Hartmann procedure)
 - D Sigmoid resection with primary anastomosis
 - E Long-term suppressive antibiotic therapy
- 7** A 72-year-old man status post-coronary artery bypass graft (CABG) 5 years ago presents with hematochezia, abdominal pain, and fevers. Colonoscopy reveals patches of dusky-appearing mucosa at the splenic flexure without active bleeding or necrosis. Which of the following is the most appropriate management of this patient?
- A Angiography with administration of intra-arterial papaverine
 - B Emergent laparotomy with left hemicolectomy and transverse colostomy
 - C Aortomesenteric bypass
 - D Exploratory laparotomy with thrombectomy of the inferior mesenteric artery
 - E Expectant management
- 8** A previously healthy 15-year-old boy is brought to the emergency room with complaints of about 12 hours of progressive anorexia, nausea, and pain of the right lower quadrant. On physical examination, he is found to have a temperature of 38.1°C (100.7°F) and direct and rebound abdominal tenderness localizing to McBurney point as well as involuntary guarding in the right lower quadrant. At operation through a McBurney-type incision, the appendix and cecum are found to be normal, but the surgeon is impressed by the marked edema of the terminal ileum, which also has an overlying fibrinopurulent exudate. Which of the following is the most appropriate next step?
- A Close the abdomen after culturing the exudate
 - B Perform a standard appendectomy
 - C Resect the involved terminal ileum
 - D Perform an ileocolic resection
 - E Perform an ileocolostomy to bypass the involved terminal ileum
- 9** A 50-year-old man who was recently diagnosed with Crohn disease asks about the need for surgery. Which of the following findings would be an indication for an immediate exploratory laparotomy?
- A Intestinal obstruction
 - B Enterovesical fistula
 - C Ileum-ascending colon fistula
 - D Enterovaginal fistula
 - E Free perforation

- 10** A 50-year-old man presents to the emergency room with a 6-hour history of excruciating abdominal pain and distention. The abdominal film shown here is obtained. Which of the following is the most appropriate next diagnostic maneuver?



- A** Emergency celiotomy
 - B** Upper GI series with small-bowel follow-through
 - C** CT scan of the abdomen
 - D** Barium enema
 - E** Sigmoidoscopy
- 11** A 75-year-old woman undergoes an uncomplicated lower extremity arterial bypass. Ten days after surgery the patient presents with sudden onset of abdominal pain and distention. An abdominal radiograph demonstrates an air-filled, kidney-bean-shaped structure in the left upper quadrant. Which of the following is the most appropriate management at this time?
- A** Decompression of the large bowel via colonoscopy
 - B** Placement of the NG tube and administration of low-dose cholinergic drugs
 - C** Administration of a gentle saline enema and encouragement of ambulation
 - D** Operative decompression with transverse colostomy
 - E** Right hemicolectomy
- 12** A 28-year-old woman who is 15 weeks pregnant has new onset of nausea, vomiting, and right-sided abdominal pain. The pain has become worse over the past 6 hours. What is the most likely diagnosis?
- A** Appendicitis
 - B** Cholecystitis
 - C** Pancreatitis
 - D** Intestinal obstruction
 - E** Acute fatty liver of pregnancy

- 13** An 80-year-old man with dementia develops abdominal distention and obstipation and is brought to the hospital from his nursing facility. The abdominal radiograph is obtained. Which of the following is the most appropriate initial management of this patient?



- A** Urgent colostomy or cecostomy
 - B** Discontinuation of anticholinergic medications and narcotics and correction of metabolic disorders
 - C** Digital disimpaction of fecal mass in the rectum
 - D** Diagnostic and therapeutic colonoscopy
 - E** Detorsion of volvulus and colectomy or resection
- 14** A 30-year-old female patient who presents with chronic, intermittent bloody diarrhea and abdominal discomfort is found at colonoscopy to have colitis confined to the transverse and descending colon. A biopsy is performed. Which of the following is a finding consistent with this patient's diagnosis?
- A** Inflammatory process confined to the mucosa and submucosa
 - B** Continuous inflammatory reaction
 - C** Superficial ulcerations
 - D** Noncaseating granulomas
 - E** Microabscesses within crypts
- 15** A 37-year-old man with a 10-year history of ulcerative colitis has a sessile polyp with high-grade dysplasia 10 cm from the anal verge. Which operation is most appropriate?
- A** Low anterior resection
 - B** Abdominoperineal resection
 - C** Subtotal colectomy with end ileostomy
 - D** Total proctocolectomy with ileoanal J-pouch
 - E** Sigmoid resection with end colostomy (Hartmann procedure)
- 16** A 33-year-old man with Crohn disease has severe abdominal pain and fever: HR 130/min, BP 105/62 mmHg, temperature 38.9 °C, WCC 32 000/mm³; plain films show a markedly dilated colon. Which operation is most appropriate?
- A** Low anterior resection
 - B** Abdominoperineal resection
 - C** Subtotal colectomy with end ileostomy
 - D** Total proctocolectomy with ileoanal J-pouch
 - E** Sigmoid resection with end colostomy (Hartmann procedure)

- 17** What are the most common causes of small bowel obstruction?
- A** Adhesions, hernias, and cancer
 - B** Adhesions, volvulus, and internal hernias
 - C** Hernias, neoplasms, and strictures
 - D** Adhesions and Crohn's disease
 - E** Incarcerated hernias, adhesions, and intussusception
- 18** What are the indications for surgery for Meckel's diverticulum?
- A** Presence of inflammation and tumor development
 - B** Bleeding, obstruction, and perforation
 - C** Asymptomatic cases in young adults
 - D** Only in cases associated with diverticulitis
 - E** Recurrent abdominal pain without complications

Answers 1-18

PRETEST MCQS

- 1** **B** **Total proctocolectomy with ileal pouch-anal anastomosis and diverting ileostomy** — For UC without rectal dysplasia, an ileal pouch-anal anastomosis is the preferred elective option.
- 2** **C** **Parastomal hernia** — Parastomal herniation is the most frequent serious complication of an end colostomy.
- 3** **A** **Simple closure with omental patch** — For perforated duodenal ulcer in patients without history of ulcer disease, simple closure with omental patch is preferred.
- 4** **B** **Pneumobilia** — Pneumobilia indicates a biliary-enteric fistula, suggestive of gallstone ileus.
- 5** **D** **Peutz-Jeghers syndrome** — Peutz-Jeghers syndrome is characterized by hamartomatous polyps and mucocutaneous pigmentation.
- 6** **D** **Sigmoid resection with primary anastomosis** — Definitive management of complicated diverticulitis often involves resection with primary anastomosis once inflammation has resolved.
- 7** **E** **Expectant management** — Ischemic colitis is typically managed conservatively with supportive care.
- 8** **B** **Perform a standard appendectomy** — Despite the inflammatory findings, a standard appendectomy is performed to avoid future diagnostic uncertainty in suspected Crohn disease.
- 9** **E** **Free perforation** — Free perforation is a surgical emergency in patients with Crohn disease.
- 10** **E** **Sigmoidoscopy** — Sigmoidoscopic decompression is the appropriate diagnostic (and potentially therapeutic) maneuver in sigmoid volvulus.
- 11** **E** **Right hemicolectomy** — A right hemicolectomy is required in cases of cecal volvulus to correct the torsion and prevent ischemia.
- 12** **A** **Appendicitis** — Appendicitis remains the most common non-obstetric surgical emergency during pregnancy.
- 13** **B** **Discontinuation of anticholinergic medications and narcotics and correction of metabolic disorders** — Management of Ogilvie syndrome begins with conservative measures including medication review and supportive care.
- 14** **D** **Noncaseating granulomas** — The presence of noncaseating granulomas supports the diagnosis of Crohn disease.
- 15** **D** **Total proctocolectomy with ileoanal J-pouch** — **UPDATED** High-grade dysplasia in UC marks a field change: total proctocolectomy, with an ileoanal J-pouch to preserve continence.
- 16** **C** **Subtotal colectomy with end ileostomy** — **UPDATED** Toxic megacolon failing medical therapy is treated by subtotal colectomy with end ileostomy, leaving the rectum for later reconstruction.
- 17** **A** **Adhesions, hernias, and cancer** — **UPDATED** Adhesions are the commonest cause of small-bowel obstruction in the West, followed by hernias and malignancy.
- 18** **B** **Bleeding, obstruction, and perforation** — **UPDATED** Symptomatic Meckel diverticulum (bleeding, obstruction from intussusception or volvulus, diverticulitis or perforation) needs resection.

16 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1** A 72-year-old woman is brought into hospital with profuse diarrhoea, abdominal pain and fever, which she has suffered for the last two days. On examination, she has generalized abdominal tenderness. Her observations include heart rate 108/min, blood pressure 110/64 mmHg and temperature 38.2°C. Abdominal and erect chest X-rays are unremarkable. Her son tells you she was in hospital two weeks ago with a chest infection. Which of the following would be the most appropriate treatment in the first instance?

 - A Intravenous co-amoxiclav
 - B Intravenous vancomycin
 - C Oral metronidazole
 - D Oral rehydration
 - E Subtotal colectomy
- 2** A 34-year-old man presents with a 10-hour history of abdominal pain associated with nausea and vomiting. On examination, he has guarding in the right iliac fossa and a temperature of 37.6°C. He has no significant past medical history. What is the most likely diagnosis?

 - A Appendicitis
 - B Crohn disease
 - C Meckel diverticulitis
 - D Mesenteric adenitis
 - E Renal colic
- 3** A 27-year-old woman undergoing surveillance colonoscopy for ulcerative colitis has multiple small projections similar in colour to normal bowel mucosa. What is the most likely morphology of the polyps?

 - A Adenomatous polyp
 - B Hamartomatous polyp
 - C Juvenile polyp
 - D Metaplastic polyp
 - E Pseudopolyp
- 4** A 64-year-old woman has a 2-day history of increasing left-sided abdominal pain and fever. On examination, she has localized peritonism in the left iliac fossa. Blood tests reveal raised WCC and CRP. Which of the following is the most likely diagnosis?

 - A Constipation
 - B Diverticular disease
 - C Diverticulitis
 - D Diverticulosis
 - E Irritable bowel syndrome
- 5** A 72-year-old woman presents with a 5-hour history of abdominal pain, bloating and vomiting. On examination, the abdomen is distended and auscultation reveals intermittent high-pitched sounds. An abdominal X-ray shows air in the biliary tree. What is the most likely diagnosis?

 - A Emphysematous cholecystitis
 - B Gallstone ileus
 - C Paralytic ileus
 - D Perforated gallbladder
 - E Pyogenic cholecystitis
- 6** A 32-year-old woman undergoing fertility treatment presents with a 2-hour history of severe right lower abdominal pain, with similar milder episodes previously. She has no other GU symptoms; pregnancy test is negative. Most likely diagnosis?

 - A Appendicitis
 - B Ectopic pregnancy
 - C Pelvic inflammatory disease
 - D Torsion of the ovary
 - E Urinary tract infection
- 7** A 54-year-old man presents with a 2-day history of upper abdominal and shoulder tip pain on the right side. He also complains of difficulty breathing and episodes of fever and sweats. He underwent a laparotomy 2 weeks ago following a perforated appendix. What is the most likely diagnosis?

 - A Appendix abscess
 - B Cholecystitis
 - C Diverticular abscess
 - D Polycystic liver disease
 - E Subphrenic abscess
- 8** A 24-year-old man, who has been suffering from intermittent fresh bleeding per rectum, presents to the emergency department with a 6-hour history of right-sided abdominal pain, fevers and nausea. On examination, he has tenderness and guarding in the right iliac fossa. His temperature is 38.2°C. What is the most likely diagnosis?

 - A Appendicitis
 - B Haemorrhoids
 - C Meckel diverticulitis
 - D Renal colic
 - E Shigella infection

- 9** A 34-year-old man presents to the emergency department with a 6-hour history of abdominal pain and vomiting. On examination, the abdomen is distended and tympanic. He tells you that for the last 6 months he has been having episodes of cramping right-sided abdominal pain with intermittent episodes of blood in his stools. Which of the following complications has resulted in his presentation today?
- A Abscess formation
 - B Fistula formation
 - C Primary sclerosing cholangitis
 - D Stricture formation
 - E Toxic megacolon
- 10** A 25-year-old woman has been investigated with lower gastrointestinal endoscopy and biopsy following complaints of frequent bloody diarrhoea which is still continuing. The results show diffuse superficial inflammation with ulceration in the rectum only. Which of the following treatment options would be most suitable in the first instance?
- A Azathioprine
 - B Colectomy
 - C Mesalazine
 - D Prednisolone
 - E Surveillance colonoscopy
- 11** An 8-year-old boy is brought into the emergency department by his father with a 1-day history of fever, right-sided abdominal pain and diarrhoea. He has no past medical history of note, but has recently recovered from a cold. On examination, he has a temperature of 39°C and is tender, but not guarding, in the right iliac fossa. What is the most likely diagnosis?
- A Appendicitis
 - B Crohn disease
 - C Coeliac disease
 - D Gastroenteritis
 - E Mesenteric adenitis
- 12** A 59-year-old woman with a long history of constipation presents with a 4-month history of episodic left-sided abdominal pain relieved by defaecation. She has also had two episodes of fresh bleeding per rectum. There is no history of weight loss and she is otherwise well. Examination is unremarkable. She is referred for a barium enema. What features would you expect to find?
- A Apple-core lesion
 - B Birds beak appearance
 - C Diverticular outpouchings
 - D Non-filling of the appendix
 - E Thumb printing of the colon
- 13** A 35-year-old woman presents to the emergency department with a 1-day history of colicky central abdominal pain and vomiting. She last opened her bowels 2 days ago. She has no past medical history of note but tells you she had an operation on her abdomen as a child although she is not sure what this was for. On examination, she looks unwell and moderately dehydrated. The abdomen is distended and tender with hyperactive bowel sounds. Which of the following is most likely to have caused her symptoms?
- A Adhesions
 - B Colorectal tumour
 - C Constipation
 - D Gallstone ileus
 - E Paralytic ileus
- 14** A 29-year-old woman presents to the emergency department with a 3-hour history of lower abdominal pain. On examination, there is marked rebound tenderness in the left iliac fossa. She has a heart rate of 92/min and a blood pressure of 134/92 mmHg. Which of the following investigations would you perform in the first instance?
- A Abdominal ultrasound
 - B Abdominal X-ray
 - C Cervical swabs
 - D Urine dipstick
 - E Urine pregnancy test
- 15** A 74-year-old man presents to the emergency department with a 6-hour history of colicky lower abdominal pain accompanied by gross abdominal distension. He has had similar episodes in the past that have been relieved by opening his bowels. Today he is not even able to pass flatus. Abdominal X-ray shows a grossly dilated loop of large bowel. Which of the following is the most likely diagnosis?
- A Adhesions
 - B Faecal impaction
 - C Intussusception
 - D Sigmoid volvulus
 - E Strangulated inguinal hernia
- 16** A 56-year-old man presents to the emergency department with a 3-hour history of severe abdominal pain and vomiting. On examination, the abdomen is rigid. A CT scan confirms free air in the peritoneum and a perforation of a sigmoid diverticulum. What is the most appropriate operation?
- A Abdominoperineal resection
 - B Extended right hemicolectomy
 - C Hartmann procedure
 - D Left hemicolectomy
 - E Right hemicolectomy

Answers 1-16

- 1** **C** Oral metronidazole — **UPDATED** Diarrhoea with fever after a recent admission and antibiotics suggests *C. difficile* colitis;

of these options, oral metronidazole (oral vancomycin is now first-line).

- 2 **A Appendicitis** — Classical presentation of periumbilical pain localizing to RIF with guarding and mild fever suggests appendicitis.
- 3 **E Pseudopolyp** — Inflammatory pseudopolyps are typical in ulcerative colitis.
- 4 **C Diverticulitis** — LLQ pain with systemic inflammatory signs suggests diverticulitis.
- 5 **B Gallstone ileus** — A gallstone erodes into the bowel causing obstruction, visible as pneumobilia on X-ray.
- 6 **D Torsion of the ovary** — Intermittent twisting of an enlarged ovary causes acute severe lower abdominal pain.
- 7 **E Subphrenic abscess** — A subphrenic abscess can follow surgery or perforation, causing fever, shoulder tip pain, and respiratory symptoms.
- 8 **C Meckel diverticulitis** — A Meckel diverticulum may mimic appendicitis but often presents with painless rectal bleeding and RIF pain.
- 9 **D Stricture formation** — Crohn's disease can cause strictures leading to obstruction.
- 10 **C Mesalazine** — Ulcerative colitis limited to the rectum can be managed initially with aminosalicylates (mesalazine).
- 11 **E Mesenteric adenitis** — Mesenteric adenitis often follows a viral infection and can mimic appendicitis in children.
- 12 **C Diverticular outpouchings** — Diverticular disease shows outpouchings on barium enema.
- 13 **A Adhesions** — Adhesions are a common cause of small bowel obstruction in someone with previous surgery.
- 14 **E Urine pregnancy test** — Always check for pregnancy in women of childbearing age with acute abdominal pain.
- 15 **D Sigmoid volvulus** — Twisting of the sigmoid colon on its mesentery causing large bowel obstruction.
- 16 **C Hartmann procedure** — For perforated sigmoid diverticulum, resection with end colostomy (Hartmann's) is used.

12 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 32-year-old man presents with a 2-day history of sudden-onset pain in the right iliac fossa. On examination you note guarding and rebound tenderness localized to an area approximately one-third of the distance from the anterior superior iliac spine and umbilicus on the right side of his abdomen. What is being described here?
 - A Grey Turner sign
 - B Cullen sign
 - C McBurney point
 - D Rovsing sign
 - E Murphy sign
- 2 A 26-year-old woman presents with sudden-onset right iliac fossa pain. On examination you note that when pressure is applied to her left iliac fossa, the patient complains of increased pain in her lower right side. What is being described here?
 - A Grey Turner sign
 - B Cullen sign
 - C McBurney point
 - D Rovsing sign
 - E Murphy sign
- 3 A 32-year-old woman presents with a 2-day history of sudden-onset lower right-sided abdominal pain. Routine blood investigations reveal a white cell count of $23 \times 10^9/L$. A urine dipstick reveals 3+ leucocytes. Which investigation is least useful in establishing the diagnosis?
 - A Abdominal X-ray
 - B Abdominal ultrasound
 - C Pregnancy test
 - D Abdominal computed tomography (CT) scan
 - E Pelvic ultrasound
- 4 A 42-year-old man presents with generalized abdominal pain and vomiting. He comments that he has not opened his bowels for the past 2 days. On examination you note severe abdominal distension and tinkling bowel sounds. An abdominal X-ray is ordered which reveals distended loops of small bowel. What is the next most appropriate step in management?
 - A Nasogastric decompression and normal saline intravenously (IV)
 - B Surgery
 - C Nasogastric decompression
 - D Nasogastric decompression and dextrose IV
 - E Normal saline IV

- 5** An elderly patient presents with left iliac fossa pain and vomiting. On examination you note tenderness in the left iliac fossa and peritonism. Routine blood investigations demonstrate a white cell count of $22 \times 10^9/L$. What is the most likely diagnosis?
- A Diverticulitis
 - B Ulcerative colitis
 - C Sigmoid volvulus
 - D Small bowel obstruction
 - E Large bowel obstruction
- 6** A 35-year-old woman presents with lower left-sided abdominal discomfort that is severe in nature. She denies any gastrointestinal symptoms. Routine investigations show a normal white cell count, CRP, and a negative serum beta human chorionic gonadotropin. What is the most likely diagnosis?
- A Diverticulitis
 - B Ovarian cyst
 - C Uterine cyst
 - D Sigmoid volvulus
 - E Ectopic pregnancy
- 7** A term baby is born with a vitelline duct still present. He remains well until 19 years of age, when he develops colicky abdominal pain. What is the likely cause?
- A Appendicitis
 - B Meckel diverticulum
 - C Exomphalos
 - D Gastroschisis
 - E Atresia
- 8** A 78-year-old female is admitted with severe sudden-onset abdominal pain for 3 days, and bloody diarrhoea for 24 hours. She has a history of atrial fibrillation. Bloods show raised lactate. What is the most likely diagnosis?
- A Ulcerative colitis
 - B Colorectal carcinoma
 - C Angiodysplasia
 - D Crohn disease
 - E Ischaemic bowel
- 9** A 21-year-old female presents with sudden-onset acute abdominal pain. She is tender in the right iliac fossa with guarding, tachycardic and hypotensive. What would be your initial management?
- A Abdominal X-ray
 - B Pregnancy test
 - C Intravenous (IV) fluids
 - D Urine dipstick
 - E Abdominal ultrasound
- 10** A 65-year-old woman has sudden left iliac fossa pain, fever, vomiting. Which investigation is most likely to lead to a diagnosis?
- A Colonoscopy
 - B Abdominal ultrasound
 - C Abdominal and pelvic CT scan
 - D Abdominal X-ray
 - E WCC & CRP
- 11** A 24-year-old obese woman has sudden severe right upper quadrant pain with guarding. Which investigation is most likely to lead to a diagnosis?
- A WCC
 - B CRP
 - C Liver function tests
 - D Abdominal X-ray
 - E Abdominal ultrasound
- 12** A 65-year-old man with sudden abdominal pain, no bowels for 3 days, vomiting. X-ray shows sigmoid volvulus. Definitive resolution next step?
- A Emergency laparotomy
 - B Flexible sigmoidoscopic decompression with flatus tube
 - C IV fluids
 - D Nasogastric tube insertion
 - E IV fluids + NG tube

Answers 1-12

- 1** **C** **McBurney point** — Typical location of the base of the appendix; suggests appendicitis.
- 2** **D** **Rovsing sign** — Pain in the right lower quadrant on palpation of the left side; appendicitis.
- 3** **A** **Abdominal X-ray** — Plain X-ray has minimal diagnostic value for suspected appendicitis.
- 4** **A** **Nasogastric decompression and normal saline intravenously (IV)** — 'Drip and suck' approach; decompress bowel and correct fluid loss.
- 5** **A** **Diverticulitis** — LIF pain, high WCC, peritonism; typical of diverticulitis.
- 6** **B** **Ovarian cyst** — **UPDATED** Severe unilateral lower abdominal pain in a young woman with normal inflammatory markers and negative β -hCG suggests an ovarian cyst (rupture or torsion); pelvic ultrasound.
- 7** **B** **Meckel diverticulum** — Remnant of vitelline duct, can mimic appendicitis in older patients.
- 8** **E** **Ischaemic bowel** — Abdominal pain out of proportion + AF risk factor + metabolic acidosis.
- 9** **C** **Intravenous (IV) fluids** — She is unstable (tachy/hypo); resuscitate first, then investigate.
- 10** **C** **Abdominal and pelvic CT scan** — Acute diverticulitis is best confirmed by CT scan
- 11** **E** **Abdominal ultrasound** — Suspected acute cholecystitis; ultrasound is first-line test.

- 12** **B** **Flexible sigmoidoscopic decompression with flatus tube**
— Best initial definitive measure for sigmoid volvulus.

Anorectal Disorders

Haemorrhoids, fissure, abscess and fistula, pilonidal disease, prolapse and anal cancer.

44

MCQS

0

EMQ SCENARIOS

7

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

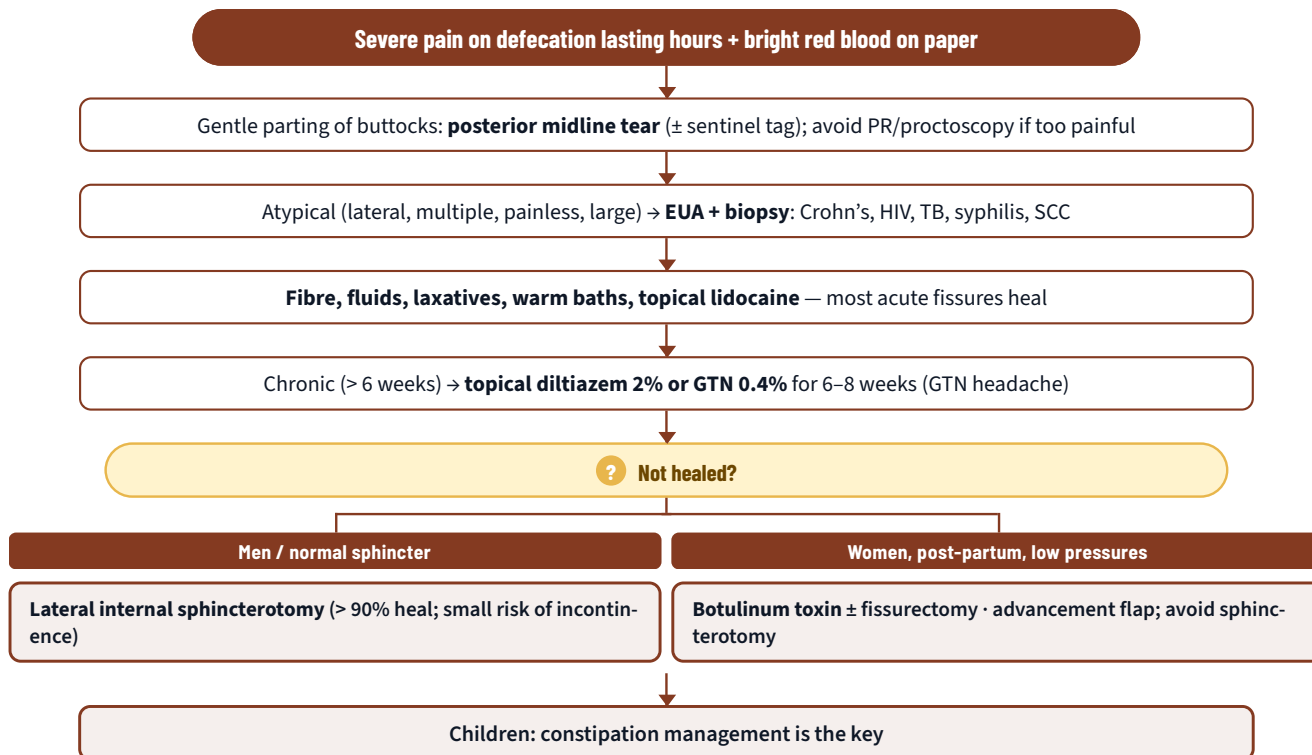
01	Management flowcharts	3 charts	page 156
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11

Step by step

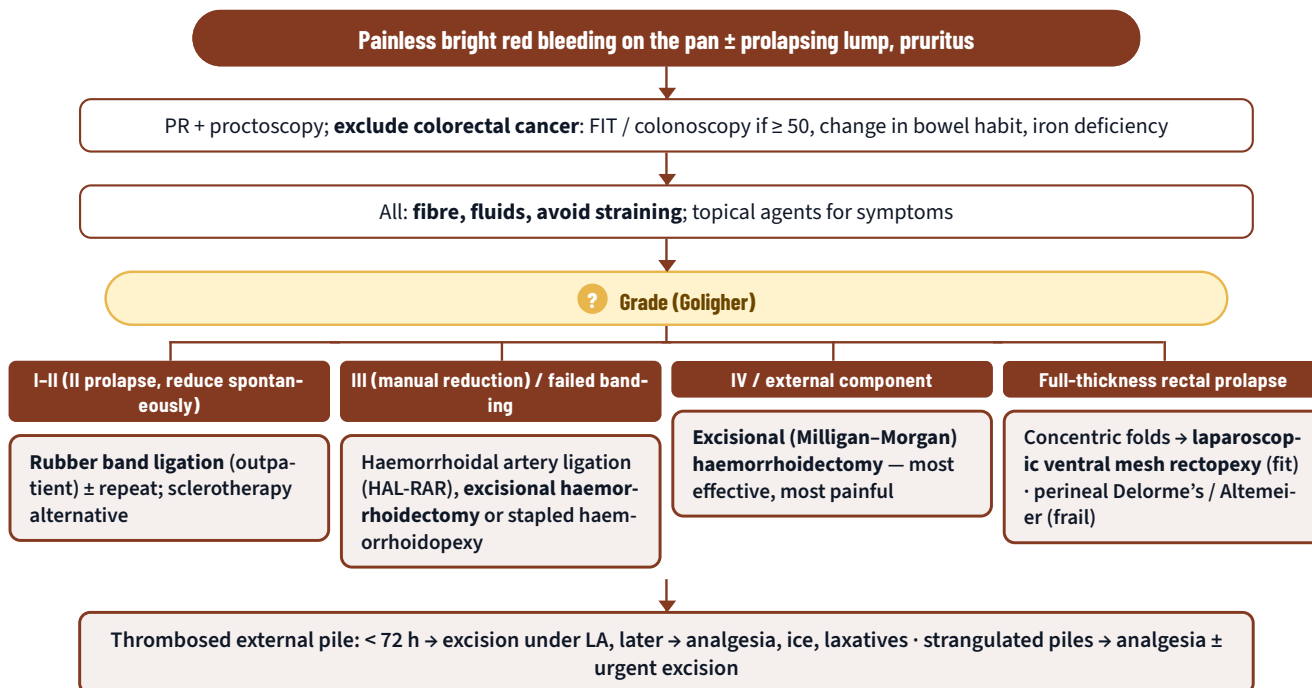
Anal fissure

ACPGBI 2017 • ASCRS 2023 ANAL FISSURE



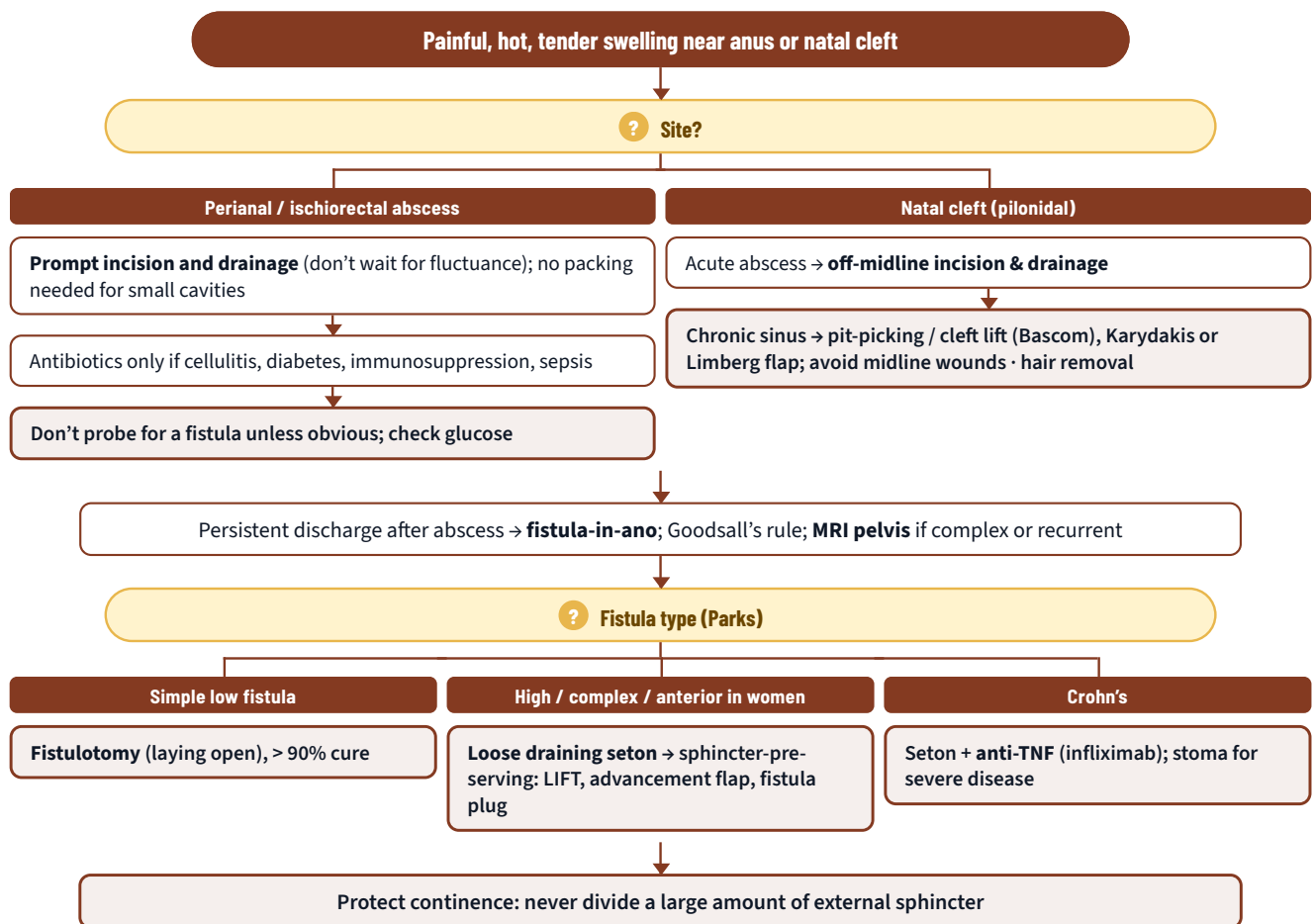
Haemorrhoids and prolapse

ESCP 2020 HAEMORRHOIDS • ACPGBI 2019 • NICE NG12



Perianal and pilonidal sepsis

ACPGBI 2016 ANAL FISTULA · ASCRS 2022 · ESCP 2024 PILONIDAL



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-2

A 21-year-old man has incapacitating perianal pain for 3 days despite local ointment. There is a tender, fluctuant right perianal swelling with overlying erythema.

1 What is the most likely diagnosis?

- A Perianal abscess
- B Perianal haematoma
- C Pilonidal abscess
- D Strangulated piles
- E Fournier gangrene

2 What is the best treatment now?

- A Antibiotics
- B Incision and drainage
- C Excision and primary repair
- D Haemorrhoidectomy
- E Local debridement

Answers 1-2

LOCAL EXAMS MCQs

1 **A Perianal abscess** — A tender, fluctuant, erythematous swelling beside the anus is a perianal abscess.

2 **B Incision and drainage** — An abscess needs prompt incision and drainage; antibiotics alone are inadequate.

19 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love MCQs

1 Which of the following statements are true?

- A The rectum is completely enveloped by peritoneum.
- B The rectovesical pouch in the male is 7.5 cm and rectouterine pouch is 5.5 cm from the anal verge.
- C The fascia in front of the rectum is the Denonvilliers' fascia and behind is the Waldeyer's fascia.
- D The mesorectum contains the superior rectal vessels and lymphatics.
- E The lymphatics from the rectum mainly drain upwards.

2 The following statements are true except:

- A Patients with rectal disease might present with extra-rectal symptoms and signs.
- B Rectal bleeding is always a sinister symptom.
- C Altered bowel habit of recent onset should be of concern.
- D Flexible sigmoidoscopy can visualise a maximum distance of 40 cm of the colon.
- E Examination of the rectum by finger or endoscope is best carried out in the left lateral position.

3 Regarding injuries and foreign bodies in the rectum, which of the following statements are true?

- A Impalement injury is the usual accidental cause.
- B Intraperitoneal rectal injuries might require a staged procedure.
- C A temporary colostomy is usually necessary.
- D In severe injuries, faecal incontinence might be the long-term outcome.
- E When a patient arrives with a self-introduced foreign body (FB), it should be removed immediately.

4 Regarding rectal prolapse, the following statements are true except:

- A In children it is usually a mucosal prolapse.
- B In children the differential diagnosis is ileocaecal intussusception.
- C In adults it is usually a full-thickness prolapse.
- D Majority of children will need an operation.
- E Majority of adults can be managed conservatively.

5 Regarding proctitis, which of the following statements are true?

- A The diagnosis is made on sigmoidoscopy followed by colonoscopy.
- B Crohn's disease is a common cause.
- C Specific infections as a cause must be excluded.
- D It is present in majority of patients with ulcerative colitis.
- E AIDS due to HIV infection can cause the condition.

- 6** Regarding rectal polyps, which of the following statements are true?
- A A polyp >1 cm is likely to be malignant.
 - B When a rectal polyp is found a full colonoscopy should be performed.
 - C In certain cases a patient might present with the effects of hypokalaemia.
 - D In villous adenoma abdomino-perineal excision is the operation of choice.
 - E In familial adenomatous polyposis the patient should be treated according to the prescribed protocol.
- 7** Regarding carcinoma of the rectum, the following statements are true except:
- A Most patients with rectal cancer will require an abdomino-perineal resection (APR).
 - B Early-morning spurious diarrhoea is a typical symptom.
 - C In locally advanced tumours preoperative chemoradiotherapy makes curative surgery possible.
 - D Following an anterior resection, a defunctioning loop ileostomy is usually performed.
 - E Endoluminal stenting can be done as a temporary or permanent procedure.
- 8** Which of the following statements are true?
- A The junction between the rectum and anal canal is the anorectal ring.
 - B The puborectalis muscle is an important component of the continence mechanism.
 - C The external sphincter forms the bulk of the anal sphincter complex.
 - D The longitudinal muscle is a direct continuation of the smooth muscle of the outer muscle coat of the rectum.
 - E The internal anal sphincter is the continuation of the circular muscle coat of the rectum.
- 9** The following statements are true except:
- A The dentate (pectinate) line divides the anal canal into an upper two-thirds and lower one-third.
 - B The mucosa and submucosa above the dentate line form anal cushions.
 - C The anal glands drain into the anal sinuses and are the source of anal sepsis.
 - D The lymphatics from the anal canal drain into the inguinal group of lymph nodes.
 - E The lower part of the anal canal is lined by stratified squamous epithelium.
- 10** Regarding special investigations on the ano-rectum, which of the following statements are true?
- A The resting tone is a reflection of internal sphincter activity.
 - B A maximal anal pressure of 60 cm H₂O is abnormal.
 - C Pelvic dyssynergia can be identified by evacuatory proctogram.
 - D Megarectum might be a finding in rectal hyposensitivity.
 - E Perineal position and descent reflect pelvic floor and pudendal nerve function.
- 11** Regarding imperforate anus, which of the following statements are true?
- A Imperforate anus is classified into high and low types.
 - B Fistula with the urethra is uncommon.
 - C The most common abnormality in girls is a fistula into the vagina.
 - D The finding of a single perineal orifice suggests a persistent cloaca.
 - E Meconium on the perineum indicates a low defect.
- 12** Regarding pilonidal sinus, the following statements are true except:
- A The condition always occurs in the natal cleft.
 - B It is a disease of the elderly.
 - C The condition predominantly affects males.
 - D Recurrent abscesses are common.
 - E Treatment is always surgical excision.
- 13** Regarding anal incontinence, which of the following statements are true?
- A Neurological pathways must be intact to maintain anal continence.
 - B Structural integrity of the gastrointestinal tract and pelvic floor musculature is essential to maintain continence.
 - C History and clinical examination might be enough to determine aetiology.
 - D Surgical treatment is tailored to the cause.
 - E A permanent stoma is a choice.
- 14** Regarding anal fissure, which of the following statement(s) is/are false?
- A Anal fissure occurs most commonly on the anterior midline at 12 o'clock position.
 - B Acute anal fissure results from trauma from the passage of a hard stool.
 - C A fissure at an unusual site might have an underlying cause.
 - D Operation is the treatment of choice.
 - E Anal dilatation should be tried as the first surgical procedure.
- 15** Regarding haemorrhoids, which of the following statements are true?
- A Haemorrhoids are symptomatic anal cushions.
 - B Traditionally, four degrees are recognised.
 - C Conservative treatment should be the first line of management.
 - D Urgent surgery is justified in certain complications.
 - E Operation when indicated is usually carried out by the 'open technique'.
- 16** Regarding pruritis ani, which of the following statements are not true?
- A Usually there is an underlying cause.
 - B Diabetes must be excluded.
 - C Treat the cause of pruritis ani.
 - D Surgical treatment is required in the majority.
 - E Hydrocortisone should be the first line of conservative treatment.

- 17** Regarding anorectal abscesses, which of the following statements are true?
- A** Anorectal abscesses might be a manifestation of a generalised disease.
 - B** Primary colorectal disease might be a cause.
 - C** The main cause is infection of the anal glands.
 - D** Presentation is always as a very tender, fluctuant lump in the perianal region.
 - E** Immediate fistulotomy should be performed if a fistula is present.
- 18** Regarding fistula-in-ano, the following statements are true except:
- A** There is always an underlying associated specific cause.
 - B** The most common cause is persistent infection of the anal glands.
 - C** MRI is an essential investigation in all cases.
 - D** Classification is into high and low-anal fistula.
 - E** The majority can be treated by fistulotomy.
- 19** Which of the following statements are true?
- A** MRI and EUS should be undertaken before surgical treatment.
 - B** Fistulotomy is the treatment for intersphincteric and trans-sphincteric types.
 - C** Setons are used in the treatment of high and complicated fistulae.
 - D** Anal advancement flaps aim to preserve anatomy and function.
 - E** Biological agents have been used with variable results.

Answers 1-19

BAILEY & LOVE MCQS

- 1** **BCDE** True: B, C, D, E — The rectum is not completely covered by peritoneum; the other statements are correct.
- 2** **BD** True: B, D — Rectal bleeding is not always sinister and flexible sigmoidoscopy can reach beyond 40 cm.
- 3** **ABCDE** True: A, B, C, D, E — **UPDATED** All statements are true, though foreign body removal should be performed under controlled conditions.
- 4** **DE** True: D, E — **UPDATED** Most children are managed conservatively and most adults require surgery.
- 5** **ACDE** True: A, C, D, E — **UPDATED** Crohn's disease is not a common cause of proctitis; the other statements are correct.
- 6** **ABCE** True: A, B, C, E — **UPDATED** Abdomino-perineal excision is not the treatment of choice for villous adenoma.
- 7** **A** Most patients with rectal cancer will require an abdomino-perineal resection (APR). — **UPDATED** APR is needed only in a minority as sphincter-sparing procedures are common.
- 8** **ABCDE** True: A, B, C, D, E — All listed statements correctly describe anorectal anatomy and continence mechanisms.
- 9** **D** The lymphatics from the anal canal drain into the inguinal group of lymph nodes. — Above the dentate line lymph drains upward to mesenteric nodes; only below it drains to inguinal nodes, making D false.
- 10** **ACDE** True: A, C, D, E — **UPDATED** Resting pressure arises from the internal sphincter; 60 cm H₂ O squeeze is normal, so B is false; the other listed statements are correct.
- 11** **ADE** True: A, D, E — **UPDATED** High vs low level guides surgery; urethral fistulae in boys are common and girls usually have vestibular fistula; cloaca gives one orifice; meconium at perineum signals low lesion.
- 12** **ABE** True: A, B, E — **UPDATED** Pilonidal disease mainly affects young hirsute men, can appear outside the cleft, and many cases are managed non-operatively.
- 13** **ABCDE** True: A, B, C, D, E — **UPDATED** Continence needs intact nerves, sphincters and rectum; careful assessment guides therapy, ranging from conservative measures to stoma diversion.
- 14** **ADE** True: A, D, E — **UPDATED** Fissures are usually posterior; initial management is medical; anal stretch is obsolete.
- 15** **ABCDE** True: A, B, C, D, E — **UPDATED** Enlarged cushions cause symptoms; Goligher I-IV guides therapy; start with life-style; strangulation may need emergency surgery; Milligan-Morgan is standard elective technique.
- 16** **DE** True: D, E — **UPDATED** Most cases managed conservatively by addressing precipitating factors; steroids reserved for proven dermatitis; surgery rarely needed.
- 17** **ABC** True: A, B, C — **UPDATED** Cryptoglandular infection is commonest; Crohn's or immunosuppression predispose; deep abscesses may lack a superficial lump; fistulotomy is delayed until sepsis resolves.
- 18** **AC** True: A, C — **UPDATED** Most fistulae are idiopathic cryptoglandular; MRI reserved for complex cases; high vs low guides therapy; most low tracts laid open.
- 19** **BCDE** True: B, C, D, E — Simple low tracts do not mandate imaging; low inter/trans-sphincteric laid open; high tracts use setons or flaps; biologics like glue or plugs show mixed success.

6 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 A 39-year-old man of Italian origin presents complaining of an exquisitely tender area over his right buttock, which has been present for several weeks. Within the past few days, the area has begun to weep profusely and his temperature at home prior to admission was 38°C. He reports having had this problem previously and was operated on during a previous admission. On examination you note that the man is extremely hairy and that there is an irregular, erythematous, warm and exquisitely tender shallow lump overlying the top of the right buttock. A small scar is seen over the lump and, on palpation, it is fluctuant and discharges purulent fluid. A pilonidal abscess is suspected. Definitive management would consist of
 - A Advising to shave the affected area
 - B Intravenous antibiotics
 - C Drainage of the abscess under local anaesthesia
 - D Drainage of the abscess under general anaesthesia
 - E Oral antibiotics
- 2 A 60-year-old homosexual man presents with a 6-month history of passing fresh rectal blood and anal pain, accompanied by loss of fecal continence. On examination, a third-degree haemorrhoid is identified along with two second-degree haemorrhoids. What is the diagnosis?
 - A Anal squamous carcinoma
 - B Haemorrhoids
 - C Sigmoid adenocarcinoma
 - D Diverticular disease
 - E Anal fissure
- 3 Which of the following statements regarding the anatomy of the anal canal is false?
 - A There is no touch or pain sensation above the dentate line
 - B The anal canal below the dentate line drains lymph to the inguinal nodes
 - C The anorectal ring is made up of the blended fibres of the puborectal muscle and internal anal sphincter
 - D The superior rectal vein drains into the portal system
 - E The pelvic diaphragm delineates the point at which the rectum becomes the anal canal
- 4 A 60-year-old diabetic patient presents with an 8-hour history of inability to pass urine, accompanied by a 3-day history of fever and throbbing perianal pain. Digital rectal examination is very painful but reveals a normal prostate. What is the diagnosis?
 - A Ischiorectal abscess
 - B Acute prostatitis
 - C Urinary tract infection
 - D Invasive pelvic malignancy
 - E Supraplevator abscess
- 5 A patient with a previous anal abscess now presents with persistent anal discharge and perianal discomfort. On examination, a fistula is identified at the 6 o'clock position. What procedure is the surgeon most likely to perform?
 - A Diversion loop colostomy
 - B Plug insertion
 - C Open exploration of tract
 - D Endoanal ultrasound
 - E Examination under anaesthetic +/- proceed
- 6 Which of the following is NOT an option when treating an anal fissure?
 - A Propranolol
 - B Botulinum toxin A injections
 - C Glyceryl trinitrate cream
 - D Diltiazem cream
 - E Laxatives

Answers 1-6

SBA MCQs

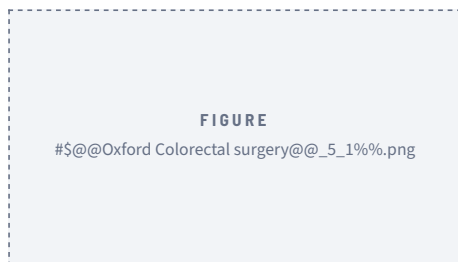
- 1 **D Drainage of the abscess under general anaesthesia** — Pilonidal abscess requires surgical drainage under general anaesthesia, frequently with wide excision.
- 2 **A Anal squamous carcinoma** — The loss of continence along with the clinical findings suggests anal squamous carcinoma.
- 3 **C The anorectal ring is made up of the blended fibres of the puborectal muscle and internal anal sphincter** — The anorectal ring actually incorporates the external sphincter, not just the internal.
- 4 **E Supraplevator abscess** — The clinical features are consistent with a supraplevator abscess.
- 5 **E Examination under anaesthetic +/- proceed** — Examination under anaesthetic enables accurate assessment and management of the fistula tract.

- 6 **A Propranolol** — Propranolol is not used in the management of anal fissures.

4 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 An 84-year-old woman has an 18-month history of intermittent swelling in the anus which appears when she strains to pass stools. Which is the single most appropriate course of initial management?



- A Offer her a laparoscopic rectopexy
 B Offer her banding of the excess mucosa
 C Offer her Delorme's operation
 D Recommend a course of injection sclerotherapy
 E Recommend a course of stool softeners, and advice on bowel habit
- 2 A 34-year-old man has a red, tender, fluctuant swelling in his natal cleft with a swelling off to the left of the midline in the tissue of the left buttock. There are visible midline pits in the natal cleft. Which is the single most appropriate treatment?
- A Fistulotomy under anaesthetic
 B Incision and drainage under anaesthetic
 C Oral antibiotics
 D Wide excision and asymmetric natal cleft closure
 E Wide excision and packing with daily dressings

- 3 A 28-year-old man has had a tender red swelling developing at his anal margin over 2 days. He has had recurrent episodes of similar symptoms with occasional discharge from the area. Examination reveals the external opening of a fistula in ano and an associated area of subcutaneous fluctuance. Which is the single most appropriate management plan?

- A Examination under general anaesthetic
 B Immediate antibiotics and review in outpatients
 C Incision and drainage on the ward
 D MRI scan
 E Needle aspiration on the ward

- 4 A 37-year-old woman has had sharp and persistent anal pain associated with defecation for 4 months. Occasionally, there are small amounts of blood on the toilet paper on wiping. There is a fissure present in the anal canal at the 6 o'clock position. Which is the single most appropriate treatment for her?

- A Anal dilatation
 B Botox injection
 C Diltiazem cream
 D Lateral internal sphincterotomy
 E Prednisolone suppositories

Answers 1-4

OXFORD MCQS

- 1 **E Recommend a course of stool softeners, and advice on bowel habit** — **UPDATED** A large cylindrical prolapse on straining in an 84-year-old is full-thickness rectal prolapse; start with stool softeners and bowel-habit advice, then Delorme or rectopexy if needed.
- 2 **B Incision and drainage under anaesthetic** — **UPDATED** A fluctuant natal-cleft swelling with midline pits is a pilonidal abscess: incision and drainage (off the midline); definitive excision later once the sepsis settles.
- 3 **A Examination under general anaesthetic** — **UPDATED** An acute perianal abscess over a known fistula needs examination under anaesthesia with incision and drainage (and laying open or seton of any fistula found).
- 4 **C Diltiazem cream** — **UPDATED** A posterior midline fissure is treated first with topical diltiazem (or GTN) plus stool softeners; botulinum toxin or lateral sphincterotomy if it fails.

1 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 An 17-year-old male high school student presents with a painful fluctuant mass in the midline between the gluteal folds. Rectal exam is unremarkable with no masses, fluctuance, or tenderness. Which of the following is the most likely diagnosis?
- A Pilonidal abscess
 - B Perianal abscess
 - C Perirectal abscess
 - D Fistula-in-ano
 - E Anal fissure

Answer 1

PRETEST MCQs

- 1 **A Pilonidal abscess** — UPDATED A midline fluctuant mass between the gluteal folds is most consistent with a pilonidal abscess.

7 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1 A 70-year-old woman presents with a 2-month history of anal pain, itching, fresh bleeding, and mucous discharge. On examination, there is an irregular tender ulceration at the anal margin extending into the anal canal. What is the most likely diagnosis?
- A Anal carcinoma
 - B Anal fissure
 - C Anal warts
 - D Fistula-in-ano
 - E Primary syphilis
- 2 A 43-year-old woman presents with bright red fresh rectal bleeding and intermittent rectal discomfort. She has recently noticed a mass in her rectum. On examination, a non-tender lump is seen emerging from the rectum. It cannot be reduced. Which of the following would be the most appropriate treatment?
- A Anal dilatation
 - B Banding
 - C Haemorrhoidectomy
 - D Injection sclerotherapy
 - E No intervention required
- 3 A 31-year-old man has episodic perianal pain for over a year with occasional discharge. A pit in the natal cleft filled with hair is found. Minimal discharge on pressure. What is the most likely diagnosis?
- A Perianal abscess
 - B Perianal haematoma
 - C Pilonidal sinus
 - D Proctalgia fugax
 - E Rectal prolapse
- 4 A 36-year-old man presents to the surgical outpatient clinic with a 3-week history of soiled underwear. He is known to have Crohn disease. On examination, an opening is seen at the posterior margin of the anus which is discharging faeculent material. Subsequent imaging demonstrates an anal fistula that tracks through the puborectalis muscle. What would be the most appropriate management of this condition?
- A Diltiazem ointment
 - B Excision of the fistula
 - C Laying open of the fistula tract
 - D Lord procedure
 - E Seton insertion

- 5** A 27-year-old man with a long history of constipation presents with a history of severe pain on defaecation and bright red blood on the tissue paper. On examination, you see a skin tag at the anal verge. Per rectum examination is not possible due to severe pain. Which of the following treatments would you suggest?
- A** Analgesic suppositories
 - B** Excision of skin tag
 - C** Haemorrhoidectomy
 - D** Incision and drainage
 - E** Topical GTN ointment
- 6** A 36-year-old man complains of a 6-month history of intermittent pain in the rectum which is unrelated to stools. The pain tends to occur at night. He denies passing any mucus or blood per rectum and examination is unremarkable. Which of the following is the most likely diagnosis?
- A** Anal fissure
 - B** Anal fistula
 - C** Perianal abscess
 - D** Perianal haematoma
 - E** Proctalgia fugax
- 7** A 49-year-old woman presents with a 4-month history of perianal itching with occasional mucous discharge. She is otherwise well. On examination, large fleshy pink lesions are seen around the anus. What is the most likely diagnosis?
- A** Condylomata lata
 - B** Condylomata acuminata
 - C** Anal carcinoma
 - D** Rectal prolapse
 - E** Haemorrhoids

Answers 1-7

GET AHEAD MCQS

- 1** **A** **Anal carcinoma** — An irregular ulcerative lesion with bleeding suggests malignant infiltration; anal carcinoma is likely.
- 2** **C** **Haemorrhoidectomy** — Irreducible (grade IV) haemorrhoids require surgical excision.
- 3** **C** **Pilonidal sinus** — A sinus in the sacrococcygeal region containing hair, often with discharge.
- 4** **E** **Seton insertion** — High trans-sphincteric fistulas in Crohn's are managed with seton insertion to avoid incontinence.
- 5** **E** **Topical GTN ointment** — Topical nitrates reduce sphincter spasm and facilitate healing of chronic anal fissures.
- 6** **E** **Proctalgia fugax** — A benign condition causing brief nocturnal rectal pain, unrelated to defecation.
- 7** **B** **Condylomata acuminata** — HPV-related anal warts causing pink fleshy lesions around the anus.

5 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1** A middle-aged woman presents with bright red bleeding on defecation, blood on toilet paper, and in the pan. Proctoscopy confirms haemorrhoids. First management option?
- A** Review in 2 months
 - B** Barron banding
 - C** Surgery
 - D** Cryotherapy
 - E** Injection sclerotherapy
- 2** A 55-year-old man has perianal discomfort. Exam shows a small dark-blue/purple swelling on the anal margin, painful on palpation, like a blackcurrant. Most likely diagnosis?
- A** Perianal haematoma
 - B** Haemorrhoid
 - C** Thrombosed haemorrhoid
 - D** Tumour
 - E** Hidradenitis suppurativa

- 3** A middle-aged man has pain on defecation that persists afterwards. Exam reveals a midline fissure for 12 weeks. Next step in management?
- A Lateral internal sphincterotomy
 - B Botulinum A toxin injection
 - C Oral diltiazem
 - D Topical glyceryl trinitrate ointment
 - E Surgical sphincter dilatation
- 4** A 40-year-old woman is suspected of faecal incontinence. Which investigation leads to a diagnosis?
- A Abdominal ultrasound scan
 - B Barium enema
 - C Anorectal physiology studies
 - D Abdominal MRI scan
 - E Colonoscopy
- 5** A middle-aged man has recurrent perianal pain and swelling with purulent discharge. Exam shows circular granulation tissue exuding pus. Most likely diagnosis?
- A Abscess
 - B Hidradenitis suppurativa
 - C Pilonidal cyst
 - D Fistula in ano
 - E Pilonidal sinus

Answers 1-5

- 1** **E Injection sclerotherapy** — First-line for early haemorrhoids with good success rates.
- 2** **A Perianal haematoma** — Small clot under skin at anal margin, very painful, black/blue lump.
- 3** **D Topical glyceryl trinitrate ointment** — First-line to relax internal sphincter and aid healing.
- 4** **C Anorectal physiology studies** — Tests sphincter function (pressure) and integrity (endoanal US).
- 5** **D Fistula in ano** — Granulation opening from rectum to perianal skin with discharge.

Gastrointestinal Bleeding

Upper and lower GI bleeding: resuscitation, risk scores, endoscopy, varices, angiodysplasia and diverticular bleeding.

49

MCQs

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

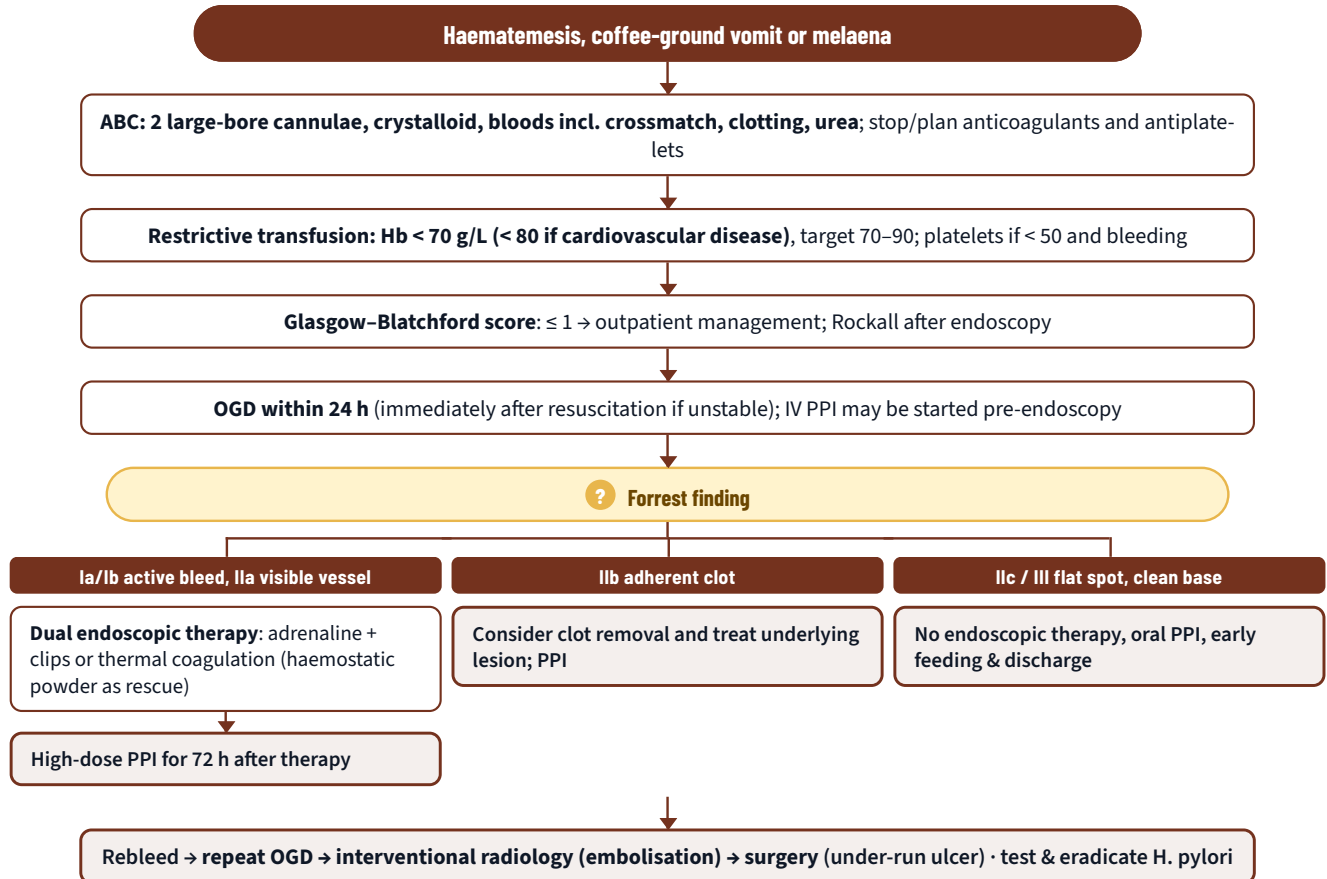
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12

Step by step

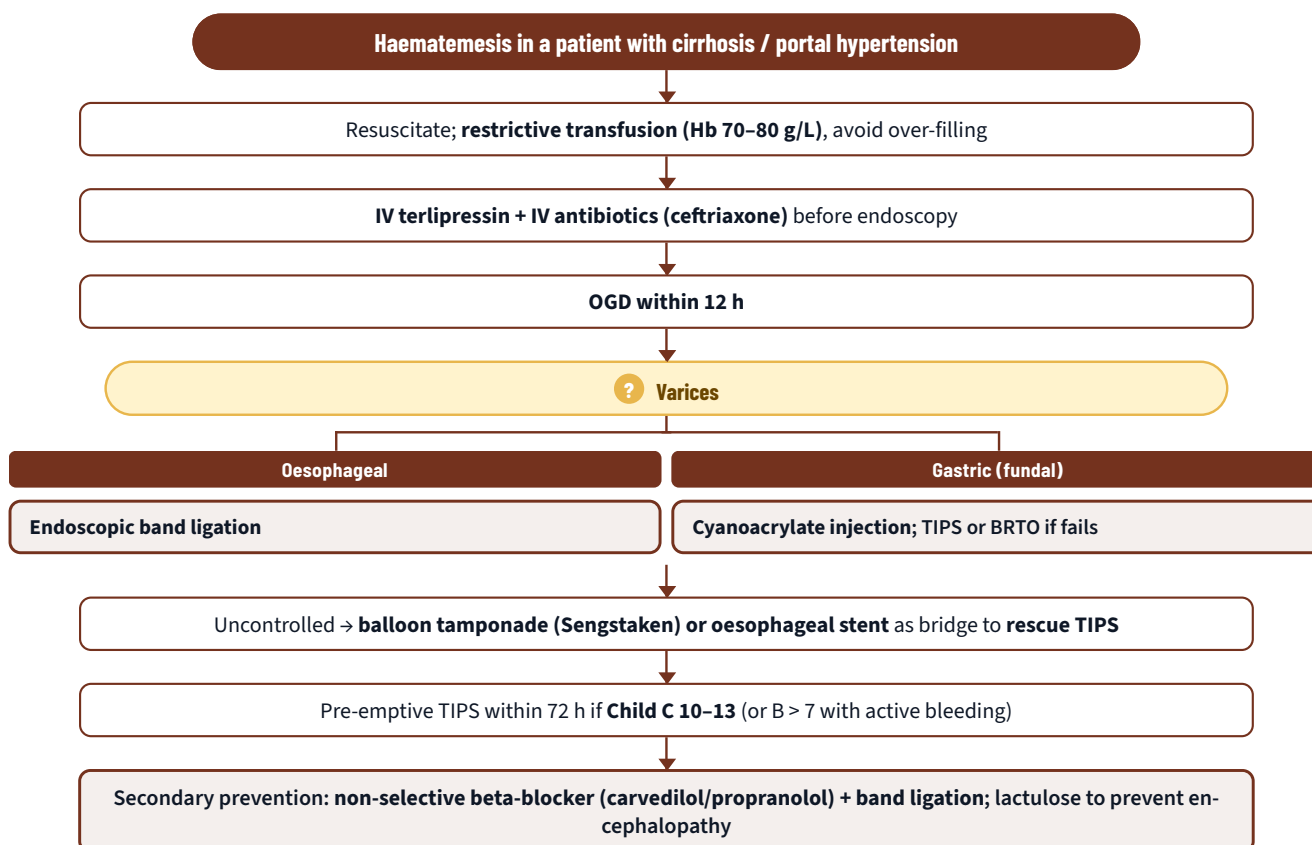
Acute upper GI bleeding (non-variceal)

NICE CG141 • ESGE 2021 NVUGIH • BSG 2022 CARE BUNDLE



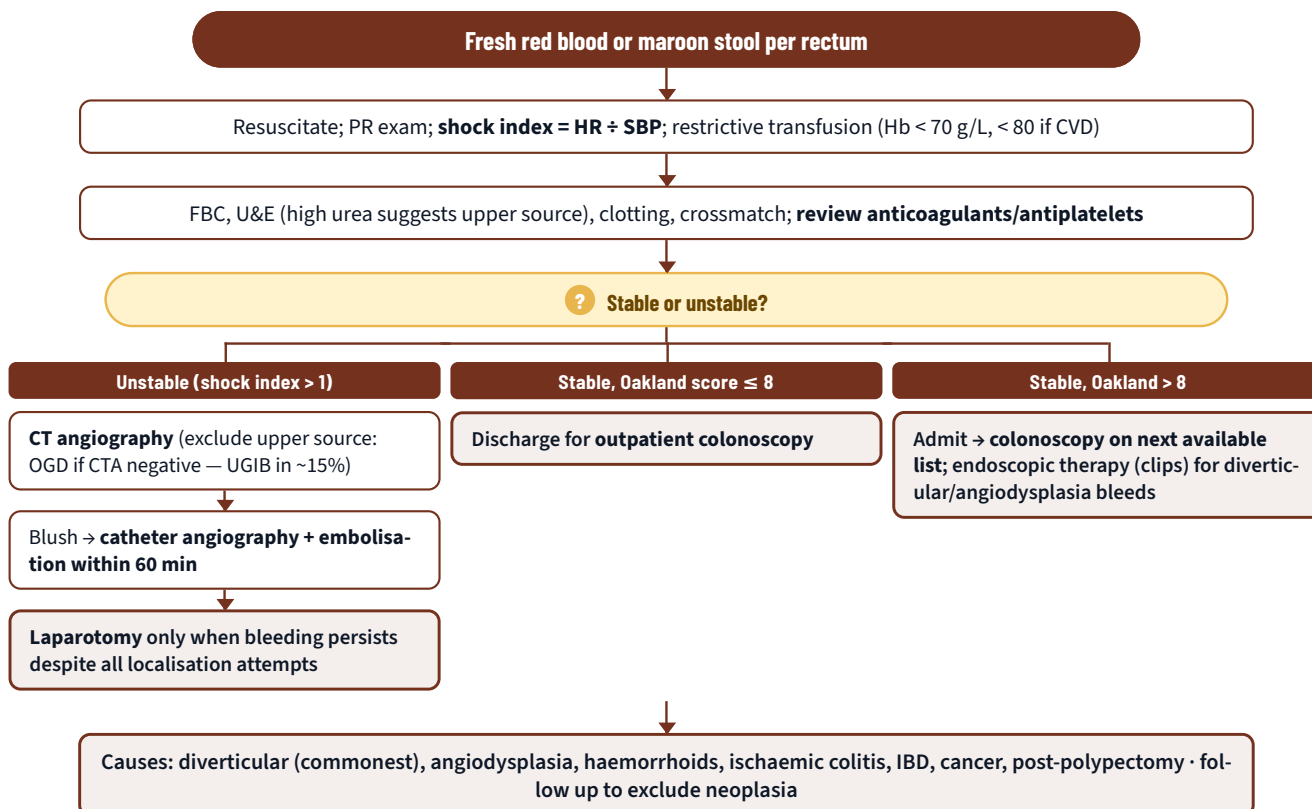
Variceal bleeding

BAVENO VII 2022 · BSG 2015 VARICEAL HAEMORRHAGE



Acute lower GI bleeding

BSG 2019 LGIB · ACG 2023 LGIB



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 58 year old woman complains of progressive weakness, easy fatiguability, and palpitation on simple exercise. She has an epigastric ulcer pain and she noticed frequent passage of black stool in the last two months. She lost appetite and weight, but she had no vomiting. On examination: she looks pale, PR 100/min, abdomen was tender at the epigastrium, and there is enlarged left supraclavicular lymph node. Ultrasound shows ascites with multiple liver lesions. The next most appropriate choice of work up is:
 - A CT abdomen with IV contrast
 - B MRI abdomen with IV contrast
 - C Diagnostic laparoscopy with ascitic fluid sample
 - D Oesophago-gastro-duodenoscopy
 - E PET scan for the whole body
- 2 A 68-year-old man presents to ER with an acute onset of vomiting containing coffee-ground material. The first endoscopy needed to be done is:
 - A Colonoscopy
 - B Sigmoidoscopy
 - C Oesophago-gastro-duodenoscopy (OGD)
 - D Endoscopic ultrasound (EUS)
 - E Endoscopic retrograde cholangiopancreatography (ERCP)
- 3 A 47-year-old alcoholic male patient presents to ER with large amount of vomiting mixed with blood. He has yellowish discoloration of his sclera. Abdomen is distended and shifting dullness sign is positive. The most likely underlying cause for this presentation is:
 - A Bleeding gastric ulcer
 - B Bleeding duodenal ulcer
 - C Acute gastric erosions
 - D GIST tumor
 - E Bleeding esophageal varices

Answers 1-3

LOCAL EXAMS MCQs

- 1 **D Oesophago-gastro-duodenoscopy** — **UPDATED** Melaena, epigastric pain, weight loss and a Virchow node suggest gastric cancer: OGD with biopsy is needed for tissue diagnosis, then CT for staging.
- 2 **C Oesophago-gastro-duodenoscopy (OGD)** — **UPDATED** Coffee-ground vomiting is upper GI bleeding: OGD (within 24 h, sooner if unstable) to find and treat the source.
- 3 **E Bleeding esophageal varices** — **UPDATED** Haematemesis with jaundice and ascites in an alcoholic suggests portal hypertension and bleeding oesophageal varices.

1 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQs

Essay to MCQ MCQs

- 1 Mention the preoperative measures that should be taken in the casualty unit before surgery.
 - A Only IV fluids
 - B Only pain management
 - C IV fluid resuscitation, nasogastric tube insertion, broad-spectrum antibiotics, and initial lab assessments including hematocrit, Hb, cross match, coagulation profile, electrolytes, and liver function tests
 - D Immediate surgery without any preparation
 - E Oral hydration and rest

Answer 1

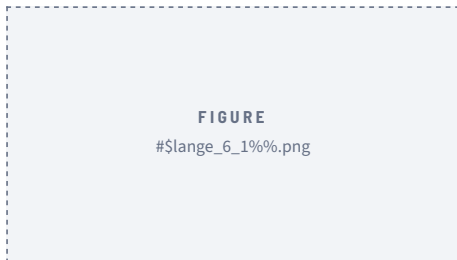
ESSAY TO MCQ MCQs

- 1 **C IV fluid resuscitation, nasogastric tube insertion, broad-spectrum antibiotics, and initial lab assessments including hematocrit, Hb, cross match, coagulation profile, electrolytes, and liver function tests** — Comprehensive preoperative measures are essential for patient stabilization.

18 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 2-year-old child presents with a 2-day history of painless rectal bleeding. On examination, the child is pale with tachycardia. The abdomen is nondistended and nontender. There is dark blood on rectal examination. Imaging suggests Meckel's diverticulum. Which of the following is the most appropriate management?



- A** surgical exploration
B aggressive resuscitation followed by surgical exploration
C colonoscopy
D acid suppression therapy
E IV steroids
- 2** A 4-year-old previously healthy girl presents with rectal bleeding and dizziness for 24 hours. She is pale and tachycardic; the abdomen is nondistended. Fresh blood is found in the rectum. Which of the following is the most likely diagnosis?
- A** a bleeding Meckel's diverticulum
B juvenile rectal polyp
C hemorrhoids
D an anal fissure
E intussusception
- 3** Which of the following is the most appropriate diagnostic study for this 4-year-old girl with suspected Meckel's diverticulum?
- A** colonoscopy
B barium enema
C technetium scan
D UGI contrast study with small-bowel follow-through
E laparoscopy
- 4** Definitive management of this child with a bleeding Meckel's diverticulum should include which of the following?
- A** immediate exploratory laparotomy
B IV fluid resuscitation, transfusion if needed, followed by laparotomy with Meckel's diverticulectomy and ileal resection
C IV fluid resuscitation, followed by colonoscopic polypectomy
D hemorrhoidectomy
E stool softeners and topical steroids

- 5** A 55-year-old man presents to the ED at 5 a.m. with vomiting blood after binge drinking. He reports multiple episodes of vomiting followed by hematemesis and melena. Which of the following is the most appropriate diagnostic test?

A barium esophagogram
B water-soluble contrast esophagogram
C esophagoscopy
D CT scan
E angiogram

- 6** A 50-year-old man with a UGI bleed from acute erosive gastritis has a hematocrit of 28% and continues to be tachycardic with signs of ischemia. What is the next appropriate step in management?

A initiation of iron supplementation therapy
B supplemental oxygen
C continued IV fluid resuscitation
D initiation of a calcium channel blocker
E blood transfusion

- 7** A 60-year-old man is brought to the emergency department after fainting at home. He has passed bright red blood per rectum for 6 hours, without abdominal pain or vomiting. Pulse 110/min, BP 95/50 mmHg; there is gross blood on rectal examination. What is the most likely diagnosis?

A diverticulosis of the colon
B ulcerative colitis
C Meckel's diverticulum
D ischemic colitis
E aortoenteric fistula
F peptic ulcer
G carcinoma of the colon
H esophageal varices
I gastric ulcer

- 8** A 56-year-old woman had an episode of melaena 48 hours ago that has not recurred. She had an abdominal aortic aneurysm repair 2 years ago. Upper GI endoscopy to the duodenal bulb and colonoscopy are normal. What is the most likely diagnosis?

A diverticulosis of the colon
B ulcerative colitis
C Meckel's diverticulum
D ischemic colitis
E aortoenteric fistula
F peptic ulcer
G carcinoma of the colon
H esophageal varices
I gastric ulcer

- 9** A 65-year-old woman has severe fatigue and weakness. She has hypertension, diabetes and coronary disease, takes daily aspirin and has a 40 pack-year smoking history. She is pale, her stool is positive for occult blood and she has a microcytic anaemia. What is the most likely diagnosis?
- A diverticulosis of the colon
 - B ulcerative colitis
 - C Meckel's diverticulum
 - D ischemic colitis
 - E aortoenteric fistula
 - F peptic ulcer
 - G carcinoma of the colon
 - H esophageal varices
 - I gastric ulcer
- 10** A 65-year-old man has haematemesis and melaena. He has had abdominal pain for 3–4 weeks and has taken ibuprofen four times daily for 3 weeks, with alcohol, for the pain. Pulse 100/min, BP 100/60 mmHg after resuscitation; there is RUQ and epigastric tenderness and melaena on rectal examination. What is the most likely diagnosis?
- A diverticulosis of the colon
 - B ulcerative colitis
 - C Meckel's diverticulum
 - D ischemic colitis
 - E aortoenteric fistula
 - F peptic ulcer
 - G carcinoma of the colon
 - H esophageal varices
 - I gastric ulcer
- 11** A 50-year-old man with chronic alcoholism has an episode of haematemesis. Pulse 120/min, BP 90/60 mmHg; there are spider naevi, palmar erythema and a distended, non-tender abdomen. What is the most likely diagnosis?
- A diverticulosis of the colon
 - B ulcerative colitis
 - C Meckel's diverticulum
 - D ischemic colitis
 - E aortoenteric fistula
 - F peptic ulcer
 - G carcinoma of the colon
 - H esophageal varices
 - I gastric ulcer
- 12** A 35-year-old man has haematemesis and black tarry stools after 2 weeks of epigastric pain partly relieved by antacids. Pulse 110/min, BP 95/60 mmHg, with epigastric tenderness and melaena. What is the most likely diagnosis?
- A diverticulosis of the colon
 - B ulcerative colitis
 - C Meckel's diverticulum
 - D ischemic colitis
 - E aortoenteric fistula
 - F peptic ulcer
 - G carcinoma of the colon
 - H esophageal varices
 - I gastric ulcer
- 13** A 35-year-old woman has a 1-week history of bloody diarrhoea, 1 month of abdominal pain and 10 lb weight loss. Temperature 38.3 °C, pulse 85/min, BP 110/60 mmHg; the abdomen is diffusely tender. What is the most likely diagnosis?
- A diverticulosis of the colon
 - B ulcerative colitis
 - C Meckel's diverticulum
 - D ischemic colitis
 - E aortoenteric fistula
 - F peptic ulcer
 - G carcinoma of the colon
 - H esophageal varices
 - I gastric ulcer
- 14** A 6-month-old boy with 1-day rectal bleeding, no pain and a normal exam, most likely has?
- A anal fissure
 - B intussusception
 - C Meckel's diverticulum
 - D milk allergy
 - E volvulus
- 15** A 70-year-old woman with vague abdominal pain and bloody diarrhea 3 days after AAA repair is most likely suffering from?
- A ischemic colitis
 - B peptic ulcer
 - C Meckel's diverticulum
 - D diverticulosis of the colon
 - E colonic polyp
 - F ulcerative colitis
 - G carcinoma of colon
- 16** A 50-year-old alcoholic man with epigastric pain and tarry, foul-smelling stools most likely has?
- A ischemic colitis
 - B peptic ulcer
 - C Meckel's diverticulum
 - D diverticulosis of the colon
 - E colonic polyp
 - F ulcerative colitis
 - G carcinoma of colon

- 17** A 60-year-old previously healthy man with massive rectal bleeding most likely has?
- A** ischemic colitis
 - B** peptic ulcer
 - C** Meckel's diverticulum
 - D** diverticulosis of the colon
 - E** colonic polyp
 - F** ulcerative colitis
 - G** carcinoma of colon

- 18** A 75-year-old woman with new-onset constipation, small-caliber stools and intermittent dark red blood per rectum is most concerning for?
- A** ischemic colitis
 - B** peptic ulcer
 - C** Meckel's diverticulum
 - D** diverticulosis of the colon
 - E** colonic polyp
 - F** ulcerative colitis
 - G** carcinoma of colon

Answers 1-18

LANGE MCQS

- 1** **B aggressive resuscitation followed by surgical exploration** — Painless rectal bleeding in a young child due to Meckel's; stabilize first, then surgical resection of the diverticulum.
- 2** **A a bleeding Meckel's diverticulum** — Painless bleeding in a child is classic for Meckel's.
- 3** **C technetium scan** — Technetium pertechnetate scan detects ectopic gastric mucosa in Meckel's diverticulum.
- 4** **B IV fluid resuscitation, transfusion if needed, followed by laparotomy with Meckel's diverticulectomy and ileal resection** — Stabilize first, then surgical resection of the diverticulum and adjacent ileum.
- 5** **C esophagoscopy** — Endoscopy is needed to determine the source of upper GI bleeding in this context.
- 6** **E blood transfusion** — Symptomatic anemia with ischemia warrants blood transfusion.
- 7** **A diverticulosis of the colon** — **UPDATED** Sudden painless massive bright-red rectal bleeding in an older patient is most often diverticular.
- 8** **E aortoenteric fistula** — **UPDATED** A 'herald' GI bleed after aortic graft surgery is an aortoenteric fistula (usually 3rd/4th part of duodenum) until proven otherwise: CT angiography.
- 9** **G carcinoma of the colon** — **UPDATED** Occult bleeding with iron-deficiency anaemia in an older patient is colorectal (often right-sided) cancer until proven otherwise: colonoscopy.
- 10** **I gastric ulcer** — **UPDATED** Regular NSAIDs and alcohol cause gastric mucosal ulceration; a bleeding gastric ulcer is most likely.
- 11** **H esophageal varices** — **UPDATED** Stigmata of chronic liver disease with haematemesis and shock point to bleeding oesophageal varices.
- 12** **F peptic ulcer** — **UPDATED** Epigastric pain relieved by antacids followed by haematemesis and melaena is a bleeding peptic (usually duodenal) ulcer.
- 13** **B ulcerative colitis** — **UPDATED** Bloody diarrhoea with fever and weight loss in a young adult suggests ulcerative colitis.
- 14** **C Meckel's diverticulum** — Painless rectal bleeding in infants is most commonly due to Meckel's diverticulum.
- 15** **A ischemic colitis** — **UPDATED** Bloody diarrhoea days after AAA repair is ischaemic colitis from IMA ligation; confirm by sigmoidoscopy.
- 16** **B peptic ulcer** — **UPDATED** Melena in an alcoholic is most commonly due to a peptic ulcer.
- 17** **D diverticulosis of the colon** — **UPDATED** Massive rectal bleeding in the elderly is most often from diverticulosis.
- 18** **G carcinoma of colon** — **UPDATED** New constipation with narrow stools and dark-red bleeding in an elderly woman is left-sided colorectal cancer until proven otherwise.

7 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 A 50-year-old man presents to the outpatient clinic with an 8-week history of bleeding from his back end. This is typically bright red and copious during or following a stool. The patient has had no change in his bowel habit, no weight loss, and has no family history of bowel cancer. What is the most appropriate course of action?
 - A Perform rubber band ligation of haemorrhoids
 - B Perform injection sclerotherapy of haemorrhoids
 - C Blood tests including CEA
 - D Examination under anaesthetic +/- proceed to treat haemorrhoids
 - E Flexible sigmoidoscopy
- 2 An 85-year-old patient in shock with a short history of large-volume fresh rectal bleeding is resuscitated but no source is identified on examination. What is the next stage of management?
 - A Laparotomy
 - B Radionuclide red cell scanning
 - C Oesophagogastrroduodenoscopy
 - D Mesenteric angiography
 - E Colonoscopy
- 3 Diagnosis of rectal bleeding A 75-year-old man, who suffers from chronic atrial fibrillation, is admitted with a sudden onset of severe central colicky abdominal pain and vomiting. The patient has been bleeding from the rectum. The blood is dark in colour and has an altered consistency. On examination the patient is pale, has cold peripheries and is tachycardic. The abdomen is diffusely tender and bowel sounds are decreased. What is the likely diagnosis?
 - A Diverticulitis
 - B Small bowel obstruction
 - C Acute mesenteric ischaemia
 - D Perforated peptic ulcer
 - E None of the above
- 4 A 45-year-old man presents with a history of coffee-ground vomiting and dark stools. Which of the following is the most sensitive guide to the severity of his gastrointestinal haemorrhage?
 - A Haemoglobin
 - B Systolic blood pressure
 - C Pulse rate
 - D Volume of vomitus/melaena
 - E Lying and standing blood pressure
- 5 A 51-year-old cirrhotic patient, post-variceal banding, rebleeds with massive haematemesis and becomes haemodynamically compromised. What is the next step?
 - A Oesophageal transection
 - B Transjugular intrahepatic portal-systemic stent shunting
 - C Sengstaken-Blakemore tube
 - D Repeat endoscopy
 - E Angiographic arterial embolisation
- 6 A patient with a posteriorly positioned duodenal ulcer that is actively bleeding on endoscopy – which vessel is responsible?
 - A Gastroduodenal artery
 - B Abdominal aorta
 - C Right gastric artery
 - D Left gastric artery
 - E Right gastroepiploic artery
- 7 A patient, 48 hours after haemostasis of a bleeding gastric ulcer, is noted to have passed black stool despite being haemodynamically stable. What is the next best course of action?
 - A Urgent endoscopy
 - B Repeat full blood count
 - C Intravenous fluids and colloid
 - D Intravenous terlipressin
 - E Nothing, this is normal following upper gastrointestinal bleeding

Answers 1-7

SBA MCQs

- 1 **E Flexible sigmoidoscopy** — Haemorrhoids are likely but a left-sided colorectal carcinoma must be excluded; flexible sigmoidoscopy is recommended.
- 2 **D Mesenteric angiography** — UPDATED Massive lower GI bleeding with shock and no source on examination: after resuscitation (and OGD if upper source possible), CT/mesenteric angiography localises and allows embolisation.
- 3 **C Acute mesenteric ischaemia** — Presentation with rectal bleeding and shock suggests acute mesenteric ischaemia.
- 4 **E Lying and standing blood pressure** — Postural changes in blood pressure are the most sensitive indicator of blood loss.
- 5 **C Sengstaken-Blakemore tube** — The use of a Sengstaken-Blakemore tube is indicated to control variceal bleeding acutely.
- 6 **A Gastroduodenal artery** — Posterior erosion of a duodenal ulcer typically involves the gastroduodenal artery.

- 7 **B Repeat full blood count** — In a stable patient, a repeat full blood count is appropriate to assess for re-bleeding.

8 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 A 52-year-old man has several years' history of intermittent bright red blood on the toilet paper when wiping after defecation. There has been no mucus or change in bowel habit. Which is the single most appropriate first examination?
 - A Colonoscopy
 - B CT colonography
 - C Endoanal ultrasound
 - D Flexible sigmoidoscopy
 - E Proctoscopy/rigid sigmoidoscopy
- 2 A 73-year-old woman has had sudden onset of acute profuse rectal bleeding 4h ago. She has had no abdominal pain. The pulse rate is 105bpm, blood pressure of 95/50mmHg, capillary refill time is 5sec and oxygen saturations are 96% on supplemental oxygen via a non-re-breathing reservoir. Which is the single most appropriate first step in her management?
 - A Catheterize the patient
 - B Insert two large bore cannulae
 - C Organize flexible sigmoidoscopy
 - D Organize rigid sigmoidoscopy
 - E Organize mesenteric angiogram
- 3 48-year-old man has been vomiting fresh blood with clots for 3h. He drinks 40 units of alcohol per week. A gastroscopy is performed within 12h and reveals evidence of bleeding from oesophageal varices. Which is the single most likely cause of his varices?
 - A Alcoholic gastritis
 - B Budd–Chiari syndrome (hepatic vein thrombosis)
 - C Essential hypertension
 - D Pancreatitis
 - E Portal hypertension
- 4 43-year-old man has had multiple previous admissions for alcohol-related problems and has had 2h of acute vomiting with copious amounts of bright and dark red blood. An urgent OGD has shown the cause to be bleeding oesophageal varices which have been banded but the vomiting of blood continues on the ward. An anaesthetist is present who has established IV access and is administering blood and fluid transfusions. Which is the single most important next step in this patient's management?
 - A Take blood for clotting and LFTs
 - B Contact endoscopy to arrange a further endoscopy
 - C Contact X-ray to arrange a portable chest-ray
 - D Contact X-ray to arrange an emergency CT scan with embolization
 - E Contact your consultant to discuss transfer to theatre for treatment
- 5 50-year-old man has a 15-pack-year history of smoking and undergoes an OGD. There is a duodenal ulcer present. Gastric biopsies are taken and undergo a rapid urease test. Which single organism is most likely to be identified?
 - A Clostridium difficile
 - B Escherichia coli
 - C Enterobacter
 - D Helicobacter pylori
 - E Staphylococcus aureus
- 6 79-year-old man has an incidental finding of a haemoglobin of 9.1g/dL and mean corpuscular volume of 72fL with mild postprandial epigastric pain. Which is the single most appropriate first-line investigation?
 - A Barium meal
 - B CT scan thorax
 - C MRI upper abdomen
 - D OGD
 - E Ultrasound scan upper abdomen

- 7 69-year-old woman with rheumatoid arthritis has had 24h of epigastric pain, abdominal tenderness, and fever. She takes regular ibuprofen and has recently been taking oral steroids. An erect chest X-ray is taken in the emergency department (see Figure 6.2). Which is the single most likely diagnosis?



- A Acute pancreatitis
- B Bleeding duodenal ulcer
- C Incarcerated hiatus hernia
- D Perforated peptic ulcer
- E Right lower lobe pneumonia

- 8 79-year-old man has been having frequent dark black/bloody semi-liquid and foul smelling stools over the last 24h associated with vague generalized abdominal pain and bloating. His haemoglobin is 12.5g/dL, white blood cell count $11 \times 10^9/L$, and serum urea 18mmol/L. Which is the single most likely diagnosis?

- A Angiodysplasia of the stomach
- B Bleeding duodenal ulcer
- C Clostridium difficile colitis
- D Colonic carcinoma
- E Gastroenteritis

Answers 1-8

OXFORD MCQS

- 1 **E Proctoscopy/rigid sigmoidoscopy** — **UPDATED** Years of bright-red blood on the paper without bowel change suggests haemorrhoids: proctoscopy/rigid sigmoidoscopy in clinic first; colonoscopy if red flags.
- 2 **B Insert two large bore cannulae** — **UPDATED** Profuse rectal bleeding with shock: resuscitation comes first — two large-bore cannulae, bloods, cross-match and fluids/blood — before investigations.
- 3 **E Portal hypertension** — Increased portal venous pressure due to liver cirrhosis causes varices.
- 4 **E Contact your consultant to discuss transfer to theatre for treatment** — **UPDATED** Variceal bleeding continuing after banding needs immediate senior help for rescue therapy (balloon tamponade, repeat endoscopy or TIPS); bloods and imaging do not stop the bleeding.
- 5 **D Helicobacter pylori** — **UPDATED** A duodenal ulcer is usually associated with H. pylori, shown by a positive rapid urease (CLO) test on gastric biopsy.
- 6 **D OGD** — **UPDATED** Microcytic anaemia in a 79-year-old suggests occult GI blood loss: OGD (with colonoscopy/CT) to find the source, starting here with OGD given the epigastric pain.
- 7 **D Perforated peptic ulcer** — **UPDATED** Free gas under the diaphragm with epigastric pain in a patient on NSAIDs and steroids is a perforated peptic ulcer; steroids can mask peritonism.
- 8 **B Bleeding duodenal ulcer** — **UPDATED** Melaena with a raised urea (absorbed blood protein) points to an upper GI source; a bleeding duodenal ulcer is the commonest.

7 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 74-year-old woman is admitted with upper gastrointestinal (GI) bleeding. She is started on omeprazole after upper endoscopy shows diffuse gastric ulcerations. Which of the following is the mechanism of action of omeprazole?
 - A Blockage of the breakdown of mucosa-damaging metabolites of nonsteroidal anti-inflammatory drugs (NSAIDs)
 - B Provision of a direct cytoprotective effect
 - C Buffering of gastric acids
 - D Inhibition of parietal cell hydrogen potassium ATPase (adenosine triphosphatase)
 - E Inhibition of gastrin release and parietal cell acid production
- 2 A 60-year-old male patient with hepatitis C with a previous history of variceal bleeding is admitted to the hospital with hematemesis. His heart rate is 110 and blood pressure is 80/60 mm Hg, physical examination reveals splenomegaly and ascites, and initial hematocrit is 25%. Prior to endoscopy, which of the following is the best initial management of the patient?
 - A Administration of intravenous octreotide
 - B Administration of a β -blocker (eg, propranolol)
 - C Measurement of prothrombin time and transfusion with cryoglobulin if elevated
 - D Empiric transfusion of platelets given splenomegaly
 - E Gastric and esophageal balloon tamponade (Sengstaken-Blakemore tube)

- 3** A 32-year-old alcoholic with end-stage liver disease has been admitted to the hospital 3 times for bleeding esophageal varices. He has undergone banding and sclerotherapy previously. He admits to currently drinking a 6 pack of beer per day. On his abdominal examination, he has a fluid wave. Which of the following is the best option for long-term management of this patient's esophageal varices?
- A Orthotopic liver transplantation
 - B Transection and reanastomosis of the distal esophagus
 - C Distal splenorenal shunt
 - D End-to-side portocaval shunt
 - E Transjugular intrahepatic portosystemic shunt (TIPS)
- 4** A 57-year-old previously alcoholic man with a history of chronic pancreatitis presents with hematemesis. Endoscopy reveals isolated gastric varices in the absence of esophageal varices. His liver function tests are normal and he has no stigmata of end-stage liver disease. Ultrasound examination demonstrates normal portal flow but a thrombosed splenic vein. He undergoes banding, which is initially successful, but he subsequently rebleeds during the same hospitalization. Attempts to control the bleeding endoscopically are unsuccessful. Which of the following is the most appropriate next step in management?
- A Transjugular intrahepatic portosystemic shunt
 - B Surgical portocaval shunt
 - C Surgical mesocaval shunt
 - D Splenectomy
 - E Placement of a tube for esophageal balloon tamponade
- 5** A 28-year-old woman presents with hematochezia. She is admitted to the hospital and undergoes upper endoscopy that is negative for any lesions. Colonoscopy is performed and no bleeding sources are identified, although blood is noted in the right colon and old blood coming from the ileocecal valve. Which of the following is the test of choice in this patient?
- A Angiography
 - B Small-bowel enteroclysis
 - C CT scan of the abdomen
 - D Technetium 99m (99mTc) pertechnetate scan
 - E Small-bowel endoscopy
- 6** A 54-year-old man presents with sudden onset of massive, painless, recurrent hematemesis. Upper endoscopy is performed and reveals bleeding from a lesion in the proximal stomach that is characterized as an abnormally large artery surrounded by normal-appearing gastric mucosa. Endoscopic modalities fail to stop the bleeding. Which of the following is the most appropriate surgical management of this patient?
- A Wedge resection of the lesion
 - B Wedge resection of the lesion with truncal vagotomy
 - C Wedge resection of the lesion with highly selective vagotomy
 - D Wedge resection of the lesion with truncal vagotomy and antrectomy
 - E Subtotal gastrectomy
- 7** A 61-year-old woman with a history of unstable angina complains of hematemesis after retching and vomiting following a night of binge drinking. Endoscopy reveals a longitudinal mucosal tear at the gastroesophageal junction, which is not actively bleeding. Which of the following is the next recommended step in the management of this patient?
- A Angiography with embolization
 - B Balloon tamponade
 - C Exploratory laparotomy, gastrotomy, and oversewing of the tear
 - D Systemic vasopressin infusion
 - E Expectant management

Answers 1-7

PRETEST MCQs

- 1** **D Inhibition of parietal cell hydrogen potassium ATPase (adenosine triphosphatase)** — Omeprazole irreversibly inhibits the proton pump, blocking the final step in acid secretion.
- 2** **A Administration of intravenous octreotide** — Initial management of variceal bleed includes volume resuscitation and octreotide administration.
- 3** **E Transjugular intrahepatic portosystemic shunt (TIPS)** — TIPS is the best long-term management option for recurrent variceal bleeding in liver disease.
- 4** **D Splenectomy** — Splenectomy is indicated for acute hemorrhage due to left-sided portal hypertension from splenic vein thrombosis.
- 5** **D Technetium 99m (99mTc) pertechnetate scan** — The pertechnetate scan is the investigation of choice for detecting Meckel diverticulum in the setting of GI bleeding.
- 6** **A Wedge resection of the lesion** — A Dieulafoy lesion causing massive bleeding is best managed with a wedge resection.
- 7** **E Expectant management** — Mallory-Weiss tears typically resolve spontaneously and are managed expectantly.

4 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 56-year-old man is brought to hospital by his wife. He has been vomiting fresh blood since earlier that morning. She tells you he has a long history of alcohol abuse and drinks at least a bottle of spirits a day. Which of the following is the most likely cause of his symptoms?
 - A Dieulafoy lesions
 - B Duodenal ulcer
 - C Gastric ulcer
 - D Oesophageal varices
 - E Oesophagitis
- 2 A 34-year-old man has presented to the emergency department following a large haematemesis. On examination, the patient appears agitated. His pulse is 120/min, blood pressure 122/84 mmHg and respiratory rate 22/min. How would you classify the patient's current condition?
 - A Class I haemorrhagic shock
 - B Class II haemorrhagic shock
 - C Class III haemorrhagic shock
 - D Class IV haemorrhagic shock
 - E Class V haemorrhagic shock
- 3 A 74-year-old woman presents to the emergency department complaining of dizziness and tiredness. On direct questioning she tells you she has recently been having black stools. She is currently awaiting a total hip replacement for arthritis. Which of the following would be most useful in establishing the primary cause of her symptoms?
 - A Barium meal
 - B Coagulation screen
 - C Faecal occult blood
 - D Full blood count
 - E Oesophagogastroduodenoscopy
- 4 A 27-year-old man is admitted to the high dependency ward having sustained a significant head injury. Two days later he starts vomiting blood. Which of the following is the most likely cause?
 - A Curling ulcer
 - B Cushing ulcer
 - C Decubitus ulcer
 - D Dendritic ulcer
 - E Martorell's ulcer

Answers 1-4

GET AHEAD MCQS

- 1 **D Oesophageal varices** — Longstanding alcohol abuse often causes cirrhosis leading to portal hypertension and varices.
- 2 **B Class II haemorrhagic shock** — Class II shock involves 15–30% blood loss with tachycardia and near-normal blood pressure.
- 3 **E Oesophagogastroduodenoscopy** — OGD directly visualizes the source of upper GI bleeding and allows therapeutic intervention.
- 4 **B Cushing ulcer** — Stress ulcer from raised intracranial pressure causing gastrin hypersecretion and GI bleeding.

1 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 An elderly man has left-sided abdominal pain and dark red rectal bleeding. He smokes 40 a day for 30 years, BP 105/65, pulse 130, no fever. Distended abdomen. Most likely diagnosis?
 - A Lymphocytic colitis
 - B Ulcerative colitis
 - C Ischaemic colitis
 - D Diverticulitis
 - E Microscopic colitis

Answer 1

CRASH COURSE MCQS

- 1 **C** Ischaemic colitis — Left-sided pain, dark red bleed, smoking, hypotension => ischaemic colitis.

Gastrointestinal Tumours

Oesophageal, gastric, small-bowel, colorectal and anal tumours: polyps, polyposis syndromes, staging and treatment.

82

MCQS

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

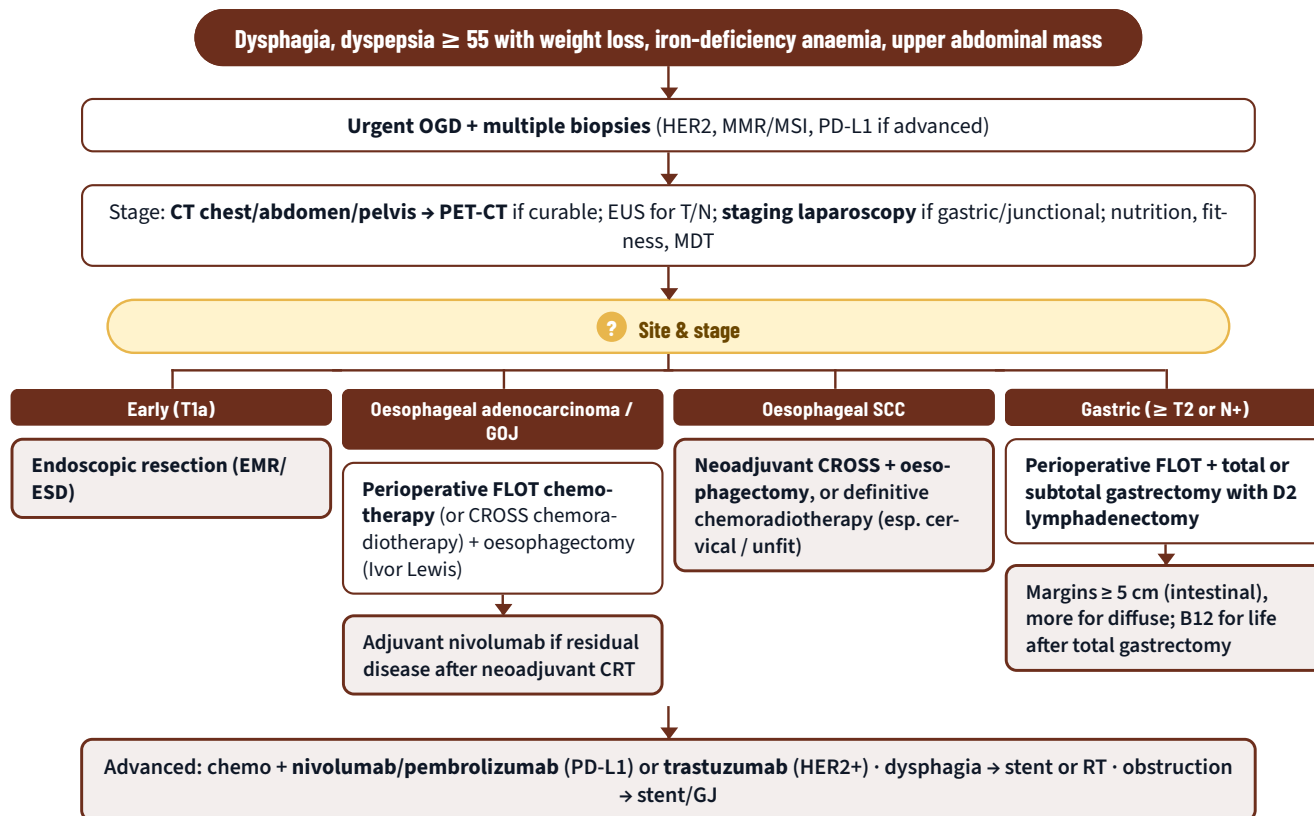
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Step by step

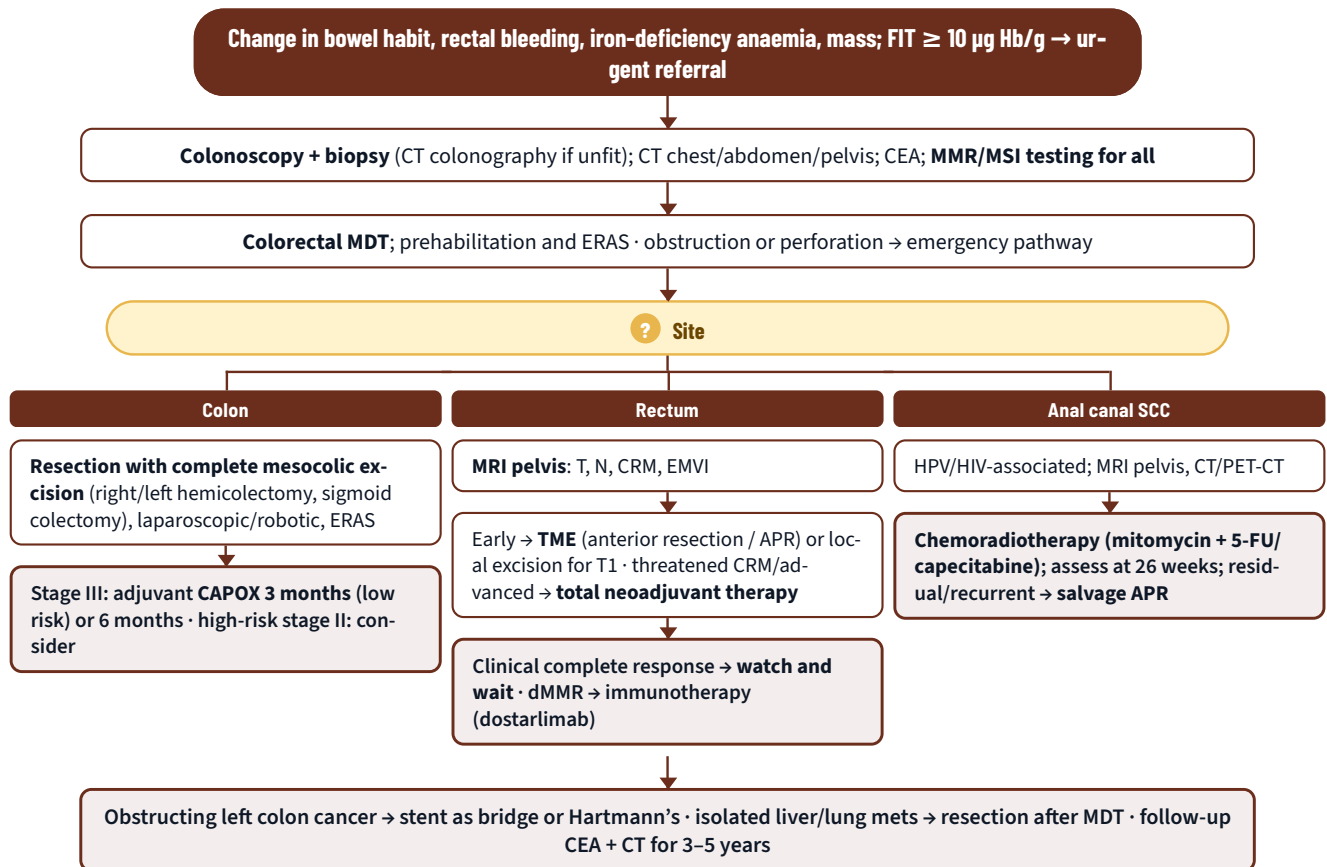
Oesophageal and gastric cancer

NICE NG83 • ESMO 2022 GASTRIC / 2023 OESOPHAGEAL • ESOPEC 2024



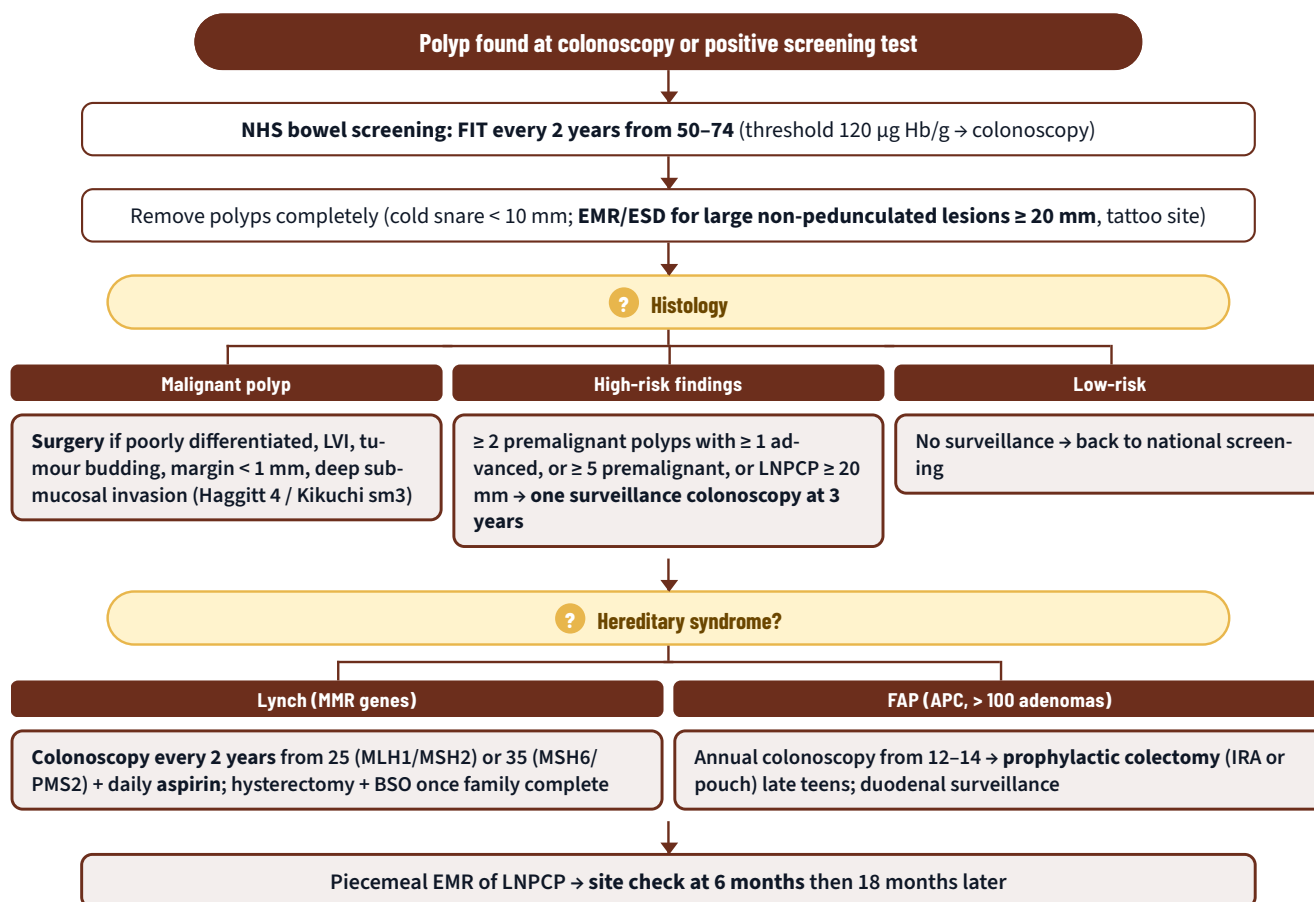
Colorectal and anal cancer

NICE NG151 · ESMO 2023 COLON / 2017 RECTAL · ESMO 2021 ANAL



Polyps, screening and hereditary syndromes

BSG/ACPGBI/PHE 2020 POST-POLYPECTOMY · BSG/ACPGBI 2020 HEREDITARY CRC

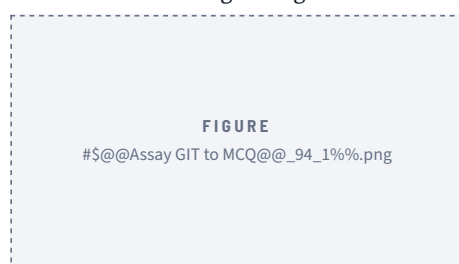


Essay to MCQ MCQs

1 What is the lifelong therapy that should be given to this patient?

- A Iron supplements
- B Calcium supplements
- C B12 vitamin supplements
- D Vitamin D supplements
- E No lifelong therapy is necessary

2 A 39-year-old female presents with generalized fatigue, weight loss over 11 kg, and episodes of maroon-colored stool. DRE reveals large hemorrhoids. CT scan shows a filling defect in the cecum. What is this study? Mention the radiological signs.



- A Ultrasound
- B Plain abdominal X-ray
- C Abdominal CT scan with IV contrast revealing a filling defect in the cecum
- D MRI of the abdomen
- E Barium enema

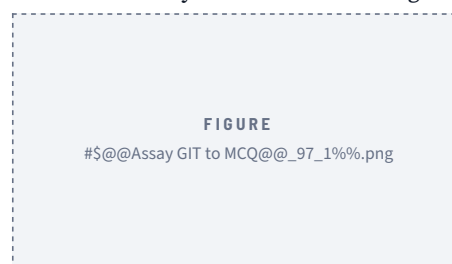
3 Suggest the next step to reach the diagnosis.

- A Observation
- B Colonoscopy & biopsy
- C Laparoscopy
- D Repeat CT scan
- E Endoscopic ultrasound

4 What kind of anaemia is seen in the above scenario? Explain.

- A Iron deficiency anemia
- B Hemolytic anemia
- C Megaloblastic anemia (macrocytic) due to side effect of metformin therapy
- D Aplastic anemia
- E Anemia of chronic disease

5 Seven years later, on a regular checkup for poor appetite, he is found to be mildly icteric. Workup shows: What is the study? Mention the findings.



- A Abdominal ultrasound showing bile duct dilation
- B CT scan of the abdomen
- C PET scan of the liver showing multiple foci of increased FDG uptake indicating liver metastasis
- D MRCP
- E Endoscopic ultrasound

6 Is ERCP beneficial for this patient? If Yes, Explain.

- A Yes, for stone removal
- B Yes, to confirm the diagnosis
- C No, ERCP is primarily used for biliary tract disorders, although it may detect biliary metastasis and obstruction
- D No, because it is contraindicated
- E Yes, for therapeutic stenting

Answers 1-6

1 **C B12 vitamin supplements** — **UPDATED** Following total gastrectomy, lifelong B12 supplementation is necessary due to malabsorption.

2 **C Abdominal CT scan with IV contrast revealing a filling defect in the cecum** — **UPDATED** The CT scan shows a filling defect in the cecum, raising suspicion for a mass.

3 **B Colonoscopy & biopsy** — **UPDATED** Colonoscopy with biopsy is the definitive step for tissue diagnosis.

4 **C Megaloblastic anemia (macrocytic) due to side effect of metformin therapy** — **UPDATED** The high MCV indicates megaloblastic anemia, a known side effect of metformin.

5 **C PET scan of the liver showing multiple foci of increased FDG uptake indicating liver metastasis** — **UPDATED** The PET scan reveals hypermetabolic foci suggestive of liver metastasis.

6 **C No, ERCP is primarily used for biliary tract disorders, although it may detect biliary metastasis and obstruction** — **UPDATED** ERCP is not particularly beneficial in this case of metastatic disease.

Bailey & Love MCQs

- 1 Which of the following statements is false?
 - A Early disease is best staged by endoscopic US (EUS).
 - B Accurate preoperative nodal involvement is possible by EUS.
 - C Liver and lung metastases are best detected by US and CT.
 - D Intraoperative US is used in liver resection.
 - E PET/CT is an important tool in oncological staging.
- 2 Regarding to Tumours of the small intestine, which of the following statement(s) are not true?
 - A Small bowel tumours account for 10% of all gastrointestinal neoplasms.
 - B Small bowel adenocarcinoma is more common in patients with Crohn's disease.
 - C Carcinoid tumour of the appendix does not give rise to carcinoid syndrome.
 - D Hodgkin's lymphoma is the most common type of lymphoma to affect the small bowel.
 - E The ileum is the most common site for GIST (gastrointestinal stromal tumour).
- 3 Which of the following statements are true of carcinoid tumour?
 - A Appendicular carcinoid are most commonly found incidentally.
 - B The substance that causes the syndrome is 5-hydroxytryptamine (serotonin).
 - C The syndrome can affect the heart in the form of mitral stenosis.
 - D Tumours in the ileum and rectum might metastasise to the liver long after removal of the primary tumour.
 - E The syndrome produces features affecting the pulmonary, cardiovascular and gastrointestinal systems.
- 4 Which of the following statements are true about familial adenomatous polyposis (FAP)?
 - A More than 80% of patients have a positive family history.
 - B The condition is inherited as an autosomal recessive condition.
 - C Family members should be screened until the age of 50 years.
 - D Restorative proctocolectomy with an ileal pouch is the preferred surgical treatment.
 - E Hereditary non-polyposis colorectal cancer (HNPCC) occurs as an autosomal dominant condition.
- 5 Regarding to Tumours of the large intestine, which of the following statements are not true?
 - A Colonic adenomas occur more on the right and carcinomas in the left colon.
 - B Sporadic adenomas are different from those of FAP.
 - C The larger the adenoma the greater the chance of dysplasia.
 - D Colorectal cancer incidence falls with screening programmes.
 - E Mutations of the APC gene occur in two-thirds of colonic adenomas and carcinomas.
- 6 Which of the following statements are true?
 - A Almost 60% occur in the rectosigmoid region.
 - B At the time of diagnosis, one-third will have liver metastases.
 - C 20% of patients present as an emergency with intestinal obstruction.
 - D Reduced dietary fibre is associated with an increased risk.
 - E All of the above.
- 7 Which one of the following statements is false?
 - A Right colonic cancers present with features of anaemia.
 - B Left colonic cancers present with rectal bleeding and obstructive symptoms.
 - C Even for an experienced colonoscopist, the failure rate to visualise the caecum is 10%.
 - D Intravenous urography (IVU) should be routinely done.
 - E Synchronous cancers occur in 5%.
- 8 Which of the following statements are true?
 - A Thorough preoperative assessment and staging should be done with colonoscopy, US and spiral CT.
 - B Resection is not done if the patient has liver metastases.
 - C If, at operation, hepatic metastases are found, biopsy should be done.
 - D Hepatic resection for metastases should be considered as a staged procedure.
 - E Left colonic carcinomas presenting with intestinal obstruction should be treated as a staged procedure.
- 9 The following statements are true except:
 - A Routine follow up after surgery for colorectal cancer is a waste of resources.
 - B In resectable hepatic metastases from colorectal cancer, the 5-year survival is 30%.
 - C In Dukes' stage A, the 5-year disease-free survival is 90%.
 - D Enhanced recovery programme (ERP) reduces hospital stay considerably.
 - E Preoperative chemotherapy in colonic cancer has a role in selected patients.

- 10** Which of the following randomised controlled trials (RCTs) demonstrate the benefits of laparoscopic surgery over traditional surgery for colorectal cancer?
- A COST.
 - B COLOR 1 and 2.
 - C CLASICC.
 - D LAFA.
 - E DREAMS.
- 11** Which of the following statements regarding laparoscopic colorectal surgery is not true?
- A Decreased surgical site infection with minimal access technique.
 - B Conversion rate of >10% is considered a good outcome.
 - C Decreased length of stay in the hospital.
 - D Decreased long-term survival.
 - E NICE recommends that cancer should be suitable for laparoscopic resection and that the surgeon should be experienced in both open and laparoscopic techniques.
- 12** Regarding to Causes and management of cancer, which of the following statements are true regarding characteristics that cells should acquire before they are fully malignant?
- A Establish an autonomous lineage
 - B Develop apoptosis
 - C Ability to invade
 - D Ability to disseminate and implant
 - E Ability to change energy metabolism
- 13** The following statements are true about cancer cells except:
- A Obtain immortality
 - B Acquire angiogenic competence
 - C Acquire genomic stability
 - D Tumour-related inflammation promotes malignant change
 - E The major part of cell growth has occurred before the tumour is clinically apparent
- 14** Which of the following statements are true?
- A Environmental factors have been the cause in >80% of cancers
 - B Genetic inheritance has a role in 20% of cancers
 - C Certain cancers can be prevented
 - D Screening aims at detecting disease in an asymptomatic stage
 - E Accurate diagnosis is the key to successful management
- 15** Regarding principles of cancer surgery, which of the following statements are not true?
- A The extent of the disease, local and distant, should be mapped by staging
 - B Once diagnosed, the cancer should be radically excised at the earliest
 - C Non-surgical treatment of cancer can be expressed in reproducible units, unlike surgery
 - D Radiotherapy works by damage to the DNA of cells and targeting gene expression
 - E Chemotherapy acts by selective toxicity
- 16** Which of the following statements are true?
- A Ultraradical surgery for local control of cancer has little effect on metastatic disease
 - B Surgery for metastatic disease is appropriate in certain circumstances
 - C Overall treatment of cancer consists of surgery, chemotherapy and radiotherapy
 - D The goal of palliative treatment is the relief of symptoms
 - E Palliative care is the same as end-of-life care

Answers 1-16

- 1** **B** **Accurate preoperative nodal involvement is possible by EUS.** — **UPDATED** EUS cannot reliably distinguish malignant from reactive nodal enlargement.
- 2** **DE** **True: D, E** — **UPDATED** Hodgkin's lymphoma is not common in the small bowel and GISTs more typically arise in the stomach.
- 3** **ABDE** **True: A, B, D, E** — **UPDATED** Carcinoid tumours are often incidentally discovered, secrete serotonin, may metastasise and produce multisystem effects; mitral stenosis is not typical.
- 4** **ACDE** **True: A, C, D, E** — **UPDATED** FAP is autosomal dominant with high positive family history; screening and pouch procedures are as stated while B is false.
- 5** **AB** **True: A, B** — **UPDATED** Adenomas and carcinomas have a similar distribution and sporadic adenomas are histologically similar to those in FAP.
- 6** **E** **All of the above.** — **UPDATED** All statements correctly describe clinical features of colorectal cancer.
- 7** **D** **Intravenous urography (IVU) should be routinely done.** — **UPDATED** Routine IVU is not indicated in colorectal cancer workup.
- 8** **ADE** **True: A, D, E** — **UPDATED** Preoperative staging is critical; liver metastases may be managed by staged resection and emergency left-sided cancers often require staged procedures.
- 9** **AE** **True: A, E** — **UPDATED** Follow-up is essential and preoperative chemotherapy is not standard in most cases.
- 10** **ABCD** **True: A, B, C, D** — **UPDATED** The COST, COLOR, CLASICC and LAFA trials have demonstrated the benefits; DREAMS focused on postoperative nausea.
- 11** **BD** **True: B, D** — **UPDATED** A high conversion rate is not acceptable and there is no evidence of reduced long-term survival with laparoscopy.

- 12** **ACDE** True: A, C, D, E — **UPDATED** Cells must acquire autonomy, invasiveness, dissemination and metabolic changes; they typically evade apoptosis.
- 13** **C** Acquire genomic stability — **UPDATED** Cancer cells are characterized by genomic instability; the other statements are true.
- 14** **ABCDE** True: A, B, C, D, E — **UPDATED** All are true: most cancers have environmental causes, inheritance contributes in a

minority, some are preventable and screening detects asymptomatic disease.

- 15** **B** Once diagnosed, the cancer should be radically excised at the earliest — **UPDATED** Radical excision without proper staging is not advised; the other statements are valid.
- 16** **ABCD** True: A, B, C, D — **UPDATED** All except E are correct; palliative care is broader than only end-of-life care.

15 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 55-year-old asymptomatic man has guaiac-positive stool. Colonoscopy shows a 3-cm pedunculated sigmoid polyp, removed by snare polypectomy. Histology shows an adenoma with a focus of invasive carcinoma in the head, a 4-mm resection margin and no tumour in the stalk. What is the most appropriate next step?
- A** CT scan
B MRI scan
C Surgical resection of the sigmoid colon
D Observation with colonoscopic surveillance
E Regular NSAID use
- 2** A 65-year-old man presents for his yearly physical examination with early fatigue and trace positive stool occult blood. Barium enema shows an “apple core” lesion. Which of the following is the most likely diagnosis?

FIGURE
 #lange_53_1%.png

- A** diverticular disease
B colon cancer
C lymphoma
D ischemia with stricture
E Crohn's colitis with stricture
- 3** Following colonoscopy in a patient with an “apple core” lesion, what is the most appropriate therapy?
- A** proximal colostomy with mucous fistula
B radiation therapy
C chemotherapy
D surgical resection and primary anastomosis
E surgical bypass (colocolostomy)

- 4** A 54-year-old woman presents for an opinion regarding additional therapy following curative sigmoid resection for colon cancer with one positive node. Which of the following is the correct stage of her colon cancer?

A stage 0
B stage I
C stage II
D stage III
E stage IV

- 5** For the same patient, what is the most appropriate recommendation regarding adjuvant therapy?

A no therapy indicated
B 5-fluorouracil chemotherapy
C 5-fluorouracil chemotherapy with leucovorin
D doxorubicin (Adriamycin) chemotherapy
E Adriamycin chemotherapy with methotrexate and cytoxan

- 6** A 62-year-old woman presents with a 6-month history of worsening constipation, pencil-thin stool, and occasional blood. A barium enema shows a near-obstructing rectal lesion with an “apple core” appearance. Which is the most likely diagnosis?

FIGURE
 #lange_57_1%.png

A Crohn's disease
B ischemia with stricture
C rectal carcinoma
D sigmoid volvulus
E diverticulitis with colovesical fistula

- 7** What is the most appropriate next step in management for the rectal lesion described above?
- A proctoscopy and passage of a rectal tube
 - B proctoscopy and biopsy
 - C colonoscopy
 - D endoscopic dilation of the stricture
 - E NPO, IV fluids, and antibiotics
- 8** A 45-year-old man presents with a 2-year history of dysphagia and retrosternal pressure. Esophagogram reveals a 6-cm, smooth, concave defect in the midesophagus with sharp borders and intact mucosa. What is the most likely diagnosis?
- A esophageal carcinoma
 - B bronchogenic carcinoma with invasion of the esophagus
 - C benign esophageal polyp
 - D leiomyoma
 - E lymphoma
- 9** What is the most appropriate next step for the esophageal lesion described above?
- A repeat esophagoscopy with biopsy
 - B thoracotomy with extramucosal resection
 - C thoracotomy with esophageal resection
 - D radiation therapy
 - E chemotherapy
- 10** A 68-year-old man presents with progressive dysphagia, mild chest discomfort, weight loss, and a 30 pack-year smoking history. A barium esophagogram shows an irregular filling defect in the distal esophagus. Which of the following is the most likely diagnosis?
- A esophagitis with stricture
 - B esophageal carcinoma
 - C lung carcinoma with invasion into the esophagus
 - D lymphoma
 - E achalasia
- 11** What is the most appropriate next step in management for the esophageal lesion described above?
- A CT scan
 - B esophagoscopy
 - C MRI scan
 - D surgical resection
 - E bronchoscopy
- 12** A 65-year-old man with a 40 pack-year smoking history and a history of colon cancer 3 years ago now presents with an elevated CEA and minimally elevated alkaline phosphatase. Which is the most appropriate next diagnostic test?
- A positron emission tomography (PET) scan
 - B radionuclide liver scan
 - C ultrasound
 - D CT scan
 - E MRI scan
- 13** CT imaging in the patient described above reveals three lesions in the right hepatic lobe, each 3–4 cm, with no extrahepatic disease. What is the most appropriate next step in management?
- A systemic chemotherapy
 - B intra-arterial chemotherapy through the hepatic artery
 - C surgical resection
 - D radiation therapy to the liver
 - E repeat imaging studies in 3 months
- 14** In discussing metastatic colon cancer resection, what 5-year survival rate should be quoted?
- A 5–10%
 - B 15–20%
 - C 25–35%
 - D 40–50%
 - E 60–70%
- 15** A 60-year-old previously healthy woman scheduled for elective colon resection for a nonobstructing sigmoid colon carcinoma undergoes an operation with gastrointestinal spillage. This operative procedure would be classified as?
- A class I clean
 - B class II clean-contaminated
 - C class II clean-contaminated with inadequate bowel preparation and gastrointestinal spillage
 - D class III contaminated
 - E class IV dirty

Answers 1–15

- 1** **D** **Observation with colonoscopic surveillance** — **UPDATED** A malignant pedunculated polyp confined to the head (Haggitt 1–2) with a clear margin (>1–2 mm), well differentiated and no lymphovascular invasion is cured by polypectomy: surveillance.
- 2** **B** **colon cancer** — **UPDATED** An “apple core” lesion on barium enema is classic for colon cancer.
- 3** **D** **surgical resection and primary anastomosis** — **UPDATED** Definitive management of colon cancer is surgical resection with primary anastomosis.
- 4** **D** **stage III** — **UPDATED** One positive node indicates stage III disease.
- 5** **C** **5-fluorouracil chemotherapy with leucovorin** — **UPDATED** Adjuvant 5-FU with leucovorin is indicated in stage III colon cancer.
- 6** **C** **rectal carcinoma** — **UPDATED** The history and radiographic “apple core” lesion are consistent with rectal carcinoma.
- 7** **B** **proctoscopy and biopsy** — **UPDATED** Tissue diagnosis is essential; proctoscopy with biopsy is indicated.

- 8** **D leiomyoma** — **UPDATED** Leiomyoma is the most common benign tumor of the esophagus presenting with these features.
- 9** **B thoracotomy with extramucosal resection** — **UPDATED** Excision of symptomatic leiomyoma is indicated without prior biopsy to avoid mucosal disruption.
- 10** **B esophageal carcinoma** — **UPDATED** The irregular filling defect with luminal distortion is most suggestive of esophageal cancer.
- 11** **B esophagoscopy** — **UPDATED** Endoscopy with biopsy is mandatory for definitive diagnosis.
- 12** **D CT scan** — **UPDATED** CT scan is the optimal imaging to evaluate for hepatic and extrahepatic metastases.
- 13** **C surgical resection** — **UPDATED** Surgical resection is the treatment of choice for isolated liver metastases from colon cancer.
- 14** **C 25–35%** — **UPDATED** After resection of colorectal liver metastases, 5-year survival is about 25–35% (up to ~40% in modern series).
- 15** **D class III contaminated** — **UPDATED** Inadequate bowel prep with spillage converts the procedure to class III contaminated.

10 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 60-year-old patient being treated for colonic carcinoma has had the primary lesion excised and chemotherapy started. However, CT scanning identifies a 1 cm metastasis in the right lobe of the liver. With no history of alcohol misuse or viral hepatitis and normal CT chest and brain, what is the most appropriate next stage of management?
- A** Liver resection
B Gadolinium-enhanced liver magnetic resonance imaging
C CT/ultrasound-guided biopsy
D Referral to a palliative care setting with counselling and support
E Monitor lesion – If size exceeds 1.5 cm, add irinotecan to 5-fluorouracil and folinic acid chemotherapy
- 2** On colonoscopy a malignant lesion is identified 5 cm proximal to the splenic flexure. With no contraindications to resection and a decision for curative intent, which procedure is most appropriate?
- A** Right hemicolectomy
B Total colectomy
C Sigmoid colectomy
D Anterior resection
E Extended right hemicolectomy
- 3** An 85-year-old female, who has not passed stool for 4 days, presents with abdominal discomfort, distension, increased bowel sounds, a firm palpable mass in the left iliac fossa and an empty rectum on digital examination. What diagnosis must be excluded?
- A** Simple constipation
B Paralytic ileus
C Sigmoid volvulus
D Peritonitis secondary to diverticular disease
E Neoplasia
- 4** Which of the following is not a syndrome associated with an increased risk of colonic adenocarcinoma?
- A** Congenital hypertrophy of retinal pigment epithelium
B Familial adenomatous polyposis coli
C Hereditary non-polyposis colon cancer
D Multiple endocrine neoplasia syndrome II
E Gardner's syndrome
- 5** Which of the following is the most common site for colorectal adenocarcinoma?
- A** Rectum
B Sigmoid colon
C Caecum and ascending colon
D Descending colon
E Transverse colon
- 6** Which of the following is not an indication for radiotherapy or chemotherapy in colonic cancer?
- A** Dukes' stage A
B Intraoperative peritoneal contamination during resection
C Dukes' stage C
D Rectal adenocarcinoma threatening the resection margin
E Dukes' stage D
- 7** Carcinoid tumours occur most commonly in which site?
- A** Large bowel
B Duodenum
C Appendix
D Stomach
E Lung
- 8** Of the following, which is not a risk factor for gastric cancer?
- A** Pernicious anaemia
B Helicobacter pylori
C Partial gastrectomy
D Blood group O
E Dried fish

- 9** Which of the following investigations are not used in staging of oesophageal malignancy?
- A Mediastinoscopy
 - B High-resolution computed tomography scanning
 - C Endoscopy
 - D Endoluminal ultrasound
 - E Laparoscopy
- 10** Which one of the following is not a recognized long-term complication of partial/total gastrectomy?
- A Gastric malignancy
 - B Obstruction
 - C Folate deficiency
 - D Iron deficiency
 - E Vitamin B1 deficiency

Answers 1-10

SBA MCQS

- 1** **B Gadolinium-enhanced liver magnetic resonance imaging** — **UPDATED** MRI is needed to assess the liver lesion before planning further management.
- 2** **E Extended right hemicolectomy** — **UPDATED** Extended right hemicolectomy achieves clear margins and is preferred for tumours near the splenic flexure.
- 3** **E Neoplasia** — **UPDATED** The presence of a palpable mass mandates exclusion of neoplasia despite features of simple constipation.
- 4** **D Multiple endocrine neoplasia syndrome II** — **UPDATED** MEN syndrome II is not linked with an increased risk of colonic adenocarcinoma.
- 5** **A Rectum** — **UPDATED** The rectum is the most common site due to prolonged exposure to carcinogens.
- 6** **A Dukes' stage A** — **UPDATED** Dukes' stage A disease does not benefit from adjuvant therapy.
- 7** **C Appendix** — **UPDATED** The appendix is the most common site for carcinoid tumours.
- 8** **D Blood group O** — **UPDATED** Blood group O is not a risk factor, unlike the other listed factors.
- 9** **C Endoscopy** — **UPDATED** Endoscopy is used for diagnosis but not for staging of oesophageal cancer.
- 10** **E Vitamin B1 deficiency** — **UPDATED** Vitamin B1 deficiency is not typically seen after gastrectomy.

4 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** A 24-year-old female undergoes a colonoscopy and is found to have multiple adenomatous polyps in the colon and rectum. Which is the single most likely genetic disorder to explain these findings?
- A Gardner's syndrome
 - B Hereditary juvenile polyposis
 - C Lynch syndrome
 - D Peutz-Jeghers syndrome
 - E Sipple's syndrome
- 2** An isolated abnormality is seen in the descending colon during a colonoscopy to investigate 5 months of change in bowel habit and blood mixed with stools in a 72-year-old man. There is no family history of colorectal disease and he is otherwise fit and well. Biopsies are taken and the colonoscopy completed successfully. Which is the single most appropriate investigation that should be arranged next?
- A Chest X-ray
 - B CT scan of chest, abdomen, and pelvis
 - C Liver ultrasound scan
 - D MRI scan of the pelvis
 - E Whole body CT positron emission tomography scan
- 3** 64-year-old woman undergoes an oesophago-gastroduodenoscopy (OGD) to investigate difficulties and discomfort with swallowing. A tumour is found in the distal oesophagus. Biopsies are taken. Which is the single most likely histological diagnosis?
- A Adenocarcinoma
 - B GI stromal tumour
 - C Lipoma
 - D Rhabdomyosarcoma
 - E Squamous cell carcinoma
- 4** 67-year-old man is found to have adenocarcinoma of the stomach. Which single blood group is this pathology associated with?
- A A
 - B AB
 - C B
 - D O
 - E None of these

Answers 1-4

- 1 **A Gardner's syndrome** — UPDATED Multiple colorectal adenomas in a young woman suggest FAP; Gardner syndrome is its variant with osteomas and soft-tissue tumours. Peutz-Jeghers and juvenile polyps are hamartomas.
- 2 **B CT scan of chest, abdomen, and pelvis** — UPDATED A colonic carcinoma on colonoscopy is staged with CT of chest, abdomen and pelvis to look for liver and lung metastases before planning resection.
- 3 **A Adenocarcinoma** — UPDATED Adenocarcinoma, arising in Barrett mucosa, is now the commonest oesophageal cancer in the West and typically lies in the distal oesophagus.
- 4 **A A** — UPDATED Gastric adenocarcinoma is associated with blood group A, H. pylori, atrophic gastritis, smoking and nitrosamine-rich diets.

15 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 45-year-old man was discovered to have a hepatic flexure colon cancer during a colonoscopy for anemia requiring transfusions. Upon exploration of his abdomen in the operating room, an unexpected discontinuous 3-cm metastasis is discovered in the edge of the right lobe of the liver. Pre-operatively, the patient was counseled of this possibility and the surgical options. Which of the following is the most appropriate management of this patient?
 - A Diverting ileostomy
 - B Right hemicolectomy
 - C Right hemicolectomy with local resection of the liver metastasis
 - D Closure of the abdomen followed by chemotherapy
 - E Right hemicolectomy with postoperative radiation therapy to the liver
- 2 A 60-year-old female comes in with 2 months of vague abdominal pain and weight loss. CT demonstrates thickening of the distal stomach and no other abnormalities. EGD shows a friable, ulcerated mass in the antrum of the stomach, biopsied as poorly differentiated adenocarcinoma with signet ring features. CT chest and PET scan are negative for distant disease. What is the next step in the workup of this patient?
 - A Carcinoembryonic antigen (CEA) level
 - B Endoscopic ultrasound (EUS)
 - C Referral to genetic counselor
 - D Diagnostic laparoscopy
 - E MRI abdomen
- 3 A 62-year-old man has been diagnosed by endoscopic biopsy as having a sigmoid colon cancer. He is otherwise healthy and presents to your office for preoperative consultation. He asks a number of questions regarding removal of a portion of his colon. Which of the following is most likely to occur after a colon resection?
 - A Chronic constipation
 - B Little long-term change in bowel habits
 - C Secretory diarrhea
 - D Osmotic diarrhea
 - E Steatorrhea
- 4 After complete removal of a 2-cm sessile polyp 1 finger length above the anal margin, the pathologist reports it to have been a villous adenoma that contained carcinoma in situ (negative margins). Which of the following is the most appropriate next step in management?
 - A Re-excision of the biopsy site with wider margins
 - B Abdominoperineal resection
 - C Low anterior resection of the rectum
 - D External radiation therapy to the rectum
 - E No further therapy
- 5 A 57-year-old woman sees blood on the toilet paper. Her doctor notes the presence of an excoriated bleeding 3-cm mass at the anus. Biopsy confirms the clinical suspicion of anal cancer. In planning the management of squamous cell carcinoma of the anus, which of the following is the best initial management strategy?
 - A Abdominoperineal resection
 - B Wide local resection with bilateral inguinal node dissection
 - C Local radiation therapy
 - D Systemic chemotherapy
 - E Combined radiation therapy and chemotherapy

- 6** An 80-year-old man is admitted to the hospital complaining of nausea, abdominal pain, distention, and diarrhea. CT scan with contrast reveals a narrowing in the rectosigmoid area. Which of the following is the most appropriate next step in his management?
- A Colonoscopic decompression and rectal tube placement
 - B Saline enemas and digital disimpaction of fecal matter from the rectum
 - C Colon resection and proximal colostomy
 - D Oral administration of metronidazole and checking a *Clostridium difficile* titer
 - E Evaluation of an electrocardiogram and obtaining an angiogram to evaluate for colonic mesenteric ischemia
- 7** A 65-year-old man who initially presented with a locally advanced adenocarcinoma of the gastric fundus has completed neoadjuvant chemotherapy. Restaging scans show improvement in the primary tumor and no metastatic disease. What is the next step in the treatment of this patient?
- A Neoadjuvant radiation therapy
 - B Distal subtotal gastrectomy
 - C Total gastrectomy with D2 lymphadenectomy
 - D Total gastrectomy with D2 lymphadenectomy, distal pancreatectomy, and splenectomy
 - E Proximal gastrectomy with D2 lymphadenectomy
- 8** A 32-year-old woman undergoes an uncomplicated appendectomy for acute appendicitis. The pathology report notes the presence of a 1-cm carcinoid tumor in the tip of the appendix with no involvement of the proximal margin. Which of the following is the most appropriate management of this patient?
- A Right hemicolectomy
 - B Right hemicolectomy and chemotherapy
 - C Chemotherapy only
 - D Radiation only
 - E No further treatment
- 9** A 60-year-old female presents with abdominal distention, early satiety, and weight gain. Ultrasound and CT demonstrate a large right retroperitoneal mass surrounding the right kidney, in close proximity to the IVC and the liver. What is the next best step in the workup of this patient?
- A MRI abdomen/pelvis
 - B Renal nuclear medicine scan to evaluate the contralateral kidney
 - C Percutaneous biopsy of the mass
 - D CT chest with IV contrast
 - E Endoscopic ultrasound
- 10** A 45-year-old man is examined for a yearly executive physical. A mass is palpated in the rectum, and a biopsy suggests carcinoid. Which of the following findings is most likely to be associated with the carcinoid syndrome?
- A Tumor <2 cm without ulceration
 - B Tumor <2 cm with ulceration
 - C Tumor >2 cm
 - D Involvement of regional lymph nodes
 - E Hepatic metastases
- 11** A 57-year-old woman presents with adenocarcinoma of the right colon. Laboratory evaluation demonstrates an elevation of carcinoembryonic antigen (CEA) to 123 ng/mL. Which of the following is the most appropriate use of CEA testing in patients with colorectal cancer?
- A As a screening test for colorectal cancer
 - B To determine which patients should receive adjuvant therapy
 - C To determine which patients should receive neoadjuvant therapy
 - D To monitor for postoperative recurrence
 - E To screen for preoperative metastatic disease
- 12** A 60-year-old woman has recurrent squamous cell carcinoma of the anus after chemoradiation. Which operation is most appropriate?
- A Low anterior resection
 - B Abdominoperineal resection
 - C Subtotal colectomy with end ileostomy
 - D Total proctocolectomy with ileoanal J-pouch
 - E Sigmoid resection with end colostomy (Hartmann procedure)
- 13** A 68-year-old woman with faecal incontinence has a large fixed adenocarcinoma 3 cm from the anal verge. Which operation is most appropriate?
- A Low anterior resection
 - B Abdominoperineal resection
 - C Subtotal colectomy with end ileostomy
 - D Total proctocolectomy with ileoanal J-pouch
 - E Sigmoid resection with end colostomy (Hartmann procedure)
- 14** What is the best method of staging of the peritoneum for gastric cancer?
- A Contrast-enhanced CT scan of the abdomen
 - B Endoscopic ultrasound evaluation
 - C MRI of the abdomen and pelvis
 - D PET scan with whole-body imaging
 - E Diagnostic laparoscopy with peritoneal washings
- 15** What resection margins are needed for colon cancer?
- A A 1-cm resection margin
 - B A 3-cm resection margin
 - C A 10-cm resection margin
 - D A 5-cm resection margin
 - E A 7-cm resection margin

Answers 1–15

PRETEST MCQS

- 1 **C Right hemicolectomy with local resection of the liver metastasis** — **UPDATED** Synchronous resection of colon cancer and resectable liver metastasis is appropriate.
- 2 **B Endoscopic ultrasound (EUS)** — **UPDATED** EUS is used to evaluate the depth of invasion and lymph node involvement in gastric cancer.
- 3 **B Little long-term change in bowel habits** — **UPDATED** The colon has a large reserve capacity, so bowel habits often remain stable post-resection.
- 4 **E No further therapy** — **UPDATED** Polypectomy with clear margins is adequate when carcinoma in situ is identified.
- 5 **E Combined radiation therapy and chemotherapy** — **UPDATED** Combined chemoradiation is now the standard initial treatment for squamous cell carcinoma of the anus.
- 6 **C Colon resection and proximal colostomy** — **UPDATED** For malignant colonic obstruction in an elderly patient, resection with proximal colostomy is indicated.
- 7 **C Total gastrectomy with D2 lymphadenectomy** — **UPDATED** After neoadjuvant chemotherapy, curative resection with total gastrectomy and D2 lymphadenectomy is indicated.
- 8 **E No further treatment** — **UPDATED** For carcinoid tumors less than 2 cm and confined to the appendix, an appendectomy is sufficient.
- 9 **C Percutaneous biopsy of the mass** — **UPDATED** A retroperitoneal mass suspicious for soft tissue sarcoma warrants a percutaneous biopsy for definitive diagnosis.
- 10 **C Tumor >2 cm** — **UPDATED** Rectal carcinoids are more likely to cause carcinoid syndrome when they are larger than 2 cm.
- 11 **D To monitor for postoperative recurrence** — **UPDATED** CEA testing is most useful for detecting postoperative recurrence in colorectal cancer.
- 12 **B Abdominoperineal resection** — **UPDATED** Recurrent anal carcinoma post-chemoradiation is managed with abdominoperineal resection.
- 13 **B Abdominoperineal resection** — **UPDATED** A low rectal adenocarcinoma with incontinence requires abdominoperineal resection.
- 14 **E Diagnostic laparoscopy with peritoneal washings** — **UPDATED** Peritoneal metastases are missed by CT and PET; staging laparoscopy with peritoneal washings is the best test before gastrectomy.
- 15 **D A 5-cm resection margin** — **UPDATED** Colon cancer resection needs about 5 cm of clear bowel on each side, with en bloc removal of the mesentery and lymph nodes (complete mesocolic excision).

10 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 65-year-old woman presents with episodes of lower abdominal pain and intermittent vaginal bleeding. On examination, a palpable mass is found in the lower abdomen. Which of the following tumour markers would be associated with this presentation?
 - A Alpha fetoprotein
 - B Beta-hCG
 - C CA 125
 - D Calcitonin
 - E Carcinoembryonic antigen
- 2 A 42-year-old woman presents to the emergency department with difficulty breathing and facial flushing which began soon after having a few drinks at the pub. She has had similar episodes recently. Examination is unremarkable; her heart rate is 82/min, blood pressure 122/86 mmHg. Which investigation will most likely confirm the underlying diagnosis?
 - A 17-hydroxyprogesterone levels
 - B 24-hour urinary vanillylmandelic acid
 - C 24-hour urinary 5-hydroxyindole acetic acid
 - D Serum calcitonin
 - E Short synacthen test
- 3 A 54-year-old man has a recent change in bowel habit and constant desire to defaecate with marked weight loss and lethargy. Which tumour marker is associated?
 - A CA 125
 - B CA 15-3
 - C CA 19-9
 - D CEA
 - E PSA
- 4 A 64-year-old woman has progressive pain and difficulty swallowing, now even liquids are hard. She is cachexic. Which is the most likely diagnosis?
 - A Oesophageal candidiasis
 - B Oesophageal carcinoma
 - C Oesophageal spasm
 - D Oesophageal stricture
 - E Oesophageal web

- 5** A 69-year-old man presents to the GP with a 2-month history of epigastric pain. He also feels full after eating small amounts of food and has lost a stone in weight over the last 2 months. Examination is unremarkable. Which of the following is the most likely diagnosis?
- A Duodenal ulcer
 - B Gastric cancer
 - C Gastric ulcer
 - D Menetrier disease
 - E Pernicious anaemia
- 6** A 73-year-old woman presents to the surgical outpatient clinic with a 3-week history of rectal bleeding, a sensation of incomplete emptying despite defaecation and rectal discomfort. She is found to have a tumour 2 cm from the anal margin. Which of the following would be the most appropriate operative intervention?
- A Abdominoperineal resection
 - B Anterior resection
 - C Hartmann procedure
 - D Left hemicolectomy
 - E Right hemicolectomy
- 7** A 73-year-old woman presents to the emergency department with abdominal pain, vomiting and an inability to open her bowels. On examination, the abdomen is distended and the rectum is empty. The patient's husband tells you that she has lost a significant amount of weight over the last month and has been having blood in her stool. Which of the following is the most likely cause of the presentation?
- A Colorectal malignancy
 - B Constipation
 - C Diverticulitis
 - D Inguinal hernia
 - E Ulcerative colitis
- 8** A 37-year-old woman presents to the GP with a 3-week history of frequent, loose motions. She is very tired and complains of muscle weakness. A subsequent colonoscopy demonstrates a 1.5 cm sessile growth that has multiple projections. The remainder of the colon was unremarkable. What is the most likely diagnosis?
- A Adenomatous polyp
 - B Colorectal carcinoma
 - C Infective colitis
 - D Pseudomembranous colitis
 - E Villous adenoma
- 9** A 59-year-old woman has recently had a resection of a sigmoid adenocarcinoma. The pathology report said that the tumour invaded through the bowel wall but there was no evidence of lymph node involvement. Which of the following best describes this stage of tumour?
- A Dukes A
 - B Dukes B
 - C Dukes C1
 - D Dukes C2
 - E Dukes D
- 10** A 27-year-old man presents to the emergency department following an episode of vomiting blood. On examination he is haemodynamically stable and is noted to have multiple freckles around his lips and mouth. A subsequent endoscopy demonstrates multiple polyps throughout the duodenum which are confirmed as benign. What is the most likely morphology of the polyps?
- A Adenomatous polyp
 - B Hamartomatous polyp
 - C Juvenile polyp
 - D Metaplastic polyp
 - E Pseudo-polyp

Answers 1-10

- 1** **C CA 125** — **UPDATED** Ovarian carcinoma often elevates CA 125.
- 2** **C 24-hour urinary 5-hydroxyindole acetic acid** — **UPDATED** Carcinoid syndrome is confirmed by elevated 5-HIAA in a 24-hour urine collection.
- 3** **D CEA** — **UPDATED** Raised carcinoembryonic antigen is linked with colorectal cancer.
- 4** **B Oesophageal carcinoma** — **UPDATED** Progressive dysphagia (solids then liquids) with weight loss strongly suggests malignancy.
- 5** **B Gastric cancer** — **UPDATED** Presents with epigastric pain, early satiety, and weight loss, often diagnosed late.
- 6** **A Abdominoperineal resection** — **UPDATED** A tumour 2 cm from the anal margin involves the sphincter complex, so abdominoperineal resection with permanent colostomy is needed (after neoadjuvant therapy as indicated).
- 7** **A Colorectal malignancy** — **UPDATED** Obstruction, rectal bleeding, weight loss are indicative of a colonic tumor.
- 8** **E Villous adenoma** — **UPDATED** Mucus-secreting colonic polyp often presenting with diarrhoea and hypokalaemia.
- 9** **B Dukes B** — **UPDATED** Tumour extends beyond the muscularis but no nodal involvement (Dukes B).
- 10** **B Hamartomatous polyp** — **UPDATED** Peutz-Jeghers syndrome features hamartomatous polyps and mucocutaneous pigmentation.

6 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 68-year-old man had a colorectal adenocarcinoma resected; histology shows spread through the muscle layer into the pericolic fat but no lymph-node involvement. What Dukes stage is this?
 - A Dukes A
 - B Dukes B
 - C Dukes C1
 - D Dukes C2
 - E Dukes D
- 2 A 46-year-old female has looser stools for 2 months with blood mixed in, no abdominal pain. First-choice investigation?
 - A Abdominal X-ray
 - B Faecal occult blood
 - C Abdominal computed tomography (CT)
 - D Ultrasound abdomen
 - E Colonoscopy
- 3 A 58-year-old male with long-standing achalasia presents with weight loss and worsening dysphagia. OGD shows a suspicious lesion in the upper third of the oesophagus. Likely histology?
 - A Adenocarcinoma
 - B Lymphoma
 - C Small cell carcinoma
 - D Squamous cell carcinoma
 - E Melanoma
- 4 A 55-year-old man has a rectal bleed and abdominal discomfort. A colonoscopy reveals a sigmoid tumour with probable local lymph node involvement (no distant mets). Next step?
 - A Surgery
 - B Chemotherapy
 - C Surgery and adjuvant chemotherapy
 - D Radiotherapy
 - E Palliative treatment
- 5 An elderly man has rectal bleeding and weight loss. Colonoscopy shows a sigmoid mass. Biopsy: cancer extends into muscularis mucosa with no nodal or distant mets. Dukes stage?
 - A Dukes A
 - B Dukes B
 - C Dukes C
 - D Dukes D
 - E Dukes E
- 6 A 50-year-old man with epigastric pain, vomiting, weight loss, heavy alcohol use (10 yrs). On exam he has a purple peri-orbital rash and a palpable left supraclavicular node. Next best investigation?
 - A Abdominal CT scan
 - B Abdominal ultrasound scan
 - C Endoscopy + biopsy
 - D Laparoscopy
 - E Abdominal X-ray

Answers 1-6

CRASH COURSE MCQS

- 1 **B Dukes B** — **UPDATED** Tumour through the muscularis propria without nodal spread is Dukes B (T3N0).
- 2 **E Colonoscopy** — **UPDATED** Rectal bleeding + changed bowel habit => urgent colonoscopy.
- 3 **D Squamous cell carcinoma** — **UPDATED** Achalasia predisposes to squamous cell carcinoma of the oesophagus.
- 4 **C Surgery and adjuvant chemotherapy** — **UPDATED** Dukes C => resection + adjuvant chemo.
- 5 **A Dukes A** — **UPDATED** Cancer confined to the bowel wall (not through the muscularis propria) with no nodes or metastases is Dukes A.
- 6 **C Endoscopy + biopsy** — **UPDATED** Weight loss with a Virchow node (and a heliotrope rash of paraneoplastic dermatomyositis) suggests gastric cancer; endoscopy with biopsy confirms it.

Gastrointestinal EMQs

Extended matching questions on the acute abdomen, dysphagia, GI bleeding, hernias, anorectal and hepatobiliary disease.

0

MCQS

511

EMQ SCENARIOS

6

SOURCES

4

REPEATED IDEAS

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Local Exams EMQs

SET 1 OF 7

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

THEME 1 Treatment of perianal disorders

- A Rubber band ligation
- B Stapled haemorrhoidopexy
- C Open (excisional) haemorrhoidectomy
- D Topical calcium channel blocker
- E Botulinum toxin injection
- F Fistulotomy
- G Incision and drainage
- H Lateral internal sphincterotomy
- I Seton placement

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A woman with an anal fissure in whom conservative management has failed.
- 2 A patient with prolapsed haemorrhoids that need manual reduction or remain prolapsed.
- 3 A man with fever and a fluctuant perianal mass.
- 4 A man with Crohn disease and a low fistula-in-ano.
- 5 A woman with painless haemorrhoids that prolapse on defecation and reduce spontaneously.

THEME 2 Upper gastrointestinal bleeding

- A Peptic ulcer disease
- B Mallory–Weiss tear
- C Oesophageal varices
- D Gastric cancer
- E Dieulafoy lesion
- F Angiodysplasia
- G Erosive gastritis
- H Aortoenteric fistula
- I Gastric antral vascular ectasia (watermelon stomach)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 52-year-old man with alcoholic cirrhosis has massive haematemesis and hypotension. He has spider naevi and ascites.
- 2 A 35-year-old man vomits blood-streaked material after several bouts of forceful retching following heavy drinking. He is stable and the bleeding stops spontaneously.
- 3 A 68-year-old man taking NSAIDs for osteoarthritis has melaena and anaemia. He is otherwise healthy.

- 4 A 65-year-old woman has fatigue and pallor and has neglected tarry stools for a year. There is a hard, non-tender lump above the left clavicle.

FORMATIVE

College formative examinations

THEME 3 Gastrointestinal Hemorrhage

- A Gastric cancer
- B Bleeding esophageal varices
- C Colonic cancer
- D Peptic ulcer disease
- E Mallory-Weiss tear

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 71-year-old female presented with severe pallor and a palpable, non-tender left supraclavicular lymph node (Virchow's node).
- 2 A 46-year-old male with a known history of chronic viral hepatitis presents with recurrent episodes of vomiting fresh blood.
- 3 A 63-year-old male presents with black tarry stools. He is a known alcoholic and has a long history of chronic NSAID use.

THEME 4 Investigation of Trauma

- A Native CT scan
- B CT scan with IV contrast
- C Cystourethrogram
- D FAST (Focused Assessment with Sonography in Trauma)
- E Diagnostic peritoneal lavage
- F Retrograde urethrogram

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A female patient with a known history of intimate partner violence presents with severe flank (loin) pain. There is suspicion of renal trauma.
- 2 A child presents after blunt trauma to the perineum. There is blood visible at the external urethral meatus. Hematuria is suspected.
- 3 A patient involved in a road traffic accident is hypotensive. You need a simple, rapid bedside test to rule out intra-abdominal bleeding.

PREVIOUS YEARS

THEME 5 Anal diseases

- A Acute anal fissure
- B Anal carcinoma
- C Chronic anal fissure
- D Fistula in ano
- E Necrotizing fasciitis
- F Perianal Crohn's disease
- G Perianal abscess
- H Perianal hematoma
- I Proctalgia fugax
- J Strangulated hemorrhoids

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 23-year-old male is presented with severe perianal pain. Examination reveals an exquisitely tender fluctuating mass near the anal verge with surrounding erythema.
- 2 A 38-year-old female is presented with sharp pain during defecation with streaks of blood on stool. She had an obstructed labor 3 days ago.
- 3 A 41-year-old male with a long history of piles is presented with sharp pain after passing a hard stool. On examination, there is a small bluish tender lump at the anal verge.

- 4 A 72-year-old female is presented with an ulcerative mass at the anal verge. On examination, it is non-tender and has a central necrosis with everted edges.

THEME 6 Intestinal obstruction

- A Adhesional obstruction
- B Colonic cancer
- C Colonic pseudo-obstruction
- D Gallstone ileus
- E Mesenteric ischemia
- F Obstructed direct inguinal hernia
- G Obstructed femoral hernia
- H Obstructed incisional hernia
- I Obstructed indirect inguinal hernia
- J Paralytic ileus

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 56-year-old female patient presents with central abdominal pain, vomiting, and a tender lump below the right inguinal ligament.
- 2 Four days post-appendectomy for perforated appendicitis, an 18 years old man is complaining from mild abdominal discomfort, vomiting and distention

Answers · Themes 1-6

Theme 1 · Treatment of perianal disorders

- 1 **E Botulinum toxin injection** — After failed topical therapy, botulinum toxin is preferred in women (obstetric sphincter risk); lateral sphincterotomy is used in men or if Botox fails.
- 2 **C Open (excisional) haemorrhoidectomy** — Grade III–IV haemorrhoids are treated by excisional haemorrhoidectomy; stapled haemorrhoidopexy is an alternative for grade III.
- 3 **G Incision and drainage** — A perianal abscess needs incision and drainage.
- 4 **I Seton placement** — Perianal Crohn fistulas are managed with a loose draining seton plus medical therapy to avoid incontinence and non-healing wounds.
- 5 **A Rubber band ligation** — Grade II haemorrhoids are treated by rubber band ligation.

Theme 2 · Upper gastrointestinal bleeding

- 1 **C Oesophageal varices** — Massive haematemesis with stigmata of chronic liver disease is bleeding oesophageal varices.
- 2 **B Mallory–Weiss tear** — Haematemesis after forceful retching that settles spontaneously is a Mallory–Weiss tear.
- 3 **A Peptic ulcer disease** — NSAID use with melaena and anaemia is a bleeding peptic ulcer.
- 4 **D Gastric cancer** — Chronic melaena with a left supraclavicular (Virchow) node is gastric cancer.

Theme 3 · Gastrointestinal Hemorrhage

- 1 **A Gastric cancer** — A left supraclavicular (Virchow's) node with anemia suggests advanced gastric malignancy.
- 2 **B Bleeding esophageal varices** — Chronic liver disease with hematemesis is most commonly due to variceal bleeding.
- 3 **D Peptic ulcer disease** — Melena in a chronic NSAID user suggests upper GI bleeding from peptic ulcers.

Theme 4 · Investigation of Trauma

- 1 **B CT scan with IV contrast** — CT with contrast is the gold standard for identifying solid organ injuries in abdominal trauma.
- 2 **C Cystourethrogram** — Blood at the meatus suggests urethral trauma, which should be evaluated by retrograde cystourethrogram.
- 3 **D FAST (Focused Assessment with Sonography in Trauma)** — FAST is a non-invasive bedside ultrasound technique for detecting free fluid in the peritoneal cavity.

Theme 5 · Anal diseases

- 1 **G Perianal abscess** — Fluctuating tender mass => Pus filled => Perianal abscess
- 2 **A Acute anal fissure** — Obstructed labor => Trauma in the perianal area => anal fissure => Pain during defecation with streaks of blood.

- 3** **H Perianal hematoma** — A small, tense, bluish, tender lump at the anal verge after straining is a perianal haematoma (thrombosed external vein); strangulated piles are larger, prolapsed and oedematous.
- 4** **B Anal carcinoma** — In an elderly patient an ulcerative mass with everted edges is highly suggestive of carcinoma.

SET 2 OF 7

THEME 7 Steatorrhea

- A Bacterial overgrowth
- B Biliary stricture
- C Chronic pancreatitis
- D Cirrhosis
- E Coeliac disease
- F Crohn's disease
- G Cystic fibrosis
- H Radiation ileitis
- I Sclerosing cholangitis
- J Short bowel syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 39-year-old illiterate male living in the countryside presents with offensive loose fatty stool. He gave a long history of cramping abdominal pain and chronic diarrhea with a history of right hemicolectomy for intestinal obstruction 2 years ago with benign pathology, although he was not compliant with medications or follow-up.
- 2** A 53-year-old male presents with loose stool that doesn't flush easily with vague midabdominal pain over the past several years. Upper and lower endoscopies were normal. Barium follow-through confirmed multiple jejunal diverticulosis.
- 3** A 67-year-old male who had a 2-year history of anterior resection of rectum with adjuvant therapy for low rectal cancer presents with vague midabdominal pain and steatorrhea.
- 4** A 72-year-old female presents with steatorrhea over the past 6 months. Her past medical history is positive for ischemic heart disease and diabetes mellitus. She underwent laparotomy for mesenteric ischemia one year ago.
- 5** An 8-year-old male brought by his parents with ill health and weight loss. Examination revealed pale conjunctiva, mouth ulceration, pigmented skin with a soft distended abdomen.

Theme 6 Intestinal obstruction

- 1** **G Obstructed femoral hernia** — Tender lump below the inguinal ligament.
- 2** **A Adhesional obstruction** — Postoperative adhesional obstruction typically develops several days after surgery (around 3-4 days), whereas paralytic ileus begins immediately and usually lasts for about 3 days.

THEME 8 Liver diseases and risk factors

- A Ascending cholangitis
- B Cholangiocarcinoma
- C Focal nodular hyperplasia
- D Haemobilia
- E Hepatic abscess
- F Hepatic adenoma
- G Hepatic hemangioma
- H Hepatic hydatid cyst
- I Hepatocellular carcinoma
- J Sclerosing cholangitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 37-year-old male with a long history of ulcerative colitis is presented with generalized itching and jaundice. MRCP revealed irregular beaded CBD.
- 2** A 30-year-old lady with a hepatic tumor that disappeared on stopping her OCP.
- 3** A 43-year-old male had fever, rigor, jaundice, and abdominal pain 72 hours after doing ERCP for an impacted CBD stone.
- 4** A 56-year-old male with chronic HBV infection, acquired it in his twenties. On examination, he had mild jaundice and gynecomastia. US shows a 5 cm liver mass.

THEME 9 Dysphagia

- A Kelly-Patterson syndrome
- B Mallory-Weiss syndrome
- C Achalasia
- D Barret's oesophagus
- E Boerhaave syndrome
- F Diffuse oesophageal spasm
- G Mega oesophagus
- H Oesophageal carcinoma
- I Oesophageal diverticulum
- J Oesophageal stricture

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** It is a complication of a parasitic infestation.
- 2** It is associated with spoon-shaped finger nails.
- 3** It means failure to relax.

- 4 It requires a strict follow up endoscopy, as it carries a recognized risk of malignant transformation.

THEME 10 RIF Masses

- A Actinomycosis
- B Appendicular mass
- C CA cecum
- D Crohn's Disease
- E Desmoid
- F Ectopic kidney
- G Germ cell tumor in undescended intraperitoneal testis
- H Ileal tuberculosis
- I Iliac LAP (lymphoma)
- J Intra-abdominal foreign body
- K Rectus sheath hematoma
- L Retroperitoneal haematoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 23-year-old female presents with night sweats, fever, and cough. An abdominal CT scan reveals a right iliac fossa (RIF) mass and adjacent LAP with no organomegaly. ESR is highly elevated.
- 2 A 26-year-old pregnant lady presents with sudden onset lower quadrant abdominal pain after lifting a heavy object. There is a tender RIF mass that doesn't change on contracting abdominal wall.
- 3 A 31-year-old male with a history of familial adenomatous polyposis (FAP) presents with a recurrent right lower abdominal wall mass despite repeated resection.
- 4 A 38 years old male with a long history of cramping abdominal pain and chronic diarrhoea, is found to have a slightly tender mass in his right lower abdomen.

- 5 A 79-year-old lady presents with pallor, fatigue, and vague abdominal pain. Examination reveals a non-tender RIF mass, and fecal occult blood test is positive.
- 6 An 18-year-old male presents with a tender mass in his right lower abdomen. He has a history of vague abdominal pain, vomiting, low-grade fever, and loss of appetite in the last week.

THEME 11 Colonic diseases and surgical treatment

- A Abdominoperineal resection
- B Anterior resection of rectum
- C Diverting colostomy
- D Extended right hemicolectomy
- E Left hemicolectomy
- F Restorative total proctocolectomy
- G Right hemicolectomy
- H Sigmoid colectomy and Hartmann's procedure
- I Sigmoid colectomy and primary anastomosis
- J Total proctocolectomy and end ileostomy
- K Watch and see

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 49-year-old patient with a history of fistulotomy is presented with a recurrent high fistula in ano.
- 2 A 52-year-old male is presented with a large malignant ulcer 2 cm above the anal verge.
- 3 A 56-year-old female had standard appendicectomy for acute appendicitis. The histopathology revealed appendicular adenocarcinoma.

Answers · Themes 7-11

Theme 7 · Steatorrhoea

- 1 **A Bacterial overgrowth** — SIBO (Small intestine bacterial overgrowth) is common following a hemicolectomy.
- 2 **A Bacterial overgrowth** — Jejunal diverticula allow stasis and small-bowel bacterial overgrowth, which deconjugates bile salts and causes steatorrhoea and B12 deficiency.
- 3 **H Radiation ileitis** — The patient is receiving adjuvant radiotherapy following resection.
- 4 **J Short bowel syndrome** — Mesenteric ischemia may necessitate resection of a significant portion of the small intestine leading to short bowel syndrome.
- 5 **E Coeliac disease** — A child with poor growth, pallor (iron deficiency), mouth ulcers and a distended abdomen suggests coeliac disease: anti-tTG IgA, then duodenal biopsy.

Theme 8 · Liver diseases and risk factors

- 1 **J Sclerosing cholangitis** — Sclerosing cholangitis is commonly associated with inflammatory bowel disease, particu-

larly ulcerative colitis. The irregular beaded appearance of the CBD on MRCP is characteristic.

- 2 **F Hepatic adenoma** — Hepatic adenomas are benign, hormone-sensitive tumors often associated with OCP use.
- 3 **A Ascending cholangitis** — Charcot triad (fever, rigor, jaundice, and abdominal pain) is classic for ascending cholangitis.
- 4 **I Hepatocellular carcinoma** — Chronic HBV infection predisposes to hepatocellular carcinoma, with the presence of a liver mass supporting the diagnosis.

Theme 9 · Dysphagia

- 1 **G Mega oesophagus** — Dilated esophagus can occur as a complication of Chagas disease, which is caused by *Trypanosoma cruzi*.
- 2 **A Kelly-Patterson syndrome** — Kelly-Patterson syndrome (Plummer Vinson syndrome) is characterized by dysphagia, esophageal webs, and iron deficiency anemia; spoon-shaped nails (koilonychia) are also seen.

- 3 **C Achalasia** — Failure of the lower esophageal sphincter to relax defines achalasia.
- 4 **D Barrett's oesophagus** — Barrett's oesophagus carries a risk of malignant transformation and therefore requires strict endoscopic follow-up.

Theme 10 · RIF Masses

- 1 **H Ileal tuberculosis** — The presence of night sweats, fever, cough, RIF mass, and elevated ESR suggests a diagnosis of ileal tuberculosis.
- 2 **K Rectus sheath hematoma** — The tender RIF mass that doesn't change with abdominal wall contraction along with sudden pain suggests a rectus sheath hematoma.
- 3 **E Desmoid** — Recurrent abdominal wall mass in a patient with FAP is suggestive of a desmoid tumor.
- 4 **D Crohn's Disease** — Chronic cramping abdominal pain and diarrhea with a slightly tender mass is indicative of Crohn's Disease.

SET 3 OF 7

THEME 12 Acute abdomen

- A Acute appendicitis
- B Acute cholangitis
- C Acute enterocolitis
- D Acute hepatitis
- E Acute gastritis
- F Acute intestinal obstruction
- G Acute pancreatitis
- H Herpes zoster
- I Mesenteric ischemia
- J Perforated viscus

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 12-year-old girl presents with vague right-sided lower abdominal pain, nausea, loss of appetite, fever, WBC 14,000, free fluid in the pelvis, and a thickened appendix wall.
- 2 A 21-year-old female has colicky central & lower abdominal pain with diarrhea and tenesmus.
- 3 A 21-year-old male presents with severe right upper abdominal pain followed by a rash on the right lower chest.
- 4 A 23-year-old male presents with vague upper abdominal pain and slight yellowish discoloration of stool.
- 5 A 25-year-old female presents with acute, severe abdominal pain without vomiting, a distended abdomen, and exaggerated bowel sounds with a history of surgery.
- 6 A 30-year-old female presents with excruciating epigastric pain radiating to the back, guarding, and ultrasound findings of gallstones.
- 7 A 45-year-old female with a history of CBD stones presents with acute upper abdominal pain, jaundice, and high-grade fever.

- 5 **C CA cecum** — Pallor, fatigue, vague abdominal pain, non-tender RIF mass and a positive fecal occult blood test raise suspicion for CA cecum.
- 6 **B Appendicular mass** — A tender mass with a history of vague pain, vomiting, fever, and appetite loss indicates an appendicular mass.

Theme 11 · Colonic diseases and surgical treatment

- 1 **C Diverting colostomy** — Diverting colostomy is used to divert the fecal stream away from the anorectum to allow healing of the fistula.
- 2 **A Abdominoperineal resection** — The malignant ulcer 2 cm above the anal verge is managed by abdominoperineal resection.
- 3 **G Right hemicolectomy** — Right hemicolectomy is recommended for appendicular adenocarcinoma to achieve oncological clearance.

- 8 A 50-year-old heavy drinker male presents with anal pain, distended abdomen, guarding, rigidity, and absent bowel sounds.
- 9 A 70-year-old male presents with severe abdominal pain, vomiting, blood-stained stool, a tender soft abdomen, and tachycardia.

THEME 13 UPPER ABDOMINAL PAIN

- A Acute cholangitis
- B Acute cholecystitis
- C Acute gastritis
- D Acute hepatitis
- E Acute pancreatitis
- F Chronic pancreatitis
- G Diabetic ketoacidosis
- H Fitz-Hugh Curtis syndrome
- I Herpes zoster
- J Liver abscess
- K Musculoskeletal spasm
- L PU

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 16-year-old male presents with high-grade swinging fever, chills, and upper abdominal pain, is jaundiced, and has a history of bloody diarrhea.
- 2 A 29-year-old lady presents with right upper abdominal pain radiating to the back, nausea, vomiting, fever, and a history of 40kg weight loss post-bariatric surgery.
- 3 A 34-year-old male presents with vague upper abdominal pain, nausea, a single episode of vomiting, and loss of appetite; later he develops blisters along his right lower chest.
- 4 A 43-year-old alcoholic male presents with vague abdominal pain for 3 months, weight loss, new-onset diabetes, and a history of similar severe attacks.

- 5 A newly married lady presents with upper abdominal pain, nausea, and vaginal discharge; ultrasound reveals right hydrosalpinx.

THEME 14 Antimicrobial therapy in surgery

- A 2nd generation cephalosporins
- B Cloxacillin
- C Direct-acting antivirals (DAAs)
- D Linezolid
- E Meropenem
- F Metronidazole
- G Vancomycin
- H Tigecycline

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 24-year-old male is found to be HCV positive on preparation for thyroid surgery. The best antiviral agent with near 100% cure is:
- 2 A 29-year-old female with a history of bloody diarrhea 2 weeks earlier now presents with upper abdominal pain, jaundice, and rigor. US revealed an 8 cm cystic mass in the liver. Drainage revealed dark brown-red thick drainage fluid. The best adjuvant treatment option would be:
- 3 A 32-year-old male underwent appendectomy for perforated gangrenous appendicitis. Which drug should be given to the patient in addition to Levofloxacin to prevent sepsis:
- 4 A 36-year-old obese lady with BMI of 43 underwent mesh repair for large ventral hernia. Postoperatively, she has severe necrotizing fasciitis with MRSA infection. She is allergic to penicillins & carbapenems. The best drug to treat MRSA in this patient would be:
- 5 A 41-year-old female will have an elective surgery. Three to five doses of this antibiotic is all that needed to prevent infection:
- 6 A 43-year-old male presents with recurrent acute pancreatitis. The best drug to counteract infection with deep penetration into pancreatic parenchyma is:
- 7 A 54-year-old male has exploratory laparotomy for perforated diverticulitis. In spite of adequate antibiotic coverage and abscess drainage, the patient is still toxic. Swab from his drainage revealed CRE or carbapenem-resistant enterococci. The best treatment now is:
- 8 An 18-year-old male is undergoing mesh repair for right inguinal hernia. The best drug prophylaxis to prevent infection is:

Answers · Themes 12-14

Theme 12 · Acute abdomen

- 1 **A Acute appendicitis** — Findings suggest acute appendicitis.
- 2 **C Acute enterocolitis** — Colicky pain with diarrhea and tenesmus is indicative of acute enterocolitis.
- 3 **H Herpes zoster** — The development of a rash following severe abdominal pain is characteristic of herpes zoster.
- 4 **D Acute hepatitis** — Slight yellow discoloration of stool with vague pain suggests acute hepatitis.
- 5 **F Acute intestinal obstruction** — A distended abdomen with exaggerated bowel sounds post-surgery indicates acute intestinal obstruction.
- 6 **G Acute pancreatitis** — Epigastric pain radiating to the back with guarding and gallstones supports acute pancreatitis.
- 7 **B Acute cholangitis** — The triad of pain, jaundice, and fever with CBD stones supports acute cholangitis.
- 8 **J Perforated viscus** — Guarding, rigidity, and absent bowel sounds in a distended abdomen suggest a perforated viscus.
- 9 **I Mesenteric ischemia** — Severe pain, vomiting, blood-stained stool, and tachycardia point toward mesenteric ischemia.

Theme 13 · UPPER ABDOMINAL PAIN

- 1 **J Liver abscess** — The presentation of upper abdominal pain, swinging fever, chills, jaundice, and a history of bloody diarrhea suggests a liver abscess.

- 2 **B Acute cholecystitis** — Right upper abdominal pain radiating to the back with nausea, vomiting, and significant weight loss post-bariatric surgery indicates acute cholecystitis.
- 3 **I Herpes zoster** — The appearance of blisters following vague upper abdominal pain is characteristic of herpes zoster.
- 4 **F Chronic pancreatitis** — A chronic history of pain, weight loss, and new-onset diabetes in an alcoholic suggests chronic pancreatitis.
- 5 **H Fitz-Hugh Curtis syndrome** — Upper abdominal pain with vaginal discharge and right hydrosalpinx supports the diagnosis of Fitz-Hugh Curtis syndrome.

Theme 14 · Antimicrobial therapy in surgery

- 1 **C Direct-acting antivirals (DAAs)** — DAA drugs directly target the replication of the Hepatitis c virus.
- 2 **F Metronidazole** — Metronidazole is commonly used for abdominal infections (And their complicated abscess).
- 3 **F Metronidazole** — After perforated gangrenous appendicitis, metronidazole is added to levofloxacin to cover anaerobes.
- 4 **D Linezolid** — Linezolid is used as an alternative to carbapenems and penicillin in patients who have drug sensitivity.
- 5 **A 2nd generation cephalosporins** — Second-generation cephalosporins, such as cefoxitin or cefotetan are commonly used as a prophylactic antibiotic prior to surgery.
- 6 **E Meropenem** — Meropenem is a broad-spectrum antibiotic with excellent penetration into pancreatic tissue.

- 7 H Tigecycline** — Tigecycline is a broad-spectrum antibiotic that is effective against multidrug-resistant organisms, including CRE.

- 8 B Cloxacillin** — Cloxacillin is a penicillinase-resistant penicillin that provides effective coverage against a wide range of bacteria, including those commonly associated with surgical site infections.

SET 4 OF 7

THEME 15 Dysphagia

- A Achalasia
- B Barrett's esophagus
- C Benign esophageal stricture
- D Boerhaave syndrome
- E Diffuse esophageal spasm
- F Esophageal carcinoma
- G Esophageal diverticulum
- H Esophageal myoma
- I Kelly-Patterson syndrome
- J Mallory-Weiss syndrome
- K Megaesophagus

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 21-year-old female presents with progressive dysphagia over the past one year. She gave a history of long duration hospitalization after suicidal ingestion of a corrosive agent one year ago.
- 2** A 33-year-old slim lady presents with progressive dysphagia. Workup with barium swallow and endoscopy revealed a hugely distended tortuous esophagus with smooth esophageal mucosa.
- 3** A 43-year-old alcoholic judge presents after self-induced vomiting with severe central chest pain, tachypnea, and tachycardia. CXR shows pneumomediastinum.
- 4** A 46-year-old obese male with a long history of using antacids for dyspepsia and heartburn presents with new onset dysphagia. Endoscopy reveals erythematous lower esophageal mucosa with goblet cells on histopathological biopsy.
- 5** A 69-year-old lady presented with progressive dysphagia and recent hoarseness of voice.

THEME 16 Abdominal pain

- A Acute appendicitis
- B Acute cholangitis
- C Acute cholecystitis
- D Acute enterocolitis
- E Acute gastritis
- F Acute hepatitis
- G Acute intestinal obstruction
- H Acute lobar pneumonia
- I Acute myocardial infarction
- J Acute pancreatitis
- K Diabetic ketoacidosis
- L Herpes zoster
- M Perforated viscus
- N Mesenteric ischaemia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 12-year-old girl presents to the emergency department with vague right-sided lower abdominal pain, with nausea and loss of appetite & low-grade fever. Examination is near normal. WBC count is 14,000. US revealed free fluid in right iliac fossa and pelvis.
- 2** A 21-year-old female has colicky central & lower abdominal pain with diarrhea (7 times) in the past 24 hours with tenesmus. The abdomen is soft and not tender. Two of her children have similar conditions 2 days earlier.
- 3** A 23-year-old male presents with vague upper abdominal pain. Abdominal examination is normal. Sclera shows slight yellowish discoloration.
- 4** A 25-year-old female presents with acute central abdominal pain with bilious vomiting. Examination shows a distended abdomen with normal shaped grid-iron scar. Bowel sounds are exaggerated.
- 5** A 28-year-old male presents with severe right upper abdominal pain over the past 7 days. Abdominal exam is completely normal apart from some rash over the right loin that appeared few days after pain. US of abdomen is normal. WBC count is 7,000.
- 6** A 36-year-old female presents with excruciating epigastric pain over the past 24 hours that radiates to the back. Examination revealed tender upper abdomen with guarding. US is nearly normal apart from several tiny gallstones with normal wall thickness of gallbladder. WBC count is 13,000.
- 7** A 45-year-old female with BMI of 52 presents with acute upper abdominal pain with jaundice and high-grade swinging fever. She declared a history of common bile duct stones the month earlier.
- 8** A 59-year-old heavy smoker male presents with exquisite diffuse abdominal pain. Abdomen is distended with generalized tenderness & rigidity. Bowel sounds are absent.
- 9** A 70-year-old male has severe abdominal pain with twice vomiting over the past 24 hours. His stool is blood-stained. Examination shows mildly tender soft abdomen. His pulse is 110 completely irregular.

THEME 17 Investigations

- A 24-hour pH monitoring
- B Barium swallow
- C CT abdomen and pelvis with IV contrast
- D Colonoscopy
- E Flexible sigmoidoscopy
- F Surveillance colonoscopy with biopsies
- G CT angiography
- H ERCP
- I EUS
- J OGD
- K PET scan
- L PTC

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 It is a highly sensitive test to identify sites of recurrence of tumor after surgery for colonic carcinoma.
- 2 A 20-year-old male presents with vague abdominal pain of 72-hour duration. His Alvarado score is 5, and US is normal. The test with near 99% certainty to diagnose acute appendicitis is:
- 3 A 29-year-old male with Bilirubin of 42 has severe retrosternal chest pain that partially responds to PPI. Upper endoscopy is near normal apart from small hiatal hernia. The next best test to reach the diagnosis is:
- 4 A 34-year-old male presents with heavy bleeding per rectum with clots. His pulse is thready (120 beats/min), and blood pressure is 70/40. The best initial test is:
- 5 A 40-year-old female presents with dyspepsia. CT scan of the abdomen revealed a cystic mass of the body of the pancreas. The best test to differentiate cystic neoplasm from pseudocysts is:
- 6 A 42-year-old male presents with obstructive jaundice. She had a history of gallstones for 3 years. The best test to reach the cause of jaundice, as well as it can be therapeutic, is:
- 7 A 44-year-old male presents with recurrent symptoms after he had fundoplication for GERD 5 years ago. The best test to detect esophageal shortening is:
- 8 A 57-year-old female presents with bleeding per rectum with her stool. DRE revealed large piles. The next test would be:
- 9 A 70-year-old male presents with jaundice and itching. Further workup revealed inoperable cholangiocarcinoma (Klatskin tumor). ERCP stenting to relieve jaundice failed. The next logical step would be:
- 10 It is a mandatory follow-up test for patients with ulcerative colitis.

Answers • Themes 15–17

Theme 15 • Dysphagia

- 1 **C Benign esophageal stricture** — Ingestion of corrosive agent leads to progressive fibrosis and narrowing of the esophagus resulting in a complete stricture.
- 2 **K Megaesophagus** — Findings are consistent with Megaesophagus.
- 3 **D Boerhaave syndrome** — Self-induced vomiting in alcoholics may result in a Mallory-Weiss tear progressing to a ruptured esophagus known as Boerhaave syndrome.
- 4 **B Barret's esophagus** — Chronic acid reflux causes metaplastic change in the lower esophagus resulting in Barret's esophagus.
- 5 **F Esophageal carcinoma** — Progressive dysphagia with compression effects (hoarseness of voice) is suggestive of Esophageal carcinoma.

Theme 16 • Abdominal pain

- 1 **A Acute appendicitis** — Presentation with vague right lower quadrant pain, low-grade fever, and free fluid is suggestive of Acute appendicitis.
- 2 **D Acute enterocolitis** — Colicky pain and diarrhea with family involvement indicate Acute enterocolitis.
- 3 **F Acute hepatitis** — Upper abdominal pain with jaundice suggests Acute hepatitis.

- 4 **G Acute intestinal obstruction** — Bilious vomiting with abdominal distension and exaggerated bowel sounds indicate Acute intestinal obstruction.
- 5 **L Herpes zoster** — The presence of a rash following severe pain in a normal exam is indicative of Herpes Zoster.
- 6 **J Acute pancreatitis** — Excruciating epigastric pain radiating to the back with guarding is consistent with Acute pancreatitis.
- 7 **B Acute cholangitis** — Jaundice with high-grade fever and a history of bile duct stones are characteristic of Acute cholangitis.
- 8 **M Perforated viscus** — Generalised tenderness, board-like rigidity and absent bowel sounds is generalised peritonitis from a perforated viscus: erect CXR/CT and laparotomy.
- 9 **N Mesenteric ischaemia** — Severe pain with few signs, bloody stool and AF in an elderly man is acute mesenteric ischaemia (embolic): lactate, CT angiography, urgent revascularisation.

Theme 17 • Investigations

- 1 **K PET scan** — PET-CT is the most sensitive test for locating recurrence when CEA rises after resection of colorectal cancer.
- 2 **C CT abdomen and pelvis with IV contrast** — With an equivocal Alvarado score and normal ultrasound, contrast CT is >95% accurate for appendicitis in adults.

- 3** **A** **24-hour pH monitoring** — Reflux-type chest pain only partly helped by PPI with a near-normal OGD needs 24-h pH monitoring to confirm reflux.
- 4** **G** **CT angiography** — Heavy rectal bleeding with shock: resuscitate and request CT angiography to localise the bleeding point and guide embolisation.
- 5** **I** **EUS** — EUS (with cyst fluid CEA and cytology) distinguishes cystic pancreatic neoplasms from pseudocysts.
- 6** **H** **ERCP** — ERCP shows and treats CBD stones (sphincterotomy and stone extraction) in obstructive jaundice; MRCP is used if therapy is not needed.
- 7** **B** **Barium swallow** — A barium swallow shows anatomy after fundoplication (slipped wrap, recurrent hernia, short oesophagus).
- 8** **D** **Colonoscopy** — Rectal bleeding at 57 needs colonoscopy even when piles are seen, to exclude proximal colorectal cancer.
- 9** **L** **PTC** — When ERCP stenting fails for an inoperable hilar (Klatskin) tumour, percutaneous transhepatic cholangiography with drainage/stenting relieves the jaundice.
- 10** **F** **Surveillance colonoscopy with biopsies** — Long-standing UC needs surveillance colonoscopy with biopsies (chromoendoscopy) to detect dysplasia, usually from 8–10 years after onset.

SET 5 OF 7

THEME 18 Lower GI Bleeding

- A Angiodysplasia
- B Bacillary dysentery
- C Colorectal cancer
- D Crohn's disease
- E Diverticular disease
- F Fissure in ano
- G Hemorrhoids
- H Ulcerative colitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 25-year-old man presents to the surgical clinic with a 3-month history of passing loose, bloody motions up to 10/day associated with fever and weight loss. Stool analysis did not show bacteria or parasites.
- 2** A 35-year-old 7-month pregnant lady complains of progressively protruding painless anal mass with passage of drops of blood with, and shortly after defecation. She has no change in bowel habit.
- 3** A 55-year-old man presents to the emergency department complaining of per rectal bleeding with passage of clots. He is medically fit apart from chronic constipation. Six months ago, he was hospitalized for an attack of left-sided lower abdominal pain with fever, for which he received antibiotics and antispasmodics.
- 4** A 70-year-old woman presents to the surgical clinic with a 3-week history of passing fresh blood and mucus per rectum. She denies a change in bowel habit but does reveal weight loss of at least 5 kg over the past month.

THEME 19 Hematemesis

- A Carcinoma of the stomach
- B Dieulafoy's syndrome
- C Duodenal ulcer
- D Esophageal varices
- E Esophagitis
- F Gastric ulcer
- G Hemophilia
- H Mallory-Weiss syndrome
- I Multiple gastric erosions
- J Osler-Weber-Rendu syndrome
- K Thrombocytopenia
- L Von Willebrand disease

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 20-year-old man presents with frequent nosebleeds and heavy bleeding after dental extraction.
- 2** A 25-year-old woman complains of difficulty swallowing and pain behind the sternum.
- 3** A 30-year-old man presents with melena and hematemesis. Endoscopy reveals an arterial bleeding lesion in the stomach.
- 4** A 35-year-old man presents with spontaneous bruising and prolonged bleeding after a minor cut.
- 5** A 40-year-old woman presents with a history of recurrent nosebleeds and telangiectasias on her face, lips, and fingers.
- 6** A 45-year-old sales executive presents with an episode of hematemesis amounting to about 500 ml in volume. He has a recent history of a gnawing pain in the upper abdomen, which radiates to the back and is relieved by eating. His weight is stable.
- 7** A 50-year-old office worker developed nausea and vomiting after a party during which he drank 10 pints of beer. Initially, he brought up gastric contents, but later he noticed a considerable amount of blood in the vomitus.

- 8** A 50-year-old woman with a history of heavy menstrual bleeding presents with easy bruising and petechiae on the skin.
- 9** A 60-year-old man presents with anemia and epigastric discomfort. Endoscopy reveals small punctate lesions in the gastric mucosa.
- 10** A 60-year-old vagrant presents with recurrent hematemesis but is hemodynamically stable. He has palmar erythema, spider naevi, splenomegaly, and ascites.
- 11** A 70-year-old retired man is admitted with a history of weight loss, poor appetite, and upper abdominal pain. He has had a few episodes of hematemesis, although the amount of blood loss is minimal.

THEME 20 GROIN SWELLINGS

- A Direct inguinal hernia
- B Ectopic testis
- C Femoral hernia
- D Indirect inguinal hernia
- E Inguinal lymphadenopathy
- F Lipoma
- G Lymphogranuloma venereum
- H Obturator hernia
- I Psoas bursa
- J Saphena varix
- K Tuberculous Psoas abscess
- L Undescended testis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 43 old pregnant lady present with painless left groin swelling that disappears on lying in bed. On exam, there are bilateral tortuous veins in her legs.

- 2** 36 year old lady with history of sleeve gastrectomy 8 months ago presents with tender swelling below her right groin crease. It is associated with central colicky abdominal pain with bilious vomiting.
- 3** A 48-year-old woman presents to the outpatient clinic with a 1-year history of a lump in the Rt. groin. On examination, the lump is above and medial to the pubic tubercle. There is an expansile impulse on coughing and the lump reduces spontaneously on lying down.
- 4** A 56-year-old woman presents to the outpatient clinic with a 3-month history of pigmented lesion (male) on her right shin that has been increasing in size. She also complains of weight loss. Examination reveals three firm round swellings below the inguinal ligament.
- 5** A 63 year farmer presents with backache, fever, night sweats and a lump in his examination, there is tenderness in the thoracolumbar spine and fluctuating groin swelling.

THEME 21 Tests for Helicobacter pylori

- A Bacterial culture
- B H pylori stool Ag
- C Serology
- D Urea breath test

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** It is a simple test for diagnosis with high sensitivity and specificity
- 2** It is the gold standard test for eradication of H pylori

Answers • Themes 18–21

Theme 18 • Lower GI Bleeding

- 1** **H Ulcerative colitis** — Three months of frequent bloody diarrhoea with fever and weight loss and negative stool studies in a young adult is ulcerative colitis.
- 2** **G Hemorrhoids** — Hemorrhoids are common in pregnancy due to the increased pressure on the pelvic veins.
- 3** **E Diverticular disease** — Clotted rectal bleeding in a constipated man with a previous attack of left-sided diverticulitis is diverticular bleeding.
- 4** **C Colorectal cancer** — Fresh blood and mucus with weight loss in a 70-year-old is colorectal cancer until proven otherwise: colonoscopy.

Theme 19 • Hematemesis

- 1** **L Von Willebrand disease** — Epistaxis and bleeding after dental extraction (mucosal bleeding) suggest von Willebrand disease.

- 2** **E Esophagitis** — Painful swallowing with retrosternal pain in a young woman suggests oesophagitis.
- 3** **B Dieulafoy's syndrome** — A large submucosal artery bleeding through normal-looking gastric mucosa is a Dieulafoy lesion.
- 4** **G Hemophilia** — Spontaneous bruising and prolonged bleeding after small cuts in a man suggests haemophilia.
- 5** **J Osler-Weber-Rendu syndrome** — Recurrent epistaxis with telangiectasia of the lips, face and fingers is hereditary haemorrhagic telangiectasia (Osler-Weber-Rendu).
- 6** **C Duodenal ulcer** — Epigastric pain relieved by eating that radiates to the back, then haematemesis, is a bleeding (posterior) duodenal ulcer.
- 7** **H Mallory-Weiss syndrome** — Haematemesis after repeated forceful vomiting during a binge is a Mallory-Weiss tear at the cardia.

- 8 **K Thrombocytopenia** — Menorrhagia with bruising and petechiae suggests thrombocytopenia.
- 9 **I Multiple gastric erosions** — Small punctate mucosal lesions with anaemia are multiple gastric erosions (erosive gastritis).
- 10 **D Esophageal varices** — Haematemesis with palmar erythema, spider naevi, splenomegaly and ascites is variceal bleeding from portal hypertension.
- 11 **A Carcinoma of the stomach** — Weight loss, anorexia and upper abdominal pain with small haematemeses in an old man suggest gastric carcinoma.

Theme 20 · GROIN SWELLINGS

- 1 **J Saphena varix** — A soft groin swelling that disappears on lying, with varicose veins, is a saphena varix (fluid thrill, cough impulse).
- 2 **C Femoral hernia** — A tender lump below the groin crease with small-bowel obstruction in a woman is a strangulated femoral hernia.

SET 6 OF 7

THEME 22 Intestinal Polyps:

- A Familial Adenomatous Polyposis (FAP)
 B Hamartomatous polyp
 C Hyperplastic (metaplastic) polyp
 D Inflammatory polyp
 E Tubular adenomatous polyp
 F Tubulovillous adenomatous polyp
 G Villous adenomatous polyp

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 24 years old male with multiple perioral bluish-black pigmented lesions is presented with multiple ileal and colonic polyps.
- 2 A 28 years old male with hundreds of polyps in his rectum and colon
- 3 A 45 years old male with a history of ulcerative colitis, is found to have multiple pseudopolyps on endoscopy.
- 4 Commonest polyp in children and adolescents.
- 5 This type of polyps carries a high risk of malignant transformation.

THEME 23 Jaundice

- A Hepatic (hepatocellular)
 B Hepatic (intrahepatic cholestasis)
 C Post-hepatic
 D Pre-hepatic

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 25-year-old mature student decides to enter a drug trial to fund his education. Routine blood tests on admission show an elevated bilirubin concentration in the absence of any other abnormality

- 3 **D Indirect inguinal hernia** — A reducible groin lump above and medial to the pubic tubercle with a cough impulse is an inguinal hernia, usually indirect in women.
- 4 **E Inguinal lymphadenopathy** — Firm nodes below the inguinal ligament with an enlarging pigmented skin lesion are inguinal nodal metastases from melanoma.
- 5 **K Tuberculous Psoas abscess** — Fever, sweats, thoracic tenderness and a fluctuant groin swelling is a tuberculous psoas abscess from spinal TB.

Theme 21 · Tests for Helicobacter pylori

- 1 **B H pylori stool Ag** — The stool antigen test is simple, non-invasive and highly sensitive and specific for active H. pylori infection.
- 2 **D Urea breath test** — The urea breath test (4 weeks off antibiotics, 2 weeks off PPI) is the reference non-invasive test of eradication.

- 2 A 39-year-old man develops jaundice after commencing rifampicin and isoniazid for tuberculosis.
- 3 A 48-year-old woman presents with right upper quadrant pain, nausea and jaundice. On further questioning, she admits to having pale stools and dark urine.
- 4 Patient with a known history of ulcerative colitis presents with jaundice

REPEATED 2x · ALSO ASKED IN
Get Ahead

THEME 24 Antimicrobial Agents

- A Amoxicillin/clavunate
 B Ampicillin/sulbactam
 C Cefotaxime
 D Ceftazidime
 E Ceftriaxone
 F Cefoxitin
 G Ertapenem
 H Direct-acting antivirals
 I Levofloxacin + metronidazole
 J Levofloxacin + amoxicillin
 K None
 L Rifampicin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 23 years old female is found to be HCV Ab +ve on regular checkup for breast lump surgery. The best treatment now is:
- 2 A 23 years old male is found to have a perforated appendicitis during surgery for acute abdomen. The best antibiotic option would be:
- 3 A 27 years old male with history of restorative proctocolectomy is enduring recurrent attacks of pouchitis. The best treatment in addition to lactobacillus is:

- 4 A 34 years old male has persistent *H. pylori* infection in spite of multiple eradication regimes. In combination with pantoprazole, the next 2nd line regime is:
- 5 A 43 years old alcoholic male presents with sudden excruciating upper abdominal pain radiating to back. CT scan shows fluid in lesser sac with enlarged pancreas. The best treatment now is:
- 6 Although it is classified as 2nd generation cephalosporin, it has an anaerobic activity.

THEME 25 Malignancies and their tumor markers:

- A Alpha-FP
- B CA 19-9
- C CEA
- D PSA

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 CA colon
- 2 CA pancreas
- 3 HCC (Hepatocellular carcinoma)

THEME 26 Salivary glands diseases

- A Bacterial sialadenitis
- B Frey's syndrome
- C Mucoepidermoid carcinoma
- D Pleomorphic adenoma
- E Preauricular lymphadenopathy
- F Salivary duct stenosis
- G Salivary fistula
- H Sialolithiasis
- I Sjogren's syndrome
- J Warthin's tumour

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 37-year-old lady with a history of left parotid tumor is presented with a left preauricular lump 2 years after she had surgery to remove it. FNAC shows epithelial and stromal cells.
- 2 A 54-year-old lady with a history of joint problem is presented with dry mouth and dry red eyes. Examination revealed non-tender mild enlargement of both parotid glands.
- 3 A 67-year-old male is on parenteral nutrition for 3 weeks, because of serious abdominal trauma in an RTA. He complains of a painful left lump below his left jaw, with a fever. Examination shows bad oral hygiene, with teeth caries. WBC count is $14,000/\text{mm}^3$.
- 4 A 68-year-old male is presented with bilateral parotid enlargement. FNAC shows neoplastic epithelial cells and lymphocytes.

Answers · Themes 22–26

Theme 22 · Intestinal Polyps:

- 1 **B Hamartomatous polyp** — Perioral pigmentation with small-bowel and colonic polyps is Peutz-Jeghers syndrome (hamartomatous polyps).
- 2 **A Familial Adenomatous Polyposis (FAP)** — Hundreds of colorectal adenomas in a young man is familial adenomatous polyposis (APC mutation).
- 3 **D Inflammatory polyp** — Inflammatory polyps can be seen in patients with inflammatory bowel disease (IBD), including ulcerative colitis. These polyps are not precancerous.
- 4 **B Hamartomatous polyp** — The commonest polyp in children is the juvenile (hamartomatous) polyp, which presents with painless rectal bleeding.
- 5 **C Vilous adenomatous polyp** — Villous adenomatous polyps have a higher risk of malignant transformation compared to other types of polyps.

Theme 23 · Jaundice

- 1 **D Pre-hepatic** — An isolated raised (unconjugated) bilirubin with otherwise normal tests is pre-hepatic, e.g. Gilbert syndrome or haemolysis.

- 2 **A Hepatic (hepatocellular)** — Rifampicin and isoniazid cause drug-induced hepatitis, a hepatocellular jaundice.
- 3 **C Post-hepatic** — Pale stools and dark urine with RUQ pain is obstructive (post-hepatic) jaundice, e.g. a CBD stone.
- 4 **B Hepatic (intrahepatic cholestasis)** — Jaundice in UC suggests primary sclerosing cholangitis, a cholestatic disease of intra- and extrahepatic ducts.

Theme 24 · Antimicrobial Agents

- 1 **H Direct-acting antivirals** — Hepatitis C is now cured in >95% by oral direct-acting antivirals (e.g. sofosbuvir-based regimens); interferon is obsolete.
- 2 **E Ceftriaxone** — Perforated appendicitis needs a third-generation cephalosporin such as ceftriaxone (with metronidazole for anaerobes).
- 3 **I Levofloxacin + metronidazole** — Pouchitis is treated with ciprofloxacin/levofloxacin plus metronidazole, with probiotics for prevention.
- 4 **J Levofloxacin + amoxicillin** — After failed clarithromycin-based eradication, second-line therapy is PPI plus amoxicillin and levofloxacin.

- 5 **K None** — Antibiotics are not given routinely in acute pancreatitis; they are reserved for proven infected necrosis or other infections.
- 6 **F Cefoxitin** — Cefoxitin (a cephamycin, grouped with second-generation cephalosporins) has useful anaerobic activity.

Theme 25 • Malignancies and their tumor markers:

- 1 **C CEA** — Elevated levels of Carcinoembryonic may indicate the presence of colorectal cancer
- 2 **B CA 19-9** — Elevated levels of CA 19-9 may indicate the presence of pancreatic cancer, but it can also be elevated in other conditions such as cholangiocarcinoma and some benign diseases.
- 3 **A Alpha-FP** — Alpha fetoprotein is associated with hepatocellular carcinoma

SET 7 OF 7

THEME 27 RIF masses

- A Abd. Actinomycosis
- B Abd. tuberculosis
- C Acute appendicitis
- D Carcinoma of caecum
- E Crohn's disease
- F Ectopic kidney
- G Germ cell carcinoma
- H Lymphoma
- I Ovarian carcinoma
- J Undescended testis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 25-year-old patient presents with diarrhea, weight loss, and a lump near the midline.

Theme 26 • Salivary glands diseases

- 1 **D Pleomorphic adenoma** — A preauricular lump recurring after parotid surgery with epithelial and stromal cells is recurrent pleomorphic adenoma (from capsule breach or enucleation).
- 2 **I Sjogren's syndrome** — Dry mouth and eyes with bilateral parotid enlargement and arthritis is Sjögren syndrome.
- 3 **A Bacterial sialadenitis** — The patient's symptoms of a painful lump below the left jaw, along with fever and bad oral hygiene, are indicative of an infection in the salivary gland.
- 4 **J Warthin's tumour** — Bilateral parotid swellings with oncocytic epithelium and lymphocytes in an older (smoking) man is Warthin tumour.

- 2 A 45-year-old man with a positive history of surgery and a right iliac fossa scar.
- 3 A 65-year-old male presents with weight loss and bloody stool. On examination, a pale face and a non-tender right iliac fossa mass are noted.
- 4 A patient presenting with multiple discharging sinuses four weeks after appendectomy surgery.
- 5 A patient presents with fever, sweating, itching, and hepatosplenomegaly.
- 6 A patient with a history of an empty scrotum for many years presents with a lump in the right iliac fossa.

Answers • Theme 27

LOCAL EXAMS EMQS

Theme 27 • RIF masses

- 1 **E Crohn's disease** — Chronic diarrhea, weight loss and lump suggests Crohn's disease.
- 2 **C Acute appendicitis** — The presence of a surgical scar in the right iliac fossa and the clinical history of surgery suggests a possible complication related to the appendix.
- 3 **D Carcinoma of caecum** — Elderly + weight loss + bleeding per rectum + RIF mass => Cancer.
- 4 **A Abd. Actinomycosis** — The presentation of multiple sinuses following an appendectomy surgery suggests the possibility of abdominal actinomycosis.
- 5 **H Lymphoma** — Fever, sweating (especially night sweats), and itching (pruritus) are known as 'B symptoms' and are characteristic features of lymphoma. Hepatosplenomegaly is also frequently observed in lymphoma
- 6 **J Undescended testis** — A long-standing empty scrotum with an iliac fossa mass suggests a tumour in an undescended (intra-abdominal) testis.

10 THEMES • 80 SCENARIOS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love EMQs

SET 1 OF 4

THEME 1 Endoscopic Diagnosis

- A Barrett's oesophagus
- B Carcinoma of the oesophagus
- C Gastrointestinal stromal tumour (GIST)
- D Hiatus hernia
- E Linitis plastica
- F Mallory-Weiss tear
- G Oesophageal varices
- H Peptic stricture
- I Rectal cancer
- J Reflux gastritis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Abnormal veins are seen in the lower end of the oesophagus, which are dilated and grape like.
- 2 A smooth, tapering stricture is seen at the lower end of the oesophagus along with inflammation and ulceration.
- 3 A mucosal tear is seen at the cardia with some fresh blood.
- 4 The pylorus is seen to be wide open with plenty of bile in the stomach. The antrum shows streaks of erythema.
- 5 The stomach is difficult to distend and appears to have low capacity. The mucosa appears stretched but no mucosal lesion is seen.
- 6 A submucosal lump is seen. The mucosa is stretched over this lump but is otherwise normal. The mucosal biopsies are normal.
- 7 There is an ulcerated and polypoidal mass at the lower end of the oesophagus causing obstruction.
- 8 There is evidence of gastro-oesophageal reflux disease (GORD) and a linear tongue of erythematous mucosa extending for a few centimetres into the oesophagus proximally.
- 9 The gastro-oesophageal junction is at 34 cm with prolapsing fundal mucosa and increased oesophageal fluid. The diaphragm is located at 39 cm.
- 10 A large polypoidal mass is noted in the lower rectum in a patient not fit for major surgery.

THEME 2 Endoscopic Complications

- A Aspiration
- B Bleeding
- C Dental injury
- D Perforation
- E Sedation overdose

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 80-year-old man who has just had an OGD after local anaesthetic spray has severe coughing and breathing problems after eating a sandwich in the recovery room.
- 2 A 71-year-old patient collapses and is found to be hypotensive after an endoscopic polypectomy.
- 3 A 70-year-old male is rushed to the emergency 8 hours after an endoscopic dilatation of an oesophageal stricture, complaining of severe chest pain and change in voice.
- 4 An 86-year-old woman has severe pain and bleeding from the mouth following an uneventful OGD.
- 5 An 80-year-old woman patient has severe bradycardia and low PO₂ and is difficult to arouse.

THEME 3 Abdominal Pain and Masses – Diagnosis Matching

- A Abdominal aortic aneurysm
- B Acute pancreatitis
- C Biliary colic
- D Carcinoma of caecum
- E Distal small bowel obstruction
- F Incisional hernia
- G Perforated duodenal ulcer
- H Secondary metastases in liver

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 58-year-old man complains of a lump in his abdomen deep to an incision for 2 years. This started 3 months after he had an emergency operation for intestinal obstruction when he underwent a right hemicolectomy. The lump is increasing in size and becomes prominent when he coughs and sits up.
- 2 A 72-year-old man complains of throbbing backache and a feeling of his heartbeat in his 'stomach'. On examination a pulsatile mass can be seen in his epigastrium, which looks full.

- 3** A 35-year-old woman complains of severe colicky abdominal pain of 6 hours duration. It is in the epigastrium and right upper quadrant and radiates to the back. The pain comes and goes, and in between the attacks she is left with a dull ache in the right subcostal region.
- 4** A 74-year-old woman complains of generalised abdominal pain and distension for 36 hours. This started with greenish vomiting, which has now been replaced with brownish faeculent material. She is dehydrated with a distended abdomen and a small tender lump in her right groin of which she is unaware.
- 5** A 62-year-old woman complains of undue shortness of breath while going about her daily activities. On examination she looks very pale with an irregular non-tender lump in her right iliac fossa.
- 6** A 45-year-old man, who admits to being a chronic alcoholic, complains of severe pain in his epigastrium radiating to the back of 8 hours duration. He is sweaty, with a pulse rate of 122 beats per minute, blood pressure of 110/80 mmHg and with a tense, tender abdomen with rigidity.
- 7** A 70-year-old woman presents with jaundice and vague abdominal pain of 6 weeks duration. On examination she is jaundiced with a long midline scar and an enlarged hard nodular liver. The midline scar is from a left hemicolectomy for carcinoma of the splenic flexure.
- 8** A 55-year-old man complains of sudden onset of very severe pain in his epigastrium, radiating to the back, of 6 hours duration. On examination his blood pressure is 130/90 mmHg, pulse rate is 90 beats per minute, the abdominal wall does not move with respiration and he has thoracic respiration. Over the past hour he has been having pain in his right shoulder tip. In the past he has suffered from 'indigestion' for a long time.

Answers • Themes 1-3

BAILEY & LOVE EMQS

Theme 1 • Endoscopic Diagnosis

- 1** **G Oesophageal varices** — Dilated, grape-like veins in the lower oesophagus are oesophageal varices from portal hypertension.
- 2** **H Peptic stricture** — A smooth tapering stricture with oesophagitis and ulceration is a peptic stricture from long-standing reflux.
- 3** **F Mallory-Weiss tear** — A mucosal tear at the cardia with fresh blood after retching is a Mallory-Weiss tear, usually self-limiting.
- 4** **J Reflux gastritis** — A wide-open pylorus with bile in the stomach and antral erythema is bile reflux gastritis.
- 5** **E Linitis plastica** — A contracted, non-distensible stomach with normal-looking mucosa is linitis plastica (diffuse submucosal gastric cancer); deep biopsies are needed.
- 6** **C Gastrointestinal stromal tumour (GIST)** — A submucosal lump with normal overlying mucosa and normal biopsies suggests a GIST; EUS-guided biopsy confirms it.
- 7** **B Carcinoma of the oesophagus** — An ulcerated, polypoid, obstructing mass at the lower oesophagus is oesophageal carcinoma.
- 8** **A Barrett's oesophagus** — A tongue of salmon-pink mucosa extending up from the GOJ in reflux disease is Barrett oesophagus; biopsies show intestinal metaplasia.
- 9** **D Hiatus hernia** — A GOJ at 34 cm with the diaphragmatic pinch at 39 cm means the stomach has slid above the diaphragm: a sliding hiatus hernia.
- 10** **I Rectal cancer** — A large polypoid mass in the lower rectum is rectal cancer until biopsy proves otherwise; in an unfit patient, local excision or radiotherapy may be used.

Theme 2 • Endoscopic Complications

- 1** **A Aspiration** — Topical pharyngeal anaesthesia suppresses the gag reflex; eating too soon after OGD causes aspiration.
- 2** **B Bleeding** — Sudden collapse and hypotension following a therapeutic procedure such as polypectomy are classic for post-procedural bleeding. It's a well-known and immediate risk.
- 3** **D Perforation** — Chest pain and voice change hours after oesophageal dilatation is oesophageal perforation: contrast CT/swallow urgently.
- 4** **C Dental injury** — Older patients with poor dentition are at high risk for dental trauma during scope insertion, particularly if they wear dentures or have brittle teeth. The lack of internal findings suggests an external cause.
- 5** **E Sedation overdose** — Bradycardia, hypoxia and drowsiness after endoscopy in an elderly patient is over-sedation: oxygen, airway support and reversal (flumazenil/naloxone).

Theme 3 • Abdominal Pain and Masses – Diagnosis Matching

- 1** **F Incisional hernia** — The description of a postoperative abdominal lump that increases with coughing or sitting up is characteristic of an incisional hernia, especially following abdominal surgery.
- 2** **A Abdominal aortic aneurysm** — A pulsatile epigastric mass with backache is a classic presentation of an abdominal aortic aneurysm. The 'heartbeat in the stomach' is a hallmark symptom.
- 3** **C Biliary colic** — Intermittent, colicky pain in the right upper quadrant radiating to the back is typical for biliary colic, usually due to gallstones obstructing the biliary tract.

- 4 **E Distal small bowel obstruction** — Foul, feculent vomiting with abdominal distension and a groin lump suggests a strangulated hernia leading to distal small bowel obstruction. The progression to dehydration supports this diagnosis.
- 5 **D Carcinoma of caecum** — Chronic anemia, shortness of breath, and a palpable mass in the right iliac fossa are indicative of carcinoma of caecum, which often presents insidiously due to occult blood loss.

SET 2 OF 4

THEME 4 Diagnoses of Abdominal Wall Hernias

- A Direct inguinal hernia
- B Divarication of recti
- C Epigastric hernia
- D Femoral hernia
- E Incisional hernia
- F Indirect inguinal hernia
- G Paraumbilical hernia
- H Port-site hernia
- I Spigelian hernia
- J Umbilical hernia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 6-month-old infant has a history of lump over the belly-button that increases in size when the baby cries.
- 2 A 26-year-old presents with a lump over his left groin. Examination reveals cough impulse and a reducible hernia extending into the scrotum. Pressure over the internal ring controls the hernia.
- 3 A 60-year-old obese woman has a painful lump just above her umbilicus. Examination confirms a round smooth tender mass with no cough impulse.
- 4 A 78-year-old woman has a suspected enlarged left groin lymph node. Examination reveals a 2-cm hard, tender and irreducible lump below and lateral to the pubic tubercle with no cough impulse.
- 5 A 40-year-old woman describes a gradually increasing lump in the lower, right abdomen. She has had three caesarean sections. There is a Pfannenstiel scar with a reducible mass at its lateral end.
- 6 A 76-year-old man has a 2-year history of small swellings in both groins, causing no real discomfort. Examination confirms reducible hernias, which are not controlled by pressure over the deep inguinal ring.
- 7 A 50-year-old man presents with a lump below but well lateral to the umbilicus. You suspect an inguinal hernia but the mass seems to be slightly too high and too lateral for this diagnosis.
- 8 A 22-year-old fit male presents with a tender, firm, 2-cm nodule in the midline just below the xiphisternum. There is no cough impulse. The patient believes that the lump increases in size on exercise.

- 6 **B Acute pancreatitis** — Severe epigastric pain radiating to the back in a chronic alcoholic with systemic distress is highly suggestive of acute pancreatitis.
- 7 **H Secondary metastases in liver** — Jaundice combined with a history of colectomy and a nodular liver on examination strongly points towards secondary metastases in the liver.
- 8 **G Perforated duodenal ulcer** — Sudden, severe epigastric pain with associated shoulder tip pain and a rigid abdomen is characteristic of a perforated duodenal ulcer, likely causing diaphragmatic irritation from free peritoneal air.

- 9 A 62-year-old labourer is referred with an abdominal hernia. When lying down, no mass is visible but on raising his head from the bed, a broadly based swelling appears between the xiphisternum and umbilicus.
- 10 A 45-year-old woman has a 3-cm firm mass at the umbilicus. She had a laparoscopic cholecystectomy 2 years ago.

THEME 5 Treatment of Abdominal Wall Hernias

- A Wait and watch
- B Laparoscopic inguinal hernia repair (TEP or TAPP)
- C Open Lichtenstein repair
- D Open or laparoscopic mesh repair
- E Non-surgical
- F Open suture or plug repair
- G Open suture or mesh repair

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Uncomplicated primary unilateral inguinal hernia.
- 2 Recurrent inguinal hernia – post-open repair.
- 3 Infantile umbilical hernia.
- 4 Recurrent inguinal hernia – post-laparoscopic repair.
- 5 Divarication of recti.
- 6 Complex incisional hernia in patient known to have extensive adhesions.
- 7 Small paraumbilical hernia.

THEME 6 Named Operations and Techniques in Hernia Surgery

- A Desarda operation
- B Lichtenstein operation
- C Mayo repair
- D Ramirez technique
- E Shouldice operation
- F Stoppa operation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Suture repair of inguinal hernia with double breasting of posterior wall of inguinal canal.

- 2 Open pre-peritoneal mesh repair for complex inguinal hernia.
- 3 Suture repair of paraumbilical hernia with double breasting of anterior sheath.
- 4 Suture repair of inguinal hernia employing external oblique aponeurosis.
- 5 Open mesh repair of inguinal hernia.
- 6 Muscle release to aid abdominal wall closure in large incisional hernia.

Answers • Themes 4–6

BAILEY & LOVE EMQS

Theme 4 • Diagnoses of Abdominal Wall Hernias

- 1 **J Umbilical hernia** — A congenital hernia at the umbilicus in infants is typically an umbilical hernia, exacerbated by crying due to increased intra-abdominal pressure.
- 2 **F Indirect inguinal hernia** — A hernia extending into the scrotum and reducible, controlled at the internal ring, confirms an indirect inguinal hernia, usually congenital.
- 3 **G Paraumbilical hernia** — A paraumbilical hernia occurs just above the umbilicus and often presents as a tender lump, common in obese women.
- 4 **D Femoral hernia** — A lump below and lateral to the pubic tubercle is classic for femoral hernia—more common in elderly women, often mistaken for a lymph node.
- 5 **E Incisional hernia** — Hernias at the site of previous incisions are incisional hernias. The mass at the lateral end of a Pfannenstiel scar supports this.
- 6 **A Direct inguinal hernia** — Bilateral, painless, reducible groin swellings not controlled at the internal ring are characteristic of direct inguinal hernias, common in older men.
- 7 **I Spigelian hernia** — A hernia lateral to the rectus muscle, near the semilunar line, indicates a Spigelian hernia—a rare type often confused with inguinal hernias.
- 8 **C Epigastric hernia** — Small, tender midline lumps between the xiphisternum and umbilicus are epigastric hernias, often fat-containing and more prominent with exertion.
- 9 **B Divarication of recti** — A midline separation of the rectus muscles (divarication of recti) presents as a visible ridge when the patient raises his head, rather than a true hernia.
- 10 **H Port-site hernia** — Post-laparoscopic surgery hernias at trocar sites (especially at the umbilicus) are known as port-site hernias.

Theme 5 • Treatment of Abdominal Wall Hernias

- 1 **C Open Lichtenstein repair** — A primary unilateral inguinal hernia is repaired by open Lichtenstein mesh repair (laparoscopic repair is an alternative).

- 2 **B Laparoscopic inguinal hernia repair (TEP or TAPP)** — A hernia recurring after open repair is best repaired laparoscopically (TEP/TAPP) through unscarred tissue planes.
- 3 **A Wait and watch** — Most infantile umbilical hernias close spontaneously by 3–4 years, so wait and watch; repair if persisting or complicated.
- 4 **C Open Lichtenstein repair** — A hernia recurring after laparoscopic repair is repaired by an open anterior (Lichtenstein) approach through virgin tissue.
- 5 **E Non-surgical** — Divarication of the recti is not a true hernia and does not strangulate, so it is managed non-surgically.
- 6 **G Open suture or mesh repair** — A complex incisional hernia with extensive adhesions is best repaired open, with mesh (or suture for small defects).
- 7 **F Open suture or plug repair** — A small paraumbilical hernia is repaired by open suture (Mayo) or a small mesh/plug.

Theme 6 • Named Operations and Techniques in Hernia Surgery

- 1 **E Shouldice operation** — The Shouldice repair is a suture repair that double-breasts (imbricates) the transversalis fascia of the posterior wall in layers.
- 2 **F Stoppa operation** — The Stoppa operation places a large preperitoneal mesh through a lower midline incision for complex, bilateral or recurrent groin hernias.
- 3 **C Mayo repair** — The Mayo repair double-breasts (overlaps) the rectus sheath above and below an umbilical/paraumbilical defect.
- 4 **A Desarda operation** — The Desarda repair reinforces the posterior wall with an undetached strip of external oblique aponeurosis, without mesh.
- 5 **B Lichtenstein operation** — The Lichtenstein operation is the standard open tension-free mesh repair of an inguinal hernia.
- 6 **D Ramirez technique** — The Ramirez technique (anterior component separation) releases the external oblique to allow midline closure of large incisional hernias.

SET 3 OF 4

THEME 7 Oesophageal Disorders – Diagnosis Matching

- A Achalasia of cardia
- B Carcinoma of oesophagus
- C Diffuse oesophageal spasm
- D Gastro-oesophageal reflux disease (GORD)
- E Pharyngeal diverticulum
- F Rolling hiatus hernia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 68-year-old woman complains of recurrent cough that is worse at night when she is woken up from her sleep. She is embarrassed by halitosis. Her general practitioner (GP) has treated her for chest infection several times with antibiotics. Examination shows some signs of basal pneumonitis.

- 2 A 48-year-old man complains of intermittent dysphagia for almost 1 year. He has some postprandial discomfort, which is relieved by belching and occasional regurgitation of food. This is associated with retrosternal pain and pressure sensation. An ECG is normal. Chest x-ray shows a gas bubble behind the heart.
- 3 A 42-year-old man complains of dysphagia with retrosternal chest pain. This has been intermittent. On one occasion the pain was very severe with pain radiating along the arms. He was admitted as an emergency to the medical ward where his ECG and cardiac enzymes were normal.
- 4 A 72-year-old man, a long-standing smoker, complains of progressive dysphagia with food sticking behind his mid-sternum for 3 months. The problem started with solids, but now he has difficulty with liquids. He has lost 14 kg of weight during this time. On examination the weight loss is obvious from his loose, ill-fitting clothes.
- 5 A 35-year-old man complains of dysphagia of almost 2 years duration. Occasionally he has regurgitation of foul-smelling stale food and frothy sputum. He has been treated for chest infection a few times by his GP. He has lost some weight and looks slightly malnourished.
- 6 A 42-year-old woman complains of retrosternal burning pain, which she refers to as heartburn, epigastric discomfort and repeated regurgitation. Her symptoms are worse after fatty and spicy foods and by stooping. She has also experienced painful swallowing when she felt an acidic taste in her throat. On examination she is obese with signs of chest infection.
- 2 A 55-year-old man complains of epigastric pain for 6 months, the pain not related to food. His appetite has been poor and he has lost 10 kg of weight during this period. There was nothing to find in his abdomen except for evidence of weight loss. An OGD showed diffuse mucosal thickening of the entire pylorus with some ulceration. Histology showed lymphoid hyperplasia with lymphoid cells expanding the lamina propria.
- 3 An 80-year-old woman presents with severe itching and her relatives noticed that she was becoming yellow over the last couple of months. Recently she has had vomiting, the vomitus not being bile stained. On examination she has lost a considerable amount of weight and has scratch marks all over her body and a smooth globular mass in the right upper quadrant of her abdomen.
- 4 A 69-year-old man presents with acute haematemesis. He has been well until his presentation, except that recently he has had a feeling of early satiety after small amounts of food. An urgent OGD showed a large submucosal mass with an ulcerated area, which was the site of bleeding.
- 5 A 52-year-old woman complains of epigastric pain on and off for 3 years for which she took medicines from across the counter. Almost 6 months ago she underwent a repeat OGD and biopsy that showed *H. pylori* infection in a gastric ulcer for which she was treated. She was well after that, but recently the pain has returned. Her haemoglobin is 11.9. Repeat OGD shows a 2-cm diameter punched-out ulcer on the lesser curve with overhanging edges.
- 6 A 65-year-old man complains of anorexia and weight loss for 3 months. He has intermittent vomiting associated with malaise. On examination he looks anaemic and has a firm mobile mass in the epigastrium and right hypochondrium.
- 7 A 35-year-old man, a bus driver, complains of epigastric pain radiating to the back on and off for several months. He has some backache, for which he took some NSAIDs, and occasionally consumes alcohol. Following a bout of drinking, he vomited and the vomitus was streaked with blood.
- 8 A 60-year-old man complains of incessant vomiting for 2 weeks. The vomitus contains stale food ingested a few days ago. There is no abdominal pain, although many years ago he suffered from prolonged 'indigestion' for which he took antacids from across the counter on and off for years. On examination he has sunken eyes, dry tongue and loss of skin turgor, and his epigastrium looks full. While being examined he had an attack of carpo-pedal spasm.

THEME 8 Gastric and Perigastric Disorders – Diagnosis Matching

- A Benign gastric ulcer
- B Carcinoma of stomach
- C Gastric lymphoma
- D Gastric outlet obstruction
- E Gastritis
- F Leiomyoma of stomach
- G Perforated duodenal ulcer
- H Periampullary carcinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 48-year-old man, a heavy smoker, has been admitted with sudden onset of very severe pain in the epigastrium radiating to the back and right shoulder tip of 3 hours duration. His pulse is 88 beats per minute and his blood pressure (BP) is 120/70 mmHg; his abdomen has board-like rigidity and does not move with respiration, which is mainly thoracic.

THEME 9 Surgical Procedures – Bariatric & Metabolic Surgery

- A** Bilio-pancreatic diversion (BPD)
- B** Gastric banding
- C** Ileal transposition
- D** Roux-en-Y gastric bypass
- E** Sleeve gastrectomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 50-year-old man weighing 105 kg and with a height of 180 cm presents with a history of progressive and poorly controlled diabetes for 15 years. He was shifted from oral hypoglycaemics to insulin but has shown poor compliance to treatment. He is now beginning to have alteration in renal functions and eye changes.
- 2** A 30-year-old married woman presents with a weight of 110 kg and a height of 155 cm. She has been diabetic for the past 10 years and is on anti-hypertensives for the past 5 years with a need to escalate the doses of both medications.

- 3** A 20-year-old man, a college student, presents with a history of weight gain over the past 5 years. He has a BMI of 40 and is particularly worried about his physique. He has tried dieting and exercise, appears well motivated, and wants a safe procedure with regular follow-up.
- 4** A 35-year-old woman who underwent a sleeve gastrectomy 3 years ago presents with weight regain and relapse of diabetes. She presently has a BMI of 52 and is otherwise fit and healthy.
- 5** A 45-year-old man presents with a BMI of 45. He has been obese for the past 10 years with no medical comorbidity except for snoring during sleep for the past 1 year.

Answers • Themes 7–9

BAILEY & LOVE EMQS

Theme 7 • Oesophageal Disorders – Diagnosis Matching

- 1** **E Pharyngeal diverticulum** — Nocturnal cough, halitosis and recurrent aspiration pneumonia in an elderly woman suggest a pharyngeal pouch (Zenker diverticulum).
- 2** **F Rolling hiatus hernia** — Postprandial fullness relieved by belching with a gas bubble behind the heart on CXR is a rolling (para-oesophageal) hiatus hernia.
- 3** **C Diffuse oesophageal spasm** — Sudden, severe chest pain radiating to the arms with normal cardiac workup points to diffuse oesophageal spasm, which can mimic angina. Dysphagia and non-cardiac chest pain are key features.
- 4** **B Carcinoma of oesophagus** — Progressive dysphagia from solids to liquids, significant weight loss, and age/smoking history are classic for oesophageal carcinoma. The mid-sternal location suggests a mid-oesophageal lesion.
- 5** **A Achalasia of cardia** — Two years of dysphagia with regurgitation of stale food, aspiration and weight loss in a young man is achalasia.
- 6** **D Gastro-oesophageal reflux disease (GORD)** — Typical heartburn, regurgitation, worse with posture and food, along with acidic taste and odynophagia (painful swallowing), point clearly to GORD. Obesity is a common risk factor.

Theme 8 • Gastric and Perigastric Disorders – Diagnosis Matching

- 1** **G Perforated duodenal ulcer** — The sudden, severe epigastric pain, board-like rigidity, and shoulder tip pain (suggesting diaphragmatic irritation) are classic for a perforated duodenal ulcer. Smoking is an important risk factor.
- 2** **C Gastric lymphoma** — Chronic epigastric pain, weight loss, and diffuse mucosal thickening of the pylorus with lymphoid hyperplasia on biopsy point towards gastric lymphoma.
- 3** **H Periampullary carcinoma** — Obstructive jaundice with severe pruritus, vomiting of non-bile-stained contents, weight

loss, and a palpable RUQ mass (suggesting Courvoisier's sign) are indicative of periampullary carcinoma.

- 4** **F Leiomyoma of stomach** — A large submucosal gastric mass with an ulcerated area that bleeds and causes early satiety is characteristic of a leiomyoma of the stomach.
- 5** **A Benign gastric ulcer** — A recurrent punched-out ulcer on the lesser curve with overhanging edges in a patient with a history of *H. pylori* infection is consistent with a benign gastric ulcer.
- 6** **B Carcinoma of stomach** — Anorexia, weight loss, intermittent vomiting, anemia, and a firm, mobile mass in the epigastrium/right hypochondrium strongly suggest carcinoma of the stomach.
- 7** **E Gastritis** — Epigastric pain radiating to the back, associated with NSAID use, occasional alcohol intake, and minor bleeding in the vomitus suggests acute gastritis.
- 8** **D Gastric outlet obstruction** — Incessant vomiting of stale food, signs of dehydration, epigastric fullness, and carpo-pedal spasm (due to electrolyte imbalance) indicate gastric outlet obstruction, likely from peptic scarring.

Theme 9 • Surgical Procedures – Bariatric & Metabolic Surgery

- 1** **C Ileal transposition** — Poorly controlled long-standing diabetes with early end-organ damage in a man who is only mildly obese suits a metabolic operation such as ileal transposition (GLP-1 effect).
- 2** **D Roux-en-Y gastric bypass** — Severe obesity (BMI ~46) with worsening diabetes and hypertension: Roux-en-Y gastric bypass gives the best metabolic remission.
- 3** **B Gastric banding** — A young, motivated patient with BMI 40 and no comorbidity wanting a reversible, adjustable option may choose gastric banding, which needs close follow-up.

- 4** **A Bilio-pancreatic diversion (BPD)** — Weight regain and relapse of diabetes after sleeve gastrectomy with BMI 52 is treated by revision to a malabsorptive procedure such as biliopancreatic diversion.

SET 4 OF 4

THEME 10 Liver and Biliary Disorders – Diagnosis Matching

- A** Amoebic liver abscess
- B** Ascending cholangitis
- C** Budd–Chiari syndrome
- D** Caroli's disease
- E** Focal nodular hyperplasia
- F** Haemangioma
- G** Hepatic adenoma
- H** Hepatocellular carcinoma (HCC)
- I** Hydatid liver disease
- J** Primary biliary cirrhosis
- K** Primary sclerosing cholangitis
- L** Pyogenic liver abscess
- M** Secondary liver metastasis
- N** Simple cystic disease
- O** Viral hepatitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 30-year-old woman has a history of recurrent attacks of fever with rigors, right upper quadrant pain and jaundice with itching. Biochemistry shows a jaundice of obstructive nature. CT scan shows intrahepatic ductal dilatation with stones.
- 2** A 40-year-old woman presents with recurrent episodes of right upper quadrant pain with jaundice. Biochemistry shows an obstructive pattern of jaundice. Five years ago she underwent a panproctocolectomy for ulcerative colitis and has been well since, except for these attacks. ERCP shows irregular narrowed intra- and extrahepatic bile ducts.
- 3** A 50-year-old woman complains of general malaise, lethargy, pruritus and jaundice, the latter being present over the last 3 months. The LFTs show a rise in bilirubin, transaminases and prothrombin time. She has had recurrent small haematemesis and has ascites.
- 4** A 30-year-old woman complains of abdominal discomfort and distension. She has had three episodes of small haematemesis in the past 6 months. On examination she has hepatomegaly and ascites. All the LFTs are deranged. CT scan of the liver shows a large congested liver.
- 5** A fit-looking 50-year-old man complains of recent onset of dull aching right upper quadrant pain of 3 to 4 months' duration. Examination shows no abnormality. Liver function tests are normal, as is an upper gastrointestinal endoscopy. An ultrasound of the liver shows a 6 cm solitary cystic lesion.
- 5** **E Sleeve gastrectomy** — BMI 45 with only snoring: sleeve gastrectomy is effective with fewer long-term nutritional problems.
- 6** A 25-year-old man complains of generally feeling unwell with fever and weight loss. He has had blood-stained motions on and off for the last 6 weeks after he returned from the Indian subcontinent where he was working for 6 months. On examination he has a tender right upper quadrant with hepatomegaly. Ultrasound shows a hypoechoic cavity in the right lobe with ill-defined borders.
- 7** A 50-year-old woman has had recurrent attacks of colicky right upper quadrant and epigastric pain with jaundice, high temperature with rigors and itching. 10 months ago she underwent an uneventful laparoscopic cholecystectomy. On examination she is jaundiced with scratch marks all over her body and hepatomegaly. She has raised serum bilirubin and the ALP is 1100 IU/L. Ultrasound shows a dilated common bile duct with stones.
- 8** A 35-year-old man presents with general ill health, weight loss, anorexia and malaise for several weeks. He has developed jaundice for the last 2 weeks; he has no pruritus. Abdominal examination shows a tender hepatomegaly. Liver function tests show raised bilirubin and transaminases.
- 9** A 75-year-old man, an insulin-dependent diabetic for 40 years, complains of anorexia, fever, malaise and right upper quadrant discomfort. On examination there is weight loss and tender hepatomegaly. Ultrasound shows a multiloculated cystic mass, a finding confirmed on CT scan.
- 10** A 50-year-old man, a native of Cyprus, presents with a painful mass in the right upper quadrant. The pain is a continuous dull ache and has the features of a mass arising from the right lobe of the liver. The blood count shows eosinophilia. CT scan shows a smooth space-occupying lesion with multiple septa within it.
- 11** A 35-year-old woman who is on the contraceptive pill presents with right upper quadrant aching pain. She is fit without any physical findings. Her LFTs and other blood tests are normal. Ultrasound of the liver shows a single well-demarcated hyperechoic mass and CT scan demonstrates a well-circumscribed vascular solid tumour.
- 12** A 40-year-old man presents with dull, persistent upper abdominal pain, weakness, weight loss and occasional fever. He had one episode of haematemesis. Abdominal examination shows an enlarged liver with a mass in the right lobe. Liver function tests show elevation of the transaminases and much-raised alpha-fetoprotein (AFP).

13 A 60-year-old man presents with dragging pain in the right upper quadrant for 3 months and weight loss. On examination he is slightly jaundiced. He has an enlarged liver. Three years ago he underwent a right hemicolectomy for cancer of the caecum. Liver function tests show elevation of all the parameters. Ultrasound shows a solid mass in the right lobe, and a CECT confirms the mass with lack of enhancement.

14 In a 30-year-old woman, during laparoscopic cholecystectomy, the surgeon noticed that the under-surface of the liver has several lesions that are blue in colour.

15 A 45-year-old woman underwent an ultrasound for suspected biliary pain. The ultrasound of the biliary tract was normal but showed a solid lesion in the liver. A subsequent CECT scan showed a vascular lesion surrounding a solid mass with central scarring.

Answers • Theme 10

BAILEY & LOVE EMQS

Theme 10 • Liver and Biliary Disorders – Diagnosis Matching

- 1** **D** **Caroli's disease** — Recurrent cholangitis with intrahepatic duct dilatation and stones is Caroli disease (congenital saccular dilatation of intrahepatic ducts).
- 2** **K** **Primary sclerosing cholangitis** — Cholestatic jaundice after colectomy for UC with beaded, narrowed intra- and extra-hepatic ducts on ERCP is primary sclerosing cholangitis.
- 3** **J** **Primary biliary cirrhosis** — Pruritus, jaundice and progression to portal hypertension in a middle-aged woman suggest primary biliary cholangitis (cirrhosis); antimitochondrial antibodies.
- 4** **C** **Budd–Chiari syndrome** — Hepatomegaly, ascites and a large congested liver in a young woman suggest hepatic venous outflow obstruction: Budd–Chiari syndrome.
- 5** **N** **Simple cystic disease** — A solitary thin-walled cystic liver lesion with normal LFTs is a simple liver cyst; treat (deroofting) only if symptomatic.
- 6** **A** **Amoebic liver abscess** — Fever, tender hepatomegaly and a hypoechoic cavity after dysentery acquired in the Indian sub-continent is an amoebic liver abscess: metronidazole.
- 7** **B** **Ascending cholangitis** — Charcot triad (pain, jaundice, rigors) with CBD stones after cholecystectomy is ascending cholangitis from retained stones: antibiotics and ERCP.

- 8** **O** **Viral hepatitis** — Malaise and anorexia followed by jaundice with tender hepatomegaly and high transaminases (no pruritus) is viral hepatitis.
- 9** **L** **Pyogenic liver abscess** — Fever and tender hepatomegaly with a multiloculated collection in an elderly diabetic is a pyogenic liver abscess: drainage and antibiotics.
- 10** **I** **Hydatid liver disease** — A septated (daughter-cyst) liver lesion with eosinophilia in someone from a sheep-farming region is a hydatid cyst.
- 11** **G** **Hepatic adenoma** — A well-circumscribed vascular solid tumour in a young woman on the pill is a hepatic adenoma (risk of bleeding and malignant change).
- 12** **H** **Hepatocellular carcinoma (HCC)** — A liver mass with very high AFP and a haematemesis (portal hypertension) is hepatocellular carcinoma.
- 13** **M** **Secondary liver metastasis** — A solid liver mass after resection of a caecal cancer is a colorectal liver metastasis.
- 14** **F** **Haemangioma** — Blue, compressible lesions on the liver surface are haemangiomas, the commonest benign liver tumour.
- 15** **E** **Focal nodular hyperplasia** — A hypervascular liver lesion with a central scar is focal nodular hyperplasia, which needs no treatment.

7 THEMES • 35 SCENARIOS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford EMQs

SET 1 OF 2

THEME 1 Investigating hepatobiliary complaints

- A** Abdominal X-ray
- B** Barium meal and follow through
- C** CT scan of the abdomen
- D** CT cholangiogram
- E** Endoscopic retrograde cholangiopancreatography (ERCP)
- F** Endoscopic ultrasound scan (biliary)
- G** OGD
- H** Magnetic resonance cholangiopancreatography (MRCP)
- I** Upper abdominal ultrasound scan

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 37-year-old woman has had 3 days of increasing epigastric pain, nausea, and fever. She is tender in the right upper quadrant of the abdomen. Investigations reveal a WBC of $17 \times 10^9/L$, a CRP of 25 mg/L, and an aspartate transaminase of 67 IU/L.
- 2** A 62-year-old man has had epigastric pain developing jaundice, pale stools, and dark urine for 1 week. An ultrasound scan has shown gallstones in the gallbladder and a common bile duct diameter of 10 mm.

- 3 A 72-year-old woman is seen in clinic following an attack of acute pancreatitis. A recent ultrasound scan has demonstrated gallstones in the gallbladder and a probable stone in the common bile duct. She is being considered for a laparoscopic cholecystectomy.
- 4 A 79-year-old man has had intermittent episodes of colicky epigastric pain, abdominal distension, and vomiting for 2 weeks. His abdomen is distended, tympanitic, but there is no evidence of tenderness or peritonism.
- 5 A 72-year-old woman has had persistent and increasing epigastric pain with a loss of appetite and some pain after eating for 5 weeks. Abdominal examination is normal.

THEME 2 Hepatobiliary anatomical structures

- A Ampulla of Vater
- B Pancreatic duct
- C Common bile duct
- D Common hepatic duct
- E Cystic duct
- F Gallbladder

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient with recurrent episodes of epigastric and right upper quadrant pain after eating, typically lasting 30 min to 1 h.
- 2 During a laparoscopic cholecystectomy the surgeon should identify and divide this structure which forms the inferior side of Calot's triangle.
- 3 During an ERCP the endoscopist identifies and passes a cannula into this structure to take the required images.
- 4 A 48-year-old woman has obstructive jaundice due to Mirizzi's syndrome. There is external compression of this structure.
- 5 A patient admitted with pancreatitis undergoes an ERCP to remove stones from this structure.

THEME 3 Hepatobiliary tumours

- A Adenocarcinoma of pancreas
- B Cholangiocarcinoma
- C Gastrinoma
- D Hepatocellular carcinoma
- E Insulinoma
- F Liposarcoma
- G Mucinous cystadenocarcinoma of pancreas
- H Pancreatic blastoma
- I VIPoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 37-year-old man has 9 months of worsening epigastric pain, weight loss, and loose stools despite acid reduction; gastroscopy shows multifocal ulceration of the stomach and duodenum without masses.
- 2 A 76-year-old man has 4 months of 5 kg weight loss and anorexia with 10 days of painless jaundice.
- 3 A 32-year-old woman has 18 months of episodic lethargy, sweating, and aggressive behaviour precipitated by hunger and relieved by eating.
- 4 A 35-year-old woman has 12 months of prolonged watery diarrhoea, lethargy, and dehydration with hyponatraemia and hypokalaemia.
- 5 A 71-year-old man drinks 35 units of alcohol per week, has hepatitis C, and has 2 months of anorexia, weight loss, and epigastric fullness.

THEME 4 Gastrointestinal investigations

- A Abdominal ultrasound scan
- B Barium enema (double contrast)
- C Capsular endoscopy
- D Colonoscopy
- E CT colonography ('CT colonoscopy')
- F CT scan of the abdomen
- G Endoscopic ultrasound scan
- H OGD with duodenal biopsy
- I OGD with gastric biopsy
- J OGD with oesophageal biopsy
- K Small bowel follow through

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 34-year-old woman has abdominal discomfort and loose stools after food with a weight loss of 5 kg over 12 months and has developed poor skin and hair texture. She has haemoglobin of 8.3 g/dL and ferritin of 7 mg/dL.
- 2 A 63-year-old man is under surveillance for Barrett's metaplasia.
- 3 A 24-year-old man has had progressive, intermittent right iliac fossa pain with variable loose stools and weight loss of 10 kg for 2 years.
- 4 An 84-year-old woman with severe chronic obstructive pulmonary disease and chronic renal impairment on long-term oral steroids has a palpable right iliac fossa mass. Her haemoglobin is 7.7 g/dL and ferritin 9 mg/dL.
- 5 A 63-year-old man admitted with upper abdominal pain and melaena for 24 h. His heart rate is 108 bpm and blood pressure is 98/45 mmHg.

THEME 5 Gastrointestinal anatomy

- A Body of stomach
- B Distal oesophagus / gastro-oesophageal junction
- C Fundus of stomach
- D Anterior wall of first part of duodenum
- E Gastric antrum (pre-pyloric)
- F Posterior wall of first part of duodenum
- G Second part of the duodenum

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 68-year-old man with a 25-pack-year history of smoking takes regular ibuprofen for headaches. He has had 2 weeks of constant, worsening epigastric pain radiating to the middle of his back and has 6h of passing dark red and black stools with 2h of vomiting clotted red blood.

- 2 A 24-year-old woman has several years of worsening difficult, painful swallowing which is worse with fluids than solids, and has recently developed nocturnal coughing and wheezing.
- 3 A 53-year-old man undergoes an OGD to investigate dyspepsia. *Helicobacter pylori* is suspected as the cause and biopsies are taken from this location.
- 4 A 28-year-old man has abdominal discomfort, weight loss, and diarrhoea. A serum tissue transglutaminase test is positive. An OGD is performed to establish a diagnosis and multiple biopsies are taken from this site.
- 5 A 45-year-old man has been vomiting violently for 3h after a take-away meal and the last three episodes of vomiting have been accompanied by streaks of fresh red blood.

Answers · Themes 1–5

OXFORD EMQS

Theme 1 · Investigating hepatobiliary complaints

- 1 **I Upper abdominal ultrasound scan** — Ultrasound scan is the investigation of choice for suspected acute cholecystitis.
- 2 **H Magnetic resonance cholangiopancreatography (MRCP)** — MRCP is non-invasive and best for imaging a dilated bile duct in obstructive jaundice.
- 3 **E Endoscopic retrograde cholangiopancreatography (ERCP)** — A probable CBD stone before cholecystectomy should be cleared first; ERCP confirms and extracts it (MRCP/EUS if the stone is uncertain).
- 4 **C CT scan of the abdomen** — CT scan of the abdomen is most appropriate to evaluate possible bowel obstruction.
- 5 **G OGD** — OGD provides direct visualization and biopsy for persistent epigastric pain.

Theme 2 · Hepatobiliary anatomical structures

- 1 **F Gallbladder** — Postprandial RUQ pain lasting 30–60 min is biliary colic from gallstones impacting in the gallbladder neck (Hartmann pouch).
- 2 **E Cystic duct** — The cystic duct forms the inferior border of Calot's triangle and must be divided.
- 3 **A Ampulla of Vater** — The ampulla of Vater is cannulated during ERCP for imaging of the biliary tree.
- 4 **D Common hepatic duct** — In Mirizzi syndrome a stone in the cystic duct or Hartmann pouch compresses the common hepatic duct from outside.
- 5 **C Common bile duct** — In gallstone pancreatitis ERCP clears stones from the common bile duct.

Theme 3 · Hepatobiliary tumours

- 1 **C Gastrinoma** — Gastrinoma causes excess gastric acid leading to multifocal peptic ulceration (Zollinger–Ellison syndrome).

- 2 **A Adenocarcinoma of pancreas** — Pancreatic adenocarcinoma presents with painless jaundice and weight loss from a head-of-pancreas mass.
- 3 **E Insulinoma** — Insulinoma causes fasting hypoglycaemia with symptoms relieved by glucose (Whipple's triad).
- 4 **I VIPoma** — VIPoma produces secretory diarrhoea, hypokalaemia, and dehydration due to excess VIP.
- 5 **D Hepatocellular carcinoma** — Hepatocellular carcinoma arises in chronic liver disease, causing anorexia, weight loss, and epigastric fullness.

Theme 4 · Gastrointestinal investigations

- 1 **H OGD with duodenal biopsy** — The clinical features in this young woman are suggestive of coeliac disease; the most appropriate investigation is an OGD with duodenal biopsy.
- 2 **J OGD with oesophageal biopsy** — Surveillance for Barrett's metaplasia is best achieved with an OGD with oesophageal biopsy.
- 3 **D Colonoscopy** — Chronic RIF pain, diarrhoea and weight loss in a young man suggest ileocaecal Crohn disease; colonoscopy with ileal intubation and biopsy confirms it.
- 4 **E CT colonography ('CT colonoscopy')** — Iron-deficiency anaemia with a RIF mass in a frail, high-risk patient: CT colonography is a safe, less invasive way to assess the colon.
- 5 **I OGD with gastric biopsy** — Melaena with tachycardia is an upper GI bleed; OGD identifies and treats the source (with gastric biopsy as needed).

Theme 5 · Gastrointestinal anatomy

- 1 **F Posterior wall of first part of duodenum** — NSAID use, back pain and a major bleed suggest a posterior duodenal ulcer eroding the gastroduodenal artery.
- 2 **B Distal oesophagus / gastro-oesophageal junction** — Progressive dysphagia worse for liquids with nocturnal cough is achalasia: failure of relaxation of the distal oesophagus.

- 3 **E Gastric antrum (pre-pyloric)** — Pylori lives beneath the mucus of the pre-pyloric antrum, so biopsies for the urease test are taken there.
- 4 **G Second part of the duodenum** — Positive tTG suggests coeliac disease; confirmation needs multiple biopsies from the second part of the duodenum (villous atrophy).

SET 2 OF 2

THEME 6 Operations for gastrointestinal complaints

- A Anti-reflux fundoplication
- B Cardiomyotomy ('Heller's operation')
- C Duodenotomy and repair
- D Gastrojejunal bypass
- E Hepaticojejunostomy
- F Oesophagectomy
- G Pancreaticoduodenectomy
- H Partial gastrectomy
- I Total gastrectomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 27-year-old woman has had several years of progressive dysphagia and a 'bird's beak' appearance on barium swallow. Balloon dilatation has failed and she requires surgery.
- 2 A 47-year-old man has longstanding proven gastro-oesophageal reflux disease and has severe persistent and debilitating symptoms despite maximal medical therapy.
- 3 A 52-year-old man has ongoing melaena in hospital. He has a proven proximal duodenal ulcer and has continued bleeding despite two episodes of endoscopic treatment.
- 4 A 67-year-old man with no other medical co-morbidity has a carcinoma of the body of the stomach staged as T2 N0.
- 5 A 74-year-old man with invasive adenocarcinoma of the head of the pancreas involving the superior mesenteric vessels.

- 5 **B Distal oesophagus / gastro-oesophageal junction** — Haematemesis after forceful vomiting is a Mallory–Weiss tear at the gastro-oesophageal junction.

THEME 7 Gastrointestinal symptoms

- A Adenocarcinoma
- B Chronic intestinal ischaemia
- C Diffuse oesophageal spasm
- D Gastro-oesophageal reflux disease
- E Laceration (traumatic)
- F Myocardial infarction
- G Neuroendocrine tumour ('carcinoid type')
- H Peptic ulcer
- I Pharyngeal pouch

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 25-year-old man undergoes an operation for suspected appendicitis. However, an inflamed Meckel's diverticulum is found and excised. Histopathological examination identifies another pathology within the diverticulum in addition to the inflammation.
- 2 A 22-year-old man with small volume haematemesis following a prolonged episode of vomiting during an evening of binge drinking.
- 3 A 76-year-old man with a history of smoking has a 2h history of acute onset epigastric and chest pain. His abdomen is non-tender and his pulse rate is 100bpm.
- 4 A 74-year-old man with months of progressive dysphagia, weight loss of 4kg, and anorexia.
- 5 A 38-year-old man who smokes ten cigarettes/day has retrosternal burning pain after meals especially when lying down. He has some pain on swallowing with a sensation of food becoming 'stuck' in his throat.

Answers · Themes 6–7

OXFORD EMQS

Theme 6 · Operations for gastrointestinal complaints

- 1 **B Cardiomyotomy ('Heller's operation')** — Failed balloon dilatation in a patient with a 'bird's beak' appearance (achalasia) necessitates a cardiomyotomy (Heller's operation).
- 2 **A Anti-reflux fundoplication** — Severe reflux disease refractory to medical therapy is best managed with an anti-reflux fundoplication.
- 3 **C Duodenotomy and repair** — A bleeding proximal duodenal ulcer that is refractory to endoscopic treatment requires duodenotomy and repair.

- 4 **I Total gastrectomy** — A T2 N0 carcinoma of the gastric body is curable by radical resection; a body tumour needs total gastrectomy with D2 lymphadenectomy.
- 5 **D Gastrojejunal bypass** — Pancreatic head cancer encasing the SMA/SMV is unresectable; palliate obstruction with a gastrojejunal (± biliary) bypass or stents.

Theme 7 · Gastrointestinal symptoms

- 1 **G Neuroendocrine tumour ('carcinoid type')** — The small bowel (including Meckel diverticulum) is the commonest site of neuroendocrine (carcinoid) tumours, often found incidentally.

- 2 E Laceration (traumatic)** — Forceful vomiting during binge drinking can result in a Mallory–Weiss tear, which is a traumatic laceration.
- 3 F Myocardial infarction** — In a smoker with acute epigastric and chest pain but a benign abdominal exam, myocardial infarction must be considered.
- 4 A Adenocarcinoma** — Progressive dysphagia with weight loss in an older patient is highly suggestive of adenocarcinoma.
- 5 D Gastro-oesophageal reflux disease** — Postprandial retrosternal burning worse lying down, with odynophagia, is gastro-oesophageal reflux disease (consider OGD for dysphagia).

26 THEMES · 130 SCENARIOS · AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead EMQs

SET 1 OF 6

THEME 1 Investigation of abdominal pain

- A CT scan
- B Erect chest X-ray
- C Faecal elastase level
- D Laparoscopy
- E Mesenteric angiography
- F Serum amylase
- G Supine chest X-ray
- H Troponin levels
- I Urgent laparotomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 29-year-old man is admitted to the emergency department with sudden-onset, severe abdominal pain radiating to the back and with vomiting. On examination, there is tenderness and guarding in the epigastrium. His pulse is 118 beats/min and his blood pressure is 96/68 mmHg.
- 2** A 74-year-old woman with a history of osteoarthritis presents with acute severe abdominal pain and vomiting. On examination, the abdomen is rigid and bowel sounds are absent.
- 3** A 24-year-old woman presents with sudden-onset right iliac fossa pain with nausea and vomiting. She was admitted last year with similar symptoms, which turned out to be torsion of an ovarian cyst. Examination reveals tenderness and guarding in the right iliac fossa.
- 4** A 69-year-old man presents with a 2-month history of severe central abdominal pain occurring 20 minutes after eating. He is reluctant to eat nowadays and has dropped a waist size. He has a medical history of intermittent claudication. Examination of the abdomen is unremarkable.
- 5** A 69-year-old man with a long history of gastro-oesophageal reflux presents with worsening dysphagia to solids, epigastric discomfort, and weight loss. An urgent endoscopy shows a mass lesion in the lower third of the oesophagus.

THEME 2 Intestinal obstruction

- A Adhesions
- B Bezoar
- C Colonic malignancy
- D Crohn's disease stricture
- E Faecal impaction
- F Gallstone ileus
- G Intussusception
- H Paralytic ileus
- I Sigmoid volvulus
- J Strangulated hernia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 72-year-old man presents with abdominal pain and distension. On examination, the pain is worse in the lower abdomen. An abdominal X-ray shows a single loop of dilated bowel that turns back on itself.
- 2** A 42-year-old woman presents with a 24-hour history of central abdominal pain and vomiting. On examination, she has a midline laparotomy scar but no other abnormality.
- 3** A 72-year-old woman presents with a 12-hour history of colicky abdominal pain and distension. An abdominal X-ray shows multiple loops of dilated small bowel and gas in the biliary tree.
- 4** A 66-year-old man underwent a semi-elective bowel resection for a caecal tumour 2 days ago. He is now vomiting and has not passed flatus or stool since the operation. On examination, his abdomen is distended but non-tender and bowel sounds are absent.
- 5** A 14-year-old girl is brought by her parents to the emergency department with a 12-month history of indigestion, nausea, and stomach upsets. She has a history of psychiatric problems and is under regular follow-up. Abdominal examination is unremarkable. A barium swallow shows an irregular, round filling defect in the duodenum.

THEME 3 Management of hernias

- A Elective repair
- B Emergency repair
- C Observation only
- D Prompt repair
- E Urgent repair

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 54-year-old man has a long-standing hernia above and medial to the pubic tubercle that does not descend into the scrotum. The hernia is easily reducible and the patient is asymptomatic.
- 2 A 75-year-old woman has a hernia in her left groin that is below and lateral to the pubic tubercle. On examination, the hernia is tender, erythematous, and irreducible, and the patient is complaining of colicky abdominal pain and vomiting.
- 3 A 47-year-old man has a hernia that is above and medial to the pubic tubercle and descends into the scrotum on standing. On examination, the hernia is reducible and the patient is asymptomatic.
- 4 A 62-year-old woman has a hernia that is below and lateral to the pubic tubercle. On examination, the patient is well and the hernia is painless but irreducible.
- 5 A 57-year-old man has a scrotal hernia that has only been present for a fortnight. On examination, the hernia is irreducible but painless.

THEME 4 Abdominal incisions

- A Battle incision
- B Gridiron incision
- C Kocher incision
- D Lanz incision
- E Midline laparotomy
- F Paramedian incision
- G Pfannenstiel incision
- H Rooftop incision
- I Rutherford Morrison incision
- J Thoracoabdominal incision

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 15-year-old girl presents with central abdominal pain that localizes to the right iliac fossa. She also complains of a loss of appetite over the last 24 hours and nausea. On examination, she has a temperature of 37.8°C and is tender in the right iliac fossa.
- 2 A 40-year-old woman is undergoing elective caesarean section for a breech presentation.

- 3 A 42-year-old man with known peptic ulcer disease presents with sudden-onset, severe epigastric pain. On examination, he has a tender, rigid abdomen and absent bowel sounds.
- 4 A 34-year-old woman is undergoing a laparoscopic cholecystectomy. Unfortunately, the surgeons are unable to clearly delineate the anatomy of the area due to the presence of adhesions.
- 5 A 64-year-old woman is about to undergo a Whipple procedure for pancreatic adenocarcinoma.

REPEATED 2× · ALSO ASKED IN
Previous-years exam

THEME 5 Investigation of gastrointestinal bleeding

- A Adrenaline injection
- B Angiography
- C Barium meal
- D Colonoscopy
- E Oesophagogastroduodenoscopy
- F Proctosigmoidoscopy
- G Sclerotherapy
- H Urgent laparotomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 76-year-old woman is admitted to the emergency department with a history of abdominal pain, passing blood per rectum, and haematemesis. An urgent upper and lower gastrointestinal endoscopy are performed, but no active bleeding is noted. The patient is showing signs of hypovolaemic shock and is still passing melaena. Her haemoglobin drops by 4 g/dL over 4 hours.
- 2 A 63-year-old woman presents with haematemesis. After initial management, she is sent for an upper gastrointestinal endoscopy. The endoscopist sees a bleeding gastric ulcer with a vessel at the base.
- 3 A 67-year-old woman presents with fresh rectal bleeding. She also admits to a history of left-sided abdominal pain that is relieved by passing stools. Examination is unremarkable and the patient is haemodynamically stable.
- 4 A 42-year-old man presents with fresh blood loss per rectum. There is no haematemesis or melaena. He has no history of similar problems. On examination, his pulse is 138 and blood pressure is 80/42.
- 5 A 45-year-old woman presents with a 4-week history of passing blood per rectum. She complains of seeing small amounts of bright red blood on the toilet paper and in the pan. The patient is otherwise well.

Answers • Themes 1-5

GET AHEAD EMQS

Theme 1 • Investigation of abdominal pain

- F Serum amylase** — Serum amylase is the initial test to confirm acute pancreatitis.
- B Erect chest X-ray** — Erect chest X-ray identifies free intraperitoneal air in suspected perforation.
- D Laparoscopy** — Laparoscopy provides direct visualisation and treatment of pelvic pathology.
- E Mesenteric angiography** — Mesenteric angiography is the investigation of choice for chronic mesenteric ischaemia.
- A CT scan** — CT scan assesses the extent and staging of an oesophageal mass.

Theme 2 • Intestinal obstruction

- I Sigmoid volvulus** — A twisted sigmoid colon loop on X-ray is characteristic of sigmoid volvulus.
- A Adhesions** — Postoperative adhesions are the most common cause of small bowel obstruction.
- F Gallstone ileus** — Pneumobilia with SBO indicates gallstone ileus.
- H Paralytic ileus** — Absent bowel sounds and distension post-laparotomy reflect paralytic ileus.
- B Bezoar** — A duodenal filling defect on barium study in a psychiatric patient suggests a bezoar.

Theme 3 • Management of hernias

- C Observation only** — Observation is appropriate for an asymptomatic, reducible hernia.
- B Emergency repair** — Emergency repair is required for a tender, irreducible strangulated hernia.

- A Elective repair** — Elective repair prevents complications of an asymptomatic inguinoscrotal hernia.
- E Urgent repair** — Urgent repair is advised for a painless but irreducible femoral hernia.
- D Prompt repair** — Prompt repair is indicated for a recent irreducible scrotal hernia.

Theme 4 • Abdominal incisions

- D Lanz incision** — Lanz incision is used for appendicectomy in the right iliac fossa.
- G Pfannenstiel incision** — Pfannenstiel incision is standard for caesarean section.
- E Midline laparotomy** — Midline laparotomy provides access for emergency peritonitis and perforated ulcer.
- C Kocher incision** — Kocher incision offers exposure for difficult gallbladder surgery.
- H Rooftop incision** — Rooftop incision provides upper abdominal access for Whipple procedure.

Theme 5 • Investigation of gastrointestinal bleeding

- B Angiography** — Angiography localizes and allows embolization in ongoing GI bleeding when endoscopy fails.
- A Adrenaline injection** — Adrenaline injection achieves haemostasis in bleeding gastric ulcer.
- D Colonoscopy** — Colonoscopy visualizes colonic source in stable lower GI bleeding.
- E Oesophagogastroduodenoscopy** — Oesophagogastroduodenoscopy excludes upper GI source in unstable patients.
- F Proctosigmoidoscopy** — Proctosigmoidoscopy evaluates distal anorectal causes in chronic bright red bleeding.

SET 2 OF 6

THEME 6 Diagnosis of jaundice

- A Alcoholic liver disease
- B Ascending cholangitis
- C Carcinoma of the head of the pancreas
- D Choledocholithiasis
- E Crigler-Najjar syndrome
- F Duodenal carcinoma
- G Gilbert syndrome
- H Hepatocellular carcinoma
- I Primary biliary cirrhosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- A 25-year-old woman presents with a 2-week history of vague abdominal pain and worsening jaundice. She also complains of passing pale motions. The patient underwent a total colectomy for familial adenomatous polyposis at the age of 20. There is no abnormality on abdominal examination.

- A 44-year-old woman presents with a 1-month history of jaundice associated with fatigue, itching, and dark urine. She has a history of early-onset rheumatoid arthritis. On examination, she appears very jaundiced and has multiple xanthelasmas around her eyes. Abdominal examination is unremarkable.
- A 65-year-old man presents with a 1-month history of worsening jaundice and itching. He has lost 2 stone in weight over this time but denies experiencing any pain. On examination, the abdomen is soft and a mass is palpable in the right upper quadrant.
- A 28-year-old man attends the emergency department with a productive cough and malaise. His routine bloods on admission show a bilirubin level of 45 µmol/L. The remainder of his liver function tests are normal.

- 5** A 48-year-old man is brought to hospital by his wife as he has developed jaundice over the last few days. He is slightly confused and is unable to give a clear history. On examination, he has a distended, tense abdomen and dilated superficial veins around his umbilicus. His liver is not palpable.

THEME 7 Gastrointestinal polyps

- A** Adenomatous polyp
- B** Hamartomatous polyp
- C** Juvenile polyp
- D** Metaplastic polyp
- E** Pseudopolyp
- F** Villous adenoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 16-year-old girl requests a colonoscopy after her older brother was recently diagnosed with colon cancer at the age of 21. Her mother died at the age of 28 from a caecal tumour. The colonoscopy shows multiple polyps throughout the colon. Histology shows these to be benign.
- 2** A 45-year-old man undergoes an upper gastrointestinal endoscopy following an episode of haematemesis. He is found to have multiple polyps in his duodenum, which are later found to be benign. It is noticed that he has multiple freckles around his lips.
- 3** A 40-year-old man presents with a 2-week history of diarrhoea and muscle weakness. A colonoscopy shows a 2 cm sessile growth that has multiple projections.
- 4** A 36-year-old man with known ulcerative colitis undergoes a routine colonoscopy. A number of small projections can be seen throughout the bowel. These are confirmed as benign.
- 5** A 12-year-old girl presents with one episode of per rectum bleeding. A lower gastrointestinal endoscopy detects the presence of a single, large lesion that appears to be on a 'stalk'.

THEME 8 CLASSIFICATION OF HERNIAS

- A** Incarcerated
- B** Obstructed
- C** Perforated
- D** Reducible
- E** Richter
- F** Sliding
- G** Strangulated

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 63-year-old man presents with a lump in his groin. On examination, there is a mass in the scrotum that disappears on lying down. There is no pain or erythema associated with the lump and the patient is asymptomatic. When the patient coughs, an impulse can be felt over the hernia.

- 2** A 34-year-old man presents with a lump in the lower midline of his abdomen. On examination, you note that the lump is in the lower end of a midline laparotomy scar. The lump is 4 cm in size, and the contents cannot be returned to the abdomen by pressure; there is no cough impulse. The lump is neither painful nor erythematous.
- 3** A 48-year-old woman presents with a small lump in the groin, below and lateral to the pubic tubercle. She also complains of vomiting and colicky abdominal pain. On examination, the lump is 1 cm in diameter, painful, and erythematous. There is no cough impulse.
- 4** A 45-year-old man who has a known direct inguinal hernia presents to the emergency department because his hernia has become painful over the last few hours. On examination, the hernia is irreducible, tender, and erythematous. The patient denies any colicky abdominal pain, vomiting, or constipation.
- 5** A 78-year-old woman with a previous epigastric hernia presents with severe abdominal pain and vomiting. On examination, she is pyrexial and tachycardic and has a diffusely painful, rigid abdomen with no bowel sounds. Her hernia is noted to be erythematous and irreducible.

THEME 9 UPPER GASTROINTESTINAL HAEMORRHAGE

- A** Aorto-enteric fistula
- B** Carcinoma of the stomach
- C** Carcinoma of the oesophagus
- D** Epistaxis
- E** Haemoptysis
- F** Mallory–Weiss tear
- G** Oesophageal varices
- H** Peptic ulceration
- I** Boerhaave syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 62-year-old man is brought to the emergency department after vomiting large amounts of fresh blood. On examination, he is drowsy with a heart rate of 120/min and blood pressure of 92/50 mmHg. An urgent full blood count shows haemoglobin 6.9 g/dL, platelets $160 \times 10^9/L$, mean cell volume 106, and an INR of 2.3.
- 2** A 25-year-old man presents with three episodes of vomiting that contained altered blood. He has recently started a busy job, has not had time to eat well, and also complains of a 6-month history of upper abdominal pain that is relieved by eating. He regularly takes ibuprofen.
- 3** A 19-year-old student presents with blood-stained vomiting. He was on a pub crawl last night and spent the latter hours vomiting and retching over a bucket.

- 4** A 59-year-old man presents vomiting copious amounts of fresh blood. On examination, his pulse is 138/min and blood pressure is 68/42 mmHg. A per rectum examination demonstrates melaena and fresh blood. His only past medical history is an abdominal aortic aneurysm repair last year. He denies any medical history.
- 5** An 83-year-old man attends the emergency department after vomiting small amounts of fresh blood. He complains of a 3-month history of aching epigastric pain and a poor appetite with a weight loss of over 2 stone. He denies difficulty swallowing, and on examination, enlarged supraclavicular lymph nodes are found.

THEME 10 MANAGEMENT OF ABDOMINAL PAIN

- A Azathioprine
B Augmentin
C Cefuroxime and metronidazole
D Lansoprazole alone
E Metoclopramide
F Ofloxacin and metronidazole
G Omeprazole, clarithromycin and metronidazole
H Opioid analgesia alone
I Steroids

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 72-year-old woman presents with a 2-day history of lower abdominal pain, vomiting, and diarrhea. Her abdomen is tender with guarding in the left iliac fossa, and her temperature is 38.2°C.
- 2** A 42-year-old woman complains of a long history of abdominal bloating. She feels full after eating small amounts and tends to vomit soon after eating. She has a history of diabetes and rheumatoid arthritis. Endoscopy shows a dilated stomach with some stale food but no evidence of ulceration or obstruction.
- 3** A 32-year-old man attends the emergency department with a short history of epigastric pain and vomiting. On examination, there is some epigastric tenderness but no guarding. Upper endoscopy reveals inflammation and ulceration in the stomach, and a CLO test is positive.
- 4** A 21-year-old woman presents with a 2-day history of lower abdominal pain, dysuria, and vomiting. On examination, there is guarding in the left iliac fossa. A per rectum examination is normal, but there is evidence of a yellow vaginal discharge.
- 5** A 25-year-old woman with previously well-controlled ulcerative colitis presents with a 2-day history of severe lower abdominal pain and bloody diarrhea. She is pyrexial on examination.

Answers • Themes 6–10

GET AHEAD EMQS

Theme 6 • Diagnosis of jaundice

- 1** **F Duodenal carcinoma** — Duodenal carcinoma causes obstructive jaundice in FAP patients.
- 2** **I Primary biliary cirrhosis** — Primary biliary cirrhosis presents with pruritus, fatigue, and xanthelasmas.
- 3** **C Carcinoma of the head of the pancreas** — Carcinoma head of pancreas causes painless obstructive jaundice with RUQ mass.
- 4** **G Gilbert syndrome** — Isolated hyperbilirubinemia with normal enzymes indicates Gilbert syndrome.
- 5** **A Alcoholic liver disease** — Portal hypertension signs in acute jaundice suggest alcoholic liver disease.

Theme 7 • Gastrointestinal polyps

- 1** **A Adenomatous polyp** — Adenomatous polyps in FAP are benign but premalignant.
- 2** **B Hamartomatous polyp** — Hamartomatous polyps with mucocutaneous pigmentation indicate Peutz-Jeghers.
- 3** **F Villous adenoma** — Villous adenoma is a large sessile polyp with malignant potential.
- 4** **E Pseudopolyp** — Pseudopolyps occur in ulcerative colitis from mucosal regeneration.

- 5** **C Juvenile polyp** — Juvenile polyps are pedunculated benign lesions in children.

Theme 8 • CLASSIFICATION OF HERNIAS

- 1** **D Reducible** — A reducible hernia can be pushed back into the abdomen and shows a cough impulse.
- 2** **A Incarcerated** — An incarcerated hernia cannot be reduced.
- 3** **G Strangulated** — A strangulated hernia presents with pain, erythema, and systemic signs of ischemia.
- 4** **E Richter** — A Richter hernia involves only a portion of the bowel wall, leading to tenderness and irreducibility.
- 5** **C Perforated** — A perforated hernia with a rigid, painful abdomen and systemic signs indicates bowel perforation.

Theme 9 • UPPER GASTROINTESTINAL HAEMORRHAGE

- 1** **G Oesophageal varices** — Massive bleeding in an unstable patient suggests variceal hemorrhage.
- 2** **H Peptic ulceration** — NSAID use with epigastric pain relieved by food indicates peptic ulceration.
- 3** **F Mallory–Weiss tear** — A history of forceful retching in a young man points to a Mallory–Weiss tear.
- 4** **A Aorto-enteric fistula** — A history of AAA repair with massive bleeding suggests an aorto-enteric fistula.

- 5 **B Carcinoma of the stomach** — Chronic epigastric pain with weight loss and supraclavicular lymphadenopathy indicates carcinoma of the stomach.

Theme 10 • MANAGEMENT OF ABDOMINAL PAIN

- 1 **C Cefuroxime and metronidazole** — Lower abdominal pain with guarding and fever indicates an infective process treated with broad-spectrum antibiotics.
- 2 **E Metoclopramide** — Gastroparesis is managed with a prokinetic agent such as metoclopramide.

SET 3 OF 6

THEME 11 DIAGNOSIS OF DYSPHAGIA

- A Bulbar palsy
- B Chagas disease
- C Diffuse oesophageal spasm
- D External oesophageal compression
- E Gastro-oesophageal reflux disease
- F Myasthenia gravis
- G Oesophageal candidiasis
- H Oesophageal carcinoma
- I Oesophageal web
- J Pseudobulbar palsy
- K Scleroderma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 46-year-old woman presents with intermittent severe retrosternal chest pains that occur soon after eating and are accompanied by dysphagia. She complains of no other symptoms. An ECG shows no acute abnormality and cardiac enzymes are normal.
- 2 A 72-year-old man presents with a 3-month history of dysphagia and odynophagia. This initially occurred with solids, but he is now finding it difficult to drink liquids. Over the past months, he has lost 12 kg in weight. Examination is unremarkable.
- 3 A 44-year-old man who has recently arrived from Brazil has a 5-month history of dysphagia for solids and liquids; he is otherwise well. A chest X-ray shows an air-fluid level behind the heart.
- 4 A 36-year-old man presents with a 4-month history of increasing difficulty in swallowing. You notice that his speech is a little slurred, and his wife tells you that this is a new development. On examination, his tongue is wasted and shows fasciculations.
- 5 A 42-year-old man taking prednisolone for an exacerbation of inflammatory bowel disease has retrosternal pain and dysphagia. There are multiple white plaques in his mouth that rub off easily.

- 3 **G Omeprazole, clarithromycin and metronidazole** — A positive CLO test confirms *H. pylori* infection, which is treated with triple therapy.
- 4 **F Ofloxacin and metronidazole** — Pelvic inflammatory disease is managed with ofloxacin and metronidazole.
- 5 **I Steroids** — A severe ulcerative colitis flare is managed with steroids.

THEME 12 MANAGEMENT OF ANORECTAL CONDITIONS

- A DeLorme procedure
- B Diltiazem cream
- C Excision of fistula
- D Incision and drainage
- E Injection sclerotherapy
- F Laying open of the fistula tract
- G Lord's procedure
- H Seton insertion
- I Surgical haemorrhoidectomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 56-year-old woman presents with a history of fresh rectal bleeding after defaecation. On examination, with a proctoscope, haemorrhoids are seen within the anal canal. The patient denies any prolapse of these haemorrhoids.
- 2 A 71-year-old woman presents with pruritus ani. On examination, she has an obvious prolapsed rectum.
- 3 A 22-year-old female with known inflammatory bowel disease presents with a constant discharge from the anus. On examination, two external openings are seen at the posterior margin of the anus, both of which are discharging faeculent material. An MRI scan shows a high-lying anal fistula that passes through the puborectalis muscle.
- 4 A 42-year-old man presents with a 2-day history of pain around his anus, difficulty sitting down and fever. On examination, there is a 1 cm tender, erythematous lump adjacent to the anal margin. No other abnormality is evident.
- 5 A 34-year-old woman presents with a 7-day history of severe stinging pain in the anus on defaecation, associated with fresh blood on the toilet paper. Examination reveals a tight anal tone and a lone skin tag on the posterior margin of the anus.

THEME 13 ANATOMY OF HERNIAS

- A Amyand
- B Gluteal
- C Littre
- D Lumbar
- E Maydl
- F Obturator
- G Pantaloon
- H Sciatic

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A hernia that has two parts, each lying on either side of the inferior epigastric artery.
- 2 A hernia that passes through the greater sciatic foramen.
- 3 A hernia that contains a 'W' loop of intestine within its sac.
- 4 A hernia that arises from the triangle of Petit.
- 5 A hernia that contains the appendix within its sac.

THEME 14 DIAGNOSIS OF ABDOMINAL PAIN

- A Acute appendicitis
- B Acute pancreatitis
- C Chronic pancreatitis
- D Diverticulitis
- E Ectopic pregnancy
- F Irritable bowel syndrome
- G Mesenteric angina
- H Pelvic abscess
- I Pelvic inflammatory disease
- J Perforated peptic ulcer
- K Subphrenic abscess

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 23-year-old woman presents with a 3-day history of left iliac fossa pain and nausea. Further questioning reveals that she is also suffering pain on micturition. On examination, there is tenderness and guarding in the left iliac fossa as well as some tenderness in the right upper quadrant. Her temperature is 37.6°C.
- 2 A 68-year-old woman with a history of osteoarthritis presents with acute severe abdominal pain and vomiting. On examination, the abdomen is rigid and bowel sounds are absent.
- 3 A 42-year-old woman presents with an 8-month history of abdominal discomfort that is improved following defaecation. She complains of a regular change in her bowel habit, alternating between constipation and loose motions. On examination, her abdomen is soft and nontender. There are no obvious masses, and bowel sounds are present.

- 4 A 44-year-old man presents with recurrent episodes of severe epigastric pain that radiates to his back and is improved on leaning forwards. On examination, he is jaundiced, with guarding in his epigastrium, but there is no evidence of haemorrhage. Observations are stable. Routine bloods are sent off, and the amylase returns at 248 IU/L.
- 5 A 68-year-old woman undergoes a laparotomy for a perforated diverticulum. A few days later, she develops pain in the left upper quadrant that radiates to the shoulder, nausea, vomiting and rigors. Her temperature is 38.9°C.

REPEATED 2* · ALSO ASKED IN
Get Ahead

THEME 15 DIAGNOSIS OF BILIARY TRACT DISEASE

- A Acute cholecystitis
- B Ascending cholangitis
- C Biliary colic
- D Cholangiocarcinoma
- E Choledocholithiasis
- F Chronic cholecystitis
- G Empyema of the gallbladder
- H Gallbladder perforation
- I Gallstone ileus
- J Mirizzi syndrome
- K Primary sclerosing cholangitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 82-year-old woman presents with worsening jaundice and weight loss. An abdominal ultrasound shows dilatation of the common bile duct, and a subsequent magnetic resonance cholangiopancreatogram (MRCP) confirms an irregular stenosing lesion in the common bile duct.
- 2 A 48-year-old woman presents with a 3-day history of abdominal pain. On examination, she is clinically jaundiced and tender in the right upper quadrant, but no masses are palpable. An ultrasound scan confirms the presence of biliary stones and intrahepatic duct dilatation. An MRCP shows a gallstone within the gallbladder pressing on the common bile duct.
- 3 A 58-year-old woman with known gallstones presents with a 12-month history of indigestion attacks and flatulence. Symptoms occur after meals and can last up to an hour. On examination, she is obese but shows no signs of jaundice or anaemia. Her abdomen is soft and non-tender.
- 4 A 32-year-old man with ulcerative colitis presents with a 4-week history of worsening jaundice. He denies any pain, fevers or nausea. On examination, his abdomen is soft. An abdominal ultrasound shows dilatation of the intrahepatic biliary tree but a normal common bile duct.

- 5 A 38-year-old woman presents with a 48-hour history of right upper quadrant pain, nausea and vomiting. On examination, she is clinically jaundiced and her temperature is 38.5°C. An abdominal ultrasound shows multiple stones in the gallbladder and a common bile duct diameter of 9 mm.

Answers · Themes 11–15

GET AHEAD EMQS

Theme 11 · DIAGNOSIS OF DYSPHAGIA

- 1 **C Diffuse oesophageal spasm** — Diffuse oesophageal spasm causes intermittent chest pain and dysphagia with normal cardiac workup.
- 2 **H Oesophageal carcinoma** — Progressive dysphagia (solids then liquids) and weight loss are red flags for oesophageal carcinoma.
- 3 **B Chagas disease** — Chagas disease causes achalasia-like dysphagia with radiographic air–fluid level.
- 4 **A Bulbar palsy** — Bulbar palsy presents with dysphagia, dysarthria, and tongue fasciculations.
- 5 **G Oesophageal candidiasis** — Oral thrush in an immunosuppressed patient indicates oesophageal candidiasis.

Theme 12 · MANAGEMENT OF ANORECTAL CONDITIONS

- 1 **E Injection sclerotherapy** — Injection sclerotherapy treats internal haemorrhoids without prolapse.
- 2 **A DeLorme procedure** — DeLorme procedure repairs full-thickness rectal prolapse.
- 3 **H Seton insertion** — Seton insertion manages complex anal fistulae in IBD to preserve continence.
- 4 **D Incision and drainage** — Incision and drainage is definitive for anal abscess.
- 5 **B Diltiazem cream** — Topical diltiazem reduces sphincter spasm in fissure healing.

Theme 13 · ANATOMY OF HERNIAS

- 1 **G Pantaloon** — Pantaloon hernia has sacs on both sides of the inferior epigastric vessels.
- 2 **B Gluteal** — Gluteal hernia protrudes through the greater sciatic foramen.

- 3 **E Maydl** — Maydl's hernia contains a 'W'-shaped loop of bowel.
- 4 **D Lumbar** — Lumbar hernia arises through the triangle of Petit.
- 5 **A Amyand** — Amyand's hernia contains the appendix in the sac.

Theme 14 · DIAGNOSIS OF ABDOMINAL PAIN

- 1 **I Pelvic inflammatory disease** — PID presents with lower abdominal pain, dysuria, and LIF tenderness.
- 2 **J Perforated peptic ulcer** — Rigid abdomen and absent bowel sounds indicate perforated peptic ulcer.
- 3 **F Irritable bowel syndrome** — Chronic discomfort with normal exam and altered bowel habit suggests IBS.
- 4 **C Chronic pancreatitis** — Recurrent epigastric pain, jaundice, and elevated amylase point to chronic pancreatitis.
- 5 **K Subphrenic abscess** — Postoperative LUQ pain with shoulder radiation and fever suggests subphrenic abscess.

Theme 15 · DIAGNOSIS OF BILIARY TRACT DISEASE

- 1 **D Cholangiocarcinoma** — Irregular biliary stenosis with weight loss suggests cholangiocarcinoma.
- 2 **J Mirizzi syndrome** — Impacted gallstone compressing CBD is characteristic of Mirizzi syndrome.
- 3 **F Chronic cholecystitis** — Recurrent dyspeptic symptoms from gallstones indicate chronic cholecystitis.
- 4 **K Primary sclerosing cholangitis** — Intrahepatic duct dilatation in UC patient indicates primary sclerosing cholangitis.
- 5 **B Ascending cholangitis** — RUQ pain, fever, and jaundice with duct dilatation define ascending cholangitis.

SET 4 OF 6

THEME 16 LOWER GASTROINTESTINAL HAEMORRHAGE

- A Anal fissure
- B Angiodysplasia
- C Colorectal cancer
- D Crohn's disease
- E Diverticular disease
- F Diverticulitis
- G Haemorrhoids
- H Infective colitis
- I Ischaemic colitis
- J Ulcerative colitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 86-year-old woman with intermittent bright-red rectal bleeding, normal barium enema and colonoscopy.
- 2 A 38-year-old woman with 2 days of loose, blood-stained motions, abdominal tenderness, and vomiting.
- 3 A 62-year-old woman with a single episode of fresh rectal bleeding, 6 months of worsening constipation and LIF pain relieved by defaecation.

- 4** A 68-year-old man with 3 weeks of alternating constipation and diarrhoea, occasional fresh blood, haemorrhoids on exam.
- 5** A 17-year-old girl with 4 weeks of RIF pain, occasional blood and mucus, palpable RIF mass.

THEME 17 DIAGNOSIS OF HERNIAS

- A** Direct inguinal
- B** Epigastric
- C** Exomphalos
- D** Gastroschisis
- E** Hiatus
- F** Incisional
- G** Indirect inguinal
- H** Paraumbilical
- I** Spigelian
- J** Umbilical hernia of infants

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A term baby is born with large amounts of visible intestine emerging from of his abdomen. The external gut is covered with a thin membrane.
- 2** A 34-year-old man presents with a lower midline mass that is 8 cm in size. He has a history of an emergency laparotomy for toxic megacolon secondary to ulcerative colitis. On examination, the mass is soft, non tender and reducible.
- 3** A 64-year-old man presents with two groin lumps, one on each side. On examination, the lumps have a cough impulse and are reducible but do not extend into the scrotum.
- 4** A 42-year-old woman presents to the emergency department with colicky abdominal pain and an abdominal lump. On examination, the lump is in the midline just above the umbilicus and is tender to touch.
- 5** A 39-year-old man presents to the emergency department with a 'tearing' pain in his upper abdomen after trying to lift weights at home. On examination, you find a 2 cm lump at the site of the pain in the midline of the abdomen, halfway between the xiphisternum and the umbilicus.

THEME 18 DIAGNOSIS OF HEPATOMEGALY

- A** Alcoholic cirrhosis
- B** Amoebic liver abscess
- C** Biliary tract obstruction
- D** Budd-Chiari syndrome
- E** Haemochromatosis
- F** Hepatocellular carcinoma
- G** Metastatic disease of the liver
- H** Polycystic disease
- I** Pyogenic liver abscess
- J** Riedel lobe

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 16-year-old girl presents with a 1-week history of right lower quadrant pain, nausea and anorexia. In the last 24 hours, she has become more unwell, developing spiking fevers and rigors. On examination, she appears mildly jaundiced. Her liver is enlarged and tender to palpation.
- 2** A 36-year-old woman presents to her general practitioner with a long history of vague abdominal pain. She denies any other symptoms, but she mentions that she was admitted to hospital last year with a bleed in her brain. On examination, there is massive, nodular hepatomegaly.
- 3** A 42-year-old man attends his general practice with a history of excessive thirst and urinary frequency. On examination, he appears to be tanned, although he denies going abroad over the winter months. The liver is palpable 4 cm below the costal margin.
- 4** A 39-year-old man from India presents with a 2-week history of aching abdominal pain and weight loss. His past medical history includes eczema and a fractured femur in a car crash 20 years ago for which he received a blood transfusion. On examination, his abdomen is tender in the right hypochondrium and is mildly distended with a fluid thrill. His liver is enlarged and nodular.
- 5** A 26-year-old woman presents with sudden-onset severe upper abdominal pain. She takes the oral contraceptive pill, but has no other medical history. On examination, she is slim but has a distended abdomen that is positive for shifting dullness. The liver is palpable three finger breadths below the costal margin.

THEME 19 INVESTIGATION OF ABDOMINAL PAIN

- A** Creatine kinase levels
- B** CT scan
- C** Erect chest X-ray
- D** Faecal elastase
- E** Laparoscopy
- F** Mesenteric angiography
- G** Serum amylase
- H** Supine chest X-ray
- I** Troponin
- J** Urgent laparotomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 54-year-old man presents with a 3-day history of left upper quadrant pain, which is worse on inspiration, malaise and fevers. He also complains of a cough productive of green sputum. Abdominal examination is unremarkable.

- 2 A 56-year-old sheep farmer presents to his general practitioner with a 2-month history of abdominal aches and worsening jaundice. He denies weight loss or any other symptoms. On examination, a smooth mass can be felt in the right upper quadrant. An abdominal X-ray shows flecks of calcification in the right upper quadrant, separate to the liver and gallbladder.
- 3 An 84-year-old woman is brought to the emergency department with a 6-hour history of severe, colicky, central abdominal pain. On arrival, she is pale and sweaty, with an irregularly irregular pulse of 125/min and a blood pressure of 105/68 mmHg. There is diffuse tenderness on abdominal examination, and per rectum exam reveals fresh blood.
- 4 A 44-year-old man presents with recurrent episodes of severe epigastric pain that radiates to his back and is improved on leaning forward. On examination, he is jaundiced, with guarding in the epigastrium. There is no evidence of bruising. Observations are stable. Routine bloods are sent off and the amylase returns at 132 IU/L.
- 5 A 62-year-old man presents with sudden-onset tight epigastric pain that radiates up to his neck. He is also nauseated and is sweating. On examination, the abdomen is soft and non-tender. Bowel sounds are present and a per rectum exam is unremarkable. The pain is only relieved by morphine.

THEME 20 DIAGNOSIS OF ANORECTAL DISEASE

- A Anal carcinoma
- B Anal fissure
- C Anal fistula
- D Condylomata acuminata
- E Condylomata lata
- F Haemorrhoids
- G Perianal abscess
- H Perianal haematoma
- I Proctalgia fugax
- J Rectal prolapse

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 62-year-old man presents with a 2-month history of pruritus ani and occasional fresh blood on the toilet paper. He denies pain on defaecation. On examination, there is evidence of ulceration at the anal margin.
- 2 A 28-year-old man presents with a long history of intermittent pain in the rectum. The pain tends to occur at night. He denies passing any blood or mucous per rectum. Examination is unremarkable.
- 3 A 46-year-old woman complains of a stinging pain that occurs on defecation. This has been occurring for over a month. Rectal examination is very painful and there is a skin tag at the anal verge.
- 4 A 39-year-old woman presents with sudden-onset anal pain that occurred while straining at the toilet. On examination, a smooth, dark lump can be seen at the anal verge.
- 5 A 32-year-old man presents with a long history of pruritus ani. On examination, large pink, confluent lesions are seen around the anus.

Answers • Themes 16–20

Theme 16 • LOWER GASTROINTESTINAL HAEMORRHAGE

- 1 **B Angiodysplasia** — Painless intermittent bleeding in elderly suggests angiodysplasia.
- 2 **H Infective colitis** — Loose, blood-stained stools with tenderness and vomiting indicate infective colitis.
- 3 **E Diverticular disease** — Fresh bleeding with left-sided discomfort relieved by defaecation suggests diverticular disease.
- 4 **C Colorectal cancer** — Change in bowel habit with bleeding in older patient raises suspicion for colorectal cancer.
- 5 **D Crohn's disease** — RIF pain with blood, mucus, and palpable mass suggests Crohn's disease.

Theme 17 • DIAGNOSIS OF HERNIAS

- 1 **C Exomphalos** — Exomphalos is a congenital midline hernia with viscera in a membranous sac.

- 2 **F Incisional** — Incisional hernia occurs at a previous surgical scar.
- 3 **A Direct inguinal** — Direct inguinal hernias are medial to epigastric vessels and do not descend into the scrotum.
- 4 **H Paraumbilical** — Paraumbilical hernias occur adjacent to the umbilicus and can be tender.
- 5 **B Epigastric** — Epigastric hernias occur between xiphisternum and umbilicus, often symptomatic with strain.

Theme 18 • DIAGNOSIS OF HEPATOMEGALY

- 1 **I Pyogenic liver abscess** — Pyogenic abscess causes fever, RUQ pain, and jaundice.
- 2 **H Polycystic disease** — Polycystic disease leads to massive nodular liver enlargement.
- 3 **E Haemochromatosis** — Haemochromatosis presents with bronze skin, diabetes, and hepatomegaly.

- 4 **F Hepatocellular carcinoma** — HCC presents with nodular liver, weight loss, and portal hypertension.
- 5 **D Budd-Chiari syndrome** — Budd-Chiari presents with ascites and hepatomegaly in hypercoagulable states.

Theme 19 · INVESTIGATION OF ABDOMINAL PAIN

- 1 **H Supine chest X-ray** — Pleuritic LUQ pain with productive cough is a left lower-lobe pneumonia mimicking abdominal pain; a chest X-ray confirms it.
- 2 **B CT scan** — A calcified RUQ cyst in a sheep farmer suggests a hydatid cyst; CT defines the cyst, its daughter cysts and biliary communication.
- 3 **F Mesenteric angiography** — AF with severe pain out of proportion and bloody stool suggests acute mesenteric ischaemia; angiography (CT or catheter) shows the occlusion.
- 4 **D Faecal elastase** — Recurrent pancreatitis-type pain with near-normal amylase suggests chronic pancreatitis; faecal elastase confirms exocrine insufficiency.

SET 5 OF 6

THEME 21 INVESTIGATION OF DYSPHAGIA

- A 24-hour lower oesophageal pH
 B Anti-cardiolipin antibody
 C Anti-centromere antibody
 D Biopsy
 E CT chest
 F Endoscopy
 G Full blood count
 H Manometry
 I Muscle biopsy
 J Serial X-rays
 K Tensilon test

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 37-year-old woman presents with dysphagia that is equally apparent with solids and liquids. She also complains of regurgitation of food, which usually occurs on lying flat. She otherwise feels well, but admits to losing 5 kg in weight over the past 3 months. A chest X-ray shows an air-fluid level behind the heart, a double right heart border and absence of the gastric air bubble.
- 2 A 60-year-old man presents with occasional retrosternal chest pain that is worse on bending and lying flat. The pain is not relieved by analgesia. He also experiences a night cough. An electrocardiogram shows no acute changes, and troponin levels after 12 hours are negative.
- 3 A 45-year-old woman presents with a 2-month history of lethargy, occasional blurred vision and difficulty swallowing that is worse at the end of the day. She has no pain on swallowing and denies any other symptoms. Examination is unremarkable.

- 5 **I Troponin** — Epigastric pain radiating to the neck with sweating and a soft abdomen may be cardiac; troponin (and ECG) excludes MI.

Theme 20 · DIAGNOSIS OF ANORECTAL DISEASE

- 1 **A Anal carcinoma** — Anal ulceration and bleeding in an elderly patient suggest carcinoma.
- 2 **I Proctalgia fugax** — Brief, recurrent nocturnal pain with normal exam indicates proctalgia fugax.
- 3 **B Anal fissure** — Painful defaecation and skin tag are classic for anal fissure.
- 4 **H Perianal haematoma** — Sudden painful dark lump at anal verge is a perianal haematoma.
- 5 **D Condylomata acuminata** — Large pink confluent warty perianal lesions are condylomata acuminata (HPV); condylomata lata are flat, moist lesions of secondary syphilis.

- 4 A 38-year-old woman presents to the general practitioner with a 2-month history of dysphagia and worsening shortness of breath on exertion. On examination, you notice that she has multiple telangiectasias on her face and tight skin around her fingers causing fixed flexion deformities.
- 5 A 6-year-old boy is brought to the emergency department with pain in his throat after swallowing a penny coin earlier in the day. An X-ray confirms the presence of a foreign body in the duodenum.

THEME 22 DIAGNOSIS OF HERNIAS

- A Direct inguinal
 B Epigastric
 C Femoral
 D Gastroschisis
 E Hiatus
 F Incisional
 G Indirect inguinal
 H Littre
 I Paraumbilical
 J Spigelian

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 57-year-old woman presents to the emergency department with a tender, erythematous swelling in the left groin. On examination, you find an irreducible, hard lump below and lateral to the pubic tubercle.
- 2 A 24-year-old man presents with a groin swelling that descends into the scrotum on standing. On examination, the swelling does not transilluminate and you cannot get above it, but it is reducible.
- 3 A 48-year-old woman presents with a 6-month history of heartburn and night cough. She denies any other symptoms. On examination, you note that she is obese, but no other abnormality is evident.

- 4** A 64-year-old man, who has a history of a laparotomy following a perforated peptic ulcer, presents with a tender mass underlying the caudal end of his midline scar. At operation, the hernial sac is found to contain a Meckel diverticulum.
- 5** A 52-year-old man presents with a lump in his abdomen that is associated with a dull ache. On examination, the lump is 2 cm in size and below the umbilicus. It is located lateral to the rectus muscles.

THEME 23 MANAGEMENT OF BILIARY TRACT DISEASE

- A** Antibiotics and elective laparoscopic cholecystectomy
- B** Endoscopic retrograde cholangiopancreatogram
- C** Laparotomy and cholecystotomy/drainage
- D** Magnetic resonance cholangiopancreatogram
- E** Medical dissolution therapy
- F** Percutaneous gallbladder drainage
- G** Ultrasound scan of abdomen

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 47-year-old woman is seen in the emergency department with a 5-day history of upper abdominal pain, nausea and worsening jaundice. She has experienced bouts of similar pains before, but has never previously had jaundice. On examination, she is afebrile and there is mild tenderness in the right upper quadrant, but no masses are palpable. An abdominal ultrasound confirms the presence of a large gallstone in the common bile duct.
- 2** A 43-year-old woman complains of her first episode of severe epigastric pain, which occurred 2 hours after eating fish and chips. On further questioning, she admits feeling nauseated. There are no other symptoms. Her observations include pulse 96, blood pressure 126/80 and temperature 37°C.
- 3** A 38-year-old woman presents with a 3-day history of worsening epigastric discomfort. On examination, the abdomen is soft, but she is tender in the epigastrium and right upper quadrant. Her pulse is 96, blood pressure 130/85 and temperature 38.5°C.
- 4** A 65-year-old woman with known gallstones presents with a 3-day history of severe right upper quadrant pain with nausea and vomiting. On examination, her temperature is 39.1°C. There is tenderness and a palpable mass in the right upper quadrant. A full blood count shows a haemoglobin 11.8 g/dL and leukocytes $27 \times 10^9/L$.
- 5** A 49-year-old woman presents with a 24-hour history of right upper quadrant pain, nausea and vomiting. On examination, she is jaundiced and her temperature is 38.5°C. An abdominal ultrasound shows multiple stones in the gallbladder and a common bile duct diameter of 9 mm.

THEME 24 COLORECTAL OPERATIONS

- A** Abdominoperineal resection
- B** Anterior resection
- C** Extended right hemicolectomy
- D** Hartmann procedure
- E** Left hemicolectomy
- F** Proctocolectomy and ileo-anal pouch
- G** Right hemicolectomy
- H** Sigmoid colectomy
- I** Strictureplasty

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 62-year-old woman with known diverticular disease of the sigmoid colon presents with sudden-onset abdominal pain and vomiting. On examination, the abdomen is rigid. An erect chest X-ray confirms air under the diaphragm.
- 2** A 64-year-old man presents with rectal bleeding and a feeling of incomplete emptying after defaecation. A per rectum examination reveals a tumour 1 cm from the anal margin.
- 3** A 65-year-old woman who presented with rectal bleeding is found to have a non-obstructing tumour of the sigmoid colon.
- 4** A 74-year-old woman presents with weight loss and rectal bleeding. An urgent colonoscopy reveals a non-obstructive tumour at the caecum.
- 5** A 70-year-old woman who was found to have anaemia on her last admission to hospital has a lower gastrointestinal endoscopy arranged. She is found to have a high rectal tumour.

THEME 25 DIAGNOSIS OF DYSPHAGIA

- A** Achalasia
- B** Boerhaave syndrome
- C** Diffuse oesophageal spasm
- D** External oesophageal compression
- E** Foreign body
- F** Hiatus hernia
- G** Mallory-Weiss tear
- H** Myasthenia gravis
- I** Oesophageal carcinoma
- J** Oesophageal web
- K** Pharyngeal pouch

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 46-year-old woman presents to the general practitioner with worsening dysphagia for solids. On examination, you notice that her tongue is smooth and she has deformed nails. Routine blood tests show a haemoglobin level of 9.0 g/dL (normal range 11.5–16.0 g/dL), with a mean cell volume of 70 fL (normal range 76–96 fL).

- 2 A 67-year-old man presents with dysphagia and a night cough – symptoms he has suffered for 6 months. He has recently been admitted to hospital with a chest infection. On examination, there is a soft swelling in the anterior triangle of the neck that gurgles on palpation.
- 3 A 72-year-old woman presents with dysphagia and worsening shortness of breath. On examination, her pulse is irregularly irregular. She has a flushed face, and auscultation reveals a mid-diastolic murmur.
- 4 A 57-year-old woman presents with occasional retrosternal pain and acid reflux. She denies other symptoms and examination is unremarkable. A chest X-ray shows an air–fluid level behind the heart.
- 5 A 48-year-old man presents with sudden-onset pain in his chest and neck with dysphagia. On examination, he appears unwell. An ECG shows no acute changes and cardiac enzymes are normal. A chest X-ray shows gas in the mediastinum.

Answers • Themes 21–25

GET AHEAD EMQS

Theme 21 • INVESTIGATION OF DYSPHAGIA

- 1 **H** **Manometry** — Manometry is diagnostic gold standard for achalasia.
- 2 **A** **24-hour lower oesophageal pH** — 24-hour pH monitoring confirms GERD.
- 3 **K** **Tensilon test** — Tensilon test confirms myasthenia gravis.
- 4 **C** **Anti-centromere antibody** — Anti-centromere antibody is positive in CREST syndrome.
- 5 **J** **Serial X-rays** — Serial X-rays track passage of ingested foreign bodies.

Theme 22 • DIAGNOSIS OF HERNIAS

- 1 **C** **Femoral** — Femoral hernia presents as irreducible lump below pubic tubercle.
- 2 **G** **Indirect inguinal** — Indirect inguinal hernia follows canal into scrotum and is reducible.
- 3 **E** **Hiatus** — Hiatus hernia causes reflux symptoms without palpable mass.
- 4 **H** **Littre** — Littre hernia contains Meckel's diverticulum in sac.
- 5 **J** **Spigelian** — Spigelian hernia occurs lateral to rectus below umbilicus.

Theme 23 • MANAGEMENT OF BILIARY TRACT DISEASE

- 1 **B** **Endoscopic retrograde cholangiopancreatogram** — ERCP removes CBD stones in obstructive jaundice.
- 2 **G** **Ultrasound scan of abdomen** — Ultrasound is first-line for suspected biliary colic.
- 3 **A** **Antibiotics and elective laparoscopic cholecystectomy** — Acute cholecystitis managed with antibiotics then elective cholecystectomy.

- 4 **F** **Percutaneous gallbladder drainage** — Percutaneous drainage indicated in high-risk acute cholecystitis with mass.
- 5 **B** **Endoscopic retrograde cholangiopancreatogram** — ERCP extracts stones and decompresses CBD in choledocholithiasis.

Theme 24 • COLORECTAL OPERATIONS

- 1 **D** **Hartmann procedure** — Hartmann procedure for perforated diverticulum with peritonitis.
- 2 **A** **Abdominoperineal resection** — APR for low rectal tumour near anal margin.
- 3 **H** **Sigmoid colectomy** — Sigmoid colectomy for non-obstructing sigmoid cancer.
- 4 **G** **Right hemicolectomy** — Right hemicolectomy for caecal tumour with lymphadenectomy.
- 5 **B** **Anterior resection** — Anterior resection for high rectal tumour preserving sphincter.

Theme 25 • DIAGNOSIS OF DYSPHAGIA

- 1 **J** **Oesophageal web** — Oesophageal web in Plummer-Vinson causes dysphagia.
- 2 **K** **Pharyngeal pouch** — Pharyngeal pouch presents with gurgling neck swelling.
- 3 **D** **External oesophageal compression** — AF, malar flush and a mid-diastolic murmur indicate mitral stenosis; the enlarged left atrium compresses the oesophagus externally.
- 4 **F** **Hiatus hernia** — Hiatus hernia causes reflux and CXR air–fluid level.
- 5 **B** **Boerhaave syndrome** — Boerhaave syndrome shows mediastinal gas after rupture.

SET 6 OF 6

THEME 26 DIAGNOSIS OF ABDOMINAL PAIN

- A Acute appendicitis
- B Acute pancreatitis
- C Acute radiation enterocolitis
- D Chronic radiation enterocolitis
- E Diverticulitis
- F Ectopic pregnancy
- G Irritable bowel syndrome
- H Mesenteric angina
- I Osler-Weber-Rendu syndrome
- J Pelvic inflammatory disease
- K Perforated peptic ulcer
- L Subphrenic abscess
- M Torsion of an ovarian cyst

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 18-year-old woman is seen with a 48-hour history of lower abdominal pain, vomiting and diarrhoea. On examination, there is tenderness and guarding in the right iliac fossa and no masses are palpable. A routine urinary β -hCG test is negative.
- 2 A 54-year-old woman presents with sudden-onset abdominal pain that she describes as moderately severe. She has a history of recurrent nose bleeds. On examination, you notice multiple telangiectasias around her mouth. Abdominal examination is unremarkable.
- 3 A 74-year-old man attends his general practice with a 3-month history of central, aching abdominal pain. The pain occurs soon after eating and can last up to half an hour. Because of this, the patient has avoided eating and has lost a stone in weight.
- 4 A 34-year-old man who is currently undergoing treatment for a testicular tumour presents with a 2-day history of cramping abdominal pain, nausea, anorexia and diarrhoea. He denies any infectious contacts or eating 'dodgy' food. A stool culture comes back as negative.
- 5 A 32-year-old woman is brought to the emergency department by her husband with a sudden-onset, severe abdominal pain. On examination, there is guarding over the lower abdomen. The patient's pulse is 125, blood pressure 86/56 and temperature 36.5°C.

Answers • Theme 26

GET AHEAD EMQS

Theme 26 • DIAGNOSIS OF ABDOMINAL PAIN

- 1 **A Acute appendicitis** — Appendicitis causes RIF pain with guarding.
- 2 **I Osler-Weber-Rendu syndrome** — Osler-Weber-Rendu syndrome presents with vascular malformations causing bleeding.
- 3 **H Mesenteric angina** — Mesenteric angina causes postprandial pain and weight loss.
- 4 **C Acute radiation enterocolitis** — Acute radiation enterocolitis occurs during radiotherapy.
- 5 **F Ectopic pregnancy** — Ectopic pregnancy presents with shock and lower abdominal guarding.

17 THEMES • 83 SCENARIOS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course EMQs

SET 1 OF 4

THEME 1 ACUTE ABDOMEN

- A Appendicitis
- B Pancreatitis
- C Renal colic
- D Abdominal aortic aneurysm (AAA)
- E Thoracic aortic aneurysm
- F Dissecting aortic aneurysm
- G Peptic ulcer disease
- H Bowel obstruction
- I Biliary colic
- J Perforated viscus

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Associated with colicky abdominal pain, vomiting and absent bowel movement.
- 2 A patient presenting with generalized abdominal tenderness with evidence of free gas on an erect chest X-ray.
- 3 Associated with colicky abdominal pain in the left flank with radiation to the groin. Abdominal signs are usually minimal. The pain may be so severe as to interrupt one's breath.

- 4** A medical student presents with severe upper abdominal tenderness and a serum amylase greater than four times the normal limit.
- 5** Associated with sudden-onset severe tearing chest pain radiating to the back. A chest X-ray demonstrates evidence of a widened mediastinum.

THEME 2 ACUTE ABDOMEN 2

- A** Acute pancreatitis
- B** Acute pyelonephritis
- C** Biliary peritonitis
- D** Mesenteric adenitis
- E** Perforated appendix
- F** Perforated peptic ulcer
- G** Perforated sigmoid diverticulum
- H** Ruptured AAA
- I** Small bowel obstruction
- J** Superior mesenteric artery occlusion

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 25-year-old man is admitted with severe upper abdominal pain radiating to the back. He has nausea and vomiting and is clearly dehydrated. He returned from holiday in Ayia Napa earlier today.
- 2** A 33-year-old woman is admitted with a 24-hour history of nausea, vomiting and absolute constipation. Her abdomen is tympanic with increased high-pitched bowel sounds.
- 3** A 57-year-old man is admitted with onset of abdominal pain initially in the left upper quadrant. He has been self-medicating with ibuprofen after spraining his left ankle playing squash. On examination the abdomen is rigid with no bowel sounds.
- 4** A 24-year-old woman is admitted with fever, rigors and severe back pain radiating to her left groin.
- 5** A 10-year-old boy is admitted with right iliac fossa pain of several hours' duration. Initially, he has guarding and rebound tenderness. Pressing in the right iliac fossa also elicits the same pain. He rapidly progresses to having a tense abdomen.

THEME 3 GASTROINTESTINAL BLEEDING (FIRST SET)

- A** Anal fissure
- B** Colonic carcinoma
- C** Crohn disease
- D** Diverticular disease
- E** Duodenal ulcer
- F** Gastric carcinoma
- G** Haemorrhoids
- H** Laryngeal carcinoma
- I** Mallory-Weiss tear
- J** Oesophageal varices

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 60-year-old man presents with altered bowel habit, crampy abdominal pain and blood mixed in with his stools.
- 2** A 40-year-old woman presents with a long history of indigestion, worse recently, black tarry stools and feeling lightheaded.
- 3** A 32-year-old woman presents to her GP following childbirth, worried about the fresh, painless red blood on the toilet paper.
- 4** A 22-year-old medical student presents to A&E with haematemesis and chest pain following celebrations on passing his finals.
- 5** A 17-year-old woman with cystic fibrosis presents to A&E vomiting profuse volumes of fresh blood.

THEME 4 HERNIA (FIRST SET)

- A** Inguinal
- B** Pantaloon
- C** Paraumbilical
- D** Sciatic
- E** Cooper
- F** Incisional
- G** Richter
- H** Littre
- I** Spigelian
- J** Diaphragmatic

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** Associated with abdominal pain, notable distension and vomiting. It may result in perforation of the bowel.
- 2** Seen commonly in newborn babies where, typically, the left side of the chest is occupied by the bowel.
- 3** A hernia occurring through a part of the abdominal wall between the lateral border of the rectus abdominis and linea semilunaris, typically diagnosed by ultrasound.
- 4** A combined indirect and direct hernia whereby the hernial sac protrudes on either side of the inferior epigastric vessels.
- 5** A lump located near the groin which protrudes straight on coughing and is not controlled by pressure over the deep ring.

THEME 5 UPPER GASTROINTESTINAL DISORDERS

- A Achalasia
- B Diffuse oesophageal spasm
- C Pharyngeal pouch
- D Zollinger-Ellison syndrome
- E Oesophagitis
- F Duodenal ulcer
- G Adenocarcinoma
- H Squamous cell carcinoma
- I Plummer-Vinson syndrome
- J Atrophic gastritis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A condition known to cause severe gastroduodenal ulceration due to excessive gastrin secretion.
- 2 A tumour of the lower third of the oesophagus thought to be the result of longstanding reflux disease.
- 3 A motility disorder of the oesophagus characterized by a "rat tail" segment at the lower oesophageal sphincter.
- 4 Associated with severe epigastric discomfort particularly at night and relieved by food.
- 5 A condition associated with iron-deficiency anaemia and swallowing difficulties.

Answers • Themes 1-5

CRASH COURSE EMQS

Theme 1 • ACUTE ABDOMEN

- 1 **H Bowel obstruction** — A classic presentation of bowel obstruction.
- 2 **J Perforated viscus** — Free subdiaphragmatic gas on an erect chest X-ray with generalised tenderness indicates a perforated viscus.
- 3 **C Renal colic** — Loin-to-groin colicky pain with minimal abdominal signs is renal colic.
- 4 **B Pancreatitis** — Severe upper abdominal pain with amylase $> 3\times$ normal is diagnostic of acute pancreatitis.
- 5 **F Dissecting aortic aneurysm** — Tearing chest pain radiating to the back with a widened mediastinum is aortic dissection.

Theme 2 • ACUTE ABDOMEN 2

- 1 **A Acute pancreatitis** — Severe epigastric pain radiating to the back with vomiting after a holiday of heavy drinking is acute pancreatitis.
- 2 **I Small bowel obstruction** — Vomiting, absolute constipation, distension and high-pitched (tinkling) bowel sounds indicate small bowel obstruction.
- 3 **F Perforated peptic ulcer** — NSAID use followed by a rigid, silent abdomen is a perforated peptic ulcer.
- 4 **B Acute pyelonephritis** — Fever, rigors and loin pain radiating to the groin in a young woman is acute pyelonephritis.
- 5 **E Perforated appendix** — RIF pain progressing rapidly to a tense, generalised peritonitic abdomen in a child is a perforated appendix.

Theme 3 • GASTROINTESTINAL BLEEDING (FIRST SET)

- 1 **B Colonic carcinoma** — Altered bowel habit with blood mixed in the stool in a 60-year-old is colorectal (colonic) carcinoma until proven otherwise.
- 2 **E Duodenal ulcer** — Long-standing dyspepsia with melaena and light-headedness is a bleeding duodenal ulcer.
- 3 **G Haemorrhoids** — Painless fresh blood on the paper after childbirth is typical of haemorrhoids.

- 4 **I Mallory-Weiss tear** — Haematemesis after a bout of vomiting during celebrations is a Mallory-Weiss tear.
- 5 **J Oesophageal varices** — Cystic fibrosis can cause biliary cirrhosis and portal hypertension; profuse fresh haematemesis is from oesophageal varices.

Theme 4 • HERNIA (FIRST SET)

- 1 **G Richter** — In a Richter hernia only part of the bowel wall is trapped, so it can strangulate and perforate without complete obstruction.
- 2 **J Diaphragmatic** — A newborn with bowel in the left chest has a congenital (Bochdalek) diaphragmatic hernia.
- 3 **I Spigelian** — A Spigelian hernia passes through the linea semilunaris at the lateral rectus border and is often only confirmed by ultrasound.
- 4 **B Pantaloon** — A pantaloon hernia has direct and indirect sacs straddling the inferior epigastric vessels.
- 5 **A Inguinal** — A groin lump protruding straight forward and not controlled by deep-ring pressure is a direct inguinal hernia.

Theme 5 • UPPER GASTROINTESTINAL DISORDERS

- 1 **D Zollinger-Ellison syndrome** — Zollinger-Ellison syndrome: a gastrinoma drives acid hypersecretion and severe, multiple or atypical ulcers.
- 2 **G Adenocarcinoma** — Adenocarcinoma of the lower third of the oesophagus arises from Barrett metaplasia after longstanding reflux.
- 3 **A Achalasia** — Achalasia gives a smooth "bird's beak / rat tail" narrowing at the lower oesophageal sphincter on barium swallow.
- 4 **F Duodenal ulcer** — Epigastric pain at night relieved by food is typical of duodenal ulcer.
- 5 **I Plummer-Vinson syndrome** — Plummer-Vinson syndrome: iron-deficiency anaemia with dysphagia from a post-cricoid web.

SET 2 OF 4

THEME 6 LOWER GASTROINTESTINAL DISORDERS

- A Perianal abscess
- B Intussusception
- C Perianal haematoma
- D Haemorrhoids
- E Crohn disease
- F Angiodysplasia
- G Collagenous colitis
- H Ulcerative colitis
- I Anal fistula
- J Ischaemic colitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Associated with vomiting and the passage of redcurrant-like stool in children.
- 2 A bluish tender perianal swelling often resulting from constipation and straining.
- 3 Typically affects the right colon and associated with severe rectal bleeding. Colonoscopy may reveal the presence of dilated tortuous vessels and "cherry red" areas.
- 4 A pan-intestinal inflammatory disorder associated with deep ulcers in the mucosa to give a cobblestone-like appearance.
- 5 Associated with left-sided abdominal pain and dark red rectal bleeding. Examination findings may demonstrate evidence of hypotension and abdominal distension. A barium enema may demonstrate thickening and blunting of the mucosal folds of the bowel.

THEME 7 HEPATOBILIARY DISORDERS

- A Cholangitis
- B Hepatic haemangioma
- C Focal nodular hyperplasia of the liver
- D Cholangiocarcinoma
- E Hepatoma
- F Cholecystitis
- G Biliary colic
- H Acalculous cholecystitis
- I Pancreatitis
- J Hepatic abscess

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient presenting with recurrent pain in the right upper quadrant radiating to the tip of the scapula. There is no associated pyrexia.
- 2 A tumour presenting typically with painless jaundice, known to result commonly from primary sclerosing cholangitis.
- 3 A patient presenting with tenderness and guarding in the right upper quadrant with evidence of hyperaesthesia below the right scapula and pyrexia.

- 4 A patient presenting with abdominal pain, fever and jaundice as a result of *Escherichia coli* infection.
- 5 A female patient presenting with anorexia, ascites and abdominal discomfort with a background history of cirrhosis and long-term use of the oral contraceptive pill.

THEME 8 ABDOMINAL PAIN (FIRST ONE)

- A Acute appendicitis
- B Acute diverticulitis
- C Colorectal cancer
- D Crohn disease
- E Familial adenomatous polyposis
- F Perforated duodenal ulcer
- G Renal colic
- H Strangulated inguinal hernia
- I Ulcerative colitis
- J Testicular torsion

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 19-year-old male who was previously fit and well presents with generalized abdominal pain for 3 days which is worsening, weight loss, diarrhoea and dysuria. He is most tender in the right iliac fossa. He has no appetite and has been feverish.
- 2 A 35-year-old man presents with 5 months of colicky abdominal pain, weight loss and bloody diarrhoea with mucus. He has also been suffering from lower back pain for the last few months.
- 3 A 37-year-old man presents with acute sudden right testicular pain which radiates to the right groin, and has blood +++ in his urine.
- 4 A 35-year-old man presents with red blood mixed in his stool and weight loss over the last 6 months. He had a colonoscopy 5 years ago, but is an infrequent attender to his GP. His father died of colorectal cancer aged 46.
- 5 A 60-year-old woman presents with a 2-day history of left iliac fossa pain and a significant (PR) bleed of bright red blood which fills the pan. She has had alternating diarrhoea and constipation over the last year.

THEME 9 HERNIAS (SECOND SET)

- A Congenital diaphragmatic hernia
- B Epigastric hernia
- C Femoral hernia
- D Incarcerated hernia
- E Incisional hernia
- F Inguinal hernia
- G Obturator hernia
- H Rolling hiatus hernia
- I Sliding hiatus hernia
- J Spigelian hernia
- K Umbilical hernia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 64-year-old labourer presents to his GP with increasing pain in his groin. He has had a lump for several years which usually disappears on lying. It no longer does so. On examination you can feel a mass in the right groin arising above the inguinal ligament. It is irreducible but bowel sounds are present.
- 2 A 57-year-old woman is referred to the surgeon who performed her laparoscopic cholecystectomy 2 years previously. She has developed a bulge, which on examination is in the midclavicular line in the right upper quadrant of the abdomen. It disappears on lying down and "gurgles."
- 3 A 92-year-old frail lady is admitted to hospital with severe abdominal pain, colicky in nature, with nausea and vomiting. A nasogastric tube is passed, but on abdominal X-ray, no bowel distension is seen. Close inspection of the film reveals the diagnosis.
- 4 A 31-year-old man presents to his GP with a mass in his left groin. It disappears on lying, is above the inguinal ligament and reappears on coughing. His brother had the same problem.
- 5 A 29-year-old woman presents to her GP with an abdominal mass following the birth of her first child. During her pregnancy, she had had an everted umbilicus and this has worsened after giving birth.

THEME 10 ACUTE ABDOMINAL PAIN (SECOND SET)

- A Acute appendicitis
- B Acute cholecystitis
- C Acute diverticulitis
- D Acute pancreatitis
- E Biliary colic
- F Large bowel obstruction
- G Perforated peptic ulcer
- H Perforated sigmoid diverticular disease
- I Small bowel obstruction
- J Urinary tract infection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 69-year-old woman presents with a 3-day history of constipation and constant pain in the left iliac fossa. The pain has suddenly become much worse and she has collapsed and been admitted to casualty. On examination she has a tachycardia and is hypotensive. There is severe lower abdominal pain with guarding throughout the mid- and lower abdomen.
- 2 A 50-year-old obese woman presents with epigastric pain. On examination, her temperature is 38.5°C. She is tender in the upper abdomen and Murphy sign is positive.
- 3 A 22-year-old woman presents with pain in the right iliac fossa. The patient is anorexic, has not vomited, but had some dysuria and frequency. Her temperature is 37.5°C. The patient is flushed and has localized guarding in the right iliac fossa and suprapubic region.
- 4 A 72-year-old man presents with increasing constipation. He has colicky lower abdominal pain and has not passed flatus for 24 hours. Examination reveals a distended abdomen which is tympanitic on percussion.
- 5 A 60-year-old man presents with a 48-hour history of sudden-onset epigastric pain radiating through to the back after an alcoholic binge. Examination reveals the patient to be afebrile, tachycardic and normotensive. The patient is diffusely tender, with guarding in the epigastrium. An erect chest X-ray is normal, but the blood gas analysis reveals hypoxia.

Answers • Themes 6–10

Theme 6 • LOWER GASTROINTESTINAL DISORDERS

- 1 **B Intussusception** — Vomiting with redcurrant-jelly stool in a child is intussusception.
- 2 **C Perianal haematoma** — A bluish, tender perianal lump after straining is a perianal haematoma.
- 3 **F Angiodysplasia** — Angiodysplasia affects the right colon; colonoscopy shows dilated tortuous vessels and cherry-red spots.

- 4 **E Crohn disease** — Crohn disease can affect any part of the gut, with deep fissuring ulcers giving a cobblestone mucosa.
- 5 **J Ischaemic colitis** — Left-sided pain with dark red bleeding and "thumbprinting" (thickened folds) in an older patient is ischaemic colitis.

Theme 7 • HEPATOBILIARY DISORDERS

- 1 **G Biliary colic** — Recurrent RUQ pain radiating to the scapula without fever is biliary colic.

- 2 **D Cholangiocarcinoma** — Cholangiocarcinoma presents with painless jaundice and is a recognised complication of primary sclerosing cholangitis.
- 3 **F Cholecystitis** — RUQ tenderness and guarding with fever and Boas sign (hyperaesthesia below the right scapula) is acute cholecystitis.
- 4 **A Cholangitis** — Pain, fever and jaundice (Charcot triad) from *E. coli* infection of an obstructed duct is ascending cholangitis.
- 5 **E Hepatoma** — Cirrhosis with ascites and anorexia suggests hepatoma (hepatocellular carcinoma).

Theme 8 · ABDOMINAL PAIN (FIRST ONE)

- 1 **A Acute appendicitis** — A pelvic appendix can cause dysuria and diarrhoea; fever, anorexia and RIF tenderness point to acute appendicitis.
- 2 **I Ulcerative colitis** — Bloody mucoid diarrhoea with weight loss and back pain (sacroiliitis/ankylosing spondylitis) suggests ulcerative colitis.
- 3 **G Renal colic** — Sudden pain radiating to the groin and testis with haematuria is a ureteric stone (renal colic), not torsion.
- 4 **E Familial adenomatous polyposis** — Rectal bleeding at 35 with a father dead of colorectal cancer at 46 suggests familial adenomatous polyposis.
- 5 **B Acute diverticulitis** — LIF pain with a large bright-red bleed and altered bowel habit in an older woman is acute diverticulitis/diverticular bleeding.

Theme 9 · HERNIAS (SECOND SET)

- 1 **D Incarcerated hernia** — A long-standing hernia that no longer reduces but has no obstruction is incarcerated; it needs urgent repair.

- 2 **E Incisional hernia** — A reducible bulge at a laparoscopic port site is an incisional (port-site) hernia.
- 3 **G Obturator hernia** — An elderly thin woman with obstruction and no visible lump suggests an obturator hernia (gas shadow in the obturator foramen).
- 4 **F Inguinal hernia** — A reducible lump above the inguinal ligament with a cough impulse and a family history is an inguinal hernia.
- 5 **K Umbilical hernia** — An everted umbilicus that worsens after pregnancy is an umbilical (paraumbilical) hernia.

Theme 10 · ACUTE ABDOMINAL PAIN (SECOND SET)

- 1 **H Perforated sigmoid diverticular disease** — Sudden collapse with peritonitis on a background of LIF pain is perforated sigmoid diverticular disease.
- 2 **B Acute cholecystitis** — Fever, upper abdominal tenderness and a positive Murphy sign is acute cholecystitis.
- 3 **A Acute appendicitis** — A pelvic appendix may cause urinary symptoms; RIF and suprapubic guarding with fever indicate acute appendicitis.
- 4 **F Large bowel obstruction** — Progressive constipation, colicky pain, no flatus and a tympanitic distended abdomen is large bowel obstruction.
- 5 **D Acute pancreatitis** — Epigastric pain to the back after a binge with hypoxia (early ARDS) and a normal erect CXR is acute pancreatitis.

SET 3 OF 4**THEME 11 · INTESTINAL OBSTRUCTION**

- A Adhesions
- B Carcinoma caecum
- C Carcinoma rectum
- D Carcinoma sigmoid
- E Gallstone ileus
- F Intussusception
- G Pseudo-obstruction
- H Sigmoid volvulus
- I Strangulated femoral hernia
- J Strangulated inguinal hernia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 80-year-old man presents with a 4-day history of abdominal distension and pain. He is vomiting faeculent fluid. He has not been feeling well for 3 months and has lost 2 stone in weight. Clinical examination reveals visible peristalsis in the midabdomen, distension and a mass in the right iliac fossa.

- 2 A 68-year-old woman presents with colicky abdominal pain and vomiting. Examination shows that she is dehydrated, with abdominal distension and a lower mid-line scar from a perforated appendix. Plain abdominal film shows multiple distended small bowel loops and the presence of air in the biliary tree.
- 3 A 60-year-old man undergoes emergency lumbar disc decompression. Postoperatively, he is immobile. He becomes increasingly constipated and develops a distended abdomen. Plain abdominal films show a grossly distended colon down to the pelvic brim.
- 4 A 78-year-old man presents with a 3-day history of vomiting faeculent fluid. He has a grossly distended abdomen and a palpable mass in the right groin. The mass is firm, slightly tender and lies below and lateral to the pubic tubercle.
- 5 An 88-year-old woman with dementia is referred by her GP because she seems to be in pain and unwell. She has a long history of constipation. On examination, she has a grossly distended tympanitic abdomen which is nontender. The plain abdominal X-ray shows a large distended loop of colon.

THEME 12 GASTROINTESTINAL BLEEDING (SECOND SET)

- A Angiodysplasia
- B Anterior duodenal ulcer
- C Caecal carcinoma
- D Crohn disease
- E Mallory-Weiss tear
- F Oesophageal varices
- G Posterior duodenal ulcer
- H Rectal carcinoma
- I Sigmoid carcinoma
- J Sigmoid diverticular disease

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 82-year-old woman suffering with osteoarthritis of the hip is admitted with haematemesis and melaena. She has been taking ibuprofen in increasing amounts to control her hip pain.
- 2 A 77-year-old man presents with his fourth episode of acute rectal bleeding. The blood is a mixture of fresh blood and clots. On this occasion, the bleeding has been severe enough to require a 4-unit blood transfusion. A barium enema is undertaken and is normal.
- 3 A 56-year-old man presents with anaemia and weight loss. Examination reveals a mass in the right iliac fossa and hepatomegaly.
- 4 A 50-year-old alcoholic presents with melaena. On examination the patient is drowsy and hypotensive. Examination of the abdomen shows splenomegaly.
- 5 A 58-year-old man is admitted with acute onset of left iliac fossa pain. Recently, he has noticed he has had some vague abdominal pain and felt more constipated. On examination he is pyrexial, pale and has localized peritonism in the left iliac fossa. His abdomen is distended. Investigations show a haemoglobin (Hb) of 7 g/dL (hypochromic, microcytic) and a WCC of $18 \times 10^9/L$.

THEME 13 OESOPHAGOGASTRIC DISORDERS

- A Achalasia
- B Duodenal ulcer disease
- C Gastric cancer
- D Gastric ulcer disease
- E GORD
- F Helicobacter atrophic gastritis
- G Mallory-Weiss tear
- H Oesophageal adenocarcinoma
- I Oesophageal squamous carcinoma
- J Pyloric stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 45-year-old man presents with severe epigastric pain. The pain is most severe when he is hungry and often wakes him from sleep at night. The pain is relieved by eating and by taking milk and snacks. The patient has gained half a stone in weight since his symptoms started.
- 2 A 19-year-old student experiences worsening dysphagia for 3 months. She has lost a stone in weight and has had two courses of antibiotics for a persistent chest infection.
- 3 A 55-year-old vagrant man has had a long history of recurrent episodes of epigastric pain. He presents with weight loss and severe vomiting. On admission, he is noted to be dehydrated and abdominal examination demonstrates a succussion splash.
- 4 A 65-year-old man presents with a long history of indigestion self-treated with antacids, but progressive dysphagia and weight loss for 3 weeks. On examination, the patient is anaemic and is markedly cachectic.
- 5 A 27-year-old man is admitted to the casualty with frank haematemesis. He has been on his stag weekend and after consuming large amounts of alcohol, has been vomiting repeatedly, and has some epigastric pain.

THEME 14 FOREGUT INVESTIGATIONS

- A Abdominal ultrasound
- B Barium meal
- C Barium swallow
- D Chest X-ray
- E CT scan
- F ERCP
- G Endoscopic ultrasonography
- H Gastroscopy
- I Laparoscopy
- J Magnetic resonance scanning

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 55-year-old man presents with flatulent dyspepsia, fatty food intolerance and right upper quadrant pain.
- 2 A 65-year-old man is diagnosed with oesophageal adenocarcinoma. The surgeon wishes to stage the local extent of the tumour and to assess whether there is local lymph node spread or invasion of the thoracic aorta.
- 3 A 70-year-old man presents with painless obstructive jaundice. He has undergone assessment by ultrasonography, showing a mass in the head of the pancreas. The surgeon wishes to stage the tumour.
- 4 An 80-year-old man presents with intermittent cervical dysphagia and has a soft swelling on the left side of his neck. He coughs and splutters after eating and often aspirates liquids.

- 5** A 59-year-old man has carcinoma of the stomach. He is suitable for surgical resection but the surgeon wants to exclude peritoneal seedlings before undertaking a laparotomy.

THEME 15 ABDOMINAL MASS

- A** Appendix mass
- B** Caecal carcinoma
- C** Crohn disease
- D** Diverticular abscess
- E** Empyema of the gallbladder
- F** Hepatomegaly
- G** Mucocoele of the gallbladder
- H** Ovarian carcinoma
- I** Ovarian cyst
- J** Sigmoid carcinoma
- K** Sigmoid volvulus
- L** Uterine fibroids

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 69-year-old woman presents to her GP with a history of fatigue for 3 months, and 4 days of colicky abdominal pain and vomiting. Examination shows that she is pale and dehydrated, with a mass in the right iliac fossa.

- 2** A 72-year-old woman presents with a 3-week history of increasing discomfort in her lower abdomen. She has not opened her bowels for 5 days but has passed flatus. On examination, she looks pale and dehydrated and the abdomen is distended and tympanitic. There is an irregular nontender mass in the left iliac fossa region.

- 3** A 25-year-old woman presents with acute onset of generalized abdominal pain. She has been unwell for about 2 months and has lost half a stone in weight. On examination, she looks unwell, has a pyrexia and a tender mass felt in the right iliac fossa with diffuse tenderness in the lower abdomen.

- 4** A 67-year-old man is seen in the outpatient clinic following an episode of severe colicky upper abdominal pain which has settled to leave some discomfort. He has a history of 'indigestion' after fatty food and alcohol intake of at least 4 units a day. On examination, he is slightly jaundiced, afebrile and, on abdominal examination, has a smooth, nontender mass in the right hypochondrium.

- 5** A 45-year-old woman complains to her GP of tiredness and a dragging sensation in the lower abdomen. She has noticed that her clothes do not fit as well. On examination, she is anaemic and her abdomen is distended with a mass consistent with the size of an 18-week pregnancy in the suprapubic region.

Answers • Themes 11–15

CRASH COURSE EMQS

Theme 11 • INTESTINAL OBSTRUCTION

- 1** **B Carcinoma caecum** — Faeculent vomiting, weight loss and a RIF mass indicate obstructing caecal carcinoma.
- 2** **E Gallstone ileus** — Small bowel obstruction with pneumobilia (air in the biliary tree) is gallstone ileus.
- 3** **G Pseudo-obstruction** — Colonic dilatation after spinal surgery in an immobile patient is acute colonic pseudo-obstruction (Ogilvie syndrome).
- 4** **I Strangulated femoral hernia** — Obstruction with a firm tender lump below and lateral to the pubic tubercle is a strangulated femoral hernia.
- 5** **H Sigmoid volvulus** — A grossly distended non-tender abdomen with a large colonic loop in an elderly constipated patient is sigmoid volvulus.

Theme 12 • GASTROINTESTINAL BLEEDING (SECOND SET)

- 1** **G Posterior duodenal ulcer** — NSAID use with haematemesis and melaena: a posterior duodenal ulcer eroding the gastroduodenal artery.
- 2** **A Angiodysplasia** — Recurrent heavy lower GI bleeding in an elderly man with a normal barium enema suggests angiodysplasia.
- 3** **C Caecal carcinoma** — Anaemia, weight loss, a RIF mass and hepatomegaly (metastases) indicate caecal carcinoma.

- 4** **F Oesophageal varices** — Melaena in an alcoholic with splenomegaly, drowsiness and hypotension is bleeding oesophageal varices.

- 5** **I Sigmoid carcinoma** — Microcytic anaemia, change in habit and LIF peritonism with fever suggest a perforated/obstructing sigmoid carcinoma.

Theme 13 • OESOPHAGOGASTRIC DISORDERS

- 1** **B Duodenal ulcer disease** — Hunger pain that wakes him at night, relieved by food, with weight gain is duodenal ulcer disease.
- 2** **A Achalasia** — Progressive dysphagia in a young person with recurrent chest infections (aspiration) is achalasia.
- 3** **J Pyloric stenosis** — Recurrent ulcer pain then vomiting, dehydration and a succussion splash is gastric outlet (pyloric) stenosis.
- 4** **H Oesophageal adenocarcinoma** — Long-standing reflux followed by rapidly progressive dysphagia and weight loss is oesophageal adenocarcinoma.
- 5** **G Mallory-Weiss tear** — Haematemesis after repeated vomiting during heavy drinking is a Mallory-Weiss tear.

Theme 14 • FOREGUT INVESTIGATIONS

- 1** **A Abdominal ultrasound** — Flatulent dyspepsia, fatty-food intolerance and RUQ pain suggest gallstones: abdominal ultrasound first.

- 2 **G Endoscopic ultrasonography** — Endoscopic ultrasound gives the best local T and N staging of oesophageal cancer.
- 3 **E CT scan** — Pancreatic cancer is staged by contrast CT (pancreatic protocol) for vascular involvement and metastases.
- 4 **C Barium swallow** — Cervical dysphagia, a soft neck swelling and aspiration suggest a pharyngeal pouch: barium swallow (not blind endoscopy).
- 5 **I Laparoscopy** — Staging laparoscopy detects peritoneal seedlings missed by CT before gastric cancer resection.

Theme 15 • ABDOMINAL MASS

- 1 **B Caecal carcinoma** — Fatigue (anaemia), subacute obstruction and a RIF mass in an older woman is caecal carcinoma.

SET 4 OF 4

THEME 16 GROIN LUMP

- A False aneurysm
- B Femoral artery aneurysm
- C Femoral hernia
- D Inguinal hernia
- E Lymphocele
- F Lymphoma
- G Malignant lymphadenopathy
- H Psoas abscess
- I Obturator hernia
- J Reactive lymphadenopathy
- K Saphena varix
- L Soft tissue sarcoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 80-year-old woman presents with a 2-day history of a lump in the left groin. It has been associated with colicky abdominal pain, vomiting and abdominal distension. Examination reveals a red, tender, irreducible lump below and lateral to the pubic tubercle. The abdomen is distended, with high-pitched 'tinkling' bowel sounds.
- 2 A 75-year-old lady presents with a 3-month history of a right groin lump. It becomes painful when she walks, but disappears when she rests. Examination reveals a right groin lump with cough impulse. It reduces to a point above and medial to the pubic tubercle. She also has long-standing varicose veins.
- 3 A 70-year-old woman presents with a pretibial laceration but also mentions that she has a 4-week history of a lump in the right groin. It is completely painless but has gradually increased in size during this period. She had a melanoma excised from her right calf 2 years previously. Examination reveals a firm, nontender, irregular lump below and lateral to the pubic tubercle.

- 2 **J Sigmoid carcinoma** — Change in habit with distension and an irregular LIF mass is sigmoid carcinoma.
- 3 **C Crohn disease** — A young woman with weight loss, fever and a tender RIF mass suggests an inflammatory Crohn mass.
- 4 **G Mucocoele of the gallbladder** — A smooth non-tender RUQ mass after biliary colic is a mucocoele (stone impacted in Hartmann pouch).
- 5 **L Uterine fibroids** — A suprapubic mass with menorrhagia-related anaemia in a 45-year-old woman is uterine fibroids.

- 4 An 80-year-old woman with recent onset of a leg ulcer presents with a painless lump in her right groin. She has recently had a femoral angiogram. On examination, there is a 3 x 2 cm pulsatile lump.

THEME 17 RECTAL BLEEDING

- A Anal carcinoma
- B Anal fissure
- C Anal skin tags
- D Angiodysplasia
- E Crohn disease
- F Haemorrhoids
- G Meckel diverticulum
- H Peptic ulceration
- I Rectal adenocarcinoma
- J Rectal villous adenoma
- K Sigmoid carcinoma
- L Sigmoid diverticular disease
- M Ulcerative colitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 60-year-old woman presents with fresh rectal bleeding, diarrhoea, profuse mucous discharge, tenesmus and lethargy. Investigations show that she is anaemic (Hb 10 g/dL) and has a potassium of 2.2 mmol/L.
- 2 A 50-year-old woman presents with a short history of painful, fresh rectal bleeding. She has a history of cervical intraepithelial neoplasia (CIN). On examination, she has a single indurated painful ulcer in the perianal region.
- 3 A 5-year-old girl's mother brings her to the GP because of increasing problems with constipation and she has noticed some fresh blood on her clothes.
- 4 An 80-year-old man is admitted as an emergency with acute onset of profuse, painless, dark red rectal bleeding. He is normally fit and well and has no history of bowel problems. On examination, he is pale, cold, clammy, hypotensive and his abdomen is nontender.

Answers • Themes 16–17

CRASH COURSE EMQs

Theme 16 • GROIN LUMP

- C Femoral hernia** — An irreducible tender lump below and lateral to the pubic tubercle with obstruction is a strangulated femoral hernia.
- D Inguinal hernia** — A lump with cough impulse that reduces above and medial to the pubic tubercle is an inguinal hernia.
- G Malignant lymphadenopathy** — A hard irregular groin node after a calf melanoma is malignant (metastatic) lymphadenopathy.
- A False aneurysm** — A pulsatile groin lump after femoral puncture is a false (pseudo-) aneurysm.

Theme 17 • RECTAL BLEEDING

- J Rectal villous adenoma** — Mucous diarrhoea with hypokalaemia is classic for a secretory rectal villous adenoma.
- A Anal carcinoma** — A painful indurated perianal ulcer in a woman with CIN (HPV) is anal carcinoma.
- B Anal fissure** — Constipation with fresh blood on clothes in a child is most often an anal fissure.
- D Angiodysplasia** — Profuse painless dark red bleeding in a fit elderly man suggests angiodysplasia (or diverticular bleeding).

10 THEMES • 44 SCENARIOS • IRFAN SURGERY EMQ COLLECTION

Irfan EMQs

SET 1 OF 2

THEME 1 Abdominal pain (i)

- A A large bowel obstruction
- B acute pancreatitis
- C perforated viscus
- D appendicitis
- E small bowel obstruction
- F acute cholecystitis
- G ulcerative colitis
- H aortic dissection
- I diverticulosis
- J duodenal ulcer
- K renal colic
- L colorectal carcinoma
- M mesenteric adenitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- A 45-year-old man with a history of gallstones presents in A&E with severe epigastric pain radiating to the back and vomiting.
- A 28-year-old man presents with sharp left loin and left upper quadrant pain radiating to the groin. He is not jaundiced.
- A 44-year-old woman presents with continuous right upper quadrant pain, vomiting and fever. There is marked right upper quadrant tenderness when palpating on inspiration.
- A 26-year-old male with a previous history of abdominal surgery presents with colicky central abdominal pain rapidly followed by production of copious bile-stained vomitus.

- A 50-year-old man with a history of epigastric pain presents with constant, severe, generalised abdominal pain. He is distressed with a rigid abdomen; pulse 110/min, BP 100/60 mmHg.

THEME 2 Abdominal pain (ii)

- A hepatitis
- B irritable bowel syndrome
- C umbilical hernia
- D primary sclerosing cholangitis
- E perforated duodenal ulcer
- F small bowel obstruction
- G ulcerative colitis
- H Crohn's disease
- I carcinoma of caecum
- J acute appendicitis
- K gastric ulcer
- L hepatocellular carcinoma
- M diverticulitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- A 21-year-old student presents with diffuse cramping abdominal pain with alternating constipation and diarrhoea; colonoscopy and inflammatory markers are normal.
- A 9-year-old girl presents with fever, nausea and right iliac fossa pain that initially started around the belly button.
- A 35-year-old man has weight loss, diarrhoea, abdominal pain and aphthous mouth ulcers; a mass is palpable in the right iliac fossa. Blood tests show low serum B12 and folate.

- 4 A 72-year-old man with a history of constipation has fever, diarrhoea and left iliac fossa pain, with tenderness in the left iliac fossa.

THEME 3 Abnormal abdominal x-rays

- A Crohn's disease
- B diverticulosis
- C chronic pancreatitis
- D acute pancreatitis
- E gluten-sensitive enteropathy
- F acute appendicitis
- G sigmoid volvulus
- H intussusception
- I ulcerative colitis
- J pyloric stenosis
- K small bowel obstruction
- L perforated duodenal ulcer

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 55-year-old patient with several months of epigastric discomfort presents acutely with free gas under the diaphragm on x-ray.
- 2 An elderly constipated woman's x-ray shows a dilated 'inverted U' loop of bowel.
- 3 A 31-year-old man presents with fever, bloody diarrhoea and a colonic dilatation of 8 cm with loss of haustral pattern.
- 4 A 26-year-old student presents with bloody diarrhoea, abdominal pain and weight loss; barium enema shows 'cobblestoning' and strictures.
- 5 A 45-year-old man presents with severe epigastric pain and vomiting; x-ray reveals an absent psoas shadow and a 'sentinel loop' of proximal jejunum.

THEME 4 Complications of gallstones

- A biliary colic
- B gallbladder carcinoma
- C acute pancreatitis
- D gallstone ileus
- E ascending cholangitis
- F acute cholecystitis
- G empyema
- H chronic cholecystitis
- I mucocele

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 45-year-old woman presents with fever and significant right upper quadrant pain. On examination, there is marked jaundice and the nurse has reported that the patient is having rigors. Blood tests show plasma bilirubin 250 mmol/L, ALT 200 U/L, ALP 800 U/L, WCC 23×10^9 .
- 2 A 55-year-old man presents with constant right upper quadrant pain associated with vomiting. He is afebrile.
- 3 A 63-year-old man presents with a 3-day history of worsening abdominal pain and vomiting. Abdominal x-ray reveals dilated loops of small bowel and air in the biliary tree.
- 4 A 52-year-old man presents with vomiting and severe epigastric pain radiating to the back. Abdominal x-ray shows a dilated proximal small bowel loop.

THEME 5 Management of inflammatory bowel disease

- A intravenous mesalazine
- B steroid enemas
- C stricturoplasty
- D high-dose intravenous steroids
- E proctocolectomy with ileal pouch
- F proctocolectomy with ileostomy
- G elemental diet
- H subtotal colectomy and ileostomy
- I fistulectomy
- J extensive small bowel resection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old patient suffering from frequent debilitating exacerbations of ulcerative colitis who desires curative surgery.
- 2 A 28-year-old patient with ulcerative colitis refractory to medical therapy who agrees on surgery but does not want a stoma.
- 3 A 30-year-old patient with ulcerative colitis presents to A&E with severe abdominal pain and is diagnosed with toxic megacolon.
- 4 A 28-year-old patient with Crohn's disease presents with complete small bowel obstruction; small bowel follow-through shows short strictures.

Answers • Themes 1-5

Theme 1 • Abdominal pain (i)

- 1 **B acute pancreatitis** — Acute pancreatitis: classic severe epigastric pain radiating to the back, often due to gallstones.
- 2 **K renal colic** — Renal colic: severe, colicky loin pain radiating to the groin, typically with nausea.

- 3 **F acute cholecystitis** — Acute cholecystitis: positive Murphy's sign with right upper quadrant tenderness.
- 4 **E small bowel obstruction** — Small bowel obstruction: early vomiting with progressive abdominal distension.
- 5 **C perforated viscus** — Perforated viscus: rigid abdomen indicating peritonitis.

Theme 2 · Abdominal pain (ii)

- 1 **B irritable bowel syndrome** — Irritable bowel syndrome: diagnosis of exclusion in young patients with alternating bowel habits.
- 2 **J acute appendicitis** — Acute appendicitis: typical shift of pain from the periumbilical region to the right iliac fossa.
- 3 **H Crohn's disease** — Crohn's disease: chronic inflammatory process with systemic features such as weight loss and oral ulcers.
- 4 **M diverticulitis** — Diverticulitis: inflammation of colonic diverticula typically in the sigmoid colon.

Theme 3 · Abnormal abdominal x-rays

- 1 **L perforated duodenal ulcer** — Perforated duodenal ulcer: free gas under the diaphragm indicates gastrointestinal perforation.
- 2 **G sigmoid volvulus** — Sigmoid volvulus: twisting of the sigmoid colon commonly seen in elderly patients.
- 3 **I ulcerative colitis** — Ulcerative colitis: radiological loss of haustra with colonic dilatation.
- 4 **A Crohn's disease** — Crohn's disease: barium enema features include cobblestoning and strictures.
- 5 **D acute pancreatitis** — Acute pancreatitis: the 'sentinel loop' suggests localized ileus near the inflamed pancreas.

Theme 4 · Complications of gallstones

- 1 **E ascending cholangitis** — Fever with rigors, RUQ pain and jaundice (Charcot triad) is ascending cholangitis: IV antibiotics and urgent biliary drainage (ERCP).
- 2 **A biliary colic** — A Biliary colic: Biliary colic is the result of gallbladder spasm due to a stone lodged in Hartmann's pouch or the cystic duct.
- 3 **D gallstone ileus** — D Gallstone ileus: Gallstone ileus is caused by mechanical obstruction from a gallstone that enters the bowel via a fistula.
- 4 **C acute pancreatitis** — C Acute pancreatitis: Acute pancreatitis in this context is due to pancreatic duct obstruction with subsequent enzyme activation.

Theme 5 · Management of inflammatory bowel disease

- 1 **F proctocolectomy with ileostomy** — F Proctocolectomy with ileostomy: A curative procedure removing the entire diseased colon, resulting in a permanent end stoma.
- 2 **E proctocolectomy with ileal pouch** — E Proctocolectomy with ileal pouch: Allows pouch formation to avoid a permanent stoma in young patients.
- 3 **H subtotal colectomy and ileostomy** — H Subtotal colectomy and ileostomy: The emergency procedure for toxic megacolon in acutely inflamed bowel.
- 4 **C stricturoplasty** — C Stricturoplasty: A bowel-conserving procedure for short strictures in Crohn's disease.

SET 2 OF 2

THEME 6 Management of colorectal cancer

- A right hemicolectomy
- B extended left hemicolectomy
- C Hartmann's procedure
- D anterior resection
- E subtotal colectomy
- F sigmoidectomy
- G extended right hemicolectomy
- H abdominoperineal resection
- I total colectomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man is found to have a rectal carcinoma invading the anal sphincter.
- 2 A 69-year-old man is brought to A&E with an acute abdomen. After resuscitation, laparotomy shows a perforated sigmoid colon secondary to a malignant-looking mass.
- 3 A 49-year-old man with weight loss and anaemia of unknown origin is found to have a large caecal tumour on colonoscopy.
- 4 A 58-year-old man with fresh rectal bleeding is diagnosed with a tumour in the middle third of the rectum.

THEME 7 Investigation of gastrointestinal bleeding

- A barium swallow
- B flexible sigmoidoscopy
- C watch and wait
- D faecal occult blood
- E CT abdomen
- F ERCP
- G oesophagogastrroduodenoscopy
- H small bowel follow-through
- I colonoscopy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 42-year-old alcoholic presents with haematemesis.
- 2 A 65-year-old man has weight loss and diarrhoea and has noticed blood mixed in with his stools.
- 3 A 74-year-old woman presents with a 3-month history of worsening constipation and rectal bleeding.
- 4 A 65-year-old man with diverticular disease is admitted with suspected diverticulitis and is not improving after 24 hours of antibiotics and intravenous fluids.

REPEATED 2x · ALSO ASKED IN
Bailey & Love

THEME 8 Small bowel obstruction

- A adhesions
- B strangulated inguinal hernia
- C small bowel atresia
- D Crohn's disease
- E irritable bowel syndrome
- F intussusception
- G intra-abdominal abscess
- H Meckel's diverticulum
- I midgut volvulus

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 54-year-old woman has a 48-hour history of colicky abdominal pain, vomiting and distension. Pulse 120/min, BP 100/75 mmHg, temperature 38 °C. There is generalised tenderness with a firm, tender 3 × 4 cm swelling in the right groin; bowel sounds are absent.
- 2 A 13-year-old boy underwent an appendicectomy 7 days ago for a suppurative appendicitis. Over the past 48 hours he has complained of right iliac fossa pain, vomiting and abdominal distension. Bowel sounds are absent. Basic observations on arrival are: pulse 110/min, BP 105/64 mmHg, temperature 37.2°C. His mother is a nurse and says that he has had spiking temperatures of above 38°C at home.
- 3 A 76-year-old man with a history of hemicolectomy 3 years ago presents to his GP with worsening colicky abdominal pain associated with vomiting, and abdominal distension. Basic observations on arrival are: pulse 98/min, BP 165/75 mmHg, temperature 37.8°C. On abdominal examination there is a distended abdomen with no tenderness, rebound or guarding. Bowel sounds are tinkling.
- 4 A 23-year-old man with 6 months of weight loss, anorexia, postprandial abdominal pain and diarrhoea now has colicky pain, vomiting, absolute constipation and distension for 3 days. Pulse 120/min, BP 89/56 mmHg, temperature 38.2 °C; central tenderness without guarding. Hb 10.1 g/dL, WCC $18.0 \times 10^9/L$, CRP 155, albumin 28.

THEME 9 Abdominal masses

- A renal cell carcinoma
- B ovarian carcinoma
- C gastric carcinoma
- D sigmoid carcinoma
- E fibroids
- F pancreatic pseudocyst
- G diverticulosis
- H hepatocellular carcinoma
- I caecal carcinoma
- J psoas abscess
- K abdominal aortic aneurysm
- L ovarian cyst

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man collapses in the street. He has an expansile, pulsatile abdominal mass above the umbilicus.
- 2 A 75-year-old man with 3 months of dyspepsia has weight loss and abdominal distension. There is a 3.5-cm hard, irregular, tender epigastric mass that moves on respiration, shifting dullness and a palpable left supraclavicular node.
- 3 A 70-year-old woman has a mass in the right iliac fossa and severe microcytic anaemia. The mass is firm, irregular and 4 cm across, and its lower edge is palpable.
- 4 A 35-year-old woman has an abdominal mass growing over 6 months with very heavy menstrual bleeding but no intermenstrual bleeding. A knobbly mass is felt in the lower midline, dull to percussion; its lower edge is not palpable.
- 5 A 70-year-old alcoholic has a tender upper abdominal mass. CT shows a thick-walled, rounded, fluid-filled mass adjacent to the pancreas.

THEME 10 Anorectal conditions

- A fissure-in-ano
- B perianal warts
- C proctalgia fugax
- D second-degree haemorrhoids
- E fistula-in-ano
- F pilonidal abscess
- G anal carcinoma
- H rectal prolapse
- I pruritus ani
- J ischiorectal abscess
- K third-degree haemorrhoids
- L syphilitic gumma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 28-year-old man with Crohn's disease complains of watery discharge from a puckered area 2cm from the anal canal.
- 2 A 32-year-old woman who has recently given birth complains of excruciating pain on defecation that persists for hours afterwards. Rectal examination is not possible due to pain.
- 3 A 30-year-old taxi driver complains of tenderness from an area in the midline of the natal cleft about 4 cm above the anus. This problem has been remitting and recurring for 2 years and has started to discharge today for the first time.
- 4 A 27-year-old pregnant woman presents with constipation and bright red blood coating her stools. On examination, two bluish tender spongy masses are found protruding from the anus. These do not reduce spontaneously and require digital reduction.

- 5 A 19-year-old woman presents with multiple papilliferous lesions around the anus.”

Answers · Themes 6–10

IRFAN EMQS

Theme 6 · Management of colorectal cancer

- 1 **H** **abdominoperineal resection** — H Abdominoperineal resection: The operation of choice for low rectal tumours when sphincter preservation is not feasible.
- 2 **C** **Hartmann's procedure** — C Hartmann's procedure: Involves resection of the diseased sigmoid colon with formation of an end colostomy, used in emergency perforated cases.
- 3 **A** **right hemicolectomy** — A Right hemicolectomy: Standard treatment for caecal tumours ensuring oncological clearance.
- 4 **D** **anterior resection** — D Anterior resection: Preferred for mid-rectal tumours with total mesorectal excision.

Theme 7 · Investigation of gastrointestinal bleeding

- 1 **G** **oesophagogastroduodenoscopy** — G Oesophagogastroduodenoscopy: Allows visualization of the upper GI tract to identify sources of bleeding (e.g. varices).
- 2 **I** **colonoscopy** — I Colonoscopy: Enables direct visualization and biopsy in suspected right-sided colonic lesions.
- 3 **B** **flexible sigmoidoscopy** — B Flexible sigmoidoscopy: Best for evaluating left-sided colonic lesions.
- 4 **E** **CT abdomen** — E CT abdomen: Helps detect complications of diverticulitis such as abscess formation.

Theme 8 · Small bowel obstruction

- 1 **B** **strangulated inguinal hernia** — B Strangulated inguinal hernia: Indicates vascular compromise of entrapped bowel needing urgent surgery.
- 2 **G** **intra-abdominal abscess** — G Intra-abdominal abscess: A complication post appendicectomy causing paralytic ileus.
- 3 **A** **adhesions** — A Adhesions: The most common cause of obstruction in patients with previous surgery.

- 4 **D** **Crohn's disease** — D Crohn's disease: Chronic inflammatory changes may lead to stricture formation and obstruction.

Theme 9 · Abdominal masses

- 1 **K** **abdominal aortic aneurysm** — Collapse with an expansile pulsatile mass above the umbilicus is a (leaking) abdominal aortic aneurysm.
- 2 **C** **gastric carcinoma** — A hard epigastric mass with ascites and a Virchow node (Troisier sign) is advanced gastric carcinoma.
- 3 **I** **caecal carcinoma** — An irregular RIF mass with iron-deficiency anaemia in an older woman is caecal carcinoma.
- 4 **E** **fibroids** — A knobbly pelvic mass arising from the pelvis (no lower edge) with menorrhagia is uterine fibroids.
- 5 **F** **pancreatic pseudocyst** — A thick-walled peripancreatic fluid collection in an alcoholic is a pancreatic pseudocyst (mature after ~4 weeks).

Theme 10 · Anorectal conditions

- 1 **E** **fistula-in-ano** — E Fistula-in-ano: Common in Crohn's disease; management is initially conservative with antibiotics and drainage if abscess forms.
- 2 **A** **fissure-in-ano** — A Anal fissure: Causes excruciating pain on defecation due to extreme tenderness.
- 3 **F** **pilonidal abscess** — F Pilonidal abscess: Occurs in the mid-line natal cleft, often presenting with recurrent discharge.
- 4 **K** **third-degree haemorrhoids** — K Third-degree haemorrhoids: Prolapsed haemorrhoids requiring digital reduction.
- 5 **B** **perianal warts** — B Perianal warts: Caused by HPV infection, presenting as multiple papilliferous lesions.

15 Paediatric Surgery

16 Paediatric Surgery EMQs



04

PAEDIATRIC SURGERY

Paediatric Surgery

Pyloric stenosis, intussusception, malrotation, Hirschsprung disease, atresias, NEC, hernias and the neonatal abdomen.

65

MCQS

0

EMQ SCENARIOS

8

SOURCES

10

REPEATED IDEAS

IN THIS CHAPTER

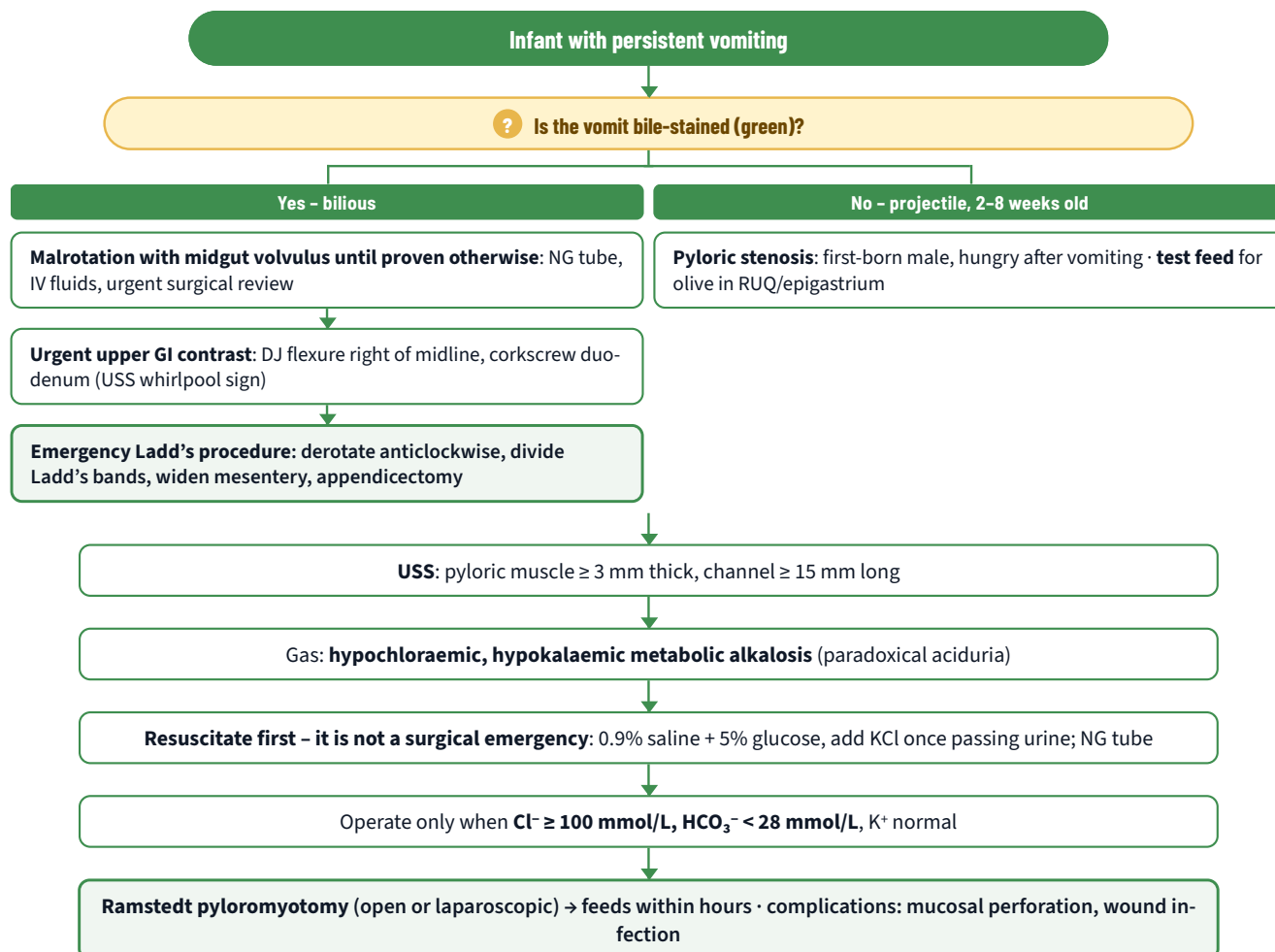
01	Management flowcharts	3 charts	page 250
02	Local Exams MCQs	15 Q	page 252
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04	Lange MCQs	12 Q	page 255
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15

Step by step

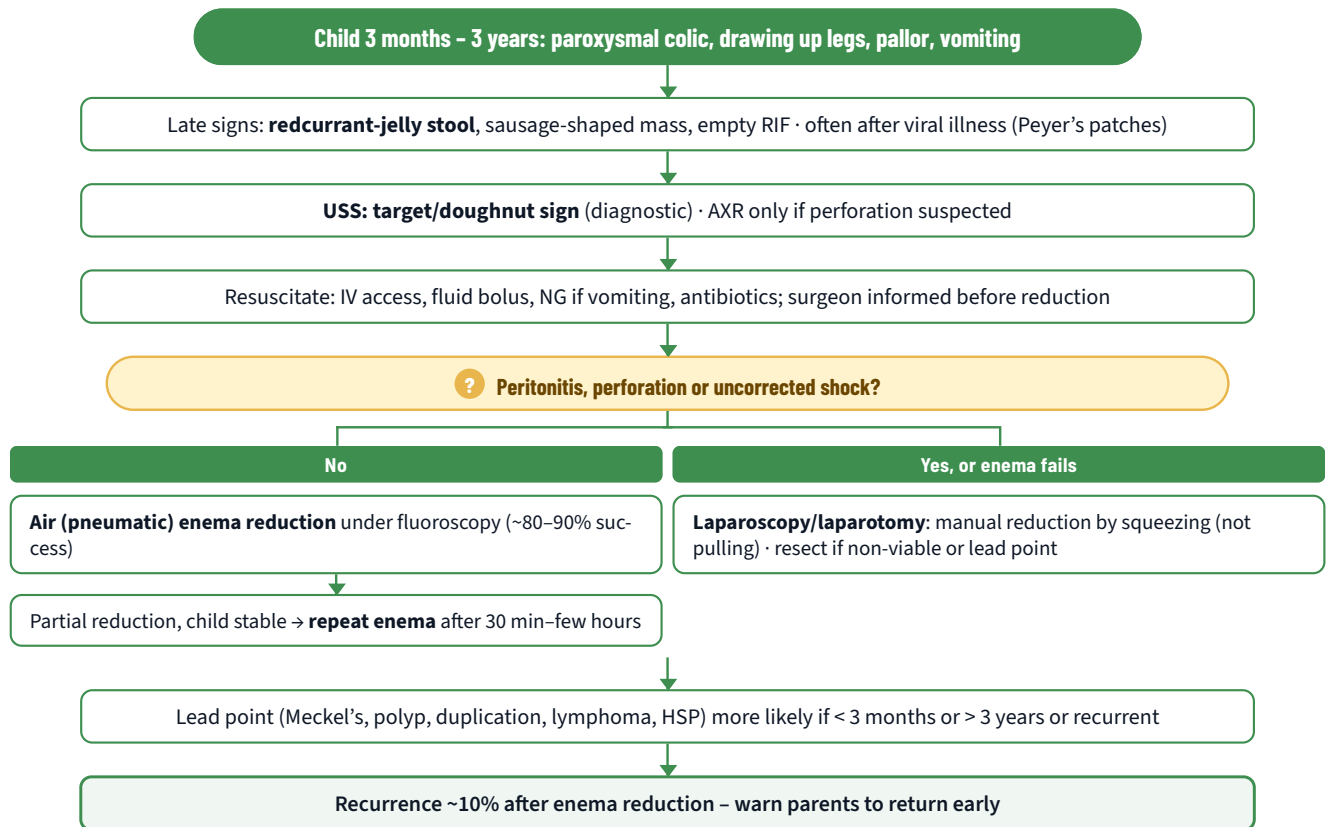
Vomiting infant: pyloric stenosis vs malrotation

NICE NG29 IV FLUIDS IN CHILDREN • BAILEY & LOVE 28E



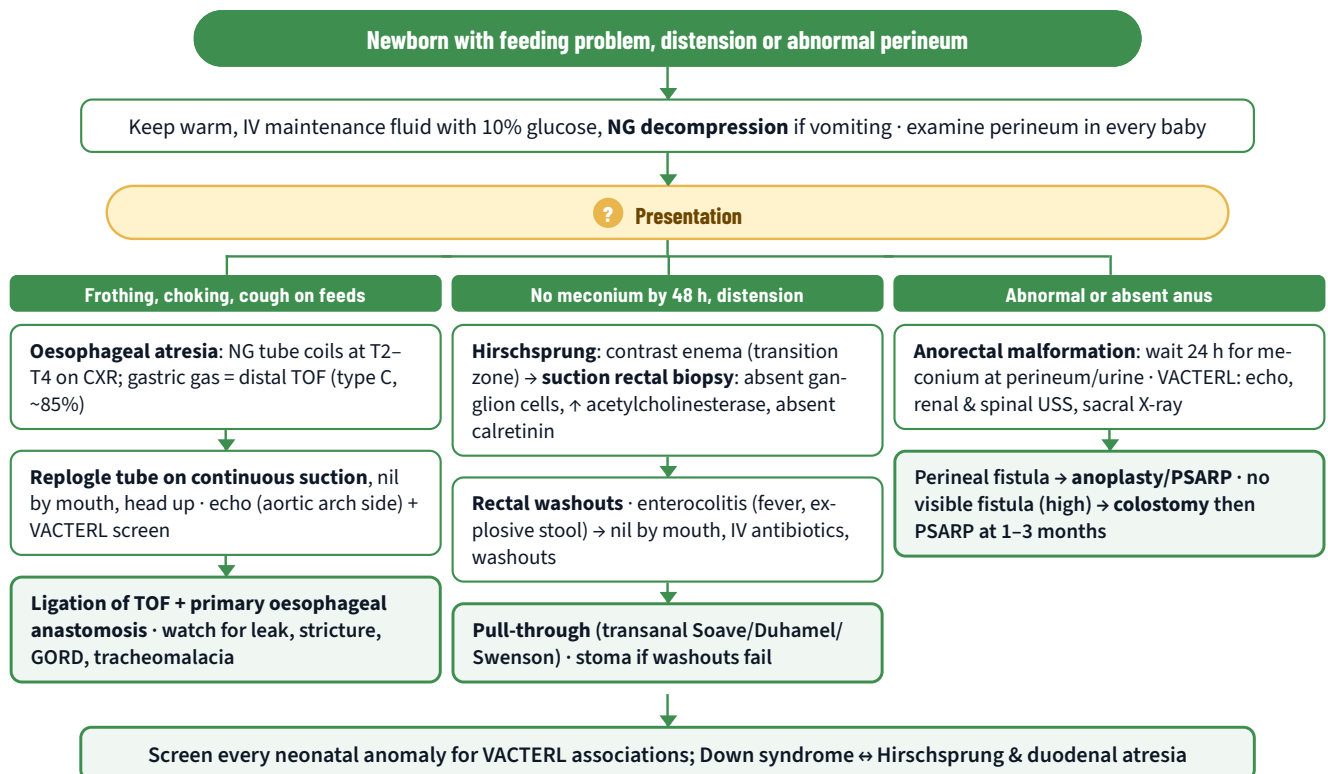
Intussusception

APSA 2021 INTUSSUSCEPTION REVIEW · BAILEY & LOVE 28E



Neonatal anomalies: OA/TOF, Hirschsprung, anorectal malformation

ERNICA 2019 OA & 2020 HIRSCHSPRUNG CONSENSUS · BAILEY & LOVE 28E



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-3

An 18-month-old child, born at 34 weeks, has had a groin lump for 1 week that enlarges on coughing and disappears spontaneously.

1 What is the most likely diagnosis?

- A Direct inguinal hernia
- B Indirect inguinal hernia
- C Femoral hernia
- D Hydrocele
- E Inguinal lymphadenopathy

2 What is the best investigation?

- A Clinical examination
- B Abdominal ultrasound
- C Doppler ultrasound
- D CT abdomen
- E MRI

3 What is the treatment?

- A Herniotomy
- B Mesh repair
- C Wait until 4 years of age
- D Hernioplasty
- E Hydrocelectomy

CASE • QUESTIONS 4-6

A baby a few months old has feeding difficulty, with milk coming out of the nose during feeds, coughing and choking. There is no excessive salivation.

4 What is the most likely diagnosis?

- A Cleft palate
- B Tracheo-oesophageal fistula
- C Gastro-oesophageal reflux disease
- D Choanal atresia
- E Laryngomalacia

5 What is the optimal time of repair?

- A At birth
- B 1-3 months
- C 6-18 months
- D 3 years
- E 7 years

6 Which is NOT a complication of this condition?

- A Feeding difficulty
- B Speech difficulty
- C Recurrent otitis media
- D Recurrent chest infection
- E Oesophageal stricture

CASE • QUESTIONS 7-8

A 12-hour-old boy has an imperforate anus with an absent anal dimple and is passing meconium in his urine.

7 What is the next step?

- A Cross-table prone lateral X-ray
- B Plain abdominal X-ray
- C Wait until 24 hours
- D Diverting colostomy
- E Immediate PSARP

8 How is the type of imperforate anus defined before definitive repair?

- A High-pressure distal colostogram
- B Invertogram
- C Abdominal ultrasound
- D MRI spine
- E Contrast enema via the perineum

CASE • QUESTIONS 9-11

A 4-month-old boy with effortless (sometimes forceful) non-bilious vomiting, fever to 39.5 °C and 4-6 green diarrhoeal stools a day.

9 All of the following are useful EXCEPT:

- A Full blood count
- B Abdominal X-ray
- C Ultrasound
- D Barium follow-through
- E Stool examination

10 What is the diagnosis?

- A Pyloric stenosis
- B Viral gastroenteritis
- C Pyloric atresia
- D Oesophageal atresia
- E Mesenteric lymphadenitis

11 What is the management?

- A Watch and wait
- B Antibiotics
- C Supportive treatment (rehydration and symptom relief)
- D Pyloromyotomy
- E Laparotomy

CASE · QUESTIONS 12–13

An 18-hour-old girl born by spontaneous vaginal delivery has an imperforate anus with a rectovestibular fistula. She weighs 2750 g, looks healthy and has no associated defects.

- 12 Initial management includes all of the following EXCEPT:
- A IV fluids
 - B IV antibiotics
 - C Incubator care
 - D Search for associated congenital anomalies (VACTERL)
 - E Immediate surgery
- 13 What is the most appropriate operation?
- A Diverting colostomy
 - B Diverting ileostomy
 - C Single-stage posterior sagittal anorectoplasty
 - D Swenson pull-through
 - E Heineke–Mikulicz pyloroplasty

FORMATIVE

College formative examinations

- 14 A 9-month-old child appears to have had a history of 48 hours of colicky abdominal pain described by his parents as episodes of crying and drawing his legs up to the abdomen. Additionally, he has passed some mucus-like blood from the rectum. What is the most likely diagnosis?

- A Intussusception
- B Acute gastroenteritis
- C Volvulus
- D Hirschsprung disease
- E Meckel's diverticulum

REPEATED 2* · ALSO ASKED IN
Formative exam

- 15 What is the most reasonable treatment for intussusception in this child?
- A Radiological (air or contrast) enema reduction
 - B Surgical resection
 - C Surgical reduction
 - D Observation
 - E Antibiotics alone

REPEATED 2* · ALSO ASKED IN
Formative exam

Answers 1–15

LOCAL EXAMS MCQS

- 1 **B Indirect inguinal hernia** — Paediatric inguinal hernias are almost always indirect, through a patent processus vaginalis; prematurity increases the risk.
- 2 **A Clinical examination** — The diagnosis is clinical (history of a reducible lump with cough impulse); imaging is not needed.
- 3 **A Herniotomy** — Herniotomy (high ligation of the sac) is done soon after diagnosis because of the high incarceration risk; mesh is not used in children.
- 4 **A Cleft palate** — Nasal regurgitation of milk without drooling suggests a cleft (often submucous) palate; TOF presents at birth with excessive salivation.
- 5 **C 6–18 months** — Cleft palate is repaired at about 6–12 months, before speech develops; cleft lip at about 3 months.
- 6 **E Oesophageal stricture** — Cleft palate causes feeding and speech problems, otitis media (Eustachian dysfunction) and aspiration; it does not cause oesophageal stricture.
- 7 **D Diverting colostomy** — Meconium in the urine means a rectourinary fistula (high lesion), so a diverting colostomy is done; imaging and waiting 24 h are for unclear cases.
- 8 **A High-pressure distal colostogram** — A high-pressure distal colostogram through the mucous fistula shows the rectal level and the fistula to the urinary tract.
- 9 **D Barium follow-through** — Fever with vomiting and diarrhoea suggests gastroenteritis; FBC, stool tests and ultrasound help, but barium follow-through has no role.
- 10 **B Viral gastroenteritis** — Fever, vomiting and diarrhoea point to infective (viral) gastroenteritis; pyloric stenosis causes vomiting without fever or diarrhoea.
- 11 **C Supportive treatment (rehydration and symptom relief)** — Viral gastroenteritis is managed supportively with oral or IV rehydration; antibiotics and surgery have no role.
- 12 **E Immediate surgery** — First 24 h: fluids, antibiotics, NG decompression and VACTERL screening; surgery is planned once the anatomy and associated anomalies are known.
- 13 **C Single-stage posterior sagittal anorectoplasty** — A healthy girl with a vestibular fistula can have a primary single-stage PSARP; a colostomy is reserved for sick babies or high defects.
- 14 **A Intussusception** — **UPDATED** The classic triad of intussusception includes colicky abdominal pain, red currant jelly stool (mucus-like blood), and vomiting.
- 15 **A Radiological (air or contrast) enema reduction** — **UPDATED** Uncomplicated intussusception in a stable child is reduced by image-guided pneumatic or hydrostatic enema (~80% success); surgery is for peritonitis, perforation or failed reduction.

Bailey & Love MCQs

- 1 Which of the following statements are true?
 - A For open surgery, vertical midline incisions give better access.
 - B The ribs are more horizontal and flexible.
 - C The liver and bladder are intra-abdominal.
 - D The infant's head constitutes 20% of the surface area.
 - E The infant has a large tongue, the larynx is high, and minimal access surgery (MAS) can be used at all ages.
- 2 Which of the following statements are false?
 - A A neonate is a baby up to 4 weeks old.
 - B The urinary bladder is an intra-abdominal organ in infants and children.
 - C An infant's head accounts for 20% of the body surface area.
 - D Gluconeogenesis in infants is as good as in adults.
 - E The infant's immune system is immature.
- 3 Which of the following statements are true?
 - A Children are regarded as small adults, and therefore their surgical management is adjusted according to their size.
 - B Attention to thermoregulation is very important.
 - C Postoperative management requires extreme care because of an inadequate stress response.
 - D Minimal access surgery (MAS) can be used at all ages.
- 4 Which of the following statements regarding paediatric trauma are true?
 - A Follow usual Advanced Trauma Life Support (ATLS) rules as in an adult.
 - B The usual rules of resuscitation apply.
 - C Blunt trauma is more common than penetrating trauma.
 - D Splenic injury in the majority should be treated by splenectomy.
 - E Consideration must be given to non-accidental injury (NAI).
- 5 Which of the following statements about inguino-scrotal swellings are false?
 - A A hydrocele is a patent processus vaginalis, as is a hernia.
 - B Hernia is always indirect.
 - C Hernia can be direct or indirect.
 - D In incarcerated hernia, reduction should be attempted by taxis followed by operation 24 hours later.
 - E A hydrocele always needs an operation.
- 6 Which of the following statements are true for undescended testis?
 - A Orchidopexy in a subdartos pouch is the treatment of choice.
 - B The operation is recommended at 6–18 months of age.
 - C When a testis is impalpable and therefore intra-abdominal, laparotomy should be done.
 - D Laparoscopy is the gold standard procedure for an intra-abdominal testis.
 - E Orchidopexy reduces the chance of malignancy.
- 7 Which of the following statements are true with regard to acute scrotal pain?
 - A Acute testicular pain can be from torsion of the testis, torsion of the hydatid of Morgagni, or acute epididymitis.
 - B Pain of testicular torsion may originate in the groin or suprapubic area.
 - C Doppler ultrasound should be done in suspected testicular torsion.
 - D Incarcerated hernia may cause similar symptoms.
 - E In case of any doubt, exploration of the scrotum must be carried out.
- 8 What is the operation for congenital hypertrophic pyloric stenosis called?
 - A Hartmann's
 - B Whipple's
 - C Heller's
 - D Ramstedt's
 - E Ivor-Lewis

Answers 1–8

BAILEY & LOVE MCQS

- 1 **BCDE** True: B, C, D, E — **UPDATED** False: A — in infants a transverse incision gives better access (the abdomen is wider than it is long). Horizontal pliable ribs, intra-abdominal liver/bladder, large head and high larynx are true.
- 2 **D** Gluconeogenesis in infants is as good as in adults. — **UPDATED** Only D is false: neonates have small glycogen stores and poor gluconeogenesis, so they become hypoglycaemic quickly. The other statements are true.
- 3 **BCD** True: B, C, D — **UPDATED** False: A — children are not small adults; their anatomy and physiology differ. Thermoregulation and postoperative care need special attention, and MAS is feasible at any age.
- 4 **BCE** True: B, C, E — **UPDATED** In paediatric trauma standard resuscitative measures apply, blunt injuries predominate, and non-accidental injury must be vigilantly considered.
- 5 **CE** True: C, E — **UPDATED** False: C — paediatric inguinal hernias are (virtually) always indirect via a patent processus vaginalis; E — most infant hydroceles resolve spontaneously and are observed until about 2 years.

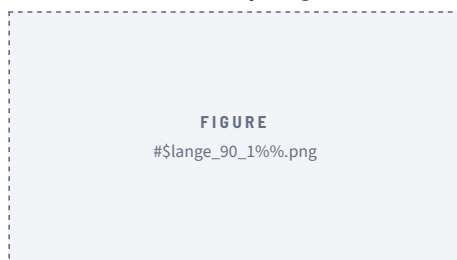
- 6** **ABD** True: A, B, D — **UPDATED** False: C — an impalpable testis may be inguinal, intra-abdominal or absent, so laparoscopy (not laparotomy) is used; E — orchidopexy aids examination but does not remove the malignancy risk.
- 7** **ABDE** True: A, B, D, E — **UPDATED** False: C — Doppler ultrasound must not delay scrotal exploration when torsion is suspected, and a normal scan does not exclude it.

- 8** **D** **Ramstedt's** — Ramstedt's operation is a pyloromyotomy that carefully splits the hypertrophied pyloric muscle while preserving the mucosa, and is the standard treatment for pyloric stenosis.

12 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 10-month-old infant presents with a 24-hour history of low-grade fever, anorexia, vomiting, and loose stools. On exam, there is fullness in the right upper quadrant and passage of a stool with blood mixed in. Which is the most likely diagnosis?



- A gastroenteritis
- B intussusception
- C midgut volvulus
- D Meckel's diverticulum
- E juvenile rectal polyp

REPEATED 2x • ALSO ASKED IN
Lange

- 2** What is the most appropriate next step in the diagnostic evaluation and management of the infant described above?

- A proctoscopy
- B oral rehydration and stool cultures
- C IV fluid rehydration and a hydrostatic barium enema
- D technetium scan
- E IV fluid rehydration, NG decompression, and a UGI contrast study

REPEATED 2x • ALSO ASKED IN
Lange

- 3** An 18-month-old boy with intermittent vomiting, RLQ fullness, and currant jelly stools most likely has?

- A constipation
- B gastroenteritis
- C intussusception
- D Meckel's diverticulum
- E volvulus

REPEATED 2x • ALSO ASKED IN
Lange

- 4** A 5-week-old infant presents with a 1-week history of progressive nonbilious vomiting, decreasing urine output, but remains eager to feed. On examination, a visible peristaltic wave is noted. Which of the following is the most likely diagnosis?

- A Viral gastroenteritis
- B Gastro-oesophageal reflux
- C Urinary tract sepsis
- D Pyloric stenosis
- E Milk protein allergy

- 5** The diagnosis of pyloric stenosis is best confirmed by which of the following?

- A Abdominal ultrasound
- B Careful clinical examination with palpation of an epigastric mass
- C Upper GI contrast study
- D Surgical exploration
- E Endoscopy

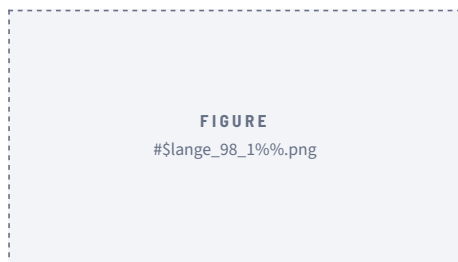
- 6** An infant with pyloric stenosis is evaluated for possible electrolyte derangements. Which of the following laboratory findings is most likely?

- A Na 145, K 3.0, Cl 110, CO₂ 17, urine pH 8.0
- B Na 130, K 3.0, Cl 80, CO₂ 36, urine pH 4.0
- C Na 135, K 4.0, Cl 104, CO₂ 23, urine pH 7.0
- D Na 140, K 5.2, Cl 100, CO₂ 16, urine pH 4.0
- E Na 132, K 3.2, Cl 96, CO₂ 25, urine pH 7.0

- 7** Which of the following is the most appropriate next step in management of this infant with pyloric stenosis?

- A Immediate surgical exploration
- B Send the child home with an oral electrolyte rehydration solution
- C Change the infant's formula and feeding regimen
- D IV fluid resuscitation, followed by surgical intervention
- E Initiate therapy with a prokinetic agent

- 8 A neonate with bile-stained vomiting, abdominal distention, and dilated loops of bowel on plain radiographs with a small-caliber colon on contrast enema is evaluated. What is the most likely diagnosis?



- A Anorectal malformation
- B Oesophageal atresia
- C Hirschsprung's disease
- D Pyloric stenosis
- E Jejunal atresia

- 9 A 1-day-old infant has persistent vomiting and no stools. An abdominal film shows a 'double bubble' sign. What is the most likely diagnosis?



- A adrenogenital syndrome
- B pyloric stenosis
- C duodenal atresia
- D esophageal atresia with distal tracheoesophageal fistula
- E cystic fibrosis

- 10 A 3-day-old term infant with progressive vomiting and abdominal distention—who passed meconium only after a glycerin suppository—requires which essential intervention before transfer?

- A nasogastric tube decompression and IV fluid resuscitation
- B sweat chloride determination
- C plain abdominal radiographs
- D blood and urine cultures, followed by initiation of broad spectrum antibiotics
- E barium enema

- 11 Upon arrival at the pediatric hospital, what is the best approach to establish a diagnosis?

- A anorectal manometry
- B plain radiographs, followed by a barium enema, and subsequent rectal biopsy
- C abdominal ultrasound
- D sweat chloride
- E laparotomy

- 12 A 3-day-old infant who has not passed stools since birth and shows abdominal distension is most likely suffering from?

- A duodenal stenosis
- B esophageal atresia
- C functional constipation
- D Hirschsprung's disease
- E breast-feeding failure

Answers 1-12

LANGE MCQs

- 1 **B Intussusception** — **UPDATED** Intussusception The infant's intermittent colicky pain and currant jelly stool are typical for intussusception.
- 2 **C IV fluid rehydration and a hydrostatic barium enema** — **UPDATED** After IV rehydration, intussusception is reduced by an image-guided enema (air/pneumatic now preferred to barium); surgery if it fails or there is peritonitis.
- 3 **C Intussusception** — **UPDATED** The classic findings indicate intussusception.
- 4 **D Pyloric stenosis** — **UPDATED** Progressive non-bilious (projectile) vomiting in a hungry 3–6-week-old with visible gastric peristalsis is hypertrophic pyloric stenosis.
- 5 **A Abdominal ultrasound** — **UPDATED** Ultrasound is the investigation of choice, showing a thickened (>3–4 mm), elongated (>15 mm) pyloric muscle; a palpable 'olive' is diagnostic but often not felt.
- 6 **B Na 130, K 3.0, Cl 80, CO₂ 36, urine pH 4.0** — **UPDATED** Loss of gastric HCl gives hypochloaemic, hypokalaemic metabolic alkalosis; the kidney then excretes H⁺ to conserve Na⁺ and K⁺, causing paradoxical aciduria.
- 7 **D IV fluid resuscitation, followed by surgical intervention** — **UPDATED** Pyloric stenosis is not a surgical emergency: correct dehydration and alkalosis with IV fluids (with potassium) before Ramstedt pyloromyotomy.
- 8 **E Jejunal atresia** — **UPDATED** Bilious vomiting with dilated small-bowel loops and an unused microcolon on contrast enema indicates jejunoileal atresia; Hirschsprung disease shows a transition zone.
- 9 **C duodenal atresia** — Double bubble sign is characteristic of duodenal atresia.
- 10 **A nasogastric tube decompression and IV fluid resuscitation** — NG decompression and IV fluids are essential before transport.
- 11 **B plain radiographs, followed by a barium enema, and subsequent rectal biopsy** — **UPDATED** This sequence confirms Hirschsprung's disease.
- 12 **D Hirschsprung's disease** — Failure to pass meconium with distension points to Hirschsprung's disease.

1 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** Management of intussusception A 13-month-old girl is diagnosed with intussusception. She was admitted 2 hours ago with vomiting, passing red mucus-like stools and persistent crying. What is the most likely first line treatment option?

- A Laparotomy and reduction
- B Radiological (air or contrast) enema reduction after an abdominal film
- C Analgesia and observation for 24 hours
- D Intravenous fluids and antibiotics
- E Laparotomy and resection

REPEATED 2* • ALSO ASKED IN
SBA

Answer 1

SBA MCQs

- 1** **B Radiological (air or contrast) enema reduction after an abdominal film** — **UPDATED** First-line treatment of uncomplicated

intussusception is image-guided air (pneumatic) or contrast enema reduction; surgery if it fails or there is peritonitis.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** 8-month-old boy is described by his mother as having 3 days of intermittent severe distress and restlessness and the recent passage of bloody motions and mucus-like material. There has been some vomiting and the abdomen is mildly distended and tender to examination. Which is the single most likely diagnosis?

- A Gastroenteritis
- B Hirschsprung's disease
- C Intussusception (ileocaecal)
- D Pyloric stenosis
- E Small bowel volvulus

REPEATED 2* • ALSO ASKED IN
Oxford

- 2** 6-week-old boy is brought to hospital with projectile and persistent vomiting. Weight gain has been poor. The child appears listless and quiet. A lump is palpable in the upper abdomen. Pulse is 100bpm, serum electrolytes are sodium 132mmol/L and potassium 3.2mmol/L. A provisional diagnosis of pyloric stenosis is made. Which is the single most appropriate initial intervention?

- A Administration of IV bicarbonate 8.4% solution (weight adjusted)
- B Administration of IV 0.9% sodium chloride (weight adjusted)
- C Administration of a test feed
- D Commencing nil by mouth
- E Insertion of a nasogastric tube

Answers 1-2

OXFORD MCQs

- 1** **C Intussusception (ileocaecal)** — **UPDATED** Intermittent inconsolable distress, vomiting and 'red-currant jelly' stools in an infant is ileocolic intussusception; ultrasound shows a target sign.

- 2** **B Administration of IV 0.9% sodium chloride (weight adjusted)** — **UPDATED** Correct dehydration and electrolytes first with weight-adjusted IV 0.9% saline (then dextrose-saline with KCl) before pyloromyotomy; bicarbonate would worsen the alkalosis.

18 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

1 A 2-month-old boy is examined because he has been straining while passing stool and has a distended abdomen. He is very low on the growth chart for age. A contrast enema is ordered by the primary care physician which shows a contracted segment of colon with dilated proximal colon. Which of the following is the best test for definitive diagnosis?

- A Colonoscopy
- B Defecography
- C Endoanal ultrasound
- D Anorectal manometry
- E Suction rectal biopsy

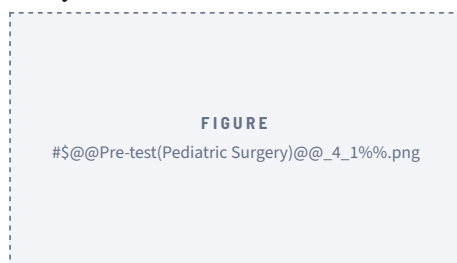
2 A newborn has an abdominal wall bulge with a 1-cm midline defect. There is no bowel present in the bulge. Which of the following is the best treatment option?

- A Observation without planned intervention
- B Observation with primary closure after age 4
- C Placement of a silo followed by primary closure
- D Emergency primary closure of the defect
- E Emergency closure of the defect with a prosthetic mesh

3 A neonate is examined in the nursery and found to have no anal orifice; only a small perineal fistulous opening is visualized. Which of the following is associated with this abnormality?

- A Congenital pulmonary airway malformation
- B Hydrocephalus
- C Duodenal atresia
- D Congenital heart disease
- E Corneal opacities

4 A 36-hour-old infant presents with bilious vomiting and an increasingly distended abdomen. At exploration, the segment pictured here is found as the point of obstruction. What is the likely etiology of this anomaly?



- A Remnant omphalomesenteric duct
- B Failure of rotation of the ventral pancreatic bud
- C Non-recanalization of the duodenum
- D Intrauterine mesenteric vascular accident
- E Obstructing peritoneal attachment

5 Approximately 2 weeks after a viral respiratory illness, an 18-month-old child complains of abdominal pain and passes some bloody mucus per rectum. A long, thin mass is palpable in the right upper quadrant of the abdomen. No peritoneal signs are present. Which of the following is the most appropriate next step in the workup?

- A Serial abdominal examinations
- B Air enema
- C Diagnostic laparoscopy
- D Exploratory laparotomy
- E Decompressive colonoscopy

6 An 18-year-old woman presents with abdominal pain, fever, and leukocytosis. With the presumptive diagnosis of appendicitis, a right lower quadrant (McBurney) incision is made and a lesion 60 cm proximal to the ileocecal valve is identified (see photo). Which of the following is the most likely diagnosis?



- A Intestinal duplication
- B Mesenteric cyst
- C Meckel diverticulum
- D Ileileal intussusception
- E "Christmas tree" type of ileal atresia

7 A newborn infant born with polyhydramnios presents with excessive salivation along with coughing and choking with the first oral feeding. An x-ray of the abdomen shows gas in stomach and a nasogastric tube coiled in the esophagus. Which of the following is the most likely diagnosis?

- A Esophageal atresia
- B Tracheoesophageal fistula
- C Esophageal atresia and tracheoesophageal fistula (TEF)
- D Omphalocele
- E Gastroschisis

- 8** A 2-week-old infant presents with sudden onset of bilious emesis. Plain films of the abdomen show evidence of an intestinal obstruction. Which of the following is the most appropriate next step in the management of this infant?
- A Observation
 - B Abdominal ultrasound
 - C Upper gastrointestinal contrast series
 - D Computed tomography of the abdomen and pelvis
 - E Contrast enema
- 9** A 29-week-old previously healthy male infant presents with fevers, abdominal distention, feeding intolerance, and bloody stools at 3 weeks of age. The patient undergoes x-ray and ultrasound examination for possible necrotizing enterocolitis. Which of the following findings on imaging is an indication for surgical management?
- A Pneumoperitoneum
 - B Ascites
 - C Portal venous gas
 - D Ileus
 - E Pneumatosis intestinalis
- 10** A newborn presents with signs and symptoms of distal intestinal obstruction. Abdominal x-rays reveal dilated loops of small bowel, absence of air-fluid levels, and a mass of meconium within the right side of the abdomen mixed with gas to give a ground-glass appearance. Which of the following should be performed as the initial management of the patient?
- A Administration of oral polyethylene glycol
 - B Performance of a contrast enema
 - C Surgical disimpaction
 - D Diverting ileostomy
 - E Resection of the dilated terminal ileum
- 11** A 4-week-old male infant presents with projectile, nonbilious emesis. Ultrasound of the abdomen reveals a pyloric muscle thickness of 8 mm (normal 3 to 4 mm). Which of the following should be administered to the patient prior to operative intervention?
- A Nothing so as not to delay intervention
 - B Normal saline
 - C Hypertonic saline
 - D Hydrochloric acid
 - E Sodium bicarbonate
- 12** A 1-month-old female infant presents with persistent jaundice. A serum direct bilirubin is 4.0 mg/dL and an ultrasound of the abdomen shows a shrunken gallbladder and inability to visualize the extrahepatic bile ducts. Which of the following is the most appropriate initial management of this patient?
- A Total parenteral nutrition
 - B Oral choleretic bile salts
 - C Methylprednisolone
 - D IV antibiotics
 - E Exploratory laparotomy
- 13** A full-term male newborn experiences respiratory distress immediately after birth. A prenatal sonogram was read as normal. An emergency radiograph is shown here. The patient is intubated and placed on 100% O₂. Arterial blood gases reveal pH 7.24, Po₂ 60, and Pco₂ 52. The baby has sternal retractions and a scaphoid abdomen. Which of the following should be performed in the management of this patient?
- FIGURE**

#\$\$@Pre-test(Pediatric Surgery)@_13_1%.png
- A Administration of intravenous steroids
 - B Placement of bilateral tube thoracostomies
 - C Immediate thoracotomy with lung resection
 - D Immediate laparotomy with repair of the diaphragm
 - E Mechanical ventilation with low tidal volumes
- 14** What is the most common type of tracheoesophageal fistula?
- A Type A: Isolated esophageal atresia
 - B Type B: Esophageal atresia with proximal trachea fistula
 - C Type C: Esophageal atresia with distal trachea fistula
 - D Type D: Esophageal atresia with proximal and distal trachea fistulas
 - E Type E: Tracheoesophageal fistula without esophageal atresia
- 15** What is the most common origin for neuroblastomas?
- A Adrenal gland
 - B Retroperitoneal nerves
 - C Paraspinal ganglia
 - D Mediastinal sympathetic chain
 - E Cervical sympathetic ganglia
- 16** What are the electrolyte abnormalities noted with hypertrophic pyloric stenosis?
- A Hyperkalemic, hyperchloremic metabolic acidosis
 - B Hypokalemic, hypochloremic metabolic alkalosis
 - C Normal electrolytes with respiratory alkalosis
 - D Hyperkalemic, hypochloremic metabolic alkalosis
 - E Hypokalemic, hyperchloremic metabolic acidosis
- 17** What diagnosis is suggested by a double bubble sign on x-ray?
- A Pyloric stenosis
 - B Duodenal atresia
 - C Intussusception
 - D Malrotation with volvulus
 - E Meckel diverticulum

18 What is the procedure for malrotation?

- A Whipple procedure
- B Nissen fundoplication
- C Ladd procedure
- D Kasai portoenterostomy
- E Roux-en-Y anastomosis

Answers 1-18

PRETEST MCQS

- 1 **E Suction rectal biopsy** — The definitive diagnosis is made by rectal biopsy.
- 2 **A Observation without planned intervention** — Umbilical hernias less than 1 cm close spontaneously by age 4.
- 3 **D Congenital heart disease** — Imperforate anus is part of VACTERL association, which includes congenital heart disease.
- 4 **D Intrauterine mesenteric vascular accident** — Ileal atresia is caused by an intrauterine mesenteric vascular accident leading to intestinal obstruction.
- 5 **B Air enema** — Intussusception is diagnosed and treated with air enema.
- 6 **C Meckel diverticulum** — Meckel diverticulum is a remnant of the omphalomesenteric duct, often located 60 cm proximal to the ileocecal valve.
- 7 **C Esophageal atresia and tracheoesophageal fistula (TEF)** — Esophageal atresia with TEF is characterized by coiled NG tube and choking on feeding.
- 8 **C Upper gastrointestinal contrast series** — Upper GI series is the test of choice for diagnosing malrotation with volvulus.
- 9 **A Pneumoperitoneum** — Pneumoperitoneum indicates bowel perforation requiring surgical intervention.
- 10 **B Performance of a contrast enema** — Contrast enema is the initial management for meconium ileus to relieve obstruction.
- 11 **B Normal saline** — Fluid resuscitation with normal saline is needed to correct electrolyte imbalances before surgery.
- 12 **E Exploratory laparotomy** — Exploratory laparotomy is indicated for suspected biliary atresia, typically involving the Kasai procedure.
- 13 **E Mechanical ventilation with low tidal volumes** — Management includes mechanical ventilation with low tidal volumes to prevent lung injury in congenital diaphragmatic hernia.
- 14 **C Type C: Esophageal atresia with distal trachea fistula** — Type C is the most common form of tracheoesophageal fistula.
- 15 **A Adrenal gland** — The adrenal gland is the most common origin for neuroblastomas.
- 16 **B Hypokalemic, hypochloremic metabolic alkalosis** — Hypertrophic pyloric stenosis leads to hypokalemic, hypochloremic metabolic alkalosis.
- 17 **B Duodenal atresia** — A double bubble sign on x-ray suggests duodenal atresia.
- 18 **C Ladd procedure** — The Ladd procedure is the standard treatment for malrotation.

7 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 6-month-old boy has had paroxysms of intense abdominal pain since this morning, drawing his legs up and passing some mucus-like blood per rectum. Temperature is 36.8°C. Which of the following would be the best treatment option in the first instance?
 - A Air or contrast enema reduction
 - B Continued observation
 - C Intravenous antibiotics
 - D Laparotomy
 - E Scrotal exploration
- 2 A 7-day-old preterm boy develops vomiting and bloody diarrhoea. On examination, his abdomen is grossly distended and discoloured. His temperature is 39.5°C. What is the most likely diagnosis?
 - A Exomphalos
 - B Gastroenteritis
 - C Gastroschisis
 - D Meconium ileus
 - E Necrotizing enterocolitis

REPEATED 2* • ALSO ASKED IN
Get Ahead

- 3** A 5-week-old boy has been having episodes of forceful vomiting after feeds for the last 2 weeks. He is always hungry and now appears lethargic. Examination reveals mild dehydration and a smooth, firm, non-tender mass in the right upper quadrant. Blood tests are sent. What biochemical abnormalities would you expect to find?
- A Hyperchloraemic, respiratory alkalosis
 - B Hyponatraemic, hyperkalaemic, metabolic alkalosis
 - C Hypochloraemic, hyperkalaemic, metabolic alkalosis
 - D Hypochloraemic, hypokalaemic, metabolic alkalosis
 - E Hyponatraemic, metabolic acidosis
- 4** A 1-day-old baby girl has repeated bile-stained vomiting. The mother had polyhydramnios. X-ray shows gas in the stomach and first part of duodenum only. What is the most likely diagnosis?
- A Choledochal cyst
 - B Duodenal atresia
 - C Oesophageal atresia
 - D Pyloric stenosis
 - E Renal agenesis
- 5** A 7-year-old boy is brought to the emergency department by his father, having told him that he had swallowed a five pence coin earlier that day. The child has not experienced any symptoms and on examination is well. Which of the following investigations would you perform in the first instance?
- A Barium swallow
 - B Laryngoscopy
 - C Neck, chest and abdominal X-rays
 - D No investigation required
 - E Upper gastrointestinal endoscopy
- 6** A 3-day-old boy is having increasing abdominal distension and vomiting. He is also noted not to have passed meconium since birth. Which of the following investigations would be most helpful in confirming the cause of obstruction?
- A Abdominal X-ray
 - B Barium enema
 - C Rectal biopsy
 - D Rectal examination
 - E Ultrasound scan
- 7** A newborn baby boy is found to have visible intestine emerging from his abdomen with no covering to the contents. Which of the following is the most likely diagnosis?
- A Epigastric hernia
 - B Exomphalos
 - C Gastroschisis
 - D Paraumbilical hernia
 - E Umbilical hernia

Answers 1-7

- 1** **A Air or contrast enema reduction** — **UPDATED** A stable infant with intussusception (colic, 'red-currant jelly' stool) is treated first by image-guided air/contrast enema reduction.
- 2** **E Necrotizing enterocolitis** — **UPDATED** Severe neonatal bowel infection/inflammation, especially in preterm infants.
- 3** **D Hypochloraemic, hypokalaemic, metabolic alkalosis** — Projectile vomiting of gastric acid leads to loss of HCl and compensatory hypokalaemia with metabolic alkalosis.
- 4** **B Duodenal atresia** — 'Double-bubble' sign with bilious vomiting in a newborn suggests duodenal atresia.
- 5** **C Neck, chest and abdominal X-rays** — Plain X-rays can confirm the position of a swallowed radio-opaque foreign body.
- 6** **C Rectal biopsy** — Hirschsprung disease is diagnosed by absent ganglion cells on rectal biopsy.
- 7** **C Gastroschisis** — **UPDATED** Abdominal wall defect with exposed bowel loops and no sac covering indicates gastroschisis.

2 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

1 A 5-week-old baby presents with projectile vomiting within 30 minutes of each feed for 3 days. On examination there is a palpable sausage-shaped mass in the right upper quadrant. What is the most likely diagnosis?

- A Pyloric stenosis
- B Neonatal jaundice
- C Duodenal atresia
- D Diaphragmatic hernia
- E Acute gastritis

REPEATED 2x • ALSO ASKED IN
Crash Course

2 A 10-month-old baby has been admitted with vomiting. The parents report he's crying more than usual and drawing his knees up. Nurses note his stools are loose and redder than normal. What is the diagnosis?

- A Pyloric stenosis
- B Gastroenteritis
- C Intussusception
- D Gastro-oesophageal reflux disease (GORD)
- E Duodenal atresia

REPEATED 2x • ALSO ASKED IN
Crash Course

Answers 1-2

CRASH COURSE MCQS

1 **A Pyloric stenosis** — **UPDATED** Typically presents 2–6 weeks old with projectile non-bile-stained vomiting.

2 **C Intussusception** — **UPDATED** Typical 'redcurrant jelly' stools; colicky pain in 5–12 months age group.

Paediatric Surgery EMQs

Extended matching questions on vomiting infants, neonatal obstruction and the paediatric acute abdomen.

0

MCQS

31

EMQ SCENARIOS

3

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

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16

Local Exams EMQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

THEME 1 The vomiting neonate and infant

- A Hirschsprung disease
- B Malrotation with midgut volvulus
- C Infantile hypertrophic pyloric stenosis
- D Meconium ileus
- E Duodenal atresia
- F Jejunoileal atresia
- G Necrotising enterocolitis
- H Intussusception

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 2-day-old infant with Down syndrome has projectile vomiting and a scaphoid abdomen. X-ray shows a double-bubble sign with no distal bowel gas.
- 2 A 2-week-old infant born at 28 weeks, in the NICU, develops sudden abdominal distension and bloody stool. X-ray shows gas in the bowel wall.
- 3 A full-term 2-day-old has not passed meconium, vomits bile and is distended. Rectal examination produces an explosive release of gas and liquid stool.
- 4 A 5-day-old who was feeding well suddenly has bilious vomiting, tenderness and haemodynamic instability. Contrast shows the duodenojejunal flexure right of the spine with a corkscrew duodenum.

Answers · Theme 1

LOCAL EXAMS EMQS

Theme 1 · The vomiting neonate and infant

- 1 **E Duodenal atresia** — Double bubble without distal gas in a baby with Down syndrome is duodenal atresia (vomiting is usually bilious if post-ampullary).
- 2 **G Necrotising enterocolitis** — Pneumatosis intestinalis with distension and bloody stool in a preterm neonate is necrotising enterocolitis.

- 3 **A Hirschsprung disease** — Delayed meconium with an explosive decompression on rectal examination is Hirschsprung disease; rectal suction biopsy confirms it.
- 4 **B Malrotation with midgut volvulus** — Bilious vomiting with an abnormally placed DJ flexure and corkscrew duodenum is malrotation with midgut volvulus: emergency Ladd procedure.

Bailey & Love EMQs

THEME 1 Vomiting

- A Congenital hypertrophic pyloric stenosis
- B Incarcerated inguinal hernia
- C Intussusception

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 2-year-old boy has been sent in as an emergency with vomiting for 24 hours. According to the parents, the vomitus was greenish to start with but over the past few hours has consisted of dirty brownish fluid. There has been no bowel action. On examination the child looks toxic and is dehydrated - sunken eyes, depressed fontanelles, loss of skin turgor. There is a tympanic abdomen with a red, irreducible swelling in the left groin not noticed by the parents.
- 2 Jonathan, a 5-week-old boy, has been vomiting intermittently for 2 weeks. This has become incessant for the past 24 hours. The vomitus is clear fluid. The infant seems hungry and takes its feed only to bring it up within a short while. The mother noticed some twitching of muscles. The baby is dehydrated. The paediatrician felt a lump in the upper abdomen when Jonathan was being fed by his mother.
- 3 Mary-Ann, a 10-month-old girl, has been brought in with occasional vomiting for a couple of days. The parents noticed that the vomitus is sometimes green and at other times brownish. They feel that Mary-Ann is in pain intermittently, because she screams, flexing her knees and elbows, denoting spasms. They noticed that the nappy has bloodstained mucus. On examination she looks ill and dehydrated, and the right side of the abdomen feels empty.

THEME 2 Acute Abdomen

- A Acute appendicitis
- B Nonspecific abdominal pain
- C Urinary tract infection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 George, a 10-year-old boy has been brought in with abdominal pain, vomiting, pyrexia and diarrhoea for the past 24 hours or so. He has been off his food for a couple of days. He has pyrexia of 39°C, tachycardia, looks toxic and has marked lower abdominal tenderness, rigidity and rebound tenderness, and the abdominal wall does not move with respiration.
- 2 Kerry, a 5-year-old girl, has been brought in by her parents with generalised abdominal pain, which has been recurrent over 6 months or so. On this occasion she looks ill, out of sorts and anorexic, and she has been vomiting. She has marked urinary frequency. On examination she has a temperature of 38°C, looks toxic, listless and dehydrated and has generalised abdominal and bilateral loin tenderness.
- 3 Millie, a 10-year-old girl, has been admitted with generalised abdominal pain, vomiting and anorexia for 2 days. She has missed school a few times because of similar attacks of abdominal pain in the past, which has subsided on its own within 24 hours. On this occasion it has persisted. Examination revealed a well-looking introspective child with generalised abdominal tenderness without any rigidity or rebound tenderness, with the abdominal wall moving freely with respiration.

THEME 3 Congenital Malformations

- A Biliary atresia
- B Congenital diaphragmatic hernia
- C Duodenal atresia
- D Hirschsprung's disease
- E Intestinal malrotation
- F Necrotising enterocolitis
- G Tracheo-oesophageal fistula

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 2-month-old Down's syndrome baby has been brought in with gradual abdominal distension and intermittent bilious vomiting with a history of delayed passage of meconium. The parents feel that the baby is unduly constipated. Examination shows a baby that has not been thriving normally, with a hugely distended abdomen and gross dehydration.
- 2 A prenatal ultrasound scan alerted the paediatricians to a congenital abnormality affecting the abdomen and chest. The premature neonate has been born with severe respiratory compromise and is on ventilatory support in the neonatal ICU.

- 3 A neonate is born with frothy saliva and episodes of cyanosis. Any attempt to feed makes the symptoms worse. A fine orogastric tube is arrested. A plain x-ray shows that the tube is curled up in the chest and there is gas in the abdomen. The condition was suspected on prenatal ultrasound.
- 4 A mother who suffered from polyhydramnios has given birth to a baby who has Down syndrome. The baby has bilious vomiting. The plain abdominal x-ray shows a 'double-bubble' appearance, and the condition was suspected on prenatal ultrasound.
- 5 A neonate has bile-stained vomiting with passage of blood-stained stools. The baby is very sick, and a contrast meal shows the bowel mostly on the right side with a subhepatic caecum.
- 6 A 2-week-old neonate who was born with jaundice has exhibited increasing yellowish discoloration of skin and conjunctiva ever since birth. There are some superficial skin bruises.
- 7 A few days after birth, a neonate has developed abdominal distension, blood-stained stools and bilious vomiting. The baby is toxic with septic shock.

THEME 4 Paediatric surgery

- A hydrocoele
- B coeliac disease
- C intussusception
- D gastroschisis
- E duodenal atresia
- F Wilms' tumour
- G Hirschsprung's disease
- H pyloric stenosis
- I necrotizing enterocolitis
- J infantile colic

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 7-month-old child is brought to A&E crying inconsolably and drawing his legs up. He has had one episode of vomiting and has been passing bloody stools. On examination a small mass is palpable in the right upper quadrant.
- 2 A 6-week-old child is brought to A&E by his mother as she has been concerned about severe forceful vomiting shortly after feeding. He is reported to be constantly hungry. There is no history of diarrhoea. On examination the child looks dehydrated and malnourished. Gastric peristalsis is visible.
- 3 A newborn has failed to pass meconium in the first 48 hours and is reluctant to feed. On examination the abdomen is distended. Plain abdominal radiography shows distended loops of bowel with absence of air in the rectum.

- 4 A newborn with trisomy 21 has suffered several episodes of bilious vomiting from a few hours since birth. On examination the baby has a scaphoid abdomen. Plain abdominal film shows a large gastric bubble proximal to an air-filled first part of duodenum.

Answers · Themes 1-4

BAILEY & LOVE EMQS

Theme 1 · Vomiting

- 1 **B Incarcerated inguinal hernia** — Bilious then faeculent vomiting with a tender, irreducible groin swelling in a toxic, dehydrated toddler is an incarcerated (obstructed) inguinal hernia.
- 2 **A Congenital hypertrophic pyloric stenosis** — Non-bilious vomiting in a hungry 5-week-old with a pyloric 'olive' felt during a feed is pyloric stenosis; the twitching (tetany) reflects metabolic alkalosis.
- 3 **C Intussusception** — Colicky screaming episodes, vomiting and redcurrant-jelly stool with an empty right iliac fossa (Dance sign) in a 10-month-old is intussusception.

Theme 2 · Acute Abdomen

- 1 **A Acute appendicitis** — High fever, toxicity, rebound and rigidity with an abdominal wall that does not move on respiration indicate perforated appendicitis with peritonitis.
- 2 **C Urinary tract infection** — Fever, vomiting, urinary frequency and bilateral loin tenderness indicate UTI (pyelonephritis); recurrent episodes warrant imaging for reflux or obstruction.
- 3 **B Nonspecific abdominal pain** — A well child with recurrent self-limiting pain and soft generalised tenderness without peritonism has non-specific abdominal pain, the commonest diagnosis in children.

Theme 3 · Congenital Malformations

- 1 **D Hirschsprung's disease** — Delayed meconium, chronic constipation, distension and failure to thrive in a Down syndrome infant suggest Hirschsprung disease; confirm by suction rectal biopsy.
- 2 **B Congenital diaphragmatic hernia** — Antenatally detected abdominal contents in the chest with severe respiratory failure

at birth is congenital diaphragmatic hernia; stabilise first, repair later.

- 3 **G Tracheo-oesophageal fistula** — Frothy saliva, choking on feeds and an orogastric tube coiled in the chest with gas in the abdomen is oesophageal atresia with distal tracheo-oesophageal fistula.
- 4 **C Duodenal atresia** — Polyhydramnios, Down syndrome, bilious vomiting and a 'double-bubble' X-ray indicate duodenal atresia, treated by duodenoduodenostomy.
- 5 **E Intestinal malrotation** — Bilious vomiting with bloody stools and a subhepatic caecum on contrast study is malrotation with midgut volvulus: emergency Ladd procedure.
- 6 **A Biliary atresia** — Jaundice persisting beyond 2 weeks with bruising (vitamin K malabsorption) suggests biliary atresia; Kasai portoenterostomy is best before 6–8 weeks.
- 7 **F Necrotising enterocolitis** — Distension, bloody stools and bilious vomiting with septic shock in a neonate is necrotising enterocolitis (pneumatosis intestinalis on X-ray).

Theme 4 · Paediatric surgery

- 1 **C intussusception** — Inconsolable colicky crying with legs drawn up, vomiting, bloody stools and a right upper quadrant mass in a 7-month-old is intussusception.
- 2 **H pyloric stenosis** — Projectile non-bilious vomiting in a hungry 6-week-old with visible gastric peristalsis is pyloric stenosis; rehydrate, then pyloromyotomy.
- 3 **G Hirschsprung's disease** — Failure to pass meconium in 48 hours with distension and no rectal gas is Hirschsprung disease; confirm by suction rectal biopsy.
- 4 **E duodenal atresia** — Bilious vomiting from birth, a scaphoid abdomen and a 'double bubble' in a trisomy 21 neonate indicate duodenal atresia.

2 THEMES · 10 SCENARIOS · AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead EMQs

THEME 1 MANAGEMENT OF PAEDIATRIC SURGICAL CONDITIONS

- A Abdominal ultrasound
- B Air enema insufflation
- C Broad spectrum antibiotics
- D Gastrografin enema
- E Observation
- F Pyloromyotomy
- G Laparotomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 7-month-old baby presents with a 2-day history of intermittent abdominal pain and crying episodes. On examination, there is a mass in the upper abdomen. The doctor notices that the baby has passed blood and mucus motions in his nappy.

- 2 A 3-week-old boy presents with a history of projectile vomiting that occurs a few minutes after every feed. There is no bile or blood in the vomit. His mother says that despite the vomiting, the baby is still hungry for feeds.
- 3 A 3-day-old baby presents with a distended abdomen and bile-stained vomit. He is known to have cystic fibrosis. On examination, there are no masses in the abdomen and the rectum is empty.
- 4 An 11-year-old boy presents with a 24-hour history of right iliac fossa pain. He has no other complaints apart from a cold that started last week. On examination, his temperature is 37.5°C and he has cervical lymphadenopathy. Routine blood tests show a white cell count of $4.7 \times 10^9/L$.
- 5 A 7-day-old infant who was born preterm presents with bile-stained vomiting, abdominal distension and bloody diarrhoea. He is tachycardic and tachypnoeic on examination. An abdominal X-ray shows distended bowel loops with some intramural air.

THEME 2 DIAGNOSIS OF PAEDIATRIC SURGICAL CONDITIONS

- A Appendicitis
- B Duodenal atresia
- C Exomphalos
- D Gastroschisis
- E Hirschsprung disease
- F Intussusception
- G Meckel diverticulum
- H Mesenteric adenitis
- I Necrotizing enterocolitis
- J Pyloric stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A newborn baby presents with bile-stained vomiting and abdominal distension on its third day of life. An abdominal X-ray reveals gas in the stomach and the first part of the duodenum, but nowhere else in the abdomen.
- 2 A 2-day-old baby presents with abdominal distension and bile-stained vomiting. It is noted that she has failed to pass meconium. An abdominal X-ray shows enlarged small and large bowel loops.
- 3 A baby is born with loops of small intestine, not covered by any membrane, herniating through a defect in the anterior abdominal wall just to the right of the umbilicus.
- 4 A 9-year-old boy presents with a 2-day history of right iliac fossa pain, nausea and vomiting, anorexia, and diarrhoea. On examination, the boy is tender in the right iliac fossa, with no masses palpable. His temperature is 37.8°C.
- 5 A 6-year-old girl presents with an episode of painless rectal bleeding. She complains of no other symptoms and she otherwise feels well. Rectal examination shows no obvious abnormality.

Answers • Themes 1-2

Theme 1 • MANAGEMENT OF PAEDIATRIC SURGICAL CONDITIONS

- 1 **B Air enema insufflation** — Intussusception in a stable infant without peritonitis is first treated by image-guided air enema reduction; laparotomy if it fails or perforation occurs.
- 2 **F Pyloromyotomy** — Pyloromyotomy treats hypertrophic pyloric stenosis.
- 3 **D Gastrografin enema** — Gastrografin enema relieves meconium ileus in cystic fibrosis.
- 4 **E Observation** — Recent URTI, cervical lymphadenopathy, low-grade fever and a normal white count suggest mesenteric adenitis; active observation is appropriate.
- 5 **C Broad spectrum antibiotics** — A preterm neonate with bloody stools and pneumatosis but no perforation has necrotizing enterocolitis: bowel rest, NG decompression and broad-spectrum antibiotics.

Theme 2 • DIAGNOSIS OF PAEDIATRIC SURGICAL CONDITIONS

- 1 **B Duodenal atresia** — Bilious vomiting with gas only in the stomach and duodenal cap ('double bubble') indicates duodenal atresia.
- 2 **E Hirschsprung disease** — Delayed meconium and distension suggest Hirschsprung disease.
- 3 **D Gastroschisis** — Gastroschisis is a para-umbilical (usually right-sided) defect with uncovered bowel; exomphalos is a central defect with a membrane-covered sac that may contain liver.
- 4 **A Appendicitis** — RIF pain with systemic features indicates appendicitis.
- 5 **G Meckel diverticulum** — Painless rectal bleeding in a well child is typically a Meckel diverticulum (ectopic gastric mucosa); confirm with a technetium-99m pertechnetate scan.

- 17 Acute Abdomen & Abdominal Trauma
- 18 Liver
- 19 Gallbladder & Biliary Tree
- 20 Pancreas
- 21 Spleen
- 22 Adrenal Gland
- 23 Hernias & Abdominal Wall
- 24 Abdomen EMQs



05

ABDOMEN

Acute Abdomen & Abdominal Trauma

Assessment of the acute abdomen, peritonitis and perforation, and blunt and penetrating abdominal trauma.

29

MCQs

0

EMQ SCENARIOS

5

SOURCES

0

REPEATED IDEAS

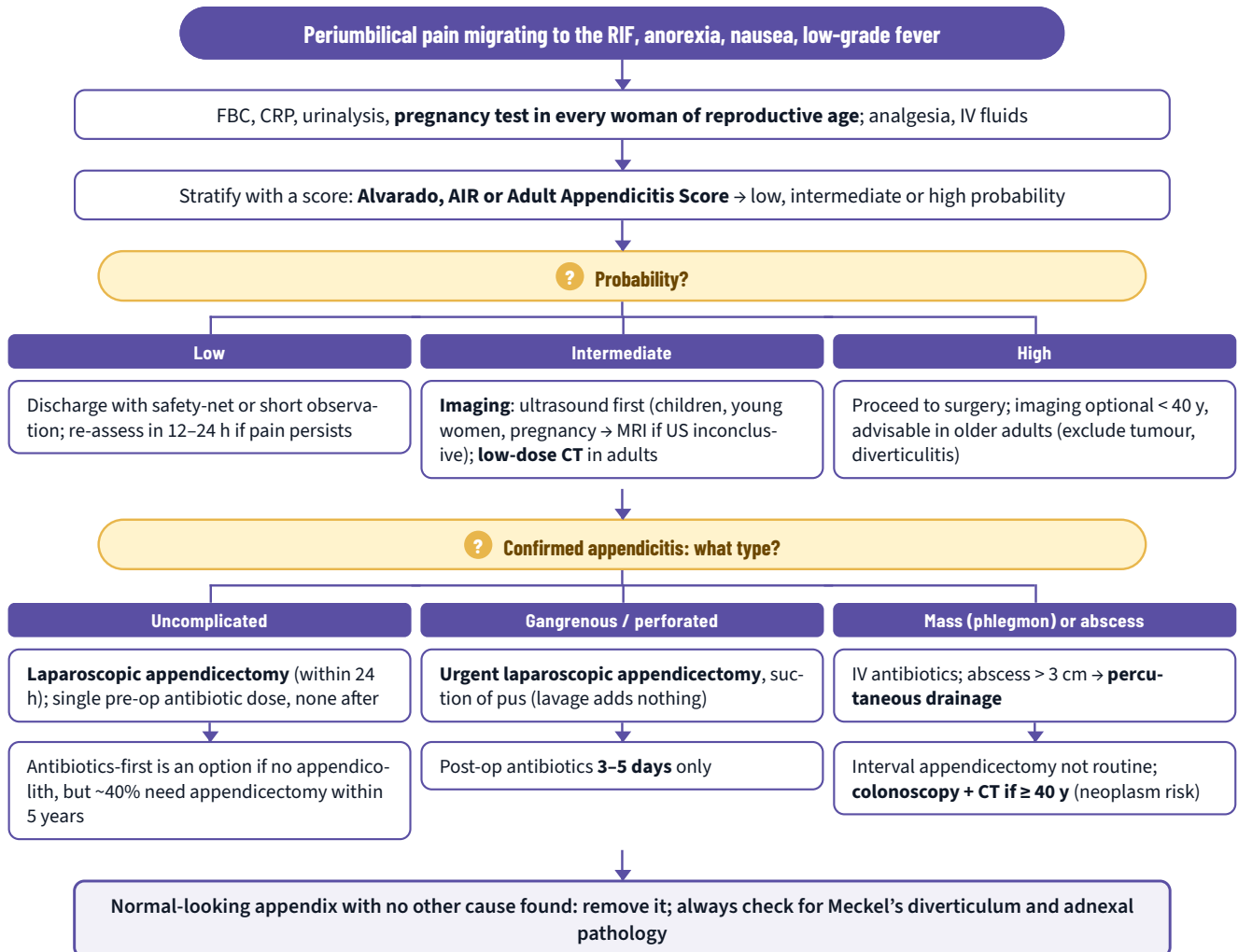
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Step by step

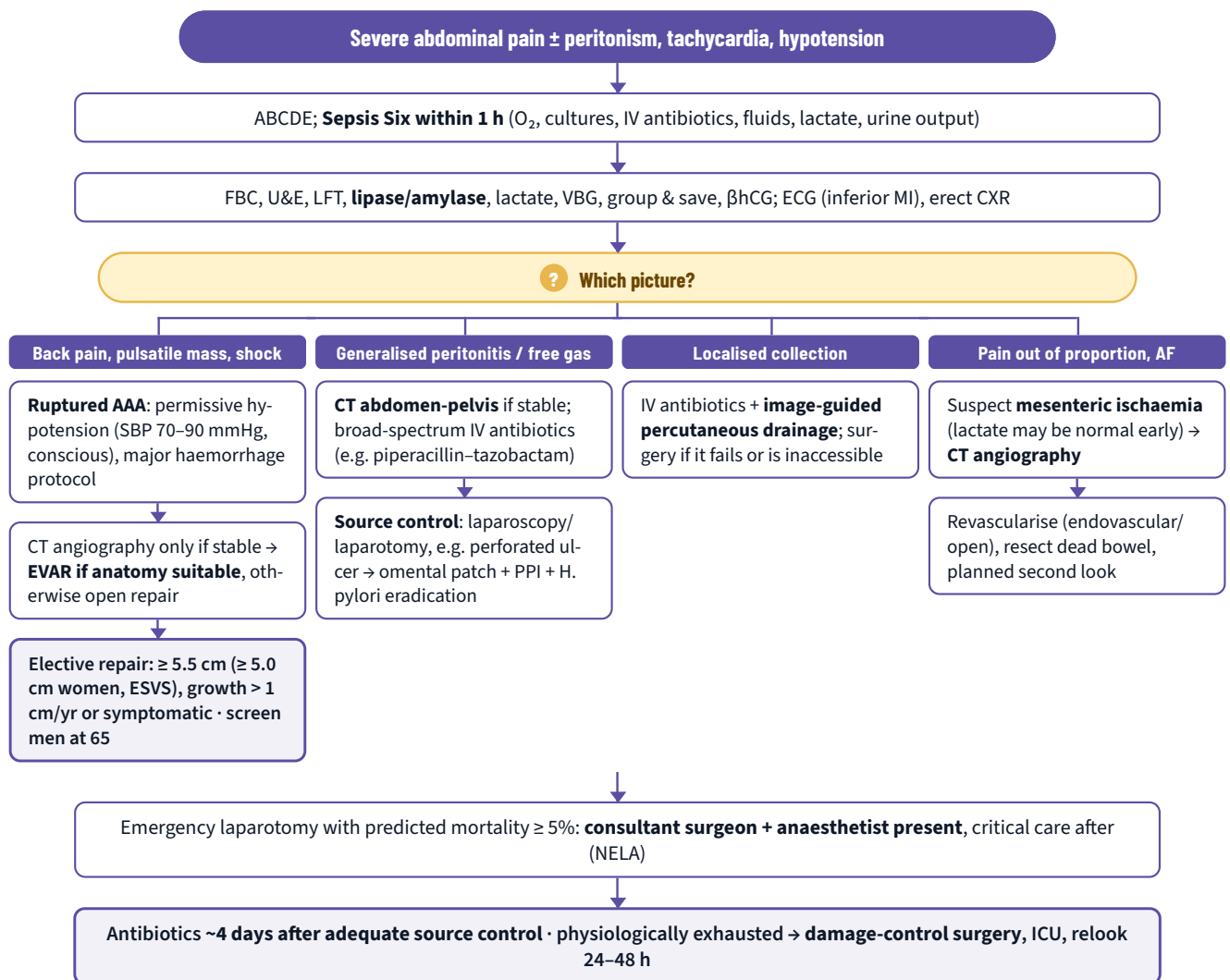
Suspected acute appendicitis

WSES 2020 JERUSALEM GUIDELINES • BAILEY & LOVE 28E



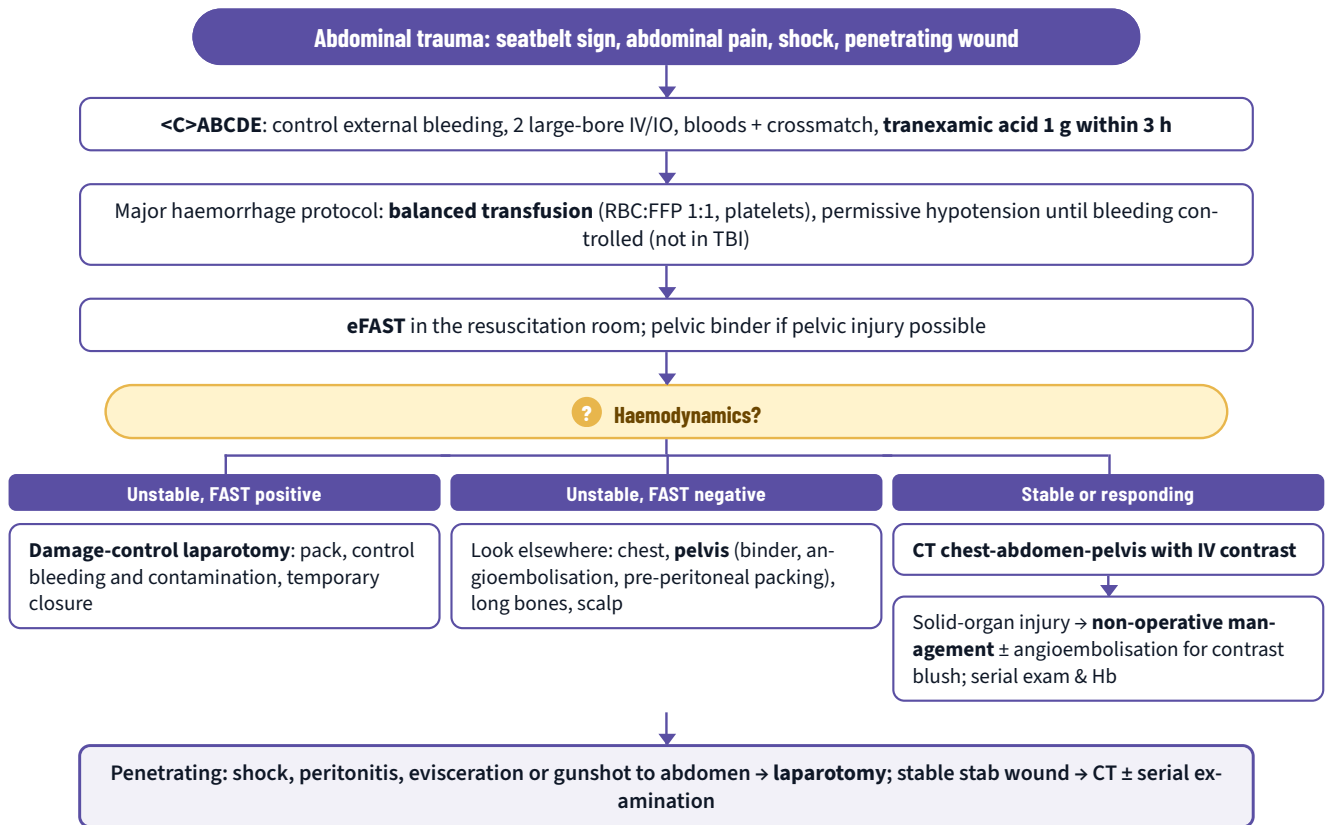
Acute abdomen with peritonitis, collection or shock

WSES 2017 INTRA-ABDOMINAL INFECTION · NELA · ESVS 2024 AAA



Blunt and penetrating abdominal trauma

ATLS 10TH ED · WSES 2020 LIVER & SPLEEN TRAUMA · NICE NG39



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-4

A 71-year-old male smoker with hypertension and type 2 diabetes has felt a pulsating abdominal mass with dull pain radiating to the back for 2 months.

- 1 What is the most likely diagnosis?
 - A Diverticulitis
 - B Renal colic
 - C Abdominal aortic aneurysm
 - D Chronic pancreatitis
 - E Mesenteric ischaemia
- 2 What is the gold-standard diagnostic test?
 - A Doppler ultrasound
 - B Abdominal X-ray
 - C CT angiography
 - D MRI
 - E ECG
- 3 He suddenly develops pain in the right leg with an absent dorsalis pedis pulse. What is the most likely diagnosis?
 - A Chronic limb ischaemia
 - B Cellulitis
 - C Acute limb ischaemia
 - D Deep vein thrombosis
 - E Compartment syndrome
- 4 What is the first-line management of this complication?
 - A Antiplatelet
 - B Anticoagulation (IV heparin)
 - C Antibiotics
 - D Leg elevation
 - E Fasciotomy

CASE • QUESTIONS 5-7

A young woman has right iliac fossa pain, low-grade fever, anorexia and dysuria, and is diagnosed with appendicitis.

- 5 Which test is 100% reliable for diagnosing appendicitis?
 - A CT
 - B Ultrasound
 - C Diagnostic laparoscopy (with histology)
 - D Alvarado score
 - E MRI

- 6 Given the presentation, where is the appendix?
 - A Retrocaecal
 - B Retroileal
 - C Pre-ileal
 - D Pelvic
 - E Subhepatic

- 7 Which investigation will you request?
 - A Complete blood count
 - B Renal function tests
 - C CRP
 - D ESR
 - E Serum amylase

- 8 Which two Alvarado (MANTRELS) criteria each score 2 points?
 - A Migration of pain and anorexia
 - B Right iliac fossa tenderness and leucocytosis
 - C Nausea/vomiting and fever
 - D Rebound tenderness and left shift
 - E Anorexia and fever

- 9 Which is NOT an advantage of laparoscopic over open appendicectomy?
 - A Fewer wound infections
 - B Less postoperative pain
 - C Shorter stay and earlier return to work
 - D Allows inspection of the pelvis and other diagnoses
 - E Lower rate of intra-abdominal abscess

- 10 An erect chest X-ray is shown. What is the most likely diagnosis on this image?



- A Pneumoperitoneum from a perforated viscus
 - B Bilateral pneumothorax
 - C Chilaiditi sign
 - D Subphrenic abscess
 - E Pleural effusion

MINISTERIAL EXAM

- 11** A 47-year-old man with a history of liver cirrhosis and well known on the general medical ward after recurrent admissions for drainage of ascites. He presents with confusion, fever (38.5°C), ascites and abdominal pain. Which of the following investigations is most important to establish the diagnosis?
- A** Stool culture and microscopy for ova, cysts and parasites
 - B** Diagnostic laparoscopy
 - C** Abdominal ultrasound scan
 - D** CT scan of abdomen
 - E** Ascitic fluid white cell count

Answers 1-11

LOCAL EXAMS MCQS

- 1** **C Abdominal aortic aneurysm** — A pulsatile, expansile abdominal mass with back pain in an elderly hypertensive smoker is an abdominal aortic aneurysm.
- 2** **C CT angiography** — Ultrasound screens and follows AAA, but CT angiography defines size, extent and neck anatomy for repair planning.
- 3** **C Acute limb ischaemia** — Sudden pain with a lost pulse is acute limb ischaemia, here from embolism of aneurysm thrombus.
- 4** **B Anticoagulation (IV heparin)** — Give IV unfractionated heparin immediately to stop clot propagation, then embolectomy or thrombolysis according to limb viability.
- 5** **C Diagnostic laparoscopy (with histology)** — Imaging and scores support the diagnosis, but only direct inspection at laparoscopy with histology confirms it.
- 6** **D Pelvic** — A pelvic appendix lies against the bladder and causes dysuria and frequency (and diarrhoea if near the rectum).
- 7** **A Complete blood count** — The white-cell count (leucocytosis with left shift) is part of the Alvarado score; urinalysis and a pregnancy test are also essential.
- 8** **B Right iliac fossa tenderness and leucocytosis** — RIF tenderness and leucocytosis score 2 each; migration, anorexia, nausea/vomiting, rebound, fever and left shift score 1 (total 10).
- 9** **E Lower rate of intra-abdominal abscess** — Laparoscopy reduces pain, wound infection and stay and is diagnostic; the intra-abdominal abscess rate is not lower (it may be slightly higher).
- 10** **A Pneumoperitoneum from a perforated viscus** — Free gas under both hemidiaphragms on an erect film means pneumoperitoneum, usually a perforated viscus.
- 11** **E Ascitic fluid white cell count** — Ascitic fluid analysis is crucial in diagnosing spontaneous bacterial peritonitis in cirrhotic patients.

13 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1** Regarding imaging in the acute abdomen, all of the following statements are true except:
 - A** US is a good first-line investigation.
 - B** CT is the best investigation for acute diverticulitis.
 - C** Plain x-ray of KUB (kidney, ureter, bladder) is the best imaging for suspected ureteric colic.
 - D** US and CT can diagnose the cause and site of intestinal obstruction.
 - E** Plain x-ray is the first-line investigation for suspected perforation or obstruction.
- 2** Regarding examination of the abdomen, which of the following statements are true?
 - A** Abdominal examination must always be preceded by general examination.
 - B** Abdominal discoloration at specific sites denotes underlying serious pathology.
 - C** Free movement of the abdominal wall on respiration makes generalised peritonitis unlikely.
 - D** Rebound tenderness is best tested by gentle percussion.
 - E** An intra-abdominal lump becomes prominent when the patient tenses the abdomen or lifts the head from the bed.

- 3** Which of the following statements are true?
- A The surface area of the peritoneum is nearly equal to that of the skin.
 - B The parietal peritoneum is poorly innervated.
 - C The peritoneum has the capacity to absorb large volumes of fluid.
 - D The peritoneum has the ability to produce fibrinolytic activity.
 - E When injured, the peritoneum produces an inflammatory exudate.
- 4** The following organisms are a gastrointestinal (GI) source of peritonitis except:
- A *Escherichia coli*
 - B *Streptococci*
 - C *Chlamydia*
 - D *Bacteroides*
 - E *Clostridium*
- 5** Regarding localised and diffuse peritonitis, which of the following statements are true?
- A Peritonitis gets localised because of anatomical factors.
 - B The greater omentum plays an important role in localising infection.
 - C Diagnostic peritoneal lavage (DPL) has a vital role in diagnosis.
 - D CT scan is useful for both diagnostic and therapeutic purposes.
 - E Operation is not necessary in all patients with peritonitis.
- 6** Regarding the management of peritonitis, which of the following statements is not true?
- A All patients should be thoroughly resuscitated.
 - B A paracolic abscess on CT scan must have open surgery.
 - C Septic shock might follow.
 - D Nutritional support must be instituted in certain patients.
 - E Young children do not have the natural barrier against the spread of peritonitis.
- 7** Regarding special forms of peritonitis, all of the following statements are true except:
- A Bile peritonitis is usually a feature following operation.
 - B A leucocytosis of $>30,000$ with 90% polymorphs is suggestive of pneumococcal peritonitis.
 - C Tuberculous ascites may be loculated.
 - D Tuberculosis (TB) spreads from the ileocaecal region.
 - E Diagnostic smear will be positive in the vast majority of cases of tuberculosis.
- 8** Regarding intraperitoneal abscess, which of the following statements are correct?
- A The pelvis is the commonest site of intraperitoneal abscess.
 - B Pus can also collect in four other intraperitoneal sites.
 - C The usual cause is postoperative.
 - D Pain may be a feature well away from the site of pathology.
 - E Open operation and drainage is the treatment of choice.
- 9** Regarding ascites, which of the following statements are not true?
- A Imbalance between plasma and peritoneal colloid osmotic and hydrostatic pressure causes ascites.
 - B Shifting dullness is a classical sign in all cases of ascites.
 - C The protein content determines if the ascitic fluid is a transudate or an exudate.
 - D A peritoneovenous shunt should be considered in all cases of ascites.
 - E Chylous ascites is most commonly caused by malignancy.
- 10** Regarding tumours of the peritoneum, which of the following statements are true?
- A Secondary tumours are more common.
 - B Carcinomatosis peritonei is usually a terminal event.
 - C Pseudomyxoma peritonei arises from a primary tumour from the appendix.
 - D Repeated 'debulking surgery' in pseudomyxoma peritonei gives relief.
 - E Peritoneal loose bodies are malignant in origin.
- 11** Regarding adhesions, all of the following statements are true except:
- A Ischaemic tissue inhibits fibrinolysis encouraging formation of adhesions.
 - B Laparoscopic surgery reduces adhesion-related complications.
 - C Adhesions are responsible for two-thirds of small bowel obstructions.
 - D Prevention of adhesions can be actively achieved.
 - E Intestinal obstruction from sclerosing encapsulating peritonitis should be treated by surgery.
- 12** Regarding the omentum and mesentery, which of the following statements are true?
- A Torsion of the omentum always presents as an emergency.
 - B Mesenteric injury occurs as a seatbelt syndrome.
 - C Acute nonspecific ileocaecal mesenteric adenitis can be confused with acute appendicitis.
 - D A mesenteric cyst clinically presents as a fixed periumbilical mass.
 - E Chylolymphatic cyst is the commonest type of mesenteric cyst.
- 13** Regarding the retroperitoneal space, which of the following statements is not true?
- A Idiopathic retroperitoneal fibrosis can present as chronic renal failure.
 - B An AAA can cause retroperitoneal fibrosis.
 - C Retroperitoneal fibrosis is always a benign condition.
 - D Retroperitoneal abscess may be a manifestation of vertebral tuberculosis.
 - E Retroperitoneal lipoma is often a sarcoma.

Answers 1-13

BAILEY & LOVE MCQS

- 1 **C** Plain x-ray of KUB (kidney, ureter, bladder) is the best imaging for suspected ureteric colic. — **UPDATED** Plain films have limited utility in ureteric colic; non-contrast CT is preferred.
- 2 **ABCD** True: A, B, C, D — **UPDATED** False: E — an intra-abdominal lump becomes less prominent when the abdominal muscles are tensed (head-raising test); a lump in the abdominal wall stays palpable.
- 3 **ACDE** True: A, C, D, E — The peritoneum is extensive, absorptive, and fibrinolytic, producing inflammatory exudate when injured; note that the parietal peritoneum is well innervated.
- 4 **C** Chlamydia — Chlamydia is not typically a GI organism, unlike the other common gut flora.
- 5 **ABDE** True: A, B, D, E — **UPDATED** Anatomical factors and the omentum help localise infection; CT scanning aids both diagnosis and management, while DPL is now rarely used.
- 6 **B** A paracolic abscess on CT scan must have open surgery. — **UPDATED** Most paracolic abscesses are managed by percutaneous drainage rather than open surgery.
- 7 **E** Diagnostic smear will be positive in the vast majority of cases of tuberculosis. — **UPDATED** Acid-fast smears are positive in only a small percentage of TB peritonitis cases; the other statements are correct.
- 8 **ABCD** True: A, B, C, D — **UPDATED** Intraperitoneal abscesses most commonly occur in the pelvis, are often postoperative, and pain may be referred; open drainage is now largely replaced by percutaneous methods.
- 9 **BD** True: B, D — **UPDATED** Shifting dullness may be absent in massive ascites and peritoneovenous shunts are reserved for refractory cases only.
- 10 **ABCD** True: A, B, C, D — **UPDATED** Secondary peritoneal tumours are most common, pseudomyxoma peritonei is typically appendiceal, and debulking can offer symptom relief; peritoneal loose bodies are benign.
- 11 **DE** True: D, E — **UPDATED** There is no guaranteed method to prevent adhesions and sclerosing encapsulating peritonitis is generally managed non-operatively.
- 12 **ABCE** True: A, B, C, E — **UPDATED** False: D — a mesenteric cyst is a mobile periumbilical mass that moves at right angles to the line of the mesentery (Tillaux sign).
- 13 **C** Retroperitoneal fibrosis is always a benign condition. — **UPDATED** Retroperitoneal fibrosis may be secondary to malignancy and is not always benign.

2 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 30-year-old man presents to the emergency department with sudden onset of severe epigastric pain and vomiting 3 hours ago. He reports a 6-month history of chronic epigastric pain occurring nearly every day and relieved by antacids. On examination, he appears sweaty and avoids movement. Vital signs reveal a temperature of 100°F, BP of 100/60 mmHg, pulse rate of 110/min, and respiratory rate of 12/min. The remainder of his examination reveals diminished bowel sounds and a markedly tender and rigid abdomen. A chest x-ray and abdominal films reveal pneumoperitoneum. For the patient with a perforated duodenal ulcer described above, what is the most appropriate next diagnostic test?
 - A CT scan
 - B Upper GI water-soluble contrast study
 - C Lower GI water-soluble contrast study
 - D Abdominal ultrasound
 - E None of the above
- 2 What is the most appropriate next step in management for the perforated duodenal ulcer patient?
 - A Immediate laparotomy
 - B Non-operative management with NG decompression and antibiotics
 - C Fluid resuscitation
 - D Administration of H2 blockers
 - E Placement of a central venous line

Answers 1-2

LANGE MCQS

- 1 **E** None of the above — **UPDATED** Peritonitis with pneumoperitoneum on plain films in a patient with ulcer symptoms is enough to proceed to surgery; further imaging only delays treatment.

- 2 **C** **Fluid resuscitation** — **UPDATED** Resuscitate with IV fluids (plus analgesia, antibiotics and a urinary catheter) before ur-

gent laparotomy and omental patch repair; a central line is not the priority.

1 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Management of an abdominal aortic aneurysm A 65-year-old man is admitted following an acute episode of abdominal pain and collapse. The pain is intermittent and radiates to the back and iliac fossae. On examination, the patient appears confused, is sweating and has tachycardia. On inspection, the abdomen appears normal, but on palpation, you discover a pulsatile, expansile swelling in the midline. What is the most important next step?
- A Establish large-bore intravenous access, cross-match blood and begin cautious (permissive hypotensive) resuscitation
 - B Send for a computed tomography scan of the abdomen
 - C Obtain blood to determine haemoglobin and amylase levels
 - D Request an abdominal plain film radiograph
 - E Perform electrocardiography

Answer 1

SBA MCQs

- 1 **A** **Establish large-bore intravenous access, cross-match blood and begin cautious (permissive hypotensive) resuscitation** — **UPDATED** A shocked patient with a pulsatile mass has

a ruptured AAA: secure IV access, cross-match and resuscitate to permissive hypotension en route to theatre; CT only if stable.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 An 83-year-old man is admitted with acute onset of severe epigastric pain and has 'board-like' rigidity of the upper abdomen. His temperature is 37.4°C, pulse rate 124bpm, blood pressure 104/75mmHg. A working diagnosis of visceral perforation is made and investigations planned. Which is the single most appropriate fluid prescription to administer?
- A IV Hartmann's solution 500mL, over 15min
 - B IV O-negative blood 2 units, run in
 - C IV sodium bicarbonate 0.13% 500mL, over 2h
 - D IV sodium chloride 0.9% 100mL, over 2h
 - E IV whole cross-matched blood 2 units, over 4h
- 2 27-year-old woman has had 4 days of gradually increasing, diffuse lower abdominal pain and 6h of acute severe worsening of the pain. Her abdomen is tender with guarding in the lower half. Her temperature is 37.2°C, pulse 104bpm, blood pressure 104/56mmHg, and capillary refill approximately 6sec. Which is the single most likely diagnosis?
- A Acute tubo-ovarian abscess
 - B Endometriosis
 - C Perforated diverticulitis
 - D Ruptured ectopic pregnancy
 - E Torted ovarian cyst

Answers 1-2

OXFORD MCQS

- 1 **A** IV Hartmann's solution 500mL, over 15min — **UPDATED** A tachycardic patient with peritonitis needs prompt balanced crystalloid resuscitation (500 mL bolus over <15 min, then re-assess); blood is for haemorrhage and bicarbonate is not indicated.
- 2 **D** Ruptured ectopic pregnancy — **UPDATED** Increasing lower abdominal pain with sudden worsening, guarding, tachycardia and poor capillary refill in a young woman is a ruptured ectopic pregnancy until a pregnancy test excludes it.

Liver

Liver abscess, hydatid cyst, benign and malignant liver tumours, portal hypertension and liver trauma.

56

MCQS

0

EMQ SCENARIOS

7

SOURCES

3

REPEATED IDEAS

IN THIS CHAPTER

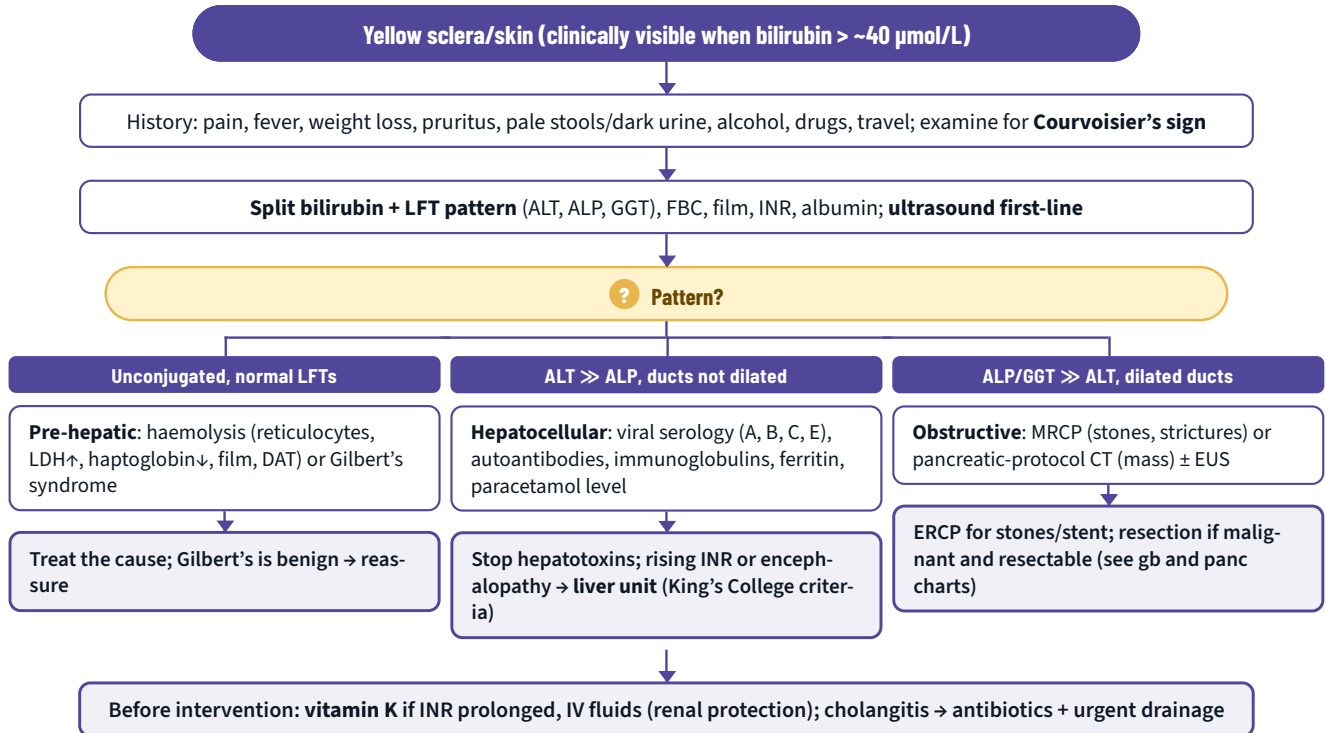
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18

Step by step

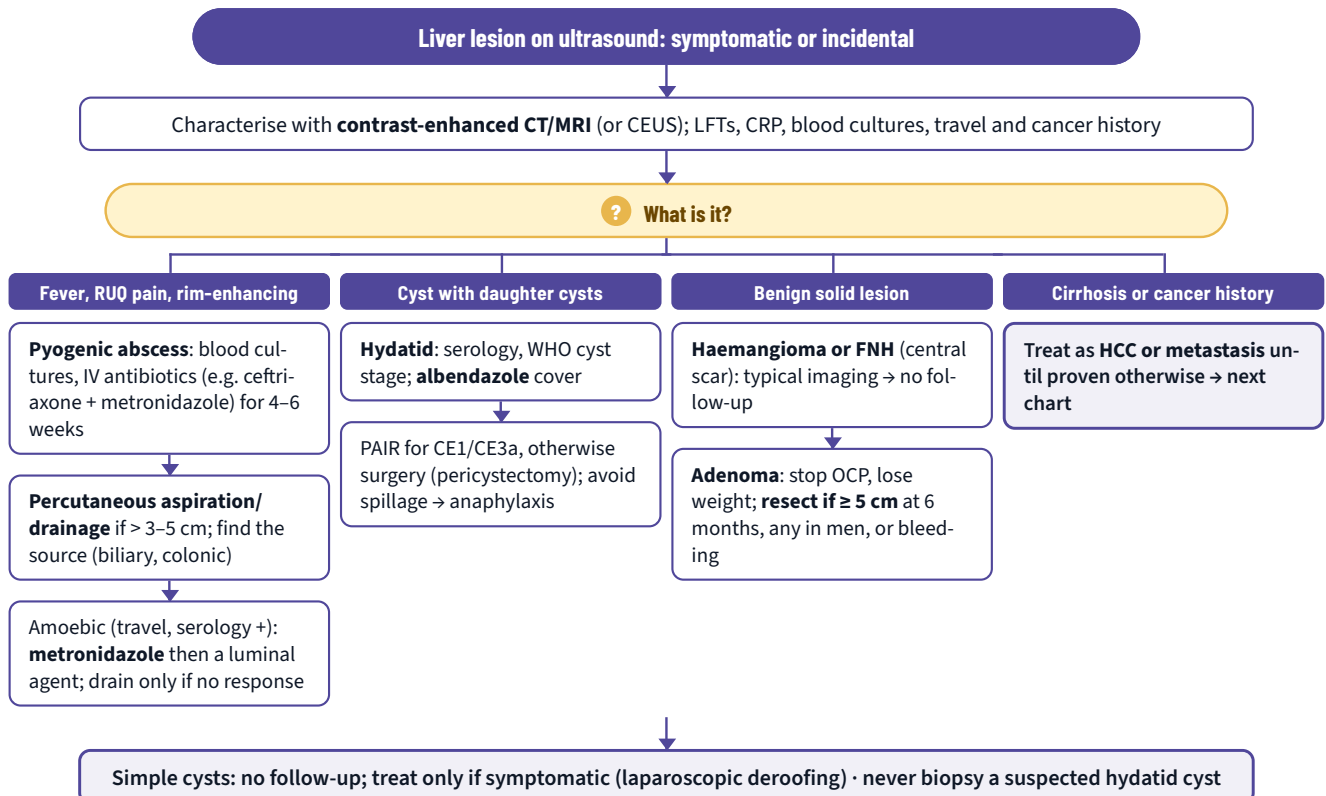
Jaundice: pre-hepatic, hepatic or post-hepatic?

BSG 2018 ABNORMAL LIVER BLOOD TESTS • BAILEY & LOVE 28E



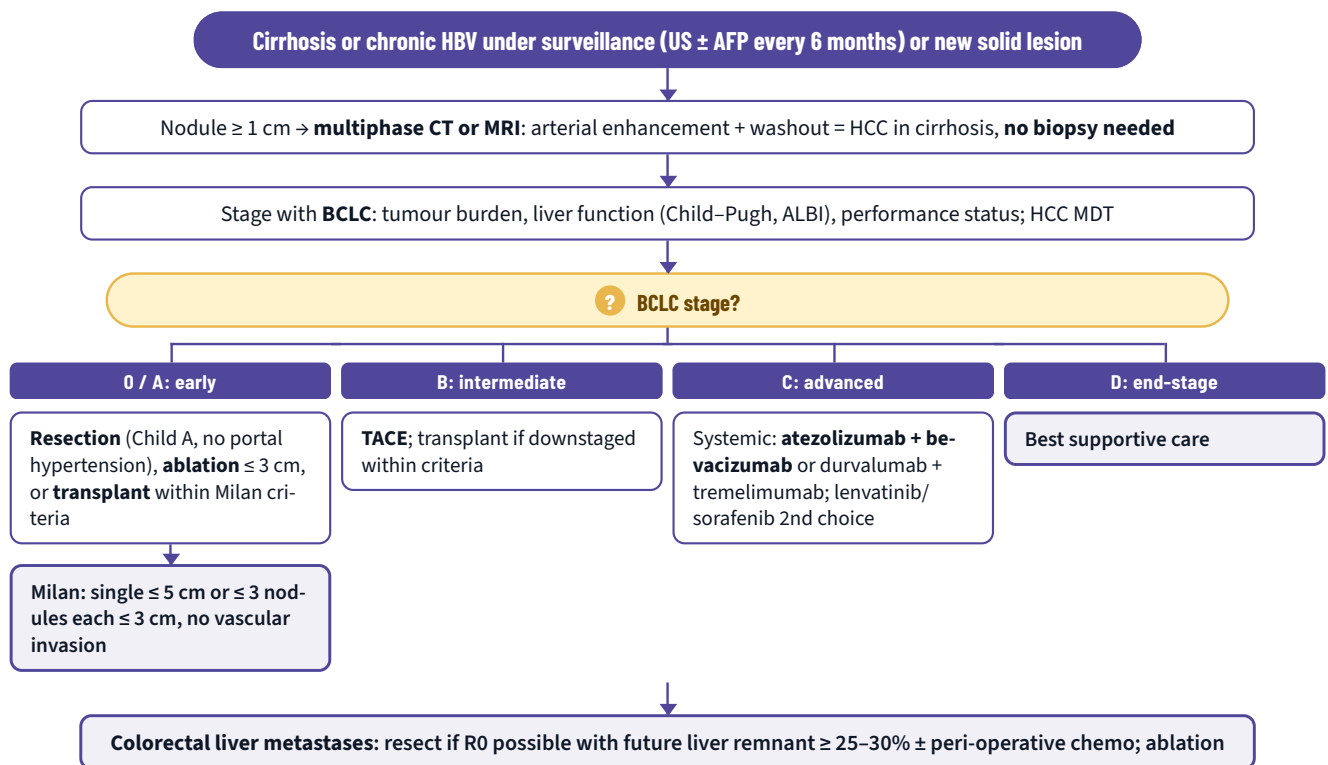
Focal liver lesion: abscess, cyst or benign tumour

EASL 2016 BENIGN LIVER TUMOURS • WHO-IWGE ECHINOCOCCOSIS



Hepatocellular carcinoma and liver metastases

EASL 2018 HCC · BCLC 2022 · ESMO 2023 MCRC



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1–4

A man has fever, malaise and tender hepatomegaly after diarrhoea following travel to Nasiriyah. Ultrasound shows a 7-cm cystic liver lesion; blood cultures are negative.

- 1 What is the most likely organism?
 - A Entamoeba histolytica
 - B Shigella
 - C Echinococcus granulosus
 - D Staphylococcus aureus
 - E Escherichia coli
- 2 How did the organism reach the liver?
 - A Direct spread from an empyema
 - B Haematogenous spread via the hepatic artery
 - C Via the portal vein from the colon
 - D Ascending infection via the bile duct
 - E Lymphatic spread
- 3 He receives a second-generation cephalosporin and metronidazole, and the cyst is aspirated. What does the aspirate show?
 - A Anchovy-sauce (paste) fluid
 - B Clear hydatid fluid
 - C Foul-smelling yellow pus
 - D Blood-stained bile
 - E Chylous fluid
- 4 Which additional treatment is essential?
 - A Diloxanide furoate
 - B Long-term metronidazole
 - C Praziquantel
 - D Albendazole
 - E Surgical drainage

- 5 A contrast CT of the upper abdomen of a febrile patient is shown. What is the most likely diagnosis on this image?



- A Liver abscess
- B Simple liver cyst
- C Hepatocellular carcinoma
- D Haemangioma
- E Hydatid cyst with daughter cysts

MINISTERIAL EXAM

Ministerial evaluation examinations

- 6 A 56-year-old man presented with abdominal pain and a mass in the right upper quadrant. He reported that he had been diagnosed with viral hepatitis several years previously. Investigations showed: Serum alpha-fetoprotein 13,500 $\mu\text{U/L}$ (normal < 10). What is the most likely underlying viral infection?
 - A Hepatitis A virus
 - B Hepatitis B virus
 - C Hepatitis C virus
 - D Hepatitis D virus
 - E Hepatitis E virus

FORMATIVE

College formative examinations

- 7 A patient is complaining of upper abdominal pain. CT reveals a cystic mass in the right lobe of the liver. Which of the following investigations is NOT used for diagnosis?
 - A ELISA for hydatid cyst
 - B Ultrasound
 - C CT scan
 - D PET scan
 - E Serology (Indirect hemagglutination test)

- 8 A patient is referred for laparoscopic resection of a hydatid cyst. Which of the following drugs is NOT used in the perioperative period?
- A Albendazole
 - B Mebendazole
 - C Praziquantel
 - D Hypertonic (20%) saline as a scolical agent
 - E Levamisole

Answers 1-8

LOCAL EXAMS MCQS

- 1 **A Entamoeba histolytica** — Fever, tender hepatomegaly and a liver abscess after dysentery with sterile blood cultures is an amoebic liver abscess.
- 2 **C Via the portal vein from the colon** — Trophozoites invade the colonic mucosa and travel in the portal venous blood to the liver (usually the right lobe).
- 3 **A Anchovy-sauce (paste) fluid** — Amoebic abscess fluid is odourless, reddish-brown "anchovy sauce" (liquefied liver); pyogenic abscesses contain yellow pus.
- 4 **A Diloxanide furoate** — After metronidazole, a luminal amoebicide (diloxanide furoate or paromomycin) is needed to clear intestinal cysts and prevent relapse.
- 5 **A Liver abscess** — A large hypodense collection with an enhancing wall and necrotic debris in the right lobe is a liver abscess.
- 6 **B Hepatitis B virus** — **UPDATED** Hepatitis B is strongly associated with hepatocellular carcinoma and elevated AFP.
- 7 **D PET scan** — PET scan is not routinely used in diagnosing hydatid cysts; US, CT, and serology are more appropriate.
- 8 **E Levamisole** — **UPDATED** Levamisole has no role in hydatid disease; albendazole/mebendazole ($\pm$ praziquantel) are given perioperatively and hypertonic saline is used intraoperatively as a scolicide.

12 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

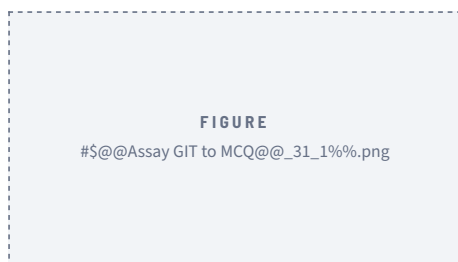
- 1 A patient presents with RUQ pain, itching, and jaundice. CT scan of the liver revealed a cystic lesion, with multiple daughter cysts. What is the diagnosis? and what is the next step in the management?
- A Hepatocellular carcinoma; immediate chemotherapy
 - B Cholangiocarcinoma; palliative care
 - C Hydatid liver cyst; prepare the patient for surgery (cystectomy) with preoperative Albendazole and Praziquantel
 - D Liver abscess; percutaneous drainage
 - E Simple liver cyst; conservative management
- 2 A patient with a hydatid liver cyst is scheduled for surgery. Is there a benefit of giving medication before surgery? Why?
- A No, medications before surgery have no impact
 - B Yes, they improve liver function
 - C Yes, albendazole ($\pm$ praziquantel) inactivates the cyst and lowers intracystic pressure, reducing the risk of recurrence
 - D No, they only increase the risk of infection
 - E Yes, they are used to anesthetize the patient
- 3 A jaundiced patient has a hydatid liver cyst with daughter cysts on CT and is planned for surgery. What further imaging would you order before surgery?
- A CT scan
 - B MRI with MRCP
 - C Ultrasound
 - D X-ray
 - E PET scan
- 4 After surgery for a hydatid liver cyst, what is your plan for minimising the risk of recurrence or reinfection?
- A Administer high-dose steroids preoperatively
 - B Use prophylactic antibiotics only during surgery
 - C Continue albendazole ($\pm$ praziquantel) postoperatively and educate the patient about hand hygiene, clean food and water and avoiding contact with dogs
 - D No additional measures are necessary
 - E Delay surgery until infection resolves

REPEATED 2* • ALSO ASKED IN
Essay to MCQ

REPEATED 2* • ALSO ASKED IN
Essay to MCQ

REPEATED 2* • ALSO ASKED IN
Essay to MCQ

- 5 A patient with right upper quadrant pain has the imaging shown. What is the imaging modality and what does it show?



- A Ultrasound showing a solid lesion
 B CT scan showing a cystic liver lesion with floating detached membranes (water-lily sign)
 C MRI indicating liver hemangioma
 D Plain X-ray with calcifications
 E PET scan showing hypermetabolic activity
- 6 What is the usual surgical option for an uncomplicated hydatid liver cyst?
- A Liver transplant
 B Laparoscopic cholecystectomy
 C Cystectomy
 D Hepatic resection
 E Percutaneous drainage
- 7 A patient with upper abdominal pain did not respond to H. pylori eradication therapy, and an ultrasound reveals a 4.5 × 3 cm cyst in the liver. What is the most likely diagnosis?
- A Hepatic hemangioma
 B Hepatocellular carcinoma
 C A liver cyst, which may be a simple cyst, an abscess, or a parasitic cyst caused by Echinococcus granulosus
 D Focal nodular hyperplasia
 E Metastatic liver disease
- 8 A patient with a hydatid liver cyst refuses surgery. What medical treatment would you offer, and for how long?
- A No treatment is available
 B Short course of antibiotics for 7 days
 C Medical treatment with albendazole or mebendazole for 3-6 months
 D High-dose steroids for 2 weeks
 E Chemotherapy for 6 months

- 9 How would you monitor the response of a hydatid liver cyst to medical treatment?
- A Clinical symptoms only
 B Liver function tests only
 C By imaging studies (Ultrasound or CT)
 D Serum tumor markers
 E Biopsy
- 10 Before starting medical treatment for a suspected hydatid liver cyst, what single test would you do, and why?
- A Liver biopsy to confirm the diagnosis
 B Serologic testing for viral hepatitis
 C Ultrasound evaluation to determine the cyst characteristics
 D Complete blood count
 E Chest x-ray
- 11 A patient with a hydatid liver cyst has a CT scan showing multiple cysts in the lung, kidney and spleen. What is the best serological test to confirm the diagnosis?
- A Western blot
 B PCR assay
 C ELISA to detect specific antibodies against echinococcus
 D Rapid antigen test
 E Serum protein electrophoresis
- 12 Two weeks later, a patient with a hydatid liver cyst develops jaundice, right upper quadrant pain, a tender mass and a swinging fever. What is the diagnosis?
- A Acute viral hepatitis
 B Biliary colic
 C Liver abscess resulting from the complication of a liver cyst
 D Cholangiocarcinoma
 E Gallstone pancreatitis

Answers 1-12

ESSAY TO MCQ MCQS

- 1 **C** Hydatid liver cyst; prepare the patient for surgery (cystectomy) with preoperative Albendazole and Praziquantel — CT findings indicate a hydatid cyst; surgery with preoperative Albendazole and Praziquantel is indicated.
- 2 **C** Yes, albendazole (± praziquantel) inactivates the cyst and lowers intracystic pressure, reducing the risk of recurrence — **UPDATED** Preoperative albendazole (± praziquantel) kills protoscolices and softens the cyst, reducing the risk of secondary hydatidosis and recurrence if spillage occurs.
- 3 **B** MRI with MRCP — **UPDATED** Jaundice suggests cyst–biliary communication; MRCP shows the biliary tree and any communication before surgery. Ultrasound is first-line but adds little after CT.
- 4 **C** Continue albendazole (± praziquantel) postoperatively and educate the patient about hand hygiene, clean food and water and avoiding contact with dogs — **UPDATED** Postoperative albendazole (usually 1–3 months) kills any spilled protoscolices, and hygiene education (dogs, unwashed vegetables) prevents reinfection.

- 5 **B** CT scan showing a cystic liver lesion with floating detached membranes (water-lily sign) — **UPDATED** Contrast CT shows a liver cyst with detached, floating endocyst membranes (water-lily sign), diagnostic of a hydatid cyst.
- 6 **C** Cystectomy — **UPDATED** Conservative surgery (cyst evacuation with partial cystectomy or pericystectomy under scolicidal cover) is standard; formal hepatectomy is reserved for selected cases.
- 7 **C** A liver cyst, which may be a simple cyst, an abscess, or a parasitic cyst caused by *Echinococcus granulosus* — **UPDATED** A cystic liver lesion may be a simple cyst, an abscess or a hydatid (*Echinococcus granulosus*) cyst; haemangioma, FNH, HCC and metastases are usually solid.
- 8 **C** Medical treatment with albendazole or mebendazole for 3-6 months — **UPDATED** For a parasitic liver cyst in a patient refusing surgery, prolonged antihelminthic therapy is appropriate.
- 9 **C** By imaging studies (Ultrasound or CT) — **UPDATED** Serial imaging is the best method to monitor treatment response.
- 10 **C** Ultrasound evaluation to determine the cyst characteristics — **UPDATED** Ultrasound stages the cyst (WHO-IWGE classification: active, transitional or inactive), which decides between drugs,PAIR, surgery or watch-and-wait.
- 11 **C** ELISA to detect specific antibodies against echinococcus — **UPDATED** ELISA confirms the diagnosis of echinococcosis.
- 12 **C** Liver abscess resulting from the complication of a liver cyst — **UPDATED** Swinging fever with a tender mass indicates secondary bacterial infection of the cyst (liver abscess); jaundice suggests rupture into the biliary tree.

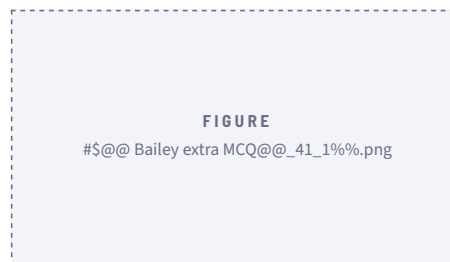
23 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love MCQs

- 1 Which of the following statements is not true?
- A The falciform ligament of the liver divides it into functional right and left units.
 - B The right and left triangular ligaments help to fix the liver and keep it in its place.
 - C The main blood supply of the liver is derived from the portal vein.
 - D The right hepatic vein has an extrahepatic course while the middle and left hepatic veins join the inferior vena cava within the liver.
 - E The left intrahepatic duct has a longer extrahepatic course than the right.
- 2 Which of the following statements are true?
- A Cantlie's line passes through the gallbladder fossa and middle hepatic vein.
 - B The functional right and left units divided by Cantlie's line are further subdivided into four segments on each side.
 - C The free edge of the lesser omentum contains the hepatic artery, bile duct and portal vein.
 - D The portal vein is formed by the union of the superior and inferior mesenteric veins.
 - E The structures at the porta hepatis are the portal vein, bile ducts, hepatic artery and hepatic veins.
- 3 Which of the following liver function tests (LFTs) are abnormal?
- A Bilirubin 5–17 $\mu\text{mol/L}$
 - B Alkaline phosphatase (ALP) 135–240 IU/L
 - C Aspartate transaminase (AST) 55–80 IU/L
 - D Alanine transaminase (ALT) 110–148 IU/L
 - E Gamma-glutamyl transpeptidase (GGT) 10–50 IU/L
- 4 Regarding liver imaging, which of the following statements are true?
- A Ultrasound (US) is the first-line investigation.
 - B Nuclear medicine scanning gives an idea of the extent of anatomical derangement.
 - C The 'gold standard' is contrast-enhanced computed tomographic scan (CECT).
 - D Magnetic resonance imaging (MRI) has advantages over CECT.
 - E Angiography is best reserved when embolisation is contemplated.
- 5 Regarding acute and chronic liver disease, which of the following statements are not true?
- A The overall mortality from acute liver failure is about 50%.
 - B Chronic congestive heart failure is a terminal event in chronic liver disease.
 - C The most useful clinical sign is flapping tremor.
 - D Model for End-Stage Liver Disease (MELD) score depends upon INR, serum bilirubin and serum creatinine.
 - E Liver transplantation has good short-term results in acute liver failure.
- 6 Regarding liver trauma, which of the following statements are not true?
- A Liver injuries are common.
 - B Blunt trauma is often associated with splenic, mesenteric and renal injuries.
 - C Penetrating trauma is often associated with pericardial or chest injuries.
 - D CECT scan must be carried out in every case of liver trauma.
 - E Interventional radiology has a role in the management of liver injury.

- 7** Which of these statements are true?
- A** Penetrating injuries with haemodynamic instability or peritonitis should be explored.
 - B** Blunt injuries are usually treated conservatively.
 - C** Exploration for a liver injury is best done by a long midline incision.
 - D** Severe crush injury is ideally treated by packing.
 - E** Venovenous bypass should be considered in major liver vascular injury.
- 8** Regarding portal hypertension, all of the following statements are true except:
- A** Acute haematemesis from portal hypertension occurs most commonly from gastric varices.
 - B** Initial endoscopic treatment of oesophageal varices with banding, as opposed to injection sclerotherapy, has less chance of oesophageal ulceration.
 - C** Long-term beta-blocker therapy coupled with sclerotherapy regime or endoscopic banding is the mainstay of treatment of portal hypertension.
 - D** In failed drug or endoscopic treatment, the ideal choice is the surgical shunt of portocaval anastomosis.
 - E** Ascites in cirrhosis can be treated by a peritoneovenous shunt.
- 9** Which of the following are true of amoebiasis?
- A** The disease is common in the Indian subcontinent, Africa and parts of Central and South America.
 - B** The majority of sufferers are symptomatic.
 - C** Amoebic liver abscess, the most common extraintestinal manifestation, occurs in more than 50% of cases.
 - D** The mode of infection is via the faeco-oral route due to poor hygiene and sanitation.
 - E** The organism enters the gut via contaminated food and water.
- 10** *Entamoeba histolytica*, the causative organism, enters the body as cysts contaminating water or food. Which of the following occurs?
- A** The cysts hatch in the large bowel and colonise the colon.
 - B** The cysts cause flask-shaped ulcers in the colon.
 - C** The trophozoites may invade the wall of the colon and pass to the liver via the portal circulation, causing focal infarction and liquefactive necrosis in the liver.
 - D** Cysts are passed in the faeces in an infective form.
 - E** The right lobe of the liver is affected in 10% of cases, the left in 80% and the rest are multiple.

- 11** A middle-aged man presents with RUQ pain, fever, night sweats, and weight loss, with imaging showing a hypoechoic/hypodense liver lesion. Which statements are true?



- A** Although the clinical features are suggestive of amoebiasis, a history of travel to an endemic area and bloody diarrhoea may also be present.
 - B** The CT scan is compatible with an amoebic liver abscess.
 - C** The abscess cavity contains blackcurrant jelly-like fluid.
 - D** The collection is normally odourless and sterile but becomes malodorous if secondarily infected.
 - E** Trophozoites may be found in the wall of the abscess cavity in a minority of cases.
- 12** Which of the following are presenting features of amoebiasis?
- A** Chronic diarrhoea, often blood-stained, with colicky abdominal pain.
 - B** Pain in the upper right abdomen and right shoulder-tip with hiccoughs and a painful dry cough.
 - C** An apple-core lesion on barium enema due to a chronic granuloma (amoeboma), most commonly in the caecum.
 - D** Features of peritonitis with shock (due to abscess rupture).
 - E** All of the above.
- 13** What is the treatment for an amoebic liver abscess?
- A** Urgent laparotomy upon diagnosis.
 - B** Medical treatment with metronidazole or tinidazole.
 - C** Medical treatment with diloxanide furoate alone is sufficient.
 - D** Aspiration may be carried out for abscesses at risk of imminent rupture.
 - E** Surgical treatment is indicated for a ruptured liver abscess, severe haemorrhage/toxic megacolon, and a suspected amoeboma not responding to treatment or where carcinoma cannot be excluded.
- 14** Which of the following are true regarding hydatid disease caused by *Echinococcus granulosus*?
- A** Man is the definitive host.
 - B** Dog is the definitive host.
 - C** Sheep and cattle are intermediate hosts.
 - D** The disease causes cystic lesions mainly in the liver but may affect any organ.
 - E** The disease causes an infiltrative process in the liver without well-defined margins.

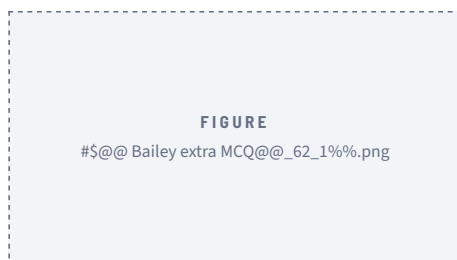
- 15** How is hydatid disease transmitted?
- A By eating infected meat (thus affecting only non-vegetarians).
 - B By the faeco-oral route through ingestion of eggs.
 - C Through skin penetration by larvae in contaminated soil.
 - D It may be transmitted by vectors.
 - E It spreads by droplet infection.

- 16** Which of the following statements regarding cystic hydatid disease are true?
- A It contains an outer pericyst formed by living parasites.
 - B It contains an intermediate ectocyst that is non-infective.
 - C It contains an inner endocyst with a germinal membrane that contains viable parasites.
 - D The germinal membrane may give rise to daughter cysts.
 - E It spreads to the liver by travelling along the intestinal lymphatics.

- 17** Which of the following features are common presentations of hydatid disease?
- A Chest symptoms of dry cough with fluffy exudates (Loeffler's syndrome).
 - B Dull aching pain in the upper right abdomen.
 - C Abdominal pain, anaphylactic shock, and collapse after trivial trauma (due to cyst rupture).
 - D Patients often complain of passing live worms in stool.
 - E Features of obstructive jaundice.

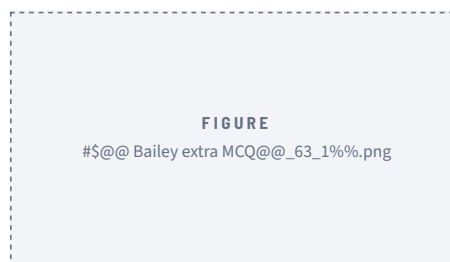
- 18** Which ultrasound features characterize a hepatic hydatid cyst?
- A A cyst with multiple septations.
 - B A cyst wall with shaggy, irregular outlines.
 - C A cyst wall with calcifications.
 - D A solid lesion with a surrounding rim of oedema.
 - E A cyst with split walls (the "double-line" or "split-wall" sign).

- 19** A young girl develops skin rash, upper abdominal pain, and eosinophilia with imaging showing a cyst with split wall. Which statements are true?



- A The findings are compatible with a ruptured simple liver cyst.
- B The features suggest a ruptured hydatid cyst.
- C Treatment is expectant.
- D Treatment is exploratory laparotomy under cover of albendazole.
- E Anaphylaxis may occur during surgical intervention.

- 20** A young woman presents with abdominal pain, jaundice, and imaging showing gallbladder calculi, dilated CBD, septated liver cyst, with ERCP delivering membranous structures. Which statements are true?



- A The likely diagnosis is cholecystitis with choledocholithiasis and an incidental liver cyst.
- B The likely diagnosis is a multiseptated hydatid cyst with biliary communication causing daughter cysts in the CBD and gallbladder calculi.
- C Treatment after endoscopic clearance of the CBD requires only a cholecystectomy.
- D Treatment should involve cholecystectomy followed by injection of sclerosants into the cyst.
- E Injection of scolicidal agents carries a risk of sclerosing cholangitis due to biliary communications.

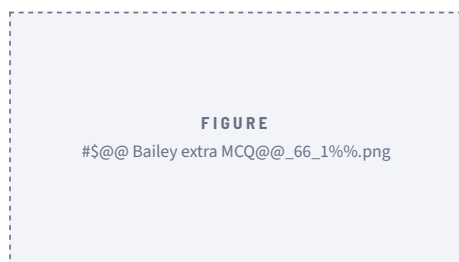
- 21** Which of the following statements are true of hydatid cysts?

- A Anaphylactic shock is more common when treating unilocular cysts than multiloculated cysts.
- B Sclerosing cholangitis is more likely following injection of scolicidal agents into multiloculated cysts.
- C Small deep cysts with calcified walls may be managed non-surgically with albendazole.
- D Surgical curettage of the cyst wall is mandatory.
- E Rupture of a cyst may lead to dissemination of the disease.

- 22** Which of the following is true of thoracic hydatid disease?

- A The lung is the most common organ affected.
- B The right lower lobe is affected slightly more than any other site.
- C The cysts are usually multiple.
- D Asymptomatic ("silent") cysts may sometimes present with an allergic reaction.
- E Pulmonary hydatid disease characteristically presents with a silent pleural effusion.

- 23** A young man develops cough, fever, and chest pain with a chest x-ray showing a rounded cyst and a second lesion with a crescent sign. Which statements are true?



- A** Air enters as the cyst erodes into an adjacent bronchus, creating a fine radiolucent line.
- B** The x-ray finding is known as the “meniscus” or “crescent sign.”
- C** Pulmonary hydatid disease never presents with pleural effusion.
- D** The “water-lily sign” on CT indicates inactive disease.
- E** Surgery remains the mainstay of treatment for pulmonary hydatid cysts.

Answers 1-23

BAILEY & LOVE MCQs

- 1** **A** The falciform ligament of the liver divides it into functional right and left units. — **UPDATED** The functional division of the liver is determined by Cantlie’s line, not the falciform ligament.
- 2** **ABC** True: A, B, C — **UPDATED** Cantlie’s line and the segmentation of the liver are as stated; note that the portal vein forms from the superior mesenteric and splenic veins and hepatic veins do not traverse the porta hepatis.
- 3** **BCD** True: B, C, D — **UPDATED** Normal adult values: bilirubin <17–21 µmol/L, ALP ~30–130 IU/L, AST/ALT <40 IU/L, GGT <50 IU/L; ranges B, C and D lie wholly above normal.
- 4** **ACDE** True: A, C, D, E — **UPDATED** US is used first, CECT is the gold standard, MRI offers high-resolution detail, and angiography is reserved for embolisation; nuclear scans provide functional rather than anatomical data.
- 5** **BE** True: B, E — **UPDATED** Chronic CHF is not the terminal event in chronic liver disease and liver transplantation in acute liver failure carries high short-term risk.
- 6** **D** CECT scan must be carried out in every case of liver trauma. — **UPDATED** False: D — CECT is only for haemodynamically stable patients; unstable patients go straight to laparotomy. The liver is one of the most commonly injured abdominal organs.
- 7** **ABCDE** True: A, B, C, D, E — **UPDATED** All true: unstable or peritonitic penetrating injuries need laparotomy via a long midline incision; most blunt injuries settle conservatively; packing controls crush injury.
- 8** **AD** True: A, D — **UPDATED** Oesophageal varices (not gastric varices) are the most common source of acute haematemesis in portal hypertension, and TIPSS is preferred to surgical shunts.
- 9** **ADE** True: A, D, E — **UPDATED** False: most infected people are asymptomatic (B), and amoebic liver abscess, although the commonest extraintestinal manifestation, occurs in a minority, not >50% (C).
- 10** **BCD** True: B, C, D — **UPDATED** False: excystation occurs in the terminal ileum, with trophozoites colonising the colon (A), and the right lobe is involved in about 80% of cases, not 10% (E).
- 11** **ABDE** True: A, B, D, E — **UPDATED** C is false: amoebic abscess fluid is classically reddish-brown ‘anchovy sauce’, not blackcurrant jelly; trophozoites are found in the wall rather than the pus.
- 12** **E** All of the above. — Amoebiasis may present with colitis, liver abscess, amoeboma and, rarely, rupture with peritonitis.
- 13** **BDE** True: B, D, E — **UPDATED** Metronidazole or tinidazole is first-line, with aspiration for impending rupture and surgery for complications. Laparotomy is not routine (A); diloxanide alone treats only luminal cysts (C).
- 14** **BCD** True: B, C, D — **UPDATED** False: man is an accidental intermediate host, not definitive (A); *E. granulosus* forms well-defined cysts, whereas *E. multilocularis* infiltrates like a tumour (E).
- 15** **B** By the faeco-oral route through ingestion of eggs. — Ingestion of eggs from infected dog faeces is the key mode of transmission.
- 16** **BCD** True: B, C, D — Hydatid cysts have a host-derived pericyst, a non-infective ectocyst, and a germinal layer that produces daughter cysts; spread is via the portal system.
- 17** **BCE** True: B, C, E — **UPDATED** False: Loeffler syndrome (A) and passing live worms (D) are features of ascariasis. Hydatid cysts cause dull RUQ pain, anaphylaxis on rupture and jaundice from biliary rupture.
- 18** **ACE** True: A, C, E — Septations, calcifications and the split-wall sign are typical ultrasound findings.

- 19 **BDE** True: B, D, E — A ruptured hydatid cyst in this setting necessitates urgent surgical management with albendazole cover due to anaphylaxis risk.
- 20 **BE** True: B, E — This presentation is typical of hydatid disease with biliary involvement; caution is needed with scolical agents.
- 21 **BCE** True: B, C, E — Multiloculated cysts risk cholangitis with scolical injection; small calcified cysts may be treated medically and rupture can disseminate the infection.
- 22 **BD** True: B, D — Pulmonary hydatid cysts are often solitary in the right lower lobe and may incite allergic reactions.
- 23 **ABE** True: A, B, E — **UPDATED** False: a ruptured cyst can cause pleural effusion or empyema (C), and the water-lily sign shows collapsed floating membranes of a ruptured cyst, not inactive disease (D).

2 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 55-year-old woman with malaise, fever, chills, anorexia, and RUQ tenderness—plus elevated WBC and mild cholestatic labs—is best evaluated with which imaging?
- A radionuclide gallium scan
 - B abdominal ultrasound
 - C computed tomography (CT) scan
 - D radionuclide technetium-99m sulfur colloid liver scan
 - E magnetic resonance imaging (MRI)
- 2 Multifocal hepatic abscesses are best initially managed with?
- A broad-spectrum antibiotics alone
 - B percutaneous drainage alone
 - C surgical drainage alone
 - D broad-spectrum antibiotics and percutaneous drainage
 - E broad-spectrum antibiotics and surgical drainage

Answers 1-2

LANGE MCQs

- 1 **C** computed tomography (CT) scan — CT is the most sensitive test for hepatic abscess.
- 2 **D** broad-spectrum antibiotics and percutaneous drainage — **UPDATED** Antibiotics plus percutaneous drainage is standard management.

7 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 60-year-old man has completed treatment for hepatitis C. CT scan done for abdominal pain shows a 3-cm mass in the right lobe of the liver with washout on delayed imaging. Alpha-Fetoprotein (AFP) is 100. Laboratory values show albumin 3.8 g/dL, bilirubin 0.9 mg/dL, creatinine 1.1 mg/dL, and platelets 200 mg/dL. What is the next best step in management of this patient?
- A Transarterial chemoembolization
 - B Radiofrequency ablation
 - C Chemotherapy
 - D Wedge resection
 - E Right hepatectomy
- 2 A 60-year-old previously healthy businessman notices that his eyes are yellow and he has been losing weight. On physical examination the patient has jaundice and scleral icterus with a palpable mass in the right upper quadrant. Abdominal ultrasound demonstrates biliary ductal dilation without gallstones. Which of the following is the most appropriate next step in the workup of this patient?
- A Esophagogastroduodenoscopy (EGD)
 - B Endoscopic retrograde cholangiopancreatography (ERCP)
 - C Acute abdominal series
 - D Computed tomography (CT) scan
 - E Positron emission tomography (PET) scan

- 3** A 32-year-old alcoholic man, recently immigrated from Mexico, presents with right upper quadrant pain and fevers for 2 weeks. CT scan of the abdomen demonstrates a non-rim-enhancing fluid collection in the periphery of the right lobe of the liver. The patient's serology is positive for antibodies to *Entamoeba histolytica*. Which of the following is the best initial management option for this patient?

 - A** Treatment with antiamebic drugs
 - B** Percutaneous drainage of the fluid collection
 - C** Marsupialization of the fluid collection
 - D** Surgical drainage of the fluid collection
 - E** Liver resection
- 4** A 43-year-old man without symptoms is incidentally noted on CT scan to have a 4-cm lesion in the periphery of the right lobe of the liver. The lesion enhances on the arterial phase of the CT scan and has a central scar suggestive of focal nodular hyperplasia (FNH). Which of the following is the recommended treatment of this lesion?

 - A** No further treatment is necessary
 - B** Wedge resection of the lesion
 - C** Formal right hepatectomy
 - D** Intra-arterial embolization of the lesion
 - E** Radiofrequency ablation of the liver lesion
- 5** A 45-year-old man presents with right upper quadrant abdominal pain. CT scan shows a large, calcified cystic mass in the right lobe of the liver. He denies any recent travel. Echinococcus is suggested by the CT findings. Which of the following is the most appropriate management of echinococcal liver cysts?

 - A** Percutaneous catheter drainage
 - B** Medical treatment with albendazole
 - C** Medical treatment with steroids
 - D** Medical treatment with metronidazole
 - E** Surgical resection of the cysts
- 6** A 45-year-old woman has an incidental finding of a liver mass on a CT scan done for workup after a motor vehicle collision. Magnetic resonance imaging (MRI) is suggestive of a hemangioma. Which of the following is the most appropriate management strategy for this patient?

 - A** Observation
 - B** Discontinuation of oral contraceptive pills
 - C** Percutaneous biopsy of the lesion to confirm the diagnosis
 - D** Resection of the hemangioma
 - E** Transarterial embolization
- 7** A 30-year-old woman with no significant past medical history and whose only medication is oral contraceptive pills (OCP) presents to the emergency room with right upper quadrant pain. CT scan demonstrates a 6-cm hepatic adenoma in the right lobe of the liver. Which of the following describes the definitive treatment of this lesion?

 - A** Cessation of oral contraceptives and serial CT scans
 - B** Intra-arterial embolization of the hepatic adenoma
 - C** Embolization of the right portal vein
 - D** Resection of the hepatic adenoma
 - E** Systemic chemotherapy

Answers 1-7

PRETEST MCQs

- 1** **D Wedge resection** — **UPDATED** For small hepatocellular carcinomas, a wedge resection achieving negative margins is preferred.
- 2** **D Computed tomography (CT) scan** — A CT scan is the best next step to evaluate suspected pancreatic or biliary neoplasm causing jaundice.
- 3** **A Treatment with antiamebic drugs** — **UPDATED** Amoebic liver abscess responds to metronidazole in >90% of cases; aspiration is reserved for non-response, large left-lobe abscesses or impending rupture.
- 4** **A No further treatment is necessary** — FNH is rarely symptomatic; resection is reserved for symptomatic lesions.
- 5** **E Surgical resection of the cysts** — Surgical resection is the treatment of choice for echinococcal cysts with measures to prevent spillage.
- 6** **A Observation** — Hepatic hemangiomas are benign and typically managed with observation.
- 7** **D Resection of the hepatic adenoma** — **UPDATED** Large hepatic adenomas warrant surgical resection due to risks of hemorrhage and malignant transformation.

3 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 12-year-old boy presents with a 2-month history of malaise and worsening jaundice. His mother notes recent behavioral changes. On examination, the liver is palpable. What is the most likely diagnosis?
 - A Haemochromatosis
 - B Hepatitis A
 - C Infectious mononucleosis
 - D Riedel lobe
 - E Wilson disease
- 2 A 27-year-old man presents with fevers and right upper abdominal pain following a 1-week history of bloody diarrhoea. He has recently returned from holiday in the Maldives, but has previously been fit and well. On examination, he has a tender palpable liver. Which of the following is the most likely cause of his symptoms?
 - A Acute cholecystitis
 - B Amoebic liver abscess
 - C Fitz-Hugh–Curtis syndrome
 - D Polycystic liver
 - E Pyogenic liver abscess
- 3 A 52-year-old man is brought into hospital by his wife with a 2-day history of confusion. On examination, he is jaundiced and has a distended abdomen with dilated veins around his umbilicus. Which of the following is the most likely cause of his jaundice?
 - A Alcoholic liver disease
 - B Autoimmune hepatitis
 - C Hepatitis A
 - D Hepatitis B
 - E Reye syndrome

Answers 1–3

GET AHEAD MCQS

- 1 **E Wilson disease** — Copper accumulation leads to hepatic dysfunction and neurological/psychiatric changes in Wilson disease.
- 2 **B Amoebic liver abscess** — **UPDATED** Bloody diarrhoea after travel followed by fever and a tender liver suggests amoebic liver abscess from *Entamoeba histolytica* colitis.
- 3 **A Alcoholic liver disease** — **UPDATED** Confusion (encephalopathy), jaundice, ascites and caput medusae indicate decompensated cirrhosis, most commonly due to alcohol in the UK.

1 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 70-year-old woman with hepatitis C has 1 week of abdominal discomfort and weight loss. Ultrasound shows an isolated liver lesion. Which additional test is most likely diagnostic?
 - A Serum bilirubin
 - B Serum alkaline phosphatase
 - C INR
 - D Prothrombin time
 - E Serum alpha-fetoprotein

Answer 1

CRASH COURSE MCQS

- 1 **E** Serum alpha-fetoprotein — **UPDATED** Elevated in hepatocellular carcinoma, likely in this scenario.

Gallbladder & Biliary Tree

Gallstones, cholecystitis, cholangitis, obstructive jaundice, bile-duct injury and cholangiocarcinoma.

81

MCQS

0

EMQ SCENARIOS

9

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

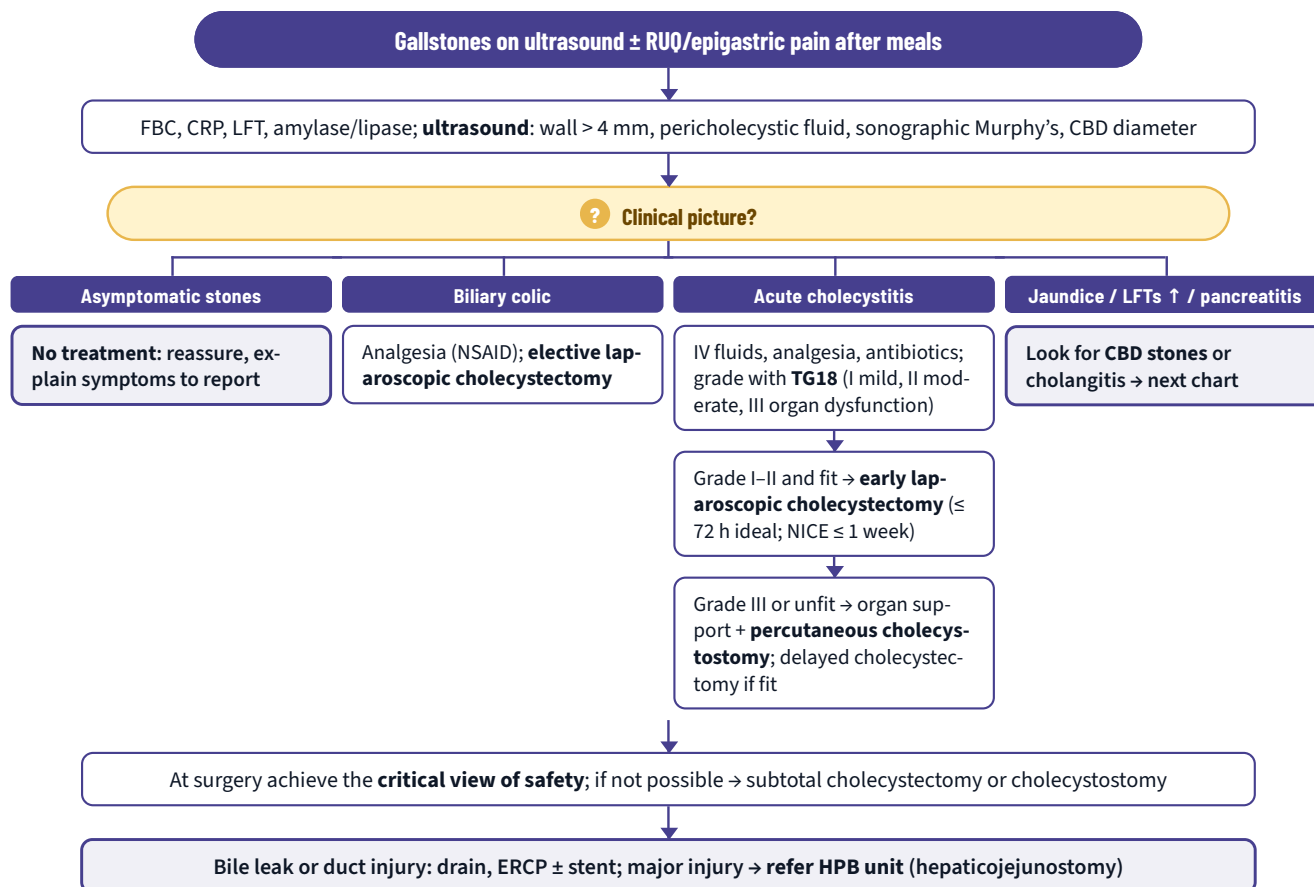
01	Management flowcharts	3 charts	page 294
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Step by step

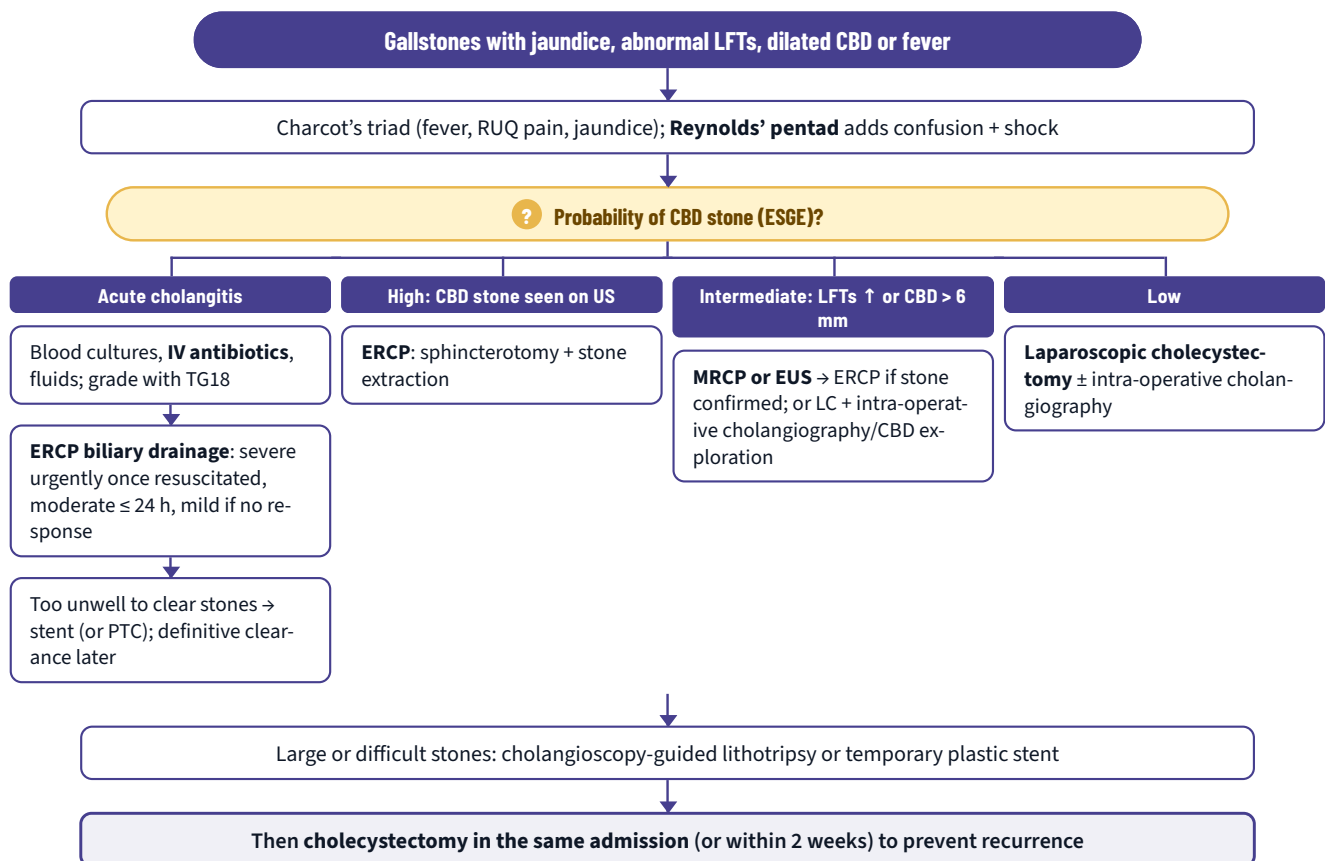
Gallstones: biliary colic and acute cholecystitis

TOKYO GUIDELINES 2018 · NICE CG188



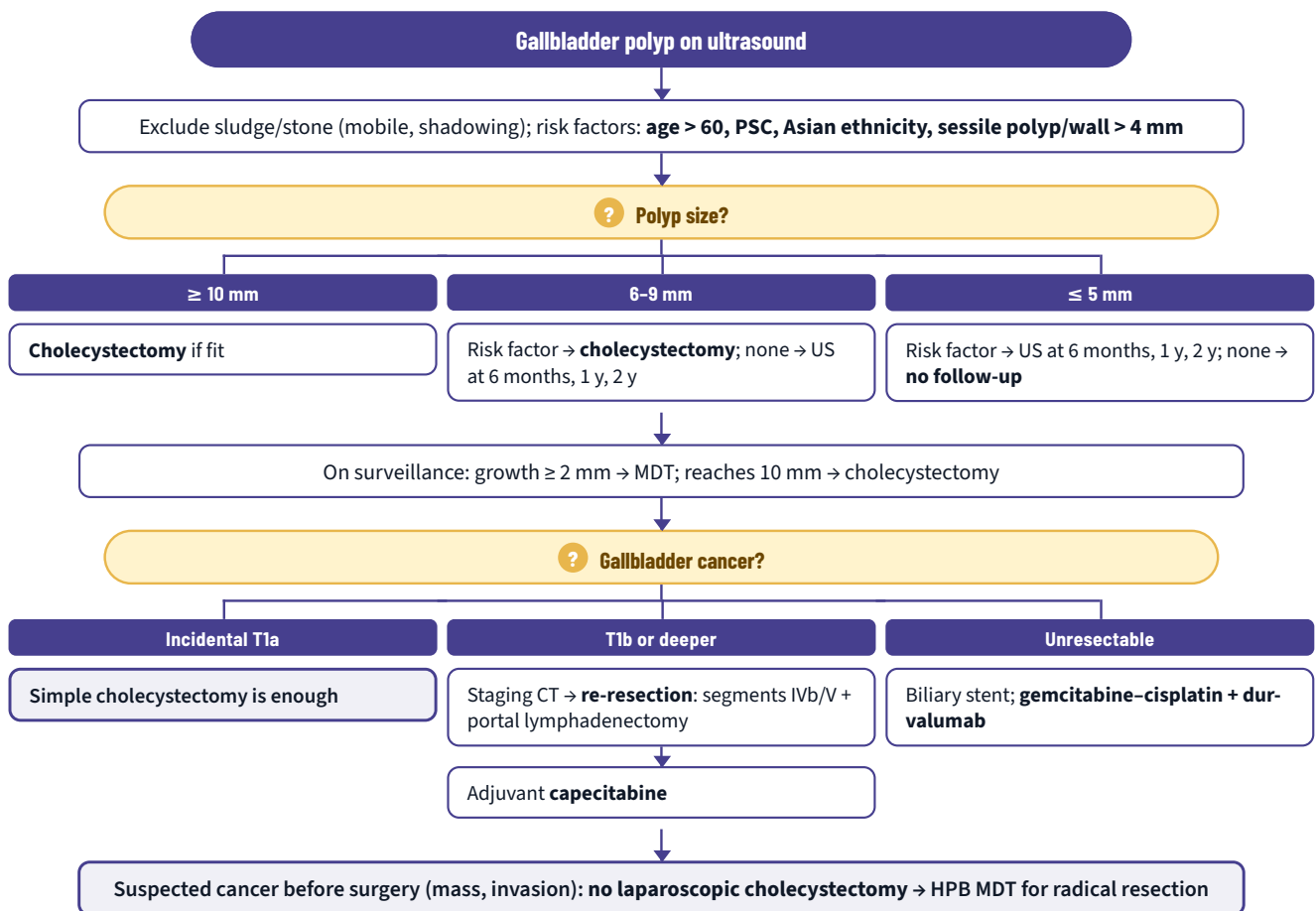
Common bile duct stones and acute cholangitis

ESGE 2019 CBD STONES · BSG 2017 · TOKYO GUIDELINES 2018



Gallbladder polyps and gallbladder cancer

ESGAR/EAES/EFISDS/ESGE 2022 · ESMO 2023 BILIARY CANCER



12 QUESTIONS • 39TH-BATCH BLOCK, INDUCTION & FINAL PAPERS • MINISTERIAL • FORMATIVE & PREVIOUS-YEARS EXAMS – ONE NUMBERING

Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1–5

A man has jaundice, weight loss and a cholestatic pattern (raised ALP and GGT). Ultrasound shows dilated intrahepatic ducts, a collapsed gallbladder and a normal pancreas.

- 1 What is the most likely diagnosis?
 - A Perihilar cholangiocarcinoma
 - B Pancreatic carcinoma
 - C Periampullary cholangiocarcinoma
 - D Hepatocellular carcinoma
 - E Acute pancreatitis
- 2 Which tumour marker is associated?
 - A AFP
 - B CEA
 - C CA 19-9
 - D CA 125
 - E PSA
- 3 What causes the itching?
 - A Bile salts
 - B Bilirubin
 - C Urea
 - D Histamine
 - E Ammonia
- 4 What is the best non-invasive investigation for diagnosis?
 - A ERCP
 - B CT scan
 - C MRCP
 - D HIDA scan
 - E Plain abdominal X-ray
- 5 The patient remains jaundiced after two ERCP attempts. What is the next step?
 - A Laparoscopic bypass
 - B Percutaneous transhepatic cholangiography and drainage
 - C Repeat ERCP
 - D Liver transplantation
 - E Observation
- 6 Which clinical triad defines ascending cholangitis?
 - A Fever with rigors, jaundice and right upper quadrant pain
 - B Jaundice, palpable gallbladder and weight loss
 - C Fever, jaundice and pruritus
 - D Right upper quadrant pain, vomiting and positive Murphy sign
 - E Jaundice, hypotension and confusion

- 7 An MRCP of a jaundiced patient is shown. What is the most likely diagnosis on this image?



- A Choledocholithiasis
- B Hilar cholangiocarcinoma
- C Choledochal cyst
- D Primary sclerosing cholangitis
- E Normal biliary tree

CASE • QUESTIONS 8–11

A 39-year-old woman with asymptomatic gallstones for 4 years has severe upper abdominal pain for 24 hours with nausea and vomiting, still painful despite NSAIDs and opiates. LFTs are normal.

- 8 All of the following are expected EXCEPT:
 - A Normal CBD
 - B Thickened gallbladder wall
 - C Leucopenia
 - D Hyperaesthesia below the right scapula (Boas sign)
 - E Raised CRP
- 9 What is the best treatment option now?
 - A Open cholecystectomy
 - B Laparoscopic cholecystectomy
 - C Tube cholecystostomy
 - D ERCP followed by surgery
 - E Conservative treatment, then surgery after 6 weeks
- 10 Five days after a difficult operation the drain shows 800 mL/day of bile-stained fluid. Which is NOT a likely cause?
 - A Slipped cystic duct clip
 - B Bile duct injury
 - C Leak from a duct of Luschka
 - D Duodenal injury
 - E Intrahepatic cholestasis

- 11** What is the best treatment now?
- A ERCP (sphincterotomy and stent)
 - B PTC
 - C IV antibiotics and anticholinergics
 - D Right hepatic lobectomy
 - E Liver transplantation

FORMATIVE

College formative examinations

- 12** Four weeks after a laparoscopic cholecystectomy, a patient's abdominal drain is still draining 900 mL of bile daily. What is the best minimally invasive treatment?
- A ERCP with sphincterotomy and stent placement
 - B Laparoscopic cholecystectomy
 - C Open drainage
 - D Antibiotics only
 - E Percutaneous drainage

Answers 1-12

LOCAL EXAMS MCQS

- 1** **A Perihilar cholangiocarcinoma** — Dilated intrahepatic ducts with a collapsed gallbladder put the block above the cystic duct: hilar (Klatskin) cholangiocarcinoma.
- 2** **C CA 19-9** — CA 19-9 is the marker for cholangiocarcinoma and pancreatic cancer (it is also raised by cholestasis itself).
- 3** **A Bile salts** — Pruritus in obstructive jaundice is due to retained bile salts (and other pruritogens) in the skin, not bilirubin.
- 4** **C MRCP** — MRCP maps the biliary tree non-invasively and defines the level and extent of a hilar stricture before any drainage.
- 5** **B Percutaneous transhepatic cholangiography and drainage** — Hilar strictures that fail endoscopic stenting are drained percutaneously (PTC/PTBD).
- 6** **A Fever with rigors, jaundice and right upper quadrant pain** — Charcot triad is fever/rigors, jaundice and RUQ pain; adding hypotension and confusion gives Reynolds pentad (suppurative cholangitis).
- 7** **A Choledocholithiasis** — Filling defects within a dilated common bile duct on MRCP are CBD stones.
- 8** **C Leucopenia** — Acute cholecystitis gives a thick-walled gallbladder, raised CRP and leucocytosis (not leucopenia); the CBD is normal.
- 9** **B Laparoscopic cholecystectomy** — Early laparoscopic cholecystectomy (within 72 h–1 week) is recommended for acute cholecystitis in fit patients (NICE).
- 10** **E Intrahepatic cholestasis** — Bile in the drain means a leak (cystic stump, duct of Luschka, bile-duct or duodenal injury); cholestasis does not cause bile leakage.
- 11** **A ERCP (sphincterotomy and stent)** — A post-cholecystectomy bile leak with a drain in place is treated by ERCP with sphincterotomy and stent; major duct injury needs MRCP and repair.
- 12** **A ERCP with sphincterotomy and stent placement** — **UPDATED** ERCP with sphincterotomy and stenting removes the sphincter pressure gradient so bile drains into the duodenum, allowing a cystic stump leak to seal.

8 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

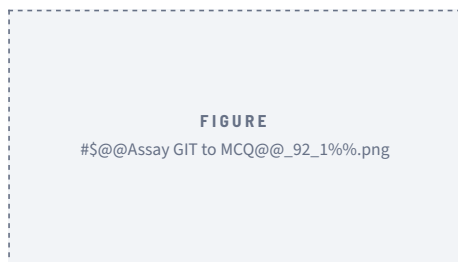
- 1 A diabetic woman has small gallstones found incidentally on ultrasound. She has never had biliary pain. Is surgery needed for her gallbladder?
 - A No, surgery is not indicated because she is asymptomatic regarding gallstones.
 - B Yes, as she is diabetic (indication in asymptomatic gallstone disease).
 - C No, because the gallstones are small and less than 7mm.
 - D Yes, because any gallstones in diabetic patients always require cholecystectomy.
 - E Yes, because all gallstones should be removed once detected
- 2 A 51-year-old male smoker presents with upper abdominal discomfort and heaviness over the past few weeks, along with a substernal burning sensation. An ultrasound shows three small stones <9 mm with normal wall thickness. He has had a partial response to antacids and paracetamol. The surgeon in the outpatient clinic recommends laparoscopic cholecystectomy. How would you counsel this patient?
 - A Explain that laparoscopic cholecystectomy is a major open surgery requiring a large incision and long recovery
 - B Advise that the procedure is minimally invasive, involves small incisions with a laparoscope, will relieve his symptoms, and prevents complications like pancreatitis and cholangitis
 - C Tell him that his symptoms are likely due to reflux and the surgery is unnecessary
 - D Recommend waiting for spontaneous passage of stones
 - E Inform him that only medication is required to manage the condition
- 3 A patient with gallstones refuses cholecystectomy. He requests a medical therapy that would reduce any future increase in the size and number of stones. Suggest a suitable option.
 - A Ursodeoxycholic acid
 - B Antibiotic prophylaxis
 - C Proton pump inhibitors
 - D Corticosteroids
 - E Bile acid sequestrants
- 4 A patient with gallstones who declined cholecystectomy develops itching two years later. The imaging shown is performed. What is the study and what are the findings?

FIGURE

#\$\$@@Assay GIT to MCQ@@_9_1%%.png

 - A CT scan revealing a normal common bile duct
 - B Ultrasound showing a dilated common bile duct and the presence of stones, confirmed by the acoustic shadow
 - C MRCP indicating no bile duct stones
 - D Plain X-ray showing calcified gallstones
 - E Endoscopic ultrasound demonstrating a pancreatic mass
- 5 Ultrasound suggests stones in a dilated common bile duct. How can you confirm these findings non-invasively?
 - A Repeat the ultrasound
 - B Perform an Endoscopic Retrograde Cholangiopancreatography (ERCP)
 - C Conduct an MRCP
 - D Use a CT scan with contrast
 - E Obtain a plain abdominal X-ray
- 6 MRCP confirms common bile duct stones in a jaundiced patient whose gallbladder is still in situ. What is the best treatment now?
 - A Laparoscopic cholecystectomy
 - B Open surgical exploration
 - C Medical management with ursodeoxycholic acid
 - D ERCP with sphincterotomy and stone extraction
 - E Conservative management with analgesics

- 7** Seven days after abdominal surgery, a patient develops high-grade fever, sweating and right upper quadrant pain. A CT scan is shown. What are the radiological findings?



- A Normal postoperative changes
- B Pneumonia
- C Abdominal CT scan showing a hypodense area (collection of fluid) below the right hemidiaphragm, consistent with a subphrenic abscess
- D Liver abscess
- E Bowel perforation

- 8** What is the treatment of a postoperative subphrenic abscess?

- A Oral antibiotics only
- B Conservative management without drainage
- C Antibiotics plus drainage (image-guided percutaneous, or surgical if this fails)
- D Immediate reoperation
- E Steroid therapy

Answers 1-8

ESSAY TO MCQ MCQS

- 1** **A** No, surgery is not indicated because she is asymptomatic regarding gallstones. — **UPDATED** Asymptomatic gallstones do not need cholecystectomy (NICE), and diabetes is no longer considered an indication; surgery is offered once symptoms develop.
- 2** **B** Advise that the procedure is minimally invasive, involves small incisions with a laparoscope, will relieve his symptoms, and prevents complications like pancreatitis and cholangitis — **UPDATED** Laparoscopic cholecystectomy is minimally invasive and prevents future complications.
- 3** **A** Ursodeoxycholic acid — **UPDATED** Ursodeoxycholic acid lowers biliary cholesterol saturation and can dissolve small radiolucent cholesterol stones in a functioning gallbladder; recurrence is common.
- 4** **B** Ultrasound showing a dilated common bile duct and the presence of stones, confirmed by the acoustic shadow —

UPDATED Ultrasound reveals CBD dilation with stones indicated by acoustic shadow.

- 5** **C** Conduct an MRCP — **UPDATED** MRCP is a non-invasive confirmatory test for bile duct stones.
- 6** **D** ERCP with sphincterotomy and stone extraction — **UPDATED** Symptomatic CBD stones are cleared by ERCP with sphincterotomy and extraction, followed by laparoscopic cholecystectomy to prevent recurrence.
- 7** **C** Abdominal CT scan showing a hypodense area (collection of fluid) below the right hemidiaphragm, consistent with a subphrenic abscess — **UPDATED** The CT findings are indicative of a subphrenic abscess.
- 8** **C** Antibiotics plus drainage (image-guided percutaneous, or surgical if this fails) — **UPDATED** A subphrenic abscess needs antibiotics plus drainage, ideally image-guided percutaneous; open drainage is reserved for failure or multiloculated collections.

Bailey & Love MCQs

- 1 Which of the following statements pertain to Asiatic cholangiohepatitis?
 - A The human acts as the intermediate host, with infected fish and snails being the definitive hosts.
 - B The parasite matures into an adult in the intrahepatic bile ducts, causing epithelial hyperplasia and periductal fibrosis.
 - C Eggs or dead worms may form the nidus for stone formation in the biliary system.
 - D Changes around the intrahepatic biliary channels may lead to dysplasia and cholangiocarcinoma.
 - E Ultrasound typically shows uniform dilatation of small peripheral intrahepatic ducts with minimal common bile duct dilatation.
- 2 What is the treatment of Asiatic cholangiohepatitis?
 - A Praziquantel and albendazole are the drugs of choice.
 - B Cholecystectomy, common bile duct exploration, and choledochoduodenostomy.
 - C Roux-en-Y choledochojejunostomy (with or without an access loop).
 - D The disease should ideally be diagnosed and treated in its subclinical form.
 - E All of the above.
- 3 In which of the following should antibiotic prophylaxis not be considered?
 - A Prosthetic heart valves.
 - B Previous history of endocarditis.
 - C Severe neutropenia.
 - D Chronic liver disease undergoing variceal sclerotherapy.
 - E Previous cholecystectomy.
- 4 Which of the following are not risk factors for post-ERCP pancreatitis?
 - A Young age.
 - B Difficult cannulation.
 - C Increased bilirubin.
 - D Pancreatic sphincterotomy.
 - E Balloon dilatation of biliary sphincter.
- 5 Anatomy and Physiology. Which of the following statements are true?
 - A The common bile duct (CBD) is traditionally divided into three parts.
 - B The cystic duct opens into the supraduodenal part of the common hepatic duct (CHD) in 80% of cases.
 - C The cystic duct junction with the CHD might vary.
 - D The cystic artery arises from the common hepatic artery.
 - E The cystic artery runs across Calot's triangle.
- 6 Which of the following statements are true regarding lymphatic drainage?
 - A The sentinel lymph node is the cystic lymph node of Lund.
 - B The cystic lymph node lies in the fork created by the cystic duct and the right hepatic duct.
 - C Efferent vessels from the cystic lymph node go to the superior mesenteric nodes.
 - D The lymphatics of the gall bladder also connect directly with the lymphatics of the liver.
 - E The lymphatic vessels are mainly in the sub-mucosal and sub-serosal layers of the wall.
- 7 Which of the following statements are true?
 - A Bile consists of 97% water.
 - B About 95% of bile salts are reabsorbed.
 - C During fasting, bile is diverted into the gall bladder.
 - D Gall bladder concentrates the bile.
 - E Gall bladder secretes mucus.
- 8 Investigations of the biliary tract. The following statements are true except:
 - A Ultrasound (US) examination shows up stones and CBD dilatation.
 - B HIDA scan is useful in diagnosing bile leak.
 - C Plain x-ray is useful to look for calcification, which is seen in 80% of gallstones.
 - D Endoscopic retrograde cholangiopancreatography (ERCP) is useful in showing up anatomy, stones and strictures.
 - E Percutaneous transhepatic cholangiogram (PTC) is ideal in high strictures.
- 9 Investigations of the biliary tract. Which of the following statements are true?
 - A Liver function tests should be routinely carried out.
 - B Magnetic resonance cholangiopancreatography (MRCP) images are comparable to images by ERCP and PTC.
 - C Peroperative cholangiography is routinely done in cholecystectomy.
 - D Choledochoscopy should be done in the appropriate patient to ensure complete removal of stones from CBD.
 - E Endoscopic and laparoscopic US are accurate techniques in staging cancers.
- 10 Extra-Hepatic Biliary Atresia. Which of the following statements are true?
 - A Associated anomalies include situs inversus.
 - B Jaundice is a late feature.
 - C Type 1 anomaly involves the common hepatic duct.
 - D Kasai procedure is best done 3 to 6 months after birth.
 - E 70%–80% patients are alive 2 to 5 years after liver transplant.

- 11** Caroli's Disease. Which of the following statements are true?
- A It involves the intra-hepatic ducts.
 - B It is a hereditary condition.
 - C The peri-portal fibrosis type presents later in life.
 - D It can be associated with cholangiocarcinoma.
 - E The mainstay of treatments is antibiotics and removal of calculi.
- 12** Choledochal Cyst. Which of the following statements are true?
- A This only involves the extra-hepatic duct.
 - B 60% of cases are diagnosed before the age of 10 years.
 - C There is an increasing risk of developing cholangiocarcinoma.
 - D MRCP is useful in the definition of the anatomy.
 - E Cholecystectomy should be a part of the surgical treatment.
- 13** Gallstones. The following statements are true except:
- A Cholecystectomy should be performed whenever gallstones are found.
 - B Types of gallstones are cholesterol, pigment and mixed.
 - C Gallstones can cause various complications including intestinal obstruction.
 - D Acute cholecystitis can occur without stones.
 - E In emergency presentation from biliary colic or acute cholecystitis, cholecystectomy can be performed at the index admission.
- 14** Complications of Cholecystectomy. Which of the following statements are true?
- A Complications can occur in 10%–15% of cases.
 - B Bile duct injury is usually recognised at the time of the operation.
 - C Post-cholecystectomy syndrome occurs in 15% of patients.
 - D Stones in the bile ducts are less often associated with infected bile than are stones in the gall bladder.
 - E Jaundice in the postoperative period is usually self-limiting and needs investigating only if it persists for more than 1 week.
- 15** Primary Sclerosing Cholangitis (PSC). Which of the following statements are true?
- A It is more common in females.
 - B It involves both intra and extra biliary ducts.
 - C It usually starts in childhood.
 - D Liver biopsy is not useful.
 - E Liver transplantation is an option if cirrhosis is not present.
- 16** Parasitic Infestation of the Biliary Tract. Which of the following statements are true?
- A The roundworm (*Ascaris lumbricoides*) enters the biliary tract through the bloodstream.
 - B Clonorchiasis is endemic in Africa.
 - C Obstructive jaundice and cholangitis are common presentations of hydatid disease.
 - D Complications include bile duct carcinoma.
 - E Recurrent stones can be a problem.

Answers 1–16

BAILEY & LOVE MCQs

- 1** **BCDE** True: B, C, D, E — *Clonorchis sinensis* causes intrahepatic duct changes leading to stone formation and possible cholangiocarcinoma; humans are definitive hosts.
- 2** **E** All of the above. — Management includes both medical and various surgical interventions with early diagnosis preferred.
- 3** **E** Previous cholecystectomy. — **UPDATED** Prophylaxis before endoscopic procedures is considered for prosthetic valves, previous endocarditis, severe neutropenia and cirrhosis with variceal therapy, not previous cholecystectomy.
- 4** **C** Increased bilirubin. — While young age, difficult cannulation, pancreatic sphincterotomy and balloon dilation increase risk, increased bilirubin may be protective.
- 5** **BCE** True: B, C, E — **UPDATED** False: the CBD is traditionally described in four parts (A), and the cystic artery usually arises from the right hepatic artery, not the common hepatic artery (D).
- 6** **ADE** True: A, D, E — **UPDATED** False: Lund's node lies in the fork between the cystic and common hepatic ducts (B), and its efferents drain to hepatic and coeliac nodes, not superior mesenteric (C).
- 7** **ABCDE** True: A, B, C, D, E — **UPDATED** All are true: bile is ~97% water, ~95% of bile salts return via the enterohepatic circulation, and the gallbladder stores and concentrates bile and secretes mucus.
- 8** **C** Plain x-ray is useful to look for calcification, which is seen in 80% of gallstones. — Only about 10% of gallstones are radiopaque, making plain x-ray of limited value.
- 9** **ABDE** True: A, B, D, E — **UPDATED** C is false: peroperative cholangiography is used selectively, not routinely, in cholecystectomy; the other statements are true.
- 10** **AE** True: A, E — A and E are correct; jaundice is an early feature, Type 1 involves the common bile duct and Kasai is best performed before 8 weeks.
- 11** **ABDE** True: A, B, D, E — C is false; the periportal fibrosis type actually presents early.
- 12** **BCDE** True: B, C, D, E — A is false; choledochal cysts often involve both intra- and extra-hepatic ducts.
- 13** **A** Cholecystectomy should be performed whenever gallstones are found. — Asymptomatic gallstones usually are managed conservatively.
- 14** **AC** True: A, C — Only A and C are correct; bile duct injuries are frequently missed, stones in the ducts are more often infected, and any postoperative jaundice warrants investigation.
- 15** **B** It involves both intra and extra biliary ducts. — Only B is correct; PSC is more common in males, typically presents in

adulthood, liver biopsy is useful for staging, and transplantation is indicated in advanced disease.

- 16 **CDE** True: C, D, E — A and B are false; entry is via the ampulla and Clonorchiasis is endemic in the Far East.

2 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 7-week-old, breast-fed, term infant presents with increasing jaundice, abdominal distention, and acholic stools. Labs show conjugated hyperbilirubinemia and elevated GGT. Which is the most sensitive imaging study in this setting?



- A Radioisotope scanning
- B Radioisotope scanning with preimaging phenobarbital administration
- C Abdominal ultrasound
- D CT scan of the abdomen
- E MRI scan of the abdomen

- 2 During an elective laparoscopic cholecystectomy, the anesthesiologist reports that the patient has developed a sudden drop in systolic blood pressure, arterial desaturation, and an increase in ventilatory pressure. Which of the following is the most appropriate step in management?

- A an IV fluid bolus
- B decompression of the pneumoperitoneum
- C insertion of a chest tube
- D re-evaluating the position of the endotracheal tube and obtaining a portable chest x-ray
- E aborting the procedure and converting to an open cholecystectomy

Answers 1-2

- 1 **B** Radioisotope scanning with preimaging phenobarbital administration — **UPDATED** Phenobarbital priming enhances hepatic excretion, so absent bowel excretion on HIDA scanning is highly sensitive for biliary atresia.
- 2 **B** decompression of the pneumoperitoneum — **UPDATED** Sudden cardiorespiratory collapse during laparoscopy (raised IAP, capnothorax, CO₂ embolism) is first managed by releasing the pneumoperitoneum, then reassessing.

11 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 A 48-year-old woman is admitted with severe epigastric pain and vomiting. The pain is continuous and worse on movement. On examination she is lying still, taking shallow breaths and sweating. There is marked tenderness in the epigastrium and right upper quadrant, Murphy's sign is positive and she is slightly pyrexial. You suspect acute cholecystitis. What is the next best step in managing this patient?
- A Keep nil by mouth, administer parenteral analgesia and systemic antibiotics
 - B Send patient for emergency laparoscopic cholecystectomy
 - C Request a plain film abdominal radiograph
 - D Administer analgesia and seek the opinion of a superior colleague
 - E Request an ultrasound of the abdomen
- 2 A 50-year-old woman presents with an acute episode of epigastric pain, vomiting and fever. The registrar, who has already clerked and examined the patient, tells you that Murphy's sign is positive. Without having seen the patient, what is the most likely diagnosis?
- A Acute appendicitis
 - B Acute cholecystitis
 - C Peritonitis
 - D Biliary colic
 - E Cholangitis

- 3 You are asked to see a 48-year-old woman admitted with sudden onset of right upper quadrant pain. Your registrar, having seen her, tells you that the patient has Charcot's triad. What is the most likely diagnosis?
 - A Biliary colic
 - B Acute cholecystitis
 - C Cholangitis
 - D Gallstone ileus
 - E Pancreatitis
- 4 A 44-year-old woman presents with acute onset of right upper quadrant pain and fever. On examination, she is lying still and has a tachycardia. The abdomen is tender in the right upper quadrant with guarding in that area. Murphy's sign is positive. What is the most useful investigation for this patient?
 - A Ultrasound
 - B Colonoscopy
 - C Barium follow-through
 - D Upper gastrointestinal endoscopy
 - E Serum amylase
- 5 The epiploic foramen of Winslow is an important anatomical landmark as it is the only communication between the greater and the lesser sac. Of the five structures listed below, which one is not a border of the epiploic foramen of Winslow?
 - A Inferior vena cava
 - B First part of the duodenum
 - C Hepatoduodenal ligament
 - D Gallbladder
 - E Caudate lobe of the liver
- 6 Which of the following is not encountered when making Kocher's (right subcostal) incision during an open cholecystectomy?
 - A Anterior rectus sheath
 - B Superior epigastric vessels
 - C Eighth intercostal nerve
 - D Falciform ligament
 - E External oblique
- 7 Which one of the following statements regarding gallstone disease is incorrect?
 - A Gallstones occur in 15% of all people over 65 in Western countries
 - B 85% of gallstones are asymptomatic
 - C 10% are identifiable using plain film radiography
 - D Gallstones are more common in males
 - E Gallstone disease is the most common cause of acute pancreatitis
- 8 Which of the following is not a risk factor for gallstone formation?
 - A Smoking
 - B Pregnancy
 - C Crohn's disease
 - D Diet high in fats
 - E Contraceptive pill
- 9 A 29-year-old patient with a short history of right upper quadrant pain, jaundice, dark urine, pale stool and a fever of 38.9 °C, with no palpable gallbladder, is most likely suffering from
 - A Ascending cholangitis
 - B Acute cholecystitis
 - C Biliary colic with duct obstruction
 - D Pancreatitis
 - E Mirizzi's syndrome
- 10 A 63-year-old patient with intermittent, colicky right upper quadrant pain and jaundice, a thickened gallbladder on ultrasound, rising bilirubin and inflammatory markers, and persistent pain with fever is best managed by
 - A Cholecystectomy
 - B Endoscopic retrograde cholangiopancreatography
 - C Open stone removal with T-tube drainage
 - D Magnetic resonance cholangiopancreatography
 - E Lithotripsy
- 11 Which of the following is not a risk factor for the development of cholangiocarcinoma?
 - A Primary sclerosing cholangitis
 - B Choledochal cysts
 - C Hepatitis B/C infection
 - D Primary biliary cirrhosis
 - E Lynch syndrome II

Answers 1-11

SBA MCQS

- 1 **A Keep nil by mouth, administer parenteral analgesia and systemic antibiotics** — **UPDATED** Suspected acute cholecystitis is managed first with nil by mouth, IV fluids, analgesia and antibiotics; ultrasound then confirms it before early cholecystectomy.
- 2 **B Acute cholecystitis** — **UPDATED** A positive Murphy's sign is highly suggestive of acute cholecystitis.
- 3 **C Cholangitis** — **UPDATED** Charcot's triad is indicative of ascending cholangitis.
- 4 **A Ultrasound** — **UPDATED** Ultrasound is the most useful investigation for suspected acute cholecystitis.
- 5 **D Gallbladder** — Of the above options, the gallbladder is the only structure that does not border the foramen of Winslow.
- 6 **D Falciform ligament** — The falciform ligament is not encountered during Kocher's incision.
- 7 **D Gallstones are more common in males** — Gallstones are more common in females.

- 8 **A Smoking** — Smoking is not known to be a risk factor for gallstone formation.
- 9 **A Ascending cholangitis** — The combination of jaundice, fever, and RUQ pain (Charcot's triad) indicates ascending cholangitis.

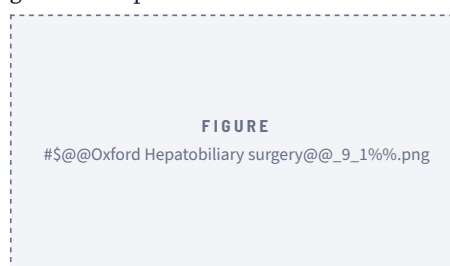
- 10 **B Endoscopic retrograde cholangiopancreatography** — ERCP is indicated to decompress the biliary tree in complicated gallstone cases.
- 11 **D Primary biliary cirrhosis** — Primary biliary cirrhosis is not associated with cholangiocarcinoma.

4 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

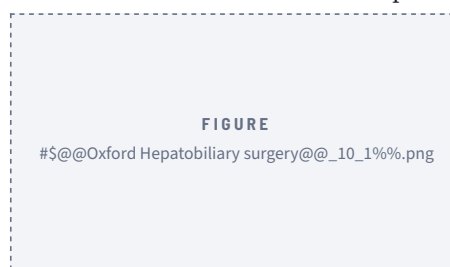
- 1 A 42-year-old woman attends the emergency department with severe epigastric pain after eating fish and chips. The pain resolves after 2h. Her temperature is 37.2°C. She has no abdominal tenderness and all blood investigations are within the normal range. Which is the single most likely diagnosis?
- A Biliary colic
B Cholangitis
C Cholecystitis
D Pancreatitis
E Perforated gallbladder
- 2 An 82-year-old man has a diagnosis of inoperable cancer of the head of the pancreas. He has developed jaundice over the last 5 days and is symptomatic with pruritus. An abdominal ultrasound scan confirms dilated common and intrahepatic bile ducts. Which is the single most suitable means of palliation for his jaundice?
- A Chlorpheniramine PO
B Choledochojunostomy
C Endoscopic retrograde cholangiopancreatography (ERCP) and internal stent
D Naloxone IV
E Percutaneous transhepatic external drain

- 3 A 23-year-old woman undergoes a laparoscopic cholecystectomy for recurrent biliary colic. On examination of the gallbladder there are stones present (see figure). Which is the single most likely underlying diagnosis to explain her condition?



- A Crigler-Najjar syndrome
B Gilbert's syndrome
C Hereditary spherocytosis
D Hypercholesterolaemia
E Pregnancy

- 4 A 47-year-old man has had malaise and anorexia for 2 weeks and has developed jaundice over 3 days. Serum testing confirms hepatitis A infection. Blood tests are taken to assess the cause of his jaundice (normal ranges are shown). Which is the single most likely combination of blood results in this patient?



- A ALP 270IU/L, ALT 27IU/L, AST 37IU/L
B ALP 330IU/L, ALT 105IU/L, CK 350IU/L
C ALP 105IU/L, ALT 75IU/L, AST 60IU/L
D ALP 450IU/L, ALT 55IU/L, γGT 120IU/L
E ALT 130IU/L, AST 25IU/L, LDH 135IU/L

Answers 1-4

- 1 **A Biliary colic** — Biliary colic is severe epigastric pain caused by gallstones that typically resolves after some time.
- 2 **C Endoscopic retrograde cholangiopancreatography (ERCP) and internal stent** — Stent placement via ERCP is effective for palliation of jaundice.

3 **C** **Hereditary spherocytosis** — **UPDATED** Chronic haemolysis from hereditary spherocytosis leads to pigment gallstones.

4 **C** **ALP 105IU/L, ALT 75IU/L, AST 60IU/L** — **UPDATED** ALT and AST elevated without significant ALP elevation is consistent with hepatitis A.

13 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1** A 60-year-old female undergoes a laparoscopic cholecystectomy for symptoms of biliary colic. Final pathology reveals a T2b adenocarcinoma, consistent with gallbladder cancer. After a complete staging workup with CT scans and labs (including tumor markers), what is the next most appropriate step?
A Observation
B Neoadjuvant chemotherapy followed by surgery
C Hepatectomy (segment IVB and V) and portal lymph node dissection
D Hepatectomy (segment IVB and V) and resection of laparoscopic port sites
E Chemotherapy only
- 2** A 29-year-old woman complains of postprandial right upper quadrant pain and fatty food intolerance. Ultrasound examination reveals no evidence of gallstones or sludge. Upper endoscopy is normal, and all of her liver function tests are within normal limits. Which of the following represents the best management option?
A Avoidance of fatty foods and reexamination in 6 months
B Repeat ultrasound immediately
C Treatment with ursodeoxycholic acid
D CCK-HIDA scan should be performed to evaluate for biliary dyskinesia
E Laparoscopic cholecystectomy for acalculous cholecystitis
- 3** A 47-year-old asymptomatic woman is incidentally found to have a 5-mm polyp and no stones in her gallbladder on ultrasound examination. Which of the following is the best management option?
A Aspiration of the gallbladder with cytologic examination of the bile
B Observation with repeat ultrasound examinations to evaluate for increase in polyp size
C Laparoscopic cholecystectomy
D Open cholecystectomy with frozen section
E En bloc resection of the gallbladder, wedge resection of the liver, and portal lymphadenectomy

- 4** A 32-year-old woman undergoes a cholecystectomy for acute cholecystitis and is discharged home on the second postoperative day. She returns to the clinic 8 months after the operation with complaints of “yellow eyes.” Laboratory values show total bilirubin 5.6 mg/dL, direct bilirubin 4.8 mg/dL, and alkaline phosphatase 250 IU. An ultrasound shows dilated intrahepatic ducts. The patient undergoes the transhepatic cholangiogram seen here. Which of the following is the most appropriate next management step?



- A** Choledochotomy with insertion of a T tube
B End-to-end choledochocolic anastomosis
C Roux-en-Y hepaticojejunostomy
D Percutaneous transhepatic dilatation
E Choledochoduodenostomy
- 5** A 48-year-old man presents with jaundice, melena, and right upper quadrant abdominal pain after undergoing a percutaneous liver biopsy. Endoscopy shows blood coming from the ampulla of Vater. Which of the following is the most appropriate first line of therapy for major hemobilia?
A Correction of coagulopathy, adequate biliary drainage, and close monitoring
B Transarterial embolization (TAE)
C Percutaneous transhepatic biliary drainage (PTBD)
D Ligation of bleeding vessels
E Hepatic resection

- 6** A 35-year-old woman presents with abdominal pain and jaundice. An ERCP reveals the congenital cystic anomaly of her biliary system. Which of the following is the most appropriate treatment?



- A** Cholecystectomy with resection of the extrahepatic biliary tract and Roux-en-Y hepaticojejunostomy
B Internal drainage via choledochoduodenostomy
C Internal drainage via choledochocystojejunostomy
D Percutaneous transhepatic biliary drainage
E Liver transplantation
- 7** A 35-year-old man presents with right upper quadrant pain, fever, jaundice, and shaking chills. Ultrasound of the abdomen demonstrates gallstones, normal gallbladder wall thickness, and a common bile duct diameter of 1.0 cm. His total bilirubin is 5. In addition to IV fluids and antibiotics, which of the following is the most appropriate next step in this patient's management?
- A** Endoscopic retrograde cholangiopancreatography (ERCP)
B Placement of a cholecystostomy tube
C Laparoscopic cholecystectomy
D Open cholecystectomy
E Emergent operation and decompression of the common bile duct with a T tube
- 8** An 88-year-old man with a history of end-stage renal failure, severe coronary artery disease, and brain metastases from lung cancer presents with acute cholecystitis. His family wants "everything done." Which of the following is the best management option in this patient?
- A** Tube cholecystostomy
B Open cholecystectomy
C Laparoscopic cholecystectomy
D Intravenous antibiotics followed by elective cholecystectomy
E Lithotripsy followed by long-term bile acid therapy

- 9** An ultrasound is performed on a patient with right upper quadrant pain. It demonstrates a large gallstone in the cystic duct but also a polypoid mass in the fundus. Which of the following is an indication for cholecystectomy for a polypoid gallbladder lesion?

- A** Size greater than 0.5 cm
B Presence of clinical symptoms
C Patient age of older than 25 years
D Presence of multiple small lesions
E Absence of shadowing on ultrasound

- 10** A 62-year-old man presents with right upper quadrant abdominal pain and jaundice. He is afebrile with normal vital signs. Blood tests show elevated bilirubin and alkaline phosphatase. Ultrasound demonstrates gallstones, normal gallbladder wall thickness, no pericholecystic fluid and a common bile duct of 1.0 cm. What is the most likely diagnosis?

- A** Symptomatic cholelithiasis
B Acute cholecystitis
C Gallstone pancreatitis
D Choledocholithiasis
E Cholangitis

- 11** A 36-year-old woman presents with right upper quadrant abdominal pain and jaundice. She is febrile and tachycardic. Blood tests show leukocytosis and elevated bilirubin and alkaline phosphatase. Ultrasound demonstrates gallstones, normal gallbladder wall thickness, no pericholecystic fluid and a common bile duct of 1.0 cm. What is the most likely diagnosis?

- A** Symptomatic cholelithiasis
B Acute cholecystitis
C Gallstone pancreatitis
D Choledocholithiasis
E Cholangitis

- 12** A 55-year-old man presents with intermittent right upper quadrant abdominal pain; each episode lasts 1 to 2 hours. He is afebrile with normal vital signs. White cell count, bilirubin, alkaline phosphatase, amylase and lipase are normal. Ultrasound demonstrates gallstones, normal gallbladder wall thickness, no pericholecystic fluid and a common bile duct of 3 mm. What is the most likely diagnosis?

- A** Symptomatic cholelithiasis
B Acute cholecystitis
C Gallstone pancreatitis
D Choledocholithiasis
E Cholangitis

- 13** What is the surgical treatment for the majority of gallbladder cancers?

- A** Simple cholecystectomy
B Extended cholecystectomy with bile duct resection
C Liver resection with portal lymph node dissection
D Palliative bypass surgery
E Laparoscopic cholecystectomy

Answers 1-13

PRETEST MCQS

- 1 **C Hepatectomy (segment IVb and V) and portal lymph node dissection** — **UPDATED** For T2b gallbladder cancer, surgical resection with hepatectomy and lymph node dissection is indicated.
- 2 **D CCK-HIDA scan should be performed to evaluate for biliary dyskinesia** — A CCK-HIDA scan is used to assess gallbladder ejection fraction in suspected biliary dyskinesia.
- 3 **B Observation with repeat ultrasound examinations to evaluate for increase in polyp size** — Small gallbladder polyps are managed conservatively with serial imaging.
- 4 **C Roux-en-Y hepaticojejunostomy** — Surgical repair via Roux-en-Y hepaticojejunostomy is indicated for iatrogenic biliary strictures.
- 5 **B Transarterial embolization (TAE)** — Transarterial embolization is the preferred first-line treatment for major hemobilia following liver biopsy.
- 6 **A Cholecystectomy with resection of the extrahepatic biliary tract and Roux-en-Y hepaticojejunostomy** — Complete excision of the choledochal cyst with biliary reconstruction is recommended to reduce cancer risk.
- 7 **A Endoscopic retrograde cholangiopancreatography (ERCP)** — ERCP is the preferred intervention to relieve biliary obstruction in cholangitis.
- 8 **A Tube cholecystostomy** — High-risk, critically ill patients benefit from a minimally invasive percutaneous tube cholecystostomy.
- 9 **B Presence of clinical symptoms** — Symptomatic gallbladder polyps require cholecystectomy regardless of size.
- 10 **D Choledocholithiasis** — **UPDATED** Obstructive jaundice with a dilated common bile duct suggests choledocholithiasis.
- 11 **E Cholangitis** — **UPDATED** Fever and tachycardia with jaundice, RUQ pain and a dilated CBD (Charcot's triad) indicate acute cholangitis needing antibiotics and biliary drainage.
- 12 **A Symptomatic cholelithiasis** — **UPDATED** Intermittent RUQ pain with normal labs and a small common bile duct is consistent with symptomatic cholelithiasis.
- 13 **C Liver resection with portal lymph node dissection** — **UPDATED** Beyond T1a, gallbladder cancer needs radical cholecystectomy: en bloc resection of liver segments IVb/V with portal lymphadenectomy; simple cholecystectomy suffices only for T1a.

10 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 45-year-old woman is due to have an elective cholecystectomy following a recent bout of acute cholecystitis (which has fully resolved). She has a past medical history of high blood pressure, which is fully controlled with tablets. Which of the following best describes her preoperative morbidity?
 - A ASA grade I
 - B ASA grade II
 - C ASA grade III
 - D ASA grade IV
 - E ASA grade V
- 2 A 45-year-old woman presents with a 2-month history of worsening jaundice, itching and malaise. On examination, she has multiple localized areas of yellow pigmentation around her eyes. She has a past history of Sjögren syndrome, but is otherwise well. What is the most likely diagnosis?
 - A Acute cholecystitis
 - B Ascending cholangitis
 - C Biliary colic
 - D Primary biliary cirrhosis
 - E Primary sclerosing cholangitis
- 3 A 55-year-old man presents with increasing itching and upper abdominal discomfort. He appears jaundiced. On examination, there is a non-tender mass in the right upper quadrant. He has a history of ulcerative colitis. What is the most likely diagnosis?
 - A Cholangiocarcinoma
 - B Gallstones
 - C Haemolysis
 - D Hepatitis
 - E Pancreatic carcinoma
- 4 A 40-year-old woman presents with a 4-hour history of right upper abdominal pain and fever. She has known gallstones but has never felt this unwell before. On examination, she is jaundiced with temperature 39°C, pulse 130/min, and localized peritonism in the RUQ. Which gallstone complication is this?
 - A Acute cholecystitis
 - B Ascending cholangitis
 - C Biliary colic
 - D Carcinoma of the gallbladder
 - E Gallstone ileus

- 5** A 27-year-old man presents to the GP with a sore throat and fevers. On examination, you notice that he is jaundiced and the tonsils are inflamed. On direct questioning the patient tells you that he has noticed this discolouration of his skin before, particularly when run down or after drinking alcohol. His blood tests are all normal except for a mildly raised bilirubin. What is the most likely diagnosis?
- A Alcoholic hepatitis
 - B Gallstones
 - C Gilbert syndrome
 - D Type 1 Crigler-Najjar syndrome
 - E Viral hepatitis
- 6** A 38-year-old woman who has a history of intermittent right upper quadrant pain associated with eating, attends the emergency department with a 3-day history of severe, unremitting abdominal pain and jaundice. A bedside ultrasound scan shows a 1 cm stone in her common bile duct. Which of the following would be most suitable management?
- A Cholecystectomy
 - B ERCP
 - C Fragmentation of the stone
 - D Laparotomy
 - E MRCP
- 7** A 40-year-old woman presents with intermittent episodes of severe right upper quadrant pain associated with meals. She is otherwise well and haemodynamically stable. Examination is unremarkable. Which of the following investigations would you perform in the first instance?
- A Abdominal ultrasound
 - B Abdominal X-ray
 - C ERCP
 - D MRCP
 - E Upper GI endoscopy
- 8** A 34-year-old woman is being investigated for jaundice. Investigations show elevations of bilirubin, aspartate transaminase and alkaline phosphatase in the blood. Gamma GT levels are normal. There is no urobilinogen in the urine. Which of the following conditions would be compatible with the above results?
- A Gallstones
 - B Gilbert syndrome
 - C Hereditary spherocytosis
 - D Liver metastasis
 - E Viral hepatitis
- 9** A 45-year-old woman presents to the emergency department with a 12-hour history of worsening right upper abdominal pain and vomiting. On examination, she is locally peritonitic in the right upper quadrant. Her temperature is 38.1°C. Which of the following treatment options would be indicated in the first instance?
- A Cholecystectomy
 - B Intravenous antibiotics
 - C Intravenous steroids
 - D Intravenous proton pump inhibitor
 - E Percutaneous drainage of the gallbladder
- 10** A 45-year-old woman has been diagnosed with gallstones. She opts to have an open cholecystectomy. Which of the following surgical incisions would be most appropriate?
- A Gridiron
 - B Lanz
 - C Right Kocher's
 - D Right paramedian
 - E Rooftop

Answers 1-10

- 1** **B ASA grade II** — Mild systemic disease without functional limitation is ASA II.
- 2** **D Primary biliary cirrhosis** — Middle-aged woman with autoimmune history, pruritus, jaundice, and xanthelasmata suggests PBC.
- 3** **A Cholangiocarcinoma** — Painless obstructive jaundice with palpable gallbladder in a patient with UC suggests cholangiocarcinoma.
- 4** **B Ascending cholangitis** — Charcot's triad (fever, jaundice, RUQ pain) suggests cholangitis requiring urgent treatment.
- 5** **C Gilbert syndrome** — Gilbert's syndrome is a benign unconjugated hyperbilirubinaemia triggered by illness or stress.
- 6** **B ERCP** — CBD stones are best managed by ERCP for stone extraction and potential stenting.
- 7** **A Abdominal ultrasound** — Gallstones are best screened by ultrasound in suspected biliary colic.
- 8** **A Gallstones** — **UPDATED** Absent urinary urobilinogen means conjugated bilirubin is not reaching the gut, i.e. biliary obstruction such as a CBD stone; haemolysis and Gilbert give unconjugated jaundice.
- 9** **B Intravenous antibiotics** — Acute cholecystitis initially managed with antibiotics and supportive care.
- 10** **C Right Kocher's** — **UPDATED** A right subcostal Kocher incision is used for open cholecystectomy.

5 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A middle-aged obese gentleman presents with severe right upper quadrant abdominal pain after a Chinese takeaway. On examination you note tenderness in his epigastrium and right upper quadrant. Which investigation is most likely to establish the actual diagnosis?
 - A White cell count
 - B Abdominal ultrasound
 - C Liver function tests
 - D Abdominal X-ray
 - E Serum amylase
- 2 A middle-aged obese gentleman presents with right upper quadrant pain. He has a fever of 38.5°C, white cell count of $16 \times 10^9/L$, and C-reactive protein level of 135 mg/L. What is the most likely diagnosis?
 - A Pancreatitis
 - B Large bowel obstruction
 - C Massive haemothorax
 - D Cholecystitis
 - E Hepatitis
- 3 A 46-year-old female presents with severe upper abdominal pain, tenderness in the right upper quadrant, fever of 38.2°C, raised alkaline phosphatase and bilirubin, plus scleral jaundice. What is the most likely diagnosis?
 - A Liver abscess
 - B Gallstone ileus
 - C Appendicitis
 - D Viral hepatitis
 - E Cholangitis
- 4 A 48-year-old woman has right upper quadrant pain and slight jaundice. Abdominal exam reveals upper tenderness. Next most appropriate initial investigation?
 - A Abdominal CT scan
 - B Abdominal X-ray
 - C Abdominal ultrasound scan
 - D Abdominal MRI scan
 - E ERCP
- 5 A 43-year-old man has right upper quadrant pain, fever of 38.5°C, WCC $17 \times 10^9/L$, CRP 98 mg/L. Which additional sign is most likely?
 - A Cullen sign
 - B Grey Turner sign
 - C Rovsing sign
 - D Boas sign
 - E Trousseau sign

Answers 1-5

CRASH COURSE MCQS

- 1 **B Abdominal ultrasound** — First-line imaging for gallbladder disease.
- 2 **D Cholecystitis** — RUQ pain, fever, raised inflammatory markers; acute cholecystitis.
- 3 **E Cholangitis** — Charcot's triad: RUQ pain, fever, jaundice; suggests cholangitis.
- 4 **C Abdominal ultrasound scan** — Jaundice + RUQ pain => check for gallstones/biliary dilation with US.
- 5 **D Boas sign** — **UPDATED** Boas sign (hyperaesthesia below the right scapula) is associated with acute cholecystitis; Cullen and Grey Turner signs suggest haemorrhagic pancreatitis.

Pancreas

Acute and chronic pancreatitis, pseudocyst, pancreatic cancer and periampullary tumours.

53

MCQS

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

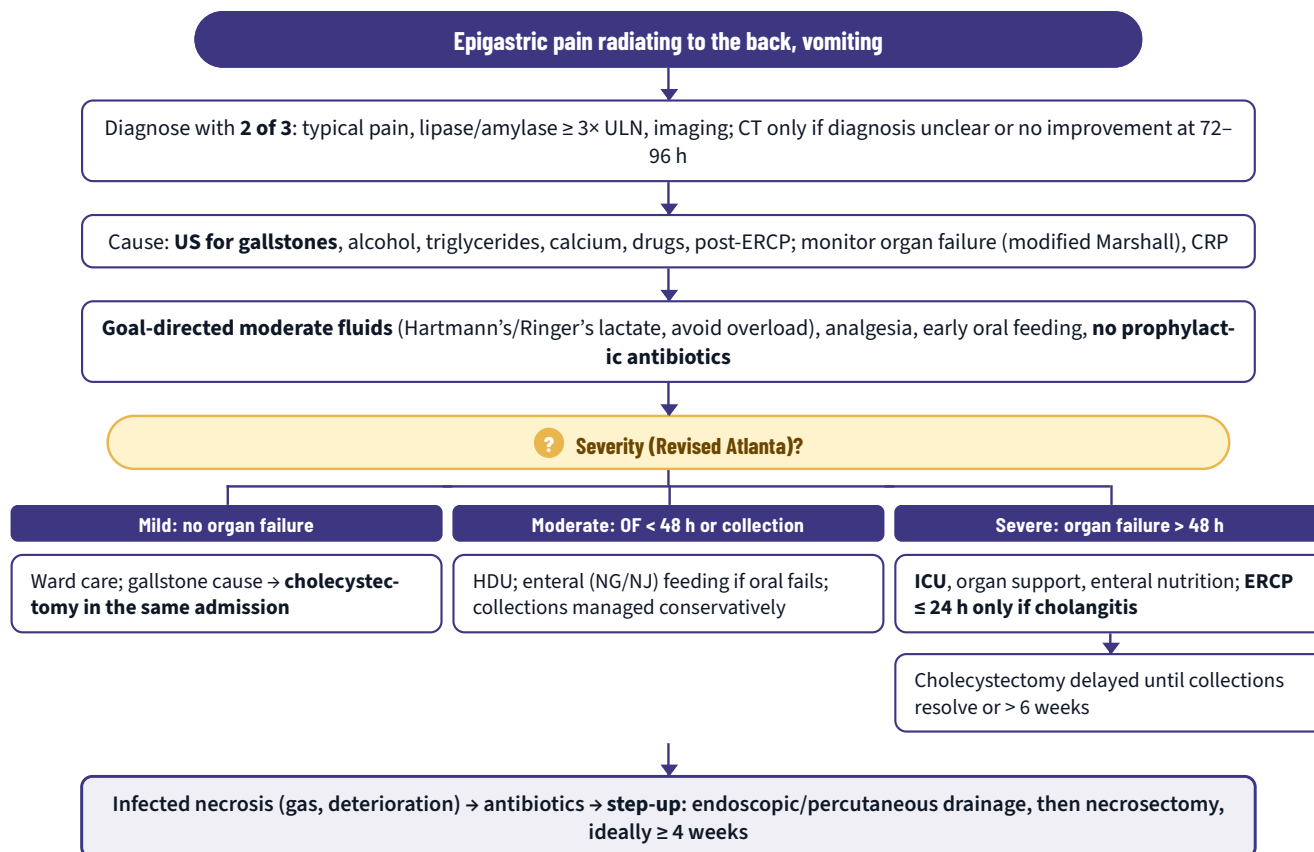
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20

Step by step

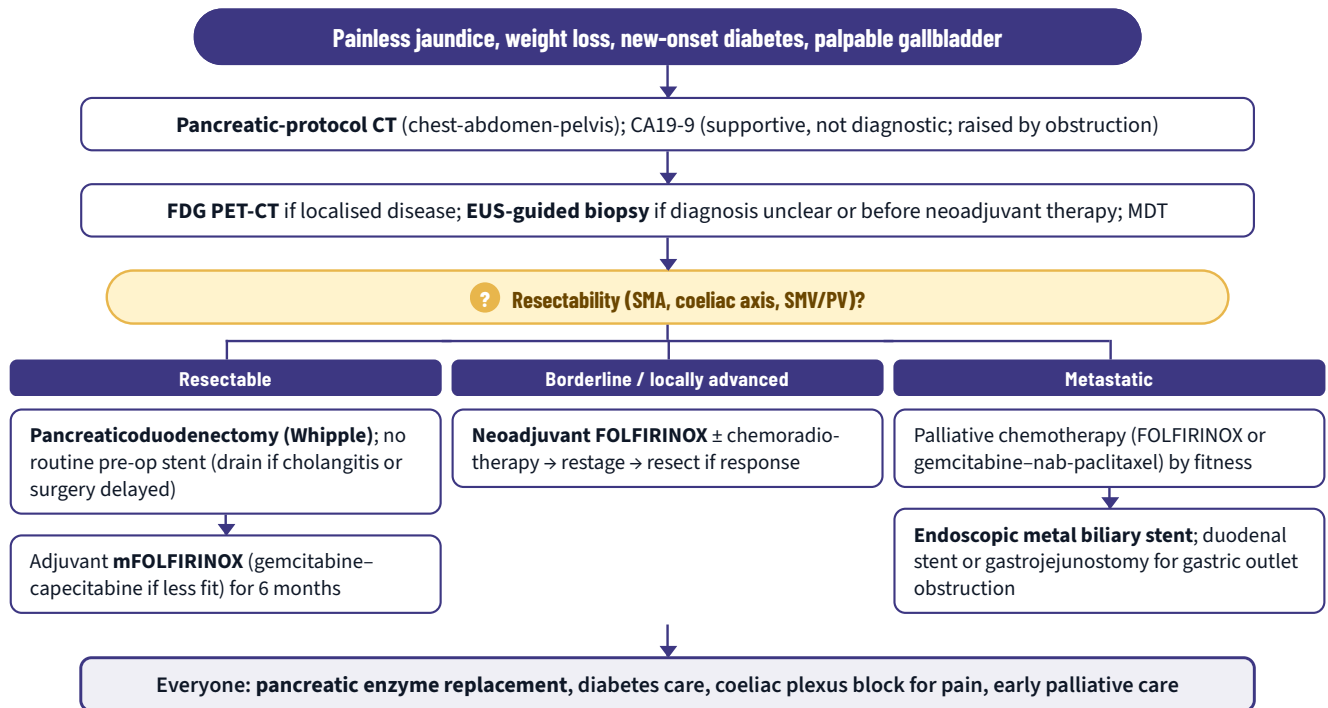
Acute pancreatitis

IAP/APA 2013 · REVISED ATLANTA 2012 · WSES 2019 · ACG 2024



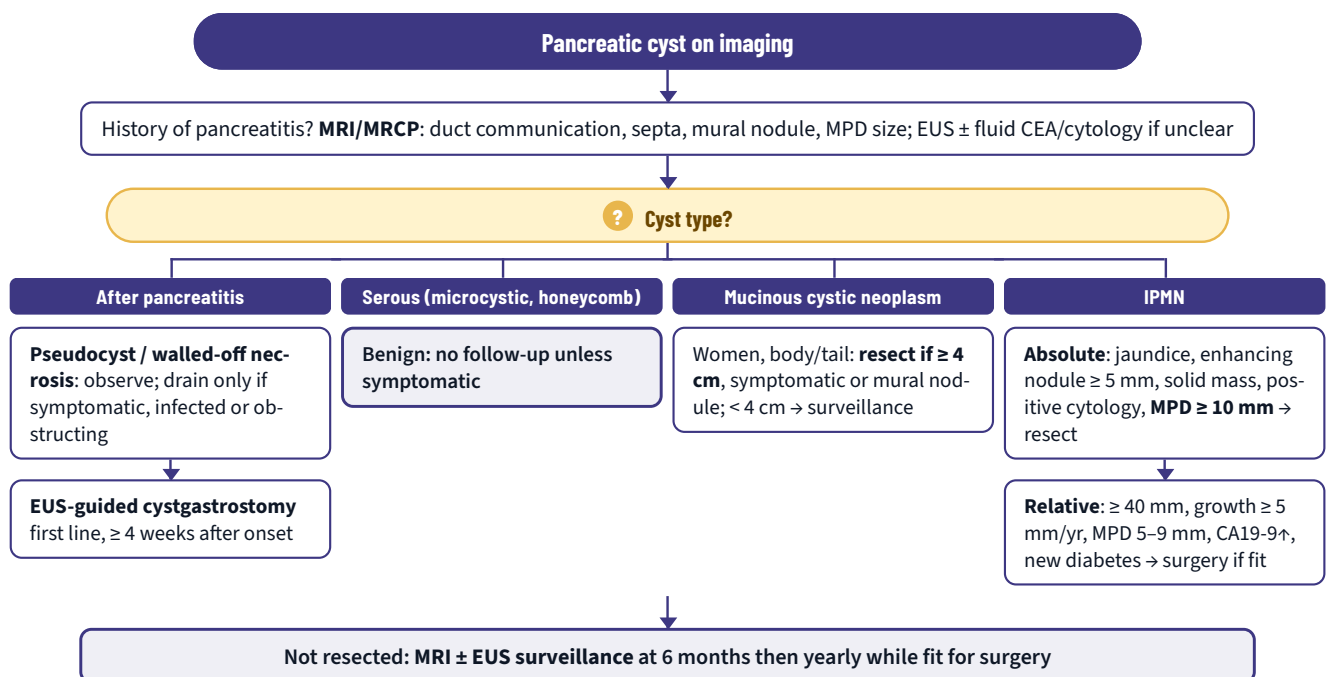
Painless obstructive jaundice: pancreatic head cancer

NICE NG85 · ESMO 2023 PANCREATIC CANCER



Pancreatic cystic lesion

EUROPEAN PCN GUIDELINES 2018 · KYOTO IAP 2024



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1 A plain abdominal X-ray is shown. What is the most likely diagnosis on this image?



- A Chronic calcific pancreatitis
- B Renal stones
- C Gallstones
- D Calcified lymph nodes
- E Splenic artery aneurysm

MINISTERIAL EXAM

Ministerial evaluation examinations

- 2 A 33-year-old woman has severe upper abdominal pain with radiation to the back within 24 hours of removing gallstones in the common bile duct by endoscopic retrograde cholangiopancreatography (ERCP). The pain is eased when she leans forward. She reports some nausea and vomiting but denies any diarrhoea. Jaundice is noted and the epigastric region is tender on palpation. Her blood pressure is 120/80 mmHg, and temperature is 37.3°C. What is the SINGLE most likely reason for her signs and symptoms?
- A Ascending cholangitis
 - B Acute pancreatitis
 - C Perforated duodenal ulcer
 - D Chronic pancreatitis
 - E Oesophageal Bleeding

Answers 1-2

LOCAL EXAMS MCQs

- 1 **A Chronic calcific pancreatitis** — Speckled calcification across the line of the pancreas (L1–L2) is chronic calcific pancreatitis.

- 2 **B Acute pancreatitis** — **UPDATED** Acute pancreatitis is a known complication following ERCP.

4 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQs

Essay to MCQ MCQs

- 1 A patient with known gallstones and mild jaundice is rushed into the emergency department with severe diffuse abdominal pain, nausea and vomiting. The abdomen is tender with some rigidity in its upper part; bowel sounds are present. Pulse 108/min, BP 105/70 mmHg, respiratory rate 32/min. What is the most likely diagnosis?
- A Acute cholecystitis
 - B Acute pancreatitis due to CBD stones
 - C Small bowel obstruction due to adhesions
 - D Acute hepatitis
 - E Peptic ulcer perforation
- 2 In this patient with suspected acute pancreatitis, which TWO initial investigations best establish the diagnosis?
- A Abdominal X-ray and ECG
 - B Complete blood count and liver function tests
 - C Pancreatic lipase & amylase measurement plus CT abdomen
 - D Endoscopic ultrasound and MRCP
 - E Urinalysis and serum electrolytes

- 3** How can acute pancreatitis be confirmed radiologically in this patient?
- A Plain X-ray of the abdomen
 - B Ultrasound with Doppler study
 - C MRCP
 - D CT scan showing diffuse enlargement and edema
 - E Endoscopic retrograde cholangiopancreatography (ERCP)
- 4** What is the treatment of this patient with severe acute biliary pancreatitis?
- A Oral antibiotics and discharge
 - B Laparoscopic cholecystectomy immediately
 - C Endoscopic stone extraction
 - D Admission to HDU/ICU with fluid resuscitation and analgesia, then ERCP (for cholangitis or persisting obstruction) and cholecystectomy for definitive treatment
 - E Conservative management with dietary restrictions only

Answers 1-4

ESSAY TO MCQ MCQS

- 1** **B Acute pancreatitis due to CBD stones** — **UPDATED** Severe upper abdominal pain with rigidity, tachycardia and tachypnoea in a jaundiced patient with gallstones indicates acute biliary pancreatitis from a CBD stone.
- 2** **C Pancreatic lipase & amylase measurement plus CT abdomen** — **UPDATED** Serum lipase or amylase >3 times normal with typical pain makes the diagnosis; CT confirms it when in doubt and later assesses necrosis.
- 3** **D CT scan showing diffuse enlargement and edema** — **UPDATED** Contrast-enhanced CT shows a diffusely enlarged, oedematous pancreas with peripancreatic stranding and, after 72 h, any necrosis.
- 4** **D Admission to HDU/ICU with fluid resuscitation and analgesia, then ERCP (for cholangitis or persisting obstruction) and cholecystectomy for definitive treatment** — **UPDATED** Severe biliary pancreatitis needs monitored resuscitation and analgesia; ERCP for cholangitis/obstruction and cholecystectomy on recovery prevent recurrence.

14 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1** Which of the following statements are true regarding tropical chronic pancreatitis?
- A It is linked to ingestion of cassava (tapioca), which contains cyanide derivatives; an insufficient intake of sulphur-containing amino acids impairs detoxification.
 - B It is caused by alcohol ingestion.
 - C Patients present with extensive pancreatic periductal fibrosis, intraductal calcium carbonate stones, and features of insulin-dependent diabetes mellitus.
 - D Pancreatic calcification appears as discrete stones on plain abdominal x-ray; ultrasound and CT confirm the diagnosis.
 - E Management includes support for exocrine and endocrine insufficiency, pain control, and sometimes surgery.
- 2** Pancreatic Anatomy and Physiology. Which of the following statements are true?
- A The pancreas weighs about 80 g.
 - B The head of the pancreas lies over the body of L2 vertebra.
 - C The uncinate process lies behind the superior mesenteric vessels.
 - D The splenic artery travels along the upper border of the pancreas.
 - E The potential space in front of the pancreas is called the greater sac.
- 3** Pancreatic Anatomy and Physiology. The following statements are true except:
- A The vast majority of the acinar tissue is endocrine.
 - B The pancreatic tail snugly fits into the splenic hilum.
 - C The pancreatic digestive enzymes are acidic in nature with a pH of 7.
 - D The islets of Langerhans contain several hormone-producing cell types (alpha, beta, delta and PP cells).
 - E The amount of pancreatic juice secreted is 1200 to 1500 mLs in 24 hours.
- 4** Investigations in Pancreatic Biochemistry. Which of the following statements are true?
- A Elevation of serum amylase is diagnostic.
 - B Estimation of serum lipase is more sensitive and specific.
 - C Urinary amylase and amylase-creatinine clearance is highly accurate.
 - D Pancreatic exocrine function can be measured.
 - E Serum elastase measurement is useful.

- 5** Imaging in Pancreatic Investigation. Which of the following statements is not true?

 - A** Ultrasonography (US) is the initial investigation of choice in the jaundiced patient.
 - B** When doing a computed tomography (CT) scan, initially an unenhanced scan must be done followed by a scan after intravenous contrast injection (CECT).
 - C** While doing a magnetic resonance cholangiopancreatography (MRCP), intravenous secretin injection helps to determine any obstruction to the pancreatic duct.
 - D** Plain abdominal x-ray showing a sentinel loop can be diagnostic.
 - E** Endoscopic retrograde cholangiopancreatography (ERCP) should always be preceded by a plain radiograph.
- 6** Pancreatic Injury. Which of the following statements are true?

 - A** Pancreatic injury is common following blunt abdominal trauma.
 - B** Pancreatic injury is often accompanied by damage to the liver, spleen and duodenum.
 - C** The serum amylase is raised in most cases of pancreatic injury.
 - D** CECT scan will delineate the damage.
 - E** In doubtful cases, urgent ERCP is helpful.
- 7** Pancreatic Injury. Which of the following statements is false?

 - A** All patients with pancreatic trauma should undergo an exploratory laparotomy.
 - B** Pancreatic duct disruption requires surgical exploration.
 - C** Severe injury to the duodenum and the head of the pancreas requires a pancreatoduodenectomy.
 - D** After conservative management for pancreatic injury, duct stricture and pseudocyst may occur as complications.
 - E** During splenectomy, iatrogenic injury to the pancreatic tail can occur.
- 8** Acute Pancreatitis. Which of the following statements is false?

 - A** Acute pancreatitis accounts for 3% of hospital admissions for abdominal pain in the United Kingdom.
 - B** Acute pancreatitis is classified into mild and severe.
 - C** 80% of cases are mild acute pancreatitis, with a mortality rate of 1%.
 - D** 20% are severe acute pancreatitis, with a mortality of 20%–50%.
 - E** In all cases of acute pancreatitis, there is a marked rise in serum amylase.
- 9** Aetiology. Which one of the following causes of acute pancreatitis is due to a congenital anatomical variation?

 - A** Gallstones
 - B** Hereditary pancreatitis
 - C** Pancreas divisum
 - D** Autoimmune pancreatitis
 - E** Hyperparathyroidism
- 10** Aetiology of Postoperative Acute Pancreatitis. Which of the following statements are true?

 - A** Following ERCP the incidence of acute pancreatitis is 10%.
 - B** Therapeutic intervention during ERCP has a higher incidence.
 - C** Patients after cardiothoracic surgery may develop acute pancreatitis.
 - D** The post-gastrectomy patient might develop acute pancreatitis.
 - E** In a post-cholecystectomy patient, acute pancreatitis may be due to a retained stone.
- 11** Severity Scoring in Acute Pancreatitis. Which of the following are not parameters in Ranson or Glasgow scores?

 - A** Age
 - B** White cell count
 - C** Serum amylase
 - D** Serum calcium
 - E** Blood urea
- 12** Signs in Acute Pancreatitis. Which of the following signs have been known to occur?

 - A** Trousseau's sign
 - B** Courvoisier's sign
 - C** Murphy's sign
 - D** Grey-Turner's sign
 - E** Cullen's sign
- 13** Pseudocysts. Which of the following statements are true?

 - A** Pseudocysts occur within the first week of onset of acute pancreatitis.
 - B** They can be confused with cystic neoplasms.
 - C** The majority of them require intervention.
 - D** They can arise after blunt trauma to the upper abdomen.
 - E** Gastrointestinal bleeding may be a complication of a pseudocyst.
- 14** Complications of Acute Pancreatitis. Which of the following statements are true?

 - A** Patients with severe acute pancreatitis require a CECT scan to detect pancreatic necrosis.
 - B** In severe acute pancreatitis, a laparotomy must be done in all cases of pancreatic necrosis.
 - C** Aneurysm of the superior mesenteric artery can occur.
 - D** The vast majority of patients with peripancreatic sepsis can be treated conservatively.
 - E** Pleural effusion is seen in 10%–20% of patients.

Answers 1–14

BAILEY & LOVE MCQs

- 1 **ACDE** True: A, C, D, E — Tropical chronic pancreatitis is related to cassava ingestion, with pancreatic fibrosis, calcifications and diabetes; alcohol is not the cause.
- 2 **ABCD** True: A, B, C, D — E is false; the correct space is the lesser sac.
- 3 **AC** True: A, C — **UPDATED** A is false because acinar cells are exocrine and C is false since the secretions are alkaline.
- 4 **BDE** True: B, D, E — **UPDATED** A and C are false: amylase is not specific (also raised in perforation, ischaemia) and urinary amylase/clearance ratio is unreliable; lipase is more sensitive and specific.
- 5 **D** Plain abdominal x-ray showing a sentinel loop can be diagnostic. — A sentinel loop is a nonspecific finding in pancreatitis.
- 6 **BCDE** True: B, C, D, E — **UPDATED** A is false: pancreatic injury is uncommon in blunt trauma; it usually accompanies liver, spleen or duodenal injury, and CECT and ERCP/MRCP define duct damage.
- 7 **A** All patients with pancreatic trauma should undergo an exploratory laparotomy. — Not all patients require surgical exploration; stable patients may be managed non-operatively.
- 8 **E** In all cases of acute pancreatitis, there is a marked rise in serum amylase. — **UPDATED** Amylase may be normal with late presentation, hypertriglyceridaemia or acute-on-chronic disease, and its level does not reflect severity.
- 9 **C** Pancreas divisum — Pancreas divisum is a congenital ductal anomaly resulting from failure of ductal fusion.
- 10 **BCDE** True: B, C, D, E — **UPDATED** A is false: pancreatitis follows about 3–5% of ERCPs (more after therapeutic procedures); cardiac surgery, gastrectomy and retained CBD stones are recognised causes.
- 11 **C** Serum amylase — Serum amylase is used for diagnosis rather than for severity assessment.
- 12 **ADE** True: A, D, E — **UPDATED** Grey Turner's (flank) and Cullen's (periumbilical) bruising occur, and hypocalcaemia may give Trousseau's sign; Courvoisier's and Murphy's signs are biliary.
- 13 **BDE** True: B, D, E — A and C are false; pseudocysts typically develop after 4 weeks and most resolve spontaneously.
- 14 **ACDE** True: A, C, D, E — B is false since not all cases of necrosis require laparotomy.

10 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Diagnosis of epigastric pain A 45-year-old Asian man is brought in with an acute onset of epigastric pain, nausea and severe vomiting. The pain is worse with movement and is only relieved slightly by leaning forward. The patient is an alcoholic and has been admitted to the emergency department on several occasions for alcohol intoxication. On examination the patient is tachycardic, pyrexial and dehydrated. The abdomen is diffusely tender and soft, and bowel sounds are normal. The patient's serum amylase is raised by six times the upper limit of normal. The most likely diagnosis is
 - A Perforated peptic ulcer
 - B Small bowel obstruction
 - C Acute cholecystitis
 - D Acute pancreatitis
 - E None of the above
- 2 Acute pancreatitis What is the least number of factors that must be present from the modified Glasgow criteria for acute pancreatitis to be classified as severe within 48 hours of admission?
 - A 2
 - B 4
 - C 3
 - D 5
 - E 1
- 3 Causes of acute pancreatitis During a ward round, you are asked to name the most common causative factor responsible for acute pancreatitis. Which option would you choose?
 - A Ethanol
 - B Steroids
 - C Gallstones
 - D Drugs
 - E Trauma
- 4 The severity of acute pancreatitis can be assessed using the Glasgow criteria. Which of the following is not used in the calculation?
 - A Serum amylase
 - B White cell count
 - C Serum calcium
 - D Lactate dehydrogenase
 - E Serum urea

- 5** A 62-year-old patient with pale stools, dark red urine and a palpable gallbladder is most likely suffering from
- A Ascending cholangitis
 - B Impacted stone in the common bile duct (choledocholithiasis)
 - C Tumour of the head of pancreas
 - D Impacted stone in the neck of the gallbladder
 - E Cholangiocarcinoma
- 6** Whipple's procedure is used in the management of pancreatic cancer and chronic pancreatitis. Which of the following structures is not resected during the classical procedure?
- A Duodenum
 - B Head of pancreas
 - C Gallbladder
 - D Common hepatic duct
 - E Pylorus
- 7** Which of the following is not secreted by the endocrine or exocrine pancreas?
- A Chymotrypsinogen
 - B Glucagon
 - C Somatostatin
 - D Lipase
 - E Gastrin
- 8** Of the following, which is not a clearly recognized cause of acute pancreatitis?
- A 5-aminosalicylic acid (mesalazine)
 - B Furosemide
 - C Coxsackie B virus
 - D Hyperlipidaemia
 - E Magnetic resonance cholangiopancreatography
- 9** A 48-year-old patient with acute-onset abdominal pain, high serum amylase and CRP, who deteriorates over 48 hours with signs of respiratory failure, is most likely suffering from
- A Septic shock
 - B Acute respiratory distress syndrome
 - C Pancreatic pseudocyst
 - D Acute alcohol withdrawal
 - E Disseminated intravascular coagulation
- 10** A patient previously treated for acute pancreatitis now presents 8 weeks later with epigastric pain radiating to the back and a markedly raised serum amylase. The most likely diagnosis is
- A Pancreatic pseudocyst
 - B Cholecystitis
 - C Chronic pancreatitis
 - D Recurrent acute pancreatitis
 - E Pancreatic abscess

Answers 1-10

SBA MCQS

- 1** **D Acute pancreatitis** — Raised amylase and relief on leaning forward indicate acute pancreatitis.
- 2** **C 3** — At least 3 positive criteria within 48 hours indicate severe acute pancreatitis.
- 3** **C Gallstones** — Gallstones are the most common cause of acute pancreatitis.
- 4** **A Serum amylase** — **UPDATED** Serum amylase is not included in the Glasgow criteria for acute pancreatitis.
- 5** **C Tumour of the head of pancreas** — A palpable gallbladder with obstructive jaundice suggests a pancreatic head tumour.
- 6** **D Common hepatic duct** — In a classical Whipple's procedure the common hepatic duct is preserved.
- 7** **E Gastrin** — Gastrin is not secreted by the pancreas in adulthood.
- 8** **E Magnetic resonance cholangiopancreatography** — **UPDATED** MRCP is an investigation tool, not a cause of acute pancreatitis.
- 9** **B Acute respiratory distress syndrome** — The patient's respiratory compromise indicates ARDS as a complication of acute pancreatitis.
- 10** **A Pancreatic pseudocyst** — A pancreatic pseudocyst is a common late complication following acute pancreatitis.

Oxford MCQs

1 46-year-old woman has a 24h history of sudden onset of constant epigastric pain radiating to the back with nausea. There is central and upper abdominal tenderness. The amylase is 1642IU. Which is the single most likely aetiology for the presenting condition?

- A Alcohol
- B Combined oral contraceptive pill
- C Gallstones
- D Hyperlipidaemia
- E Systemic viral infection

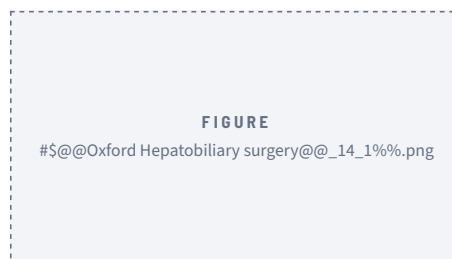
2 52-year-old man has been drinking up to 30 units of alcohol per week and has longstanding epigastric pain. He has had multiple admissions to hospital with pain. He is admitted with further epigastric pain, mild epigastric tenderness, a white cell count of $7 \times 10^9/L$, and an amylase of 102IU. Which is the single most likely diagnosis?

- A Acute pancreatitis
- B Biliary colic
- C Chronic pancreatitis
- D Hiatus hernia
- E Intestinal ischaemia

3 43-year-old man is referred by his GP with ongoing abdominal pain, vomiting, and a persistent raised temperature several weeks after an episode of acute pancreatitis. There is a palpable upper abdominal mass. Which is the single most likely diagnosis?

- A Abdominal aortic aneurysm
- B Carcinoma of the pancreas
- C Empyema of the gallbladder
- D Pseudocyst of the lesser sac
- E Small bowel obstruction

4 78-year-old man has acute severe pancreatitis caused by alcohol. His results are shown in Table 7.2. Which is his single correct Glasgow severity score?



- A 1
- B 2
- C 3
- D 4
- E 5

5 62-year-old man has an adenocarcinoma of the head of the pancreas at the ampulla of Vater. There is no evidence of distant disease and no obvious loco-regional involvement on CT scanning. He is considered for potentially curative treatment. Which is the single most appropriate operation for him to be offered?

- A Cholecystectomy
- B Distal pancreatectomy
- C Gastrojejunostomy
- D Pancreaticoduodenectomy
- E Radical gastrectomy

Answers 1-5

1 **C Gallstones** — Gallstones are the most common cause of acute pancreatitis.

2 **C Chronic pancreatitis** — Longstanding alcohol use and recurrent pain suggest chronic pancreatitis.

3 **D Pseudocyst of the lesser sac** — Persistent pain and palpable mass after pancreatitis suggest pancreatic pseudocyst.

4 **C 3** — **UPDATED** One point per Glasgow criterion (age >55, WCC >15, glucose >10, urea >16, PaO₂ <8 kPa, Ca <2, albumin <32, LDH >600); this patient meets three.

5 **D Pancreaticoduodenectomy** — Whipple's operation is the curative surgery for pancreatic head carcinoma.

PreTest MCQs

- 1 A 39-year-old previously healthy male is hospitalized for 2 weeks with epigastric pain radiating to his back, nausea, and vomiting. Initial laboratory values revealed an elevated amylase level consistent with acute pancreatitis. Five weeks following discharge, he complains of early satiety, epigastric pain, and fevers. On presentation, his temperature is 38.9°C (102°F) and his heart rate is 120 beats/minute; his white blood cell (WBC) count is 24,000/mm³ and his amylase level is normal. He undergoes a CT scan demonstrating a 6 cm by 6 cm rim-enhancing fluid collection in the body of the pancreas. Which of the following would be the most definitive management of the fluid collection?
 - A Antibiotic therapy alone
 - B CT-guided aspiration
 - C Antibiotics and CT-guided aspiration
 - D Antibiotics and percutaneous catheter drainage
 - E Surgical internal drainage with a cyst-gastrostomy
- 2 A previously healthy 79-year-old woman presents with early satiety and abdominal fullness. CT scan of the abdomen, pictured here, reveals a cystic lesion in the body and tail of the pancreas. CT-guided aspiration demonstrates an elevated carcinoembryonic antigen (CEA) level. Which of the following is the most appropriate treatment option for this patient?

FIGURE

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 - A Distal pancreatectomy
 - B Serial CT scans with resection if the lesion increases significantly in size
 - C Internal drainage with Roux-en-Y cyst-jejunostomy
 - D Percutaneous drainage of the fluid-filled lesion
 - E Endoscopic retrograde cholangiopancreatography (ERCP) with pancreatic stent placement
- 3 A 53-year-old woman with a history of a vagotomy and antrectomy with Billroth II reconstruction for peptic ulcer disease presents with recurrent abdominal pain. An esophagogastroduodenoscopy (EGD) demonstrates a gastric ulcer. Serum gastrin levels are greater than 1000 pg/mL on three separate determinations. Which of the following is the best test for confirming the diagnosis?
 - A A 24-hour urine gastrin level
 - B A secretin stimulation test
 - C A serum glucagon level
 - D A 24-hour urine secretin level
 - E A serum glucose to insulin ratio
- 4 A 52-year-old man with a family history of multiple endocrine neoplasia type 1 (MEN1) has an elevated gastrin level and is suspected to have a gastrinoma. Which of the following is the most likely location for his tumor?
 - A Fundus of the stomach
 - B Antrum of the stomach
 - C Within the triangle formed by the junction of the second and third portions of the duodenum, the junction of the neck and body of the pancreas, and the junction of the cystic and common bile duct
 - D Tail of the pancreas
 - E Within the triangle formed by the inferior edge of the liver, the cystic duct, and the common hepatic duct
- 5 A 53-year-old woman presents to the emergency room complaining of severe epigastric pain radiating to her back, nausea, and vomiting. CT scan of the abdomen demonstrates inflammation and edema of the pancreas. A right upper quadrant ultrasound demonstrates the presence of gallstones in the gallbladder. Which of the following is an important prognostic sign in acute pancreatitis according to Ranson's criteria?
 - A Amylase level
 - B Age
 - C Total bilirubin level
 - D Albumin level
 - E Lipase level

- 6** A 55-year-old man who reports weakness, sweating, tachycardia, confusion, and headache whenever he fasts for more than a few hours. He has prompt relief of symptoms when he eats. Physical exam reveals a BMI of 41 kg/m² and a benign abdominal exam. Laboratory examination reveals an inappropriately high level of serum insulin during the episodes of fasting. Which of the following is the most appropriate treatment for this condition?
- A Diet modification to include frequent meals
 - B Long-acting somatostatin analogue octreotide
 - C Simple excision of the tumor
 - D Total pancreatectomy
 - E Chemotherapy and radiation
- 7** A 54-year-old man complains that his eyes are yellow. His bilirubin is 8. A CT of the abdomen and pelvis shows a 3-cm mass in the head of the pancreas encasing the superior mesenteric artery. Cytology from the ERCP is positive for cancer. Which of the following is the most appropriate treatment for this patient?
- A Pancreaticoduodenectomy
 - B Pancreaticoduodenectomy with reconstruction of the superior mesenteric artery
 - C Total pancreatectomy
 - D Total pancreatectomy with reconstruction of the superior mesenteric artery
 - E Placement of metallic biliary stent and chemotherapy
- 8** After a weekend drinking binge, a 45-year-old man presents to the hospital with abdominal pain, nausea, and vomiting. On physical examination, the patient is noted to have tenderness to palpation in the epigastrium. Laboratory tests reveal an amylase of 25,000 U/dL (normal <180 U/dL). An ultrasound is negative for gallstones. The patient is medically managed and sent home after 1 week. A CT scan done 4 weeks later is pictured here. Currently the patient is asymptomatic. Which of the following is the most appropriate initial management of this patient?
- FIGURE**
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- A Distal pancreatectomy
 - B Percutaneous catheter drainage
 - C Endoscopic drainage
 - D Surgical drainage
 - E No intervention is warranted at this time
- 9** An alcoholic man has been suffering excruciating pain from chronic pancreatitis recalcitrant to analgesics and splanchnic block. A surgeon recommends total pancreatectomy. A patient who has a total pancreatectomy might be expected to develop which of the following complications?
- A Diabetes mellitus and steatorrhea
 - B Diabetes mellitus and constipation
 - C Diabetes mellitus only
 - D Hypoglycemia and steatorrhea
 - E Hypoglycemia and constipation
- 10** A 23-year-old woman presents with epigastric abdominal pain and nausea. She is afebrile with normal vital signs. On laboratory results she has no leukocytosis with normal levels of bilirubin and alkaline phosphatase. The amylase and lipase are elevated. Ultrasound demonstrates gallstones, normal gallbladder wall thickness, no pericholecystic fluid, and a common bile duct of 3 mm. What is the most likely diagnosis?
- A Symptomatic cholelithiasis
 - B Acute cholecystitis
 - C Gallstone pancreatitis
 - D Choledocholithiasis
 - E Cholangitis

Answers 1-10

- 1** **D** Antibiotics and percutaneous catheter drainage — Infected pancreatic pseudocysts are definitively managed with percutaneous drainage and antibiotics.
- 2** **A** Distal pancreatectomy — **UPDATED** Raised cyst-fluid CEA indicates a mucinous (pre-malignant) cystic neoplasm; a symptomatic body/tail lesion in a fit patient is treated by distal pancreatectomy.

- 3 **B** **A secretin stimulation test** — A secretin stimulation test confirms Zollinger-Ellison syndrome by demonstrating a rise in gastrin levels.
- 4 **C** **Within the triangle formed by the junction of the second and third portions of the duodenum, the junction of the neck and body of the pancreas, and the junction of the cystic and common bile duct** — Gastrinomas are most commonly located within the gastrinoma triangle.
- 5 **B** **Age** — Age is one of the initial criteria in Ranson's score predicting the severity of acute pancreatitis.
- 6 **C** **Simple excision of the tumor** — Insulinomas are typically benign and are best managed by simple surgical excision.
- 7 **E** **Placement of metallic biliary stent and chemotherapy** — Unresectable pancreatic cancer is managed palliatively with biliary stenting and chemotherapy.
- 8 **E** **No intervention is warranted at this time** — Pancreatic pseudocysts often resolve spontaneously; intervention is deferred to allow cyst wall maturation.
- 9 **A** **Diabetes mellitus and steatorrhea** — **UPDATED** Total pancreatectomy removes all endocrine and exocrine tissue, causing brittle diabetes and steatorrhea needing insulin and enzyme replacement.
- 10 **C** **Gallstone pancreatitis** — **UPDATED** Elevated pancreatic enzymes with gallstones indicate gallstone pancreatitis.

5 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 50-year-old woman with known gallstones presents with severe epigastric pain radiating to the back together with nausea and vomiting. Examination reveals localised epigastric peritonitis and investigations reveal an amylase of 650 IU/L. Which of the following would indicate a poor prognosis in this condition?
A Amylase of 650 IU/L
B Arterial pO₂ of 7.0 kPa
C Patient age of 50 years
D Pyrexia of 38.5°C
E White cell count of 10 × 10⁹/L
- 2 A 27-year-old man presents with a 6-hour history of upper abdominal pain radiating to the back with vomiting. On examination, there is marked epigastric tenderness and a bluish discoloration around his umbilicus. Heart rate is 118/min, BP 108/76 mmHg. Which of the following blood analytes would be most useful in identifying a diagnosis?
A Amylase
B C-reactive protein
C Haemoglobin
D Sodium
E Urea
- 3 A 42-year-old woman presents with episodes of severe hunger despite regular meals. She states that if she does not eat she becomes sweaty and agitated. Her symptoms are exacerbated by exercise. As a result of needing to eat so regularly she has gained a lot of weight. Which of the following blood results would correlate with her symptoms?
A Decreased insulin levels
B Increased C-peptide levels
C Increased glucagon levels
D Increased somatostatin levels
E Raised fasting blood glucose
- 4 A 65-year-old man is on the surgical ward being treated for pancreatitis. Initially he appeared to be improving, however on day 4 he develops fever and increasing epigastric pain. His serum C-reactive protein is noted to be rising. Which of the following is the most likely cause?
A Ascites
B Gastrointestinal haemorrhage
C Ileus
D Infective pancreatic necrosis
E Pancreatic pseudocyst
- 5 A 65-year-old man has been brought into hospital by his wife. She is concerned as he has become increasingly yellow in colour and has unintentional weight loss. The patient denies any pain. On examination he is jaundiced and has a palpable mass in the right upper quadrant. What is the most likely diagnosis?
A Cholangiocarcinoma
B Gallstones
C Gastric carcinoma
D Hepatocellular carcinoma
E Pancreatic carcinoma

Answers 1-5

GET AHEAD MCQS

- 1 **B Arterial pO₂ of 7.0 kPa** — **UPDATED** A low arterial pO₂ (<8 kPa) is a poor prognostic marker in acute pancreatitis.
- 2 **A Amylase** — Elevated serum amylase is highly suggestive of acute pancreatitis.
- 3 **B Increased C-peptide levels** — Insulinoma causes elevated insulin and C-peptide levels with hypoglycaemia.
- 4 **D Infective pancreatic necrosis** — Worsening fever/inflammation suggests infected necrosis in pancreatitis.
- 5 **E Pancreatic carcinoma** — **UPDATED** Painless progressive jaundice with weight loss and a palpable gallbladder (Courvoisier's law) suggests carcinoma of the pancreatic head.

3 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 An 80-year-old woman has pancreatic cancer with distant metastases on CT. Next management step?
 - A Surgery
 - B Palliative therapy
 - C Radiotherapy
 - D Chemotherapy
 - E Surgery + chemotherapy
- 2 A 35-year-old chronic alcoholic complains of upper abdominal pain radiating to his back. On examination you note tenderness and guarding in the epigastrium. Which investigation is most likely to lead to a definitive diagnosis?
 - A Abdominal CT scan
 - B Abdominal X-ray
 - C Chest X-ray
 - D Abdominal ultrasound
 - E Liver function tests
- 3 A 32-year-old man presents with severe abdominal pain radiating to his back. On examination you note tenderness and guarding in the epigastric region, and bruising around the umbilicus. What is the term most likely used to describe this finding?
 - A Grey Turner sign
 - B Cullen sign
 - C McBurney point
 - D Rovsing sign
 - E Murphy sign

Answers 1-3

CRASH COURSE MCQS

- 1 **B Palliative therapy** — **UPDATED** Metastatic disease => palliative treatment for symptom control.
- 2 **A Abdominal CT scan** — Likely acute pancreatitis; CT is key for definitive diagnosis.
- 3 **B Cullen sign** — **UPDATED** Periumbilical bruising (Cullen's sign) reflects retroperitoneal haemorrhage, e.g. haemorrhagic pancreatitis; flank bruising is Grey Turner's sign.

Spleen

Splenic trauma, splenomegaly, indications for splenectomy and post-splenectomy sepsis.

12

MCQS

0

EMQ SCENARIOS

5

SOURCES

0

REPEATED IDEAS

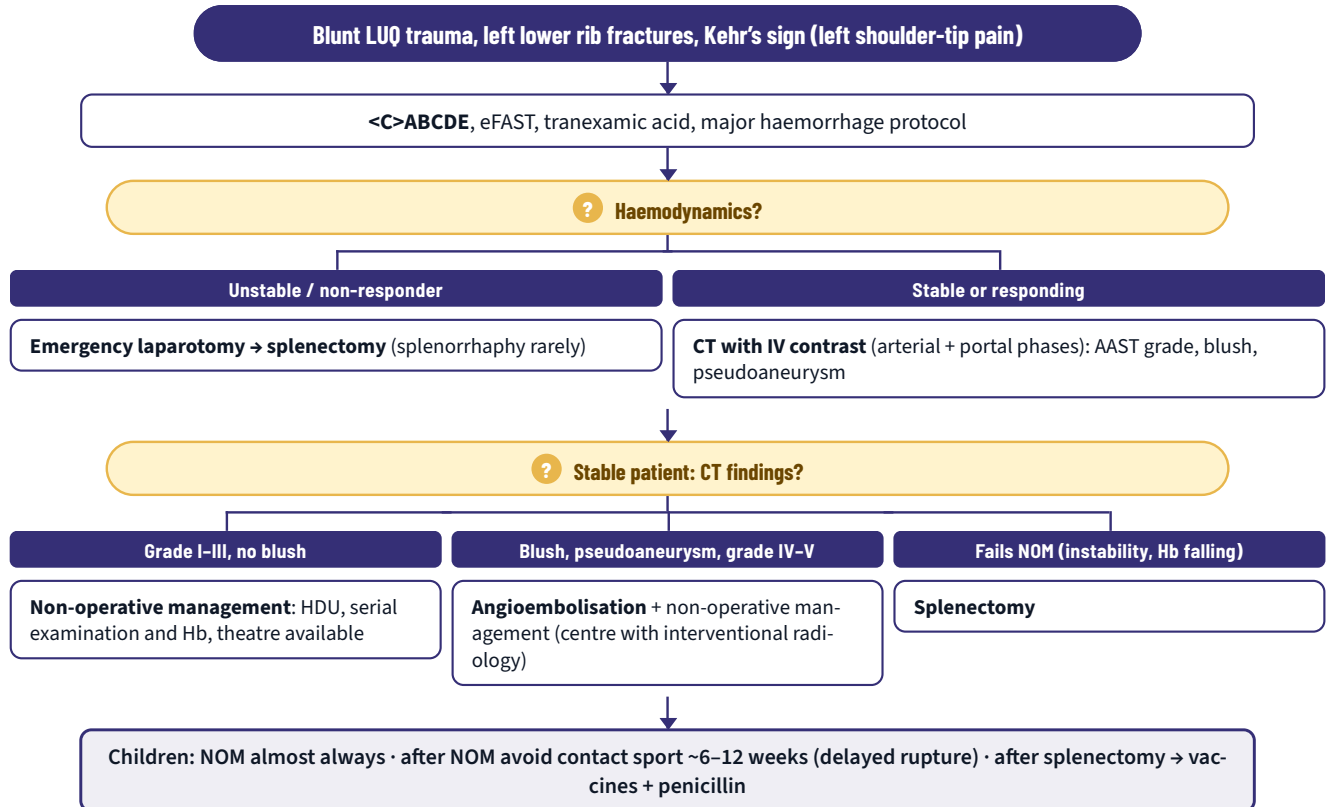
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02	Local Exams MCQs	1 Q	page 327
03	Bailey & Love MCQs	5 Q	page 327
04	Lange MCQs	2 Q	page 328
05	SBA MCQs	3 Q	page 328
06	PreTest MCQs	1 Q	page 329

Step by step

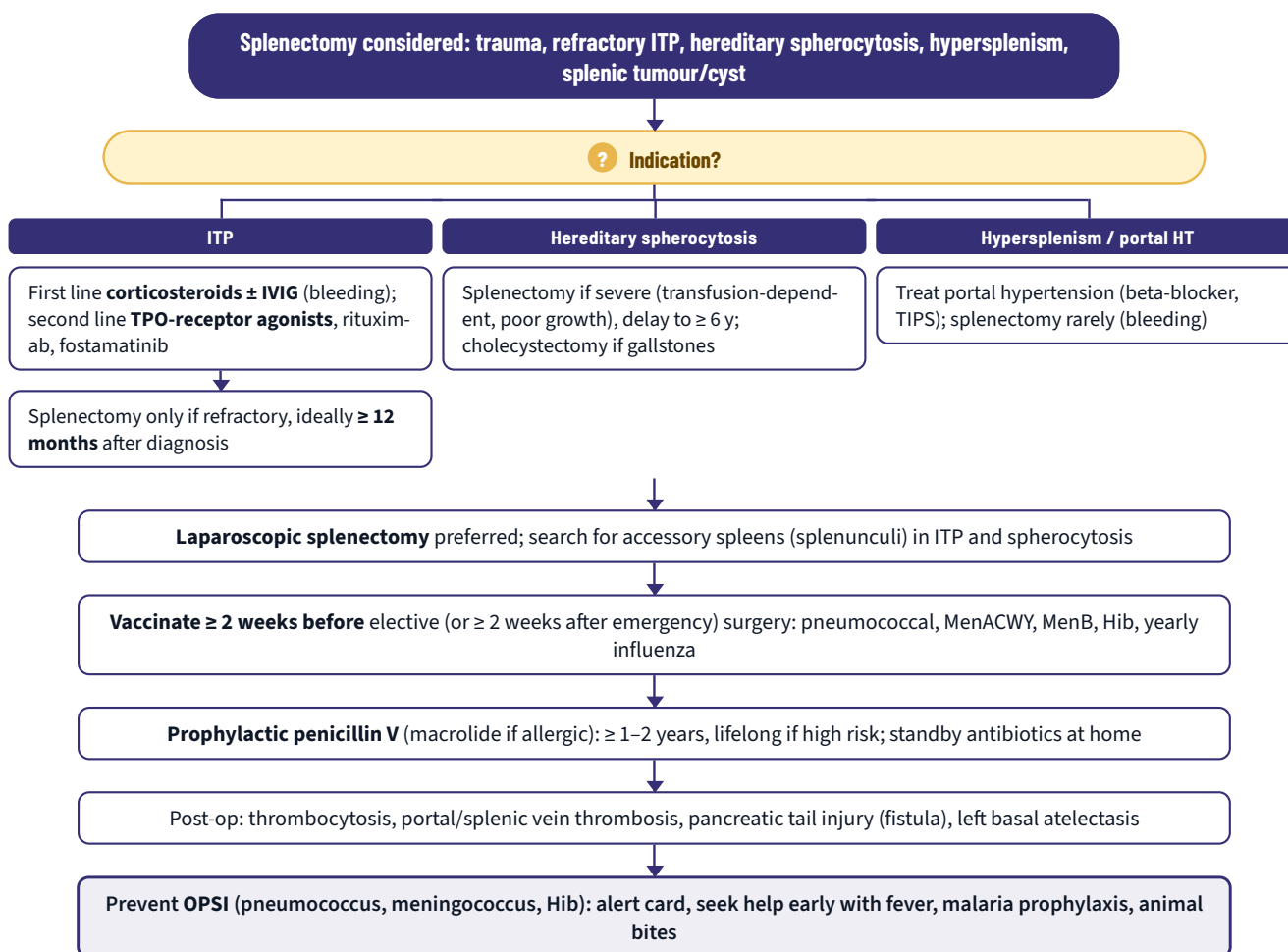
Splenic trauma

WSES 2017 SPLEEN TRAUMA • ATLS 10TH ED • AAST 2018



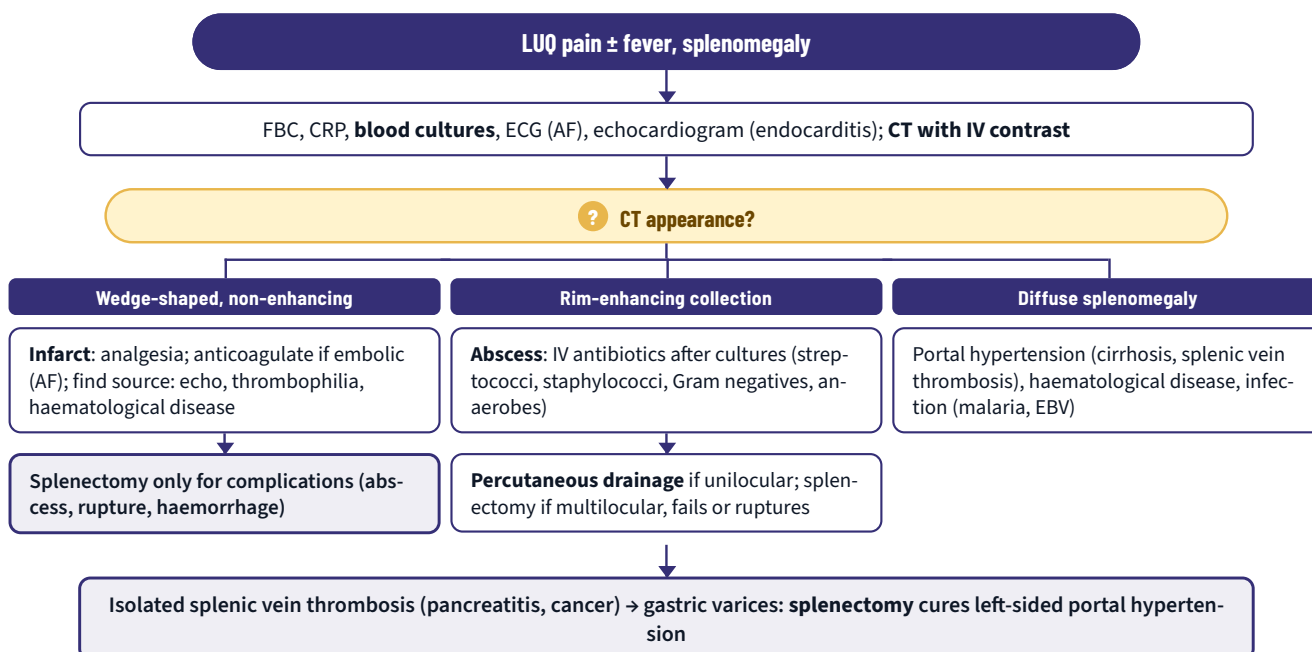
Splenectomy and post-splenectomy prophylaxis

BSH 2011 ASPLENIA · UKHSA GREEN BOOK · ICR 2019 ITP



LUQ pain with fever: splenic infarct or abscess

BAILEY & LOVE 28E · ESC 2023 ENDOCARDITIS



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1 Which vaccines should be given around splenectomy?
- A Pneumococcal, Haemophilus influenzae b, meningococcal (ACWY and B) and annual influenza
 - B BCG and oral polio
 - C Hepatitis B and tetanus only
 - D Live typhoid and yellow fever
 - E No vaccines if lifelong penicillin is taken

Answer 1

LOCAL EXAMS MCQS

- 1 **A** Pneumococcal, Haemophilus influenzae b, meningococcal (ACWY and B) and annual influenza — Give pneumococcal,

Hib, meningococcal ACWY and B plus yearly influenza, 2 weeks before elective or 2 weeks after emergency splenectomy.

5 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Anatomy of the spleen. Which of the following statements are true?
- A The splenic vessels are contained within the splenorenal ligament.
 - B The splenic artery arises from the left gastric artery.
 - C The short gastric branches of the splenic artery lie within the gastrosplenic ligament.
 - D The spleen is a major site of immune function.
 - E Haemopoiesis occurs in the foetal spleen.
- 2 Anatomy of the spleen. All of the following statements are true except:
- A The spleen lies in front of the left seventh, eighth, and ninth ribs.
 - B The tail of the pancreas lies in the splenorenal ligament.
 - C The lymphatics of the spleen ultimately drain into the coeliac nodes.
 - D The left extremity of the lesser sac lies against the splenic hilum.
 - E The visceral surface has impressions for the stomach, left kidney and splenic flexure of the colon.
- 3 Physiology of the spleen. Which of the following statements is not true?
- A The splenic parenchyma consists of white and red pulp.
 - B Pitting is a specific function of the spleen.
 - C The spleen plays a role in immune function.
 - D Haemopoiesis occurs in the foetal spleen.
 - E The spleen has a major reservoir function of red cell mass.
- 4 Investigations of splenic disease. Which of the following statements are true?
- A Plain abdominal x-ray is a common modality of imaging in splenomegaly.
 - B Splenomegaly with portal hypertension requires upper GI endoscopy (OGD).
 - C Lymph node biopsy may be required in some cases of splenomegaly.
 - D CECT is the best imaging modality.
 - E 99mTc-labelled colloid is useful in certain cases.
- 5 Splenectomy. Which of the following statements is false?
- A Partial splenectomy can result in splenic regeneration.
 - B Before splenectomy for haemolytic anaemia, abdominal ultrasound is essential.
 - C A pancreatic or gastric fistula can occur as a post-splenectomy complication.
 - D In a left hemicolectomy the spleen can be in danger of iatrogenic injury.
 - E The risk of opportunist post-splenectomy infection (OPSI) is greatest after the first 6 months of splenectomy.

Answers 1-5

BAILEY & LOVE MCQS

- 1 **ACDE** True: A, C, D, E — **UPDATED** B is false: the splenic artery arises from the coeliac trunk, not the left gastric artery. The other statements are true.
- 2 **A** The spleen lies in front of the left seventh, eighth, and ninth ribs. — **UPDATED** A is false: the spleen lies deep to the left 9th–11th ribs, its long axis along the 10th rib.
- 3 **E** The spleen has a major reservoir function of red cell mass. — While the spleen contributes to immunity and foetal haemopoiesis, it does not serve as a major reservoir of red blood cells.
- 4 **BCDE** True: B, C, D, E — The plain x-ray is not routinely useful while the other investigations are appropriate.
- 5 **E** The risk of opportunist post-splenectomy infection (OPSI) is greatest after the first 6 months of splenectomy. — OPSI risk is highest during the first 2–3 years after surgery, making statement E false.

2 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 14-year-old girl with menorrhagia, frequent nosebleeds, and iron-deficiency anemia is found to have a low platelet count with normal coagulation and abundant megakaryocytes on bone marrow biopsy. What is the most appropriate initial therapy?
 A Splenectomy
 B Platelet transfusion when the count drops below 50 000/ μ L
 C Systemic steroids
 D Chemotherapy
 E Expectant management with intervention only if significant bleeding occurs
- 2 A patient with ITP initially responds to steroids, but over 6–12 months the responses become less marked and shorter, with increasing side effects. What is the next step in management?
 A Partial splenectomy
 B Splenectomy
 C Increase in steroid dose and frequency
 D Bone marrow transplant
 E Plasmapheresis

Answers 1-2

LANGE MCQS

- 1 **C** Systemic steroids — **UPDATED** ITP with significant bleeding is treated first with corticosteroids ($\pm$ IVIg); splenectomy is reserved for refractory or relapsing disease.
- 2 **B** Splenectomy — **UPDATED** Steroid-dependent ITP with waning responses and rising side effects is an indication for splenectomy, which gives durable remission in about two-thirds.

3 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Which of the following abnormal findings on peripheral blood film is not typical of hyposplenism?
 A Bite cells
 B Howell–Jolly bodies
 C Target cells
 D Echinocytes
 E Pappenheimer bodies
- 2 Which is the most common early complication following splenectomy?
 A Pancreatitis
 B Gastric dilatation
 C Atelectasis
 D Thrombosis
 E Overwhelming post-splenectomy sepsis

- 3 Splenectomy patients should be immunized against all the following organisms, except
- A Haemophilus influenzae
 - B Clostridium tetani
 - C Meningococcus C
 - D Streptococcus pneumoniae
 - E Influenza virus

Answers 1-3

SBA MCQS

- 1 **A Bite cells** — **UPDATED** Bite cells occur in oxidative haemolysis (e.g. G6PD deficiency) as the spleen removes Heinz bodies; Howell-Jolly bodies, target cells and Pappenheimer bodies indicate hyposplenism.
- 2 **C Atelectasis** — Atelectasis is the most common early post-operative complication after splenectomy.
- 3 **B Clostridium tetani** — Vaccination against tetanus is not routinely required for splenectomy patients.

1 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 35-year-old woman presents with frequent and multiple areas of cutaneous ecchymosis. Workup demonstrates a platelet count of 15,000/ μ L, and evaluation of the bone marrow reveals a normal number of megakaryocytes. Ultrasound examination demonstrates a normal-sized spleen. Based on the exclusion of other causes of thrombocytopenia, she is given a diagnosis of immune (idiopathic) thrombocytopenic purpura (ITP). Which of the following is the most appropriate treatment upon diagnosis?
- A Expectant management with close follow-up of platelet counts
 - B Immediate platelet transfusion to increase platelet counts to greater than 50,000/ μ L
 - C Glucocorticoid therapy
 - D Intravenous immunoglobulin (IVIG) therapy
 - E Referral to surgery for laparoscopic splenectomy

Answer 1

PRETEST MCQS

- 1 **C Glucocorticoid therapy** — **UPDATED** Initial treatment of ITP is with glucocorticoids; splenectomy is reserved for refractory cases.

Adrenal Gland

Phaeochromocytoma, Conn and Cushing syndromes, incidentaloma and adrenal surgery.

16

MCQs

0

EMQ SCENARIOS

4

SOURCES

0

REPEATED IDEAS

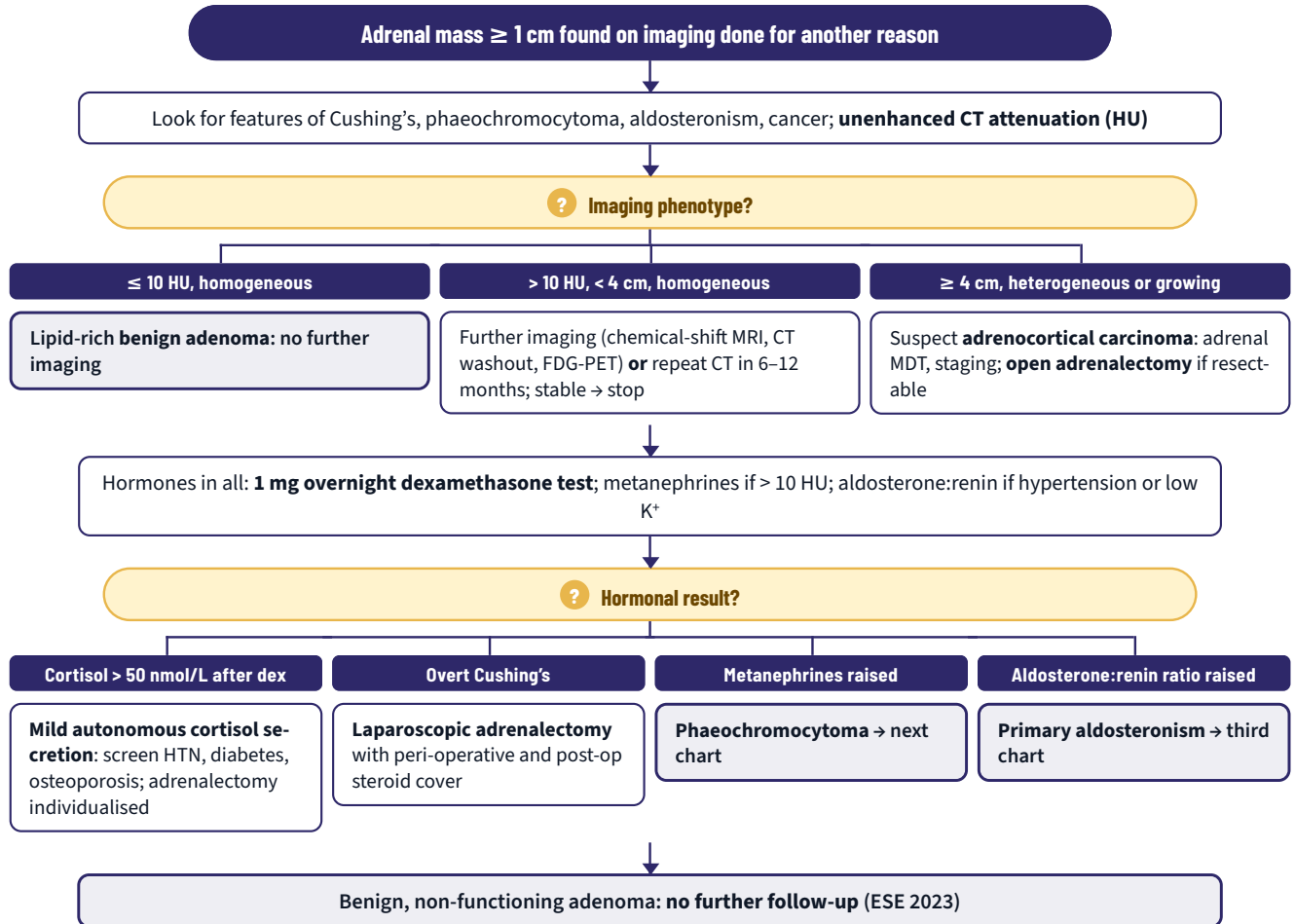
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02	Local Exams MCQs	3 Q	page 333
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04	PreTest MCQs	7 Q	page 334
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Step by step

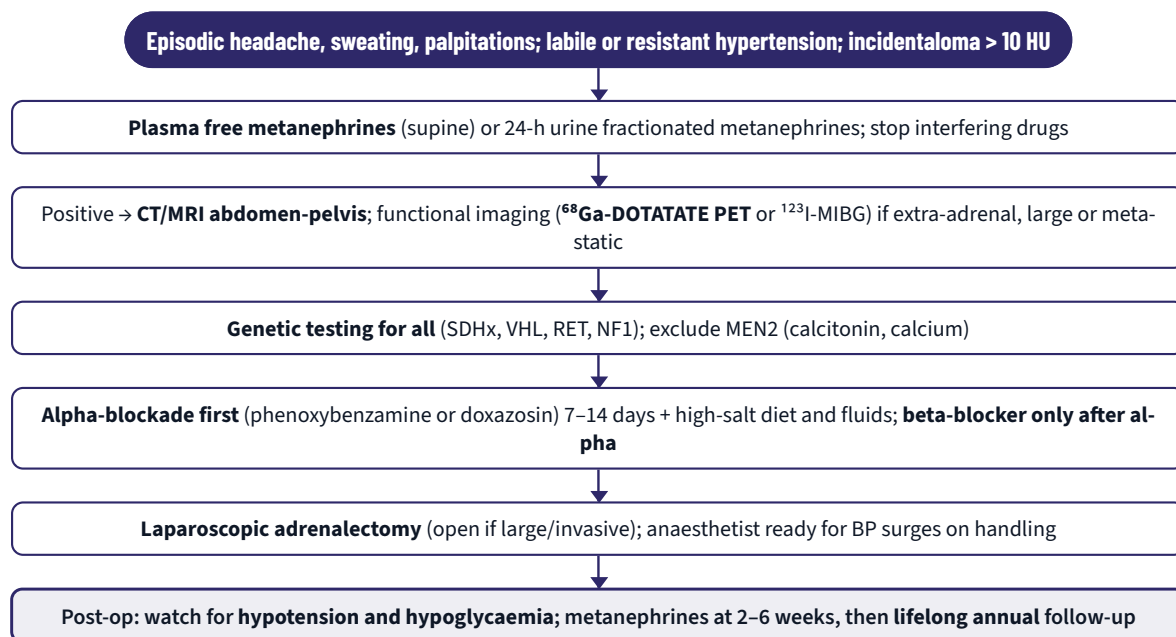
Adrenal incidentaloma

ESE/ENSAT 2023 ADRENAL INCIDENTALOMA



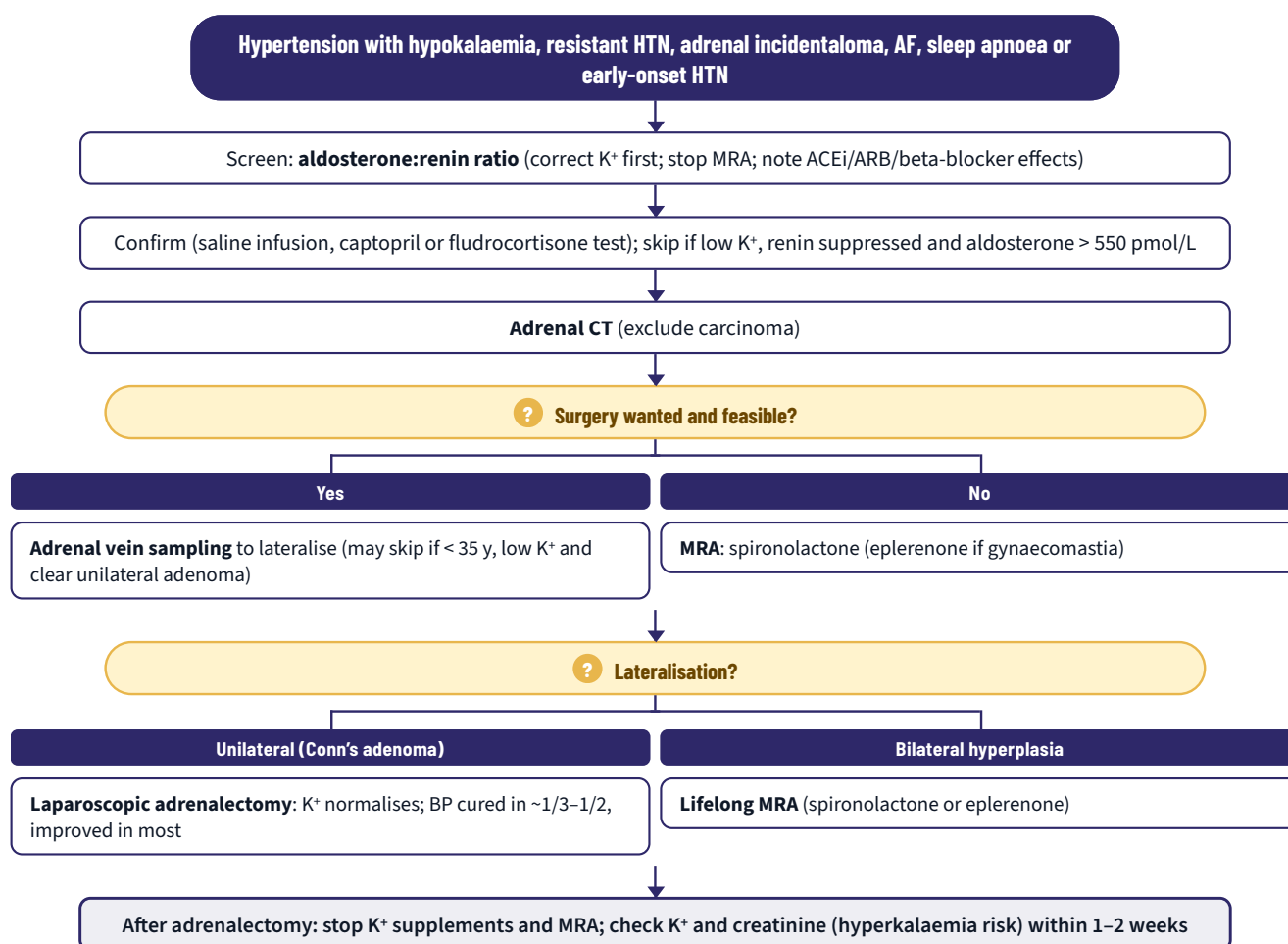
Phaeochromocytoma and paraganglioma

ENDOCRINE SOCIETY 2014 PPGL · ESE 2023 · ESH 2023



Primary aldosteronism (Conn's)

ENDOCRINE SOCIETY 2016 · ESH 2023 HYPERTENSION



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-3

A young woman has recurrent palpitations, headache and sweating. Her BP is 185 mmHg systolic despite a calcium channel blocker, an ACE inhibitor and a diuretic.

1 What is the most likely diagnosis?

- A Pheochromocytoma
- B Conn syndrome
- C Hyperthyroidism
- D Renal artery stenosis
- E Cushing syndrome

2 Which test confirms the diagnosis?

- A Plasma free metanephrines
- B Thyroid function tests
- C Plasma renin
- D Overnight dexamethasone suppression test
- E Serum calcium

3 Which drug should NOT be used (first) in this condition?

- A Labetalol
- B Metoprolol
- C Prazosin
- D Sodium nitroprusside
- E Hydralazine

Answers 1-3

LOCAL EXAMS MCQS

1 **A** **Pheochromocytoma** — Paroxysms of headache, palpitations and sweating with resistant hypertension is pheochromocytoma.

2 **A** **Plasma free metanephrines** — Plasma free (or 24-h urinary fractionated) metanephrines are the biochemical test; imaging (CT/MRI) follows.

3 **B** **Metoprolol** — A beta-blocker alone before alpha-blockade leaves alpha stimulation unopposed and can cause a hypertensive crisis; alpha-block first.

2 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

1 Which of the following statements are true? (Adrenal Incidentalomas)

- A A diagnostic work up is unnecessary if it is <6 cm.
- B Most incidentalomas are nonfunctioning.
- C A 1-mg overnight dexamethasone suppression test is helpful in diagnosis.
- D An adrenal biopsy is always indicated.
- E Adrenal metastasis is more likely in patients with a history of cancer elsewhere.

2 The following statements are true except: (Adrenal Glands Anatomy and Hormones)

- A The weight of a normal adrenal gland is 15 g.
- B Adrenals are retroperitoneal organs.
- C The right adrenal vein drains into the right renal vein.
- D Dehydroepiandrosterone is produced in the zona glomerulosa.
- E Aldosterone is produced in the zona reticularis.

Answers 1-2

BAILEY & LOVE MCQS

1 **BCE** **True: B, C, E** — **UPDATED** Most incidentalomas are non-functioning; the dexamethasone test is useful and metastasis is more likely in patients with known cancers.

2 **ACDE** **True: A, C, D, E** — **UPDATED** Normal adrenal weight is ~4 g, the right adrenal vein drains into the IVC, DHEA is produced in the zona reticularis and aldosterone in the zona glomerulosa.

7 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 34-year-old woman presents with a chief complaint of “feeling weak and thirsty as well as urinating a lot.” On presentation, her blood pressure is 165/89, heart rate is 75, and respiratory rate is 14. Her electrolyte panel is as follows: Na 146, K 2.9, Cl 95, HCO₃ 28. Which of the following is the best initial test given her presentation and laboratory findings?
 - A Plasma renin activity and plasma aldosterone concentration
 - B Urine electrolytes
 - C Plasma cortisol level
 - D Overnight low-dose dexamethasone suppression test
 - E Twenty-four-hour urinary aldosterone level
- 2 A 52-year-old woman presents with hypertension, obesity, and new skin striae. You are concerned about possible Cushing syndrome. Which of the following is the most common cause of Cushing syndrome?
 - A Adrenocortical hyperplasia
 - B Adrenocorticotrophic hormone (ACTH)–producing pituitary tumor
 - C Primary adrenal neoplasms
 - D Ectopic ACTH-secreting carcinoid tumor
 - E Pharmacologic glucocorticoid use
- 3 A patient with mild skin pigmentation is admitted emergently because of sudden abdominal pain, fever and a rigid abdomen. Her blood pressure is 88/58 mm Hg. Blood tests show WBC 25 000/μL, glucose 55 mg/dL, sodium 119 mEq/L and potassium 6.2 mEq/L. Exploratory laparotomy shows no intra-abdominal pathology. Which of the following is the definitive treatment for her primary condition?
 - A 10% dextrose infusion
 - B Antibiotics
 - C Hypertonic saline
 - D Corticosteroids
 - E Vasopressors
- 4 A 49-year-old obese man has become irritable, his face has changed to a round configuration, he has developed purplish lines on his flanks, and he is hypertensive. A 24-hour urine collection demonstrates elevated cortisol levels. This is confirmed with bedtime cortisol measurements of 700 ng/mL. Which of the following findings is most consistent with the diagnosis of Cushing disease?
 - A Decreased ACTH levels
 - B Glucocorticoid use for the treatment of inflammatory disorders
 - C A 3-cm adrenal mass on computed tomography (CT) scan
 - D Suppression with high-dose dexamethasone suppression testing
 - E A 1-cm bronchogenic mass on magnetic resonance imaging (MRI)
- 5 A 45-year-old woman presents with hypertension, development of facial hair and a 7-cm suprarenal mass. Which of the following is the most likely diagnosis?
 - A Myelolipoma
 - B Cushing disease
 - C Adrenocortical carcinoma
 - D Pheochromocytoma
 - E Carcinoid
- 6 A 36-year-old woman presents with palpitations, anxiety and hypertension. Workup reveals a pheochromocytoma. Which of the following is the best approach to optimising the patient preoperatively?
 - A Fluid restriction 24 hours preoperatively to prevent intraoperative congestive heart failure
 - B Initiation of an α-blocker 24 hours prior to surgery
 - C Initiation of an α-blocker at 1 to 3 weeks prior to surgery
 - D Initiation of a β-blocker 1 to 3 weeks prior to surgery
 - E Escalating antihypertensive drug therapy with β-blockade followed by α-blockade starting at least 1 week prior to surgery
- 7 What is the size threshold for resection of an adrenal incidentaloma?
 - A 2 to 3 cm
 - B 6 to 8 cm
 - C 4 to 6 cm
 - D >8 cm only
 - E <3 cm if symptomatic

Answers 1-7

PRETEST MCQs

- 1 **A** Plasma renin activity and plasma aldosterone concentration — **UPDATED** Biochemical diagnosis of hyperaldosteronism

requires demonstration of elevated plasma aldosterone concentration with suppressed plasma renin activity.

- 2 **E Pharmacologic glucocorticoid use** — **UPDATED** The most common cause of Cushing syndrome is iatrogenic, due to administration of exogenous glucocorticoids.
- 3 **D Corticosteroids** — **UPDATED** This patient has adrenal insufficiency and needs treatment with corticosteroids.
- 4 **D Suppression with high-dose dexamethasone suppression testing** — Cushing disease is cortisol excess caused by an ACTH-hypersecreting pituitary adenoma. In these patients, the ACTH level is normal or elevated and cortisol is suppressed with high-dose dexamethasone.
- 5 **C Adrenocortical carcinoma** — **UPDATED** The symptoms and large adrenal mass are consistent with adrenocortical carcinoma.
- 6 **C Initiation of an α -blocker at 1 to 3 weeks prior to surgery** — **UPDATED** Patients with pheochromocytomas should be treated with α -blockade 1-3 weeks before surgery to control blood pressure.
- 7 **C 4 to 6 cm** — Adrenal incidentalomas larger than 4 to 6 cm are generally considered for surgical resection due to the increased risk of malignancy.

4 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 32-year-old woman reports recurring episodes of panic, sweating, and palpitations. They occur randomly and are increasing in frequency. Examination is unremarkable. Which of the following investigations will be the most helpful in confirming the diagnosis?
- A 17-hydroxyprogesterone levels
B 24-hour urinary vanillylmandelic acid
C 24-hour urinary 5-hydroxyindole acetic acid
D Serum calcitonin
E Short synacthen test
- 2 A 37-year-old man presents to the GP with a 2-month history of thirst and frequency of urination. He has no significant past medical history and examination is unremarkable. Routine bloods demonstrate the following: random glucose 5.6 mmol/L, sodium 142 mmol/L and potassium 2.9 mmol/L. What is the most likely diagnosis?
- A Addison disease
B Congenital adrenal hyperplasia
C Conn syndrome
D Multiple endocrine neoplasia
E Pheochromocytoma
- 3 A 6-year-old girl is referred by her GP to the paediatric clinic with precocious puberty. On examination, she is found to have clitoromegaly and some pubic hair. She is on the 98th centile for height and weight. Which of the following will be most helpful in determining the underlying diagnosis?
- A 17-hydroxyprogesterone levels
B 24-hour urinary vanillylmandelic acid
C 24-hour urinary 5-hydroxyindole acetic acid
D Serum calcitonin
E Short synacthen test
- 4 A 37-year-old woman presents to the emergency department following a faint after standing up from a sitting position. She did not lose consciousness during this episode but reported feeling 'dizzy and light-headed'. She has no significant past medical history. On examination you notice multiple areas of skin depigmentation but increased pigmentation in the palmar creases and on the elbows. Which of the following is the best management for her condition?
- A Desmopressin
B Hydrocortisone and fludrocortisone
C Octreotide
D Spironolactone
E Thyroidectomy

Answers 1-4

GET AHEAD MCQS

- 1 **B 24-hour urinary vanillylmandelic acid** — Pheochromocytoma secretes catecholamines; VMA is a key breakdown product in urine.
- 2 **C Conn syndrome** — **UPDATED** Hypokalaemia causes polyuria and thirst; with high-normal sodium in a young adult this suggests primary hyperaldosteronism (Conn syndrome), usually with hypertension.
- 3 **A 17-hydroxyprogesterone levels** — Elevated 17-hydroxyprogesterone suggests congenital adrenal hyperplasia (21-hydroxylase deficiency).
- 4 **B Hydrocortisone and fludrocortisone** — **UPDATED** Primary adrenal insufficiency (Addison's) requires glucocorticoid/mineralocorticoid replacement.

Hernias & Abdominal Wall

Inguinal, femoral, umbilical, incisional and rare hernias: anatomy, complications and repair.

51

MCQs

0

EMQ SCENARIOS

9

SOURCES

1

REPEATED IDEAS

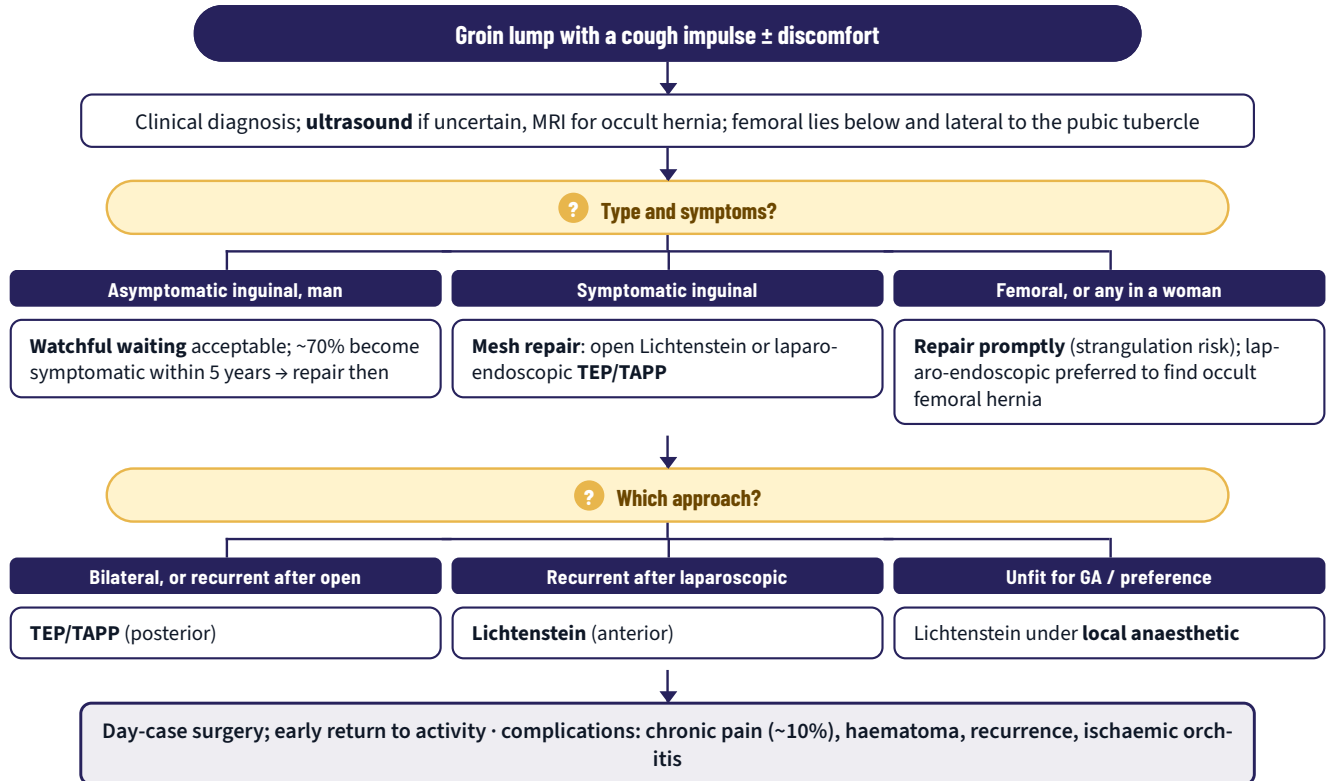
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Step by step

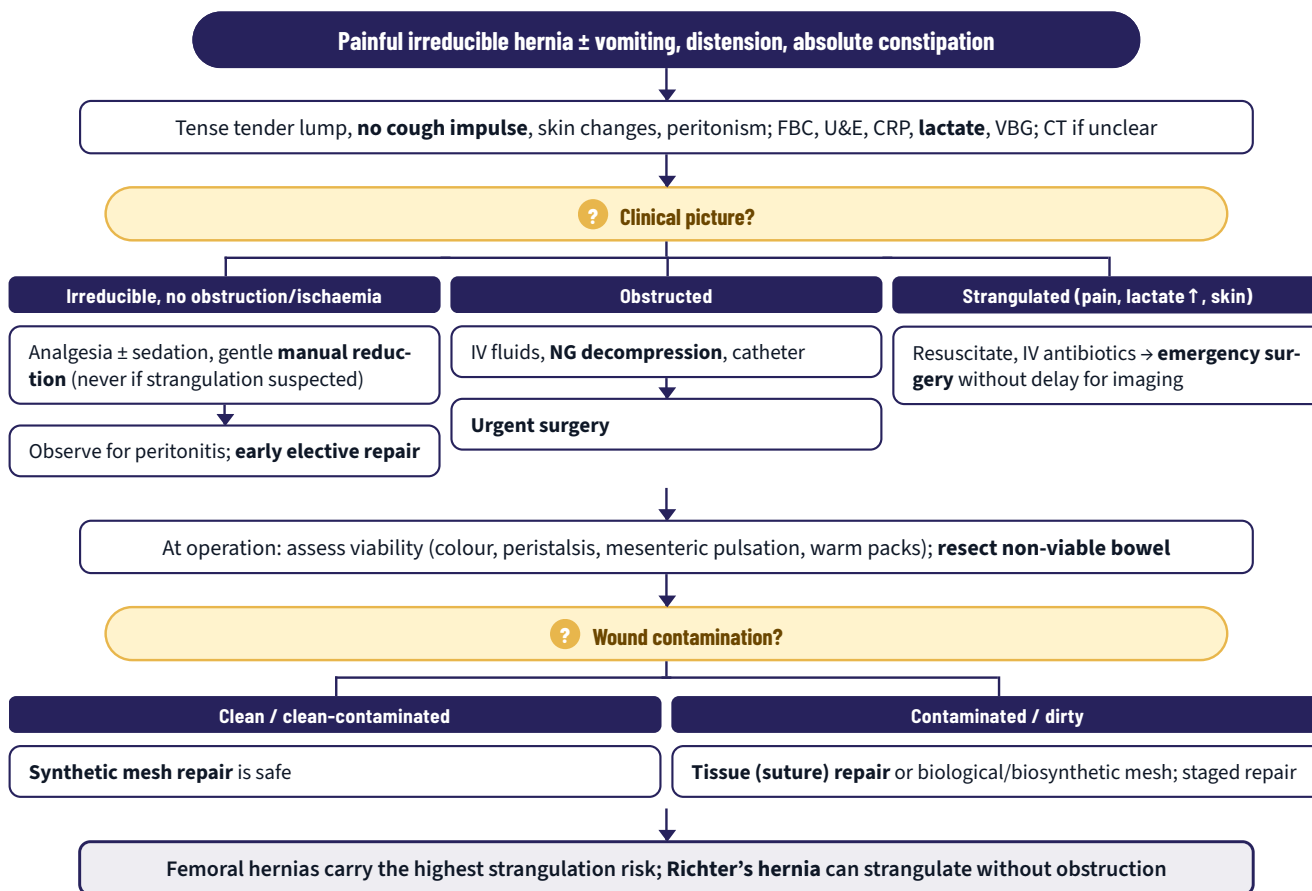
Inguinal and femoral hernia

HERNIASURGE 2018 INTERNATIONAL GROIN HERNIA GUIDELINES (2023 UPDATE)



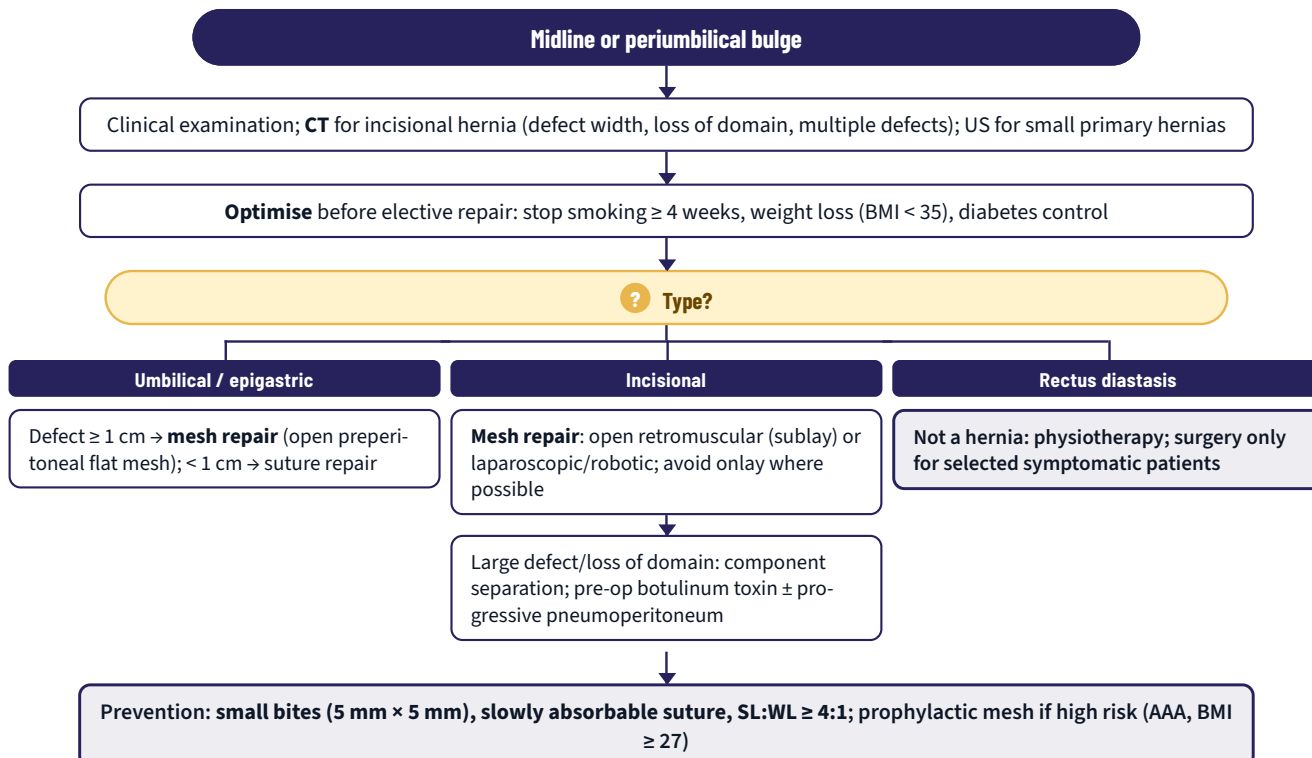
Irreducible, obstructed or strangulated hernia

WSES 2017 INCARCERATED/STRANGULATED HERNIA · HERNIASURGE 2018



Umbilical, epigastric and incisional hernia

EHS/AHS 2020 UMBILICAL · EHS 2023 INCISIONAL · EHS/AHS 2022 CLOSURE



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 33-year-old woman is scheduled for an elective paraumbilical hernia repair. Preoperative evaluation shows the presence of moderate hypothyroidism. Select the most appropriate plan of action:
 - A Proceed with surgery knowing that minor perioperative complications could develop
 - B Postpone surgery until a euthyroid state is achieved
 - C Proceed with surgery while beginning treatment with levothyroxine
 - D Proceed with surgery while beginning treatment with thioamides
 - E Proceed with surgery if severe clinical symptoms are not present
- 2 An otherwise well 78 year-old woman presents with a left groin lump for two months. On examination, there is a unilateral, reducible lump in the left femoral triangle with a cough impulse. Which is the single most appropriate piece of advice which should be given to this patient?
 - A Discharge with advice to seek her GP advice if symptoms occur
 - B Recommend an outpatient review in 3 months to assess her hernia status
 - C Recommend investigation with groin ultrasound to confirm the diagnosis
 - D Recommend surgical exploration and repair as soon as possible
 - E Recommend wearing a supportive truss

REPEATED 2* • ALSO ASKED IN
Oxford

- 3 A 45-year-old multiparous lady noticed a painless lump in her right groin. On examination, the mass was irreducible and located in the upper thigh just below and lateral to the pubic tubercle. Which one among the following conditions is the least likely diagnosis?
 - A Inguinal hernia
 - B Saphena varix
 - C Femoral hernia
 - D Enlarged lymph node
 - E Spigelian hernia

Answers 1-3

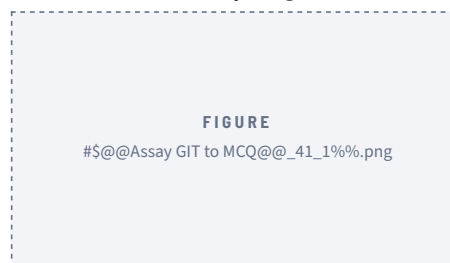
- 1 **B Postpone surgery until a euthyroid state is achieved** — **UPDATED** Moderate hypothyroidism raises anaesthetic risk (hypothermia, hypotension, slow recovery), so elective surgery is postponed until euthyroid.
- 2 **D Recommend surgical exploration and repair as soon as possible** — **UPDATED** A femoral hernia (femoral triangle lump in an elderly woman) has a high strangulation risk through its narrow neck, so prompt repair is advised.
- 3 **E Spigelian hernia** — **UPDATED** A lump below and lateral to the pubic tubercle may be a femoral hernia, saphena varix or node; a Spigelian hernia lies far higher, at the lateral rectus border.

4 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

- 1 An obese woman with a strangulated paraumbilical hernia causing small bowel obstruction undergoes emergency surgery, and a segment of gangrenous small bowel is resected. What is the best surgical option for the hernia now, and why?
 - A Laparoscopic cholecystectomy
 - B Exploratory laparotomy
 - C Herniotomy with suture repair (herniorrhaphy); mesh avoided because of the high infection risk in a contaminated field
 - D Incision and drainage
 - E Laparoscopic hernia repair with mesh
- 2 An obese middle-aged woman presents with a reducible swelling just above the umbilicus that has slowly enlarged over several months. What is the cause of this mass?
 - A Lipoma
 - B Paraumbilical hernia due to obesity and raised intra-abdominal pressure
 - C Omental cyst
 - D Gastrointestinal stromal tumor
 - E Renal mass

- 3 Which THREE clinical signs best confirm that this supraumbilical swelling is a paraumbilical hernia?
 - A Fever, cough, and dyspnea
 - B Palpable lump above the umbilicus, reducibility and an expansile cough impulse
 - C Nausea, vomiting, and diarrhea
 - D Weight loss, anemia, and fatigue
 - E Hypertension, tachycardia, and palpitations
- 4 The same woman later develops bilious vomiting and central colicky abdominal pain with distension; the hernia is now tense, tender and irreducible. Her X-ray is shown. What is the study, what does it show, and what is the most likely diagnosis?



- A Upright chest X-ray showing free air under the diaphragm indicating perforation
- B Abdominal ultrasound showing cholelithiasis
- C Erect Abdominal X-ray showing dilated small bowel loops, indicating small bowel obstruction due to a strangulated umbilical hernia
- D CT scan showing bowel ischemia
- E Barium study showing a filling defect

Answers 1-4

ESSAY TO MCQ MCQS

- 1 **C** Herniotomy with suture repair (herniorrhaphy); mesh avoided because of the high infection risk in a contaminated field — **UPDATED** After resecting gangrenous bowel the field is contaminated, so a sutured repair is preferred; a permanent synthetic mesh carries a high infection risk.
- 2 **B** Paraumbilical hernia due to obesity and raised intra-abdominal pressure — **UPDATED** A reducible swelling beside the umbilicus in an obese adult is a paraumbilical hernia through the linea alba, promoted by raised intra-abdominal pressure.
- 3 **B** Palpable lump above the umbilicus, reducibility and an expansile cough impulse — **UPDATED** A hernia is a lump that reduces and has an expansile cough impulse; redness and tenderness suggest strangulation rather than confirm a hernia.
- 4 **C** Erect Abdominal X-ray showing dilated small bowel loops, indicating small bowel obstruction due to a strangulated umbilical hernia — **UPDATED** Dilated central small-bowel loops with air-fluid levels on an erect abdominal film, with a tender irreducible hernia, indicate obstruction from a strangulated paraumbilical hernia.

6 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 The following statements are true except:
 - A Abdominal pain might be a symptom from non-surgical conditions.
 - B Altered bowel habit usually indicates obstructive and inflammatory pathology.
 - C A hernia causing abdominal symptoms will always be obvious.
 - D An inebriated person with abdominal pain should be diagnosed as alcoholic gastritis.
 - E Abdominal pain related to food habits and associated with other symptoms usually denote pathology in the biliary tract or stomach and duodenum.
- 2 Inguinal and femoral hernia : Which of the following statements are true?
 - A Inguinal hernia is more common in men than in women.
 - B Femoral hernia is more common than inguinal hernia in women.
 - C Femoral hernia is felt below and medial to the pubic tubercle.
 - D The risk of bowel strangulation is greatest in femoral hernia, less in indirect inguinal hernia, and least in direct inguinal hernia.
 - E A 'sliding' inguinal hernia can pass either medial or lateral to the inferior epigastric vessels.
- 3 Surgery for inguinal hernia : Which of the following statements are true?
 - A Open repair of inguinal hernia should always involve placement of a mesh.
 - B The Lichtenstein Open Repair has the lowest recurrence rate of any type of repair.
 - C Laparoscopic repair has been shown to have less postoperative pain, less chronic pain, accelerated postoperative recovery and fewer wound complications.
 - D Approximately 5% of cases of inguinal hernia will be bilateral and 10% will be on patients with recurrent inguinal hernia after previous surgery.
 - E Chronic postoperative pain occurs in more than 20% of cases after surgery.
- 4 Mesh for hernia repair: Which of the following are true?
 - A Mesh should not be used in the presence of infection.
 - B Mesh is mainly used to induce fibrosis, thus making a strong repair.
 - C Polypropylene mesh is hydrophobic and monofilament.
 - D Simple polypropylene is ideal for intraperitoneal use.
 - E Synthetic, nonabsorbable meshes can contract in size by up to 50% over time.
- 5 Surgery for femoral hernia : Which of the following are true?
 - A There is no place for laparoscopic surgery in femoral hernia.
 - B Skin discolouration over the hernia suggests possible strangulation.
 - C There is a decreased risk of femoral hernia after open inguinal hernia repair.
 - D A high, abdominal, McEvedy approach is best in patients presenting with a strangulating femoral hernia.
 - E The femoral nerve is at risk due to its proximity to the neck of a femoral hernia.
- 6 Incisional hernia and its repair : Which of the following are not true?
 - A Incisional hernias never develop more than 2 years after the original surgery.
 - B Jenkins Rule states that in closing an abdominal incision the length of suture used should be four times the length of the incision. However, this has been disproved.
 - C Transverse abdominal incisions are less likely to herniate than vertical incisions.
 - D Non-absorbable meshes placed at initial surgery will reduce the risk of subsequent incisional herniation.
 - E Laparoscopic mesh repair of incisional hernia has a lower wound infection rate than open mesh repair.

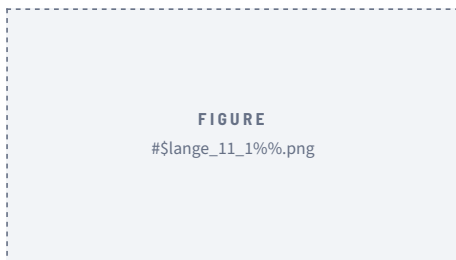
Answers 1-6

BAILEY & LOVE MCQS

- 1 **CD** True: C, D — **UPDATED** C and D are false: a hernia (e.g. femoral, obturator) can easily be missed, and an inebriated patient may still have serious surgical pathology.
- 2 **AD** True: A, D — **UPDATED** B and C are false: inguinal hernia remains the commonest hernia in women, and a femoral hernia lies below and lateral to the pubic tubercle.
- 3 **CDE** True: C, D, E — **UPDATED** A and B are false: mesh is standard but not mandatory (e.g. Shouldice in selected cases), and Lichtenstein does not have the lowest recurrence of all repairs.
- 4 **CE** True: C, E — **UPDATED** A, B and D are false: mesh may be used selectively in contaminated fields, it acts as a scaffold for tissue ingrowth, and plain polypropylene causes intraperitoneal adhesions.
- 5 **BD** True: B, D — **UPDATED** A, C and E are false: laparoscopic repair is used, femoral hernia risk rises after open inguinal repair, and the femoral vein (not nerve) is at risk.
- 6 **AB** True: A, B — **UPDATED** A and B are false: incisional hernias can appear many years later, and Jenkins' 4:1 suture-length rule remains valid (now with small bites).

Lange MCQs

- 1** An 85-year-old man with a long-standing hiatal hernia presents with acute onset of midepigastria pain, vomiting, and inability to keep food down. Which of the following is the most likely diagnosis?



- A Sliding hiatal hernia
- B Hernia of Bochdalek
- C Hernia of Morgagni
- D Paraesophageal hernia
- E Eventration of the diaphragm

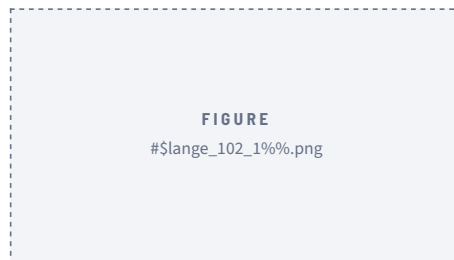
- 2** Which of the following is the most appropriate next step in management of a paraesophageal hernia with obstruction?

- A Laparotomy or laparoscopy and operative repair
- B Continued NG tube decompression and total parenteral nutrition (TPN)
- C Thoracotomy or thoracoscopy and operative repair
- D Endotracheal intubation and ventilatory support
- E Upper endoscopy

- 3** A 75-year-old woman admitted from a nursing home for abdominal pain and pneumonia develops a tender bulge in the medial left thigh below the inguinal ligament on hospital day 3. Which is the most likely diagnosis?

- A Incarcerated direct inguinal hernia
- B Lymph node with abscess
- C Femoral artery aneurysm
- D Incarcerated indirect inguinal hernia
- E Incarcerated femoral hernia

- 4** In a 6-month-old previously healthy male infant, an abnormality is noted during routine diaper changes. The finding is a reducible inguinal bulge that comes and goes. Which is the most likely diagnosis?



- A Non-communicating hydrocele
- B Inguinal adenitis
- C Reducible inguinal hernia
- D Incarcerated inguinal hernia
- E Undescended testes

- 5** For the infant described above with an inguinal hernia, what is the most appropriate management at this time?

- A Antibiotics
- B Reassurance that the abnormality will resolve without intervention
- C Referral to the ED for immediate surgical consultation
- D Referral for elective surgical repair
- E Scrotal support

- 6** Several weeks later, the same infant presents with 4 hours of irritability, nonbilious vomiting, and refusal to feed, with the same inguinal abnormality noted. What is the most appropriate management at this time?

- A Urgent surgical exploration
- B Systemic antibiotics
- C Elective surgical repair
- D Sedation with manual reduction and arrangements for elective surgical repair
- E Sedation with manual reduction, admission, rehydration, and surgical repair within 24–48 hours

Answers 1-6

LANGE MCQS

- 1** **D** Paraesophageal hernia — **UPDATED** Acute pain and vomiting in an elderly patient with a hiatus hernia suggest an obstructed or volvulised paraesophageal (rolling) hernia; sliding hernias rarely obstruct.
- 2** **A** Laparotomy or laparoscopy and operative repair — **UPDATED** An obstructed paraesophageal hernia risks strangulation and needs prompt (usually laparoscopic) reduction, sac excision and crural repair after decompression.

- 3** **E** Incarcerated femoral hernia — **UPDATED** A tender bulge below the inguinal ligament in an elderly woman with coughing (pneumonia) is an incarcerated femoral hernia.

- 4** **C** Reducible inguinal hernia — **UPDATED** An intermittent, reducible inguinal bulge in an infant is an indirect inguinal hernia through a patent processus vaginalis.

- 5 **D Referral for elective surgical repair** — **UPDATED** Infant inguinal hernias do not resolve and carry a high incarceration risk, so prompt elective herniotomy is arranged.
- 6 **E Sedation with manual reduction, admission, rehydration, and surgical repair within 24–48 hours** — **UPDATED** An incar-

cerated infant hernia without peritonitis is reduced under sedation, then repaired within 24–48 h once oedema settles, as re-incarceration is common.

5 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

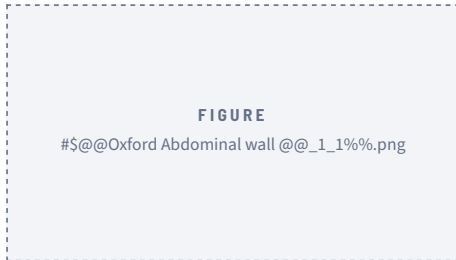
- 1 Which one of the following muscles lies closest to the peritoneal cavity?
 - A Rectus abdominis
 - B External oblique
 - C Internal oblique
 - D Transversus abdominis
 - E Cremaster
- 2 Regarding hernias in females, the most common is
 - A Epigastric
 - B Umbilical
 - C Femoral
 - D Inguinal
 - E Incisional
- 3 Which of the following structures form a part of both the inguinal and femoral canals?
 - A Transversalis fascia
 - B Internal oblique muscle
 - C Transversus abdominis muscle
 - D Inguinal ligament
 - E Pectineal ligament
- 4 A 35-year-old professional weightlifter presents with a red and swollen lump in the left groin. An inguinal hernia is suspected and, at the time of operation, the lump is found to contain a small loop of necrotic bowel. This type of hernia is best described as
 - A Irreducible
 - B Strangulated
 - C Obstructed
 - D Sliding
 - E Richter's
- 5 Management of an inguinal hernia A 78-year-old African Caribbean man presents with severe pain arising from his hernia in the left groin. The patient is also experiencing central colicky abdominal pain. On examination, the abdomen is generally tender and distended and bowel sounds are raised. Examination of the hernial orifices reveals a left-sided, irreducible, tense and extremely tender inguinal hernia. The overlying skin of the hernia is warm and erythematous. What is the most appropriate course of action in managing this patient?
 - A Alert theatre and send patient for emergency surgery
 - B Request a computed tomography scan of the abdomen
 - C Request an ultrasound
 - D Attempt to reduce the hernia
 - E None of the above

Answers 1–5

- 1 **D Transversus abdominis** — The transversus abdominis is the deepest abdominal muscle and lies adjacent to the peritoneum.
- 2 **D Inguinal** — Inguinal hernia is the most common type in females.
- 3 **D Inguinal ligament** — The inguinal ligament forms part of both the inguinal and femoral canals.
- 4 **B Strangulated** — **UPDATED** Necrotic bowel in the sac means its blood supply has been cut off: a strangulated hernia. Irreducible or obstructed hernias have viable contents.
- 5 **A Alert theatre and send patient for emergency surgery** — **UPDATED** A tense, tender, red, irreducible hernia with obstruction is strangulated: resuscitate and operate urgently; do not attempt reduction or delay for imaging.

Oxford MCQs

- 1** Figure 8.1 shows several key landmarks in the right groin of a man. The bony prominences are labelled. The structure labelled A provides an important landmark in inguinal surgery. Figure 8.1 Key landmarks in the right groin of a man Which single term correctly describes structure A?



- A Cremasteric artery
- B External iliac artery
- C Inferior epigastric artery
- D Inferior pudendal artery
- E Superficial circumflex artery

- 2** A 32-year-old man has a modest sized, freely reducible, indirect inguinal hernia. The hernia can be held in reduction by maintaining pressure over one particular location. Which single term most likely describes this location?

- A Arcuate line
- B Deep inguinal ring
- C McBurney's point
- D Mid-inguinal point
- E Superficial inguinal ring

- 3** A 47-year-old woman has had a swelling in the right side of the abdomen for 2 months. A hernia is suspected and the lump's physical characteristics are being assessed. Which single clinical feature would confirm the clinical suspicion?

- A Fluctuance
- B Incompressibility
- C Presence of fluid thrill
- D Reduction on palpation
- E Tenderness to palpation

Answers 1-3

OXFORD MCQS

- 1** **C Inferior epigastric artery** — The inferior epigastric artery marks the medial limit of indirect hernias.
- 2** **B Deep inguinal ring** — Pressure at the deep inguinal ring maintains hernia reduction.

- 3** **D Reduction on palpation** — Only a hernia will truly reduce on palpation.

4 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 48-year-old woman develops pain in the right lower quadrant while playing tennis. The pain progresses and the patient presents to the emergency room later that day with a low-grade fever, a WBC count of 13,000/mm³, and complaints of anorexia and nausea as well as persistent, sharp pain of the right lower quadrant. On examination, she is tender in the right lower quadrant with guarding and a bulge in that location. An ultrasound is ordered and shows an apparent mass in the abdominal wall. Which of the following is the most likely diagnosis?
 - A Acute appendicitis
 - B Cecal carcinoma
 - C Hematoma of the rectus sheath
 - D Torsion of an ovarian cyst
 - E Cholecystitis
- 2 A 22-year-old college student notices a bulge in his right groin. It is accentuated with coughing and straining, but is easily reducible. Which of the following hernias follows the path of the spermatic cord within the cremaster muscle?
 - A Femoral
 - B Direct inguinal
 - C Indirect inguinal
 - D Spigelian
 - E Obturator
- 3 A 22-year-old woman is seen in a surgery clinic for a bulge in the right groin. She denies pain and is able to make the bulge disappear by lying down and putting steady pressure on the bulge. She has never experienced nausea or vomiting. On examination she has a reducible hernia below the inguinal ligament. Which of the following is the most appropriate management of this patient?
 - A Observation for now and follow-up in surgery clinic in 6 months
 - B Observation for now and follow-up in surgery clinic if she develops further symptoms
 - C Elective surgical repair of hernia
 - D Emergent surgical repair of hernia
 - E Emergent surgical repair of hernia with exploratory laparotomy to evaluate the small bowel
- 4 A 58-year-old man presents with a bulge in his right groin associated with mild discomfort. On examination the bulge is easily reducible and does not descend into the scrotum. Which of the following changes is most concerning for possible strangulation requiring emergent repair of the hernia?
 - A Increase in size of the hernia
 - B Descent of hernia into the scrotum
 - C Development of a second hernia in the left groin
 - D Inability to reduce hernia
 - E Worsening pain over the hernia with walking

Answers 1-4

PRETEST MCQS

- 1 **C Hematoma of the rectus sheath** — UPDATED Sudden RIF pain on exertion with a tender abdominal wall mass on ultrasound suggests rectus sheath haematoma from inferior epigastric vessel rupture.
- 2 **C Indirect inguinal** — Indirect inguinal hernia follows the path of the spermatic cord through the internal inguinal ring.
- 3 **C Elective surgical repair of hernia** — Elective repair is appropriate for a reducible femoral hernia.
- 4 **D Inability to reduce hernia** — Inability to reduce the hernia indicates incarceration and a risk of strangulation.

Get Ahead MCQs

- 1** A 62-year-old man presents to the GP with a lump in the left groin which has been present for over 2 months. On examination, the lump is above the inguinal ligament. It is reducible and has a cough impulse, but does not extend into the scrotum. Which of the following is the most likely diagnosis?

 - A Direct inguinal hernia
 - B Femoral hernia
 - C Gluteal hernia
 - D Indirect inguinal hernia
 - E Obturator hernia
- 2** Which term best fits the description of the hernia given below? This hernia arises from a triangle bounded by the external oblique muscle, the latissimus dorsi and the iliac crest below.

 - A Amyand
 - B Littre
 - C Lumbar
 - D Obturator
 - E Sciatic
- 3** A 42-year-old woman presents with an upper midline mass that has been present for over a year. She has a history of a partial gastrectomy for a perforated ulcer. On examination, the mass is 5 cm in size, soft, non-tender and reducible. Which of the following is the most likely diagnosis?

 - A Direct inguinal hernia
 - B Epigastric hernia
 - C Hiatus hernia
 - D Incisional hernia
 - E Paraumbilical hernia
- 4** Which term best fits the description of the hernia given below? A hernia that has two parts, each lying either side of the inferior epigastric artery.

 - A Gluteal
 - B Lumbar
 - C Maydl
 - D Obturator
 - E Pantaloon
- 5** A 43-year-old woman presents to the GP with a lump in the right groin, which she says has been there on and off for a few months, but is now persistent. She has no other symptoms of note. On examination, there is a grape-sized lump below and lateral to the pubic tubercle. It is not tender and there is no cough impulse felt. What is the most likely diagnosis?

 - A Incarcerated femoral hernia
 - B Incarcerated inguinal hernia
 - C Obstructed femoral hernia
 - D Strangulated femoral hernia
 - E Strangulated inguinal hernia
- 6** A 58-year-old man is admitted to hospital with a strangulated paraumbilical hernia that requires urgent operative intervention. He has a history of severe depression and is currently under Section for treatment of his mental condition. The patient does not consent to treatment even though he understands his condition, the benefits of having an operation and the potential outcomes of not having it. Which is the most appropriate course of action?

 - A Doctor can consent for the patient
 - B Doctor can treat the patient under duty of care
 - C Doctor can treat the patient under the Mental Health Act
 - D Patient's refusal to treatment is valid and cannot be overridden
 - E Patient's refusal to treatment is invalid and can be overridden
- 7** An 87-year-old woman with a history of chronic obstructive pulmonary disease with recurrent episodes of bronchitis comes to see you at the GP practice. She has noticed a lump in the lower abdomen when she stands. It does not cause her any concern but she just wanted to get it checked out. On examination, she is obese and there is a soft, non-tender reducible mass in the lower abdomen. You note lower midline scarring from previous caesarean sections overlying the lump. What would be the best course of management?

 - A Conservative treatment unless symptoms change
 - B Elective laparoscopic repair
 - C Elective open repair
 - D Emergency open repair
 - E Prompt laparoscopic repair

- 8 A 56-year-old woman presents to the emergency department with an acutely painful lump in the right groin. She has vomited six times in the last hour and is also complaining of worsening abdominal pain. On examination, there is a grape-sized lump below and lateral to the pubic tubercle. It is warm, red and exquisitely tender. No cough impulse is felt. What would be the best course of management?
- A Elective repair
 - B Emergency repair
 - C Observation alone
 - D Prompt repair
 - E Urgent repair
- 9 Which term best fits the description of hernia given below? A hernia that contains a 'W' loop of intestine within its sac.
- A Gluteal
 - B Inguinal
 - C Maydl
 - D Obturator
 - E Sciatic
- 10 A 39-year-old man presents to the emergency department with a painful lump in the groin. He tells you that this lump has been present on and off for a few months and has previously been asymptomatic. Today, the lump has become painful. On examination, there is an erythematous, tender, irreducible lump above and medial to the pubic tubercle. The patient denies abdominal pain, vomiting or constipation. Which is the best descriptive term for the hernia above?
- A Incarcerated hernia
 - B Perforated hernia
 - C Richter hernia
 - D Sliding hernia
 - E Strangulated hernia

Answers 1-10

GET AHEAD MCQS

- 1 **A Direct inguinal hernia** — Protrusion through Hesselbach's triangle above the inguinal ligament, not reaching scrotum.
- 2 **C Lumbar** — Petit's (lumbar) hernias occur through the inferior lumbar triangle bounded by these muscles.
- 3 **D Incisional hernia** — Incisional hernias protrude through a previous scar, often soft and reducible with a wide neck.
- 4 **E Pantaloon** — Pantaloon hernia straddles the inferior epigastric vessels with both direct and indirect components.
- 5 **A Incarcerated femoral hernia** — Irreducible painless lump below and lateral to pubic tubercle indicates an incarcerated femoral hernia.
- 6 **D Patient's refusal to treatment is valid and cannot be overridden** — He has capacity despite mental illness, so his informed refusal must be respected.
- 7 **A Conservative treatment unless symptoms change** — Asymptomatic incisional hernias in high-risk patients can be managed conservatively.
- 8 **B Emergency repair** — Acute obstruction/strangulation risk in a femoral hernia mandates immediate surgical repair.
- 9 **C Maydl** — **UPDATED** Maydl's hernia has two loops (W) in the sac; the intervening intra-abdominal loop may strangulate unseen, so it must be checked at surgery.
- 10 **C Richter hernia** — Only part of the bowel wall is entrapped, causing pain but no complete obstruction.

10 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 You are a house officer in general surgery clerking a 65-year-old man (with COPD) for elective inguinal hernia repair. He says he had a chest X-ray 6 weeks ago. Which investigation is most highly recommended prior to his surgery?
- A Full blood count
 - B ECG
 - C Chest X-ray
 - D Urea and electrolytes
 - E Clotting screen
- 2 A 32-year-old man is complaining of increasing pain 2 days post inguinal hernia repair. He is currently on paracetamol 1 g four times a day. What is the next most appropriate step in management?
- A Codeine
 - B Tramadol
 - C Morphine
 - D Co-codamol
 - E Diclofenac

- 3** A 67-year-old male has a swelling in the right groin that appears and disappears. It lies above and medial to the pubic tubercle and protrudes on coughing. It reappears despite pressure over the deep inguinal ring. What hernia is this most likely to be?
A Incisional hernia
B Direct inguinal hernia
C Indirect inguinal hernia
D Femoral hernia
E Umbilical hernia
- 4** A 25-year-old weightlifter has a lump near his right groin. On exam it reduces upwards and laterally, moves down medially, and is controlled by pressure over deep ring. Most likely diagnosis?
A Indirect inguinal hernia
B Direct inguinal hernia
C Femoral hernia
D Richter hernia
E Sliding hernia
- 5** A 53-year-old man has a lump near his groin. It reduces on lying down and immediately reaches full size on standing. On coughing it protrudes straight and is not controlled by pressure over the deep ring. Most likely diagnosis?
A Indirect inguinal hernia
B Direct inguinal hernia
C Femoral hernia
D Richter hernia
E Sliding hernia
- 6** A 35-year-old woman has a lump in her groin. It is below and lateral to the pubic tubercle. Most likely diagnosis?
A Indirect inguinal hernia
B Direct inguinal hernia
C Femoral hernia
D Richter hernia
E Sliding hernia
- 7** A 25-year-old woman complains of a painful swelling at the navel since birth. On exam there is an obvious swelling at the umbilicus. Most likely diagnosis?
A Epigastric hernia
B Spigelian hernia
C Umbilical hernia
D Femoral hernia
E Paraumbilical hernia
- 8** A 43-year-old man has epigastric pain and a tender, irreducible midline swelling. Most likely diagnosis?
A Epigastric hernia
B Umbilical hernia
C Paraumbilical hernia
D Richter hernia
E Sliding hernia
- 9** A 42-year-old woman has a lump at the lateral edge of the rectus sheath. Most likely diagnosis?
A Epigastric hernia
B Umbilical hernia
C Spigelian hernia
D Paraumbilical hernia
E Richter hernia
- 10** A 65-year-old woman has 2 days of abdominal discomfort and vomiting, plus right-sided hip pain. An abdominal X-ray confirms small bowel obstruction. Emergency laparotomy is performed. Most likely diagnosis?
A Epigastric hernia
B Obturator hernia
C Umbilical hernia
D Paraumbilical hernia
E Richter hernia

Answers 1-10

CRASH COURSE MCQS

- 1** **B ECG** — NICE guidelines recommend ECG in older patients with cardiorespiratory disease; chest X-ray not repeated if recent.
- 2** **E Diclofenac** — According to WHO analgesic ladder, adding an NSAID is next step.
- 3** **B Direct inguinal hernia** — **UPDATED** Protrudes through posterior wall medial to inferior epigastric vessels; typical direct hernia.
- 4** **A Indirect inguinal hernia** — Enters internal ring lateral to epigastric vessels, typical indirect path.
- 5** **B Direct inguinal hernia** — Bulges medially, not controlled by deep ring pressure.
- 6** **C Femoral hernia** — Femoral hernias present below and lateral to pubic tubercle.
- 7** **C Umbilical hernia** — **UPDATED** A swelling through the umbilical scar present since birth is a persistent congenital umbilical hernia; paraumbilical hernias are acquired, beside the umbilicus.
- 8** **A Epigastric hernia** — Occurs in midline above umbilicus; often tender and irreducible.
- 9** **C Spigelian hernia** — Protrusion through Spigelian fascia at the lateral rectus border.
- 10** **B Obturator hernia** — Elderly women; small bowel obstruction passing through obturator canal.

Abdomen EMQs

Extended matching questions on abdominal pain, masses, splenomegaly, pancreatic and hepatobiliary disease.

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MCQS

79

EMQ SCENARIOS

5

SOURCES

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REPEATED IDEAS

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Local Exams EMQs

SET 1 OF 2

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

THEME 1 Surgical imaging

- A Abdominal ultrasound
- B Intravenous urography (IVU)
- C Dynamic (triphasic) CT
- D Endoscopic ultrasound
- E Barium swallow
- F Barium meal
- G Barium (contrast) enema
- H Barium follow-through

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A child has bile-stained vomiting. What is the best initial imaging?
- 2 A patient has a hydroureter but no stone is seen. Which study is best for diagnosis?
- 3 A patient has rectal bleeding, weight loss and a rectal mass on DRE. Which test best shows the depth of invasion?
- 4 A 36-year-old woman has a 6-cm solitary liver mass found on an infertility ultrasound.
- 5 A 2-month-old has episodes of crying, an empty right iliac fossa and a sausage-shaped mass. Which test is diagnostic and therapeutic?

THEME 2 Pancreatic disorders

- A Acute pancreatitis
- B Chronic pancreatitis
- C Carcinoma of the pancreas
- D Intraductal papillary mucinous neoplasm
- E Gastrinoma
- F Somatostatinoma
- G Glucagonoma
- H Ampullary tumour
- I Pancreatic pseudocyst
- J Pancreatic hydatid cyst
- K Insulinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 50-year-old with vague abdominal pain has been treated for multiple gastric and duodenal ulcers.
- 2 A patient has recurrent episodes of loss of consciousness, sweating and dizziness, and eats large amounts of food.
- 3 A patient has chronic malnutrition and says food always passes completely undigested.

- 4 A patient with a previously treated liver hydatid cyst has upper abdominal pain; imaging shows an additional cyst in the pancreas.
- 5 A patient with gallstones for 10 years has recurrent upper abdominal pain and maldigestion.
- 6 A patient has upper abdominal pain and fever. CT shows a markedly dilated main pancreatic duct with a filling defect.

THEME 3 Adrenal disorders

- A Adrenal metastasis
- B Adrenal adenoma (non-functioning)
- C Cushing syndrome
- D Angiomyolipoma
- E Conn syndrome
- F Pheochromocytoma
- G Adrenal haemorrhage

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Episodes of hypertension, sweating and palpitations.
- 2 Resistant hypertension with hypokalaemia.
- 3 Central obesity, striae and a buffalo hump.
- 4 A patient with lung cancer has bilateral adrenal masses on CT; they secrete no hormones.
- 5 A woman has an incidental 2-cm non-secreting adrenal mass on CT.

THEME 4 Obstructive jaundice

- A Cholangiocarcinoma
- B Ascending cholangitis
- C Drug-induced hepatitis
- D Choledocholithiasis
- E Mirizzi syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Fever, jaundice, right upper quadrant pain and a dilated bile duct.
- 2 Jaundice with a dilated bile duct on ultrasound, but the stone is in the gallbladder.
- 3 Jaundice with dilated intrahepatic ducts and a contracted gallbladder on ultrasound.
- 4 A patient on chemotherapy develops jaundice, fever and right upper quadrant pain.

- 5 A patient has jaundice and a dilated common bile duct on ultrasound.

Answers · Themes 1-4

LOCAL EXAMS EMQS

Theme 1 · Surgical imaging

- 1 **F** Barium meal — Bilious vomiting needs an urgent upper GI contrast study to show the DJ flexure and exclude malrotation with volvulus.
- 2 **B** Intravenous urography (IVU) — Urography (IVU, or CT urography today) shows the level and cause of ureteric obstruction such as a stricture or filling defect.
- 3 **D** Endoscopic ultrasound — Endorectal ultrasound (or pelvic MRI) gives local T staging of rectal cancer.
- 4 **C** Dynamic (triphasic) CT — A solid liver lesion is characterised by multiphase (triphasic) contrast CT or MRI.
- 5 **G** Barium (contrast) enema — Intussusception is confirmed on ultrasound and reduced by air or contrast enema (diagnostic and therapeutic).

Theme 2 · Pancreatic disorders

- 1 **E** Gastrinoma — Multiple or recurrent peptic ulcers suggest Zollinger–Ellison syndrome from a gastrinoma.
- 2 **K** Insulinoma — Fasting hypoglycaemia relieved by eating (Whipple triad) with weight gain is an insulinoma.
- 3 **B** Chronic pancreatitis — Steatorrhoea and maldigestion are exocrine failure from chronic pancreatitis.
- 4 **J** Pancreatic hydatid cyst — A pancreatic cyst in a patient with liver hydatid disease is a pancreatic hydatid cyst.
- 5 **B** Chronic pancreatitis — Recurrent attacks of gallstone pancreatitis can progress to chronic pancreatitis with exocrine insufficiency.

- 6 **D** Intraductal papillary mucinous neoplasm — A markedly dilated main duct with filling defects (mucin) is a main-duct IPMN; it has high malignant potential and needs resection.

Theme 3 · Adrenal disorders

- 1 **F** Pheochromocytoma — Paroxysmal hypertension, sweating and palpitations is pheochromocytoma.
- 2 **E** Conn syndrome — Resistant hypertension with hypokalaemia is primary hyperaldosteronism (Conn syndrome).
- 3 **C** Cushing syndrome — Central obesity, purple striae and a buffalo hump are Cushing syndrome.
- 4 **A** Adrenal metastasis — Bilateral non-functioning adrenal masses in lung cancer are metastases (the adrenal is a common site).
- 5 **B** Adrenal adenoma (non-functioning) — A small (< 4 cm), non-functioning, homogeneous adrenal incidentaloma is an adenoma; follow-up only.

Theme 4 · Obstructive jaundice

- 1 **B** Ascending cholangitis — Charcot triad (fever, jaundice, RUQ pain) with a dilated duct is ascending cholangitis.
- 2 **E** Mirizzi syndrome — A stone impacted in the gallbladder neck compressing the common hepatic duct is Mirizzi syndrome.
- 3 **A** Cholangiocarcinoma — Intrahepatic dilatation with a collapsed gallbladder means a hilar block: cholangiocarcinoma.
- 4 **C** Drug-induced hepatitis — Jaundice during chemotherapy without biliary dilatation suggests drug-induced hepatitis.
- 5 **D** Choledocholithiasis — Jaundice with a dilated CBD most commonly means a CBD stone (choledocholithiasis).

SET 2 OF 2

THEME 5 Liver and biliary disease

- A Choledocholithiasis
- B Non-alcoholic fatty liver disease
- C Viral hepatitis
- D Cholangiocarcinoma
- E Liver metastases
- F Alcoholic liver disease
- G Mirizzi syndrome
- H Stauffer syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 30-year-old nurse has jaundice, hepatomegaly and ALT and AST > 1000 IU/L.
- 2 A woman with gallstones for 4 years has acute abdominal pain for 24 h; the CBD is normal but ALP is raised.

- 3 A 40-year-old diabetic man has slightly raised ALT and AST (about 93 IU/L).
- 4 A patient with renal cell carcinoma has a 7-cm liver mass on ultrasound.
- 5 A middle-aged man has painless jaundice, weight loss and a palpable, non-tender, distended gallbladder.

THEME 6 Causes of jaundice

- A Viral hepatitis
- B Chronic liver disease
- C Choledocholithiasis
- D Carcinoma of the pancreas
- E Gilbert syndrome
- F Drug-induced hepatitis
- G Hepatocellular carcinoma
- H Liver metastases
- I Primary sclerosing cholangitis
- J Common bile duct ligation
- K Mirizzi syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 22-year-old student develops mild jaundice during exam stress. LFTs and haemoglobin are normal; bilirubin is unconjugated and rises with fasting.
- 2 A 60-year-old man has painless progressive jaundice, dark urine, pale stools and weight loss, with a palpable non-tender gallbladder.
- 3 A 30-year-old man has fatigue, nausea and yellow eyes for a week; ALT and AST exceed 1000 IU/L. Two colleagues have a similar illness.

PREVIOUS YEARS

Previous-years college examinations

THEME 7 Complications of splenectomy

- A Atelectasis
- B OPSI
- C Pancreatic fistula
- D Pancreatitis
- E Pleural effusion
- F Pneumonia
- G Pneumothorax
- H Pulmonary embolism

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 5-year-old boy with sore throat, high-grade fever, and fatigue, which rapidly progresses to confusion post-splenectomy.

- 2 A 52-year-old male develops shortness of breath immediately post-difficult splenectomy; breath sounds are absent on the left with hyperresonance.
- 3 A 61-year-old male, 24 hours post-splenectomy, presents with low-grade fever and reduced left chest breath sounds.
- 4 A 63-year-old patient, 5 days post-splenectomy, develops severe dyspnea, tachypnea, tachycardia with hemoptysis and a tender swollen left leg.

THEME 8 Pancreatic endocrine tumours

- A Insulinoma
- B Carcinoid
- C Gastrinoma
- D Glucagonoma
- E Non-functioning tumour
- F PPoma
- G Somatostatinoma
- H VIPoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 24-year-old presented with severe watery diarrhea that was unresponsive to antimicrobial and antispasmodic therapy. Ultrasound showed a 1.7 cm solid mass in the body of the pancreas with no liver metastasis.
- 2 A 35-year-old mentally challenged male presented with recurrent syncopal episodes. His BMI is 42, and a CT scan showed a 2 cm mass in the body of the pancreas.
- 3 A 42-year-old female presented with gallstones and greasy loose bowel movements, along with a recent history of diabetes mellitus. MRI revealed a 2.5 cm mass in the head of the pancreas.
- 4 A 46-year-old male presented with migratory erythema with necrosis in his trunk and a 2-year history of uncontrolled diabetes mellitus. An ultrasound revealed a 3 cm mass in the tail of the pancreas.
- 5 A 50-year-old male with a history of recurrent duodenal ulceration presented with hematemesis and melena despite adequate acid suppression. Somatostatin scintigraphy revealed a 1 cm mass in the duodenal wall adjacent to the pancreas.

Answers • Themes 5–8

LOCAL EXAMS EMQS

Theme 5 • Liver and biliary disease

- 1 **C Viral hepatitis** — Transaminases > 1000 in a healthcare worker suggest acute viral hepatitis (hepatitis B from needle-stick).
- 2 **G Mirizzi syndrome** — A raised ALP with a normal CBD in a patient with gallstones suggests extrinsic CHD compression by a stone in Hartmann pouch (Mirizzi).

- 3 **B Non-alcoholic fatty liver disease** — Mildly raised transaminases in a diabetic (metabolic syndrome) is most often non-alcoholic fatty liver disease.
- 4 **E Liver metastases** — A liver mass in a patient with a known primary cancer is most likely a metastasis; Stauffer syndrome causes abnormal LFTs without metastases.

- 5 **D Cholangiocarcinoma** — Painless jaundice with a palpable gallbladder means distal biliary obstruction by tumour (distal cholangiocarcinoma or pancreatic head cancer).

Theme 6 · Causes of jaundice

- 1 **E Gilbert syndrome** — Isolated unconjugated hyperbilirubinaemia worsened by fasting or stress is Gilbert syndrome; no treatment needed.
- 2 **D Carcinoma of the pancreas** — Painless jaundice with a palpable gallbladder (Courvoisier law) suggests carcinoma of the pancreatic head.
- 3 **A Viral hepatitis** — Very high transaminases with a cluster of cases is acute viral hepatitis (hepatitis A).

Theme 7 · Complications of splenectomy

- 1 **B OPSI** — The presentation is indicative of Overwhelming Post-Splenectomy Infection (OPSI).
- 2 **G Pneumothorax** — Postoperative shortness of breath with absent breath sounds and hyperresonance suggests a pneumothorax.

- 3 **A Atelectasis** — Reduced breath sounds and low-grade fever post-splenectomy are characteristic of atelectasis.
- 4 **H Pulmonary embolism** — Dyspnoea, tachycardia and haemoptysis with a swollen leg 5 days after surgery indicate DVT with pulmonary embolism.

Theme 8 · Pancreatic endocrine tumours

- 1 **H VIPoma** — Secretory watery diarrhoea unresponsive to treatment with a pancreatic mass is a VIPoma (Verner–Morris–on syndrome, pancreatic cholera).
- 2 **A Insulinoma** — Recurrent faints from fasting hypoglycaemia, with obesity from frequent eating and a small pancreatic mass, suggest an insulinoma.
- 3 **G Somatostatinoma** — Diabetes, gallstones and steatorrhoea form the inhibitory triad of a somatostatinoma, usually in the pancreatic head.
- 4 **D Glucagonoma** — Glucagonoma is associated with necrolytic migratory erythema and uncontrolled diabetes mellitus.
- 5 **C Gastrinoma** — Refractory duodenal ulceration with a duodenal wall mass in the gastrinoma triangle indicates a gastrinoma (Zollinger–Ellison syndrome).

1 THEMES · 6 SCENARIOS · AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love EMQs

THEME 1 Retroperitoneal and Abdominal Pathologies

- A Idiopathic retroperitoneal fibrosis
- B Mesenteric cyst
- C Retroperitoneal liposarcoma
- D Right subhepatic abscess
- E Small bowel obstruction from adhesions
- F Tuberculous ascites

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 62-year-old man presents with generalised colicky abdominal pain with bilious vomiting for 48 hours. His symptoms have been intermittent. He has a distended abdomen with generalised tenderness without any rigidity or rebound tenderness. Following an operation for left hemicolectomy 3 years ago through a long midline incision, he has had several such episodes, some of them requiring admission.
- 2 A 58-year-old man complains of malaise, bilateral backache, nausea and occasional vomiting for 2 months. Clinically there is nothing significant except for bilateral hydroceles, of which the patient was not aware. Routine blood tests reveal: Hb 10 g/dL, ESR 88, CRP 292, urea 22 mmol/L, creatinine 256 µmol/L, K 6.1 mmol/L, Na 135 mmol/L, Cl 98 mmol/L and HCO₃ 18 mmol/L.

- 3 A 45-year-old woman complains of an abdominal lump that she noticed in the bath 6 weeks ago. She experiences some generalised discomfort, and on examination she appears well with a smooth, fluctuant periumbilical mass that is freely mobile along a diagonal plane from the right subcostal to the left iliac fossa.
- 4 A 50-year-old man underwent a laparoscopic closure of a perforated duodenal ulcer. His postoperative period during the first 4–5 days was uneventful; thereafter, he developed hiccoughs, pain in the right subcostal area and right shoulder tip, swinging pyrexia, toxicity, tachypnoea, and tenderness in the right upper quadrant with overlying skin oedema.
- 5 A 38-year-old woman, who arrived from the Indian subcontinent 1 year ago, complains of general malaise, cough, and low-grade evening temperatures for 6 months. She has experienced weight loss and on examination has central abdominal distension with a sensation of an encysted periumbilical mass.
- 6 A 47-year-old woman complains of feeling generally unwell with weight loss for 6 months. She also reports discomfort over her entire left side of the abdomen accompanied by constipation. On examination, there is marked fullness in her left lumbar region and loin.

Answers • Theme 1

BAILEY & LOVE EMQs

Theme 1 • Retroperitoneal and Abdominal Pathologies

- 1 **E Small bowel obstruction from adhesions** — A history of previous abdominal surgery with recurrent, intermittent symptoms is highly suggestive of adhesive small bowel obstruction, supported by the absence of peritoneal irritation.
- 2 **A Idiopathic retroperitoneal fibrosis** — The combination of renal impairment, markedly elevated inflammatory markers, and bilateral hydroceles suggests a compressive retroperitoneal process, typical of idiopathic retroperitoneal fibrosis.
- 3 **B Mesenteric cyst** — A freely mobile, smooth, and fluctuant periumbilical mass in an otherwise well patient is characteristic of a mesenteric cyst.
- 4 **D Right subhepatic abscess** — The onset of systemic toxicity, hiccoughs, right subcostal and shoulder tip pain after upper abdominal surgery strongly indicates the development of a right subhepatic abscess.
- 5 **F Tuberculous ascites** — Chronic constitutional symptoms, weight loss, and abdominal distension in a patient from a TB-endemic region point to tuberculous ascites with peritoneal involvement.
- 6 **C Retroperitoneal liposarcoma** — A deep-seated, slowly enlarging mass in the left abdomen accompanied by systemic symptoms is highly suggestive of a retroperitoneal liposarcoma.

4 THEMES • 20 SCENARIOS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford EMQs

THEME 1 Abdominal masses

- A Appendix mass
- B Colonic carcinoma
- C Diverticular abscess
- D Femoral hernia
- E Inguinal hernia
- F Lymphoma
- G Palpable bladder
- H Pelvic abscess
- I Renal transplant

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old diabetic man has a painless, smooth, firm mass in the right iliac fossa. He also has a radiocephalic fistula.
- 2 A 75-year-old woman has a haemoglobin of 9.8 g/dL, mean corpuscular volume of 72 fL, and ferritin of 6 mcg/L. There is a palpable mass in the right iliac fossa.
- 3 A 10-year-old boy with 1 week of vague abdominal pains, 24 h of swinging pyrexia, and a tender, firm mass in the right iliac fossa.
- 4 A 75-year-old woman with 48 h of vomiting, abdominal distension, and a new swelling in the left groin which appears to arise below the inguinal ligament.
- 5 A 70-year-old man with 5 weeks of intermittent left iliac fossa pain, anorexia, and anergia. He has lost 2 kg of weight and has had low-grade fevers for the last 3 nights. There is a tender mass in the left lower quadrant.

THEME 2 Abdominal incisions

- A Gridiron
- B Groin
- C Kocher right subcostal
- D Lanz
- E Lower midline
- F Paramedian
- G Pfannenstiel
- H Rooftop
- I Subumbilical

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 12-year-old boy undergoing open excision of an appendix mass secondary to acute appendicitis through an oblique muscle-splitting incision centred on McBurney's point.
- 2 A 28-year-old woman with acute lower abdominal pain from a suspected ruptured ovarian cyst is undergoing a diagnostic procedure to confirm the diagnosis.
- 3 A 43-year-old woman undergoing hysterectomy for intermenstrual bleeding secondary to uterine leiomyomas ('fibroids').
- 4 A 55-year-old man with a past history of aortic aneurysm surgery undergoing elective open surgery for symptomatic gallstones.
- 5 A 72-year-old man undergoing radical gastrectomy for gastric carcinoma.

THEME 3 Abdominal wall muscles

- A External oblique
- B Iliacus
- C Internal oblique
- D Psoas major
- E Pyramidalis
- F Quadratus lumborum
- G Rectus abdominis
- H Transversus abdominis
- I Serratus anterior

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 The muscle aponeurosis first encountered in the dissection during an open appendicectomy using a right iliac fossa incision.
- 2 The muscle most likely to be involved with retroperitoneal abscess (e.g. retrocaecal abscess) formation.
- 3 The muscle of the anterior abdominal wall which has muscle fibres running medially and superiorly.
- 4 The muscle with a thickened aponeurosis where it forms the inguinal ligament.
- 5 The muscle that must be divided during surgical retroperitoneal approaches to the kidney.

THEME 4 Acute abdominal pain

- A Abdominal aortic aneurysm (leaking)
- B Acute appendicitis
- C Acute cholecystitis
- D Acute mesenteric ischaemia
- E Acute pancreatitis
- F Acute sigmoid diverticulitis
- G Acute urinary retention
- H Biliary colic
- I Renal colic

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 52-year-old woman with 48h of epigastric and right-sided abdominal pain, no jaundice, a temperature of 37.8°C, a pulse of 100bpm, and a white blood cell count of $14.7 \times 10^9/L$.
- 2 A 72-year-old man with 2h of severe central and left-sided abdominal pain radiating to the left flank and back. He is feeling faint with mild abdominal tenderness.
- 3 A 47-year-old woman with 3 days of increasing left lower abdominal pain, mild dysuria, and left iliac fossa and left flank tenderness. Her temperature is 37.7°C, pulse is 82bpm, blood pressure is 135/76mmHg, and her urine is positive for blood and protein.
- 4 A 52-year-old man with 12h of intermittent colicky right-sided abdominal pain, nausea, and retching. His temperature is 37.4°C, pulse is 82bpm, and blood pressure is 135/85mmHg.
- 5 An 81-year-old man with cognitive impairment is reported by his carers to have had 6h of progressively increasing lower abdominal, left iliac fossa, and suprapubic pain. There is mild lower abdominal tenderness, a temperature of 37.2°C, pulse of 88bpm, and blood pressure of 155/85mmHg.

Answers · Themes 1-4

Theme 1 · Abdominal masses

- 1 **I Renal transplant** — Radiocephalic fistula indicates haemodialysis; a transplanted kidney is palpable in the iliac fossa.
- 2 **B Colonic carcinoma** — Right-sided colonic carcinoma commonly presents with iron deficiency anaemia and a RIF mass.
- 3 **A Appendix mass** — Appendiceal abscess presents in children with tender RIF mass and swinging pyrexia.
- 4 **D Femoral hernia** — Femoral hernia presents below the inguinal ligament with obstructive symptoms in elderly women.
- 5 **C Diverticular abscess** — Diverticular abscess causes a septic presentation with a tender LLQ mass.

Theme 2 · Abdominal incisions

- 1 **A Gridiron** — The gridiron is an oblique muscle-splitting incision through McBurney's point; the Lanz is a more transverse skin-crease version.
- 2 **I Subumbilical** — A subumbilical incision is used for laparoscopic access and pneumoperitoneum.
- 3 **G Pfannenstiel** — Pfannenstiel incision provides transverse pelvic access for hysterectomy with minimal muscle division.
- 4 **C Kocher right subcostal** — The Kocher right subcostal incision offers optimal exposure for gallbladder surgery.
- 5 **H Rooftop** — The rooftop incision gives wide upper abdominal access for gastrectomy.

Theme 3 • Abdominal wall muscles

- 1 **A External oblique** — The external oblique aponeurosis is the first layer encountered in a right iliac fossa incision.
- 2 **D Psoas major** — The psoas major lies retroperitoneally and is prone to abscess formation.
- 3 **C Internal oblique** — The internal oblique fibres run medially and superiorly from the iliac crest.
- 4 **A External oblique** — The external oblique's inferior margin thickens to form the inguinal ligament.
- 5 **H Transversus abdominis** — The transversus abdominis must be divided to access the retroperitoneum for kidney surgery.

Theme 4 • Acute abdominal pain

- 1 **C Acute cholecystitis** — Epigastric/right-sided pain for 48 h with fever, tachycardia and leucocytosis but no jaundice indicates acute cholecystitis.

- 2 **A Abdominal aortic aneurysm (leaking)** — Sudden severe pain radiating to the flank and back with faintness in an elderly man is a leaking AAA until proven otherwise; renal colic does not cause collapse.
- 3 **F Acute sigmoid diverticulitis** — Three days of LIF pain with low-grade fever suggests sigmoid diverticulitis; adjacent bladder inflammation explains dysuria and urinary blood/protein.
- 4 **H Biliary colic** — Intermittent colicky right-sided pain with nausea is typical of biliary colic.
- 5 **G Acute urinary retention** — In an elderly patient with cognitive impairment and lower abdominal pain, acute urinary retention should be considered.

1 THEMES • 5 SCENARIOS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead EMQs

THEME 1 Pancreatic tumours

- A Gastrinoma
- B Glucagonoma
- C Insulinoma
- D Non-secreting islet cell tumour
- E Somatostatinoma
- F VIPoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 42-year-old man presents with episodes of faintness and tiredness. The symptoms worsen after exercise and are relieved by eating. On examination, he is pale and sweaty.
- 2 A 48-year-old woman presents with a 4-week history of watery diarrhea, with up to 15 episodes a day. On examination, she is severely dehydrated; laboratory tests show a low potassium level and mild renal impairment, and her stool culture is negative.
- 3 A 38-year-old woman presents with a 2-week history of thirst and urinary frequency, and she admits to losing 3 kg in weight. On examination, she has an erythematous, blistering rash over her buttocks.
- 4 A 54-year-old man presents with coffee-ground vomiting. He describes intermittent epigastric pain and diarrhea over the last month. Endoscopy shows multiple, large ulcers and erosions over the stomach, duodenum, and proximal jejunum.
- 5 A 48-year-old woman presents with a long history of foul-smelling, bulky stools that are becoming more frequent and severe. She also reports increased thirst and passing large amounts of dilute urine. On examination, she appears thin and wasted but there is no other abnormality.

Answers • Theme 1

GET AHEAD EMQS

Theme 1 • Pancreatic tumours

- 1 **C Insulinoma** — Fasting/exercise-induced faintness and sweating relieved by eating (Whipple's triad) suggests an insulinoma.
- 2 **F VIPoma** — Profuse watery diarrhoea with hypokalaemia and dehydration despite negative stool cultures suggests a VIPoma.
- 3 **B Glucagonoma** — Diabetes, weight loss and necrolytic migratory erythema (blistering rash) indicate a glucagonoma.
- 4 **A Gastrinoma** — Multiple peptic ulcers indicate Zollinger-Ellison syndrome due to a gastrinoma.
- 5 **E Somatostatinoma** — Steatorrhoea plus diabetes (thirst, polyuria) and weight loss form the inhibitory syndrome of a somatostatinoma.

Irfan EMQs

THEME 1 Abdominal masses

- A renal cell carcinoma
- B ovarian carcinoma
- C gastric carcinoma
- D sigmoid carcinoma
- E fibroids
- F pancreatic pseudocyst
- G diverticulosis
- H hepatocellular carcinoma
- I caecal carcinoma
- J psoas abscess
- K abdominal aortic aneurysm
- L ovarian cyst

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man collapses with an expansile, pulsatile mass above the umbilicus.
- 2 A 75-year-old man presents with dyspepsia, weight loss and a firm, irregular epigastric mass with a palpable left supraclavicular node.
- 3 A 70-year-old woman presents with a firm, irregular mass in the right iliac fossa and severe microcytic anaemia.
- 4 A 35-year-old woman presents with a progressively enlarging abdominal mass and heavy menstrual bleeding; the mass is knobbly and dull on percussion.
- 5 A 70-year-old alcoholic presents with a tender upper abdominal mass; CT reveals a thick-walled, rounded, fluid-filled mass adjacent to the pancreas.

THEME 2 Splenomegaly

- A malaria
- B pernicious anaemia
- C sarcoidosis
- D idiopathic thrombocytopenic purpura
- E Gaucher's disease
- F infective endocarditis
- G spherocytosis
- H infectious mononucleosis
- I cutaneous leishmaniasis
- J Budd–Chiari syndrome
- K myeloma
- L Felty's syndrome
- M acute lymphoblastic leukaemia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 75-year-old woman has noticed a fullness in the left upper quadrant. On examination, the GP feels that there is a smooth moderately enlarged spleen. Past history includes rheumatoid arthritis and high blood pressure. Routine blood tests show normocytic anaemia and low WCC.
- 2 A 23-year-old man presents with a week's history of fever and sore throat. He developed a macular rash after receiving ampicillin. On examination he has enlarged posterior cervical nodes, palatal petechiae and splenomegaly.
- 3 A 21-year-old female backpacker returning from India presents with flu-like symptoms followed by a periodic fever. She is anaemic, jaundiced and has moderate splenomegaly.
- 4 A 28-year-old woman presents with abdominal pain, vomiting and jaundice. On examination, tender hepatosplenomegaly and ascites are found. She has a history of recurrent miscarriages.
- 5 A 16-year-old child is being investigated for gallstones. Routine blood tests show mild anaemia; clinically there is smooth, painless splenomegaly. A direct Coomb's test is negative and blood film shows reticulocytes and spherocytes.

Answers • Themes 1-2

Theme 1 • Abdominal masses

- 1 **K** abdominal aortic aneurysm — Collapse with an expansile, pulsatile mass above the umbilicus is a ruptured abdominal aortic aneurysm until proven otherwise.

- 2 **C** gastric carcinoma — Dyspepsia, weight loss, an epigastric mass and a left supraclavicular (Virchow) node indicate gastric carcinoma.

- 3 **I** **caecal carcinoma** — Iron-deficiency (microcytic) anaemia with a right iliac fossa mass in an elderly patient is caecal carcinoma.
- 4 **E** **fibroids** — Menorrhagia with a knobbly mass arising from the pelvis indicates uterine fibroids (leiomyomas).
- 5 **F** **pancreatic pseudocyst** — A thick-walled fluid collection next to the pancreas in an alcoholic with pancreatitis is a pseudocyst.

Theme 2 • Splenomegaly

- 1 **L** **Felty's syndrome** — Rheumatoid arthritis with splenomegaly and neutropenia (low WCC) is Felty's syndrome.

- 2 **H** **infectious mononucleosis** — Fever, sore throat, posterior cervical nodes, palatal petechiae and a rash after ampicillin indicate EBV infectious mononucleosis.
- 3 **A** **malaria** — Periodic fever, anaemia, jaundice and splenomegaly after travel to India suggest malaria.
- 4 **J** **Budd–Chiari syndrome** — Tender hepatosplenomegaly, ascites and jaundice with recurrent miscarriage (antiphospholipid syndrome) suggest Budd–Chiari hepatic vein thrombosis.
- 5 **G** **spherocytosis** — Gallstones, anaemia, splenomegaly and spherocytes with a negative direct Coombs test indicate hereditary spherocytosis.

- 25 Urinary Tract Infection
- 26 Urinary Stones
- 27 Renal Tumours
- 28 Urological Investigation & Haematuria
- 29 Urological Emergencies & Trauma
- 30 Prostate
- 31 Congenital Urology
- 32 Scrotum & Testis
- 33 Bladder Tumours
- 34 Urology EMOs

06

UROLOGY



Urinary Tract Infection

Cystitis, pyelonephritis, prostatitis, renal abscess and urinary tuberculosis.

19

MCQs

0

EMQ SCENARIOS

6

SOURCES

0

REPEATED IDEAS

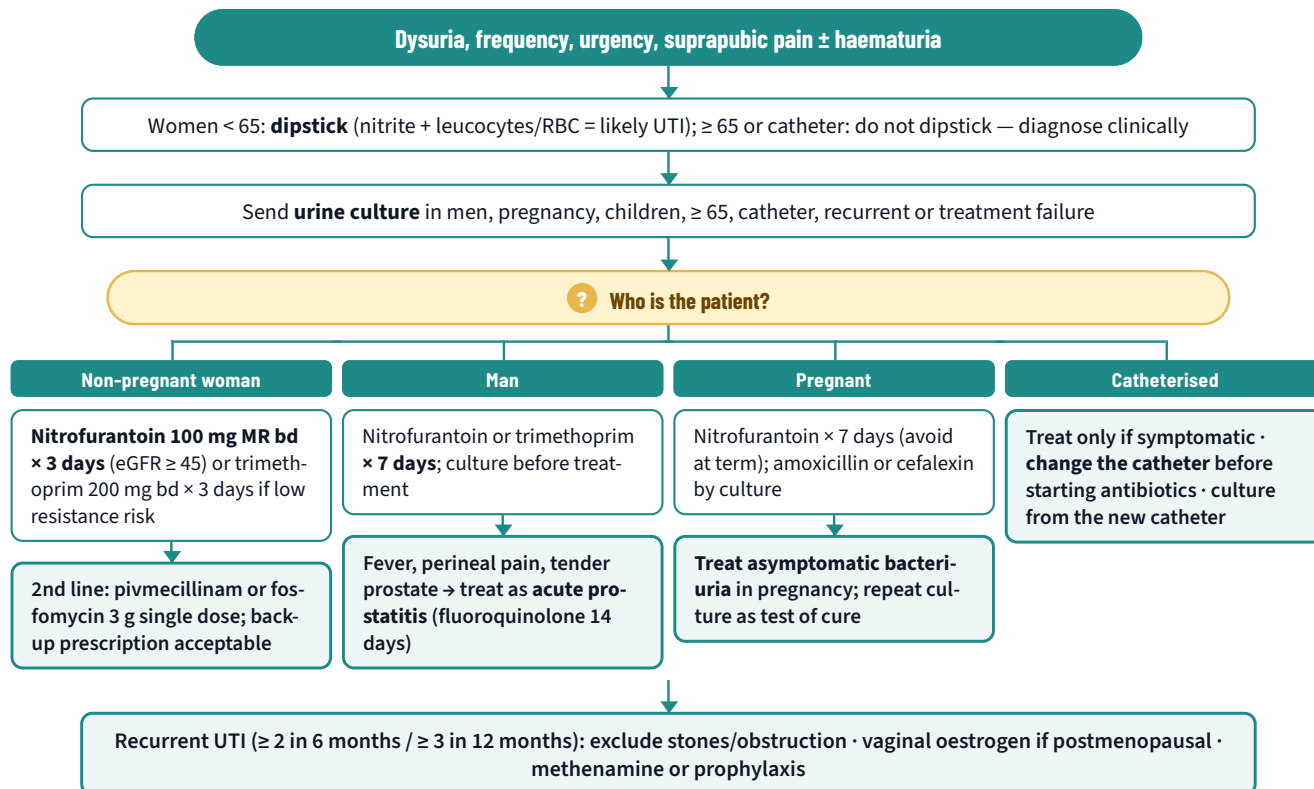
IN THIS CHAPTER

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02	Local Exams MCQs	1 Q	page 364
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Step by step

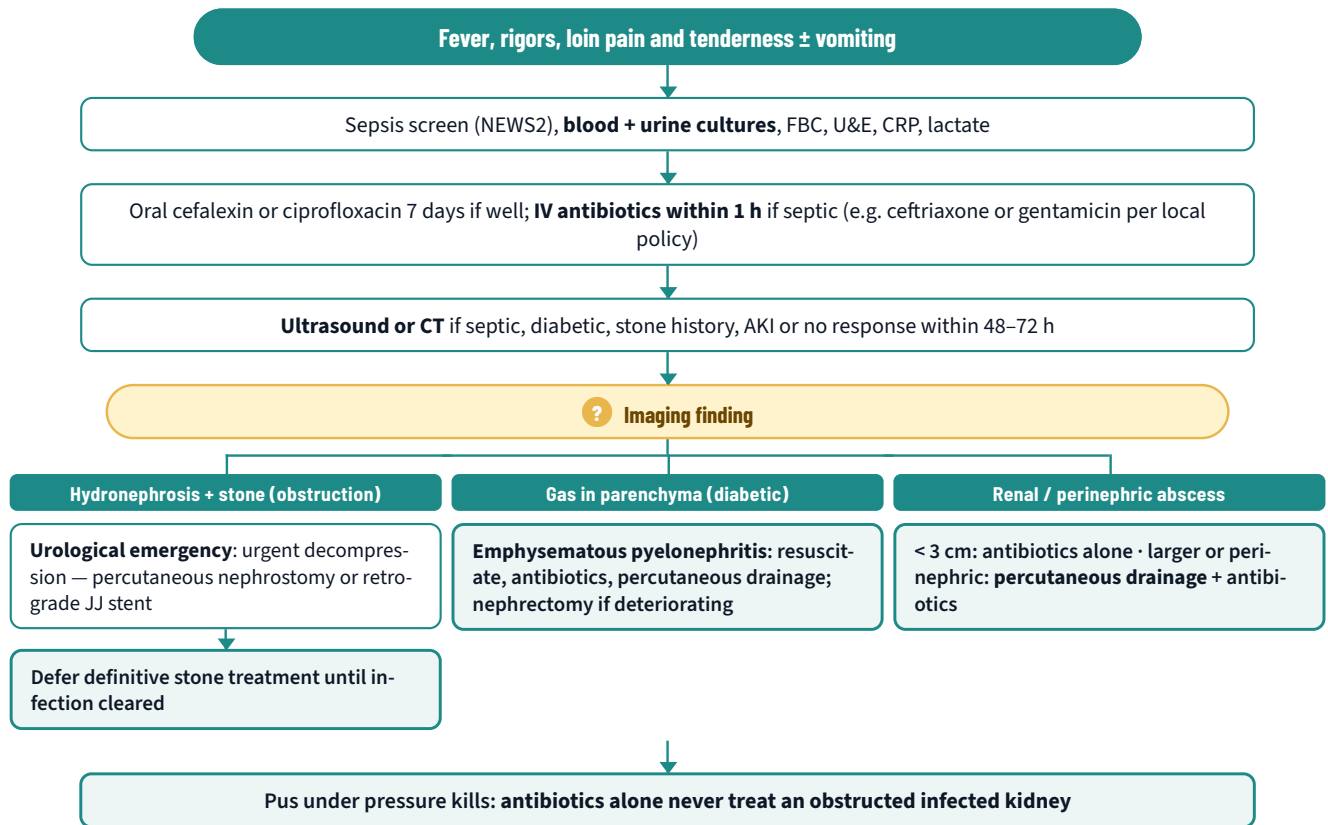
Lower UTI (cystitis): who gets what

NICE NG109 • EAU 2024 UROLOGICAL INFECTIONS



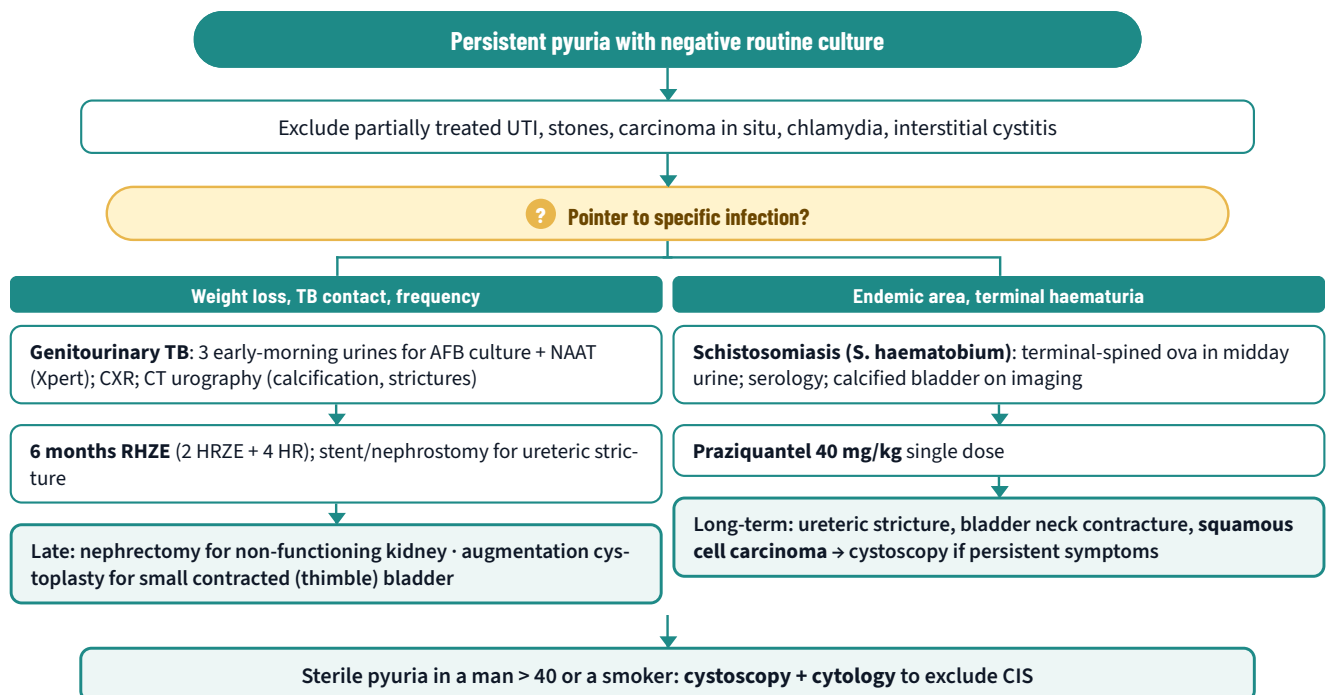
Acute pyelonephritis and the obstructed infected kidney

NICE NG111 · EAU 2024 UROLOGICAL INFECTIONS



Sterile pyuria and specific urinary infections

WHO 2022 TB · EAU 2024 UROLOGICAL INFECTIONS · BAILEY & LOVE 28E



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 22-year-old previously healthy male complains of dysuria. Genital examination: there is urethral discharge. Urinalysis shows 30 to 40 white blood cells per high power field. First void urine sample is positive for *Chlamydia trachomatis*. Which of the following antibiotic regimens is most appropriate?
- A Azithromycin 1 g orally once and ofloxacin 400 mg orally once.
 - B Trimethoprim-sulfamethoxazole two tablets orally twice a day for 3 days.
 - C Ciprofloxacin 500 mg orally twice a day for 2 weeks.
 - D Cephalexin 500 mg orally for 7 days.
 - E Ciprofloxacin 500 mg orally once.

Answer 1

LOCAL EXAMS MCQs

- 1 **A** Azithromycin 1 g orally once and ofloxacin 400 mg orally once. — **UPDATED** Single-dose azithromycin treats *Chlamydia*

(7 days of doxycycline is now first line) and the added single dose of ofloxacin covers co-infecting gonorrhoea.

Bailey & Love MCQs

- 1 Regarding preliminary investigations of the urinary tract, the following statements are true except:
- A If 30% of kidney function is lost, renal failure becomes evident by blood results.
 - B Negative dipstick usually indicates a negative urine culture.
 - C Organisms of $>10^5$ /mL on a midstream specimen of urine (MSU) is due to contamination.
 - D Biochemical examination of urine helps in calculus disease.
 - E Cytological examination of urine is sensitive and specific for poorly differentiated transitional cell cancer.
- 2 Regarding urinary tract infection, the following statements are true except:
- A Ascending infection in the urinary tract is the most common route.
 - B *E. coli* and other Gram-negative organisms are most commonly responsible.
 - C *Proteus* and staphylococci thrive in acidic urine.
 - D Up to 50% of children with urinary infection have an underlying abnormality.
 - E All patients present electively with dysuria and frequency of micturition.
- 3 Regarding upper urinary tract infection, which of the following statements are true?
- A Sterile pyuria should alert one to the possibility of renal tuberculosis.
 - B Chronic pyelonephritis is often associated with vesicoureteric reflux.
 - C Pyonephrosis is an infected hydronephrosis and is most commonly due to a stone-causing obstruction.
 - D Renal carbuncle is most commonly seen in the immunocompromised.
 - E Perinephric abscess might occur from extension of a retrocaecal appendix abscess.
- 4 Regarding urinary fistulae, which of the following statements is not true?
- A Most urinary fistulae are vesicovaginal.
 - B A combined ureterovaginal fistula might be present in 10% of patients.
 - C Examination under anaesthesia (EUA), IVU and cystoscopy are necessary to evaluate them.
 - D When there are multiple tracts, the causes might be radiation, malignancy, or sepsis.
 - E Conservative management by catheter drainage of the bladder is usually successful in a post-hysterectomy fistula.

- 5** Regarding UTI and cystitis, which of the following statements are true?
- A** The condition is more common in women.
 - B** In recurrent infections, haematuria and rigors, cystoscopy and imaging are essential.
 - C** Sterile pyuria is a sinister finding.
 - D** In tuberculous cystitis, the route of infection is usually haematogenous or lymphogenous.
 - E** Carcinoma in situ might present as recurrent abacterial cystitis.
- 6** Regarding inflammation of the urethra, which of the following statements are true?
- A** Meatal ulcer occurs in middle-aged men from sexually-transmitted disease.
 - B** Gonorrhoeal urethritis might lead to urethral stricture.
 - C** The causative agent in nonspecific urethritis (NSU) is unknown in 50% of patients.
 - D** In Reiter's disease conjunctivitis is a feature in 50% of patients.
 - E** Bulbar periurethral abscess might cause urinary extravasation.
- 7** Regarding epididymo-orchitis, which of the following statements are true?
- A** The causes are Chlamydia and gonococci as sexually-transmitted infections.
 - B** The initial symptoms are those of urinary tract infection (UTI).
 - C** In mumps, 18% of men might develop this condition.
 - D** It is a common postoperative complication after prostatectomy.
 - E** Treatment is doxycycline for Chlamydia or a broad-spectrum antibiotic.

Answers 1-7

BAILEY & LOVE MCQs

- 1** **AC** True: A, C — **UPDATED** Renal failure typically manifests after >70% loss of function, and a count of >10⁵/mL indicates infection rather than contamination.
- 2** **CE** True: C, E — **UPDATED** C is false: urease-producing *Proteus* and staphylococci make and thrive in alkaline urine. E is false: UTI often presents acutely, e.g. as pyelonephritis or septicemia.
- 3** **ABCDE** True: A, B, C, D, E — **UPDATED** All are true: sterile pyuria suggests renal TB, pyonephrosis usually follows stone obstruction and a retrocaecal appendix abscess can track to the perinephric space.
- 4** **E** Conservative management by catheter drainage of the bladder is usually successful in a post-hysterectomy fistula.
- **UPDATED** Catheter drainage heals only a minority of small, early post-hysterectomy vesicovaginal fistulae; most need formal vaginal or abdominal repair.
- 5** **ABCE** True: A, B, C, E — **UPDATED** D is false: tuberculous cystitis is almost always secondary to renal TB, spreading down the ureter (descending infection), not by haematogenous or lymphatic routes.
- 6** **BCDE** True: B, C, D, E — **UPDATED** A is false: meatal ulcer is typically seen in circumcised infants (ammoniacal dermatitis from wet napkins), not as an STD in middle-aged men.
- 7** **ABCE** True: A, B, C, E — **UPDATED** A, B, C and E are true; epididymo-orchitis is rare after prostatectomy.

3 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 24-year-old sexually active medical student is diagnosed with a urinary tract infection. Which organism is most commonly associated with community-acquired UTI?
- A** *Escherichia coli*
 - B** *Pseudomonas*
 - C** *Staphylococcus saprophyticus*
 - D** *Staphylococcus aureus*
 - E** *Streptococcus faecalis*
- 2** A 40-year-old female lawyer with recurrent UTIs and chronic pyelonephritis is evaluated. Which sonographic feature is diagnostic?
- A** Absent kidney
 - B** Hydronephrotic kidney
 - C** Multiple renal stones
 - D** Atrophic kidney
 - E** Poor urinary concentration

- 3** Urethral discharge: A 28-year-old student with a 2-day history of dysuria and mucopurulent urethral discharge is confirmed to have *Neisseria gonorrhoeae*. What type of organism is *N. gonorrhoeae*?
- A Lactose-fermenting Gram-negative rod
 - B Maltose-fermenting Gram-negative coccus
 - C Coagulase-positive Gram-positive coccus
 - D Glucose-fermenting Gram-negative coccus
 - E Lactose-non-fermenting Gram-negative rod

Answers 1-3

SBA MCQS

- 1** **A** **Escherichia coli** — *Escherichia coli* is responsible for most community-acquired UTIs.
- 2** **D** **Atrophic kidney** — An atrophic kidney on ultrasound is diagnostic of chronic pyelonephritis.
- 3** **D** **Glucose-fermenting Gram-negative coccus** — **UPDATED** *Gonorrhoeae* is a glucose-fermenting Gram-negative diplococcus.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** 42-year-old woman with type 1 diabetes develops groin and loin pain. Her temperature is 38.5°C. Examination reveals right renal angle and suprapubic tenderness. Her urine dip shows blood, nitrites, white cells, protein, and ketones. It has been sent for culture. Which is the single most appropriate next investigation?
- A Arterial blood gases
 - B Blood cultures
 - C Blood glucose
 - D Renal tract ultrasound scan
 - E X-ray KUB
- 2** 55-year-old woman develops constant right renal angle tenderness, worse on movement. Her temperature is 37.5°C; urine analysis demonstrates haematuria 1+ and white cells 1+, but no nitrites. Her white cell count is $12.3 \times 10^9/L$. Which is the single most appropriate initial investigation?
- A Abdominal CT
 - B CT KUB
 - C Cystoscopy
 - D Intravenous urogram
 - E Renal ultrasound scan

Answers 1-2

OXFORD MCQS

- 1** **C** **Blood glucose** — **UPDATED** Ketonuria in a febrile type 1 diabetic suggests developing ketoacidosis, so check blood glucose (and ketones) first; imaging follows.
- 2** **E** **Renal ultrasound scan** — **UPDATED** Loin tenderness with low-grade fever and leucocytosis suggests pyelonephritis or a perinephric collection; renal ultrasound is the first-line test for obstruction or abscess.

3 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- A 24-year-old woman has a 2-day history of lower abdominal pain with foul-smelling vaginal discharge. She is tender and guarding in the suprapubic region and right iliac fossa. Urine dipstick shows leukocytes. What is the most likely diagnosis?

 - A Appendicitis
 - B Ectopic pregnancy
 - C Mittelschmerz
 - D Pelvic inflammatory disease
 - E Urinary tract infection
- A 4-year-old girl is referred to the urology clinic by the GP as she has been having recurrent urinary tract infections. Examination of the child is unremarkable. Which of the following is most likely to be causing the problem?

 - A Hydronephrosis
 - B Pelvic kidney
 - C Posterior urethral valve
 - D Renal agenesis
 - E Vesicoureteric reflux
- A 24-year-old woman with no past medical history of note comes to see you at the GP practice complaining of a 2-day history of urinary frequency together with a stinging sensation on passing urine. She is systemically well. Which of the following organisms is most likely to have caused her symptoms?

 - A Candida albicans
 - B Escherichia coli
 - C Herpes simplex
 - D Proteus mirabilis
 - E Pseudomonas aeruginosa

Answers 1-3

GET AHEAD MCQs

- D Pelvic inflammatory disease** — PID can present with lower abdominal pain, guarding, discharge, and urinary findings.
- E Vesicoureteric reflux** — **UPDATED** Abnormal retrograde flow of urine from bladder to ureter predisposes to UTIs.
- B Escherichia coli** — **UPDATED** Coli is the most common cause of uncomplicated UTIs in young women.

3 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQs

Crash Course MCQs

- A 62-year-old man has sudden onset loin pain, frequency, dysuria, confusion, fever 38.2°C. Next most appropriate initial investigation?

 - A Urinalysis and culture
 - B Abdominal ultrasound
 - C IV urogram
 - D Abdominal X-ray
 - E PSA
- A 23-year-old woman with frequency, dysuria, T=38.7°C, blood cultures show gram-negative organisms. First management?

 - A Oral antibiotics
 - B IV antibiotics
 - C Oral fluids
 - D Urinary catheterization
 - E Analgesia
- A 45-year-old man has frequency, dysuria, haematuria, suprapubic tenderness. Most likely diagnosis?

 - A Pyelonephritis
 - B Interstitial cystitis
 - C Prostatitis
 - D Acute cystitis
 - E Epididymo-orchitis

Answers 1-3

CRASH COURSE MCQS

- 1 **A Urinalysis and culture** — Suspect complicated UTI or pyelonephritis; check urine first.
- 2 **B IV antibiotics** — **UPDATED** Gram-negative bacteraemia from a urinary source is urosepsis and needs prompt IV antibiotics (with fluids), not oral therapy.
- 3 **D Acute cystitis** — **UPDATED** Frequency, dysuria and haematuria with suprapubic tenderness but no fever or loin pain indicate lower urinary tract infection (acute cystitis).

Urinary Stones

Renal colic, imaging, medical expulsive therapy, ESWL, ureteroscopy and PCNL.

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MCQS

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

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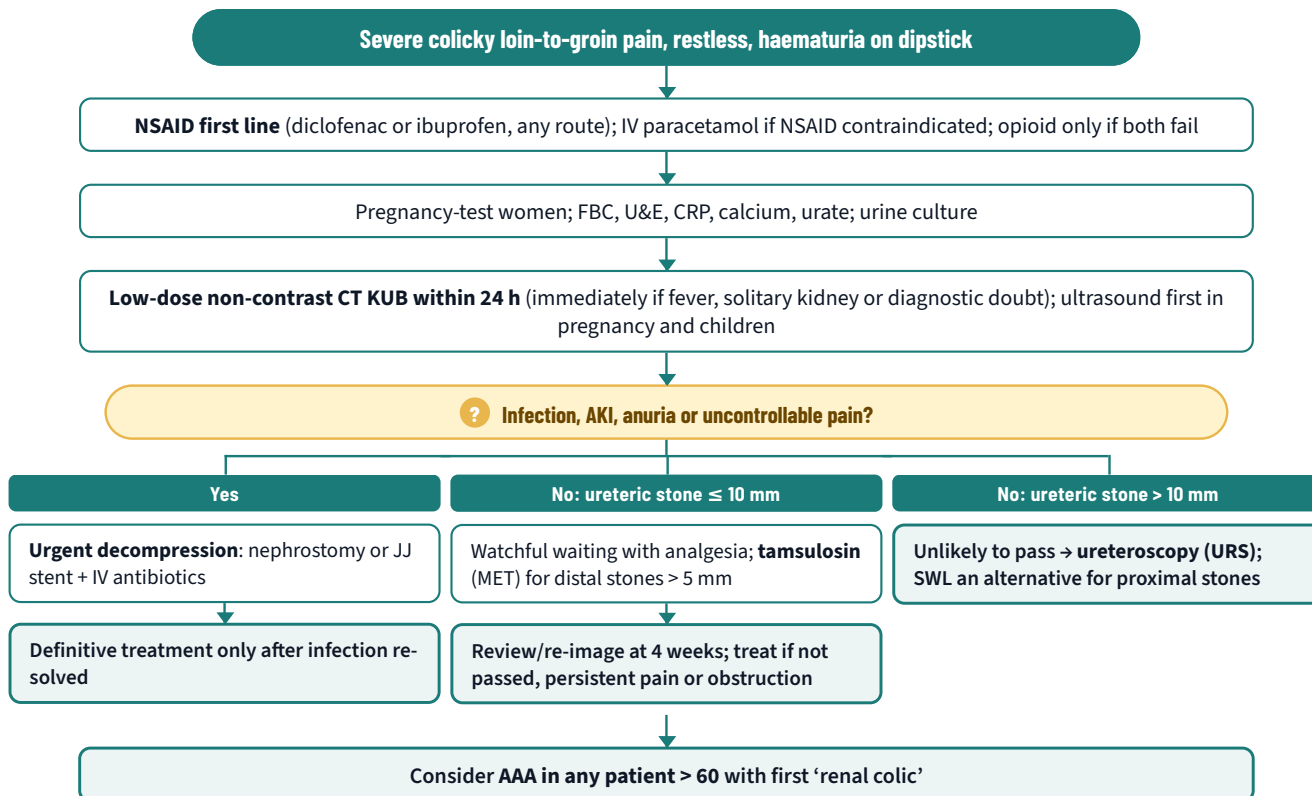
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Step by step

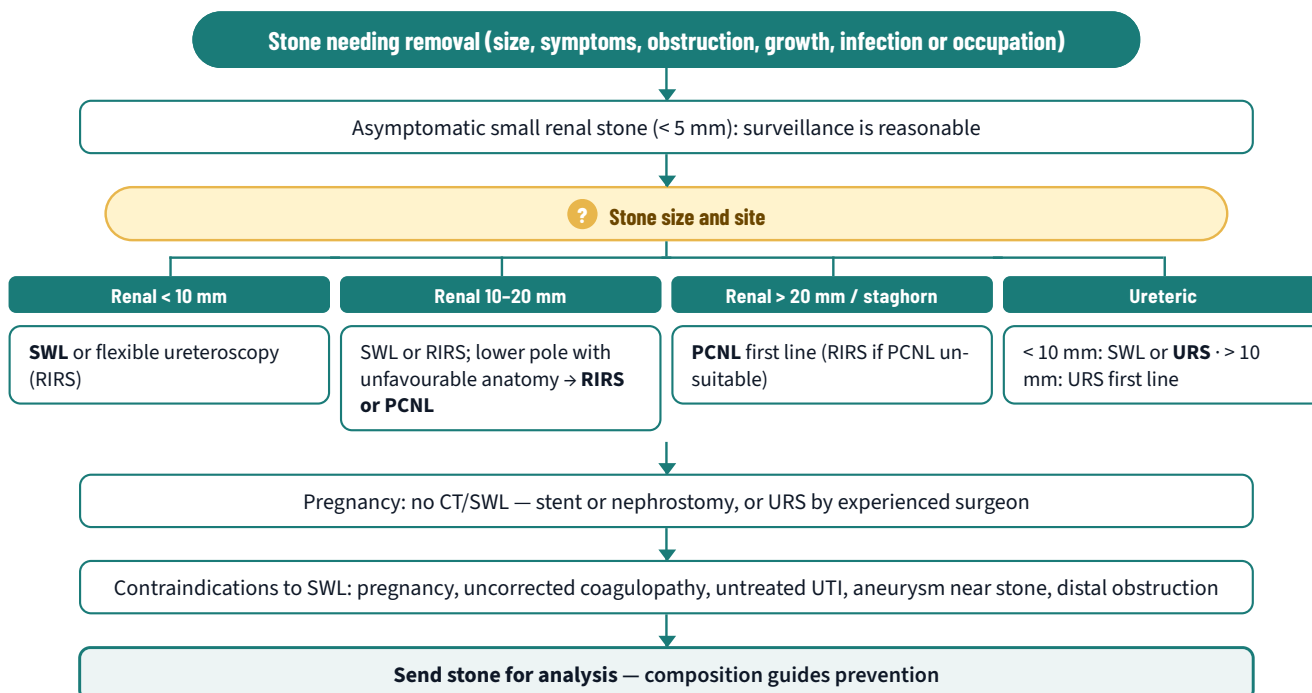
Acute renal colic

EAU 2024 UROLITHIASIS · NICE NG118



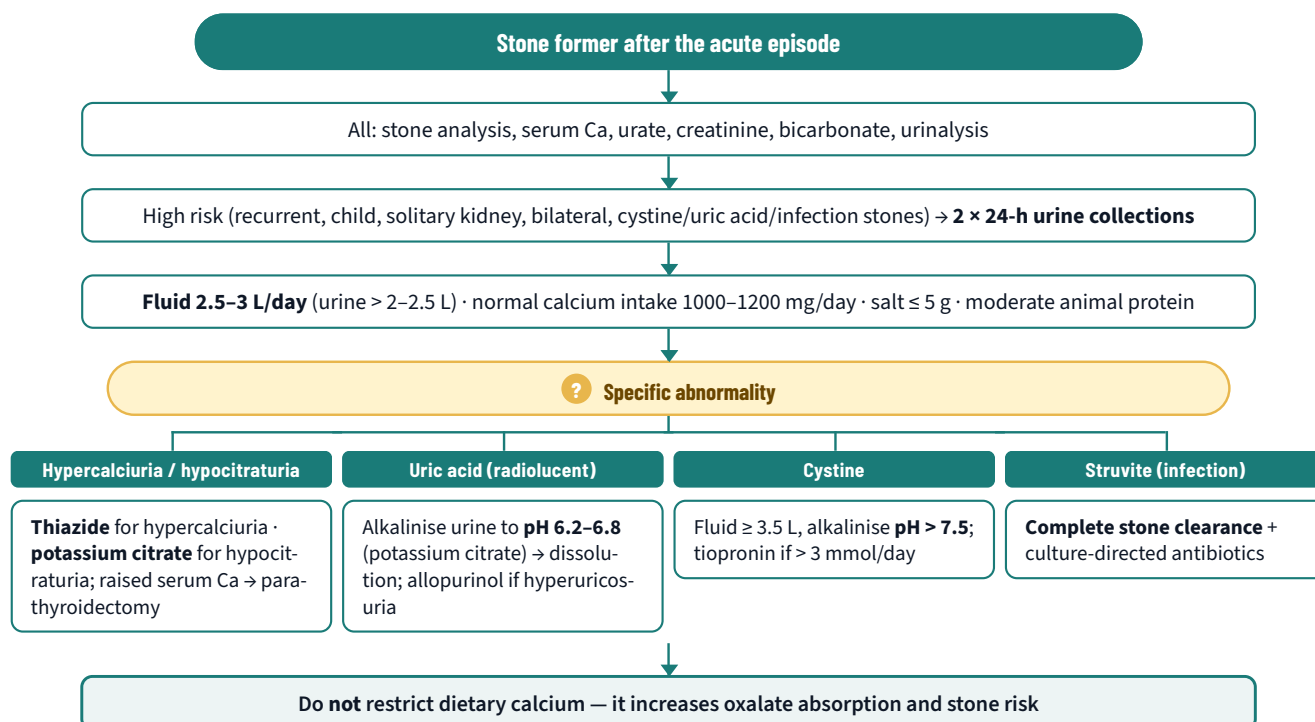
Choosing active stone removal

EAU 2024 UROLITHIASIS · NICE NG118



Metabolic work-up and prevention

EAU 2024 UROLITHIASIS · NICE NG118



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 35-year-old healthy patient presented with left loin pain and microscopic hematuria. K.U.B. revealed radio-opacity at the left flank. Native CT scan revealed a stone with a maximum diameter 12 mm in the left renal pelvis. Which among the following is the most appropriate treatment option?
 - A Percutaneous nephrolithotomy (PCNL).
 - B Extracorporeal shock wave lithotripsy (ESWL).
 - C Alkalinization of urine by using potassium citrate.
 - D Trial of tamsulosin 0.4 mg nocte for 2 weeks.
 - E Open pyelolithotomy.
- 2 A 42-year-old male presents to the emergency unit with severe left loin pain and high-grade fever for the past three days. His native CT shows severe left hydroureteronephrosis ending at an impacted 8-mm lower ureteric stone. The next most appropriate line of treatment for this scenario is:
 - A Antibiotic until fever subsides
 - B Antibiotics and analgesia until the fever subsides.
 - C Antibiotic, analgesia, and ESWL.
 - D Analgesia, antibiotics, and stenting of left ureter.
 - E Analgesia, antibiotics, and ureterolithotomy.

Answers 1-2

LOCAL EXAMS MCQs

- 1 **B Extracorporeal shock wave lithotripsy (ESWL).** — UPDATED
For a renal pelvic stone under 2 cm, ESWL (or flexible ureteroscopy) is first line; PCNL is for stones over 2 cm, and tamsulosin helps only distal ureteric stones.
- 2 **D Analgesia, antibiotics, and stenting of left ureter.** — UPDATED
An infected obstructed kidney is a urological emergency: antibiotics plus urgent decompression (ureteric stent or nephrostomy); definitive stone treatment is delayed.

9 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love MCQs

- 1 Regarding pain of urological origin, which of the following statements are true?
 - A Pain of renal origin is a deep-seated sickening ache and 'bursting' in nature.
 - B Pain from a ureteric stone is colicky and radiates to the groin, scrotum, or labium.
 - C Pain from the urinary bladder is a suprapubic discomfort.
 - D Perineal pain is a penetrating ache and can be referred to the rectum.
 - E Large, slow-growing, space-occupying lesions produce a constant ache.
- 2 Regarding imaging in urological conditions, which of the following statements are true?
 - A All urinary calculi are radiodense.
 - B Pelvic phleboliths cannot be distinguished from lower ureteric calculi on a plain x-ray.
 - C Ultrasound scanning (US) provides the same information as an intravenous urogram (IVU).
 - D IVU can be dangerous.
 - E The procedure of antegrade pyelography can be used therapeutically.
- 3 Regarding urological imaging, the following statements are true except:
 - A Retrograde ureteropyelography can cause septicaemia.
 - B Venography should be used when tumour invasion into the inferior vena cava (IVC) is suspected.
 - C Non-contrast spiral CT should be routine for diagnosis of urinary calculi.
 - D CT scan is crucial in the staging of renal carcinoma.
 - E Radioisotope scanning has a role to ascertain selective renal function.
- 4 Regarding renal calculi, which of the following statements are true?
 - A The common bacteria found as a nidus for urinary stones are staphylococci, streptococci and *Proteus*.
 - B Hyperparathyroidism is found in 5% or less of those presenting with radio-opaque calculi.
 - C If a parathyroid adenoma is found to be the cause of renal calculi, it should be removed before treatment of the calculus.
 - D Pure uric acid stones are radio-opaque.
 - E A staghorn calculus is composed of calcium-ammonium-magnesium phosphate.

- 5** Regarding renal and ureteric calculi, which of the following statements are false?
- A Renal stones are usually visible on a plain x-ray.
 - B The severity of pain of ureteric colic is related to the size of stone.
 - C Hydronephrosis or pyonephrosis with a palpable loin swelling is a common clinical presentation.
 - D Haematuria is common.
 - E There are very few physical signs.
- 6** Which of the following imaging techniques are used for urinary calculus disease?
- A Plain kidney, ureter and bladder (KUB) x-ray
 - B Spiral CT scan
 - C IVU
 - D Ultrasound (US)
 - E Retrograde pyelography
- 7** Regarding management of urinary calculi, the following statements are true except:
- A Ureteric calculi smaller than 1 cm will pass spontaneously.
 - B Infection in the presence of upper urinary tract obstruction due to a stone requires surgical intervention.
 - C Most urinary calculi can be treated by minimal-access surgical techniques.
 - D In bilateral renal stones, the kidney with poorer function is treated first.
 - E Open operations for renal stones are performed through an extraperitoneal loin approach.
- 8** Which of the following statements are true in ureteric stones?
- A The majority pass spontaneously.
 - B If a stone has not passed after 6 weeks, intervention is required.
 - C Stones are commonly arrested at the two sites of ureteric narrowing.
 - D Most stones are treated by ureteroscopic electrohydraulic fragmentation.
 - E Severe renal pain subsiding after a day or two suggests that the stone has passed.
- 9** Regarding bladder stones, which of the following statements are true?
- A Men are more often affected than women.
 - B An oxalate calculus develops in sterile urine.
 - C A cystine calculus is radio-opaque.
 - D Treatment is crushing with an optical lithotrite in all cases.
 - E In men with stones and outflow obstruction from enlarged prostate, both can be dealt with at the same time.

Answers 1-9

- 1** **ABCD** True: A, B, C, D — **UPDATED** E is false: large, slow-growing space-occupying lesions (e.g. hydronephrosis, renal tumours) are often painless or cause only vague discomfort.
- 2** **BDE** True: B, D, E — **UPDATED** Phleboliths may mimic calculi on plain films; IVU carries contrast risks, and antegrade pyelography can be used therapeutically.
- 3** **B** **Venography should be used when tumour invasion into the inferior vena cava (IVC) is suspected.** — **UPDATED** B is false: tumour extension into the IVC is assessed by CT or MRI; venography is now obsolete.
- 4** **ABCE** True: A, B, C, E — **UPDATED** D is false: pure uric acid stones are radiolucent on plain X-ray (visible on CT); staghorn calculi are usually struvite from urease-producing infection.
- 5** **BC** True: B, C — **UPDATED** Pain intensity does not correlate with stone size and palpable swellings are rare.
- 6** **ABCDE** True: A, B, C, D, E — **UPDATED** All are used: non-contrast CT KUB is now the investigation of choice; KUB, US and IVU still have roles, and retrograde pyelography is used in selected cases.
- 7** **AD** True: A, D — **UPDATED** A is false: stones under 5 mm usually pass but many 5–10 mm stones do not. D is false: with bilateral stones the better-functioning kidney is usually treated first.
- 8** **AB** True: A, B — **UPDATED** C is false: stones lodge at several narrowings (PUJ, pelvic brim, VUJ). D is false: most pass, the rest mainly by laser ureteroscopy/ESWL. E is false: pain may settle despite obstruction.
- 9** **ABCE** True: A, B, C, E — **UPDATED** D is false: most bladder stones are fragmented endoscopically, but large or hard stones may need percutaneous or open cystolithotomy.

2 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 40-year-old previously healthy man presents with sudden onset of severe right flank pain radiating to the groin, associated with nausea, sweating, and urinary urgency. Examination is normal except for distress. Which of the following is the most likely diagnosis?
 - A torsion of the right testicle
 - B pyelonephritis
 - C appendicitis
 - D right ureteral calculus
 - E acute urinary retention
- 2 Which of the following is the most appropriate next step in management of a suspected ureteral stone?
 - A insertion of a urethral catheter
 - B IV fluid hydration, IV analgesics, and nonenhanced computed tomography (CT) scan
 - C IV fluid hydration, IV analgesics, and arrangements for lithotripsy
 - D cystoscopy and retrograde pyelogram
 - E urine culture, followed by antibiotic therapy

Answers 1-2

LANGE MCQs

- 1 **D right ureteral calculus** — Classic colicky flank-to-groin pain suggests a ureteric stone.
- 2 **B IV fluid hydration, IV analgesics, and nonenhanced computed tomography (CT) scan** — Noncontrast CT is the best

imaging to confirm and locate the stone; treat pain and hydrate.

6 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 A 40-year-old office executive presents with excruciating left loin pain and a 4 mm radio-opaque stone in the left ureter. What type of stone is more likely?
 - A Xanthine
 - B Uric acid
 - C (Triple) phosphate
 - D Calcium oxalate
 - E Cystine
- 2 A 30-year-old PhD student has acute colicky left loin pain suggestive of a urinary calculus. Which initial investigation should he have?
 - A Dimercaptosuccinic acid (DMSA) scan
 - B Kidney ureter bladder (KUB) radiograph
 - C Intravenous urogram/pyelogram
 - D Flexible cystoscopy
 - E Diethylene triamine pentaacetic acid (DTPA) scan
- 3 A 35-year-old housewife with dysuria, frequency and a large left pelvicalyceal calculus is evaluated; which microorganism is NOT associated with such stones?
 - A Proteus
 - B Klebsiella
 - C Pseudomonas
 - D Mycoplasma
 - E Mycobacteria
- 4 A 75-year-old man with a 24-hour history of left loin pain and a history of gout is suspected to have a urinary calculus. Which substance is most likely the major component?
 - A Xanthine
 - B Uric acid
 - C (Triple) phosphate
 - D Calcium oxalate
 - E Cystine
- 5 A 49-year-old postmenopausal woman is admitted following severe attacks of abdominal pain, nausea and vomiting. The pain is colicky in nature, starting from the left flank and radiating to the left groin. You are unable to obtain a history as she is writhing in pain. On examination, she is sweating profusely. What is the most likely diagnosis?
 - A Diverticulitis
 - B Ruptured ectopic pregnancy
 - C Renal colic
 - D Small bowel obstruction
 - E Ruptured abdominal aortic aneurysm

- 6** A 49-year-old woman has severe colicky left flank pain radiating to the left groin, with nausea and vomiting; renal colic is suspected. Which investigation will be of most diagnostic value?
- A Computed tomography scan of the abdomen
 - B KUB (kidneys, ureters and bladder)
 - C Intravenous urogram
 - D Ultrasound
 - E Abdominal plain film radiography

Answers 1-6

SBA MCQS

- 1** **D Calcium oxalate** — **UPDATED** Calcium oxalate (alone or mixed with calcium phosphate) forms most urinary stones (about 75–80%), and these are radio-opaque.
- 2** **B Kidney ureter bladder (KUB) radiograph** — **UPDATED** Of these options, a plain KUB film is the initial test (about 90% of stones are radio-opaque); non-contrast CT KUB is now the definitive investigation.
- 3** **E Mycobacteria** — Mycobacteria are not typically implicated in infection-related renal stones.
- 4** **B Uric acid** — **UPDATED** Gout (hyperuricosuria and persistently acidic urine) predisposes to uric acid stones, which are radiolucent on plain film but seen on CT.
- 5** **C Renal colic** — **UPDATED** Colicky pain radiating from the loin to groin with diaphoresis suggests renal colic.
- 6** **A Computed tomography scan of the abdomen** — **UPDATED** Non-contrast (low-dose) CT KUB is the gold standard for diagnosing urinary stones, showing size and site and excluding other causes such as an aortic aneurysm.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** 45-year-old man has severe colicky right-sided pain radiating from loin to groin, with nausea and vomiting. He is restless and tender in the right renal angle. Urinalysis shows blood +. Which is the single most appropriate investigation?
- A FBC, CRP, U&Es
 - B Intravenous urogram
 - C Plain abdominal X-ray (kidneys, ureter, bladder—KUB)
 - D Renal tract CT scan
 - E Renal ultrasound scan
- 2** 47-year-old man develops sudden onset of colicky pain in his right loin, radiating into his testicle. The pain is severe and is associated with vomiting. A junior colleague asks for teaching about calculi. Which single statement with respect to urinary calculi is correct?
- A Calcium phosphate stones are associated with renal tubular alkalosis
 - B Cystine stones are associated with urinary tract infection
 - C Staghorn calculi are associated with high urinary uric acid
 - D Struvite stones are the most common in UK patients
 - E There is seasonal variation in presentation of urinary tract lithiasis

Answers 1-2

OXFORD MCQS

- 1** **D Renal tract CT scan** — Renal tract CT scan is the gold standard for imaging urinary stones.
- 2** **E There is seasonal variation in presentation of urinary tract lithiasis** — Urinary stone presentation varies seasonally, often peaking in summer due to dehydration.

1 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 36-year-old man presents to the emergency room with renal colic. His vital signs are normal and a urinalysis shows microscopic hematuria. A radiograph reveals a 1.5-cm stone. Which of the following is the most appropriate management of this patient?
- A Hydration and analgesics
 - B α -Adrenergic blocker
 - C Extracorporeal lithotripsy
 - D Percutaneous nephrostomy tube
 - E Open surgery to remove the stone

Answer 1

PRETEST MCQs

- 1 **C Extracorporeal lithotripsy** — Extracorporeal shock-wave lithotripsy is appropriate for a 1.5-cm renal stone.

3 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1 A 73-year-old man presents with sudden-onset right-sided back pain radiating to the abdomen. He is stable, afebrile, tender in the right abdomen. Trace blood on dipstick. Which investigation next?
- A CT abdomen
 - B Intravenous urogram
 - C Laparotomy
 - D MRI spine
 - E Ultrasound abdomen
- 2 A 27-year-old man has severe left-sided loin pain for 2 hours, constant but fluctuates, with vomiting. Mild left loin tenderness. Dipstick shows 2+ blood. Most likely diagnosis?
- A Biliary colic
 - B Dissecting aortic aneurysm
 - C Pyelonephritis
 - D Renal cell carcinoma
 - E Renal colic
- 3 A 65-year-old man presents to the emergency department with a 3-hour history of left loin pain radiating to the left groin. He is known to have benign prostatic hypertrophy and recurrent urinary tract infections. The intravenous urogram (IVU) shows a standing column on the left. Which of the following types of stone is most likely to have caused his symptoms?
- A Calcium
 - B Cholesterol
 - C Cystine
 - D Magnesium ammonium phosphate
 - E Uric acid

Answers 1-3

GET AHEAD MCQs

- 1 **A CT abdomen** — To exclude AAA or renal stone; CT is definitive to distinguish these pathologies.
- 2 **E Renal colic** — Flank pain radiating anteriorly plus haematuria is characteristic of ureteric stone.
- 3 **D Magnesium ammonium phosphate** — **UPDATED** Struvite stones are associated with chronic UTIs by urease-producing organisms.

7 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQs

Crash Course MCQs

- 1 A 68-year-old man has urinary urgency/frequency worse on exercise over weeks. Most likely diagnosis?
 - A Bladder cancer
 - B Bladder stone
 - C Renal stone
 - D Prostatitis
 - E Urinary infection
- 2 A 42-year-old man presents with severe colicky pain in his right loin radiating to his groin. On examination you note tenderness in the right loin. A urine dipstick is positive for blood. What is the next most appropriate initial investigation?
 - A Abdominal ultrasound
 - B Abdominal X-ray
 - C Abdominal CT scan
 - D 24-Hour urine collection
 - E IV urogram
- 3 A 75-year-old male with gout and recent chemotherapy has right flank pain radiating to groin, onset was rapid. Microscopic haematuria. USS KUB negative but CT confirms the diagnosis. What to prevent this in future?
 - A Citrate
 - B Sodium restriction
 - C Penicillamine
 - D Thiazide diuretics
 - E Surgery
- 4 A previously fit 45-year-old with severe left loin-to-groin colicky pain. Plain abdominal X-ray shows a left ureteric calculus, urine dipstick trace of blood only. Stone composition?
 - A Cystine
 - B Uric acid
 - C Calcium oxalate
 - D Magnesium-ammonium phosphate
 - E Iron deposits
- 5 A 31-year-old man has acute colicky right flank-to-groin pain, frequency, dysuria, haematuria. Most common metabolic abnormality with this condition?
 - A Hypercalciuria
 - B Increased urine citrate
 - C Hyperuricaemia
 - D Alkaline urine pH
 - E Hypernatraemia
- 6 A 26-year-old man with severe colicky right loin-to-groin pain, urine dipstick positive for blood. IVU shows a 3 mm ureteric stone. After analgesia, what next?
 - A Extracorporeal shockwave lithotripsy
 - B Percutaneous nephrolithotomy
 - C Open surgical removal
 - D Ureteric stenting
 - E Allow spontaneous passage
- 7 A 26-year-old man with severe right loin-to-groin colic, dipstick positive for blood/nitrites, IVU shows a 4 mm ureteric stone. What first?
 - A Analgesia
 - B Start an α -blocker
 - C Open surgical removal
 - D Arrange ESWL
 - E Antibiotics

Answers 1-7

CRASH COURSE MCQs

- 1 **B Bladder stone** — **UPDATED** Irritative voiding, especially with movement, can indicate bladder calculi.
- 2 **C Abdominal CT scan** — **UPDATED** Loin-to-groin colic with haematuria suggests a ureteric stone; non-contrast CT KUB is the first-line imaging test (sensitivity about 97%).
- 3 **A Citrate** — **UPDATED** Uric acid stones (gout, tumour lysis after chemotherapy) are prevented by urinary alkalinisation with potassium citrate, plus hydration and allopurinol.
- 4 **C Calcium oxalate** — Most common type, typically radio-opaque on X-ray.
- 5 **A Hypercalciuria** — Most calcium stones are linked to elevated urinary calcium.
- 6 **E Allow spontaneous passage** — <5 mm stone often passes spontaneously.
- 7 **A Analgesia** — Priority is pain control; then consider tamsulosin if indicated.

Renal Tumours

Renal cell carcinoma, Wilms tumour, angiomyolipoma and the renal mass.

13

MCQS

0

EMQ SCENARIOS

7

SOURCES

0

REPEATED IDEAS

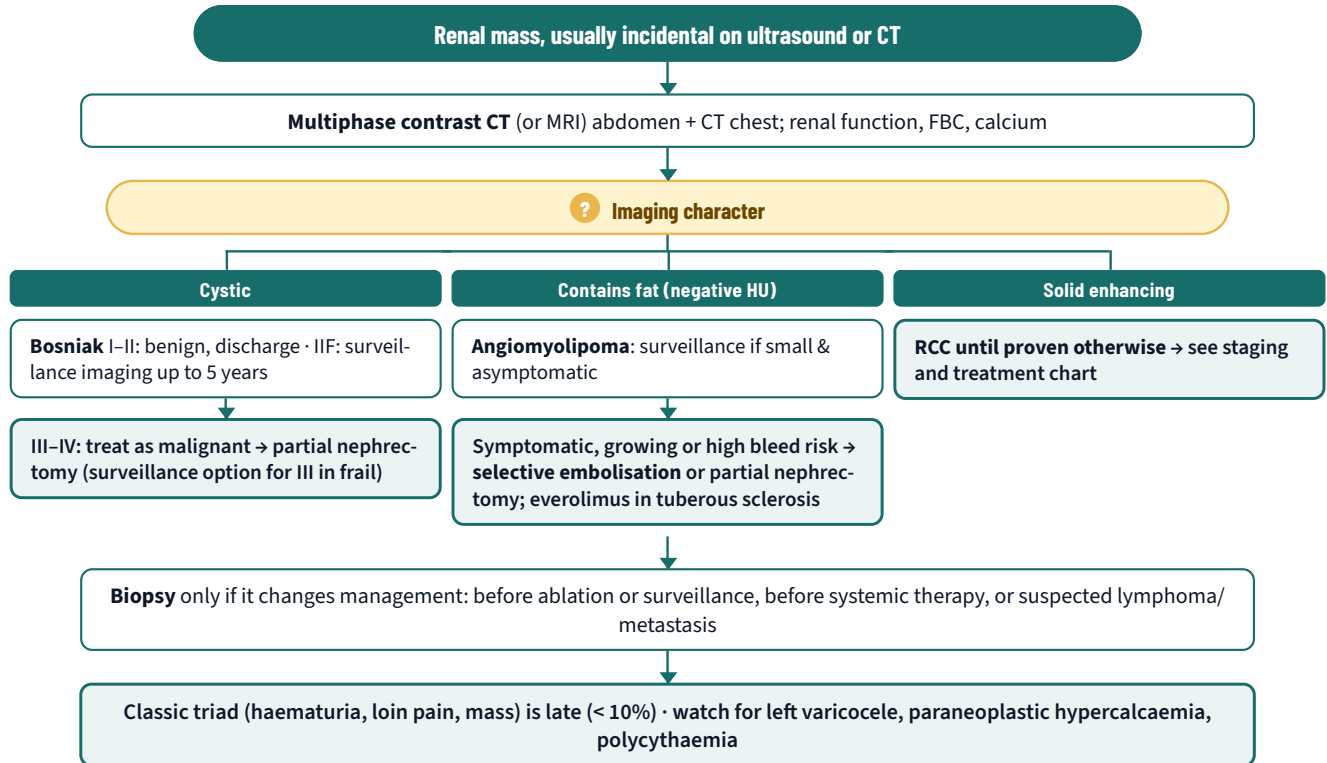
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Step by step

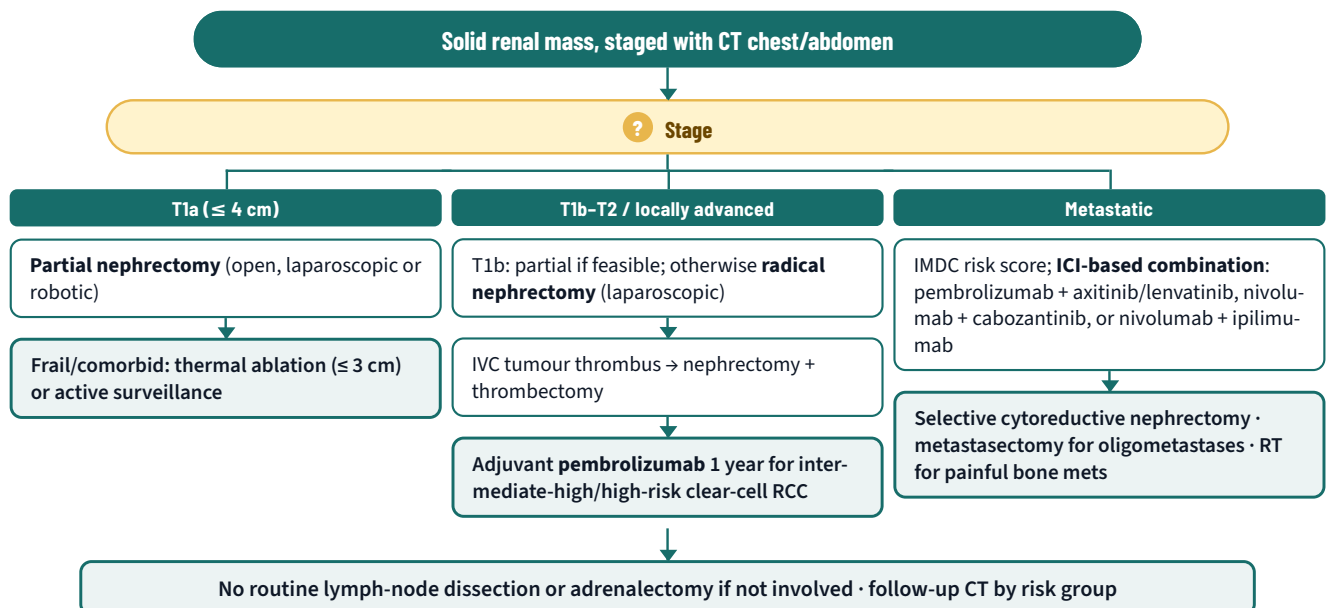
Work-up of a renal mass

EAU 2024 RENAL CELL CARCINOMA • BOSNIAK 2019



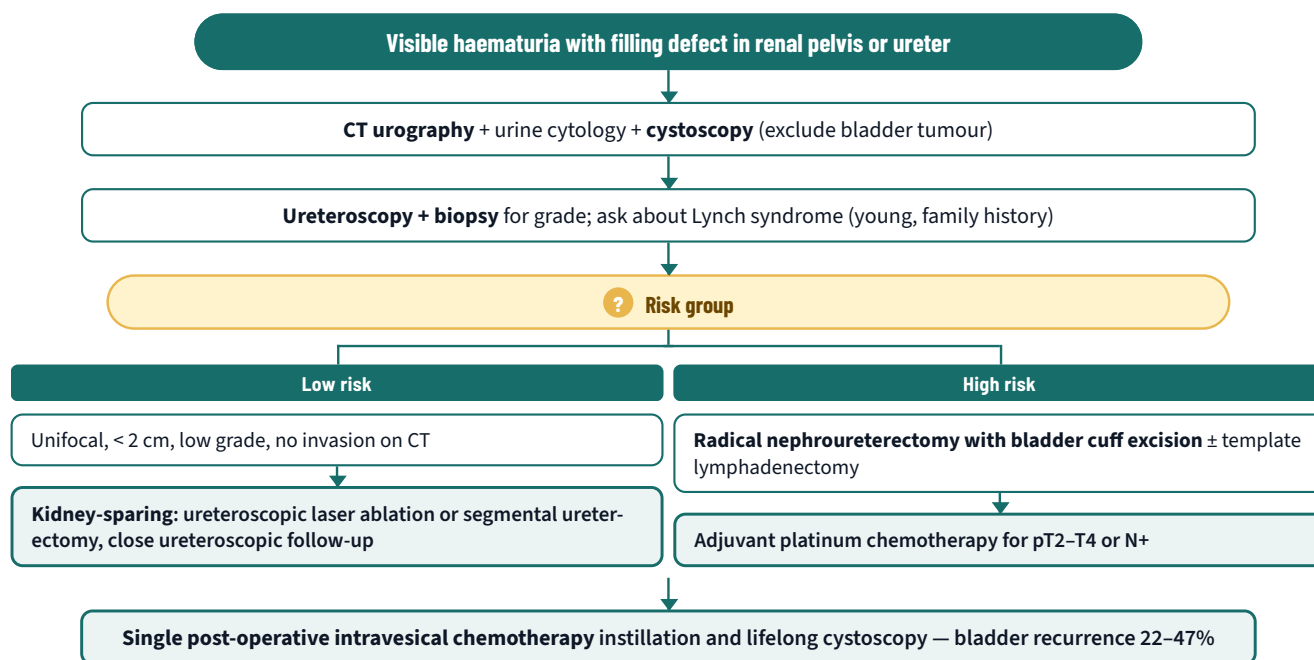
Renal cell carcinoma: stage-based treatment

EAU 2024 RENAL CELL CARCINOMA



Upper tract urothelial carcinoma

EAU 2024 UTUC



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 45-year-old male is planning a kidney donation for his wife who has an end stage renal disease. Which of the following reasons listed below would most likely prevent the husband from donating his kidney?
 - A The donor has a body mass index (BMI) of 28 kg/m².
 - B The donor has hypertension controlled by one medication.
 - C The donor's blood group is O negative.
 - D The donor has a family history of benign prostatic hyperplasia.
 - E The donor has a GFR of 40 ml/min per 1.73 m².
- 2 A 68-year-old female with unexplained persistent asymptomatic microscopic hematuria underwent CT abdomen and pelvis with IV contrast; there was a 3x2 cm solid irregular enhancing mass arising from the parenchyma of the right kidney. The patient is asymptomatic and vitally stable. What is the most likely diagnosis?
 - A Transitional cell cancer.
 - B Renal cell carcinoma.
 - C Renal abscess.
 - D Oncocytoma.
 - E Angiomyolipoma.

Answers 1-2

LOCAL EXAMS MCQS

- 1 **E** The donor has a GFR of 40 ml/min per 1.73 m². — **UPDATED** Living donors need adequate renal reserve; a GFR of 40 mL/min/1.73 m² excludes donation, whereas BMI 28, single-drug-controlled hypertension and group O are acceptable.
- 2 **B** Renal cell carcinoma. — **UPDATED** A solid enhancing parenchymal renal mass in an adult is renal cell carcinoma until proven otherwise; TCC arises from the collecting system and AML contains fat.

Bailey & Love MCQs

- 1 Which of the following statements about Wilms' tumour is false?
 - A It is a tumour of embryonic nephrogenic tissue occurring below the age of 5 years.
 - B Haematuria and fever are the most common presentations.
 - C Haematogenous spread is mainly to the lungs.
 - D Imaging modalities are US, CT and MRI.
 - E Treatment is by chemotherapy, surgery and radiotherapy.
- 2 Which of the following statements about hypernephroma (renal cell carcinoma) are true?
 - A In 25% of patients there are no local symptoms.
 - B A left-sided varicocele might be a presenting feature in a male.
 - C Haematuria and clot colic are the most common presentations.
 - D US and contrast-enhanced CT are the imaging methods of choice.
 - E Following nephrectomy, chemotherapy and radiotherapy should be given.

Answers 1-2

BAILEY & LOVE MCQS

- 1 **B** Haematuria and fever are the most common presentations. — **UPDATED** Wilms' tumour most commonly presents as an abdominal mass.
- 2 **ABCD** True: A, B, C, D — **UPDATED** E is false: renal cell carcinoma is resistant to chemotherapy and radiotherapy, so these are not given after nephrectomy; a left varicocele may signal renal vein involvement.

2 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Which one of the following statements regarding renal cell carcinoma is true?
 - A It is an incidental finding in fewer than 10% of patients
 - B Women are more commonly affected than men
 - C Approximately 20% of cases are familial
 - D Renal cell carcinoma accounts for around 3% of adult cancers
 - E Metastases are poorly visualized by standard imaging techniques
- 2 A young girl with fever, left loin swelling and microscopic haematuria is ultimately diagnosed with Wilms tumour. Which statement about Wilms tumour is true?
 - A The tumour may be associated with anophthalmia
 - B 5-year survival in stage IV disease is approximately 65%
 - C Children most commonly present at the age of 8–10 years
 - D Some cases are associated with a gene mutation on chromosome 13
 - E It commonly metastasizes to the brain

Answers 1-2

SBA MCQs

- 1 **D** Renal cell carcinoma accounts for around 3% of adult cancers — **UPDATED** Renal cell carcinoma represents about 3% of adult cancers.
- 2 **B** 5-year survival in stage IV disease is approximately 65% — **UPDATED** By elimination: Wilms tumour is linked to aniridia and WT1 on chromosome 11, peaks at 3–4 years and spreads to the lungs; stage IV survival is roughly 65–80%.

1 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 An ultrasound scan in a 50-year-old blind man with painless haematuria suggests the presence of adenocarcinoma of the kidney. A subsequent CT scan of his chest, abdomen, and pelvis confirms the diagnosis and shows renal vein invasion by the tumour (see Figure 9.1). The patient is anaemic but serum creatinine is 92µmol/L, potassium is 4.8mmol/L, calcium is 2.4mmol/L, and alkaline phosphatase is 33U/L. Which is the single most appropriate next step?

FIGURE

#\$@@@Oxford Urological_surgery@@_10_1%%.png

- A Biological (interferon/interleukin) therapy
- B Genetic counselling
- C Isotope bone scan
- D Neoadjuvant chemotherapy
- E Partial nephrectomy

Answer 1

OXFORD MCQS

- 1 **B Genetic counselling** — **UPDATED** Renal cancer in a blind man suggests von Hippel–Lindau disease (retinal angiomas), so genetic counselling and family screening are indicated.

3 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 45-year-old otherwise healthy woman presents with a 7-cm renal cell carcinoma without radiologic evidence of abdominal lymph node involvement and with no distant metastases. Which of the following is the most appropriate next step in the management of this patient?
- A Chemotherapy
 - B Radiation therapy
 - C Immunotherapy
 - D Radiofrequency ablation
 - E Radical nephrectomy
- 2 What is the most common type of renal cell carcinoma?
- A Papillary renal cell carcinoma
 - B Chromophobe renal cell carcinoma
 - C Collecting duct carcinoma
 - D Clear cell renal cell carcinoma
 - E Oncocytoma
- 3 What syndrome associated with renal cell carcinoma is caused by a mutation in a tumor suppressor gene on chromosome 3?
- A Beckwith-Wiedemann syndrome
 - B Turcot syndrome
 - C Von-Hippel Lindau disease
 - D Li-Fraumeni syndrome
 - E Tuberous sclerosis

Answers 1–3

PRETEST MCQS

- 1 **E Radical nephrectomy** — Surgery with radical nephrectomy is the main treatment for localized renal cell carcinoma.
- 2 **D Clear cell renal cell carcinoma** — Clear cell carcinoma is the most common histological subtype of renal cell carcinoma.
- 3 **C Von-Hippel Lindau disease** — **UPDATED** Von Hippel–Lindau disease is caused by mutation of the VHL tumour suppressor gene on chromosome 3p and predisposes to clear cell renal cell carcinoma.

1 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 3-year-old boy is brought to the GP practice by his mother. He has been complaining intermittently about abdominal pain. His mother feels that his abdomen is distended. On examination, there is a large mass on the left side of the abdomen which does not cross the midline. What is the most likely diagnosis?
- A Hepatoblastoma
 - B Lipoma
 - C Nephroblastoma
 - D Neuroblastoma
 - E Polycystic kidney

Answer 1

GET AHEAD MCQS

- 1 **C Nephroblastoma** — Wilms tumour arises from embryonic renal tissue and typically doesn't cross midline.

2 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 49-year-old woman has loin pain and haematuria. On exam, a mass in the right loin. Investigation?
- A Urea and electrolytes
 - B Abdominal CT scan
 - C Chest X-ray
 - D IV urogram
 - E Urinalysis
- 2 A middle-aged man diagnosed with renal cell carcinoma of the left kidney, no distant metastasis. Next step?
- A Embolization of renal artery
 - B Radiotherapy
 - C Chemotherapy
 - D Radical nephrectomy
 - E Hormonal treatments

Answers 1-2

CRASH COURSE MCQS

- 1 **B Abdominal CT scan** — **UPDATED** A loin mass with haematuria suggests a renal tumour; contrast-enhanced CT of the abdomen characterises the mass and stages it.
- 2 **D Radical nephrectomy** — Potentially curative resection for localized RCC.

Urological Investigation & Haematuria

Haematuria work-up, urinalysis, imaging and endoscopy of the urinary tract.

22

MCQS

0

EMQ SCENARIOS

6

SOURCES

1

REPEATED IDEAS

IN THIS CHAPTER

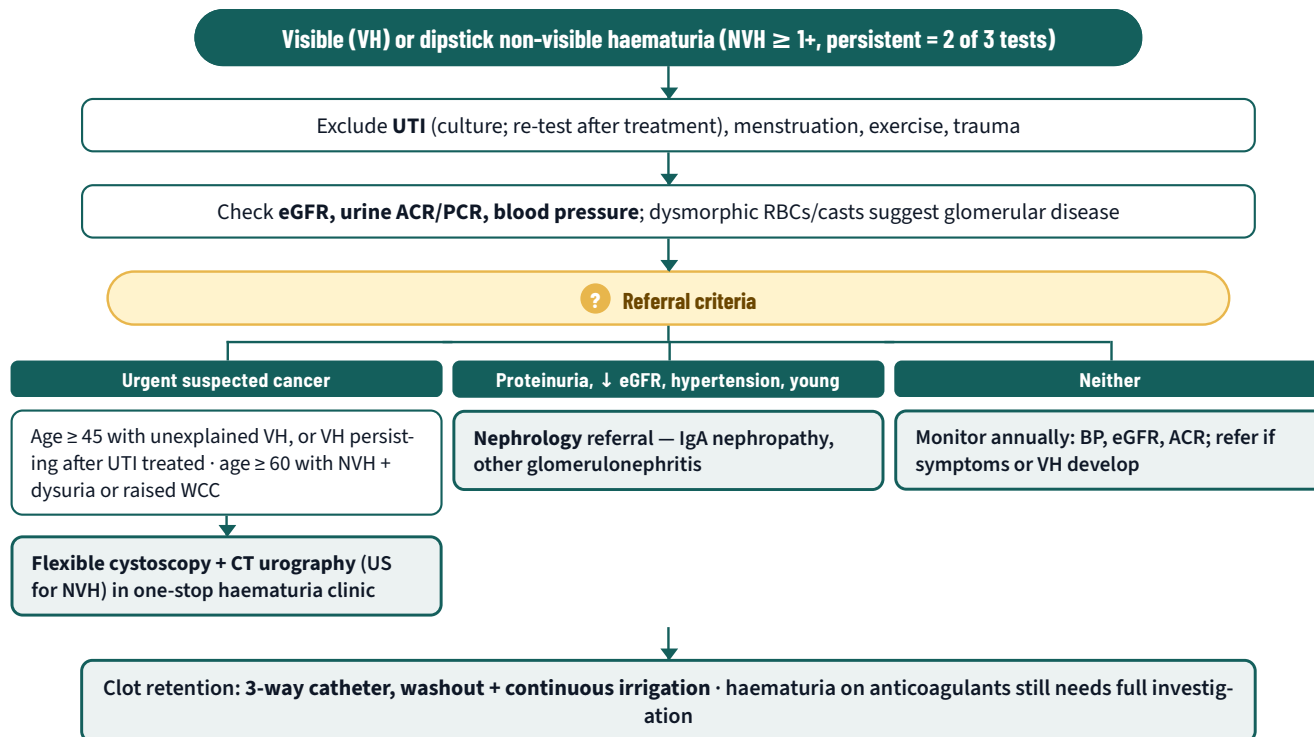
01	Management flowcharts	3 charts	page 386
02	Local Exams MCQs	3 Q	page 388
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Step by step

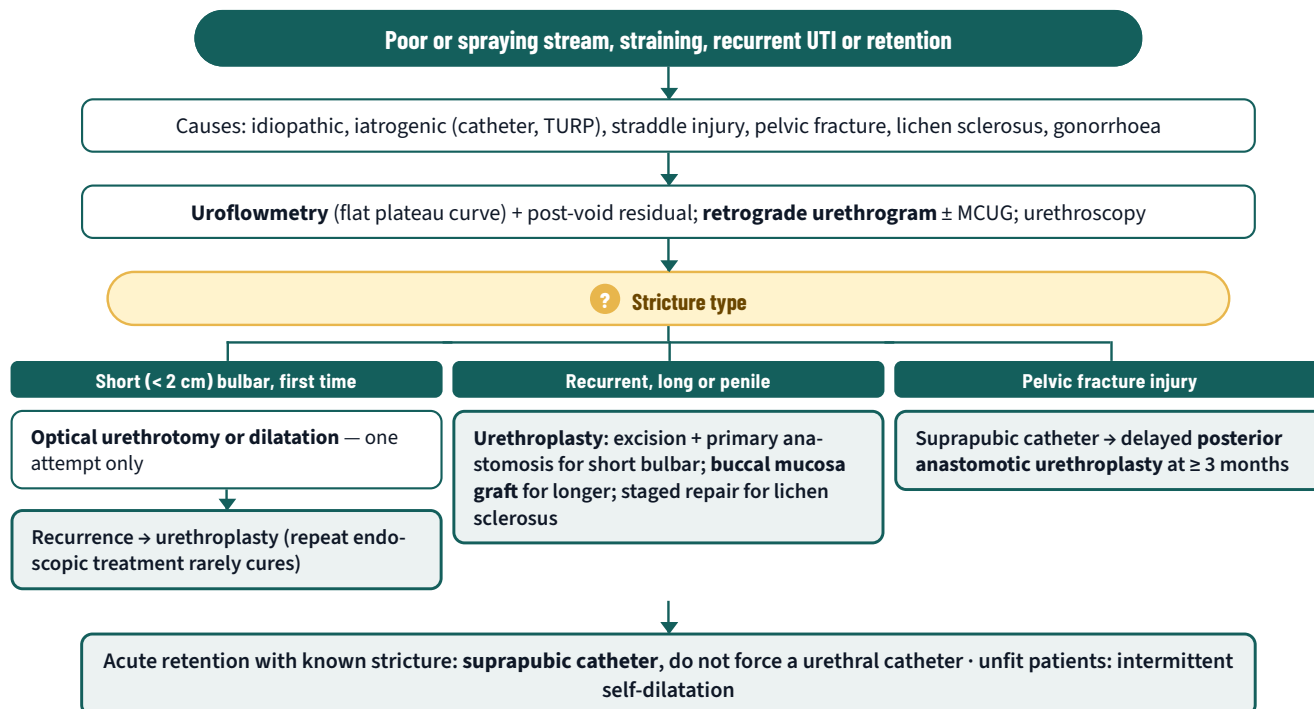
Haematuria: assessment and referral

NICE NG12 • BAUS • EAU 2024



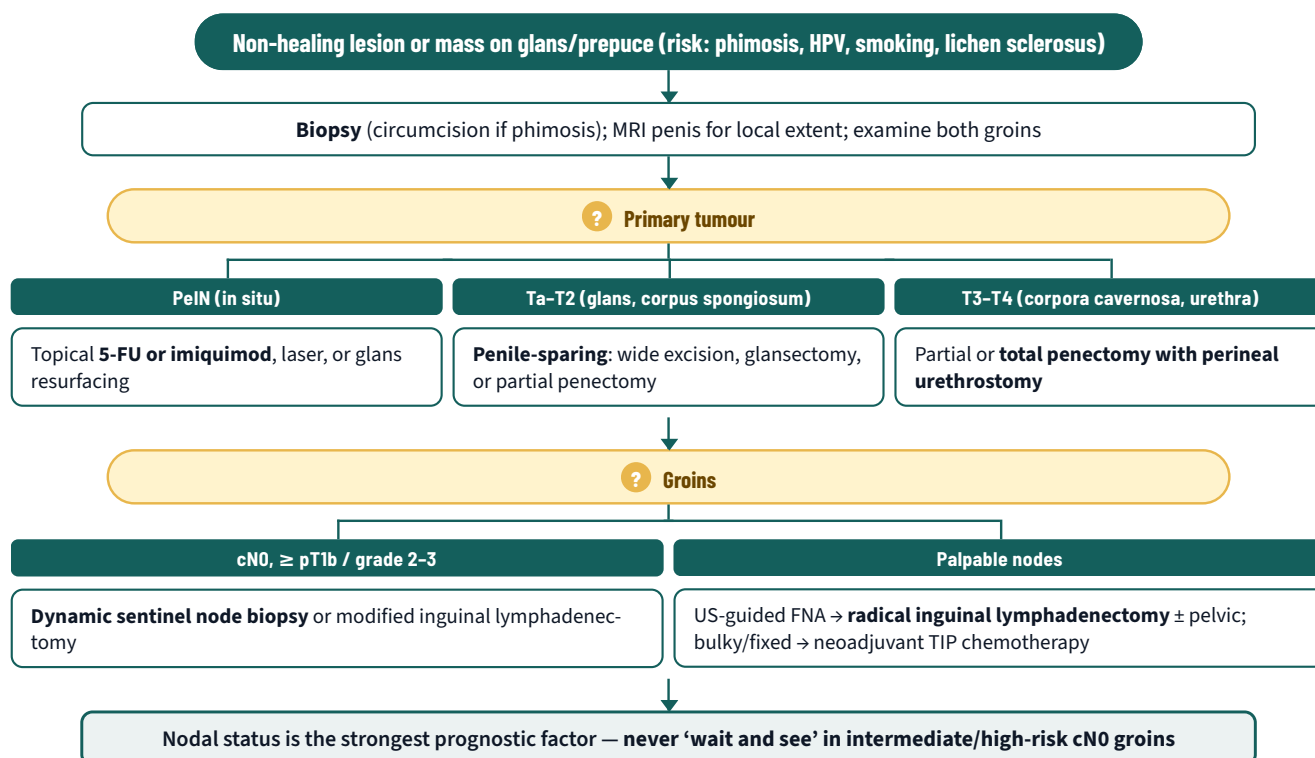
Male urethral stricture

EAU 2024 URETHRAL STRICTURES • BAILEY & LOVE 28E



Penile lesion and penile cancer

EAU-ASCO 2023 PENILE CANCER



Local Exams MCQs

MINISTERIAL EXAM

Ministerial evaluation examinations

- 1 A 55-year-old male patient with multiple comorbidities presents with an erectile dysfunction. Which of the following of his medications listed below is least likely to cause an erectile dysfunction?
 - A Metformin.
 - B Beta-blocker.
 - C Fluoxetine.
 - D Thiazide.
 - E Finasteride.
- 2 A 35-year-old man on an alpha blocker for hypertension is undergoing an infertility evaluation for secondary infertility. Has a normal physical examination, including genitals and DRE. Two semen analyses both demonstrate semen volume of less than 1 ml; with normal sperm count, motility, and morphology. Which of the following is the most appropriate test to be recommended next in this case?
 - A Trans-rectal ultrasound.
 - B Post-ejaculatory urinalysis.
 - C Serum testosterone.
 - D Serum FSH and LH.
 - E Serum prolactin.
- 3 A 75-year-old lady with a long history of uncontrolled DM presented with chronic retention of urine and frequent unaware leakage of urine with wetting of her clothes. Ultrasound revealed a post void residue (PVR) of 200 ml on two separate occasions. This condition is most appropriately described as:
 - A Stress incontinence.
 - B Overflow incontinence.
 - C Urge incontinence.
 - D Nocturnal enuresis.
 - E Total incontinence.

Answers 1-3

LOCAL EXAMS MCQS

- 1 **A Metformin.** — **UPDATED** Beta-blockers, thiazides, SSRIs such as fluoxetine and 5 α -reductase inhibitors such as finasteride can all cause erectile dysfunction; metformin does not.
- 2 **B Post-ejaculatory urinalysis.** — **UPDATED** Low-volume ejaculate with normal sperm in a man on an α -blocker suggests retrograde ejaculation, confirmed by finding sperm in post-ejaculatory urine.
- 3 **B Overflow incontinence.** — **UPDATED** Diabetic autonomic neuropathy causes a hypotonic bladder with chronic retention and dribbling overflow incontinence, as shown by the large post-void residue.

3 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Regarding urethral stricture, the following statements are true except:
 - A Urinary flow trace will show a prolonged flow with a plateau-shaped curve.
 - B In acute retention suprapubic catheterisation is required.
 - C Causes are inflammatory, traumatic and iatrogenic.
 - D Traumatic strictures are best treated by urethroplasty.
 - E The ideal treatment in the majority is urethral dilatation.
- 2 Regarding female urethra, which of the following statements are true?
 - A The bladder neck is the most important entity for urinary continence.
 - B Urethral prolapse occurs as a result of childbirth.
 - C Fowler's syndrome is best treated by urethral dilatation.
 - D Urethral caruncle is premalignant.
 - E Urethral diverticulum is more common in women than men.

- 3** Regarding surgical conditions of the foreskin, the following statements are true except:
- A The most common indication for circumcision is cultural reasons.
 - B In children circumcision is the treatment for non-retractile foreskin due to adhesions.
 - C Circumcision is indicated in true phimosis with balanoposthitis.
 - D Balanoposthitis is precancerous.
 - E Monopolar diathermy should never be used during a circumcision.

Answers 1-3

BAILEY & LOVE MCQs

- 1** **E** The ideal treatment in the majority is urethral dilatation. — **UPDATED** Urethral dilatation is not the ideal treatment in most cases; the other statements accurately describe urethral stricture.
- 2** **BE** True: B, E — **UPDATED** A false: in women the urethral (mid-urethral) sphincter matters most; C false: Fowler syndrome is treated by sacral neuromodulation; D false: a caruncle is benign.
- 3** **B** In children circumcision is the treatment for non-retractile foreskin due to adhesions. — **UPDATED** Non-retractile foreskin due to adhesions is usually physiological; the other statements are correct.

9 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** All of the following statements regarding carcinoma of the penis are true, except
- A It is a squamous cell carcinoma
 - B Incidence is lower in men infected with the human papilloma virus
 - C Incidence is lower in men circumcised at birth
 - D Staging of disease is by the tumour node metastasis (TNM) system
 - E 5-year survival in patients with localized disease is around 80%
- 2** A 55-year-old man with an angulated penis on erection is diagnosed with Peyronie's disease. Which statement is NOT true?
- A It is a connective tissue disorder of unknown origin
 - B Around 1% of men are affected
 - C It is the result of a gene mutation on chromosome 5
 - D Surgical treatment is considered only 1 year after initial presentation
 - E There is an association with Dupuytren's contracture
- 3** All of the following form a covering layer over the kidney, except:
- A Gerota's fascia (peri-renal fascia)
 - B Peri-renal fat
 - C Liver
 - D Para-renal fat
 - E Fibrous capsule
- 4** The posteromedial aspect of the kidney is related to which one of the following structures?
- A Psoas muscle
 - B Ilio-inguinal nerve
 - C Diaphragm
 - D Pancreas
 - E 12th rib
- 5** In its course from the kidney to the bladder, the ureter runs on top of which muscle?
- A Quadratus lumborum
 - B Transversus abdominis
 - C Psoas
 - D Iliacus
 - E Latissimus dorsi
- 6** At which intervertebral level do the renal arteries arise from the aorta?
- A T11/T12
 - B T12/L1
 - C L1/L2
 - D L2/L3
 - E L3/L4

- 7** In a non-exercising individual, approximately what proportion of the cardiac output is supplied to the kidneys?
- A 5%
 - B 10%
 - C 15%
 - D 20%
 - E 40%
- 8** A 34-year-old father of five children requests a vasectomy. Which of the following layers lies closest to the vas deferens?
- A Superficial scrotal fascia (Dartos fascia)
 - B Internal spermatic fascia
 - C Tunica vaginalis
 - D External spermatic fascia
 - E Preperitoneal fat
- 9** With regard to the male urethra, which one of the following segments is the narrowest?
- A Pre-prostatic
 - B Prostatic
 - C Membranous
 - D Bulbar
 - E Penile

Answers 1-9

SBA MCQS

- 1** **B Incidence is lower in men infected with the human papilloma virus** — **UPDATED** B is false: HPV infection (types 16 and 18) increases the risk of penile squamous cell carcinoma; neonatal circumcision reduces it.
- 2** **C It is the result of a gene mutation on chromosome 5** — **UPDATED** Peyronie's disease is not caused by a gene mutation on chromosome 5.
- 3** **C Liver** — **UPDATED** The kidney is covered by its fibrous capsule, perirenal fat, Gerota fascia and pararenal fat; the liver is only an anterior relation of the right kidney.
- 4** **A Psoas muscle** — **UPDATED** The posteromedial aspect of the kidney is related to the psoas muscle.
- 5** **C Psoas** — **UPDATED** The ureter runs on the psoas muscle.
- 6** **C L1/L2** — **UPDATED** Renal arteries arise from the aorta at the L1/L2 level.
- 7** **D 20%** — **UPDATED** At rest the kidneys receive about 20% of the cardiac output (1–1.2 L/min), despite being less than 1% of body weight.
- 8** **B Internal spermatic fascia** — **UPDATED** The vas runs in the spermatic cord, whose innermost covering is the internal spermatic fascia; the tunica vaginalis surrounds the testis, not the vas.
- 9** **C Membranous** — **UPDATED** The membranous urethra is the narrowest segment of the male urethra.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1** 50-year-old stonemason develops erectile dysfunction. Which of the single points arising from his history is least likely to be related to his erectile problem?
- A He has a Dupuytren's contracture of his right palmar fascia
 - B He has become progressively deaf during the last 3 years
 - C He has had an inguinal hernia repair complicated by ilioinguinal nerve damage
 - D He reports that his GP is struggling to control his hypertension
 - E He suffered a slipped intervertebral disc at work last year
- 2** 55-year-old man with type 2 diabetes has an inability to maintain an erection during intercourse. He reports morning erections. His only medication is for glycaemic control. He has a normal neurological examination. Which is the single most appropriate initial investigation?
- A Angiography
 - B Check blood HbA1c
 - C Check hormone profile (testosterone, FSH/luteinizing hormone, prolactin, thyroid function)
 - D Dynamic cavernosometry
 - E Psychological validated questionnaire

Answers 1-2

- 1 **C** He has had an inguinal hernia repair complicated by ilioinguinal nerve damage — **UPDATED** Damage to the ilioinguinal nerve is not typically associated with erectile dysfunction.

- 2 **E** Psychological validated questionnaire — **UPDATED** Morning erections suggest a psychogenic cause; a validated questionnaire is appropriate for initial assessment.

3 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 18-year-old high school football player is kicked in the left flank. Three hours later he develops hematuria. His vital signs are HR 80, BP 123/67, and RR 14. A CT scan demonstrates a nonexpanding perirenal hematoma confined to the renal retroperitoneum. Which of the following is the most appropriate next step in the management of this patient?

A Observation
B Angiography
C Ureteral stent placement
D Nephrostomy tube placement
E Exploratory laparotomy

- 2 A 32-year-old man is in a high-speed motorcycle collision and presents with an obvious pelvic fracture. On examination, he has a scrotal hematoma and blood at his urethral meatus. Which of the following is the most appropriate next step in his management?

A Placement of a Foley catheter
B Cystoscopy
C CT of the pelvis
D Retrograde urethrogram
E Nephrostomy tube placement

- 3 A 26-year-old man complains of pelvic pain after a motorcycle collision. Physical and radiologic examinations confirm a pelvic fracture. Urologic workup reveals a normal urethrogram and an extraperitoneal bladder injury. Which of the following is the most appropriate next step in the treatment of his bladder injury?

A Immediate surgical repair
B Placement of a suprapubic tube
C Catheter drainage
D Cystoscopy
E Bilateral nephrostomy tubes

REPEATED 2x • ALSO ASKED IN
PreTest

Answers 1-3

- 1 **A** **Observation** — **UPDATED** A non-expanding perirenal haematoma in a stable patient is a low-grade (AAST grade II) renal injury, managed non-operatively with bed rest and monitoring.
- 2 **D** **Retrograde urethrogram** — **UPDATED** Blood at the meatus with pelvic fracture and scrotal haematoma suggests urethral

injury; retrograde urethrogram before any catheter; suprapubic catheter if disrupted.

- 3 **C** **Catheter drainage** — **UPDATED** Uncomplicated extraperitoneal bladder rupture heals with urethral catheter drainage for about 10–14 days; intraperitoneal rupture needs surgical repair.

2 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 33-year-old man has increasing difficulty passing urine with poor flow. He had gonorrhoea thrice. Which investigation best establishes the cause?
- A Intravenous urogram
 - B Ultrasound of renal tract
 - C Urethral swabs
 - D Urethrography
 - E Urinary flow rates
- 2 A 27-year-old man who has just returned from a holiday in Asia presents to the GP surgery having noticed a painless hard ulcer over the penile glans. He is otherwise well. On examination, the patient has generalized painless lymphadenopathy. Which of the following is the most likely cause of his symptoms?
- A *Escherichia coli*
 - B *Haemophilus ducreyi*
 - C Herpes simplex virus
 - D Human papilloma virus
 - E *Treponema pallidum*

Answers 1-2

GET AHEAD MCQS

- 1 **D Urethrography** — UPDATED A history of gonococcal infection suggests urethral stricture investigated by urethrography.
- 2 **E *Treponema pallidum*** — UPDATED A painless ulcer (chancre) and painless lymphadenopathy is classic for primary syphilis.

Urological Emergencies & Trauma

Urinary retention, testicular torsion, priapism, Fournier gangrene and renal, bladder and urethral injury.

41

MCQs

0

EMQ SCENARIOS

7

SOURCES

2

REPEATED IDEAS

IN THIS CHAPTER

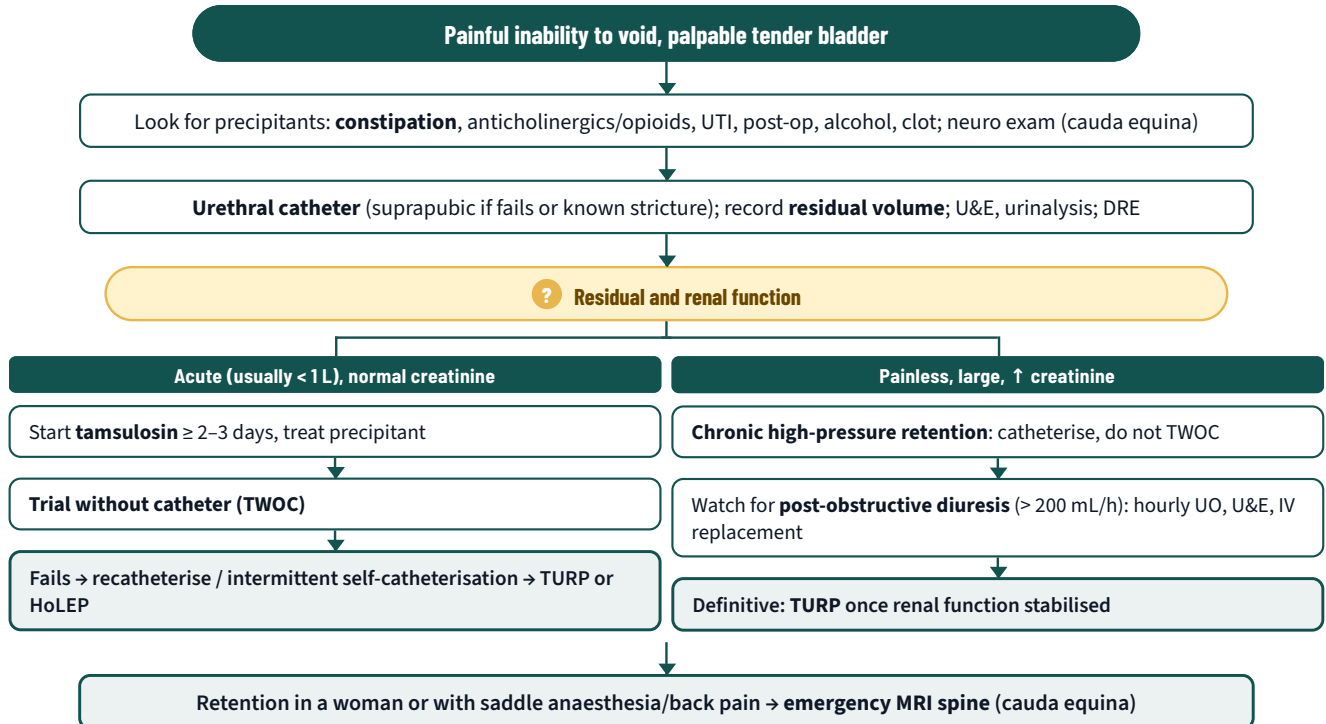
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07	Get Ahead MCQs	6 Q	page 401
08	Crash Course MCQs	5 Q	page 402

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Step by step

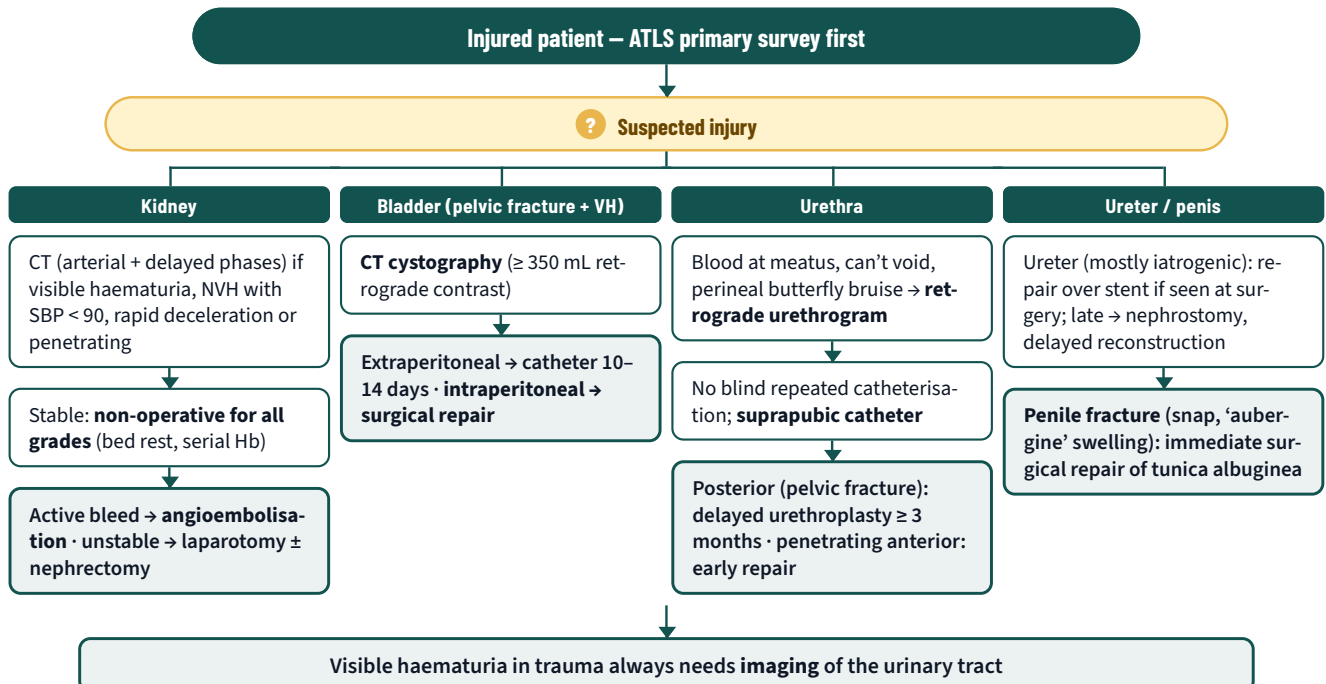
Acute urinary retention

EAU 2024 MALE LUTS • NICE CG97 • BAUS



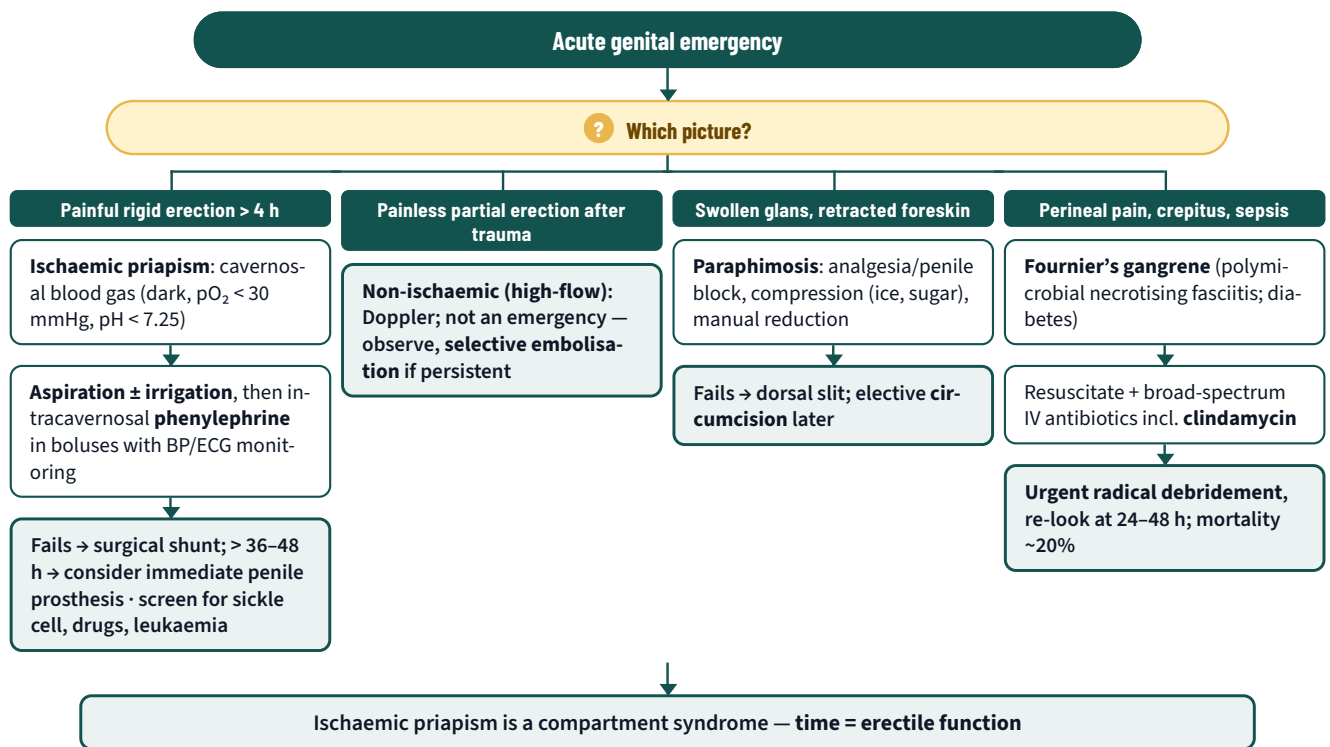
Genitourinary trauma

EAU 2024 UROLOGICAL TRAUMA • ATLS 10TH ED



Priapism, paraphimosis and Fournier's gangrene

EAU 2024 SEXUAL & REPRODUCTIVE HEALTH · BAILEY & LOVE 28E



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-2

A 68-year-old man arrives in the emergency department with abdominal pain and a distended bladder.

- 1 Which is the correct technique for male urethral catheterisation?
 - A Aseptic technique, lidocaine gel, penis held upright, insert to the hub and see urine before inflating the balloon
 - B Inflate the balloon as soon as resistance is felt
 - C Clean technique without lubrication
 - D Use the largest catheter to overcome resistance
 - E Insert only half the catheter length, then inflate
- 2 The catheter is passed but no urine drains. What is the most appropriate next step?
 - A Inflate the balloon and wait
 - B Check the position: advance fully, flush with sterile saline and confirm a full bladder (scan) before inflating
 - C Remove it and force a larger catheter
 - D Give IV furosemide
 - E Insert a suprapubic catheter at once
- 3 A patient needs a Foley catheter and develops haematuria during insertion. What is the most likely cause of the bleeding?
 - A Urethral trauma (false passage or balloon inflated in the urethra)
 - B Bladder carcinoma
 - C Urinary tract infection
 - D Ureteric stone
 - E Renal trauma
- 4 A patient needs a Foley catheter and develops haematuria during insertion. How should it be managed?
 - A Continue and inflate the balloon
 - B Stop, deflate and remove; then one gentle attempt by a senior/urologist (coudé tip) or a suprapubic catheter
 - C Repeated forceful attempts
 - D Start anticoagulation
 - E Immediate open urethroplasty

REPEATED 2* • ALSO ASKED IN
39th batch Block 2 exam

REPEATED 2* • ALSO ASKED IN
39th batch Block 2 exam

CASE • QUESTIONS 5-6

A 72-year-old man with benign prostatic hyperplasia has acute urinary retention and severe suprapubic pain; bladder scan shows 900 mL.

- 5 Which is the most common complication of a long-term indwelling catheter?
 - A Catheter-associated urinary tract infection
 - B Bladder cancer
 - C Renal stones
 - D Urethral stricture
 - E Bladder perforation
- 6 Which is an absolute contraindication to urethral catheterisation?
 - A Suspected urethral injury (blood at the meatus, pelvic fracture)
 - B Acute urinary retention
 - C Benign prostatic hyperplasia
 - D Diabetes mellitus
 - E Previous TURP

CASE • QUESTIONS 7-8

A 25-year-old woman had a difficult caesarean section for bleeding. Ten days later she has severe right flank pain; ultrasound shows severe hydroureteronephrosis down to the pelvis with no stones.

- 7 Which test best establishes the cause of obstruction?
 - A CT urography
 - B MRI
 - C GUE
 - D KUB
 - E CBC
- 8 Ureteric ligation is suspected. All of the following are initial options EXCEPT:
 - A IV fluids
 - B IV antibiotics
 - C Percutaneous nephrostomy
 - D Ureteroscopy (attempted stenting)
 - E Nephrectomy

MINISTERIAL EXAM

- 9 A 35-year-old male, a victim of RTA, was brought to ER with multiple injuries including a fractured pelvis. Resuscitation was performed. The patient was conscious and haemodynamically stable. Upon secondary survey, fresh blood was noticed at the external urethral meatus. The next most appropriate step in the approach for the Urology team is:
- A Explorative laparotomy.
 - B Retrograde urethrography.
 - C CT Cystography.
 - D Insert a 3-way Foley's catheter.
 - E Diagnostic cystourethroscopy.

- 10 You are consulting at a general practitioner's (GP) clinic and a 15-year-old boy was brought by his father stating that his son was woken up from sleep with severe pain in the left testis. There was no history of trauma. On examination, the left testis is swollen and extremely tender on palpation. Temperature: 37.1°C. What is the most appropriate next step in management?
- A Request a urethral swab.
 - B Start immediate IV antibiotics.
 - C Refer urgently to the urologist.
 - D Start ice packs, follow-up in 3 days.
 - E Discharge with oral analgesia.

Answers 1-10

LOCAL EXAMS MCQS

- 1 **A Aseptic technique, lidocaine gel, penis held upright, insert to the hub and see urine before inflating the balloon** — Insert the catheter fully to the bifurcation and inflate only once urine drains, so the balloon cannot be inflated in the urethra.
- 2 **B Check the position: advance fully, flush with sterile saline and confirm a full bladder (scan) before inflating** — No urine means the catheter is in the urethra, blocked by gel, or the bladder is empty; check, flush and scan before inflating.
- 3 **A Urethral trauma (false passage or balloon inflated in the urethra)** — Bleeding at catheterisation is usually urethral trauma (false passage) or a balloon inflated in the prostatic urethra.
- 4 **B Stop, deflate and remove; then one gentle attempt by a senior/urologist (coudé tip) or a suprapubic catheter** — Stop, deflate and withdraw; if retention persists use a suprapubic catheter or urology-guided catheterisation, and check Hb and urine.
- 5 **A Catheter-associated urinary tract infection** — Bacteriuria and CAUTI are universal with long-term catheters; encrustation, blockage, stricture and stones also occur.
- 6 **A Suspected urethral injury (blood at the meatus, pelvic fracture)** — Suspected urethral injury (blood at the meatus, high-riding prostate) contraindicates urethral catheterisation; a suprapubic catheter is used.
- 7 **A CT urography** — CT urography (or antegrade/retrograde study) shows the level of a ligated or injured ureter and any urinoma.
- 8 **E Nephrectomy** — Initial care is resuscitation, antibiotics and drainage (nephrostomy or retrograde stent); delayed reimplantation repairs it. Nephrectomy is not indicated.
- 9 **B Retrograde urethrography.** — **UPDATED** Blood at the meatus after a pelvic fracture suggests posterior urethral injury; perform retrograde urethrography before any urethral catheter is passed.
- 10 **C Refer urgently to the urologist.** — **UPDATED** Sudden severe testicular pain waking an adolescent is torsion until proven otherwise; urgent urological referral for scrotal exploration (within 6 hours) saves the testis.

10 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Regarding renal failure, which of the following statements is false?
- A Anuria is defined as complete absence of urine production.
 - B Oliguria is defined as urinary output of less than 400 mL in 24 hours.
 - C Certain drugs can cause renal failure.
 - D All patients with renal failure will require renal replacement therapy.
 - E Indwelling stents can be used to relieve ureteric obstruction.
- 2 Regarding renal injury, which of the following statements are true?
- A Haematuria following trivial injury to the kidney indicates damage to a pathological kidney.
 - B Closed renal injury is usually extraperitoneal.
 - C In closed renal injury, surgical exploration is necessary in the majority.
 - D Intravenous urography or contrast-enhanced computed tomography (CECT) scan should be performed urgently in suspected renal injury.
 - E Hypertension might be a long-term complication of renal injury.

- 3** Regarding ureteric injury, which of the following statements are false?
- A Most ureteric injuries are due to surgical trauma during pelvic surgery.
 - B Preoperative ureteric catheterisation helps to protect them from injury during operation.
 - C When recognised during operation, an injured ureter should be repaired immediately.
 - D When ureteric injury is diagnosed postoperatively, delayed repair is undertaken.
 - E Urinary fistula through an abdominal or vaginal wound indicates a damaged ureter.
- 4** Regarding rupture of the bladder, which of the following statements are true?
- A Extraperitoneal rupture is more common than intraperitoneal rupture.
 - B Extraperitoneal rupture mimics rupture of membranous urethra.
 - C CT cystography is the investigation of choice.
 - D Retrograde cystography is useful in making a diagnosis.
 - E Laparotomy is required in all cases of bladder rupture.
- 5** In acute retention of urine, the following statements are true except:
- A Bladder outlet obstruction is the most common cause in the male.
 - B Retroverted gravid uterus is a cause in the female.
 - C Non-urological conditions might cause acute retention.
 - D Suprapubic puncture should be done if urethral catheterisation fails.
 - E Once urine drains on passing the catheter, the balloon should be immediately inflated.
- 6** In chronic retention of urine, which of the following statements are true?
- A The patient complains of very severe pain from his distended bladder.
 - B Diuresis might follow catheterisation.
 - C Spinal cord injury might result in acute neuropathic bladder.
 - D Intermittent self-catheterisation is a form of treatment for patients with spinal cord injury.
 - E Bladder dysfunction can occur after radical rectal excision.
- 7** Regarding incontinence of urine, which of the following statements is false?
- A Urodynamic studies have an important role in the investigation of urinary incontinence.
 - B In men with bladder outflow obstruction, 50% have detrusor instability.
 - C Genuine stress incontinence (GSI) is from increased true detrusor pressure.
 - D Vesicovaginal fistula might be a cause in the female.
 - E Intravesical Botox injection has a role in treatment.
- 8** In bulbar urethral rupture, which of the following statements are true?
- A Urethral rupture is caused by direct blow to perineum.
 - B Acute retention occurs.
 - C There is perineal haematoma with blood at the urethral meatus.
 - D In a full bladder, suprapubic catheterisation is carried out.
 - E Urethral catheterisation might be attempted as an alternative.
- 9** In rupture of the membranous urethra, the following statements are true except:
- A It is almost always associated with a pelvic fracture.
 - B In fractured pelvis 5% will have associated urethral injury.
 - C It is usually a part of multiple trauma.
 - D The prostate might be high-riding and out of reach on rectal examination.
 - E A urethral catheter is inserted as a part of initial resuscitation.
- 10** Regarding testicular torsion, which of the following statements are true?
- A The typical symptom is sudden agonising pain in the scrotum.
 - B It is most common between 10 and 25 years of age.
 - C Inversion of the testis and a transverse lie are the most common causes.
 - D Doppler ultrasound (US) scan should be done to confirm the diagnosis.
 - E The emergency operation of fixation of the testis to the tunica albuginea with non-absorbable sutures should be carried out on the affected side.

Answers 1-10

BAILEY & LOVE MCQS

- 1** **D** All patients with renal failure will require renal replacement therapy. — **UPDATED** Not every case of renal failure necessitates replacement therapy; management depends on the underlying cause and severity.
- 2** **ABDE** True: A, B, D, E — **UPDATED** C is false: most closed renal injuries settle with conservative management; exploration is reserved for haemodynamic instability or an expanding haematoma.
- 3** **D** When ureteric injury is diagnosed postoperatively, delayed repair is undertaken. — **UPDATED** Early repair is recommended to prevent fibrosis in postoperative ureteric injuries.
- 4** **ABCD** True: A, B, C, D — **UPDATED** E is false: extraperitoneal rupture usually heals with catheter drainage; laparotomy and repair are needed for intraperitoneal rupture.
- 5** **E** Once urine drains on passing the catheter, the balloon should be immediately inflated. — **UPDATED** E is false: advance the catheter fully to its hub after urine appears before inflating the balloon, so it is not inflated in the prostatic urethra.
- 6** **BCDE** True: B, C, D, E — **UPDATED** A is false: chronic retention is typically painless, often presenting with overflow incontinence or renal impairment.

- 7 **C** Genuine stress incontinence (GSI) is from increased true detrusor pressure. — **UPDATED** C is false: genuine stress incontinence is leakage with raised intra-abdominal pressure without detrusor contraction, owing to sphincter weakness.
- 8 **ABCD** True: A, B, C, D — **UPDATED** E is false: urethral catheterisation is avoided as it may convert a partial into a complete tear; a full bladder is drained by suprapubic catheter.
- 9 **E** A urethral catheter is inserted as a part of initial resuscitation. — Avoid urethral catheterisation in membranous ruptures; the other statements correctly describe its context.
- 10 **BC** True: B, C — **UPDATED** Only B and C are true; note that pain may not be confined to the scrotum, US should not delay surgery and bilateral fixation is recommended.

2 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 A 69-year-old diabetic man presents to the acute surgery unit with a 5-day history of mild dysuria, frequency and feeling generally unwell. On examination, he is found to be pyrexial and tachycardic. A genital examination reveals both the penis and the scrotum to be swollen, red and tender to touch, with erythema also extending into the groin bilaterally. Of note, the examining surgeon believes that there is palpable crepitus in the perineum. Routine bloods and cultures are taken (which later grow both aerobic and anaerobic organisms), and fluid resuscitation and broad-spectrum antibiotics are commenced. Following further discussion with a urologist, he is taken promptly into the operating theatre for definitive management. The most likely diagnosis in this patient is
- A Fournier's gangrene
B Epididymo-orchitis
C Testicular tumour
D Testicular torsion
E Prostatitis
- 2 A 45-year-old railway worker with a crush injury displays red-brown urine containing myoglobin. Which metabolic abnormality is NOT associated with rhabdomyolysis?
- A Hypocalcaemia
B Hyperkalaemia
C Hypernatraemia
D Hyperphosphataemia
E Hyperuricaemia

Answers 1-2

SBA MCQs

- 1 **A** Fournier's gangrene — Fournier's gangrene is a necrotizing fasciitis of the perineum seen in diabetics.
- 2 **C** Hypernatraemia — Hypernatraemia is not typically seen in rhabdomyolysis unlike other electrolyte disturbances.

7 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 60-year-old man is seen by the surgeon prior to discharge, following bilateral inguinal hernia repair earlier that day. He has worsening dull lower abdominal pain and distension. Prior to surgery he was fit and well. Which is the single most likely diagnosis?
- A Ilioinguinal nerve damage
B Scrotal haematoma
C Strangulated omental fat within repair
D Urinary retention
E Wound infection
- 2 51-year-old man has a painless persistent penile erection, unrelated to sexual stimulation, which has been present for some hours. Which single statement is most appropriate?
- A Aspiration of the corpora cavernosa is used to treat high-flow priapism
B High-flow priapism is commonly caused by blunt force trauma to the penis
C Low-flow priapism is characterized by involvement of the glans and corpus spongiosum
D Priapism involves damage to the tunica albuginea penis
E Serum electrophoresis to detect myeloma is diagnostic for high-flow priapism

- 3** 30-year-old African-Caribbean man with sickle cell disease is seen in the emergency department with a persistent penile erection which has been present for 5h. His haemoglobin is 7.4g/dL. He has been started on IV fluids and supplementary oxygen. Which is the single next most appropriate management step?
- A Aspiration of the corpora cavernosa and manual pressure
 - B Blood transfusion
 - C Intracavernosal phenylephrine
 - D Oral beta-agonists
 - E Oral beta-antagonists
- 4** 13-year-old boy, who had mumps 4 months ago, develops sudden onset of pain in the right testicle, radiating to the right iliac fossa. He vomits twice. There is no tenderness or rebound on deep palpation of the abdomen. The right testicle is tender and high in the scrotum with a transverse lie. The patient is afebrile, with white cell count $7 \times 10^9/L$. Which is the single most appropriate subsequent management?
- A Active observation
 - B Doppler assessment of testicular blood flow
 - C Measurement of viral titres
 - D Mid-stream urine and urethral swab
 - E Urgent surgical exploration of scrotum
- 5** During the night, a 69-year-old man with haematuria develops acute urinary retention and presents to the emergency department. He denies any trauma. Several attempts to pass a transurethral catheter have failed. Which is the single most appropriate initial management?
- A Analgesia and check FBC, U&Es, creatinine
 - B Analgesia and fluid restriction
 - C Attempt to pass an irrigating three-way catheter
 - D Further attempt to pass single-lumen transurethral catheter
 - E Suprapubic catheterization
- 6** Nursing staff ask you to see a 65-year-old man, 6h following catheterization for retention, because he is producing 300mL of urine per hour. Which is the single most appropriate management?
- A Check U&Es
 - B Check U&Es and clamp his catheter
 - C Check U&Es and consider IV fluids
 - D Check U&Es and restrict his oral fluids
 - E Check U&Es and stop bendroflumethiazide and angiotensin-converting enzyme inhibitor
- 7** A 12-year-old boy has had a painful and swollen scrotum for 3h. There is no history of trauma, upper respiratory tract symptoms, or urinary symptoms. The testicle is very tender to palpation making clinical examination very difficult and unhelpful. Which is the single most appropriate next step?
- A Arrange colour flow Doppler assessment
 - B Arrange for surgical exploration of the testicle
 - C Obtain a urine sample for urinalysis
 - D Prescribe IV analgesia and reassess
 - E Prescribe oral ciprofloxacin

Answers 1-7

OXFORD MCQs

- 1** **D Urinary retention** — **UPDATED** Urinary retention is a recognised complication of bilateral hernia repair, precipitated by pain, anaesthetic drugs and prostatic enlargement.
- 2** **B High-flow priapism is commonly caused by blunt force trauma to the penis** — **UPDATED** High-flow (arterial) priapism usually follows blunt perineal or penile trauma; it is painless and non-ischaemic. Low-flow priapism spares the glans and spongiosum.
- 3** **A Aspiration of the corpora cavernosa and manual pressure** — **UPDATED** Ischaemic priapism over 4 h needs corporal aspiration first, then intracavernosal phenylephrine; in sickle cell disease transfusion is only an adjunct.
- 4** **E Urgent surgical exploration of scrotum** — Clinical signs suggest testicular torsion, requiring immediate surgical intervention.
- 5** **A Analgesia and check FBC, U&Es, creatinine** — **UPDATED** After failed catheterisation give analgesia, check renal function and call urology; repeated attempts risk false passage and SPC is avoided with haematuria.
- 6** **C Check U&Es and consider IV fluids** — High urine output may indicate post-obstructive diuresis; check electrolytes and consider IV fluids.
- 7** **B Arrange for surgical exploration of the testicle** — Urgent surgical exploration is mandatory to prevent testis infarction due to torsion.

1 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 35-year-old man is in a high-speed motorcycle collision and is initially transported to a small community hospital. Workup reveals a significant pelvic ring fracture. The hospital plans to insert a urinary catheter prior to transport to a level 1 trauma center. Which of the following is an indication to perform a retrograde urethrogram prior to insertion of a urinary catheter?
- A Microscopic hematuria on urinalysis
 - B Gross hematuria on urinalysis
 - C Suprapubic pain on abdominal exam
 - D High-riding prostate on digital rectal exam
 - E Pelvic fluid on focused assessment with sonography for trauma (FAST)

Answer 1

PRETEST MCQs

- 1 **D High-riding prostate on digital rectal exam** — High-riding prostate on DRE indicates possible urethral injury; perform retrograde urethrogram.

6 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1 A 17-year-old boy presents at 1 am with a 3-hour history of severe right testicular pain. No history of trauma, but he runs long distances. Examination is difficult due to pain. The testicle is swollen, slightly red, and extremely tender. What do you do next?
- A Admit for urgent ultrasound scan the next morning
 - B Contact the urologist on call and organize urgent ultrasound scan
 - C Contact the urologist on call and prepare for theatre
 - D Obtain a urine sample and prescribe antibiotics
 - E Prescribe strong analgesia and advise rest
- 2 You are called to see a 62-year-old man on the ward who was admitted earlier in the day with frank haematuria with the passage of clots. He is now complaining of lower abdominal pain and an inability to pass urine. On examination the bladder is distended. Which of the following would you perform in the first instance?
- A Catheterization using a Foley catheter
 - B Catheterization using a three-way catheter
 - C Cystoscopy
 - D Suprapubic catheterization
 - E Ultrasound of the bladder
- 3 A 34-year-old man with a history of renal stones presents with a 3-hour history of right-sided loin pain. On examination, he looks unwell and his right kidney is palpable. The presence of a large stone is confirmed on X-ray. Blood results show a urea of 25 mmol/L and a creatinine of 400 μ mol/L. On his previous admission last month, his urea and electrolytes were normal. Which of the following would be indicated in the first instance?
- A Extracorporeal shock wave lithotripsy
 - B Nephrectomy
 - C Percutaneous nephrolithotomy
 - D Percutaneous nephrostomy
 - E Ureteroscopy
- 4 A 69-year-old man with poorly controlled insulin-dependent diabetes mellitus comes to see you in the urology outpatient department complaining of troublesome incontinence. He tells you he does not feel the urge to pass urine. What is the most likely cause?
- A Atonic bladder
 - B Benign prostatic hypertrophy
 - C Bladder calculi
 - D Detrusor instability
 - E Urinary tract infection

- 5** A 20-year-old man presents to the GP with a 2-day history of severe pain and swelling in his right testicle. On examination, there is a purulent discharge present at the urethral meatus and the right testicle is tender to palpation. What is the most likely diagnosis?
- A Epididymo-orchitis
 - B Mumps orchitis
 - C Testicular cancer
 - D Testicular torsion
 - E Varicocele
- 6** A 21-year-old man is brought into the emergency department with penile pain and swelling following a fall onto the crossbar of a bicycle. On examination, the penis is oedematous and bruised and there is blood at the urethral meatus. He is otherwise stable. Which of the following investigations is indicated?
- A Bladder scan and urethral catheterization
 - B CT abdomen and pelvis
 - C Cystoscopy
 - D Intravenous urogram
 - E Retrograde urethrogram

Answers 1-6

GET AHEAD MCQS

- 1** **C Contact the urologist on call and prepare for theatre** — Suspected testicular torsion must be explored surgically without delay to save the testis.
- 2** **B Catheterization using a three-way catheter** — A three-way catheter allows irrigation to remove clots and relieve retention.
- 3** **D Percutaneous nephrostomy** — Urgent decompression of obstruction is needed to manage obstructive uropathy and prevent renal damage.
- 4** **A Atonic bladder** — Autonomic neuropathy in diabetes leads to a neurogenic atonic bladder and overflow incontinence.
- 5** **A Epididymo-orchitis** — Bacterial infection ascending from the urethra causing painful, swollen testis.
- 6** **E Retrograde urethrogram** — Suspected urethral injury must be confirmed by retrograde study before any catheter attempt.

5 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1** A 55-year-old woman with severe intermittent claudication is admitted for an elective arteriogram. She has mildly impaired renal function. After the procedure her urea and creatinine levels rise from 12.8 mmol/L and 163 µmol/L to 32.8 mmol/L and 363 µmol/L, respectively. What is the most likely cause of her acute renal failure?
- A Age
 - B Minimal fluid intake
 - C Radiological contrast nephropathy
 - D Penicillamine
 - E Past medical history of renal failure
- 2** A 47-year-old man has right testicular pain of quick onset. No fever, dysuria, or discharge. Enlarged tender right hemi-scrotum. Urinalysis normal, WCC $13 \times 10^9/L$. Doppler shows increased flow to testis/epididymis. Best initial treatment?
- A Ceftriaxone
 - B Clindamycin
 - C Famciclovir
 - D Ofloxacin
 - E Urgent urology review
- 3** A 25-year-old sexually active man has testicular pain + vomiting. On exam, the scrotum is tender, dipstick 3+ leucocytes + nitrites. Most likely diagnosis?
- A Pyelonephritis
 - B Interstitial cystitis
 - C Prostatitis
 - D Acute cystitis
 - E Epididymo-orchitis
- 4** A 65-year-old man post-TURP becomes unconscious with tonic-clonic seizures. Most likely cause?
- A Hypernatraemia
 - B Hyperkalaemia
 - C Hyponatraemia
 - D Hypokalaemia
 - E Withdrawal of anaesthetic
- 5** A 76-year-old man presents with a nontender, dome-shaped cystic mass up to the umbilicus, dull to percussion. Labs: Hb 96 g/L, urea 15 mmol/L, creatinine 160 µmol/L. Next step?
- A Abdominal CT scan
 - B Sigmoidoscopy
 - C Suprapubic aspiration
 - D Abdominal X-ray
 - E Urinary catheterization

Answers 1-5

CRASH COURSE MCQS

- 1 **C Radiological contrast nephropathy** — Contrast is nephrotoxic; her mild renal impairment put her at risk.
- 2 **D Ofloxacin** — Epididymo-orchitis in older men often from coliforms, covered by fluoroquinolones.
- 3 **E Epididymo-orchitis** — **UPDATED** Testicular pain with a tender scrotum and pyuria in a young sexually active man suggests epididymo-orchitis (Chlamydia, gonorrhoea or coliforms).
- 4 **C Hyponatraemia** — **UPDATED** TURP syndrome: absorption of hypotonic glycine irrigation causes dilutional hyponatraemia, leading to confusion, seizures and coma.
- 5 **E Urinary catheterization** — **UPDATED** A painless, non-tender dull suprapubic mass with raised urea and creatinine is chronic retention: catheterise, then watch for post-obstructive diuresis.

Prostate

Benign prostatic hyperplasia, LUTS, PSA and prostate cancer.

36

MCQs

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

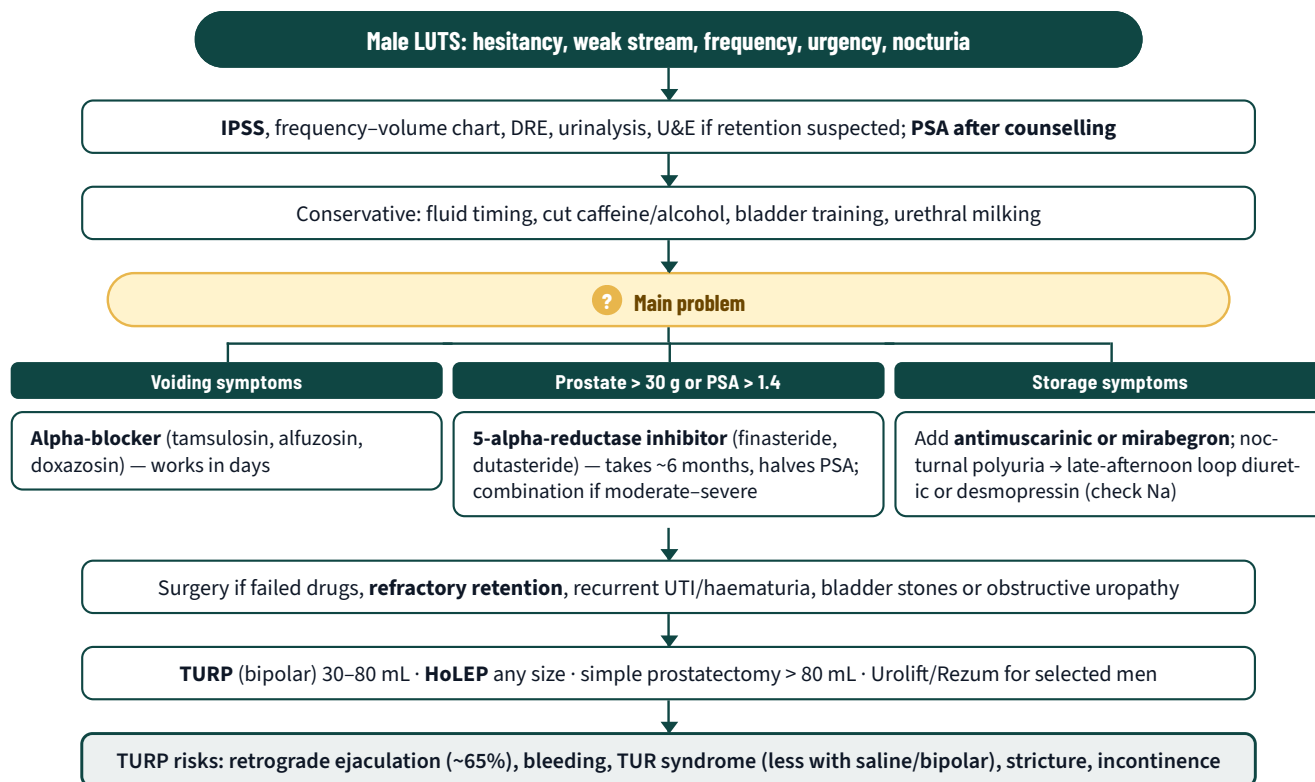
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Step by step

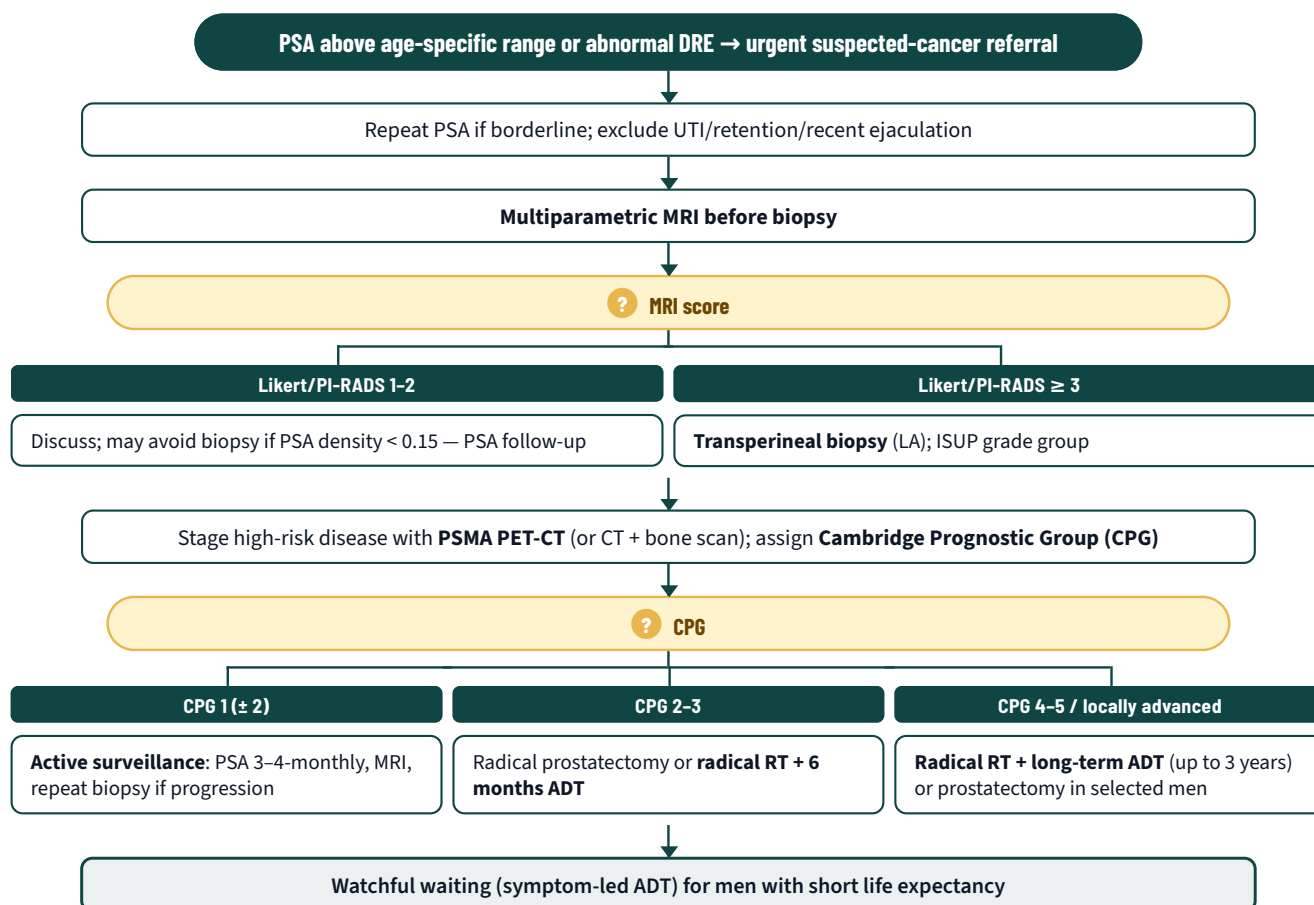
Benign prostatic enlargement and male LUTS

NICE CG97 • EAU 2024 MALE LUTS



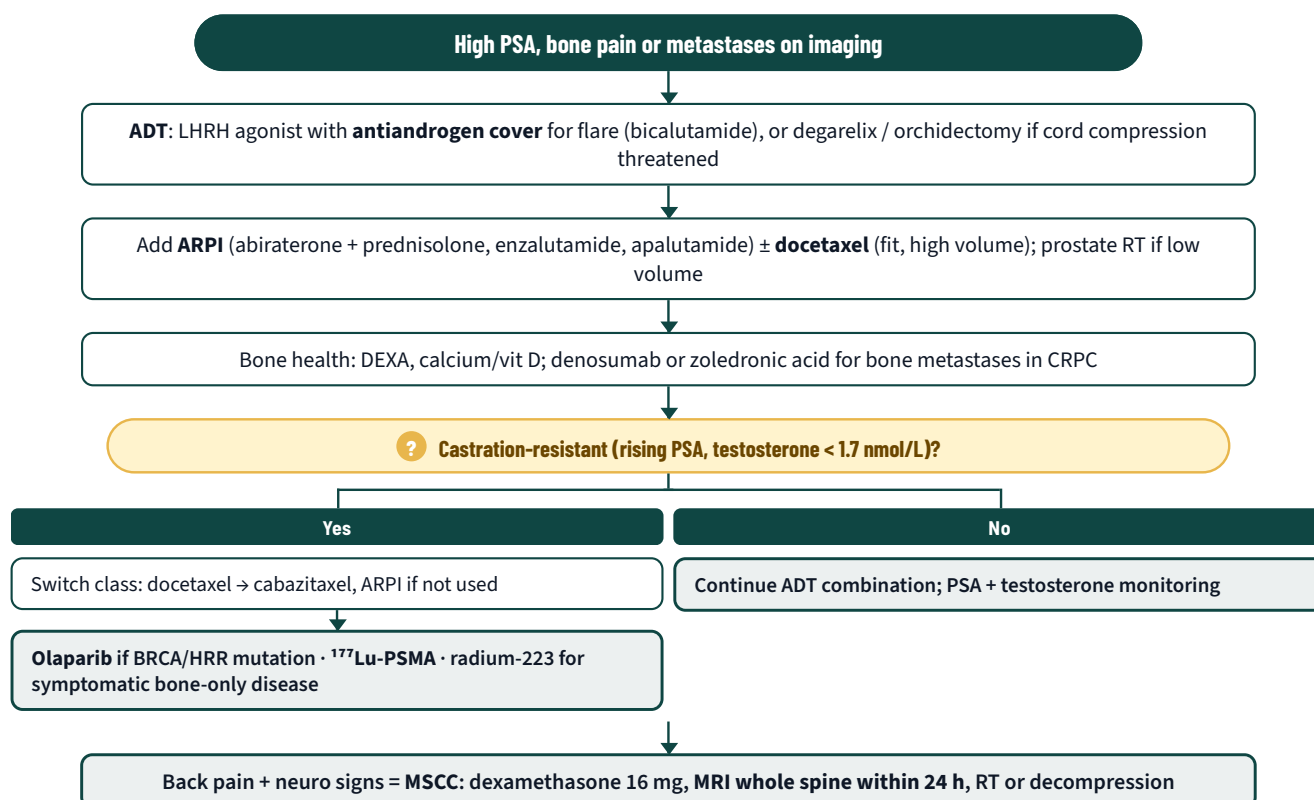
Raised PSA to localised prostate cancer

NICE NG131 · NICE NG12 · EAU 2024 PROSTATE CANCER



Metastatic prostate cancer

EAU 2024 PROSTATE CANCER · NICE NG131



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1 A 61-year-old man has frequency, hesitancy and a weak stream. DRE shows a smooth, symmetrically enlarged prostate. He has also had erectile dysfunction for 6 months. What is the first-line management?
 - A Tamsulosin
 - B Finasteride
 - C Doxazosin
 - D Tadalafil
 - E Prazosin
- 2 Eight years later his PSA is 16 ng/mL and the prostate is nodular and irregular. He has localised adenocarcinoma, Gleason 4+4, with no metastasis. What is the best treatment?
 - A TURP
 - B Radiotherapy
 - C Radical prostatectomy
 - D Chemotherapy
 - E Bilateral orchidectomy

MINISTERIAL EXAM

Ministerial evaluation examinations

- 3 A 62-year-old male with history of bothersome LUTS, mainly in the form of weak and interrupted stream, for the past 2 years came to the Urology clinic. He stated that despite being on tamsulosin 0.4 mg, his symptoms worsened and he had a Foley's catheter inserted in two previous occasions due to his inability to pass urine. DRE: moderately enlarged, yet benign-feeling prostate gland. Ultrasound: prostate size 52 grams. The most appropriate treatment option among the following is:
 - A Add finasteride 5 mg.
 - B Add an anticholinergic medication.
 - C Insert a permanent suprapubic catheter.
 - D Recommend a TURP procedure.
 - E Recommend an open prostatectomy.
- 4 A 72-year-old male patient with a known history of cancer presented with severe back pain for the past few months. A lateral view of the lumbar spine showed multiple dense sclerotic vertebral body lesions. Which of the following is the most likely primary cancer?
 - A Brain tumor.
 - B Renal carcinoma.
 - C Thyroid carcinoma.
 - D Prostate cancer.
 - E Mesothelioma.

Answers 1-4

LOCAL EXAMS MCQs

- 1 **D Tadalafil** — Tadalafil 5 mg daily treats both LUTS from BPH and erectile dysfunction; an alpha-blocker is first-line when ED is absent.
- 2 **C Radical prostatectomy** — High-risk localised prostate cancer in a fit man is treated radically: radical prostatectomy, or radiotherapy with long-term ADT.
- 3 **D Recommend a TURP procedure.** — **UPDATED** Recurrent retention despite α -blocker therapy is an absolute indication for surgery; TURP suits a 30–80 mL prostate.
- 4 **D Prostate cancer.** — **UPDATED** Osteosclerotic (osteoblastic) bone metastases in an elderly man are characteristic of prostate cancer; renal and thyroid metastases are typically lytic.

Bailey & Love MCQs

- 1 Regarding the anatomy of the prostate, the following statements are true except:
 - A The prostate is anatomically divided into a peripheral zone, a central zone and a transitional zone.
 - B The glands of the peripheral zone are lined by transitional epithelium.
 - C Benign prostatic hypertrophy occurs in the transitional zone.
 - D Most carcinomas arise in the peripheral zone.
 - E Denonvilliers' fascia separates the prostate from the rectum.
- 2 Regarding the hormonal control of the prostate, which of the following statements are true?
 - A LH from the anterior pituitary controls testosterone secretion.
 - B Leydig cells of the testes secrete 90% of testosterone.
 - C 5 α -reductase type II is found in high concentration in the prostate.
 - D 5 α -reductase converts testosterone into 5 α -dihydrotestosterone (DHT).
 - E Prostate-specific antigen (PSA) is both specific and sensitive in the diagnosis of prostate cancer.
- 3 Which of the following statements are true in carcinoma of the prostate?
 - A It is the most common cancer in men over the age of 65 years.
 - B In the younger age group who develop prostate cancer, 10%–15% have a positive family history.
 - C A patient who has had a prostatectomy for BPH can develop prostate cancer.
 - D Carcinoma mostly originates in the peripheral zone of the prostate.
 - E Prostate-specific antigen (PSA) is an extremely reliable method of screening.
- 4 Regarding carcinoma of the prostate, the following statements are true except:
 - A The histological type is an adenocarcinoma.
 - B Gleason scoring system is based on the degree of glandular differentiation.
 - C The Gleason score correlates well with spread and prognosis.
 - D Prostate is the most common site of origin for skeletal metastasis.
 - E Skeletal metastases from prostate cancer are mostly osteolytic.
- 5 On rectal examination, which of the following features does not suggest carcinoma?
 - A Nodules within the prostate.
 - B Obliteration of the median sulcus.
 - C Irregular stony hard induration.
 - D Mobile rectal mucosa over the prostate.
 - E Extension of a hard nodule beyond the capsule into the bladder base.
- 6 In the assessment of prostate cancer, which of the following investigations are necessary?
 - A Transrectal ultrasound-guided biopsy (TRUS).
 - B Cross-sectional MRI.
 - C Bone scan.
 - D General blood tests, PSA and liver function tests.
 - E IVU.
- 7 Regarding the treatment of prostate cancer, the following statements are true except:
 - A Radical prostatectomy is only suitable for T1 and T2 disease (early cancer).
 - B Radical external-beam radiotherapy is an alternative to radical prostatectomy.
 - C T3 patients are treated by surgery, radical radiotherapy, or androgen ablation.
 - D Incidentally diagnosed disease is treated by radical prostatectomy.
 - E The incidence of severe stress incontinence after radical prostatectomy is about 2%.
- 8 Regarding benign prostatic hyperplasia (BPH), which of the following statements are true?
 - A It is the most common cause of bladder outflow obstruction (BOO) in men >70 years of age.
 - B Decrease in serum testosterone levels and therefore relative increase in serum oestrogens cause BPH.
 - C The condition affects the transitional zone and the central zone.
 - D All lower urinary tract symptoms (LUTS) in men >70 years are due to BPH.
 - E The prostatic urethra is elongated in BPH.
- 9 In urodynamic studies, which of the following statements are true?
 - A A voided volume >200 mL with a peak flow rate >15 mL/s is normal.
 - B A voiding pressure of <60 cm H₂O is normal.
 - C Increase in voiding pressures with a low flow rate is diagnostic of BOO.
 - D In patients with a residual volume of 250 mL or more, bilateral hydronephrosis will result.
 - E Urodynamically proven BOO only occurs from BPH.

- 10** In assessing LUTS in men, which of the following statements are false?
- A Charting a patient's symptoms according to the IPSS is routinely used.
 - B Flow-rate measurement studies and pressure-flow urodynamic studies are done.
 - C Thorough nervous system examination is mandatory.
 - D The upper urinary tract should always be imaged.
 - E Cystourethroscopy is a good guide as an indication for surgery.
- 11** Regarding management of BPH causing bladder out-flow obstruction, which of the following statements are true?
- A All patients with acute retention ultimately require prostatectomy.
 - B Chronic retention accounts for 15% of prostatectomies.
 - C Severe symptoms account for 60% of prostatectomies.
 - D Drug therapy is with α -blockers and/or 5-reductase inhibitor.
 - E A low maximum flow rate of <10 mL/s and a residual volume of 100–250 mL are indications for surgery.
- 12** The following statements are true except:
- A α -adrenergic blocking agents inhibit the contraction of prostatic smooth muscle.
 - B 5-reductase inhibitors inhibit the conversion of testosterone to DHT.
 - C Drug therapy results in 20% improvement in symptom scores.
 - D Transurethral resection of the prostate (TURP) results in 75% improvement in symptom scores.
 - E Retrograde ejaculation occurs in all patients after TURP.

Answers 1–12

- 1** **B** The glands of the peripheral zone are lined by transitional epithelium. — **UPDATED** In reality, the peripheral zone is lined by columnar epithelium.
- 2** **ABCD** True: A, B, C, D — **UPDATED** E is false: PSA is prostate-specific but not cancer-specific (raised in BPH, prostatitis and retention) and can be normal in some cancers.
- 3** **ABCD** True: A, B, C, D — **UPDATED** E is false: PSA lacks specificity and sensitivity, causing overdiagnosis and missed cancers, so it is not a reliable screening test.
- 4** **E** Skeletal metastases from prostate cancer are mostly osteolytic. — **UPDATED** Prostate cancer bone metastases are typically osteoblastic.
- 5** **D** Mobile rectal mucosa over the prostate. — A mobile rectal mucosa is more suggestive of benign conditions.
- 6** **ABCD** True: A, B, C, D — **UPDATED** E is false: IVU has no routine role; assessment uses biopsy (TRUS or MRI-guided), MRI, bone scan for higher-risk disease, and PSA and blood tests.
- 7** **D** Incidentally diagnosed disease is treated by radical prostatectomy. — **UPDATED** Incidentally discovered prostate cancer is usually managed conservatively or with active surveillance.
- 8** **ABCE** True: A, B, C, E — **UPDATED** D is false: LUTS in older men also arise from detrusor over- or underactivity, stricture, neurological disease and cancer, not only BPH.
- 9** **ABC** True: A, B, C — **UPDATED** D is false: a large residual may, but need not, cause hydronephrosis. E is false: BOO also arises from stricture, bladder neck stenosis and cancer.
- 10** **DE** True: D, E — **UPDATED** D is false: upper tracts are imaged only with retention, renal impairment, haematuria or infection. E is false: cystoscopic findings correlate poorly with obstruction.
- 11** **BCDE** True: B, C, D, E — **UPDATED** A is false: many men with acute retention void after catheter removal on an α -blocker (trial without catheter), so not all need prostatectomy.
- 12** **E** Retrograde ejaculation occurs in all patients after TURP. — Retrograde ejaculation is common after TURP but does not occur in every case.

1 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 70-year-old man with back pain, difficulty initiating stream, hard irregular prostate, elevated PSA, and blastic bony lesions. Biopsy shows well-differentiated adenocarcinoma. Which of the following is the treatment of choice?
- A Radical prostatectomy
 - B Transurethral prostatectomy
 - C Cytotoxic chemotherapy
 - D Hormonal manipulation
 - E Radiotherapy

Answer 1

LANGE MCQs

- 1 **D** **Hormonal manipulation** — UPDATED Metastatic prostate cancer (blastic bone lesions) is treated with androgen deprivation

(orchidectomy or LHRH analogue); radical treatment is for localised disease.

5 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Which of the following vessels contributes to the blood supply of the prostate gland?
- A Coeliac plexus
 - B Superior mesenteric artery
 - C Inferior mesenteric artery
 - D Internal iliac artery
 - E External iliac artery
- 2 An 80-year-old man with urinary frequency, slow stream and terminal dribbling is found to have a smoothly enlarged prostate. Which zone of the prostate is most likely enlarged?
- A Transition zone
 - B Central zone
 - C Peripheral zone
 - D Anterior fibromuscular stroma
 - E All of the above

- 3 A 75-year-old man presents to the urology outpatient clinic with a 6-month history of urinary frequency associated with difficulty initiating micturition, dribbling on reaching the end of his stream and nocturia. A digital rectal examination reveals a hard prostate gland. Core biopsy is performed on the gland and the pathology report indicates the presence of a tumour involving most of the right lobe. According to the TNM classification, the tumour stage is
- A T2 N1 M0
 - B T3 Nx Mx
 - C T3 N1 M0
 - D T2 Nx Mx
 - E T1 N1 M0
- 4 A 75-year-old man is admitted with acute-onset suprapubic pain and inability to pass urine for 2 days. On examination, the patient is in discomfort, neurologically intact, and the abdomen is particularly tender in the suprapubic region. A digital rectal examination reveals a smooth, enlarged prostate. What is the most likely diagnosis?
- A Bladder outflow obstruction due to prostate cancer
 - B Bladder outflow obstruction due to benign prostatic hypertrophy
 - C Bladder outflow obstruction due to a urethral stricture
 - D Bladder outflow obstruction due to a spinal cord lesion
 - E None of the above

- 5 A 75-year-old man has acute suprapubic pain and inability to pass urine for 2 days; rectal examination shows a smooth, enlarged prostate. What is the most appropriate course of action in managing him?
- A Ask the urology registrar to see the patient
 - B Obtain blood for urea and electrolytes sampling
 - C Request an abdominal plain film radiograph
 - D Urinary catheterisation
 - E Request an intravenous urogram

Answers 1-5

SBA MCQS

- 1 **D Internal iliac artery** — **UPDATED** The prostate gland is mainly supplied by branches of the internal iliac artery.
- 2 **A Transition zone** — **UPDATED** Benign prostatic hyperplasia typically involves enlargement of the transition zone.
- 3 **D T2 Nx Mx** — **UPDATED** A palpable tumour confined to the prostate involving more than half of one lobe is T2 (T2b); without staging scans, nodes and metastases are unassessed (Nx, Mx).
- 4 **B Bladder outflow obstruction due to benign prostatic hypertrophy** — **UPDATED** DRE findings indicate benign prostatic hypertrophy causing urinary retention.
- 5 **D Urinary catheterisation** — **UPDATED** Immediate catheterisation is essential to relieve acute urinary retention.

3 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 58-year-old man has unremitting right-sided chest pain, worse on deep inspiration and coughing. He has urinary hesitancy, diminished stream, and post-micturition dribbling. He is tender in the right upper quadrant of the abdomen, and in the lower chest on palpation of the costal margin and lower ribs. His abdomen is soft. He has a hard enlarged prostate. His alkaline phosphatase is 260U/L. Which is the single most appropriate investigation?
- A Abdominal ultrasound
 - B Abdominal X-ray
 - C Chest X-ray
 - D Serum prostate-specific antigen
 - E Transrectal ultrasound scan
- 2 66-year-old man has episodes of recurrent suprapubic pain, with haematuria and dysuria. He describes a poor urinary stream, difficulty in initiating micturition, and has to get up frequently during the night to void. Which is the single most likely diagnosis?
- A Benign prostatic hyperplasia
 - B Bladder calculi
 - C Duplex system
 - D Neuropathic bladder
 - E Transitional cell carcinoma
- 3 93-year-old man has symptoms of prostatism, a craggy asymmetric prostate on rectal examination, and an elevated serum prostate-specific antigen. Transrectal ultrasound suggests the presence of a carcinoma localized to the prostate. Which is the single most appropriate next step in management?
- A Active monitoring
 - B External beam radiotherapy
 - C Hormone therapy
 - D Radical prostatectomy
 - E Transurethral resection of prostate

Answers 1-3

OXFORD MCQS

- 1 **C Chest X-ray** — **UPDATED** Pleuritic pain with rib tenderness, a hard prostate and raised ALP suggest rib metastases or pathological fracture; a chest X-ray assesses ribs, lungs and pleura first.
- 2 **B Bladder calculi** — **UPDATED** Recurrent suprapubic pain, haematuria and dysuria with outflow symptoms suggest bladder calculi forming from stasis behind BPH; BPH alone does not explain the pain.

- 3 **C** **Hormone therapy** — **UPDATED** At 93, radical treatment is inappropriate; androgen deprivation controls symptomatic cancer and relieves outflow symptoms.

3 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 58-year-old man is found to have an enlarged prostate on examination. His serum prostate-specific antigen (PSA) concentration is elevated. Which of the following is the most appropriate next step in the evaluation of this patient?
 - A Prostate biopsy
 - B Computed tomography of the abdomen and pelvis
 - C Magnetic resonance imaging of the abdomen and pelvis
 - D Positron emission tomography scan
 - E Bone scan
- 2 A 60-year-old man seeks medical attention because of recurrent urinary tract infections. The patient also reports a history of increasing difficulty in urination (decreased flow, straining, and hesitancy) over the last several months. A PSA level is mildly elevated and a prostate biopsy proves benign. Which of the following is the most appropriate initial management of this patient with benign prostatic hyperplasia (BPH)?
 - A α -Adrenergic blocker
 - B 5-Alpha reductase inhibitor
 - C α -Adrenergic blocker and 5-alpha reductase inhibitor
 - D Transurethral resection of the prostate (TURP)
 - E Open prostatectomy
- 3 A 55-year-old man presents with fever and pain in the perineal region. Upon further questioning he also complains of frequency, urgency, dysuria, and a decreased urinary stream. On physical examination, his abdomen is soft, nondistended, and nontender. Digital rectal examination demonstrates exquisite tenderness on the anterior aspect. Laboratory examination reveals leukocytosis and findings on urinalysis are consistent with a bacterial infection. Which of the following is the most likely diagnosis?
 - A Urinary tract infection
 - B Benign prostatic hyperplasia
 - C Prostatitis
 - D Pyelonephritis
 - E Nephrolithiasis

Answers 1-3

PRETEST MCQS

- 1 **A** **Prostate biopsy** — **UPDATED** Raised PSA with an abnormal prostate needs histological confirmation by biopsy (now usually preceded by prostate MRI); staging scans follow a positive biopsy.
- 2 **D** **Transurethral resection of the prostate (TURP)** — Recurrent urinary tract infection is an indication for TURP.
- 3 **C** **Prostatitis** — The patient has signs and symptoms consistent with acute prostatitis.

3 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1 A 72-year-old man complains of a 6-month history of urinary hesitancy, poor stream, and occasional incontinence. He has a PSA of 30 ng/mL. Which of the following is the most appropriate next step?
 - A Antibiotics
 - B Medical management with alpha-blockers
 - C Referral for transurethral resection of prostate
 - D Urgent referral for renal tract ultrasound
 - E Urgent referral for transrectal ultrasound and prostate biopsy
- 2 A 69-year-old man presents with a 6-week history of lower back and right hip pain. An X-ray of the hip shows an osteosclerotic lesion in the proximal femur. On direct questioning he tells you he has been having problems passing urine over the past few months. Which of the following is the most likely cause of his symptoms?
 - A Benign prostatic hypertrophy
 - B Paget disease
 - C Prostatic carcinoma
 - D Prostatitis
 - E Testicular carcinoma
- 3 A 59-year-old man presents with an 8-month history of urinary hesitancy, poor stream and the sensation of incomplete bladder emptying. He is otherwise well. On examination, his prostate is homogeneously enlarged and smooth. Which of the following interventions would be indicated?
 - A Bladder neck incision
 - B Brachytherapy
 - C External beam radiotherapy
 - D Radical prostatectomy
 - E Transurethral resection of prostate

Answers 1-3

GET AHEAD MCQs

- 1 **E Urgent referral for transrectal ultrasound and prostate biopsy** — High PSA and obstructive symptoms require urgent biopsy to exclude prostate carcinoma.
- 2 **C Prostatic carcinoma** — Osteosclerotic bone lesions with urinary symptoms suggest metastatic prostate cancer.
- 3 **E Transurethral resection of prostate** — UPDATED Of these options only TURP treats outflow obstruction from benign enlargement (usually after failed medical therapy); bladder neck incision suits small prostates.

5 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQs

Crash Course MCQs

- 1 An 82-year-old man with a 3-month history of weak stream, straining, hesitancy. Severely enlarged prostate, normal urinalysis, PSA 4.2 µg/L. Which is correct for BPH treatment?
 - A α-adrenergic antagonists reduce smooth muscle tone
 - B β-reductase agonists that increase smooth muscle tone
 - C β-reductase agonists that reduce smooth muscle tone
 - D 5α-reductase inhibitors that reduce smooth muscle tone
 - E 5α-reductase inhibitors that increase smooth muscle tone
- 2 A 62-year-old man has frequency, dysuria, and a tender, boggy prostate on rectal exam. Most likely diagnosis?
 - A Pyelonephritis
 - B Interstitial cystitis
 - C Prostatitis
 - D Acute cystitis
 - E Epididymo-orchitis

- 3** A 55-year-old man is worried about prostate cancer like his father and uncle. Asks for a PSA. Which statement is not true about PSA?
- A PSA is a protein from prostate cells
 - B PSA is sensitive & specific for prostate cancer
 - C PSA is elevated in prostatitis
 - D PSA is elevated in BPH
 - E PSA monitors recurrence of prostate cancer
- 4** A 72-year-old man with mild BPH symptoms per IPSS. Next step?
- A Watchful waiting
 - B Radical prostatectomy
 - C Radiotherapy
 - D 5 α -reductase inhibitor
 - E α -blockers
- 5** A 56-year-old man has fever, backache, poor urinary stream. PR: tender, boggy, regular-size prostate. Which drug?
- A Amoxicillin
 - B Ciprofloxacin
 - C Doxazosin
 - D Finasteride
 - E Trimethoprim

Answers 1-5

CRASH COURSE MCQS

- 1** **A** α -adrenergic antagonists reduce smooth muscle tone — First-line in BPH to improve urine flow (e.g. tamsulosin).
- 2** **C** Prostatitis — Tender, boggy prostate with LUTS => acute prostatitis.
- 3** **B** PSA is sensitive & specific for prostate cancer — PSA is neither fully sensitive nor specific; can be raised in other conditions.
- 4** **A** Watchful waiting — Mild symptoms => conservative management first.
- 5** **B** Ciprofloxacin — Acute bacterial prostatitis; fluoroquinolone penetrates prostate well.

Congenital Urology

Undescended testis, hypospadias, PUJ obstruction, vesicoureteric reflux and posterior urethral valves.

22

MCQs

0

EMQ SCENARIOS

6

SOURCES

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REPEATED IDEAS

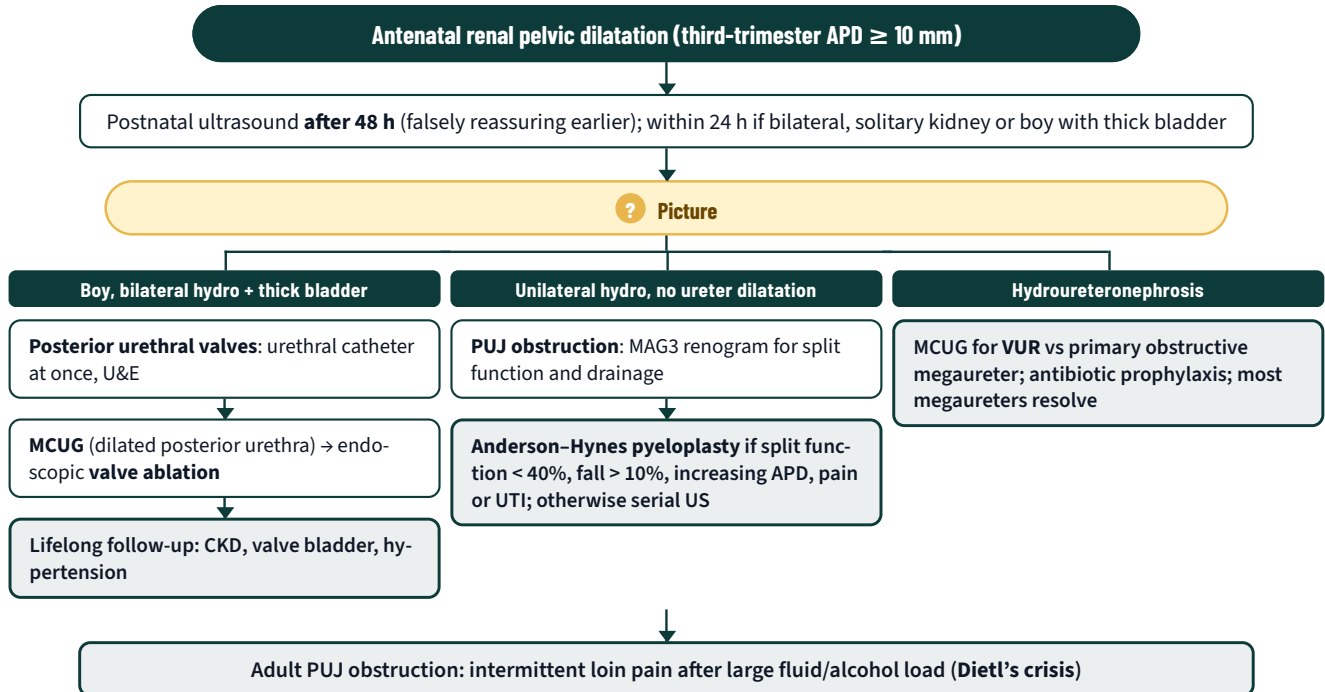
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Step by step

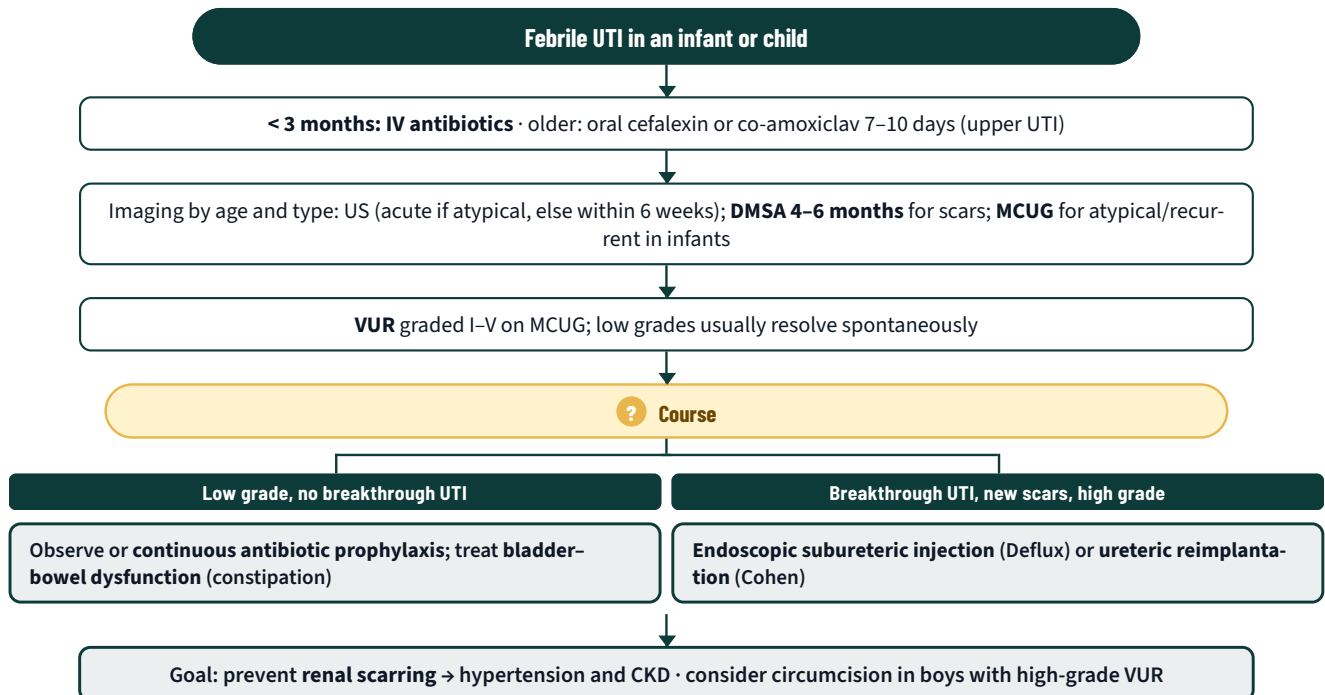
Antenatal hydronephrosis, PUJ obstruction and posterior urethral valves

EAU/ESPU 2024 PAEDIATRIC UROLOGY



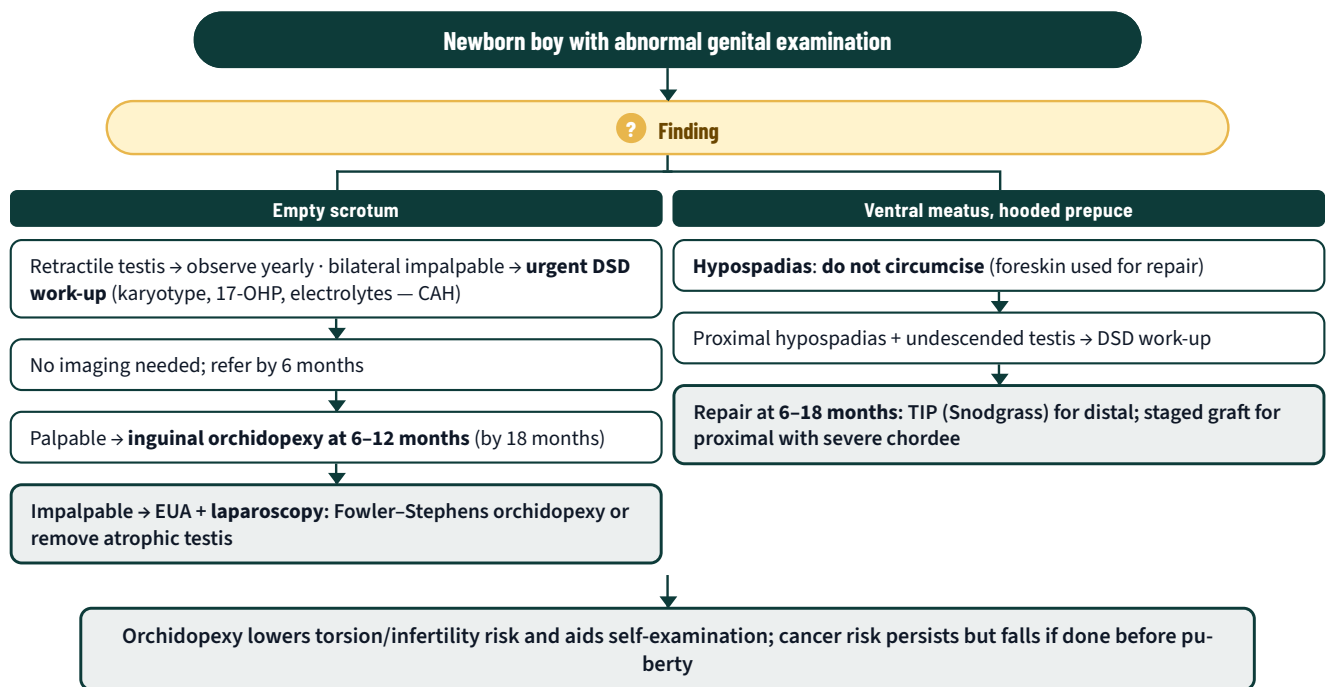
Childhood UTI and vesicoureteric reflux

NICE NG224 • EAU/ESPU 2024 PAEDIATRIC UROLOGY



Undescended testis and hypospadias

EAU/ESPU 2024 PAEDIATRIC UROLOGY · BAPS



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1–3

A mother brings her baby boy because of an abnormal downward urinary stream. The urethral opening is not in its normal position.

- 1 What is the most likely diagnosis?
 - A Phimosis
 - B Hypospadias
 - C Epispadias
 - D Meatal stenosis
 - E Posterior urethral valves
- 2 Which associated abnormality is most common?
 - A Undescended testis
 - B Polydactyly
 - C Coarctation of the aorta
 - D Inguinal hernia
 - E Renal agenesis
- 3 What is the most common complication after surgical repair?
 - A Penile fracture
 - B Urethrocutaneous fistula
 - C Renal failure
 - D Erectile dysfunction
 - E Urinary retention

CASE • QUESTIONS 4–6

A 4-year-old boy has left lower abdominal pain and is underweight. Ultrasound shows hydronephrosis and urinalysis shows mild painless pyuria.

- 4 Which investigation confirms the diagnosis?
 - A IVU
 - B Contrast CT
 - C MRI (MR urography)
 - D KUB
 - E Ureteroscopy

- 5 Which is the best test of split renal function and drainage?
 - A DMSA scan
 - B MAG3 diuretic renogram
 - C Ultrasound
 - D eGFR
- 6 What is the best long-term management?
 - A Percutaneous nephrostomy
 - B Long-term antibiotics
 - C Dismembered pyeloplasty
 - D Ureteric stent
 - E Nephrectomy

MINISTERIAL EXAM

Ministerial evaluation examinations

- 7 A 2-year-old male presents with recurrent febrile UTI. Ultrasound showed severe dilatation of the left renal pelvis and calyces with normal-looking bladder. Voiding cystourethrography (VCUG) was unremarkable. What is the most likely provisional diagnosis?
 - A Left-sided obstructive ureterocele.
 - B Pelvi-ureteric junction obstruction.
 - C Posterior urethral valve.
 - D Wilms tumor of the left kidney.
 - E Left-sided vesico-ureteric reflux.

FORMATIVE

College formative examinations

- 8 A 3-year-old boy has a history of recurrent UTIs. Ultrasound shows mild bilateral hydronephrosis. What is the most likely diagnosis?
 - A Vesicoureteric reflux
 - B Posterior urethral valve
 - C Bladder outlet obstruction
 - D Urolithiasis
 - E Neurogenic bladder
- 9 What is the best diagnostic tool for confirming vesicoureteric reflux?
 - A Intravenous urogram (IVU)
 - B CT scan
 - C Voiding cystourethrogram (VCUG)
 - D MRI
 - E Ultrasound

Answers 1–9

LOCAL EXAMS MCQs

- 1 **B Hypospadias** — A ventral ectopic meatus with a downward stream is hypospadias (often with chordee and a dorsal hooded foreskin).
- 2 **A Undescended testis** — Undescended testis ($\approx 10\%$) and inguinal hernia are the commonest associations; with both hypospadias and impalpable testes, exclude a DSD.

- 3** **B Urethrocutaneous fistula** — Urethrocutaneous fistula is the commonest complication of hypospadias repair; meatal stenosis and stricture also occur.
- 4** **A IVU** — PUJ obstruction is confirmed on urography (IVU, CT or MR urography) and a diuretic renogram; ureteroscopy has no role.
- 5** **B MAG3 diuretic renogram** — A MAG3 diuretic renogram measures differential function and confirms obstruction; DMSA shows scarring and static function.
- 6** **C Dismembered pyeloplasty** — Symptomatic PUJ obstruction or falling function is treated by dismembered (Anderson–Hynes) pyeloplasty; nephrectomy only for a non-functioning kidney.
- 7** **B Pelvi-ureteric junction obstruction.** — **UPDATED** Marked unilateral pelvicalyceal dilatation with a normal bladder and a normal VCUG (no reflux, no valves) indicates pelvi-ureteric junction obstruction.
- 8** **A Vesicoureteric reflux** — Recurrent UTI with hydronephrosis in a child strongly suggests vesicoureteric reflux.
- 9** **C Voiding cystourethrogram (VCUG)** — VCUG is the gold standard for diagnosing vesicoureteric reflux.

8 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1** Which of the following statements are true?
- A The incidence of horseshoe kidney is one in 1000.
 - B A horseshoe kidney is liable to pelviureteric junction obstruction.
 - C Once the diagnosis of a horseshoe kidney is made the isthmus should be divided.
 - D Polycystic kidneys can be transmitted by either parent as an autosomal dominant trait.
 - E Polycystic kidneys present in childhood as renal failure.
- 2** The following statements are true except:
- A Duplication of the renal pelvis is found in 4% and ureter in 3% of patients.
 - B The previously mentioned abnormalities are usually incidental findings.
 - C Retrocaval ureter causes bilateral hydronephrosis.
 - D Ureterocele is a cystic enlargement of the intramural ureter.
 - E All ureteroceles must be treated by endoscopic diathermy incision.
- 3** In pelviureteric junction (PUJ) obstruction (idiopathic or congenital hydronephrosis) the following statements are true except:
- A The condition might be asymptomatic.
 - B It can be diagnosed in utero.
 - C Ultrasound scanning is the least invasive method of imaging.
 - D An intravenous urogram (IVU) is the ideal imaging.
 - E The kidney should be preserved if it has more than 5% of renal function.
- 4** Regarding anatomy of Urethra and penis, which of the following statements are true?
- A The verumontanum is situated in the membranous urethra.
 - B The membranous urethra is the primary location of continence.
 - C The anterior urethra is composed of a bulbar and a penile part.
 - D The posterior urethra comprises of preprostatic, prostatic and membranous segments.
 - E The major portion of the female urethra is lined by stratified squamous epithelium.
- 5** Which of the following statements is false?
- A Posterior urethral valves causing obstruction can be diagnosed antenatally.
 - B The incidence of hypospadias is 1 in 200–300.
 - C Hypospadias is the most common congenital urethral anomaly.
 - D Glandular hypospadias should be surgically treated early in life.
 - E Circumcision must be avoided in boys with hypospadias.
- 6** Regarding embryology and anatomy of testis & scrotum, which of the following statements are true?
- A The processus vaginalis accompanies the testis as it descends.
 - B The testis is at the deep inguinal ring by the seventh month of fetal life.
 - C The testicular veins drain into the inferior vena cava.
 - D The lymphatics of the testes drain into the para-aortic lymph nodes.
 - E The testicular arteries arise from the abdominal aorta.

- 7** Regarding incompletely descended testis, the following statements are true except:
- A The incidence is 4%.
 - B A testis absent from the scrotum after 3 months is unlikely to descend.
 - C An incompletely descended testis tends to atrophy as puberty approaches.
 - D Retractable testis is best treated by orchidopexy.
 - E Orchidopexy reduces the chances of developing a testicular tumour.
- 8** The hazards of incomplete descent of the testis include the following:
- A Impaired fertility.
 - B Hernia.
 - C Torsion.
 - D Increased liability to malignant change in later life.
 - E Epididymo-orchitis.

Answers 1-8

BAILEY & LOVE MCQS

- 1** **ABD** True: A, B, D — **UPDATED** C is false: the isthmus is not divided routinely (only if needed during pyeloplasty); E is false: adult polycystic kidney disease usually presents in adult life.
- 2** **CE** True: C, E — Retrocaval ureter affects only the right side (causing unilateral hydronephrosis) and not all ureterocele require treatment.
- 3** **DE** True: D, E — Isotope renography is preferred for evaluation and kidney preservation generally requires >20% function.
- 4** **BCDE** True: B, C, D, E — **UPDATED** B, C, D, E are true; verumontanum is actually in the prostatic urethra.
- 5** **D** Glandular hypospadias should be surgically treated early in life. — Glandular hypospadias is mild and usually does not require early surgery; the other statements are true.
- 6** **ABDE** True: A, B, D, E — **UPDATED** A, B, D and E are true; note that testicular venous drainage is asymmetrical (C is false).
- 7** **DE** True: D, E — **UPDATED** Retractable testis is a normal variant and orchidopexy does not reduce cancer risk.
- 8** **ABCDE** True: A, B, C, D, E — **UPDATED** All are recognised hazards (Bailey & Love): infertility, associated hernia, torsion, epididymo-orchitis (mimics appendicitis on the right) and malignant change.

2 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A 41-year-old man with multiple kidney cysts and hepatomegaly is suspected of having adult polycystic kidney disease. What is the most common mode of inheritance?
- A Autosomal recessive gene chromosome 16
 - B Autosomal dominant gene chromosome 16
 - C Autosomal dominant gene chromosome 8
 - D Autosomal recessive gene chromosome 4
 - E Autosomal dominant gene chromosome 4
- 2** A 6-week-old boy presents with a urethral meatus located ventrally just below the glans. What is the most likely diagnosis?
- A Hypospadias
 - B Epispadias
 - C Phimosis
 - D Chordee
 - E Perineal urethra

Answers 1-2

SBA MCQS

- 1** **B** Autosomal dominant gene chromosome 16 — Adult polycystic kidney disease is most commonly autosomal dominant on chromosome 16.
- 2** **A** Hypospadias — A ventrally displaced urethral meatus is characteristic of hypospadias.

1 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 In clinic, the parents of a 5-year-old boy report that his foreskin often appears ballooned and appears to have a white tip. The boy's father had suffered from the same condition as a teenager. Which is the single most appropriate advice?
- A A course of antibiotics is required
 - B Foreskin biopsy should be undertaken
 - C The definitive treatment is circumcision
 - D The symptoms should settle without treatment
 - E Topical steroid ointment should be prescribed

Answer 1

OXFORD MCQS

- 1 **C** The definitive treatment is circumcision — UPDATED A white, scarred preputial tip suggests balanitis xerotica obliter-

ans (pathological phimosis), for which circumcision is definitive; simple ballooning alone is physiological.

1 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 What congenital problem predisposes to testicular cancer?
- A Hydrocele
 - B Phimosis
 - C Cryptorchidism
 - D Hypospadias
 - E Inguinal hernia

Answer 1

PRETEST MCQS

- 1 **C** Cryptorchidism — Cryptorchidism increases the risk of testicular cancer, especially if not corrected early.

1 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 4-year-old boy is brought to the urology clinic by his mother. She is concerned as he is having difficulty urinating properly and is causing a mess. He is otherwise growing and developing normally. On examination, you see the urethral meatus is present on the shaft of the underside of the penis. Which of the following treatment options would you recommend?
- A Antibiotics
 - B Circumcision
 - C Intracavernosal injection of phenylephrine
 - D Referral to a specialist for corrective surgery
 - E Steroid cream

Answer 1

GET AHEAD MCQS

- 1 **D** Referral to a specialist for corrective surgery — This is hypospadias, requiring corrective surgery if symptomatic.

Scrotum & Testis

Hydrocele, varicocele, epididymo-orchitis, scrotal swellings and testicular tumours.

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MCQS

0

EMQ SCENARIOS

7

SOURCES

1

REPEATED IDEAS

IN THIS CHAPTER

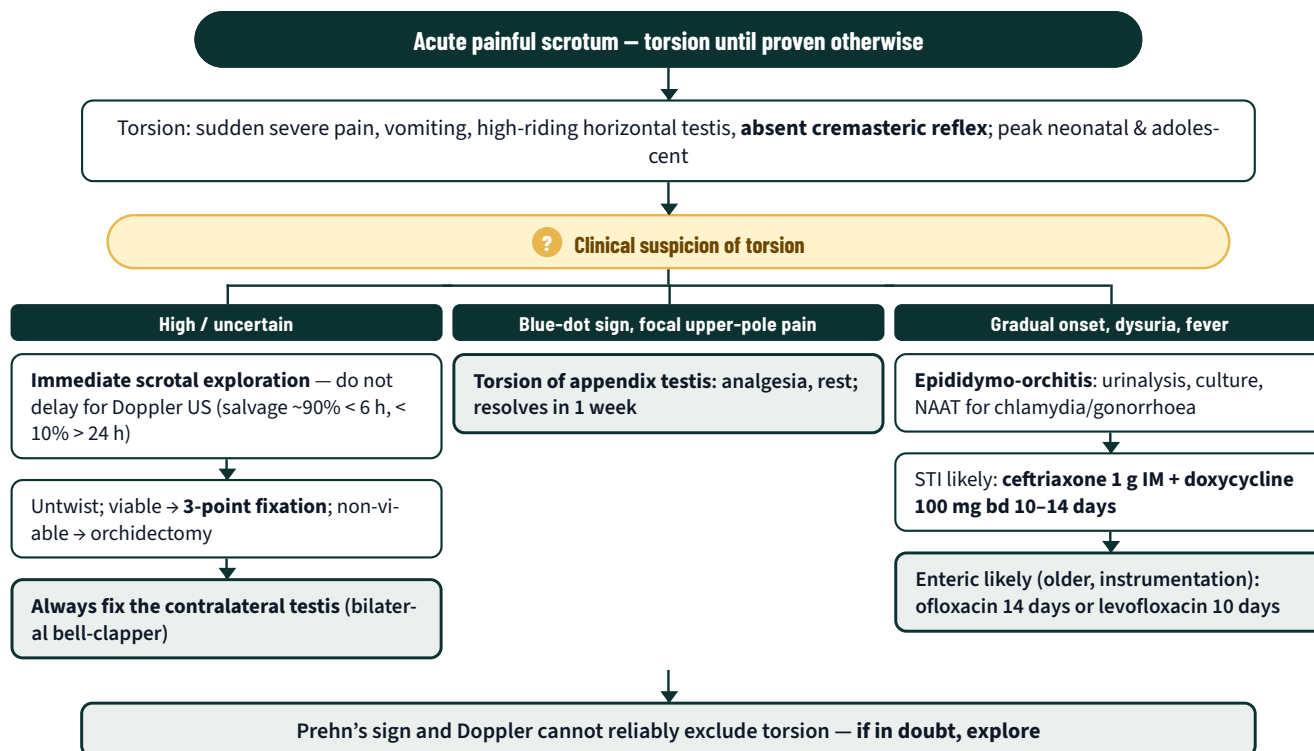
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Step by step

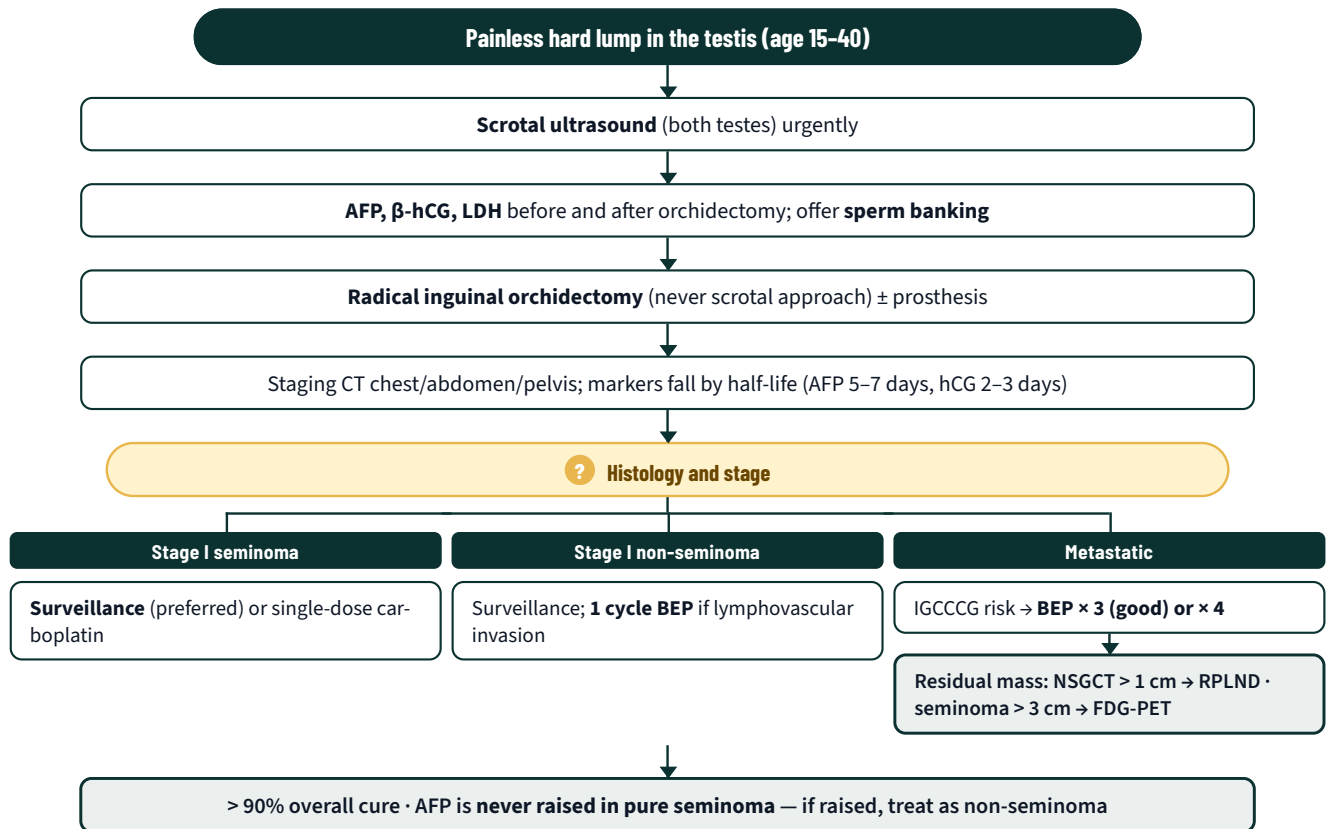
Acute scrotal pain

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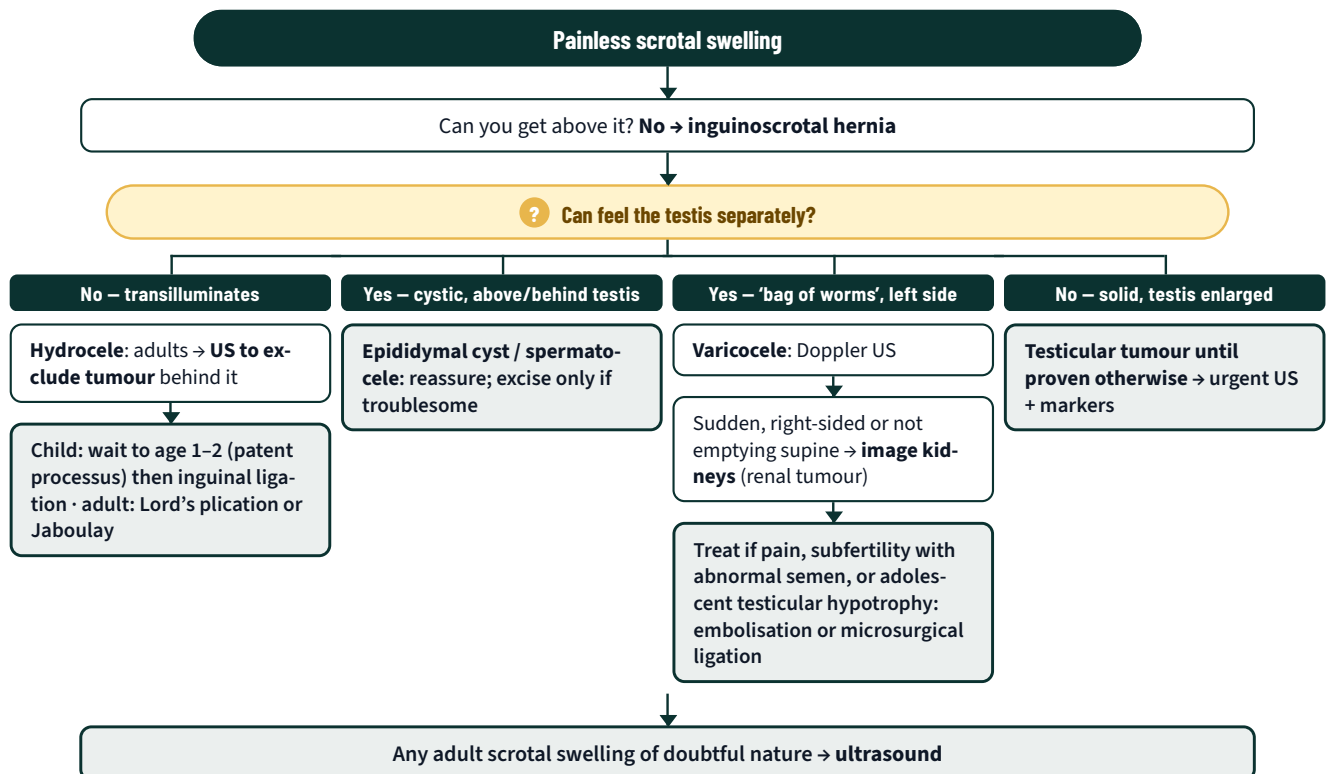
Testicular lump and testicular cancer

EAU 2024 TESTICULAR CANCER · ESMO 2022



Non-tender scrotal swelling

EAU 2024 · BAILEY & LOVE 28E



Bailey & Love MCQs

- 1 Regarding varicocele, which of the following statements are false?
 - A The usual cause is absence or incompetence of valves in the testicular vein.
 - B The vast majority occur on the right side.
 - C When it occurs on the left side it might indicate a left renal carcinoma.
 - D In a person with a varicocele and reduced sperm count, the latter can be improved by an operation on the varicocele.
 - E In symptomatic patients, the first-line treatment should be embolisation.
- 2 Regarding hydrocele, the following statements are true except:
 - A A hydrocele is a collection of fluid within the tunica vaginalis.
 - B A congenital hydrocele causes an intermittent swelling.
 - C An acute hydrocele in a young man might be a sinister finding.
 - D Testicular pathology might cause a hydrocele.
 - E Aspiration is an effective treatment.
- 3 Regarding testicular tumour, which of the following statements are false?
 - A A testicular lump that is inseparable from the testis is likely to be a tumour.
 - B Lymphatic spread is usually to the inguinal lymph nodes.
 - C Teratomas occur in the third decade and seminomas in the fourth decade.
 - D Seminomas usually spread via the lymphatics.
 - E Diagnosis is confirmed by US-guided fine-needle aspiration cytology (FNAC).
- 4 Which of the following statements are true with regard to the management of testicular tumours?
 - A Tumour markers are measured and a chest x-ray carried out.
 - B Initial surgical treatment is orchidectomy through the groin.
 - C Staging is done by CT scan of the abdomen and chest.
 - D Seminomas are radiosensitive.
 - E Secondaries from teratoma are always treated by surgery, as they are radio-resistant.

Answers 1-4

BAILEY & LOVE MCQS

- 1 **B** The vast majority occur on the right side. — **UPDATED** Only B is false: about 90% of varicoceles are left-sided. Repair can improve semen parameters in oligozoospermic men (EAU), and embolisation is a first-line option.
- 2 **E** Aspiration is an effective treatment. — **UPDATED** Aspiration is not effective and usually leads to recurrence.
- 3 **BE** True: B, E — **UPDATED** Testicular tumours spread to para-aortic nodes and FNAC is contraindicated due to seeding risk.
- 4 **ABCD** True: A, B, C, D — **UPDATED** Management involves assessment with markers and imaging, groin orchidectomy and CT staging; seminomas are radiosensitive while teratoma secondaries may be chemosensitive.

Lange MCQs

- 1 A 13-year-old boy with a 4-hour history of right scrotal pain, sudden in onset with nausea. Exam shows a high-riding tender testis. Normal WBC/urinalysis. Which of the following is the most appropriate next step?
 - A Admit for bed rest
 - B Analgesics and a scrotal support
 - C Antibiotic therapy
 - D Schedule a testicular isotope scan
 - E Urgent surgical exploration
- 2 A 28-year-old man with a history of bilateral orchidopexy for cryptorchidism presents with a painless, unilateral right scrotal mass and enlarged inguinal nodes. Ultrasound shows a heterogeneous testis with a hydrocele and CT reveals right retroperitoneal adenopathy. Which of the following would help confirm the diagnosis?
 - A Transscrotal needle biopsy
 - B Transscrotal aspiration of the hydrocele for cytology
 - C Radical orchidectomy through an inguinal incision
 - D Transscrotal exploration and orchidectomy
 - E Laparotomy with pelvic and retroperitoneal node dissection

- 3 Staging workup in the patient described above reveals seminoma with positive inguinal and retroperitoneal nodes. What is the therapeutic management?
- A External beam radiotherapy
 - B Multidrug combination chemotherapy
 - C Combination radiotherapy and multidrug chemotherapy
 - D Clinical surveillance
 - E Laparotomy with pelvic and retroperitoneal node dissection

Answers 1-3

LANGE MCQs

- 1 **E Urgent surgical exploration** — **UPDATED** Sudden scrotal pain with a high-riding tender testis in a teenager is torsion until proven otherwise; explore urgently without waiting for imaging.
- 2 **C Radical orchidectomy through an inguinal incision** — **UPDATED** A suspected testicular tumour is diagnosed by radical inguinal orchidectomy with high cord clamping; scrotal biopsy or aspiration risks tumour seeding.
- 3 **B Multidrug combination chemotherapy** — **UPDATED** Seminoma with inguinal and retroperitoneal nodal spread needs cisplatin-based combination chemotherapy (BEP/EP); adding radiotherapy adds toxicity without benefit.

3 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Which one of the following statements regarding testicular tumours is true?
- A They account for 5%–10% of male malignancy in the United Kingdom
 - B Seminomas most commonly present between the ages of 15 and 25 years
 - C Teratomas commonly present between the ages of 30 and 50 years
 - D Seminomas secrete α -fetoprotein and β -human chorionic gonadotrophin
 - E 5-year survival for seminomas is over 95%
- 2 A 36-year-old man diagnosed with seminoma: To which lymph nodes do seminomas most commonly spread first?
- A Para-aortic lymph nodes
 - B Superficial inguinal lymph nodes
 - C Anterior cervical chain
 - D Posterior cervical chain
 - E Deep inguinal lymph nodes
- 3 A 12-year-old boy is admitted to the emergency department with sudden onset of severe right testicular and lower abdominal pain during athletic training. He has had one episode of vomiting and constantly feels nauseous. On examination, the patient is sweating and in unbearable pain. There is marked tenderness and swelling of the right testicle which is observed to be lying horizontally. What is the most appropriate next step in this patient's management?
- A Order a Doppler ultrasound of the testicular arteries
 - B Send the patient immediately for emergency surgical exploration of the scrotum
 - C Perform urine dipstick
 - D Manage the patient with analgesia and observe
 - E Obtain a second opinion from your senior colleague, who will only be able to see the patient in an hour

Answers 1-3

SBA MCQs

- 1 **E 5-year survival for seminomas is over 95%** — **UPDATED** Seminoma is cured in over 95%; testicular tumours are 1–2% of male cancers, seminoma peaks at 30–40 years, teratoma at 20–30, and seminomas do not secrete AFP.
- 2 **A Para-aortic lymph nodes** — **UPDATED** Seminomas typically spread first to the para-aortic lymph nodes.
- 3 **B Send the patient immediately for emergency surgical exploration of the scrotum** — **UPDATED** Presentation is consistent with testicular torsion, a surgical emergency.

Oxford MCQs

- 1** 25-year-old man notices a painless lump in his testicle on self-examination. An ultrasound scan confirms the presence of a testicular tumour. Tumour markers β -HCG and AFP are raised. A CT confirms the presence of lung metastases (stage 4 disease). Which is the single most appropriate treatment for the likely tumour type?



- A Inguinal orchidectomy and chemotherapy
- B Inguinal orchidectomy and radiotherapy
- C Inguinal orchidectomy, retroperitoneal lymph node dissection and radiotherapy
- D Transscrotal orchidectomy and chemotherapy
- E Transscrotal orchidectomy and radiotherapy

- 2** 38-year-old man is found to have an irregular, firm, non-transilluminating, painless testicular mass at an insurance medical examination. He has no other symptoms or signs. Which is the single most appropriate plan of investigation?
- A Testicular ultrasound scan; serum AFP, β -HCG and LDH; CT chest, abdomen and pelvis
 - B Testicular ultrasound scan; CT chest, abdomen and brain
 - C Testicular ultrasound scan; CT chest, abdomen; and bone scan
 - D Testicular ultrasound scan; CT chest, abdomen, and brain; and bone scan
 - E Testicular ultrasound scan; serum β -HCG and AFP
- 3** A 45-year-old man develops a left scrotal mass with the appearance of distended tortuous veins. Scrotal and groin ultrasound confirms a varicocele and a shrunken left testis. Which is the single most appropriate next step?
- A Interventional radiological embolization
 - B Intravenous urogram
 - C Renal ultrasound
 - D Sclerotherapy under local anaesthesia
 - E Surgical ligation

Answers 1-3

OXFORD MCQS

- 1** **A Inguinal orchidectomy and chemotherapy** — **UPDATED**
 Raised AFP excludes pure seminoma; metastatic non-seminomatous germ cell tumour is treated by inguinal orchidectomy plus cisplatin-based chemotherapy (BEP).
- 2** **A Testicular ultrasound scan; serum AFP, β -HCG and LDH; CT chest, abdomen and pelvis** — **UPDATED**
 Work-up is scrotal ultrasound, tumour markers (AFP, β -hCG, LDH) and staging CT

of chest, abdomen and pelvis; brain CT and bone scan only if clinically indicated.

- 3** **C Renal ultrasound** — **UPDATED**
 A new varicocele in an older man may signal a renal tumour obstructing the testicular vein, so the kidney is imaged first (renal ultrasound).

3 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 On which side are varicoceles more common?
 - A Right side
 - B Equally on both sides
 - C Midline
 - D Left side
 - E Bilateral with left dominance
- 2 A 32-year-old man presents with an asymptomatic mass in his right testicle. On examination, the mass cannot be transilluminated. Ultrasound shows a solid mass in the right testicle. Chest, abdomen, and pelvis CT scans are negative for metastases. Serum levels of alpha-fetoprotein and beta human chorionic gonadotropin are elevated. Which of the following is the next best step in his diagnostic workup?
 - A No further testing is necessary
 - B Percutaneous biopsy of the testicular mass
 - C Incisional biopsy of the testicular mass through a scrotal incision
 - D Excisional biopsy of the testicular mass through a scrotal incision
 - E Radical inguinal orchiectomy
- 3 A 10-year-old boy presents to the emergency room with pain in the left testicle. The pain was acute in onset and began 1 hour ago. On physical examination, he is noted to have a high-riding, firm, and markedly tender left testis. The right testicle is normal. Doppler ultrasound shows decreased intratesticular blood flow of the left as compared to the right testicle. He is taken emergently to surgery where his left testicle is detorsed, after which it appears viable. Which of the following is the most appropriate next step in the management of this patient?
 - A No additional intervention is required
 - B Orchiopexy of the left testicle
 - C Orchiopexy of bilateral testicles
 - D Orchiectomy of the left testicle and orchiopexy of the right testicle
 - E Orchiectomy of bilateral testicles

Answers 1-3

PRETEST MCQs

- 1 **D Left side** — **UPDATED** Varicoceles are more common on the left due to drainage into the left renal vein.
- 2 **E Radical inguinal orchiectomy** — **UPDATED** Radical inguinal orchiectomy gives the histological diagnosis and treats the
- 3 **C Orchiopexy of bilateral testicles** — Orchiopexy should be performed on both testes to prevent recurrence of torsion.

4 QUESTIONS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead MCQs

- 1 A 44-year-old man presents to the GP having noticed a painless swelling in the right side of the scrotum. On examination, the swelling transilluminates and the testis is not palpable. What is the most likely diagnosis?
 - A Epididymo-orchitis
 - B Hydrocele
 - C Inguinal hernia
 - D Testicular torsion
 - E Varicocele
- 2 An 8-year-old boy has a 3-day history of right-sided testicular pain. He was unwell with jaw pain on eating a week ago. There is unilateral swelling and tenderness of the right testis. Most appropriate treatment?
 - A Analgesia and bed rest
 - B Antibiotics
 - C Hydrocelectomy
 - D Orchidectomy
 - E Reduction and fixation of testis

- 3** A 27-year-old man presents to the GP practice having noticed a painless swelling of his right testicle. He is otherwise well. On examination, the testis is enlarged, firm and has a nodular texture. What is the most likely diagnosis?
- A Epididymal cyst
 - B Gumma
 - C Haematocele
 - D Orchitis
 - E Testicular cancer
- 4** A 40-year-old man is being investigated for an enlarging painless lump in the right testicle. Blood results show a significantly raised alpha fetoprotein and a normal beta-hCG. Which of the following testicular tumours is most likely to be the cause of his symptoms?
- A Choriocarcinoma
 - B Leydig cell tumour
 - C Seminoma
 - D Testicular lymphoma
 - E Yolk sac carcinoma

Answers 1-4

GET AHEAD MCQS

- 1** **B Hydrocele** — A fluid collection in the tunica vaginalis that transilluminates and obscures the testis.
- 2** **A Analgesia and bed rest** — Likely mumps orchitis, managed supportively with analgesia and rest.
- 3** **E Testicular cancer** — A painless, irregular, hard testicular swelling is highly suggestive of testicular cancer.
- 4** **E Yolk sac carcinoma** — **UPDATED** Raised AFP with normal β -hCG indicates a yolk sac (endodermal sinus) tumour element; pure seminoma never raises AFP.

7 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1** A 15-year-old boy presents with sudden-onset severe pain in his left scrotum and vomiting. On examination there is no relief of pain on elevation of the scrotum, and a urine dipstick is normal. What is the most likely diagnosis?
- A Epididymitis
 - B Hydrocele
 - C Orchitis
 - D Appendicitis
 - E Testicular torsion
- REPEATED 2x • ALSO ASKED IN**
Crash Course
- 2** A 25-year-old sexually active male presents with scrotal swelling and pain. On examination there is scrotal erythema. A urine dipstick is positive for leucocytes. What is the most likely diagnosis?
- A Epididymitis
 - B Hydrocele
 - C Orchitis
 - D Appendicitis
 - E Testicular torsion
- 3** A 22-year-old student, previously healthy, has an incidental painless testicular lump. Past history of undescended testis. On exam, a hard mass that does not transilluminate. Which tests needed?
- A Testicular ultrasound
 - B CT lungs/abdomen/pelvis
 - C Alpha-fetoprotein (AFP)
 - D β -HCG
 - E All of the above
- 4** A 50-year-old man has a dull ache/dragging sensation in right scrotum for 1 month. Exam: twisted, 'bag of worms' mass along spermatic cord not changing in supine position. Main investigation?
- A PSA
 - B Chest radiograph
 - C Plain abdominal radiograph
 - D Urinalysis
 - E CT abdomen
- 5** A 35-year-old man has a translucent, non-tender scrotal swelling that you cannot separately feel the testis or epididymis. Most likely diagnosis?
- A Hydrocele
 - B Epididymo-orchitis
 - C Epididymal cyst
 - D Testicular tumour
 - E Testicular torsion
- 6** A 24-year-old man has a non-translucent, extremely tender scrotal lump, cannot identify testis or epididymis. Most likely diagnosis?
- A Hydrocele
 - B Epididymo-orchitis
 - C Epididymal cyst
 - D Testicular tumour
 - E Testicular torsion

- 7 A 32-year-old man has a translucent, separate-from-testis, non-tender scrotal swelling. Most likely diagnosis?
- A Hydrocele
 - B Epididymo-orchitis
 - C Epididymal cyst
 - D Testicular tumour
 - E Testicular torsion

Answers 1-7

- 1 **E Testicular torsion** — **UPDATED** Sudden severe scrotal pain with vomiting, no relief on elevation (negative Prehn sign) and a normal dipstick in a teenager indicates testicular torsion.
- 2 **A Epididymitis** — **UPDATED** Scrotal pain and erythema with pyuria in a sexually active young man indicate epididymitis, usually due to chlamydia or gonorrhoea.
- 3 **E All of the above** — Testicular cancer staging: US + tumour markers + imaging.
- 4 **E CT abdomen** — **UPDATED** A right-sided varicocele, or one that does not empty when supine, in an older man may signal a renal or retroperitoneal mass, so CT the abdomen.
- 5 **A Hydrocele** — **UPDATED** A painless transilluminating swelling enveloping the testis, which cannot be felt separately, is a hydrocele.
- 6 **B Epididymo-orchitis** — **UPDATED** An extremely tender, non-transilluminating swelling of the testis and epididymis is epididymo-orchitis; torsion must still be excluded in young men.
- 7 **C Epididymal cyst** — **UPDATED** A non-tender transilluminating swelling felt separately from the testis is an epididymal cyst.

Bladder Tumours

Haematuria, transitional cell carcinoma, TURBT, cystectomy and urinary diversion.

22

MCQs

0

EMQ SCENARIOS

8

SOURCES

0

REPEATED IDEAS

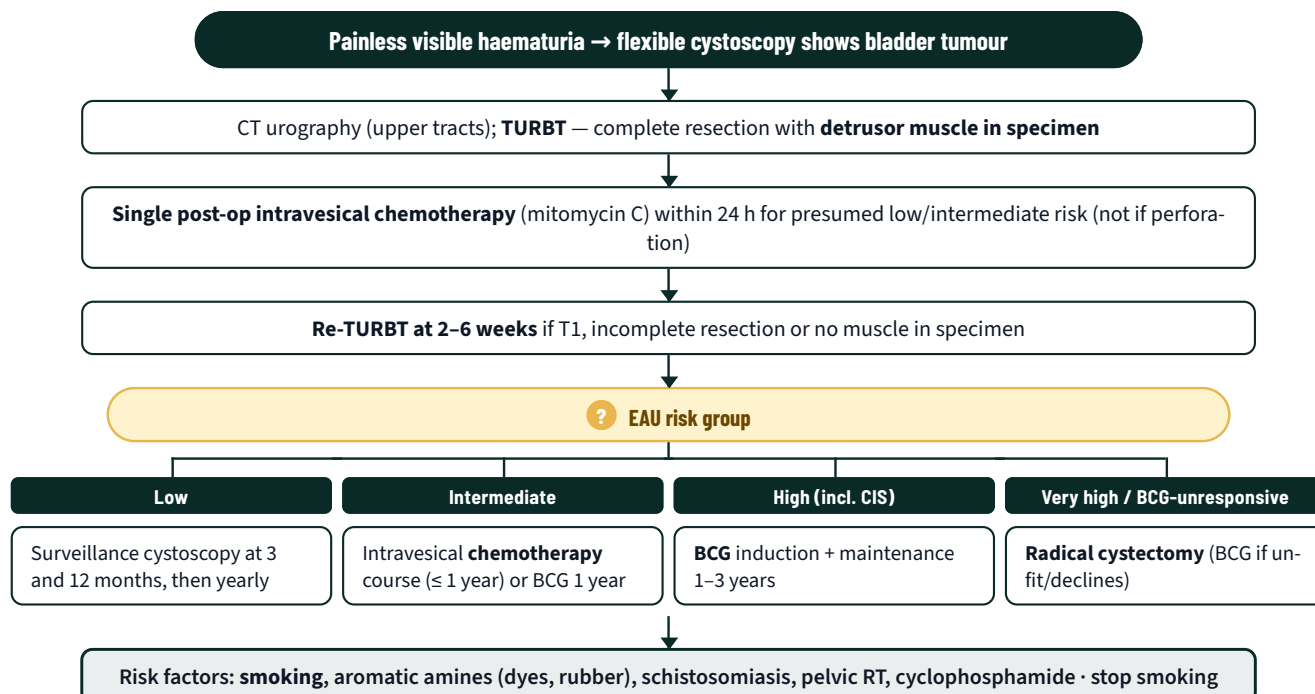
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Step by step

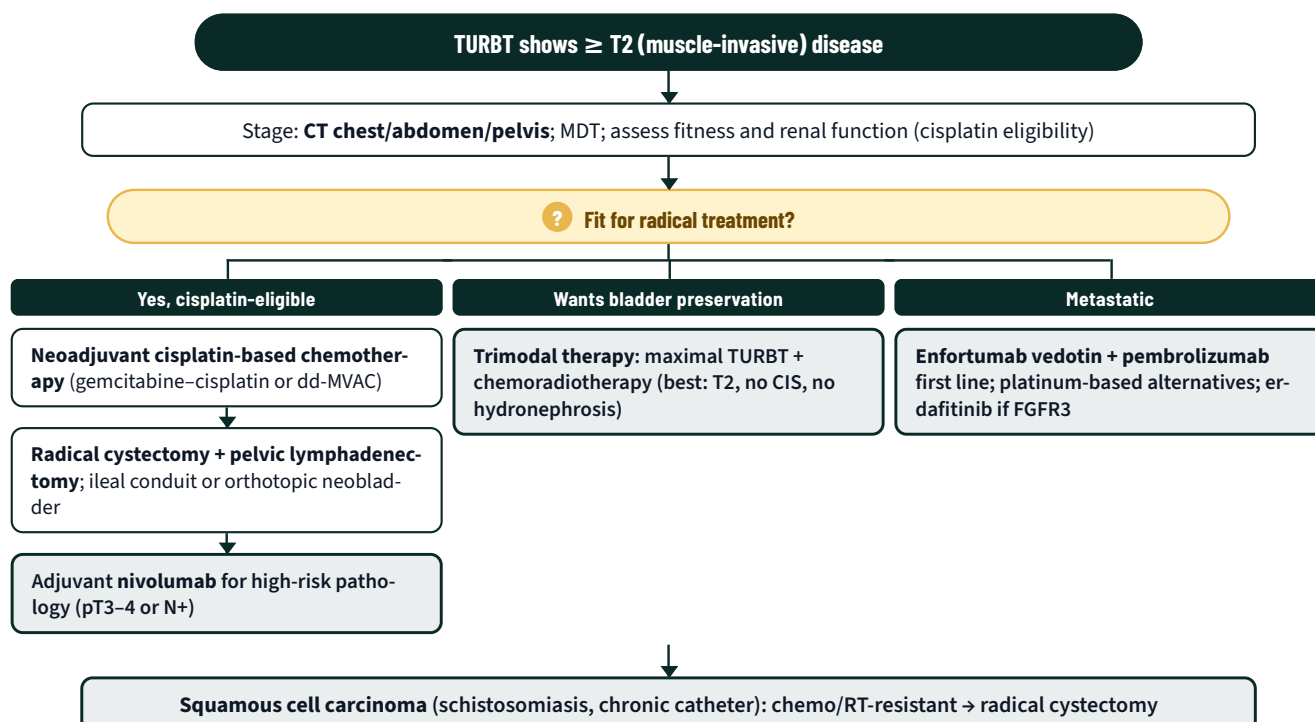
Non-muscle-invasive bladder cancer

EAU 2024 NMIBC • NICE NG2



Muscle-invasive bladder cancer

EAU 2024 MIBC • NICE NG2



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1 An elderly man has haematuria; ultrasound shows a 2-cm tumour in the bladder. What is the next step?
 - A Cystoscopy (with biopsy/TURBT)
 - B CT scan
 - C Urine cytology only
 - D MRI pelvis
 - E IVU
- 2 An elderly man has haematuria; ultrasound shows a 2-cm tumour in the bladder. What is the treatment?
 - A TURBT
 - B TURP
 - C Radical cystectomy
 - D Intravesical BCG alone
 - E Radiotherapy

MINISTERIAL EXAM

Ministerial evaluation examinations

- 3 A 64-year-old smoker male presented with intermittent painless hematuria for 6 weeks duration. CT-scan with IV contrast revealed a 2-cm enhancing papillary mass at the right lateral wall of the urinary bladder. The next most appropriate step in the management is:
 - A Radical cystectomy
 - B Partial cystectomy
 - C MRI abdomen and pelvis with IV contrast
 - D Transurethral resection of bladder tumor.
 - E Intravesical mitomycin chemotherapy

Answers 1-3

LOCAL EXAMS MCQs

- 1 **A** Cystoscopy (with biopsy/TURBT) — A bladder mass needs cystoscopy and transurethral resection for histology and staging; CT urography assesses the upper tracts.
- 2 **A** TURBT — TURBT is diagnostic and therapeutic for non-muscle-invasive tumours; intravesical therapy follows according to risk.
- 3 **D** Transurethral resection of bladder tumor. — **UPDATED** A bladder tumour on imaging needs transurethral resection first: it gives histology, grade and depth of invasion and treats non-muscle-invasive disease.

6 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love MCQs

- 1 Regarding haematuria, the following statements are true except:
 - A Microscopic haematuria is not always abnormal.
 - B Haematuria at the start of urinary stream indicates a cause in the lower urinary tract.
 - C Haematuria where the urine is uniformly mixed with the urine points to a cause in the upper urinary tract.
 - D Terminal haematuria is caused by bladder irritation or infection.
 - E Painful haematuria indicates malignant pathology.
- 2 Which of the following statements are true?
 - A The bladder is lined by transitional epithelium.
 - B Hypertrophy of the detrusor muscle results in bladder trabeculation.
 - C The epithelium of the trigone extends into the lower ends of ureters and proximal urethra.
 - D The internal sphincter prevents urinary incontinence.
 - E The distal urethral sphincter is supplied by S2–S4 fibres via the pudendal nerves.
- 3 The following statements are true except:
 - A Pubovesical ligaments anchor the bladder neck to the pubis.
 - B The bladder is supplied to a minor extent by the obturator and inferior gluteal arteries.
 - C The principal lymphatic drainage is to the external iliac nodes.
 - D Parasympathetic fibres provide the main motor innervation.
 - E The sympathetic input is from L1 to L4 segments.
- 4 Regarding bladder diverticula, which of the following statements are true?
 - A They are diagnosed when they produce double micturition and repeated urinary tract infections.
 - B All bladder diverticula must be excised.
 - C Trabeculation and sacculation precede diverticula formation.
 - D The normal intravesical pressure during voiding is 35–50 cm H₂O.
 - E Haematuria is a symptom in 30%.

- 5** In a patient with suspected bladder cancer, which of the following investigations are routinely carried out?
- A Urine cytology.
 - B IVU and US.
 - C Contrast-enhanced computed tomography (CECT) and MRI.
 - D Bone scan.
 - E Cystourethroscopy and bimanual examination.
- 6** In the treatment of bladder cancers, which of the following statements are true?
- A Non-muscle invasive (superficial) tumours are treated by endoscopic resection and a single dose of mitomycin instillation.
 - B Grade 3 superficial disease is best managed by BCG immunotherapy.
 - C External beam radiotherapy should be the first-line treatment in muscle invasive disease.
 - D When the cancer has invaded muscle (pT2 and pT3) radical cystectomy and lymphadenectomy is the treatment of choice.
 - E Neoadjuvant cisplatin-based chemotherapy improves survival in muscle-invasive tumours.

Answers 1-6

BAILEY & LOVE MCQS

- 1** **E** Painful haematuria indicates malignant pathology. — **UPDATED** E is false: painful haematuria suggests infection, stones or inflammation, whereas cancer bleeds painlessly. Transient microscopic haematuria (e.g. after exercise) can be benign.
- 2** **ABCE** True: A, B, C, E — The internal sphincter actually functions to prevent retrograde ejaculation rather than urinary incontinence.
- 3** **E** The sympathetic input is from L1 to L4 segments. — Sympathetic innervation actually originates from T10 to L2 segments.
- 4** **CDE** True: C, D, E — **UPDATED** A is false: diverticula are often found incidentally; B is false: only symptomatic or complicated diverticula (stones, infection, tumour) are excised.
- 5** **ABCE** True: A, B, C, E — **UPDATED** Only D is false: a bone scan is not routine and is reserved for bone pain, raised alkaline phosphatase or advanced disease.
- 6** **ABDE** True: A, B, D, E — **UPDATED** Only C is false: radical cystectomy (after neoadjuvant cisplatin chemotherapy) is first-line for muscle-invasive disease; radical radiotherapy is an alternative.

2 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 70-year-old man with a 50 pack-year smoking history presents with a 6-week history of painless, gross haematuria and urinary frequency. Which is the most appropriate initial diagnostic evaluation?
- A CT urography
 - B Voiding cystourethrogram
 - C MRI of the pelvis
 - D Abdominal ultrasound
 - E Urine for cytology
- 2** In the patient above, CT urography is normal. What is the best next step?
- A Abdominal CT scan
 - B Cystourethroscopy and urinary cytology
 - C Transrectal ultrasound
 - D Exploratory laparoscopy
 - E Re-evaluation in 2–4 weeks with repeat urinalysis and culture

Answers 1-2

LANGE MCQS

- 1** **A** CT urography — **UPDATED** Visible haematuria in an older smoker is investigated by CT urography to image the upper tracts and bladder, together with cystoscopy.
- 2** **B** Cystourethroscopy and urinary cytology — **UPDATED** Imaging misses many bladder tumours, especially flat carcinoma in situ, so cystoscopy (with cytology) is mandatory in every patient with visible haematuria.

2 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 With regard to carcinoma of the bladder, which statement is NOT correct?
 - A Epithelial tumours account for the majority of bladder tumours
 - B Smoking is the most important risk factor in the United Kingdom
 - C Males are more commonly affected than females
 - D There is a strong correlation with exposure to industrial dyes
 - E It frequently presents with painful haematuria
- 2 A 69-year-old retired canal engineer who has previously worked in North Africa presents to the urology outpatients department with a 2-month history of intermittent painless haematuria. A cystoscopy is performed showing a sessile mass on the posterior bladder wall. A biopsy is taken of this mass, which confirms transitional cell carcinoma invading the bladder muscle, but no local nodes are involved. A further staging computed tomography scan shows no distant metastases. According to the TNM classification, the tumour stage in this individual is
 - A Tis Nx Mx
 - B T2 N0 M0
 - C T2 N1 M0
 - D T3 N1 M1
 - E T3 N0 M1

Answers 1-2

SBA MCQs

- 1 **E** It frequently presents with painful haematuria — **UPDATED** Bladder cancer usually presents with painless haematuria.
- 2 **B** T2 N0 M0 — **UPDATED** Invasion into the detrusor muscle is T2; no involved nodes is N0 and no distant spread on CT is M0.

2 QUESTIONS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford MCQs

- 1 67-year-old man develops acute urinary retention after having painless haematuria for 3 months. Bladder irrigation has been established via a three-way catheter. Which is the single most appropriate next step in management?
 - A CT scan
 - B Flexible cystoscopy
 - C Rigid cystoscopy
 - D Ultrasound scan
 - E Urine cytology
- 2 70-year-old man, who worked in a petrochemical plant for many years, develops painless haematuria and undergoes transurethral endoscopic resection of an in situ transitional cell carcinoma of the bladder. Which is the single most appropriate next step in the patient's management?
 - A Intravesical BCG
 - B Intravesical mitomycin C
 - C Partial cystectomy
 - D Radical cystectomy
 - E Regular check cystoscopy

Answers 1-2

OXFORD MCQs

- 1 **C** Rigid cystoscopy — **UPDATED** Clot retention from haematuria needs rigid cystoscopy under anaesthesia to evacuate clot, find the source and resect or biopsy; cytology is unreliable in bloody urine.
- 2 **A** Intravesical BCG — Intravesical BCG is effective in preventing progression of transitional cell carcinoma.

1 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 60-year-old man sees a urologist for what he describes as bloody urine. A urine sample is positive for cytologic evidence of malignancy. Cystoscopy confirms the presence of superficial transitional cell carcinoma. Which of the following is the recommended treatment for stage A (superficial and submucosal) transitional cell carcinoma of the bladder?
- A Intravesical chemotherapy
 - B Radical cystectomy
 - C Radiation therapy
 - D Transurethral resection and intravesical chemotherapy
 - E Systemic chemotherapy

Answer 1

PRETEST MCQS

- 1 **D** Transurethral resection and intravesical chemotherapy —
UPDATED Non-muscle-invasive urothelial cancer is treated by

transurethral resection plus intravesical chemotherapy (BCG for high-risk disease).

1 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 63-year-old man has two episodes of painless frank haematuria. He has no past medical history but is a lifelong smoker. Which is the most likely cause?
- A Cystitis
 - B Diverticulitis
 - C Renal calculi
 - D Transitional cell carcinoma
 - E Urethral injury

Answer 1

GET AHEAD MCQS

- 1 **D** Transitional cell carcinoma — Painless haematuria in an older smoker strongly suggests bladder TCC.

5 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 41-year-old man has frequency, dysuria, urgency, but negative urine cultures. Most likely diagnosis?
 - A Pyelonephritis
 - B Interstitial cystitis
 - C Prostatitis
 - D Acute cystitis
 - E Epididymo-orchitis
- 2 A 70-year-old man has a 1-month history of painless haematuria, frequency, mild pelvic/back discomfort. He recalls haematuria in childhood in West Africa. Most likely diagnosis?
 - A Transitional cell carcinoma of bladder
 - B Adenocarcinoma of bladder
 - C Acute bladder infection
 - D Squamous cell carcinoma of bladder
 - E Bladder stone
- 3 A 65-year-old smoker with painless haematuria, normal exam, raised urea 15.4 mmol/L, creatinine 154 µmol/L. Suspect bladder cancer. Next initial investigation?
 - A Cystoscopy + biopsy
 - B Chest X-ray
 - C IV urogram
 - D Abdominal MRI
 - E Abdominal CT
- 4 A 60-year-old man has painless haematuria. Transurethral resection shows urothelial carcinoma invading the superficial detrusor muscle (T2a), and staging CT shows no nodal or distant metastases. He is fit for surgery. What is the next step in management?
 - A Radical cystectomy
 - B Radiotherapy
 - C Systemic chemotherapy
 - D Transurethral resection
 - E Watchful waiting
- 5 A 60-year-old man with painless haematuria, positive urinary cytology. Most likely diagnosis?
 - A Clear cell renal cancer
 - B Renal transitional cell carcinoma
 - C Bladder adenocarcinoma
 - D Prostatic adenocarcinoma
 - E Transitional cell carcinoma of bladder (TCC)

Answers 1-5

CRASH COURSE MCQS

- 1 **B Interstitial cystitis** — **UPDATED** Frequency, urgency and dysuria with persistently sterile urine suggest interstitial cystitis (bladder pain syndrome).
- 2 **D Squamous cell carcinoma of bladder** — **UPDATED** Childhood haematuria in West Africa implies Schistosoma haematobium infection, whose chronic irritation predisposes to squamous cell carcinoma of the bladder.
- 3 **A Cystoscopy + biopsy** — **UPDATED** Cystoscopy and biopsy is the key test for suspected bladder cancer; with impaired renal function, iodinated contrast (IVU/CT) is best avoided initially.
- 4 **A Radical cystectomy** — **UPDATED** Muscle-invasive (T2) bladder cancer in a fit patient is treated by radical cystectomy, ideally after neoadjuvant cisplatin chemotherapy; radical radiotherapy is the alternative.
- 5 **E Transitional cell carcinoma of bladder (TCC)** — Painless haematuria + positive cytology => TCC of bladder is classic.

Urology EMQs

Extended matching questions on haematuria, LUTS, scrotal swellings, urological investigation and emergencies.

0

MCQS

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EMQ SCENARIOS

6

SOURCES

1

REPEATED IDEAS

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Local Exams EMQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

THEME 1 Urological operations

- A Partial nephrectomy
- B Simple nephrectomy
- C Radical nephrectomy
- D Nephroureterectomy
- E Percutaneous nephrostomy
- F Ureteric (JJ) stent
- G Ureteric reimplantation
- H Pyeloplasty

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 55-year-old man has an incidental 3-cm enhancing mass in the upper pole of the kidney on CT; PET is negative.
- 2 A 48-year-old woman with chronic infections and recurrent stones has a left kidney contributing < 5% of total function on MAG3.
- 3 A 70-year-old man on IV antibiotics for 5 days has fever, rigors and flank pain; HR 120, BP 80/60. CT shows hydroureteronephrosis to the mid-ureter.
- 4 A 4-year-old girl has recurrent febrile UTIs despite long-term antibiotics; micturating cystogram shows a dilated, tortuous left ureter.

- 5 A 44-year-old man has painless haematuria; CT shows a tumour of the left renal pelvis without metastasis.

THEME 2 Urinary tract infection

- A Acute pyelonephritis
- B Acute cystitis
- C Asymptomatic bacteriuria
- D Acute prostatitis
- E Urethritis
- F Pyonephrosis
- G Emphysematous pyelonephritis
- H Vesicoureteric reflux

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 25-year-old woman has dysuria, frequency and suprapubic pain without fever or flank pain. Urine is cloudy with leucocytes and nitrites.
- 2 A 60-year-old man has fever, chills and perineal pain; DRE is difficult because of severe anterior rectal tenderness. Urine grows *E. coli*.
- 3 A 28-year-old diabetic woman has high fever, flank pain, vomiting and renal angle tenderness. CT shows pockets of gas and fluid in the left kidney.

Answers • Themes 1-2

LOCAL EXAMS EMQS

Theme 1 • Urological operations

- 1 **A** Partial nephrectomy — A small (T1a, ≤ 4 cm) renal cell carcinoma is treated by nephron-sparing partial nephrectomy.
- 2 **B** Simple nephrectomy — A non-functioning, chronically infected kidney is removed by simple nephrectomy.
- 3 **E** Percutaneous nephrostomy — An infected obstructed kidney with septic shock needs urgent decompression; percutaneous nephrostomy is safest in the unstable patient.
- 4 **G** Ureteric reimplantation — High-grade vesicoureteric reflux with breakthrough febrile UTIs is treated by ureteric reimplantation.

- 5 **D** Nephroureterectomy — Upper-tract urothelial carcinoma is treated by nephroureterectomy with a bladder cuff.

Theme 2 • Urinary tract infection

- 1 **B** Acute cystitis — Lower urinary symptoms without fever or loin pain are acute (uncomplicated) cystitis.
- 2 **D** Acute prostatitis — Fever, perineal pain and an exquisitely tender prostate is acute bacterial prostatitis (avoid prostatic massage).
- 3 **G** Emphysematous pyelonephritis — Gas within the renal parenchyma in a diabetic is emphysematous pyelonephritis: antibiotics, drainage, sometimes nephrectomy.

5 THEMES · 38 SCENARIOS · AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love EMQs

SET 1 OF 2

THEME 1 Diagnoses

- A Carcinoma of kidney
- B Carcinoma of ureter
- C Congenital hydronephrosis
- D Horseshoe kidney
- E Pyonephrosis
- F Staghorn calculus
- G Transitional cell carcinoma (TCC) of renal pelvis
- H Trauma to kidney
- I Ureteric calculus
- J Wilms tumour

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 40-year-old man has come to A&E complaining of a sudden onset of very severe pain in his left loin of 4 hours duration, radiating to the front of the lumbar area and groin and the left testis. He has vomited a couple of times and has urinary frequency. On examination he is writhing in pain and cannot find a comfortable position.
- 2 A 60-year-old man complains of haematuria associated with pain in his left loin radiating to the lower abdomen for about 2 months. He passes clots that are 'worm-like'. On examination he has an enlarged left kidney, and scrotal examination reveals a varicocele of which he was unaware.
- 3 A 45-year-old female patient complains of nagging pain in her right loin and urinary frequency for several months. On examination she has tenderness in her loin over the kidney. Urine examination shows red blood cells and a growth of *Proteus* and *staphylococci*.
- 4 A fit 20-year-old young man had a game of rugby, following which he passed frank blood in his urine in the changing room. He felt some discomfort in his left loin. After returning home, his haematuria became worse and he noticed some fullness in his left loin. He then came to A&E. His blood pressure was 110/60 mmHg, his pulse was 110/minute and he had fullness in his left loin.
- 5 A 60-year-old female patient was seen as an emergency with high temperature, rigors, and pain in her right loin. She had a blood pressure of 160/70 mmHg, bounding pulse of 90/minute, and extreme tenderness in her right loin.

- 6 A 62-year-old man complains of left loin ache with intermittent painless haematuria in the form of clots, occasional high temperature and rigors, some dysuria, and frequency for 6 months. An IVU shows an irregular filling defect in the lower one-third of the left ureter. There is nothing to find on clinical examination. His Hb is 10.2 gm/dL.
- 7 A 35-year-old woman presents with recurrent attacks of urinary tract infection. On every occasion this was successfully treated with antibiotics. In view of several such episodes in the past, she underwent an IVU, which showed the renal calyces to be pointing toward the vertebrae.
- 8 A male child of 3 years has been brought by his mother, who noticed while bathing him that the right side of his abdomen felt larger than the left and was asymmetrical; she thought that there was probably a lump in his abdomen.
- 9 A 10-year-old boy has been brought in as an emergency by his parents as they noticed that he had bright red haematuria for the past 24 hours or so. About 6 hours prior to that he was climbing a tree and accidentally fell on his left side.
- 10 A 72-year-old woman complains of intermittent haematuria associated with a dull ache in her left loin. She has some dysuria and urine specimen showed *E. coli* infection. She has no physical findings. An IVU showed a persistent irregular filling defect in the left renal pelvis.

THEME 2 Bladder diagnoses

- A Bladder calculus
- B Bladder carcinoma
- C Bladder diverticulum
- D Schistosomiasis of bladder
- E Tuberculosis of the bladder

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 35-year-old male, a recent visitor from Egypt, attended the emergency department with painless terminal haematuria. He had a few similar episodes while at home in Egypt. Clinical examination revealed no abnormality. On flexible cystoscopy at the one-stop haematuria clinic, he was found to have scattered tubercles and islands of pale patches resembling sand.

- 2 A 70-year-old male, a heavy smoker for more than 50 years, complains of painless, profuse and periodic haematuria for 6 weeks. The blood is uniformly mixed with the urine. He has frequency of micturition and some retropubic discomfort. Clinical examination reveals no abnormality.
- 3 A 70-year-old male complains of poor micturition stream, post-micturition dribbling and a feeling of insufficient emptying of his bladder. He has some dysuria. On occasions he has found that shortly after micturition he again passes a large amount of urine. Clinical examination reveals no abnormality.
- 4 A 60-year-old man complains of haematuria and painful micturition, and he occasionally finds that his micturition stream suddenly stops. He has learned to restart his stream by changing position. Clinical examination reveals no abnormality.
- 5 A 35-year-old female patient, a recent visitor from the Indian subcontinent, complains of frequency of micturition and bilateral loin discomfort. She has evening pyrexia, general malaise and weight loss. Clinical examination reveals no abnormality. Urine examination carried out by her doctor shows sterile pyuria.

THEME 3 Prostate and Urinary Retention Diagnoses

- A Acute prostatitis
- B Acute retention of urine
- C Benign prostatic hypertrophy (BPH) causing BOO
- D Carcinoma of prostate
- E Chronic urinary retention with overflow

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 40-year-old man complains of feeling ill with pyrexia of 39°C, rigors with aches all over his body, and flu-like symptoms in general. He has rectal irritation, pain on defecation and perineal discomfort. He has pain on micturition and passes threads in his urine. Rectal examination reveals a tender, swollen prostate.

- 2 A 70-year-old man complains of poor stream, frequency of micturition both in the day and at night, hesitancy and intermittent stream, a feeling of incomplete emptying and terminal haematuria for about 4 months; recently he has had backache. Clinical examination shows no abnormality. On rectal examination he has a hard, nodular prostate with overlying fixed rectal mucosa.
- 3 A 72-year-old man has come to the emergency department with severe acute pain in his lower abdomen not having passed urine for almost 10 hours. He has been on some medicines, which he bought across the counter for cold and flu. This has also made him constipated. On examination he is in severe pain from a large mass in his suprapubic region arising from the pelvis. The mass is dull on percussion.
- 4 A 70-year-old man complains of urgency, frequency and hesitancy of micturition with a feeling of incomplete emptying. He also has poor flow, intermittent stream and post-micturition dribbling. These symptoms have been going on for 2 years and are gradually getting worse. Clinical examination reveals no abnormality. On rectal examination he has an enlarged smooth prostate with overlying mobile rectal mucosa, and the median vertical groove is easily felt.
- 5 A 75-year-old male complains of general malaise, lethargy, abdominal distension and urinary incontinence. On examination he has a large painless mass in his suprapubic region arising from the pelvis and has continuous urinary dribbling. The mass is dull on percussion. He is a type 2 diabetic on oral medication.

Answers · Themes 1-3

BAILEY & LOVE EMQS

Theme 1 · Diagnoses

- 1 **I Ureteric calculus** — Sudden severe loin pain radiating to the groin and testis, with vomiting and restlessness, is ureteric colic from a ureteric calculus.
- 2 **A Carcinoma of kidney** — Haematuria with worm-like clots, loin pain, a renal mass and a new left varicocele (renal vein invasion) indicate renal cell carcinoma.
- 3 **F Staghorn calculus** — Chronic loin pain with Proteus infection (urease-splitting, alkaline urine) suggests a struvite staghorn calculus.
- 4 **C Congenital hydronephrosis** — Disproportionate haematuria and loin swelling after minor trauma suggest

bleeding from an abnormal kidney, classically congenital hydronephrosis (PUJ obstruction).

- 5 **E Pyonephrosis** — High fever, rigors and an exquisitely tender loin indicate an infected obstructed kidney (pyonephrosis) needing urgent drainage.
- 6 **B Carcinoma of ureter** — Haematuria with an irregular filling defect in the lower ureter is urothelial (transitional cell) carcinoma of the ureter until proven otherwise.
- 7 **D Horseshoe kidney** — Calyces pointing medially towards the spine on IVU are characteristic of a horseshoe kidney, which predisposes to infection and stones.

- 8 **J Wilms tumour** — An asymptomatic flank mass noticed by a parent in a 3-year-old is classic for Wilms tumour (nephroblastoma).
- 9 **H Trauma to kidney** — Frank haematuria after a fall on to the loin indicates blunt renal trauma; children's kidneys are relatively unprotected.
- 10 **G Transitional cell carcinoma (TCC) of renal pelvis** — A persistent irregular filling defect in the renal pelvis with haematuria in an older patient indicates transitional cell carcinoma of the renal pelvis.

Theme 2 • Bladder diagnoses

- 1 **D Schistosomiasis of bladder** — Sandy patches and tubercles on cystoscopy with terminal haematuria in someone from Egypt are typical of urinary schistosomiasis (*S. haematobium*).
- 2 **B Bladder carcinoma** — Painless total haematuria in an elderly heavy smoker is bladder carcinoma until proven otherwise.
- 3 **C Bladder diverticulum** — Passing a large volume again shortly after voiding (double micturition) is typical of a bladder diverticulum emptying back into the bladder.
- 4 **A Bladder calculus** — A stream that stops suddenly and re-starts with a change of posture, with painful haematuria, is classic for a mobile bladder calculus.

SET 2 OF 2

THEME 4 Urethra and Penis Diagnoses

- A Balanitis xerotica obliterans (BXO)
- B Balanoposthitis
- C Carcinoma of penis
- D Chordee
- E Persistent priapism
- F Peyronie's disease
- G Rupture of bulbous urethra
- H Rupture of membranous urethra

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 35-year-old man complains of deformity of his penis during erection. He has increasing difficulty in sexual intercourse. In the past he had urethritis, which was treated successfully with antibiotics.
- 2 A 68-year-old man complains of bloody, foul-smelling discharge from his penis. He has a hard ulcerated lesion under his prepuce. He also complains of lumps in his groin.
- 3 A 45-year-old man complains of progressive deformity of his penis, which is very pronounced during erection; this has been going on for the past 2 years. On examination, indurated plaques can be palpated around the penile shaft. On questioning he has thickening of his palmar fascia on both hands.
- 4 A 60-year-old man who is under treatment for leukaemia recently started developing penile erection without any reason. This is prolonged and painful and is now distressing him.

- 5 **E Tuberculosis of the bladder** — Sterile pyuria with frequency, evening fever and weight loss in a patient from an endemic area suggests genitourinary tuberculosis.

Theme 3 • Prostate and Urinary Retention Diagnoses

- 1 **A Acute prostatitis** — Fever, rigors, perineal pain, dysuria and an exquisitely tender, swollen prostate on rectal examination indicate acute bacterial prostatitis.
- 2 **D Carcinoma of prostate** — LUTS with a hard, nodular prostate fixed to the rectal mucosa and new back pain (bone metastases) indicate prostate carcinoma.
- 3 **B Acute retention of urine** — Painful inability to void with a tender, dull suprapubic mass, precipitated by cold remedies and constipation, is acute urinary retention.
- 4 **C Benign prostatic hypertrophy (BPH) causing BOO** — Progressive LUTS with a smooth, enlarged prostate and a preserved median groove indicate BPH causing bladder outlet obstruction.
- 5 **E Chronic urinary retention with overflow** — A painless distended bladder with continuous dribbling and malaise (possible uraemia) is chronic retention with overflow incontinence.

- 5 A 22-year-old man complains of itching around his glans penis and prepuce. Recently, he has developed purulent discharge from the subpreputial area. His glans and foreskin look red and inflamed. He is a type I diabetic on insulin.
- 6 A 32-year-old man was involved in a road-traffic accident in which he sustained polytrauma, chief among which was fracture of pelvis with fractured femoral shaft. After stabilisation according to the ATLS protocol, it became apparent that he has a distended bladder and is unable to pass urine.
- 7 A 24-year-old man while performing on a Pommel Horse in a gymnastics competition, slipped and fell astride on the Pommel Horse. He complains of severe perineal pain in the penoscrotal junction where there is a haematoma. He has blood in his external urinary meatus.
- 8 A 52-year-old man complains of thickening of his foreskin over a period of 2 years. During this period he has had difficulty in retracting his foreskin, as a result of which he has been having difficulty in maintaining good hygiene. He finds that his urinary stream tends to spray around.

THEME 5 Scrotal Swellings and Testicular Conditions

- A Acute epididymo-orchitis
- B Encysted hydrocele of the cord
- C Epididymal cyst
- D Idiopathic oedema of the scrotum
- E Spermatocele
- F Testicular tumour
- G Torsion of the hydatid of Morgagni
- H Torsion of the testis
- I Vaginal hydrocele
- J Varicocele

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 22-year-old man complains of acute pain in the top of his left testis for 16 hours. On examination the scrotum looks normal and the testis can be felt with care without undue tenderness. There is a small, blue, pea-sized tender mass on top of the testis.
- 2 A 30-year-old man complains of a swollen scrotum for almost 1 year. This is not painful but recently has caused discomfort. On examination he has a tense fluctuant lump in his right scrotum where the testis cannot be felt separately. The swelling transilluminates.
- 3 A 45-year-old man complains of a painful left testis of 4 days' duration. This is associated with fever and rigors. He has painful micturition and considerable frequency. On examination he looks unwell, has pyrexia of 39°C and has a swollen, red, oedematous and shiny left hemiscrotum; the testis and epididymis feel indurated and tender.
- 4 An 18-year-old man complains of sudden onset of agonising pain in his right groin and suprapubic area of 6 hours' duration. He has no urinary symptoms. On examination he is in agony, the right testis is drawn up and the scrotum is red. The testis is impossible to feel because of pain.
- 5 A 25-year-old man complains of feeling heavy in his left scrotum where he noticed a lump while in the shower. A few days before, he felt a lump in the periumbilical region of his abdomen. At the same time he noticed another lump on the left side of his neck. The abdominal and neck lumps do not give him any symptoms. On questioning he admits to coughing up some blood in his sputum 1 month ago. On examination he has a 2 cm irregular lump over his left supraclavicular area and a 5-cm irregular firm lump in his umbilical region. Examination of the left testis reveals a large lump, which is hard and heavy.
- 6 A 50-year-old man complains of a lump in his right scrotum of 1 year's duration. This is gradually growing larger in size and gives him some discomfort. Examination of his scrotum reveals a cystic lump, about 7 cm in diameter, above and behind the right testis, which is minimally tender. Transillumination is positive and shows septa within the lump. There is a smaller similar lump on the other side of which he is unaware.
- 7 A 4-year-old boy has been brought by his parents to A&E with a red, swollen scrotum. On examination the child is not in any pain and there is redness and oedema around the penis and perineum.
- 8 A 40-year-old man presents with a swelling in his right scrotum, which he says feels like 'a third testis'. He noticed it in the shower and is worried that he has got cancer. He has no symptoms. On examination he has a lax, cystic swelling on top of his right testis. It is not tender and transillumination is hazy.
- 9 A 30-year-old man presents with a swelling in his left hemiscrotum, which he discovered by chance in the bath. On examination the testis feels separate from the swelling, which is about 2 cm in diameter and above the testis. It transilluminates and comes down on traction to the testis.
- 10 A 45-year-old man complains of a dragging discomfort in the left side of his scrotum. He noticed an irregular lump in his scrotum separate from the testis. He thinks he has had this for many years but only recently he noticed discomfort in the lump. On examination he has a bunch of veins in the scrotum and the testis feels normal. On lying down the veins disappear. Abdominal examination in the left renal area does not show any abnormality.

Answers • Themes 4–5

BAILEY & LOVE EMQS

Theme 4 • Urethra and Penis Diagnoses

- 1 **D Chordee** — Penile curvature on erection after urethritis reflects chordee from periurethral fibrosis (stricture); congenital chordee accompanies hypospadias.
- 2 **C Carcinoma of penis** — A hard ulcerated subpreputial lesion with bloody, foul discharge and inguinal nodes in an elderly man is squamous carcinoma of the penis.
- 3 **F Peyronie's disease** — Plaques in the tunica albuginea causing curvature on erection, with Dupuytren contracture of the palms, indicate Peyronie disease.

- 4 **E Persistent priapism** — A prolonged painful erection unrelated to sexual stimulation in leukaemia is ischaemic (low-flow) priapism, a urological emergency.
- 5 **B Balanoposthitis** — Itching, redness and purulent subpreputial discharge in a diabetic indicate balanoposthitis (often candidal or bacterial).
- 6 **H Rupture of membranous urethra** — A pelvic fracture with a distended bladder and inability to void indicates rupture of the membranous (posterior) urethra.
- 7 **G Rupture of bulbous urethra** — A fall-astride (straddle) injury with perineal haematoma and blood at the meatus indicates rupture of the bulbar urethra.
- 8 **A Balanitis xerotica obliterans (BXO)** — Progressive thickening and non-retractility of the foreskin with a spraying stream in a middle-aged man indicates BXO (lichen sclerosus).

Theme 5 • Scrotal Swellings and Testicular Conditions

- 1 **G Torsion of the hydatid of Morgagni** — A tender blue pea-sized nodule at the upper pole ('blue dot' sign) with a non-tender testis is torsion of the hydatid of Morgagni.
- 2 **I Vaginal hydrocele** — A tense, fluctuant, transilluminating swelling in which the testis cannot be felt separately is a vaginal hydrocele.

- 3 **A Acute epididymo-orchitis** — Fever, urinary symptoms and a red, oedematous hemiscrotum with tender, indurated testis and epididymis indicate acute epididymo-orchitis.
- 4 **H Torsion of the testis** — Sudden agonising pain with a drawn-up testis and red scrotum in a young man is testicular torsion; explore urgently.
- 5 **F Testicular tumour** — A hard testicular mass with para-aortic and supraclavicular nodes and haemoptysis (lung metastases) indicates metastatic testicular tumour.
- 6 **C Epididymal cyst** — A transilluminating, septated cyst above and behind the testis, often bilateral, is an epididymal cyst.
- 7 **D Idiopathic oedema of the scrotum** — A painless red, oedematous scrotum spreading to the penis and perineum in a young boy is idiopathic scrotal oedema; it settles spontaneously.
- 8 **E Spermatocele** — A lax cyst above the testis felt as 'a third testis', with hazy transillumination (milky fluid containing sperm), is a spermatocele.
- 9 **B Encysted hydrocele of the cord** — A transilluminating swelling above and separate from the testis that moves down on traction of the testis is an encysted hydrocele of the cord.
- 10 **J Varicocele** — A bag of veins above the testis that empties on lying down is a varicocele, usually left-sided.

3 THEMES • 15 SCENARIOS • AFTER THE OXFORD SPECIALTY TRAINING / OXFORD SURGERY QUESTIONS

Oxford EMQs

THEME 1 Causes of scrotal swelling

- A Epididymal cyst
- B Epididymo-orchitis
- C Hydrocoele
- D Inguinal hernia
- E Lipoma
- F Non-seminomatous germ cell tumour
- G Scrotal haematoma
- H Seminoma
- I Torsion of the hydatid of Morgagni
- J Varicocele

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 23-year-old man has a firm, painless irregular mass in his right testicle which has enlarged over 4 months.
- 2 A 34-year-old man has a non-tender, fluctuant scrotal mass which causes discomfort when he wears tight pants. The scrotum transilluminates and the testis is not palpable.
- 3 A 45-year-old man has an intermittent lump in his left scrotum which is associated with groin discomfort. It is not possible to feel the upper limit of the mass within the scrotum.

- 4 A 37-year-old man notices a 1cm diameter lump in his scrotum which on examination is adjacent to, but feels separate from, the testicle.
- 5 A 62-year-old man has an acutely tender mass in the right side of his scrotum which has been associated with episodic dysuria during the previous 6 months. He is tender to testicular palpation and there is a mass adjacent to and apparently involving the testicle.

THEME 2 Causes of scrotal/testicular pain

- A Epididymo-orchitis
- B Glandular fever
- C Mumps orchitis
- D Ruptured testicle
- E Scrotal haematoma
- F Testicular torsion
- G Testicular haematoma
- H Thrombosed varicocele
- I Torsion of the hydatid of Morgagni

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 28-year-old man has scrotal pain in both testes, associated with a purulent urethral discharge. The pain is made marginally more comfortable by a scrotal support.

- 2 A 16-year-old boy with maldescent of the testicle has sudden onset of right testicular pain. On palpation the testicle is in a high-lying position in the scrotum.
- 3 A 22-year-old man develops sudden onset of testicular pain. On transillumination of the scrotum there is the appearance of a localized dark region adjacent to the testicle.
- 4 A 13-year-old boy has pain and swelling in his scrotum 24 h after being hit by a cricket ball.
- 5 A 12-year-old boy with neck swelling develops a unilateral swollen tender testicle.

THEME 3 Causes of haematuria/loin pain

- A Bladder transitional cell carcinoma
- B Glomerulonephritis
- C Pelvic fracture
- D Porphyria
- E Prostate carcinoma
- F Pyelonephritis
- G Renal cell carcinoma
- H Renal colic
- I Schistosomiasis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 35-year-old motorcyclist with bilateral fractured femurs has haematuria following a road traffic collision. There is blood at the penile meatus and a high-riding prostate on rectal examination.
- 2 Painless haematuria, with frequency of micturition, in a 28-year-old man who spent 4 months trekking in Tanzania and Rwanda 1 year ago. FBC reveals an eosinophilia.
- 3 Haematuria, weight loss, and mild renal angle tenderness in a 55-year-old woman with recent onset of back pain and white cell count of $5 \times 10^9/L$.
- 4 A 45-year-old woman with type 1 diabetes and a history of recurrent urinary tract infections has microscopic haematuria, loin pain, and tenderness of the renal angle. Her temperature is $38.2^\circ C$ and white cell count is $21 \times 10^9/L$.
- 5 A 55-year-old man develops sudden onset of severe colicky left loin pain radiating to the groin. Treatment with NSAIDs and drinking several glasses of water appears to relieve the pain overnight.

Answers • Themes 1–3

OXFORD EMQS

Theme 1 • Causes of scrotal swelling

- 1 **F** Non-seminomatous germ cell tumour — Non-seminomatous germ cell tumours peak in men aged 20–30, whereas seminomas occur in older men.
- 2 **C** Hydrocoele — Hydrocoeles are fluctuant, transilluminating scrotal masses in which the testis is not separately palpable.
- 3 **D** Inguinal hernia — Inguinal hernias are intermittent, reducible masses; the examiner cannot get above the mass.
- 4 **A** Epididymal cyst — Epididymal cysts are retention cysts adjacent to but separate from the testicle.
- 5 **B** Epididymo-orchitis — Epididymo-orchitis presents with an inflammatory mass, tenderness, and urinary symptoms.

Theme 2 • Causes of scrotal/testicular pain

- 1 **A** Epididymo-orchitis — Bilateral scrotal pain with purulent urethral discharge, eased by scrotal support (Prehn sign), indicates sexually transmitted epididymo-orchitis.
- 2 **F** Testicular torsion — Sudden testicular pain with a high-lying testis in a teenager (maldescent predisposes) is testicular torsion.
- 3 **I** Torsion of the hydatid of Morgagni — A localised dark spot by the testis on transillumination ('blue dot' sign) is torsion of the hydatid of Morgagni.

- 4 **E** Scrotal haematoma — Pain and swelling after a direct blow to the scrotum is a scrotal haematoma; ultrasound is used to exclude testicular rupture.
- 5 **C** Mumps orchitis — A swollen tender testis after parotid ('neck') swelling indicates mumps orchitis.

Theme 3 • Causes of haematuria/loin pain

- 1 **C** Pelvic fracture — Blood at the meatus and a high-riding prostate after major trauma indicate pelvic fracture with posterior urethral disruption; do not catheterise per urethra.
- 2 **I** Schistosomiasis — Painless haematuria with eosinophilia after travel in East Africa suggests *Schistosoma haematobium* infection.
- 3 **G** Renal cell carcinoma — Haematuria, weight loss and loin tenderness with new back pain (bone metastasis) suggest renal cell carcinoma.
- 4 **F** Pyelonephritis — Fever, loin pain, renal angle tenderness and leucocytosis in a diabetic with recurrent UTIs indicate acute pyelonephritis.
- 5 **H** Renal colic — Sudden severe colicky loin-to-groin pain settling with NSAIDs and fluids is renal colic from a ureteric stone.

6 THEMES • 30 SCENARIOS • AFTER GET AHEAD! SURGERY MCQs & EMQs

Get Ahead EMQs

SET 1 OF 2

THEME 1 Management of testicular conditions

- A Antibiotics
- B Aspiration of fluid
- C Chemotherapy alone
- D Excision of cyst
- E Herniorrhaphy
- F Orchiectomy alone
- G Orchiectomy + chemotherapy
- H Orchiectomy + radiotherapy
- I Orchiectomy + radiotherapy + chemotherapy
- J Radiotherapy alone
- K Reassurance and simple analgesia
- L Reduction and fixation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 18-year-old man presents with a swelling in his left scrotum. On examination, there is a smooth, fluctuant swelling above and behind the testis that transilluminates; the testis can be felt separately. The patient wants treatment for this swelling.
- 2 A 10-year-old boy presents with sudden-onset pain and swelling in his left testicle. On examination, the left testicle lies higher than the right, is erythematous, and is exquisitely tender.
- 3 A 42-year-old man presents with a painless lump in his right scrotum that he first noticed last week. On examination, the swelling is hard and irregular and does not transilluminate. A CT scan rules out lymph node involvement.
- 4 A 37-year-old man presents with a 2-day history of pain and swelling in his right testicle and pain on passing water. On examination, his right scrotum is swollen, erythematous, and tender. Urine dipstick shows white cells.
- 5 A 23-year-old man presents with pain and swelling in his scrotum. On examination, a hard, irregular mass is felt in the left scrotum that does not transilluminate. A CT scan demonstrates the presence of enlarged para-aortic lymph nodes.

THEME 2 Management of prostate disease

- A Active surveillance
- B Brachytherapy
- C Doxazosin
- D External-beam radiotherapy
- E Finasteride
- F Radical prostatectomy
- G Transurethral resection of prostate

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 87-year-old man has a transrectal ultrasound-guided biopsy following admission for acute urinary retention. Histology shows a T1 prostate cancer.
- 2 A 68-year-old man with known metastatic prostate cancer presents with worsening bony pain in his left hip. He is currently on a course of hormonal treatment.
- 3 A 57-year-old man is diagnosed with a T2 prostate cancer. He expresses a wish to have curative therapy but mentions that he would like to reduce the risk of developing impotence as a result of therapy.
- 4 A 63-year-old man presents with a 6-month history of poor stream and dribbling at the end of micturition, which is bothering him. He is found to have smooth prostatic enlargement on digital rectal examination. His PSA is 7.5 ng/mL.
- 5 A 58-year-old man who has benign prostatic hyperplasia has been treated with finasteride for over a year. He presents to the clinic because he has found no benefit in his symptoms since being started on medical management.

THEME 3 Investigation of urological disease

- A Retrograde urethrogram
- B Bladder scan and urethral catheterisation
- C CT scan
- D Cystoscopy and retrograde ureteric stent
- E Intravenous urogram
- F Percutaneous nephrostomy
- G Three-way catheter and irrigation
- H Transrectal ultrasound and biopsy
- I Ultrasound scan of abdomen
- J Urgent surgical exploration

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 22-year-old man presents to the emergency department after being hit by a reversing car. He has an episode of macroscopic haematuria, but abdominal and per rectum examinations show no other abnormality. His blood pressure is 114/74 mmHg and his pulse 86/min.
- 2 A 74-year-old man presents with acute-onset suprapubic pain and an inability to pass urine. On examination, he is visibly uncomfortable and the bladder is percussable 6 cm above the pubis.

- 3** A 37-year-old man was cycling home when his bike skidded on ice and he crashed into a post box. On admission to the emergency department, he has complained of an inability to pass urine and on examination there is blood at the external urethral meatus.
- 4** A 61-year-old man presents with sudden-onset left loin pain that radiates to his groin. He has been unable to pass urine over the last 24 hours. He has a history of a right nephrectomy for a renal cell carcinoma. A spiral CT scan confirms the presence of a 6 mm stone at the left vesicoureteric junction and a left hydronephrosis.
- 5** A 58-year-old man is referred to the clinic by his general practitioner with a prostate-specific antigen level of 14.0 ng/mL. He has no other symptoms. Abdominal examination shows no abnormality, but an enlarged, hard, irregular prostate gland is palpable per rectum.

REPEATED 2* • ALSO ASKED IN
Get Ahead

THEME 4 Management of urological disease

- A Analgesia alone
- B Cystoscopy
- C Extracorporeal shock wave lithotripsy
- D Intravenous antibiotics
- E Open pyelolithotomy
- F Nephrectomy
- G Percutaneous nephrolithotomy
- H Percutaneous nephrostomy
- I Suprapubic catheterisation
- J Three-way catheter and irrigation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 38-year-old woman presents with left-sided loin pain, fever and rigors. On examination, there is a palpable mass on the left side. Her pulse is 106/min and her blood pressure 104/64 mmHg. A KUB film done in the emergency department shows a 7 mm stone at the vesicoureteric junction.
- 2** A 56-year-old male who has a 3-day history of frank haematuria presents with sudden-onset suprapubic discomfort and an inability to pass urine.
- 3** A 32-year-old man presents to the emergency department with severe right loin pain that radiates to the groin. On examination, there is microscopic haematuria on dipstick but no other abnormality. A KUB X-ray confirms the presence of a 4 mm stone in the ureter.
- 4** A 54-year-old man presents with loin-to-groin pain. An intravenous urogram shows a 1 cm stone in the renal collecting system.

- 5** A 62-year-old man presents with a 3-month history of left loin pain. Examination yields no abnormality. A KUB X-ray demonstrates a staghorn calculus in the left renal collecting system. A DMSA (dimercaptosuccinic acid) scan shows a split function of 10% in the left kidney and 90% in the right.

THEME 5 Management of urinary retention

- A Check urea and electrolytes
- B Cystoscopy
- C Intravenous antibiotics
- D Lithotripsy
- E Percutaneous nephrostomy
- F Reassurance alone
- G Suprapubic catheterisation
- H Transurethral resection of prostate
- I Trial without catheter
- J Urethral catheterisation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 64-year-old man presents with suprapubic pain and a sudden inability to pass urine. On examination, there is a palpable suprapubic mass that is dull to percussion. A bladder scan demonstrates 750 mL in the bladder.
- 2** A 64-year-old man presents with suprapubic pain and a sudden inability to pass urine. On examination, there is a palpable suprapubic mass that is dull to percussion. This was preceded by a 4-week history of passage of frank haematuria, including the occasional clot. Three failed attempts have been made at urethral catheterization.
- 3** A 64-year-old man presents with suprapubic pain and a sudden inability to pass urine. On examination, there is a palpable suprapubic mass that is dull to percussion. There is no significant past medical or surgical history. Three failed attempts have been made at urethral catheterization.
- 4** A 64-year-old man presents with suprapubic pain and a sudden inability to pass urine. On examination, there is a palpable suprapubic mass that is dull to percussion. A urethral catheter is passed successfully and a 500 mL residual volume is recorded. You see the patient the next day on the ward round, and he asks you what will happen next.
- 5** A 64-year-old man presents with a history of overflow incontinence. He does not complain of any pain and is still able to empty his bladder, although this takes some time and there is dribbling at the end of the stream. On examination, the bladder is palpable 5 fingerbreadths above the pubis.

Answers • Themes 1-5

GET AHEAD EMQS

Theme 1 • Management of testicular conditions

- D Excision of cyst** — A transilluminating swelling above and separate from the testis is an epididymal cyst; a symptomatic cyst is excised (aspiration recurs).
- L Reduction and fixation** — Reduction and fixation is required for acute testicular torsion.
- H Orchidectomy + radiotherapy** — A hard testicular lump at 42 is likely a seminoma; stage I disease is treated by inguinal orchidectomy with adjuvant radiotherapy (or carboplatin/surveillance).
- A Antibiotics** — Antibiotics treat epididymo-orchitis presenting with infection signs.
- G Orchidectomy + chemotherapy** — Orchidectomy and chemotherapy are indicated for testicular cancer with nodal spread.

Theme 2 • Management of prostate disease

- A Active surveillance** — Active surveillance avoids overtreatment in elderly T1 prostate cancer.
- D External-beam radiotherapy** — External-beam radiotherapy palliates bone pain in metastatic disease.
- B Brachytherapy** — Brachytherapy provides curative treatment with lower sexual side effects.
- E Finasteride** — Bothersome LUTS with a smooth enlarged prostate (PSA raised by gland size) respond to finasteride, a 5 α -reductase inhibitor that shrinks large glands.
- G Transurethral resection of prostate** — TURP is indicated when medical therapy fails for BPH.

Theme 3 • Investigation of urological disease

- C CT scan** — CT scan is the investigation of choice for stable blunt renal trauma with haematuria.
- B Bladder scan and urethral catheterisation** — Bladder scan and catheterisation manage acute urinary retention.

- A Retrograde urethrogram** — Blood at the meatus with inability to void after trauma suggests urethral injury; a retrograde urethrogram is done before any urethral catheter is passed.
- D Cystoscopy and retrograde ureteric stent** — An obstructing stone in a solitary kidney with anuria needs urgent decompression, here by cystoscopy and retrograde ureteric stenting.
- H Transrectal ultrasound and biopsy** — Transrectal ultrasound and biopsy confirm prostate cancer.

Theme 4 • Management of urological disease

- H Percutaneous nephrostomy** — Nephrostomy decompresses infected obstructing stones.
- J Three-way catheter and irrigation** — Visible haematuria with clot retention is treated with a three-way catheter and continuous bladder irrigation.
- A Analgesia alone** — Analgesia is sufficient for small ureteric stones.
- C Extracorporeal shock wave lithotripsy** — ESWL fragments a 1 cm renal calculus.
- F Nephrectomy** — Nephrectomy is indicated for a poorly functioning staghorn kidney.

Theme 5 • Management of urinary retention

- J Urethral catheterisation** — Acute retention is relieved first by urethral catheterisation, recording the drained (residual) volume.
- B Cystoscopy** — After haematuria with clots and failed catheterisation, cystoscopy is safer than a suprapubic catheter, which could puncture or seed a bladder tumour.
- G Suprapubic catheterisation** — When urethral catheterisation fails in uncomplicated retention, a suprapubic catheter is inserted.
- I Trial without catheter** — Trial without catheter assesses return of normal voiding after decompression.
- A Check urea and electrolytes** — Checking urea and electrolytes evaluates renal function in chronic retention.

SET 2 OF 2

THEME 6 Urological conditions

- A Acute cystitis
- B Anticoagulation therapy
- C Benign prostatic hyperplasia
- D Bladder neck stenosis
- E Nephroblastoma
- F Polycystic kidney disease
- G Prostate carcinoma
- H Pyelonephritis
- I Renal cell carcinoma
- J Renal tract calculi
- K Transitional cell carcinoma of the bladder

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- A 67-year-old man presents with sudden-onset back pain that occurred while walking. He has been having difficulty passing urine over the past few months and admits to terminal dribbling. On examination, there is dullness to percussion in the suprapubic region.
- An 18-year-old woman presents with a 24-hour history of burning sensation on micturition. She has also been passing water more frequently than usual. On examination, she is tender suprapubically, but there is no tenderness in either renal angle.

- 3** A 76-year-old man presents with malaise and a long history of difficulty in voiding urine. Abdominal examination is normal except for a suprapubic mass that is dull to percussion. A per rectum exam reveals a large, smooth prostate gland.
- 4** A 64-year-old man presents with difficulty voiding and a weak stream. He had similar symptoms 6 years ago, and the underlying cause was found to be benign prostatic hyperplasia, for which he had a transurethral resection of the prostate. Abdominal examination was normal and per rectum examination demonstrated no mass.
- 5** A 12-month-old boy is brought to the general practice by his mother with irritability, fever and occasional blood in the nappy. On examination, a large mass is felt on the left side of the abdomen. The child has dropped two centile lines for weight and height over the last 6 months.

Answers • Theme 6

GET AHEAD EMQS

Theme 6 • Urological conditions

- 1** **G Prostate carcinoma** — Prostate carcinoma causes obstruction and back pain.
- 2** **A Acute cystitis** — Acute cystitis presents with dysuria and frequency.
- 3** **C Benign prostatic hyperplasia** — BPH leads to bladder outlet obstruction and retention.
- 4** **D Bladder neck stenosis** — Bladder neck stenosis is a complication of TURP.
- 5** **E Nephroblastoma** — An abdominal mass with haematuria and failure to thrive in an infant suggests nephroblastoma (Wilms tumour).

3 THEMES • 14 SCENARIOS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course EMQs

THEME 1 Lower urinary tract symptoms

- A Acontractile bladder
- B Benign prostatic hypertrophy
- C Bladder neck stenosis
- D Low-pressure, low-flow voiding
- E Multiple sclerosis
- F Prolapsed intervertebral disc
- G Prostatic cancer
- H Spinal canal stenosis
- I Stress incontinence
- J Urethral stricture

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 70-year-old man presents with a 5-year history of a poor flow, hesitancy, frequency, nocturia and postmicturition dribbling. A flow rate showed a Qmax of 7 mL/s with a voided volume of 270 mL and a residual of 100 mL.
- 2** A 40-year-old woman presents with urinary incontinence when coughing, sneezing and during her fitness class. She has had three children delivered vaginally. The first labour was prolonged and she had a repair of a second-degree tear.

- 3** A 16-year-old man presents with a long history of hesitancy and worsening urinary stream. He also complains of postmicturition dribbling, which may occur some minutes after he completes his normal void. He had a significant fall astride a bar injury as a child.
- 4** A 40-year-old woman presents with urinary frequency, urgency and episodes of urge incontinence. She had a transient episode of unilateral blindness 2 years prior.
- 5** A 40-year-old man has a long history of intermittent low back pain but develops acute onset of difficulty passing urine. On examination he has a palpable bladder and an area of saddle anaesthesia.

THEME 2 Urological investigations

- A Urine dipstick
- B Cystoscopy
- C Serum PSA
- D Renal ultrasound scan
- E Non-contrast CT of kidneys, ureters and bladder (CT KUB)
- F Serum urea and creatinine
- G Urodynamic studies (cystometrogram)
- H Abdominal X-ray of kidneys, ureters and bladder (KUB)
- I Intravenous urography
- J Renal aortography

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient presenting with dysuria and frequency. Routine observations reveal a temperature of 38.6°C.
- 2 A 25-year-old long-distance runner presenting with severe right-sided loin pain and vomiting.
- 3 A middle-aged man presenting with urinary hesitancy, a poor stream and terminal dribbling postmicturition with evidence of renal dysfunction.
- 4 A 75-year-old man presenting with urinary frequency, nocturia and weight loss. An irregular mass is evident on the prostate on digital examination.
- 5 A middle-aged female who has had five vaginal deliveries presenting with urinary leakage when coughing or laughing.

THEME 3 Urgent urological referrals

- A Benign prostatic hypertrophy
- B Cystitis
- C Epididymitis
- D Lymphoma
- E Priapism
- F Prostate cancer
- G Pyelonephritis
- H Seminoma of testis
- I Teratoma of testis
- J Testicular torsion
- K Ureteric colic
- L Urinary tract infection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 19-year-old man presents with a firm, nontender lump in his right testicle.
- 2 A 23-year-old woman presents with right-sided loin pain which radiates to the groin. The pain comes and goes in waves. She also complains of frequency of micturition. Dipstick urine test reveals 4+ blood. The only abnormality on examination is tenderness in the right loin.
- 3 An 88-year-old man presents via his GP with suprapubic pain and a history of frequency and passing only very small amounts of urine for 1 week. He has had long-standing problems with nocturia and poor stream. He has a tender palpable mass arising from the pelvis. His PSA is 5 ng/mL (NR 0-4 ng/mL).
- 4 A 52-year-old man presents with a 3-day history of dysuria and frequency and a gradual onset of left testicular pain and swelling. Dipstick urine reveals 3+ blood, 2+ leukocytes and nitrites.

Answers • Themes 1-3

Theme 1 • Lower urinary tract symptoms

- 1 **B Benign prostatic hypertrophy** — Long-standing voiding LUTS with a low Qmax (under 10 mL/s) and a residual in an older man indicate bladder outlet obstruction from BPH.
- 2 **I Stress incontinence** — Leakage on coughing, sneezing and exercise after vaginal deliveries is stress urinary incontinence.
- 3 **J Urethral stricture** — Progressive poor stream and post-void dribbling after a childhood fall-astiride injury indicate a bulbar urethral stricture.
- 4 **E Multiple sclerosis** — Urge incontinence in a young woman with previous transient monocular blindness (optic neuritis) suggests multiple sclerosis causing detrusor overactivity.
- 5 **F Prolapsed intervertebral disc** — Acute retention with saddle anaesthesia and back pain is cauda equina syndrome from a prolapsed disc, needing urgent MRI and decompression.

Theme 2 • Urological investigations

- 1 **A Urine dipstick** — Dysuria and frequency with fever suggest UTI; the first test is a urine dipstick (then MSU culture).
- 2 **E Non-contrast CT of kidneys, ureters and bladder (CT KUB)** — Suspected renal colic is best confirmed by low-dose non-

contrast CT KUB (NICE), which detects almost all stones; plain KUB misses many.

- 3 **D Renal ultrasound scan** — Voiding LUTS with renal impairment needs renal ultrasound to look for hydronephrosis and a high residual from chronic retention.
- 4 **C Serum PSA** — An irregular prostate on rectal examination with weight loss suggests prostate cancer; check serum PSA (then MRI and biopsy).
- 5 **G Urodynamic studies (cystometrogram)** — Of these, urodynamics best characterises stress versus urge incontinence; NICE reserves it for mixed, unclear or pre-surgical cases.

Theme 3 • Urgent urological referrals

- 1 **I Teratoma of testis** — A firm, painless testicular lump in a 19-year-old is most likely a teratoma (NSGCT, peak 20-30 years) and needs urgent referral.
- 2 **K Ureteric colic** — Colicky loin-to-groin pain with haematuria on dipstick and loin tenderness is ureteric colic.
- 3 **A Benign prostatic hypertrophy** — Long-standing LUTS now with a tender palpable bladder is retention from BPH; a PSA of 5 at 88 years fits a large benign gland.

- 4 **C** **Epididymitis** — Gradual testicular pain and swelling with dysuria and a positive dipstick (leucocytes, nitrites) indicate epididymitis.

9 THEMES • 44 SCENARIOS • IRFAN SURGERY EMQ COLLECTION

Irfan EMQs

SET 1 OF 2

THEME 1 Haematuria

- A Wilms' tumour
- B neuroblastoma
- C renal cell carcinoma
- D fibroepithelial polyp
- E squamous cell carcinoma of the bladder
- F polycystic kidney disease
- G adrenal carcinoma
- H rhabdomyolysis
- I renal calculus
- J transitional cell carcinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 75-year-old woman has had a UTI treated but has persistent microscopic haematuria seen on urine dipstick. She's not worried but her GP refers her to the one-stop haematuria clinic.
- 2 A 50-year-old man with weight loss, loss of appetite and shortness of breath has recently noticed a left-sided varicocele which does not disappear on lying supine.
- 3 A 35-year-old man with new-onset hypertension is admitted to hospital for investigation. His creatinine is 134 $\mu\text{mol/L}$. After imaging he is advised to tell his brother to undergo family screening and genetic counselling.
- 4 An 18-year-old is travelling in the Australian outback during his gap year. He returns to London to run the marathon which he does successfully. The next day he attends A&E concerned about his dark red urine. He is otherwise completely asymptomatic but is admitted to hospital.
- 5 A 29-year-old man from Egypt presents with weight loss and haematuria. Pseudotubercles and nodules seen on cystoscopy are biopsied.

THEME 2 Testicular conditions

- A varicocele
- B chronic orchitis
- C ectopic testis
- D hydrocele
- E non-seminomatous testicular cancer
- F seminoma
- G epididymal cyst
- H indirect inguinal hernia
- I acute epididymo-orchitis
- J testicular atrophy
- K testicular torsion

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 52-year-old man presents with a testicular swelling that has increased in size gradually over a period of years. On examination the swelling transilluminates and the testis cannot be felt separate from the swelling.
- 2 A 40-year-old man complains of severe pain and swelling over the last 48 hours in his right scrotum. Testis and epididymis are very tender. He reports unprotected intercourse.
- 3 A 20-year-old man with a history of undescended testes presents with weight loss and a hard, painless testicular lump.
- 4 A 9-year-old boy wakes at night crying from severe testicular pain associated with vomiting.
- 5 A 45-year-old man presents with a left-sided fluctuating scrotal swelling that worsens on standing.

THEME 3 Bladder outlet obstruction

- A phimosis
- B urethral stricture
- C pregnancy
- D prostate cancer
- E benign prostatic hypertrophy
- F clot retention
- G pelvic mass
- H bladder stones
- I urethral diverticula
- J urethral tumour
- K bladder tumour

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 75-year-old man describes difficulty passing urine with a white scar and ballooning foreskin.
- 2 A 58-year-old woman awaiting hysterectomy for a large fibroid uterus has bilateral hydronephrosis and chronic renal failure.
- 3 A 35-year-old man with a history of recurrent urine infections and vesico-ureteric reflux presents with voiding difficulty.
- 4 A 63-year-old man, 1 week post-TURP, reattends in discomfort unable to pass urine.
- 5 An 84-year-old man complains of difficulty passing urine, nocturia, hesitancy, and terminal dribbling.

THEME 4 Management of prostate cancer

- A radical prostatectomy
- B bilateral orchiectomy
- C transurethral resection of prostate
- D watch and wait
- E radiotherapy
- F bone scan and dexamethasone
- G anti-androgen medication
- H luteinizing hormone releasing hormone analogue
- I long-term catheter and hospice
- J family screening
- K repeat PSA

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 64-year-old man with a history of hesitancy, nocturia, frequency and terminal dribbling fails a trial without catheter despite being on an alpha-blocker. Rectal examination reveals a smooth enlarged prostate. He is otherwise fit and healthy and his biopsy results are negative for cancer.
- 2 A 66-year-old man has recently had a routine blood screen and is reported to have an elevated PSA of 5.5 ng/mL. He is otherwise asymptomatic.

- 3 A 34-year-old man attends your clinic reporting that his twin brother has been diagnosed with prostate cancer.
- 4 A 73-year-old man with prostate cancer attends your clinic complaining of new-onset back pain.
- 5 A 52-year-old man is diagnosed with prostate cancer and staging confirms the absence of secondary spread of disease.

THEME 5 Urinary incontinence (diagnosis/management)

- A mid-stream urine MC&S
- B cystoscopy
- C flow rate
- D ultrasound of kidneys and bladder residual
- E video urodynamics
- F DMSA
- G intravenous urogram
- H MAG3 scan
- I urethrogram
- J pelvic floor exercises
- K bladder augmentation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 35-year-old woman presents 2 years after the birth of her child with urinary leakage on coughing, standing and sneezing.
- 2 A 25-year-old man presents with slowing of his urinary flow and terminal urethral dribbling. He has a past history of a treated sexually transmitted disease.
- 3 A 75-year-old woman presents with recent onset of severe urinary urgency and bladder pain. Urinary dipstick shows no nitrites, protein 1+ and blood 2+.
- 4 An 80-year-old man presents to the urology clinic with poor flow, frequency and nocturia. He is noted to have an elevated creatinine.
- 5 A 40-year-old woman who has had two previous colposuspensions for urinary incontinence now complains of mixed urge and stress leakage.

Answers • Themes 1-5

Theme 1 • Haematuria

- 1 **J transitional cell carcinoma** — Persistent haematuria after a treated UTI in an older patient needs investigation for urothelial (transitional cell) carcinoma, the commonest cause at this age.
- 2 **C renal cell carcinoma** — A left varicocele that does not empty on lying flat, with weight loss and dyspnoea (lung metastases), suggests renal cell carcinoma invading the renal vein.
- 3 **F polycystic kidney disease** — Young-onset hypertension, renal impairment and advice for sibling screening point to autosomal dominant polycystic kidney disease.

- 4 **H rhabdomyolysis** — Dark urine after a marathon in an otherwise well young man is myoglobinuria from exertional rhabdomyolysis.
- 5 **E squamous cell carcinoma of the bladder** — Bladder pseudotubercles in an Egyptian man indicate chronic schistosomiasis, which predisposes to squamous cell carcinoma of the bladder.

Theme 2 • Testicular conditions

- 1 **D hydrocele** — A long-standing transilluminating swelling in which the testis cannot be felt separately is a hydrocele.

- 2 **I acute epididymo-orchitis** — Acute painful swelling with a tender testis and epididymis after unprotected sex is acute epididymo-orchitis (chlamydia/gonorrhoea).
- 3 **E non-seminomatous testicular cancer** — A hard painless testicular lump at 20 years with previous maldescent is most likely a non-seminomatous germ cell tumour (peak 20–30 years).
- 4 **K testicular torsion** — Sudden severe testicular pain waking a boy from sleep, with vomiting, is testicular torsion until proven otherwise.
- 5 **A varicocele** — A scrotal swelling that enlarges on standing and feels like a bag of worms is a varicocele.

Theme 3 • Bladder outlet obstruction

- 1 **A phimosis** — A white, scarred, ballooning foreskin with voiding difficulty is phimosis (here due to BXO).
- 2 **G pelvic mass** — A large pelvic mass such as a fibroid uterus can compress the bladder neck and ureters, causing outflow obstruction and obstructive renal failure.
- 3 **B urethral stricture** — Recurrent urinary infection and its instrumentation predispose a young man to urethral stricture, causing slow, difficult voiding.
- 4 **F clot retention** — Retention a week after TURP is typically clot retention from secondary haemorrhage; treat with a three-way catheter and irrigation.
- 5 **E benign prostatic hypertrophy** — Hesitancy, nocturia and terminal dribbling in an elderly man are most commonly due to benign prostatic hyperplasia.

Theme 4 • Management of prostate cancer

- 1 **C transurethral resection of prostate** — BPH with a failed trial without catheter despite an alpha-blocker, and a negative biopsy, is an indication for TURP.

- 2 **K repeat PSA** — A mildly raised PSA in an asymptomatic man is first repeated (excluding UTI, recent ejaculation or exercise) before MRI or biopsy.
- 3 **J family screening** — A first-degree relative (especially a twin) with prostate cancer raises the risk; offer counselling and early PSA screening.
- 4 **F bone scan and dexamethasone** — New back pain in prostate cancer suggests bone metastases or cord compression: give dexamethasone and image urgently (bone scan/MRI spine).
- 5 **A radical prostatectomy** — Localised prostate cancer in a fit 52-year-old is suitable for curative radical prostatectomy.

Theme 5 • Urinary incontinence (diagnosis/management)

- 1 **J pelvic floor exercises** — Leakage on coughing and sneezing after childbirth is stress incontinence; first-line treatment is supervised pelvic floor muscle training for at least 3 months.
- 2 **I urethrogram** — A slowing stream with post-void dribbling after an STI in a young man suggests urethral stricture, best shown by urethrography.
- 3 **B cystoscopy** — New severe urgency and bladder pain with haematuria and no infection in an elderly woman need cystoscopy to exclude carcinoma in situ or bladder cancer.
- 4 **D ultrasound of kidneys and bladder residual** — Outflow symptoms with a raised creatinine need ultrasound of kidneys and bladder to look for hydronephrosis and a high residual (chronic retention).
- 5 **E video urodynamics** — Recurrent mixed incontinence after previous continence surgery needs video urodynamics before any further intervention.

SET 2 OF 2

THEME 6 Management of renal stones

- A simple analgesia and discharge home with outpatient follow-up
- B emergency nephrectomy
- C MAG3 scan
- D extracorporeal shock wave lithotripsy (ESWL)
- E ureteroscopy and laser fragmentation
- F percutaneous nephrolithotomy (PCNL)
- G embolization
- H nephrostomy
- I ultrasound kidneys
- J intravenous antibiotics
- K retrograde ureteric stenting

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 50-year-old man has heavy haematuria and a falling haemoglobin 2 days after percutaneous nephrolithotomy. Bleeding continues despite clamping the nephrostomy tube.
- 2 A 28-year-old woman who is 24 weeks pregnant presents with right loin-to-groin pain and microscopic haematuria.

- 3 A 60-year-old woman with a known obstructing left ureteric stone has a temperature of 39.5 °C, rigors and hypotension. Ultrasound shows a dilated left collecting system containing debris. She is too unwell for a general anaesthetic.
- 4 A 30-year-old man with right ureteric colic is pain-free after diclofenac. CT KUB shows a 4 mm stone in the distal ureter; he is afebrile with normal renal function.
- 5 A 55-year-old woman has recurrent loin pain and urinary infections. CT shows a 3 cm staghorn calculus in the left kidney, which has good function on renography.

THEME 7 Lumps in the groin

- A maldescended testis
- B femoral aneurysm
- C hydrocoele
- D lipoma of the cord
- E inguinal hernia
- F psoas abscess
- G varicocele
- H seminoma
- I scrotal carcinoma
- J saphena varix
- K inguinal lymphadenopathy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An incidental lump found during inguinal hernia repair that is benign.
- 2 A 60-year-old smoker with a throbbing lump in the right groin showing an expansile pulsation.
- 3 A 9-year-old boy with a new significant groin lump post-unsuccessful laparoscopy for undescended testis.
- 4 An 80-year-old man with a right-sided scrotal mass that reduces on lying flat.
- 5 A 69-year-old woman with right leg pain and firm, matted groin masses.

THEME 8 Urological imaging/intervention

- A Transurethral resection of the prostate
- B Retrograde ureterogram
- C Nephrostomy
- D Micturating cystourethrogram
- E Ultrasound of urinary tract
- F Intravenous urogram (IVU)
- G Furosemide
- H High-intensity focused ultrasound (HIFU) prostate ablation
- I Radical prostatectomy
- J Nephroureterectomy, cystectomy and ileal conduit formation
- K Active surveillance

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 45-year-old man has muscle-invasive transitional cell carcinoma of the bladder extending to multiple sites along the right ureter and involving the renal pelvis.

- 2 An 83-year-old man with a PSA of 10 ng/mL and positive TRUS biopsy results denies lower urinary tract symptoms. He does not want any surgical intervention.
- 3 A 45-year-old man has a raised PSA of 29 ng/mL and positive TRUS biopsy results. His brother died of prostate cancer aged 60 years.
- 4 A 30-year-old man with a history of renal stones has been on a business trip and taken diclofenac and codeine throughout his meetings. He returns to attend A&E with further pain.
- 5 A 15-year-old girl complains of severe loin pain when voiding.

THEME 9 Renal masses/tumours

- A lymphoma
- B polycystic kidney
- C simple cyst
- D sarcoma
- E adrenal mass
- F renal abscess
- G staghorn calculus
- H perirenal haematoma
- I squamous cell carcinoma
- J medullary sponge kidney
- K renal cell carcinoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 57-year-old man with acute cholecystitis undergoes ultrasound, which incidentally detects a renal lesion that is benign.
- 2 A 46-year-old woman with a 3-month history of haematuria, weight loss, fevers, lethargy, loin pain, and a palpable mass.
- 3 A 23-year-old man after a motorbike accident presents with an expanding left loin mass, bruising, and haematuria.
- 4 A 37-year-old HIV-positive man presents with haematuria, swinging pyrexia, raised white cell count and a tender left loin mass.

Answers • Themes 6–9

Theme 6 • Management of renal stones

- 1 **G embolization** — Persistent bleeding after PCNL despite tamponade suggests an arteriovenous fistula or pseudoaneurysm; selective angioembolisation is the next step.

- 2 **I ultrasound kidneys** — In pregnancy, ultrasound is the first-line imaging for suspected stones or obstruction because it avoids ionising radiation.
- 3 **H nephrostomy** — An infected obstructed kidney (pyonephrosis) needs urgent decompression plus antibiotics; a percu-

taneous nephrostomy under local anaesthetic suits an unstable patient.

- 4 **A** **simple analgesia and discharge home with outpatient follow-up** — Ureteric stones of 5 mm or less usually pass spontaneously; a pain-free, apyrexial patient with normal renal function can go home with analgesia and follow-up.
- 5 **F** **percutaneous nephrolithotomy (PCNL)** — Large (over 2 cm) or staghorn renal stones in a functioning kidney are best treated by percutaneous nephrolithotomy.

Theme 7 • Lumps in the groin

- 1 **D** **lipoma of the cord** — An incidental benign fatty mass in the spermatic cord found at hernia repair is a lipoma of the cord.
- 2 **B** **femoral aneurysm** — An expansile (not merely transmitted) pulsatile groin mass in an elderly smoker is a femoral artery aneurysm.
- 3 **A** **mal descended testis** — A groin lump after a failed attempt to bring down an undescended testis is the mal descended testis lying in the inguinal canal.
- 4 **E** **inguinal hernia** — A scrotal mass in an elderly man that reduces on lying flat (you cannot get above it) is an indirect inguinal hernia.
- 5 **K** **inguinal lymphadenopathy** — Firm, matted groin masses with leg pain are inguinal lymphadenopathy (e.g. metastases or lymphoma) needing a search of the drainage area.

Theme 8 • Urological imaging/intervention

- 1 **J** **Nephroureterectomy, cystectomy and ileal conduit formation** — Multifocal muscle-invasive urothelial cancer of bladder,

ureter and renal pelvis needs cystectomy with nephroureterectomy and urinary diversion (ileal conduit).

- 2 **K** **Active surveillance** — An asymptomatic elderly man with low-volume prostate cancer who declines intervention is best managed conservatively by surveillance.
- 3 **I** **Radical prostatectomy** — A young, fit man with prostate cancer and a strong family history should be offered curative radical prostatectomy (assuming localised disease).
- 4 **E** **Ultrasound of urinary tract** — Heavy NSAID use may impair renal function, so ultrasound (no contrast) is the safest initial test for obstruction.
- 5 **D** **Micturating cystourethrogram** — Loin pain on voiding suggests vesicoureteric reflux, which is demonstrated by a micturating cystourethrogram.

Theme 9 • Renal masses/tumours

- 1 **C** **simple cyst** — An incidental benign renal lesion on ultrasound is most often a simple cyst (thin-walled, anechoic, Bosniak I).
- 2 **K** **renal cell carcinoma** — Haematuria, loin pain and a mass with fever, weight loss and lethargy (paraneoplastic features) suggest renal cell carcinoma.
- 3 **H** **perirenal haematoma** — An expanding loin mass with bruising and haematuria after a motorbike accident is a perirenal haematoma from renal injury.
- 4 **F** **renal abscess** — Swinging fever, leucocytosis and a tender loin mass in an immunocompromised patient indicate a renal (perinephric) abscess.

- 35 Trauma
- 36 Metabolic Response to Trauma
- 37 Shock
- 38 Obesity & Bariatric Surgery
- 39 Trauma & Perioperative EMQs



07

TRAUMA & CRITICAL CARE

Trauma

ATLS primary survey, airway, chest, head and limb injuries, and burns.

79

MCQs

0

EMQ SCENARIOS

6

SOURCES

1

REPEATED IDEAS

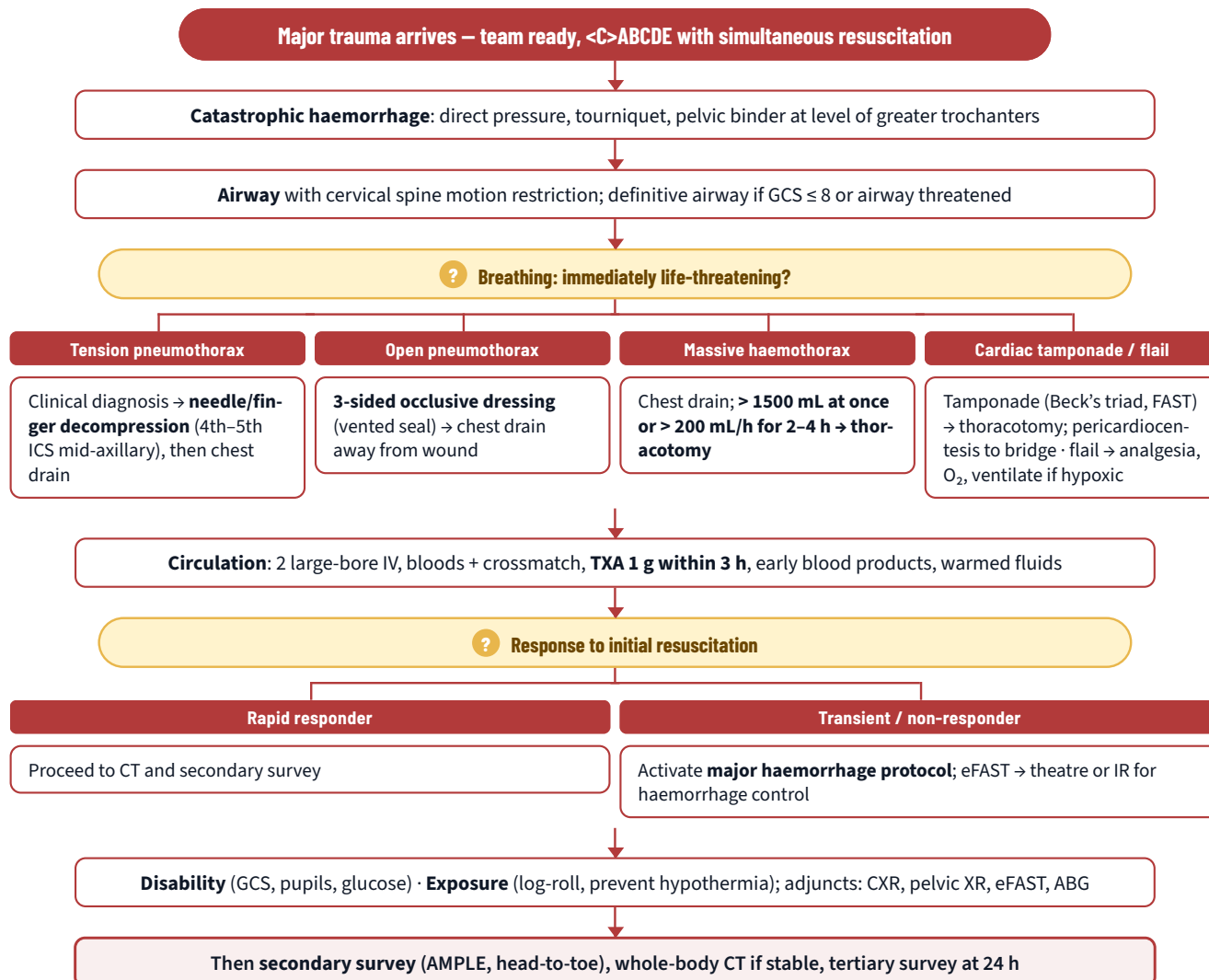
IN THIS CHAPTER

01	Management flowcharts	3 charts	page 460
02	Local Exams MCQs	13 Q	page 463
03	Bailey & Love MCQs	26 Q	page 465
04	Lange MCQs	6 Q	page 469
05	SBA MCQs	7 Q	page 470
06	PreTest MCQs	25 Q	page 471
07	Get Ahead MCQs	2 Q	page 475

Step by step

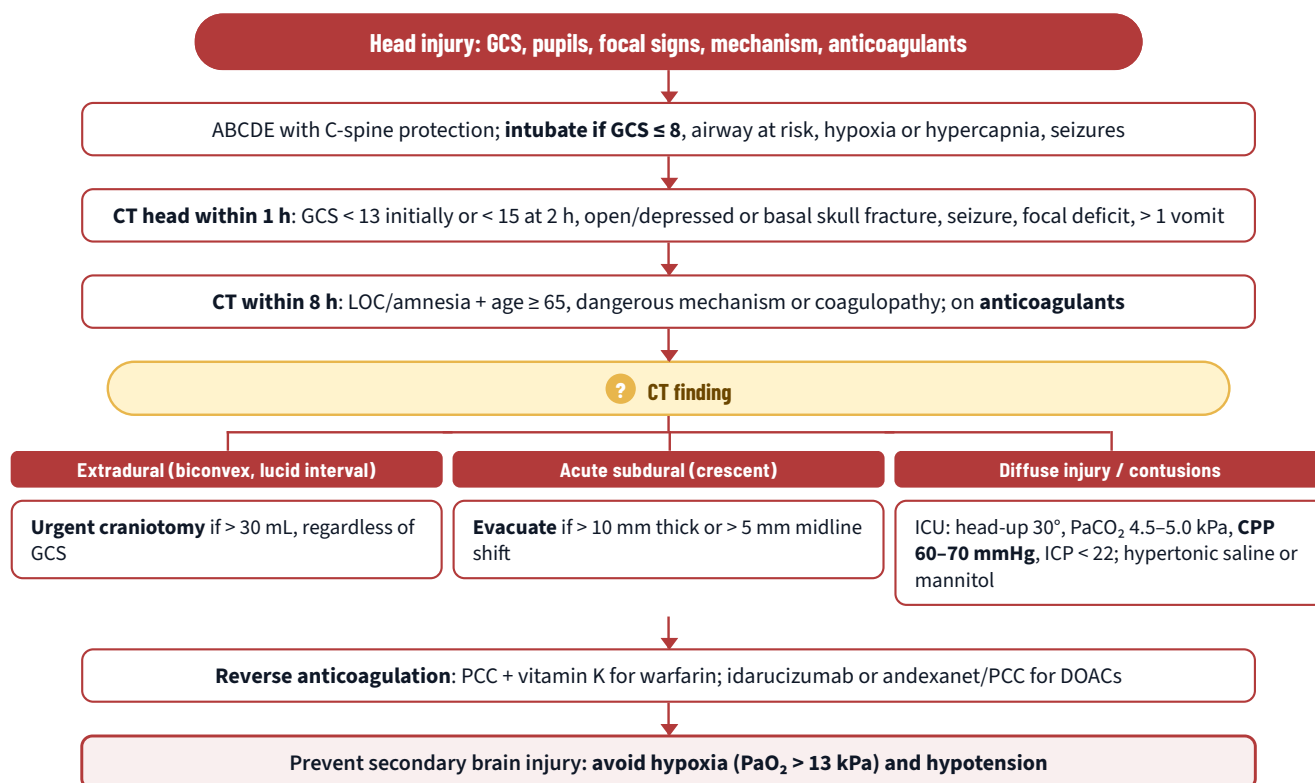
Primary survey of the injured patient

ATLS 10TH ED • NICE NG39 MAJOR TRAUMA



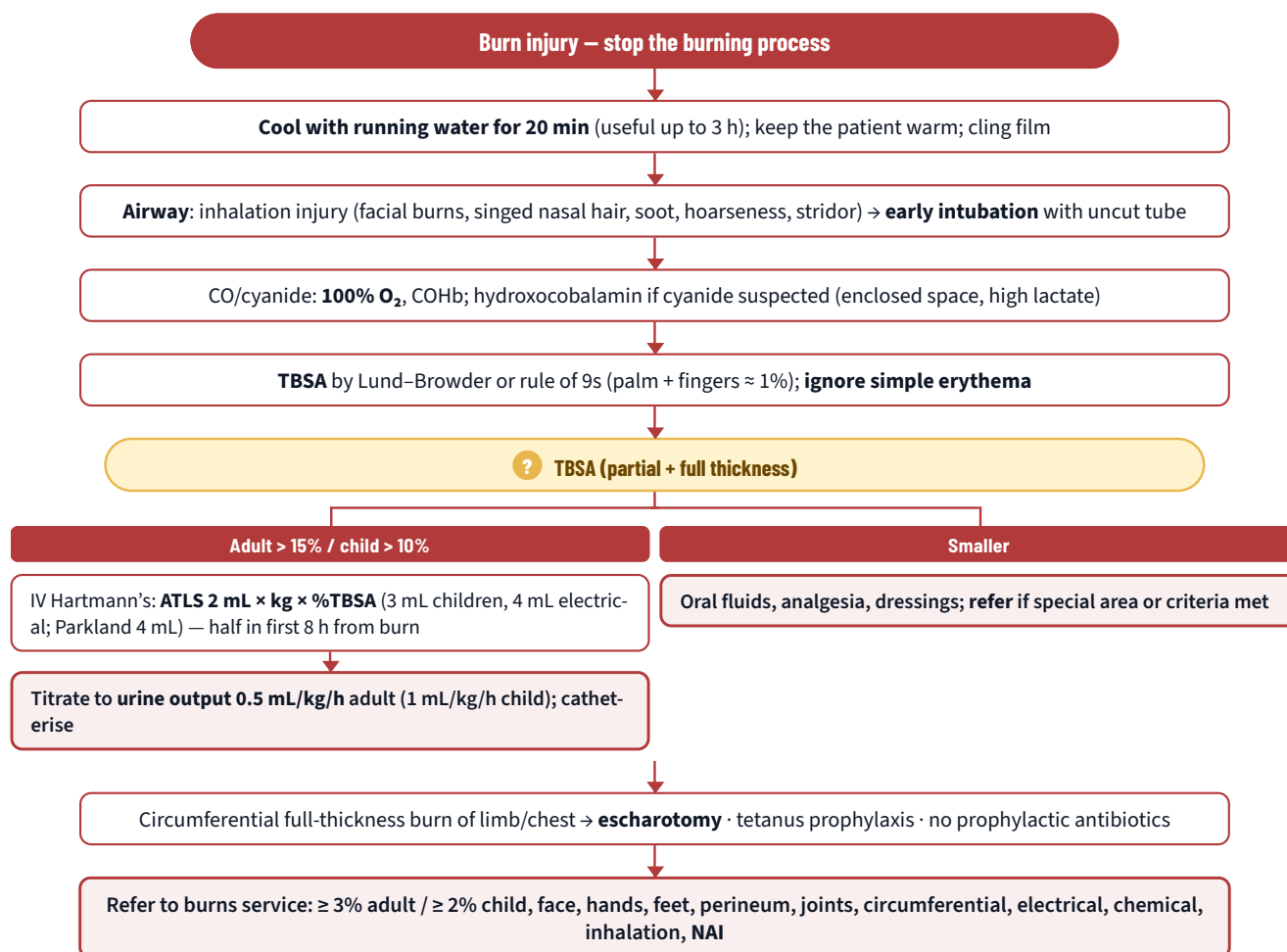
Head injury

NICE NG232 (2023) · BRAIN TRAUMA FOUNDATION 4TH ED



Major burn

ATLS 10TH ED · BBA / NATIONAL BURN CARE REFERRAL GUIDANCE



13 QUESTIONS • 39TH-BATCH BLOCK, INDUCTION & FINAL PAPERS • MINISTERIAL • FORMATIVE & PREVIOUS-YEARS EXAMS – ONE NUMBERING

Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1–4

A 75-year-old man with type 2 diabetes, hypertension and heart failure (EF 40%) was caught in a house fire when his clothes ignited. He has extensive white, leathery burns of the arms and legs with red blistered margins; pulse 110/min, BP 90/60 mmHg.

- 1 What is the best initial management?
 - A Apply silver sulfadiazine cream
 - B Immediate grafting in theatre
 - C Escharotomy only
 - D Airway assessment and IV fluid resuscitation
 - E Oral fluids and analgesia
- 2 What is the depth of these burns?
 - A Superficial (first degree)
 - B Superficial partial thickness
 - C Deep partial thickness
 - D Full thickness (third degree)
 - E Fourth degree
- 3 What is the most common complication in the first 24 hours?
 - A Hypovolaemic (burn) shock
 - B Sepsis
 - C Contracture
 - D Contour deformity
 - E ARDS
- 4 Given his diabetes and heart failure, what is the key consideration in fluid resuscitation?
 - A Use the standard formula without change
 - B Avoid fluids; use vasopressors only
 - C Start Parkland cautiously and titrate to urine output
 - D Replace fluids only after 24 hours
 - E Give colloids only

CASE • QUESTIONS 5–8

A 45-year-old man is brought after a car crash. The right chest moves paradoxically with bruising, reduced expansion and diminished breath sounds. He is conscious but distressed; BP 95/60 mmHg, SpO₂ 86%.

- 5 What is the most likely diagnosis?
 - A Tension pneumothorax
 - B Flail chest
 - C Massive haemothorax
 - D Pulmonary contusion
 - E Lung collapse

- 6 Which mechanism mainly compromises his lungs?
 - A Hyperventilation from pain
 - B Underlying pulmonary contusion
 - C Mediastinal shift
 - D Air trapping on expiration
 - E Diaphragmatic rupture

- 7 Chest X-ray and CT show a massive haemothorax. What is the next immediate step?
 - A Urgent thoracotomy
 - B Intubation and positive-pressure ventilation
 - C Chest drain insertion
 - D Needle decompression
 - E Pericardiocentesis

- 8 He improves but remains hypoxic with paradoxical movement. What is the next definitive step?
 - A Analgesia and oxygen only
 - B Intubation and mechanical ventilation
 - C Surgical rib fixation
 - D Repeat chest X-ray after 24 hours
 - E Chest strapping

- 9 An elderly patient has headache 3 weeks after a minor head injury; CT shows a crescent-shaped collection. What is the diagnosis?
 - A Acute subdural haematoma
 - B Chronic subdural haematoma
 - C Idiopathic intracranial hypertension
 - D Subarachnoid haemorrhage
 - E Cerebral venous sinus thrombosis
 - F Intracerebral haemorrhage
 - G Temporal arteritis
 - H Brain tumour

- 10 A man cuts his forearm on glass; the wound is not deep and its edges are everted. Which closure and immediate drug are most appropriate?
 - A Simple interrupted non-absorbable monofilament (e.g. 4/0 nylon) sutures and tetanus prophylaxis
 - B Absorbable braided sutures and IV antibiotics
 - C Leave open and give oral steroids
 - D Staples and antiplatelet
 - E Skin glue and anticoagulant

- 11 A man cuts his forearm on glass; the wound is not deep and its edges are everted. If the wound is deep, how should it be closed?
 - A Layered closure: absorbable subcutaneous sutures, then skin sutures
 - B Skin sutures only
 - C Leave open to heal by secondary intention
 - D Tight single-layer closure
 - E Skin glue

MINISTERIAL EXAM

12 A 30-year-old driver sustained an abdominal trauma after a road traffic accident. He was transferred to the emergency department in a haemodynamically unstable state. He was resuscitated with an intravenous fluid and a blood transfusion, after which vital signs became normal. An abdominal CT scan revealed a capsular laceration of 6 cm in depth in the liver. Which of the following grades of liver injury does this patient suffer from?

- A Grade1
- B Grade2
- C Grade3
- D Grade4
- E Grade5

13 A 35-year-old man is involved in a motor vehicle collision in which he was thrown against the steering column, sustaining blunt trauma to his left upper abdomen. On arrival at the emergency department he complains of abdominal pain but is haemodynamically stable. CT shows a splenic tear with a small intra-abdominal haematoma; the tear extends through the splenic capsule but not to the hilum. Which one of the following treatment options is least likely to be indicated at this time?

- A Cross-match, group and save.
- B Serial haemoglobin measurement and abdominal examination.
- C Explorative laparotomy.
- D Serial CT scanning.
- E 24-hour monitoring in ICU.

REPEATED 2* · ALSO ASKED IN
SBA

Answers 1-13

LOCAL EXAMS MCQS

- 1** **D Airway assessment and IV fluid resuscitation** — Burns follow ATLS: secure the airway (inhalation risk) and start IV resuscitation; dressings and surgery come later.
- 2** **D Full thickness (third degree)** — White, leathery, painless skin is full-thickness burn; the red blistered margins are partial thickness.
- 3** **A Hypovolaemic (burn) shock** — Capillary leak causes hypovolaemic burn shock in the first 24–48 h; sepsis and contractures come later.
- 4** **C Start Parkland cautiously and titrate to urine output** — Parkland is only a starting point; titrate to urine output 0.5 mL/kg/h to avoid both under-resuscitation and fluid overload in heart failure.
- 5** **B Flail chest** — Paradoxical movement of a chest segment after blunt trauma is flail chest (two or more ribs broken in two or more places).
- 6** **B Underlying pulmonary contusion** — Hypoxia in flail chest is caused mainly by the underlying pulmonary contusion, aggravated by pain-limited ventilation.
- 7** **C Chest drain insertion** — Massive haemothorax is treated with a large-bore chest drain and resuscitation; thoracotomy if > 1500 mL initially or > 200 mL/h.

- 8** **B Intubation and mechanical ventilation** — Persistent hypoxia despite analgesia and oxygen needs intubation and ventilation (internal splinting); rib fixation is increasingly used to wean.
- 9** **B Chronic subdural haematoma** — A crescentic (hypodense) collection weeks after minor trauma in an elderly patient is a chronic subdural haematoma: burr-hole drainage.
- 10** **A Simple interrupted non-absorbable monofilament (e.g. 4/0 nylon) sutures and tetanus prophylaxis** — Superficial lacerations are closed with interrupted non-absorbable monofilament sutures; check tetanus status and give prophylaxis.
- 11** **A Layered closure: absorbable subcutaneous sutures, then skin sutures** — Deep wounds need layered closure with absorbable (e.g. polyglactin) deep sutures to close dead space and reduce tension, haematoma and infection.
- 12** **C Grade3** — **UPDATED** AAST grade III liver injury is a parenchymal laceration >3 cm deep (or subcapsular haematoma >50% of surface); grade II is 1–3 cm deep.
- 13** **C Explorative laparotomy.** — **UPDATED** A haemodynamically stable patient with a non-hilar splenic laceration is managed non-operatively (HDU monitoring, cross-match, serial examination/imaging); laparotomy is for instability or peritonitis.

26 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love MCQs

SET 1 OF 2

- 1 Regarding trauma, which of the following statements is true?
 - A It is the fifth most common cause of death overall.
 - B It is the second most common cause of death and disability below 40 years of age.
 - C One-year mortality following a proximal femoral fracture in patients over 65 years is 30%.
 - D Trauma is predominantly secondary to high-energy injuries.
 - E Less than 25% of children who die of abuse have had previous contact with health or social services.
- 2 Regarding trauma assessment, which of the following statements is false?
 - A The patient, mechanism of injury, and injury are the keystones of management.
 - B It is as essential to identify obvious injuries as it is to identify the occult (hidden) injuries.
 - C High-velocity bullet injuries are associated with cavity formation.
 - D Blunt injuries can be direct (impact) or indirect (force transmission).
 - E Knife injury over a limb is easy to evaluate.
- 3 In relation to blunt injuries, which of the following statements is false?
 - A The mechanisms are direct or indirect.
 - B In indirect injury, associated injuries may be present and should be sought.
 - C In chest injuries, damage to abdominal organs is rare.
 - D Overt injury should lead the clinician to look for covert (hidden) injuries.
 - E Significant internal injury may be present with little external evidence.
- 4 In relation to covert injuries, which of the following statements is true?
 - A A deductive approach is not necessary.
 - B A look-everywhere approach is not helpful.
 - C A focused approach that limits the scope of your assessment is not useful.
 - D Head to pelvis CT is part of a look-everywhere approach.
 - E Screen patients where clinical signs are obvious.
- 5 In relation to covert injuries, which of the following statements is false?
 - A Electrocution = burn (obvious) and posterior dislocation of the shoulder (covert).
 - B Dashboard = patella fracture (obvious) and posterior dislocation of the hip (covert).
 - C Inhalational injury = facial burns (obvious) and laryngeal oedema (covert).
 - D Left-side impact RTA = left pneumothorax with chest wall contusions (obvious) and liver laceration (covert).
 - E Flexion-distraction = head injury (obvious) and cervical spine fracture (covert).
- 6 In relation to NAI in children, which of the following statements is false?
 - A History is inconsistent with the injury sustained.
 - B There is a changing history.
 - C There are likely to be injuries of differing ages or duration.
 - D There may be aggressive behaviour from the carers.
 - E A femoral shaft fracture in a 6-month-old is not suspicious of NAI.
- 7 Which of the following statements is false with regard to a polytrauma patient?
 - A A drop in body temperature occurs.
 - B There is a generalised immune response.
 - C The patient has compensatory mechanisms to blood loss.
 - D The early management is protocol-driven.
 - E Diversion from protocol should be avoided.
- 8 Regarding mechanisms of trauma, which of the following statements is false?
 - A The most common cause of blunt trauma is a motor vehicle accident (MVA).
 - B A 10% increase in impact speed equates to a ~40% increase in pedestrian fatality.
 - C Seat belts reduce the risk of death or serious injury for front-seat occupants by ~20%.
 - D For head-on collisions, airbags reduce the risk of fatality by ~30%.
 - E Motorcyclists have a higher rate of mortality than car occupants involved in a MVA.
- 9 Regarding airway assessment, which of the following statements is false?
 - A If the patient is talking, he or she does not have an imminently compromised airway.
 - B Suction of the airway may be required.
 - C Nasopharyngeal and Guedal airways are useful adjuncts.
 - D GCS of 7 is not an indication for a definitive airway.
 - E Cervical spine control is required throughout airway assessment.

- 10** Regarding assessing disability, which of the following can be responsible for a reduced consciousness level?
- A Head injury
 - B Hypovolaemia
 - C Hypoglycaemia
 - D Alcohol
 - E All of the above
- 11** Which of the following is not a routine adjunct to the primary survey?
- A Pulse oximetry
 - B Urinary catheter
 - C Bloods including FBC, U&Es, clotting screen and glucose
 - D Radiographs of the C-spine, chest and pelvis
 - E CT with or without angiography
- 12** Regarding the secondary survey, which of the following is not part of the AMPLE assessment?
- A Allergies
 - B Medication
 - C Pain assessment
 - D Last meal
 - E Events of the incident
- 13** Regarding paediatric trauma, which of the following combination of observations is optimal for a preschool child (<5 years)?
- A Pulse 160 bpm, SBP 90, RR 40, UO 1.5 mL/kg/hour
 - B Pulse 160 bpm, SBP 90, RR 40, UO 1 mL/kg/hour
 - C Pulse 140 bpm, SBP 80, RR 30, UO 1 mL/kg/hour
 - D Pulse 140 bpm, SBP 90, RR 40, UO 1.5 mL/kg/hour
 - E Pulse 140 bpm, SBP 90, RR 30, UO 1.5 mL/kg/hour
- 14** Regarding the hazards of imaging and radiation, which of the following statements are true?
- A Low-osmolality contrast media (LOCM) are safer than their higher-osmolality counterparts.
 - B Routine steroid prophylaxis is recommended before use of contrast in the high-risk patient.
 - C In the diabetic, metformin should be stopped before using contrast.
 - D The majority of ionising radiation comes from medical exposure from investigations.
 - E Radionuclide imaging allows function to be studied.
- 15** Regarding diagnostic imaging, which of the following statements is false?
- A Conventional x-rays will delineate different soft tissue reliably.
 - B Conventional x-rays can be manipulated.
 - C Dedicated transducers can help in endocavitary ultrasound (US).
 - D Change in the frequency of an US wave can be caused by red blood cells.
 - E The higher the frequency of the US wave, the greater the resolution of the image.
- 16** Regarding imaging modalities, which of the following statements are true?
- A US has no disadvantages.
 - B Computed tomography (CT) scan has a higher resolution than plain radiographs.
 - C Magnetic resonance imaging (MRI) scans give excellent contrast resolution.
 - D MRI scan has no disadvantages.
 - E Radionuclide imaging allows function to be studied.
- 17** The following statements are true except:
- A The clinician referring the patient must balance risk against benefit.
 - B The clinician must give as much information as possible.
 - C Use portable machines and fluoroscopy wherever possible.
 - D Take all precautions when using an image intensifier.
 - E The lungs, heart and pancreas are especially vulnerable to irradiation.
- 18** Regarding orthopaedic imaging, which of the following statements is false?
- A Synovitis can be detected by plain x-ray.
 - B MR arthrography is the ideal imaging for articular cartilage damage.
 - C MRI is the ideal method for local staging of a malignant bone tumour.
 - D X-ray is the first investigation in destructive bone lesions.
 - E US is used to examine mass lesions of soft tissues.
- 19** Regarding imaging in trauma, which of the following statements is false?
- A In a multiply injured patient, CT of head and spine should be the first line of imaging.
 - B Focused assessment with sonography for trauma (FAST) helps in detecting intraperitoneal fluid and cardiac tamponade.
 - C CT should not be used when a patient is unstable.
 - D US is useful for diagnosing occult pneumothorax.
 - E CT is the main imaging method for intracranial, intra-abdominal and vertebral injuries.
- 20** Which of the following statements are true about skin incisions?
- A Skin incisions are best made along tension lines.
 - B Incisions should avoid bony prominences and skin creases.
 - C The 22-blade scalpel is ideal for all incisions.
 - D Incisions should be made bearing in mind cosmesis.
 - E Transverse abdominal incisions have fewer respiratory complications.
- 21** Regarding wound closure, all of the following statements are true except:
- A Mass closure technique is the usual method used for abdominal wall.
 - B Sutures lose 50% of their tensile strength at the knot.
 - C Non-absorbable suture material should be used in the biliary and urinary tract.
 - D Synthetic suture materials elicit minimal tissue reaction.
 - E Round-bodied needles are used in intestinal surgery.

- 22** Regarding the principles of anastomosis, which of the following statements are true?
- A** The current favoured technique of bowel anastomosis is the one-layer extramucosal interrupted method.
 - B** The bowel must be prepared prior to any anastomosis.
 - C** It is essential that in end-to-end anastomosis the bowel ends should match the diameter.
 - D** In vascular anastomosis, polypropylene is the suture material to be used.
 - E** A transverse arteriotomy should always be used to prevent stenosis after closure.
- 23** Regarding surgical drains, which of the following statements are false?
- A** A drain is best inserted through the end of a wound so as not to have another scar.
 - B** 'When in doubt drain' should be the motto.
 - C** Drains are mandatory after a thoracotomy.
 - D** Drains are usually used following exploration of the common bile duct.
 - E** During removal of a chest drain, the patient should be asked to do a Valsalva manoeuvre.
- 24** Regarding the principles of diathermy, which of the following statements are true?
- A** The two types of diathermy in use are monopolar and bipolar.
 - B** Monopolar diathermy is ideal for microsurgery and delicate procedures.
 - C** Bipolar diathermy is ideal for patients who have pacemakers.
 - D** Accidental burns are more common when monopolar diathermy is used.
 - E** Capacitance coupling can occur when a metal port is used in laparoscopic surgery.
- 25** Which of the following statements about filariasis are true?
- A** It is mainly caused by the parasites *Brugia malayi* and *Brugia timori*; in 10% of cases, *Wuchereria bancrofti* is the cause.
 - B** Infective larvae inoculated by a mosquito bite migrate to the lymphatics, where they mature into adult worms.
 - C** The adult worms cause lymphatic blockage leading to massive limb oedema; secondary bacterial infections may worsen lymphatic fibrosis and lead to elephantiasis.
 - D** Chyluria and chylous ascites seen in cancers blocking lymphatics are never seen in filariasis.
 - E** A mild form affecting the respiratory tract, presenting with dry cough, is called tropical pulmonary eosinophilia.

Answers 1-25

- 1** **C** One-year mortality following a proximal femoral fracture in patients over 65 years is 30%. — **UPDATED** About 30% of over-65s die within a year of a proximal femoral fracture; trauma is the leading (not second) cause of death under 40 and most injuries are low-energy.
- 2** **E** Knife injury over a limb is easy to evaluate. — Knife injuries may be deceptively complex despite an unassuming appearance.
- 3** **C** In chest injuries, damage to abdominal organs is rare. — **UPDATED** Blunt injury to the lower chest commonly damages the liver and spleen lying under the lower ribs, so abdominal injury must always be sought.
- 4** **D** Head to pelvis CT is part of a look-everywhere approach. — A comprehensive CT from head to pelvis aids in detecting hidden injuries.
- 5** **D** Left-side impact RTA = left pneumothorax with chest wall contusions (obvious) and liver laceration (covert). — Left-side impact typically causes splenic rather than liver injury.
- 6** **E** A femoral shaft fracture in a 6-month-old is not suspicious of NAI. — Femoral shaft fractures in non-ambulatory infants are highly suspicious for non-accidental injury.
- 7** **E** Diversion from protocol should be avoided. — **UPDATED** Early polytrauma care is protocol-driven, but the experienced clinician may need to deviate from the protocol when the clinical situation demands it.
- 8** **C** Seat belts reduce the risk of death or serious injury for front-seat occupants by ~20%. — **UPDATED** Seat belts reduce death or serious injury in front-seat occupants by about 45–50%, not 20%; airbags reduce fatality in head-on collisions by about 30%.
- 9** **D** GCS of 7 is not an indication for a definitive airway. — **UPDATED** A GCS of 8 or less mandates a definitive airway, so a GCS of 7 is an indication; a patient who is talking has a patent airway for now.
- 10** **E** All of the above — Head injury, hypovolaemia, hypoglycaemia, and alcohol can all lower consciousness.
- 11** **E** CT with or without angiography — **UPDATED** CT ($\pm$ angiography) follows in the stable patient after the primary survey; pulse oximetry, catheter, bloods and chest/pelvic films are primary-survey adjuncts.
- 12** **C** Pain assessment — Pain assessment is important clinically but not part of the AMPLE mnemonic.
- 13** **E** Pulse 140 bpm, SBP 90, RR 30, UO 1.5 mL/kg/hour — **UPDATED** Bailey & Love quotes normal values for a preschool child of pulse 140/min, systolic BP 90 mmHg, respiratory rate 30/min and urine output 1.5 mL/kg/h.
- 14** **AE** True: A, E — **UPDATED** B false: routine steroid prophylaxis is not recommended; C false: metformin need not be stopped if eGFR >30; D false: most exposure is natural background radiation.
- 15** **A** Conventional x-rays will delineate different soft tissue reliably. — **UPDATED** Plain radiographs do not reliably differentiate soft tissues.
- 16** **BCE** True: B, C, E — **UPDATED** A and D are false: US is operator-dependent and blocked by bone and gas, and MRI is slow,

costly and contraindicated with some implants; radionuclide imaging shows function.

- 17 **CE** True: C, E — **UPDATED** Portable films and fluoroscopy give higher doses and should be minimised; the most radiosensitive tissues are the gonads, lens, thyroid, breast and marrow.
- 18 **A** Synovitis can be detected by plain x-ray. — **UPDATED** Plain radiographs cannot reliably detect synovitis; advanced imaging is needed.
- 19 **A** In a multiply injured patient, CT of head and spine should be the first line of imaging. — **UPDATED** Initial management relies on plain films and FAST; CT is reserved for stable patients.
- 20 **ADE** True: A, D, E — **UPDATED** B is false: incisions should avoid bony prominences but are best placed in skin creases; C is false: no single blade suits all incisions (the 10 and 15 are standard).
- 21 **C** Non-absorbable suture material should be used in the biliary and urinary tract. — **UPDATED** Non-absorbable sutures in

the biliary or urinary tract act as a nidus for stone formation, so absorbable sutures are used; the other statements are true.

- 22 **AD** True: A, D — **UPDATED** B false: routine bowel preparation is not required; C false: size mismatch is overcome by a Cheatle slit or side-to-side join; E false: a longitudinal arteriotomy with patch is often used.
- 23 **AB** True: A, B — **UPDATED** Drains should exit via a separate incision and their use is now selective; post-thoracotomy drainage and Valsalva on removal are standard.
- 24 **ACDE** True: A, C, D, E — **UPDATED** B is false: bipolar (not monopolar) suits microsurgery and delicate work; bipolar is safer with pacemakers, monopolar causes more burns and capacitance coupling can occur.
- 25 **BCE** True: B, C, E — **UPDATED** A false: *Wuchereria bancrofti* causes ~90% of cases and *Brugia* ~10%; D false: filariasis is a classic cause of chyluria and chylous ascites.

SET 2 OF 2

- 26 Which of the following statements about filariasis are true?
- A Blood tests often reveal an elevated lymphocyte count.
 - B Immature worms can be seen in nocturnal peripheral blood smears.
 - C Medical treatment with diethylcarbamazine is effective even in cases of advanced elephantiasis.
 - D Hydroceles are treated in the usual way, although excess skin may need surgical trimming.
 - E Elephantiasis can be treated easily with operations to reduce limb size.

Answer 26

BAILEY & LOVE MCQs

- 26 **BD** True: B, D — **UPDATED** A false: blood shows eosinophilia; C false: diethylcarbamazine kills microfilariae but cannot re-

verse established elephantiasis; E false: limb-reduction surgery is difficult.

6 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 A 45-year-old man in an automobile crash is hypotensive (BP 80/40), tachycardic, with a distended abdomen, and shows no response to 2 L of saline. Which of the following is the most appropriate next step?
 - A Abdominal CT scan
 - B Insertion of a Swan–Ganz catheter
 - C Exploratory laparotomy
 - D Focused assessment with sonography for trauma (FAST)
 - E Diagnostic peritoneal lavage
- 2 A 23-year-old man presents to the emergency department after sustaining a single gunshot wound to the right chest. He is awake and alert and complaining of significant abdominal pain and chest pain. He has a pulse of 110, blood pressure (BP) of 90/60, respirations of 24. On examination, his breath sounds are decreased on the right. His chest is dull to percussion on the right. His abdomen is slightly distended and diffusely tender. His pulse oximeter shows 88% on 6 L oxygen by face mask. A right chest drain is inserted and he responds to resuscitation. Which of the following is the most appropriate next step?
 - A CT scan of the abdomen
 - B FAST examination of the abdomen
 - C exploratory laparotomy
 - D local wound exploration
 - E diagnostic peritoneal lavage
- 3 A 17-year-old teenager who suffers a direct blow to the upper abdomen and develops progressive bilious emesis 48 hours later is most likely to have?
 - A pancreatitis
 - B gastritis
 - C acute cholecystitis
 - D duodenal hematoma
 - E small-bowel perforation
- 4 Which is the treatment of choice for a post-traumatic duodenal haematoma causing obstruction?
 - A immediate exploration
 - B nasogastric decompression and parenteral alimentation
 - C retrocolic gastrojejunostomy
 - D antacids and H₂ blockers
 - E cholecystectomy
- 5 A 26-year-old man with a stab wound to the fifth left intercostal space has tachycardia, hypotension, midline trachea, distant heart sounds, and equal breath sounds. What is the most likely diagnosis?
 - A transected descending aorta
 - B cardiac tamponade
 - C massive left hemothorax
 - D tension pneumothorax
 - E phrenic nerve paralysis
- 6 A 26-year-old man with a stab wound to the left fifth intercostal space is tachycardic and hypotensive with distant heart sounds, a midline trachea and equal breath sounds. Which is the most appropriate next step in management?
 - A insertion of a left chest tube
 - B emergency department thoracotomy
 - C pericardiocentesis
 - D rapid infusion of an IV fluid bolus
 - E intubation and assisted ventilation

Answers 1–6

LANGE MCQS

- 1 **C Exploratory laparotomy** — **UPDATED** A hypotensive non-responder with a distended abdomen after blunt trauma has intra-abdominal haemorrhage and needs immediate laparotomy; CT is only for stable patients.
- 2 **C exploratory laparotomy** — **UPDATED** A gunshot wound to the lower chest with abdominal distension and tenderness implies transdiaphragmatic injury and mandates laparotomy once the haemothorax is drained.
- 3 **D duodenal hematoma** — **UPDATED** Delayed bilious vomiting after trauma is characteristic of a duodenal hematoma.
- 4 **B nasogastric decompression and parenteral alimentation** — **UPDATED** Most duodenal haematomas resolve within 2–3 weeks with NG decompression and parenteral nutrition; surgery only if obstruction persists or perforation is suspected.
- 5 **B cardiac tamponade** — **UPDATED** A precordial stab wound with hypotension, distant heart sounds, midline trachea and equal breath sounds is tamponade; tension pneumothorax or haemothorax alters breath sounds.
- 6 **C pericardiocentesis** — **UPDATED** Pericardiocentesis relieves tamponade as a temporising measure; definitive repair is by sternotomy or thoracotomy in theatre. ED thoracotomy is for traumatic arrest.

7 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 Which one of the following statements regarding diagnostic peritoneal lavage is not true?
 - A A positive test would follow injury to spleen, liver, pancreas or intestine
 - B It is more sensitive than computed tomography and focused assessment with sonography for trauma (FAST) scanning
 - C It is the technique of choice when attempting to confirm the hollow viscus injury
 - D Urinary catheterization and nasogastric tube insertion is required prior to diagnostic peritoneal lavage
 - E Diagnostic peritoneal lavage is contraindicated in the presence of an indication for explorative laparotomy
- 2 Which of the following is not an independent indication for laparotomy following trauma?
 - A Evisceration of healthy bowel
 - B Evisceration of omentum
 - C Stab wound to anterior abdomen
 - D Gunshot to abdomen
 - E Blunt abdominal trauma with free intraperitoneal air on erect chest radiograph
- 3 The following factors detract from the diagnostic sensitivity of focused assessment with sonography for trauma (FAST) scanning, except:
 - A Surgical emphysema
 - B A patient with high body mass index
 - C Operator inexperience
 - D Previous surgery
 - E Large-volume intraperitoneal blood loss
- 4 As well as measuring oxygen saturation, a pulse oximeter also gives useful information regarding what other factor in the initial assessment of the traumatised patient?
 - A Blood pressure
 - B Partial pressure of oxygen
 - C Partial pressure of carbon dioxide
 - D Peripheral perfusion
 - E Acid–base balance
- 5 A 42-year-old construction worker is admitted following a crush injury. The patient is in great distress, complaining of chest pain and working hard to breathe, with paradoxical movement of the chest wall. Arterial blood gases show pO_2 7.5 kPa and pCO_2 8.2 kPa. A chest radiograph shows multiple rib fractures. The life-saving intervention is
 - A High-flow oxygen
 - B Cricothyroidotomy
 - C Endotracheal tube insertion
 - D Aggressive fluid resuscitation
 - E Adequate analgesia to allow effective respiration
- 6 A male patient is admitted following a fall from height. On arrival his Glasgow Coma Scale score is 5/15 and he is therefore intubated. During primary resuscitation a chest film is taken which shows a widened mediastinum and right-sided deviation of the trachea. The diagnosis is
 - A Tension pneumothorax
 - B Ruptured oesophagus
 - C Cardiac tamponade
 - D Right lobe collapse
 - E Aortic rupture
- 7 A 20-year-old woman was resuscitated in the emergency department and required the insertion of a chest drain. The drain was removed 2 days later before she was discharged. She re-presents 10 days later complaining of chest pain associated with high fever and sweats. An empyema is suspected and a chest radiograph confirms a collection. The most appropriate next course of action is
 - A Intravenous antibiotics for 6 weeks
 - B Needle tap and aspiration
 - C Chest drain reinsertion
 - D Computed tomography scan of the thorax
 - E Ultrasound scan

Answers 1–7

SBA MCQs

- 1 **A A positive test would follow injury to spleen, liver, pancreas or intestine** — **UPDATED** DPL misses retroperitoneal injuries (pancreas, duodenum, kidney), so a pancreatic injury may give a negative lavage; it is very sensitive for intraperitoneal blood.
- 2 **C Stab wound to anterior abdomen** — **UPDATED** Not every stab wound penetrates the peritoneum, so exploratory laparotomy is not automatically indicated.
- 3 **E Large-volume intraperitoneal blood loss** — **UPDATED** A large volume of free fluid actually enhances the sensitivity of FAST scanning.
- 4 **D Peripheral perfusion** — **UPDATED** The plethysmographic trace and a reliable signal depend on pulsatile flow, so a poor trace indicates poor peripheral perfusion; oximetry gives no CO_2 or acid–base information.
- 5 **C Endotracheal tube insertion** — **UPDATED** Flail chest with type 2 respiratory failure (hypoxia and hypercapnia) needs in-

tubation and positive-pressure ventilation; oxygen and analgesia alone are insufficient.

- 6** **E Aortic rupture** — A widened mediastinum with tracheal deviation is highly suggestive of an aortic injury.

25 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1** A teenage boy falls from his bicycle and is run over by a truck. On arrival in the emergency room (ER), he is awake and alert and appears frightened but in no distress. The chest radiograph suggests an air-fluid level in the left lower lung field and the nasogastric tube seems to coil upward into the left chest. Which of the following is the next best step in his management?
A Placement of a left chest tube
B Thoracotomy
C Laparotomy
D Esophagogastrosocopy
E Diagnostic peritoneal lavage
- 2** A 10-year-old boy was the backseat belted passenger in a high-speed motor vehicle collision. On presentation to the ER, he is awake and alert. His heart rate is 110 beats/minute. He is complaining of abdominal pain and has an ecchymosis on his anterior abdominal wall where the seatbelt was located. A CT scan of the abdomen and pelvis demonstrates free fluid in the absence of solid organ injury. Which of the following is the next best step in his management?
A Serial amylase levels
B Repeat CT scan in 24 hours
C Upright chest x-ray
D Angiography of mesenteric vessels
E Diagnostic laparoscopy
- 3** A 36-year-old man who was hit by a car presents to the ER complaining of abdominal pain. His vital signs are HR 92, BP 127/65, and RR 16. On examination, he has tenderness to palpation in the right upper quadrant. An ultrasound examination is performed and reveals fluid between the liver and the kidney in Morison's pouch. Which of the following is the most appropriate next step in his management?
A CT scan of the abdomen/pelvis
B Intravenous pyelogram
C Angiography
D Diagnostic laparoscopy
E Exploratory laparotomy
- 4** A 25-year-old woman arrives in the ER following an automobile accident. Her heart rate is 125 beats/minute, blood pressure is 140/90, respirations are 40 breaths/minute, pulse oximetry is 94% on 4 L/min via nasal cannula, and temperature is 99°F. Breath sounds are markedly diminished on the right side. Which of the following is the best first step in the management of this patient?
A Take a chest x-ray
B Draw arterial blood for blood-gas determination
C Decompress the right pleural space
D Perform pericardiocentesis
E Administer intravenous fluids
- 5** A 17-year-old adolescent boy is stabbed in the left seventh intercostal space, midaxillary line. He presents to the ER with a heart rate of 86 beats/minute, blood pressure of 125/74 mm Hg, and oxygen saturation of 98%. Breath sounds are equal bilaterally. Which of the following is the most appropriate next step in his workup?
A Local exploration of the wound
B Left tube thoracostomy
C Diagnostic peritoneal lavage
D Diagnostic laparoscopy
E Pericardiocentesis
- 6** An 18-year-old man sustains a right-sided neck laceration after an assault. Which of the following is a relative, rather than an absolute, indication for neck exploration?
A Expanding hematoma
B Dysphagia
C Dysphonia
D Pneumothorax
E Hemoptysis
- 7** **E Ultrasound scan** — **UPDATED** Ultrasound characterises the collection (fluid, loculation, septation) and guides aspiration and chest-drain placement; blind drain insertion is avoided.

- 7** Following blunt abdominal trauma, a 12-year-old girl develops upper abdominal pain, nausea, and vomiting. An upper gastrointestinal series reveals a total obstruction of the duodenum with a coiled spring appearance in the second and third portions. In the absence of other suspected injuries, which of the following is the most appropriate management of this patient?
- A Observation
 - B Esophagogastroduodenoscopy
 - C Diagnostic peritoneal lavage
 - D Diagnostic laparoscopy
 - E Exploratory laparotomy
- 8** A 45-year-old man presents after a high-speed motor vehicle collision. He has weakness of his left arm and leg. His head CT is negative for intracranial hemorrhage and a cervical spine CT is negative for fracture. Face CT shows a mandibular fracture. What is the most appropriate next test in his evaluation?
- A Brain MRI
 - B Neck CT angiography
 - C Cerebral angiography
 - D Carotid duplex
 - E CT myelogram
- 9** An 18-year-old man was assaulted and sustained significant head and facial trauma. Which of the following is the most common initial manifestation of increased intracranial pressure?
- A Change in level of consciousness
 - B Ipsilateral pupillary dilation
 - C Contralateral pupillary dilation
 - D Hemiparesis
 - E Hypertension
- 10** A 28-year-old man is brought to the ER for a severe head injury after a fall. He was intubated in the field for his decreased level of consciousness. He is tachycardic and hypotensive. On examination, he is noted to have an obvious skull fracture and his right pupil is dilated. Which of the following is the most appropriate method for initially reducing his intracranial pressure?
- A Elevation of the head of the bed
 - B Saline-furosemide infusion
 - C Mannitol infusion
 - D Intravenous dexamethasone
 - E Hyperventilation
- 11** A 31-year-old man is brought to the ER following an automobile accident in which his chest struck the steering wheel. His vital signs are heart rate of 78 beats/minute, blood pressure of 129/65, respirations of 14 breaths/minute, and oxygen saturation of 98% on room air. Examination reveals no evidence of respiratory distress, but the patient exhibits multiple right-sided palpable rib fractures. A right-sided chest wall segment appears to move inward rather than outward during inspiration and outward rather than inward during expiration. Chest x-ray shows no evidence of pneumothorax or hemothorax. Which of the following is the most appropriate initial management of this patient?
- A Pain control
 - B Placement of a chest binder
 - C Placement of right chest tube
 - D Intubation
 - E Operative rib fixation
- 12** A 23-year-old, previously healthy man presents to the ER after sustaining a single gunshot wound to the right chest. The entrance wound is 3 cm inferior to the nipple and the exit wound is just below the scapula. A chest tube is placed that drains 400 mL of blood and continues to drain 50 to 75 mL/h during the initial resuscitation. His blood pressure is 100/70 mm Hg after 1 U packed red blood cells and 1 U fresh frozen plasma. Abdominal examination is unremarkable. Chest x-ray reveals a reexpanded lung and no free air under the diaphragm. Which of the following is the best next step in his management?
- A Admission and observation
 - B Peritoneal lavage
 - C Exploratory thoracotomy
 - D Exploratory laparotomy
 - E Local wound exploration
- 13** A patient is brought to the ER after a motor vehicle accident. He is unconscious and has a deep scalp laceration and one dilated pupil. His heart rate is 120 beats/minute, blood pressure is 80/40 mm Hg, and respiratory rate is 35 breaths/minute. Despite initiation of massive transfusion protocol after 1 L normal saline via two large-bore IVs, the patient's vital signs do not change significantly. Which of the following is the most appropriate next step in the workup of his hypotension?
- A Intracranial pressure monitor placement
 - B Central venous catheter placement
 - C Direct antiglobulin (Coombs) testing
 - D CT scan of the chest, abdomen, and pelvis
 - E Focused assessment with sonography in trauma (FAST)

- 14** A 25-year-old man is involved in a gang shoot-out and sustains multiple abdominal gunshot wounds. At laparotomy, he has a liver laceration requiring packing and multiple small-bowel and colon injuries that are resected. The patient has a persistent coagulopathy despite ongoing transfusions. The anesthesiologist notes that the patient's temperature is 33.5°C (92.3°F). Intra-operative ABG: pH 7.19, P_{CO2} 42, P_{O2} 80, and lactate of 9 mmol/L. In addition to rewarming, which of the following is the most appropriate next step in the management of this patient?
- A Administration of bicarbonate
 - B Hyperventilation
 - C Hemodialysis
 - D Intra-aortic balloon pump placement
 - E Transfusion of blood products
- 15** A 27-year-old man presents to the ER after a high-speed motor vehicle collision with chest pain and marked respiratory distress. On physical examination, he is hypotensive with distended neck veins and absence of breath sounds in the left chest. Which of the following is the proper initial treatment?
- A Intubation
 - B Chest x-ray
 - C Pericardiocentesis
 - D Chest decompression with a needle
 - E Emergent thoracotomy
- 16** A 22-year-old woman who is 21-weeks pregnant presents after a high-speed motor vehicle collision complaining of abdominal pain and right leg pain. She has an obvious deformity of her right femur. Her heart rate is 125 beats/minute and blood pressure is 118/52 mm Hg. FAST exam shows fluid in the pelvis. Which of the following is the best next step in her management?
- A Observation with serial abdominal exams
 - B Diagnostic peritoneal lavage
 - C CT scan of the abdomen and pelvis
 - D MRI of the abdomen and pelvis
 - E Diagnostic laparoscopy
- 17** A 56-year-old woman sustains blunt abdominal trauma from an assault. Her blood pressure is 107/56 mm Hg and her pulse is 92. She complains of abdominal pain. She undergoes CT scanning of the abdomen and pelvis, which demonstrates a splenic injury without evidence of extravasation. Which of the following is the most appropriate next step in the management of this patient?
- A Observation
 - B Angiography
 - C Diagnostic laparoscopy
 - D Splenectomy
 - E Splenorrhaphy
- 18** A 22-year-old man undergoes an exploratory laparotomy after a gunshot wound to the abdomen. The patient has multiple injuries, including a significant liver laceration, multiple small-bowel and colon injuries, and an injury to the infrahepatic vena cava. The patient receives 35 U of packed RBCs, 15 L of crystalloid, 12 U of fresh frozen plasma (FFP), and 12 packs of platelets. The patient's abdomen is packed closed and he is taken to the intensive care unit for further resuscitation. Which of the following warrants a decompressive laparotomy?
- A Increased peak airway pressure
 - B Increased cardiac output
 - C Decreased systemic vascular resistance
 - D Decreased plasma renin and aldosterone
 - E Increased cerebral perfusion pressure
- 19** A 10-year-old girl is the unrestrained backseat passenger in a high-speed motor vehicle collision. She is intubated in the field for unresponsiveness and on presentation to the ER, her heart rate is 160 beats/minute, and her blood pressure is 60/35 mm Hg. She weighs 30 kg. Which of the following is the most appropriate recommendation for her initial fluid resuscitation?
- A 300 cc of normal saline
 - B 600 cc of normal saline
 - C 300 cc of packed RBCs
 - D 600 cc of packed RBCs
 - E 300 cc of fresh frozen plasma
- 20** A 58-year-old man presents to the ER after falling 10 ft from a ladder. Examination reveals stable vital signs, no evidence of respiratory distress, and multiple right-sided rib fractures. Chest x-ray shows a hemothorax on the right side and a right tube thoracostomy is performed in the ER. Approximately 700 mL of blood is immediately drained with placement of the thoracostomy tube. Over the next 4 hours he continues to drain 300 mL/h after the original evacuation. His coagulation profile is normal. Which of the following is the definitive treatment for this patient?
- A Observation
 - B Second tube thoracostomy
 - C Angiography
 - D Video-assisted thoracoscopic surgery (VATS)
 - E Thoracotomy
- 21** Which of the following clinical triads is indicative of cardiac tamponade and includes muffled heart tones, jugular venous distention, and hypotension?
- A Charcot's triad
 - B Virchow's triad
 - C Beck's triad
 - D Cushing's triad
 - E Whipple's triad

- 22** According to Advanced Trauma Life Support (ATLS) guidelines, in the context of a traumatic hemothorax, thoracotomy is indicated if the initial output from chest tube insertion reaches or exceeds what volume?
- A 500 mL
 - B 1000 mL
 - C 1500 mL
 - D 2000 mL
 - E 2500 mL
- 23** What are the anatomical landmarks used to define the three zones of the neck used in trauma assessment?
- A Zone I is below the clavicles, zone II is between the clavicles and cricoid, and zone III is above the cricoid
 - B Zone I is from the base of the skull to the cricoid cartilage, zone II is below the cricoid to the clavicle, zone III is the thoracic inlet
 - C Zone I is inferior to the cricoid cartilage, zone II is between the cricoid cartilage and the angle of the mandible, and zone III is between the angle of the mandible and the skull
 - D Zone I includes the thoracic outlet, zone II is the mid-neck, zone III is the cranial vault
 - E Zone I is from the thoracic inlet to the mandible, zone II is from mandible to skull base, zone III is the brainstem region
- 24** In the context of retroperitoneal hemorrhage, what are the anatomical zones used to categorize the bleeding areas?
- A Zone I includes pelvis, Zone II includes kidneys laterally, Zone III includes upper abdominal structures
 - B Zone I includes kidneys laterally, Zone II includes pelvis, Zone III includes diaphragmatic structures
 - C Zone I includes diaphragm to aortic bifurcation, Zone II includes kidneys laterally to paracolic gutters, Zone III includes the pelvis
 - D Zone I includes the pelvis and iliac arteries, Zone II includes the upper mesentery, Zone III includes lower retroperitoneum
 - E Zone I includes liver and pancreas, Zone II includes kidneys and adrenals, Zone III includes ureters
- 25** What is the most likely diagnosis in a trauma patient presenting with a liver injury, jaundice, and gastrointestinal bleeding?
- A Hepatic artery pseudoaneurysm
 - B Portal vein thrombosis
 - C Hemobilia
 - D Hepatocellular carcinoma
 - E Budd-Chiari syndrome

Answers 1-25

PRETEST MCQs

- 1** **C Laparotomy** — **UPDATED** An NG tube coiling into the left chest after blunt trauma means acute diaphragmatic rupture; repair via laparotomy, which also allows assessment of associated abdominal injuries.
- 2** **E Diagnostic laparoscopy** — **UPDATED** A seatbelt sign with free fluid but no solid-organ injury on CT suggests hollow viscus or mesenteric injury; diagnostic laparoscopy (or laparotomy) is indicated.
- 3** **A CT scan of the abdomen/pelvis** — **UPDATED** A haemodynamically stable patient with a positive FAST should have CT of the abdomen and pelvis to define the injury; laparotomy is for instability or peritonitis.
- 4** **C Decompress the right pleural space** — **UPDATED** Severe respiratory distress with markedly reduced breath sounds on one side after trauma needs immediate pleural decompression (then chest drain) before any imaging.
- 5** **D Diagnostic laparoscopy** — **UPDATED** A stable patient with a stab wound to the left lower chest (thoracoabdominal zone) needs diagnostic laparoscopy to exclude a diaphragmatic injury, which CT may miss.
- 6** **D Pneumothorax** — **UPDATED** Expanding haematoma, dysphagia, dysphonia and haemoptysis signal major vascular or aerodigestive injury and mandate exploration; pneumothorax needs a chest drain and is only relative.
- 7** **A Observation** — **UPDATED** Coiled-spring duodenal obstruction after blunt trauma is an intramural duodenal haematoma; manage non-operatively with NG decompression and parenteral nutrition.
- 8** **B Neck CT angiography** — **UPDATED** Lateralising neurology with a normal head CT and a high-risk mechanism (mandibular fracture) suggests blunt cerebrovascular injury; CT angiography of the neck is the screening test.
- 9** **A Change in level of consciousness** — **UPDATED** A falling level of consciousness (GCS) is the earliest sign of rising ICP; pupillary dilatation, hemiparesis and the Cushing response are late signs.
- 10** **E Hyperventilation** — **UPDATED** In a hypotensive patient mannitol and head-up tilt worsen hypotension, so brief hyperventilation (PaCO_2 ~4–4.5 kPa) is the initial temporising measure.
- 11** **A Pain control** — **UPDATED** Flail chest without respiratory compromise is managed with good analgesia (e.g. epidural or regional block) and physiotherapy; ventilation only if respiratory failure develops.
- 12** **D Exploratory laparotomy** — **UPDATED** A gunshot wound below the nipple line may cross the diaphragm and needs laparotomy; thoracotomy is reserved for massive or persistent chest drain output.
- 13** **E Focused assessment with sonography in trauma (FAST)** — **UPDATED** Hypotension after trauma is haemorrhage until proven otherwise, not head injury; FAST rapidly identifies abdominal bleeding in the unstable patient.
- 14** **E Transfusion of blood products** — **UPDATED** Hypothermia, acidosis and coagulopathy form the lethal triad; correct with rewarming and balanced blood product transfusion (damage-control resuscitation), not bicarbonate.
- 15** **D Chest decompression with a needle** — **UPDATED** Hypotension, distended neck veins and absent unilateral breath sounds are tension pneumothorax: immediate needle decom-

pression (4th/5th intercostal space, mid-axillary line), then chest drain.

- 16 **C CT scan of the abdomen and pelvis** — **UPDATED** A normotensive pregnant patient with a positive FAST needs CT abdomen/pelvis; maternal stabilisation and diagnosis take priority over fetal radiation risk.
- 17 **A Observation** — **UPDATED** A haemodynamically stable patient with splenic injury and no contrast extravasation is managed non-operatively with observation; angioembolisation if there is a blush.
- 18 **A Increased peak airway pressure** — **UPDATED** Abdominal compartment syndrome raises peak airway pressure and lowers cardiac and urine output; with bladder pressure >20–25 mmHg it warrants decompressive laparotomy.
- 19 **B 600 cc of normal saline** — **UPDATED** ATLS: the initial bolus for a hypotensive child is 20 mL/kg warmed isotonic crystalloid (600 mL for 30 kg); give 10 mL/kg packed red cells if there is no response.
- 20 **E Thoracotomy** — **UPDATED** Initial drainage ≥1500 mL or ongoing loss of >200 mL/h for 2–4 hours (here 300 mL/h) indicates thoracotomy to control the bleeding.
- 21 **C Beck's triad** — Beck's triad is a classic presentation of cardiac tamponade, consisting of muffled heart tones, jugular venous distention, and hypotension.
- 22 **C 1500 mL** — Per ATLS, an immediate thoracotomy is warranted if the initial chest tube output is ≥1500 mL, indicating massive hemothorax.
- 23 **C Zone I is inferior to the cricoid cartilage, zone II is between the cricoid cartilage and the angle of the mandible, and zone III is between the angle of the mandible and the skull** — **UPDATED** Zone I: clavicles/sternal notch to cricoid; zone II: cricoid to angle of the mandible; zone III: angle of the mandible to the skull base.
- 24 **C Zone I includes diaphragm to aortic bifurcation, Zone II includes kidneys laterally to paracolic gutters, Zone III includes the pelvis** — The zones of retroperitoneal hemorrhage are: Zone I from diaphragm to aortic bifurcation, Zone II includes kidneys laterally, Zone III includes the pelvis.
- 25 **C Hemobilia** — Hemobilia refers to bleeding into the biliary tree, often seen in trauma involving the liver. It classically presents with the triad of right upper quadrant pain, jaundice, and gastrointestinal bleeding.

2 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 19-year-old man is brought into the emergency department following a gunshot wound to the abdomen. Despite resuscitation he remains haemodynamically unstable. The decision is made to take him to theatre for exploration. Which of the following surgical incisions would be most appropriate?
- A Battle incision
 - B Median thoracotomy
 - C Midline laparotomy
 - D Pfannenstiel incision
 - E Thoraco-abdominal incision
- 2 A 33-year-old man is brought into the emergency department following an assault with a baseball bat. He has bruising over the abdomen and right loin and complains of abdominal pain. On examination, his chest is clear with good air entry but his abdomen is rigid. On arrival, his heart rate is 140/min and blood pressure 90/60 mmHg. He is given O-negative blood in the resuscitation room but his observations remain unchanged. What would be the next step in his management?
- A Diagnostic peritoneal lavage
 - B Laparotomy
 - C Urgent CT scan
 - D Urgent intravenous urogram
 - E Urgent ultrasound scan

Answers 1-2

- 1 **C Midline laparotomy** — **UPDATED** A midline laparotomy is rapid, provides excellent access, and is easily extended for trauma exploration.
- 2 **B Laparotomy** — **UPDATED** Haemodynamic instability with suspected intra-abdominal bleed necessitates urgent surgical exploration.

Metabolic Response to Trauma

The stress response, fluids, electrolytes and nutrition in the surgical patient.

15

MCQS

0

EMQ SCENARIOS

5

SOURCES

2

REPEATED IDEAS

IN THIS CHAPTER

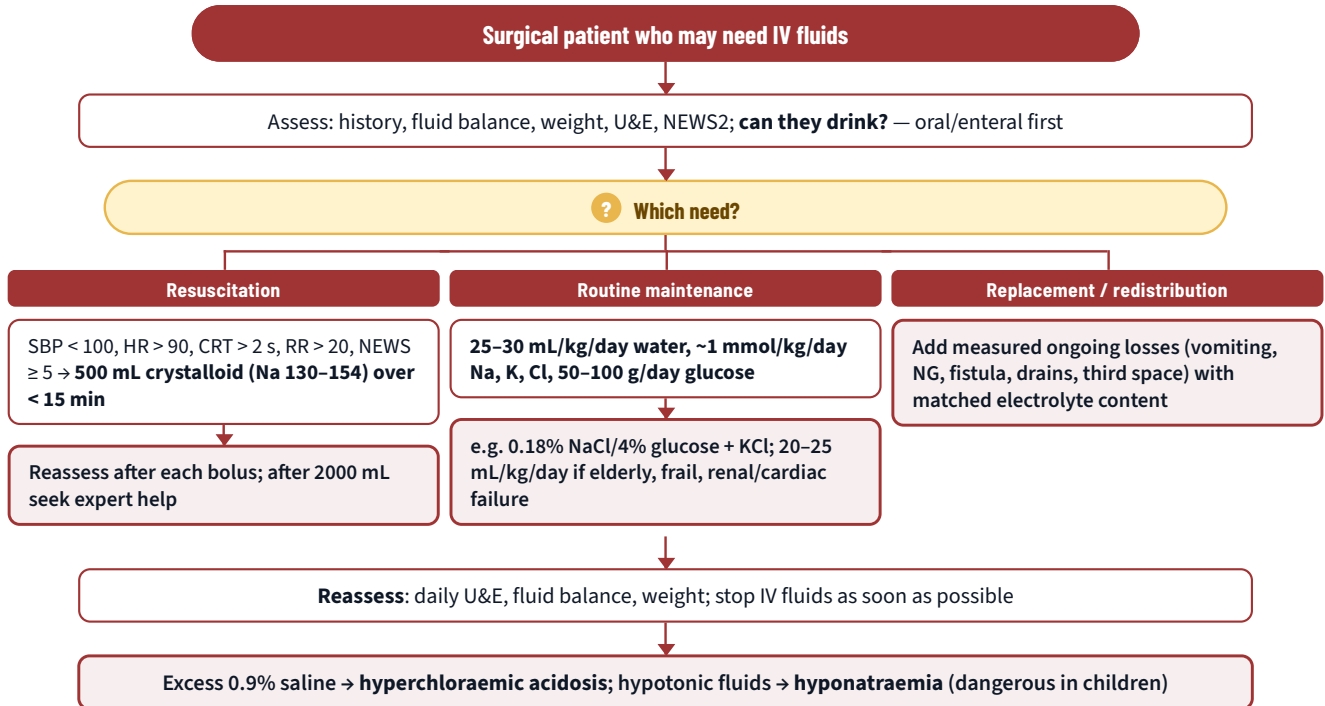
01	Management flowcharts	3 charts	page 477
02	Local Exams MCQs	2 Q	page 479
03	Essay to MCQ MCQs	1 Q	page 479
04	Bailey & Love MCQs	6 Q	page 480
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Step by step

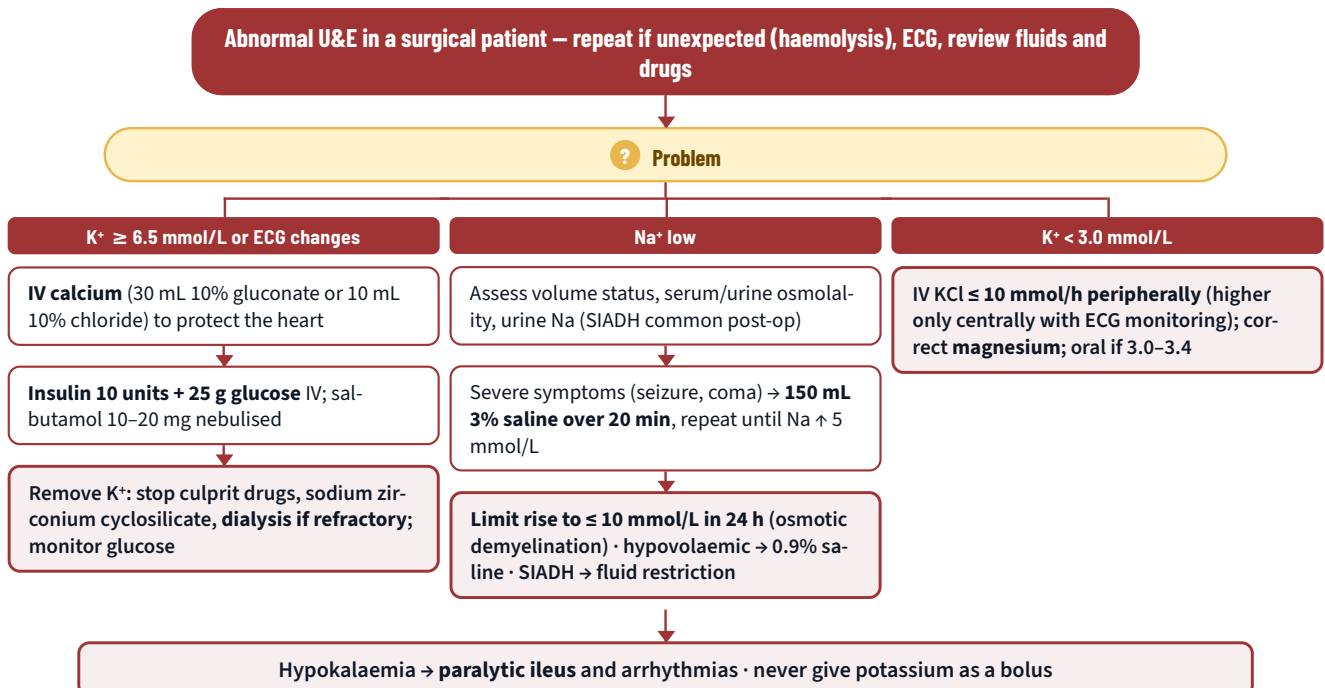
IV fluid prescribing: the 5 Rs

NICE CG174 IV FLUIDS (2017)



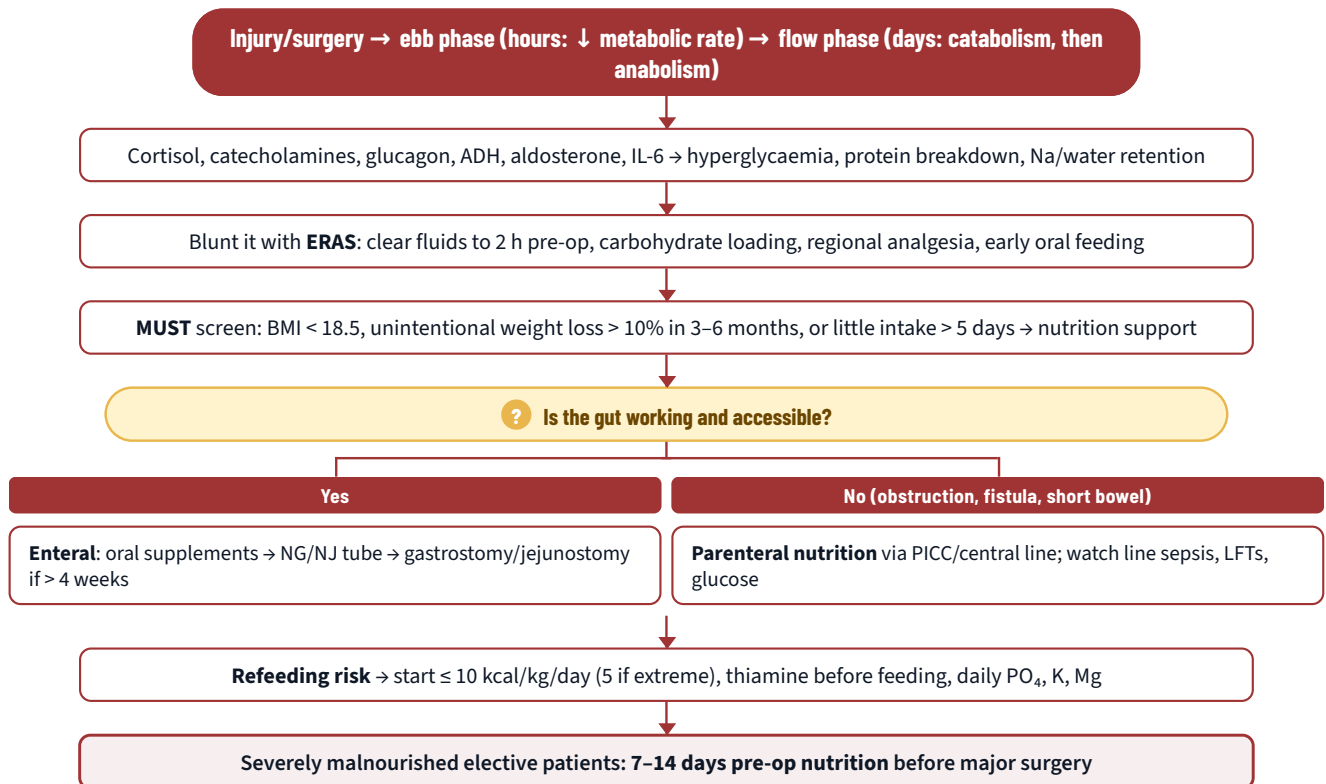
Peri-operative electrolyte emergencies

UKKA 2023 HYPERKALAEMIA • ESE/ERA-EDTA 2014 HYPONATRAEMIA



Metabolic response to surgery and nutritional support

NICE CG32 · ESPEN 2021 CLINICAL NUTRITION IN SURGERY · ERAS SOCIETY



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

- 1 A sedated, intubated patient in the ICU needs a nasogastric tube. What is the most dangerous complication of insertion?

- A Misplacement into the tracheobronchial tree
- B Epistaxis
- C Sinusitis
- D Nasal discomfort
- E Hiccups

REPEATED 2* • ALSO ASKED IN
39th batch Block 2 exam

- 2 A sedated, intubated patient in the ICU needs a nasogastric tube. After NG insertion and feeding he becomes hypotensive and desaturates with secretions around the mouth and tube. What happened?

- A Aspiration of feed into the lungs
- B Anaphylaxis
- C Pneumothorax
- D Septic shock
- E Myocardial infarction

REPEATED 2* • ALSO ASKED IN
39th batch Block 2 exam

Answers 1-2

LOCAL EXAMS MCQS

- 1 **A Misplacement into the tracheobronchial tree** — In a sedated, intubated patient the tube can pass into the airway unnoticed (no cough reflex); feeding into the lung is the danger.

- 2 **A Aspiration of feed into the lungs** — This is aspiration: stop feeds and suction. Prevent it by checking aspirate pH ≤ 5.5 or X-ray before feeding and nursing 30–45° head-up.

1 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

- 1 A 58-year-old woman with type 2 diabetes controlled on oral hypoglycaemic agents is to undergo a major abdominal operation. What is the best way to control her blood glucose perioperatively?

- A Increase the dose of her oral hypoglycaemic agents
- B Continue oral agents on the morning of surgery and check glucose after the operation
- C Use long-acting insulin preoperatively
- D Switch to intermittent subcutaneous sliding-scale insulin
- E Stop oral agents and give an IV infusion of soluble insulin with glucose and potassium (KCl)

Answer 1

ESSAY TO MCQ MCQS

- 1 **E Stop oral agents and give an IV infusion of soluble insulin with glucose and potassium (KCl)** — **UPDATED** For major surgery oral agents are stopped and a variable-rate IV soluble in-

sulin infusion with glucose and potassium (GKI) is given, adjusted to hourly capillary glucose.

6 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Which one of the following statements about homeostasis is false?
 - A It is defined as a stable state of the normal body.
 - B The central nervous system, heart, lungs, kidneys and spleen are the essential organs that maintain normal homeostasis.
 - C Elective surgery should cause little disturbance to homeostasis.
 - D Emergency surgery should cause little disturbance to homeostasis.
 - E Return to normal homeostasis after an insult (operation, injury, infection) would depend upon the presence of comorbid conditions.
- 2 Neuroendocrine pathways of the stress response consist of the following except:
 - A Spinal cord
 - B Thalamus
 - C Hypothalamus
 - D Pituitary
 - E Thyroid
- 3 Changes in body metabolism that occur in response to trauma are the following except:
 - A Lipolysis
 - B Gluconeogenesis
 - C Protein breakdown
 - D Hypoglycaemia
 - E Hypermetabolism
- 4 Which of the following statements about mediators are true?
 - A They consist of neural, endocrine and inflammatory.
 - B Every endocrine gland plays a part.
 - C They play an important role in the flow phase.
 - D These mediators are released over several days.
 - E They play an important role in the recovery process.
- 5 Which of the following statements are true of optimal perioperative care?
 - A Surgery should be carried out by the use of adequate large incisions to give good exposure.
 - B Adequate pain relief is essential.
 - C Early mobilisation.
 - D Avoid ongoing insults and secondary trauma.
 - E Maintain good fluid load with several litres of normal saline.
- 6 The following statements are true regarding Enhanced Recovery After Surgery (ERAS) programmes except:
 - A It optimises rehabilitation following major surgery.
 - B It reduces hospital stay.
 - C Patient engagement is an integral part.
 - D It is only used in colorectal surgery.
 - E Blocking afferent painful stimuli is important in reducing stress response.

Answers 1-6

BAILEY & LOVE MCQS

- 1 **D Emergency surgery should cause little disturbance to homeostasis.** — **UPDATED** Emergency surgery markedly disrupts homeostasis due to unprepared physiology and severe pathology, unlike elective procedures.
- 2 **E Thyroid** — **UPDATED** The thyroid is not directly involved; spinal cord, thalamus, hypothalamus and pituitary mediate the response.
- 3 **D Hypoglycaemia** — **UPDATED** Trauma induces lipolysis, gluconeogenesis, protein breakdown and hypermetabolism—not hypoglycaemia.
- 4 **ABCE True: A, B, C, E** — **UPDATED** Mediators are rapidly released and jointly modulate the stress response; they are not released over several days.
- 5 **BCD True: B, C, D** — **UPDATED** Effective pain control, early mobilisation and prevention of secondary trauma are key; large incisions and excess fluids are not recommended.
- 6 **D It is only used in colorectal surgery.** — **UPDATED** ERAS programmes are now applied across many specialties, not solely in colorectal surgery.

5 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1 Select the most likely diagnosis for this patient with polyuria: A 25-year-old man with severe head injury and basal skull fracture. Urine osmolality 150 mOsm/L, serum osmolality 350 mOsm/L, no change after fluid restriction, but urine osmolality rises to 300 mOsm/L after IV vasopressin.
 - A central diabetes insipidus
 - B nephrogenic diabetes insipidus
 - C water intoxication
 - D solute overload
 - E diabetes mellitus
- 2 A 70-year-old man in the ICU with multiple renal insults and amphotericin therapy develops polyuria; no change with vasopressin. Serum osmolality is high, urine remains dilute. Which is the most likely diagnosis?
 - A central diabetes insipidus
 - B nephrogenic diabetes insipidus
 - C water intoxication
 - D solute overload
 - E diabetes mellitus
- 3 A 25-year-old woman with severe head injury, pulmonary contusion, and pelvic fracture is unresponsive in the ICU. Which of the following is the best initial approach to her nutritional needs?
 - A insertion of a subclavian venous catheter and initiation of central IV hyperalimentation
 - B wait for extubation and improvement of neurologic status, allowing institution of oral intake
 - C early institution of NG or nasojejun tube feeding with an elemental formulation
 - D wait for resolution of the associated ileus, followed by delayed hypercaloric regimen
 - E peripheral IV hyperalimentation
- 4 A 65-year-old diabetic man presents with a penetrating buttock wound from a wooden stump. The wound shows violaceous discoloration, bullae, crepitus, and grayish drainage. What is the optimal treatment?
 - A high-dose IV penicillin G and broad-spectrum antibiotics
 - B high-dose IV penicillin G, broad-spectrum antibiotics, and local wound care with unroofing of bullae and culture
 - C high-dose IV penicillin G, broad-spectrum antibiotics, with surgical debridement only if no improvement
 - D radical surgical debridement
 - E high-dose IV penicillin G, broad-spectrum antibiotics, radical surgical debridement, and hyperbaric oxygen therapy
- 5 A 45-year-old woman involved in a motor vehicle crash with prolonged parenteral nutrition develops diarrhea, mental depression, alopecia, and perioral/periorbital dermatitis. Which trace element supplementation is most likely to reverse these complications?
 - A iodine
 - B zinc
 - C selenium
 - D silicon
 - E tin

Answers 1-5

LANGE MCQs

- 1 **A central diabetes insipidus** — UPDATED Head trauma damaging posterior pituitary; responds to vasopressin.
- 2 **B nephrogenic diabetes insipidus** — UPDATED Acquired from nephrotoxic injury; unresponsive to ADH.
- 3 **C early institution of NG or nasojejun tube feeding with an elemental formulation** — Early enteral feeding aids GI mucosal integrity and is preferable to parenteral nutrition.
- 4 **E high-dose IV penicillin G, broad-spectrum antibiotics, radical surgical debridement, and hyperbaric oxygen therapy** — Necrotizing soft-tissue infection requires aggressive surgical debridement plus antibiotics and adjunctive hyperbaric oxygen.
- 5 **B zinc** — Zinc deficiency from prolonged PN causes these symptoms.

1 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1 A 65-year-old hypertensive man scheduled for TURP shows a serum potassium of 2.7 mmol/L. Which ECG change is NOT characteristic of hypokalaemia?
- A Left bundle branch block
 - B Flattened T waves
 - C U waves
 - D ST segment depression
 - E Prolonged QT interval

Answer 1

SBA MCQS

- 1 **A Left bundle branch block** — UPDATED Left bundle branch block is not a typical ECG finding in hypokalaemia.

Shock

Classes of haemorrhage, types of shock, resuscitation and transfusion.

31

MCQS

0

EMQ SCENARIOS

7

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

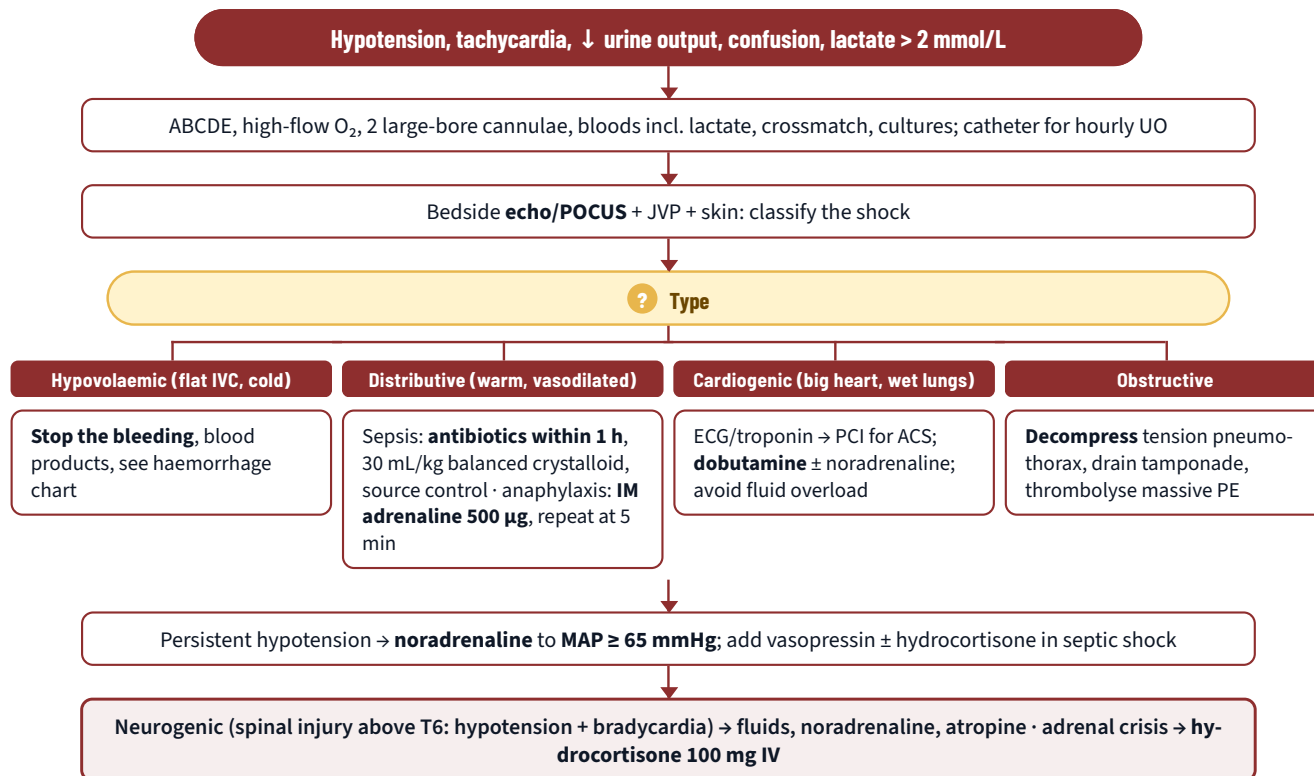
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Step by step

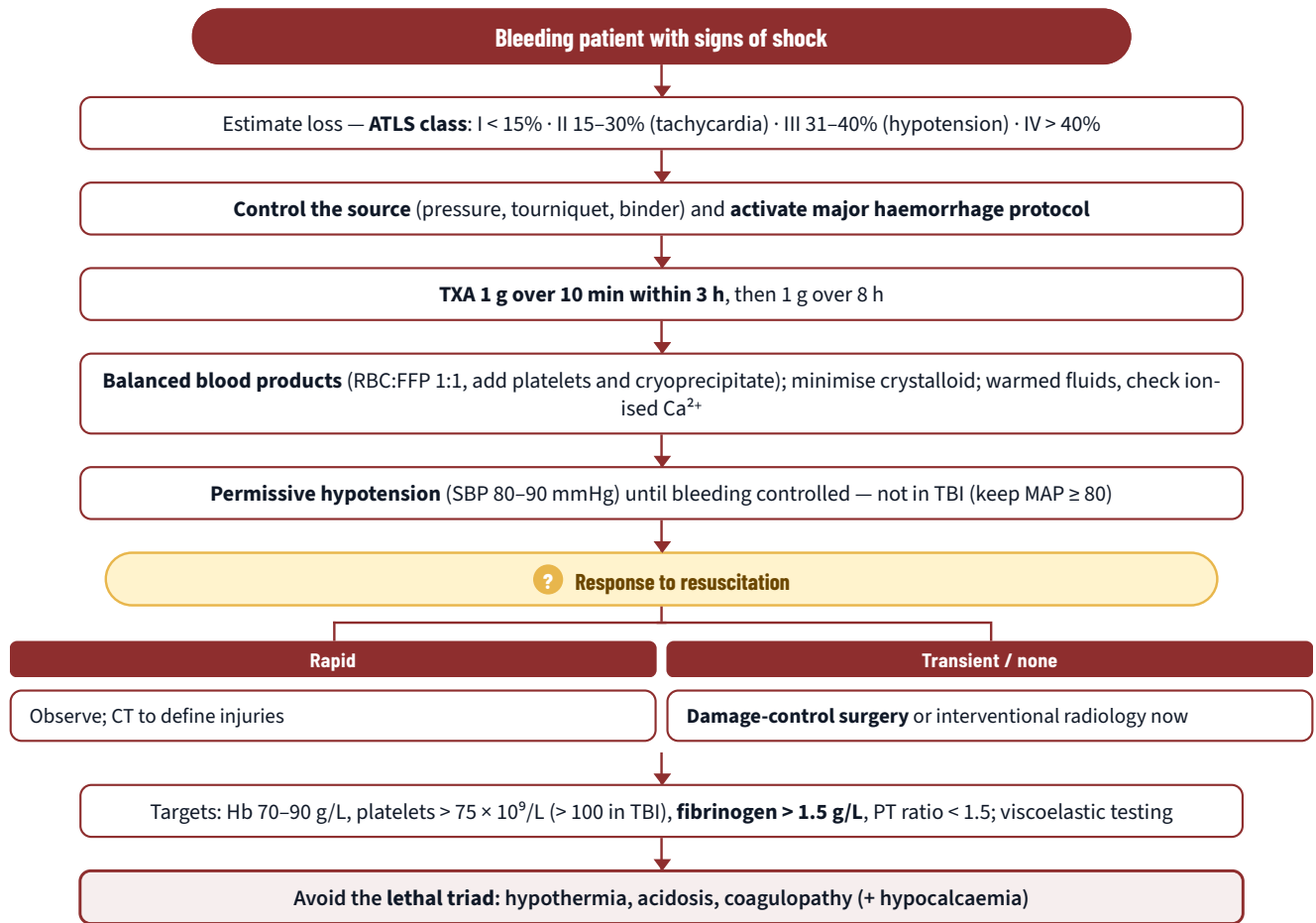
Undifferentiated shock

SURVIVING SEPSIS CAMPAIGN 2021 • NICE NG51 • RCUK ANAPHYLAXIS 2021



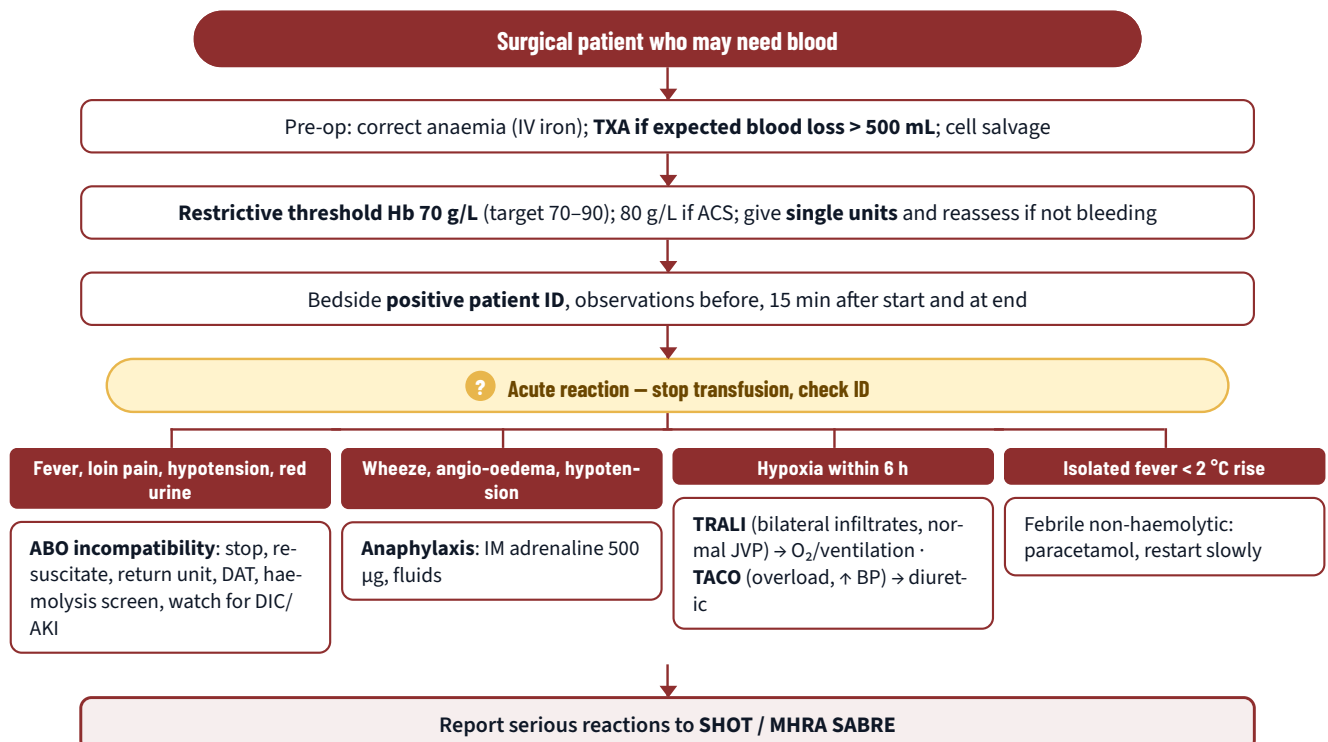
Haemorrhagic shock and major haemorrhage

ATLS 10TH ED · BSH/JPAC MAJOR HAEMORRHAGE · ESA/EUROPEAN TRAUMA 2023



Blood transfusion and transfusion reactions

NICE NG24 · BSH 2023 · SHOT



Local Exams MCQs

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

CASE • QUESTIONS 1-2

A 28-year-old man is brought after a road traffic accident with a deep right-thigh laceration and pulsatile bleeding. He is pale and anxious; BP 85/60 mmHg, HR 130/min.

- 1** What type of bleeding is this?
- A** Arterial: bright red and spurting in time with the pulse
 - B** Venous: dark red, steady flow
 - C** Capillary: oozing
 - D** Reactionary haemorrhage
 - E** Secondary haemorrhage

- 2** What is the maximum safe tourniquet time?

- A** About 2 hours (note the time applied)
- B** 15 minutes
- C** 6 hours
- D** 12 hours
- E** No limit if the limb is cooled

Answers 1-2

LOCAL EXAMS MCQs

- 1** **A** Arterial: bright red and spurting in time with the pulse — Arterial bleeding is bright red and pulsatile; venous bleeding is dark and flows steadily. Control with direct pressure or a tourniquet.

- 2** **A** About 2 hours (note the time applied) — Limit tourniquet time to about 2 h (record it); complications are nerve palsy, ischaemia, compartment syndrome and reperfusion injury.

7 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQs & EMQs

Bailey & Love MCQs

- 1** Which of the following statements are true in the pathophysiology of shock?
- A** In shock, cells switch from aerobic to anaerobic metabolism.
 - B** The product of anaerobic respiration is carbon dioxide, resulting in respiratory acidosis.
 - C** Hypoxia and acidosis generate oxygen free radicals and cytokine release.
 - D** Renal ischaemia leads to decreased glomerular filtration (GFR), thus activating the renin-aldosterone-angiotensin system (RAAS).
 - E** Once shock has been treated and circulation restored, the physiological disturbances return to normal.

- 2** Regarding the severity of shock, which of the following statements are false?
- A** In compensated shock, blood flow is maintained to the brain, lungs and kidneys.
 - B** Loss of 15% of circulating blood volume will overload the body's compensatory mechanisms.
 - C** Capillary refill time is an accurate clinical parameter in all types of shock.
 - D** Hypotension is an early sign of shock.
 - E** Prolonged systemic ischaemia and reperfusion injury contribute to multi-organ failure.

- 3** Regarding resuscitation, which of the following statements are true?

- A** In a patient who is shocked from a leaking abdominal aortic aneurysm or a bleeding peptic ulcer, time should not be wasted in instituting high-volume fluid resuscitation.
- B** In a patient with intestinal obstruction with hypovolaemic shock, time should be spent in optimising with adequate fluid resuscitation.
- C** In hypovolaemia, inotropes must be started as soon as possible as first-line therapy.
- D** The ideal fluid is colloid in hypovolaemia.
- E** The shock status is best determined by an initial fluid challenge.

- 4** Regarding monitoring in shock, which of the following statements are true?

- A** CVP monitoring is a dynamic reflection on fluid challenge.
- B** Measurement of cardiac output (CO) helps in distinguishing the type of shock.
- C** Mixed venous oxygen saturation is an indicator of systemic perfusion.
- D** Pulmonary artery wedge pressure (PAWP) measurement should be instituted in all shocked patients in the ICU.
- E** It is mandatory to perform Allen's test prior to insertion of an arterial line.

- 5** Regarding haemorrhage, all of the following statements are true except:
- A** Haemorrhage with hypovolaemic shock results in a state of coagulopathy.
 - B** Haemoglobin level is an accurate indicator of the degree of haemorrhage.
 - C** The acid-base disturbance is metabolic acidosis.
 - D** Permissive hypotension is a strategy to be followed until haemorrhage is controlled.
 - E** Haemorrhagic shock can be classified into three groups – mild, moderate and severe.
- 6** Regarding blood transfusion, which of the following statements are true?
- A** Blood group 'O' is regarded as universal donor.
 - B** Blood group 'AB' is regarded as universal recipient.
 - C** The vast majority of the population is Rhesus negative.
 - D** In coagulopathy packed red blood cells should be transfused.
 - E** Fresh frozen plasma (FFP) has a shelf life of 2 years.
- 7** Regarding conscious sedation, which of the following statements is false?
- A** Sedation has no significant dangers and can be used without restrictions.
 - B** All sedated patients require secure intravenous access.
 - C** Co-administration of opiates and benzodiazepines has a synergistic effect and opiates administered wherever possible before benzodiazepines.
 - D** Use of supplemental oxygen is essential in sedated patients.
 - E** All sedated patients require pulse oximetry to monitor oxygen saturations.

Answers 1-7

BAILEY & LOVE MCQs

- 1** **ACD** True: A, C, D — **UPDATED** B is false: anaerobic metabolism produces lactate (metabolic acidosis); E is false: reperfusion injury and organ dysfunction may persist after circulation is restored.
- 2** **BCD** True: B, C, D — **UPDATED** Up to 15% blood loss is compensated and neither capillary refill nor early hypotension are reliable indicators.
- 3** **ABE** True: A, B, E — **UPDATED** Bleeding (AAA, ulcer) needs haemorrhage control, not prolonged fluid loading; obstruction needs full resuscitation first. Inotropes are not first-line and crystalloid is preferred.
- 4** **ABC** True: A, B, C — **UPDATED** D false: PAWP catheters are not routine in shocked ICU patients; E false: Allen's test has poor predictive value and is not mandatory before radial arterial cannulation.
- 5** **BE** True: B, E — **UPDATED** Acute blood loss does not immediately change haemoglobin and shock is classically divided into four classes.
- 6** **ABE** True: A, B, E — **UPDATED** C false: about 85% of people are RhD positive; D false: coagulopathy needs FFP, platelets and cryoprecipitate, not red cells. FFP keeps about 2 years frozen.
- 7** **A** Sedation has no significant dangers and can be used without restrictions. — **UPDATED** Sedation is not without risk – it may cause hypoventilation and airway obstruction, so proper monitoring (IV access, oxygen, pulse oximetry) is essential.

6 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 30-year-old man with a pelvic fracture is hypotensive and tachycardic. Arterial blood gas shows pH 7.24, PaCO₂ 32 mmHg, HCO₃⁻ 14 mmol/L and base deficit -10. What is the most appropriate management of his acid-base derangement?
- A** Administration of sodium bicarbonate to correct the base deficit
 - B** Restoration of blood volume with aggressive IV fluid and blood resuscitation
 - C** IV hydrochloric acid
 - D** Intubation and hyperventilation
 - E** No treatment; this patient has no acid-base abnormality
- 2** A 70-year-old man admitted after AAA repair has low pulmonary wedge pressure, tachycardia, and low urine output. What is the most appropriate next step in management?
- A** IV furosemide
 - B** A bolus of IV crystalloid
 - C** A dopamine infusion
 - D** A nitroprusside infusion
 - E** IV digoxin

- 3** A 70-year-old man is given IV fluids for hypovolaemia after AAA repair. Several hours later his BP is 150/85 mmHg with distended neck veins and an elevated pulmonary wedge pressure, and urine output remains low. What is the next step?
- A** IV furosemide
 - B** A bolus of IV crystalloid
 - C** A dopamine infusion
 - D** A nitroprusside infusion
 - E** IV digoxin
- 4** A 39-year-old multiparous woman has a retained placenta 60 minutes after vaginal birth of a 3650-g healthy boy. There was no episiotomy or lacerations. She now has profuse vaginal bleeding with BP 80/50 mmHg and pulse 120/min; her uterine fundus is firm. Which treatment is most appropriate?
- A** Immediate blood transfusion (volume resuscitation)
 - B** uterine packing
 - C** intravenous pitocin
 - D** uterine artery embolization
 - E** intravenous administration of methylergonovine
- 5** A 65-year-old diabetic male, postoperative day 3 after duodenal ulcer surgery, has a large amount of salmon colored fluid draining from his midline wound. What is the next appropriate step?
- A** open wound at the bedside
 - B** ultrasound-guided drainage of fluid collection
 - C** CT scan of the abdomen
 - D** immediate esophagogastroduodenoscopy (EGD)
 - E** immediate exploration of wound in the operating room (OR)
- 6** A 65-year-old diabetic man develops spiking fever on the 6th day after laparotomy for duodenal ulcer, with incisional discomfort and a small amount of wound drainage. Appropriate evaluation and management should include which of the following?
- A** blood, urine, and sputum cultures; chest x-ray and abdominal CT scan
 - B** discharging the patient on oral antibiotics
 - C** culture of the wound drainage, and initiation of IV antibiotics
 - D** opening the incision, culture of the wound drainage, and initiating local wound care
 - E** opening the incision, culture of the wound drainage, and initiating IV antibiotics

Answers 1-6

LANGE MCQS

- 1** **B Restoration of blood volume with aggressive IV fluid and blood resuscitation** — **UPDATED** Lactic (metabolic) acidosis from hypoperfusion is corrected by restoring circulating volume and perfusion; bicarbonate does not treat the cause and may worsen intracellular acidosis.
- 2** **B A bolus of IV crystalloid** — **UPDATED** A low wedge pressure with tachycardia and oliguria after AAA repair means hypovolaemia, so a crystalloid bolus is given; diuretics or vasodilators would worsen it.
- 3** **D A nitroprusside infusion** — **UPDATED** Hypertension with high filling pressures and poor output after fluid loading responds to afterload reduction with nitroprusside, improving cardiac output and renal perfusion.
- 4** **A Immediate blood transfusion (volume resuscitation)** — **UPDATED** She is in haemorrhagic shock, so blood volume is restored first, then the placenta is removed manually; with a firm fundus, uterotonics or packing for atony will not help.
- 5** **E immediate exploration of wound in the operating room (OR)** — **UPDATED** Salmon-coloured (serosanguinous) wound discharge on day 3 heralds fascial dehiscence, so the wound is explored and the abdomen closed in theatre.
- 6** **D opening the incision, culture of the wound drainage, and initiating local wound care** — **UPDATED** A superficial wound infection is treated by opening the incision, sending pus for culture and local wound care; antibiotics only if there is cellulitis or systemic sepsis.

8 QUESTIONS • SINGLE BEST ANSWERS IN SURGERY

SBA MCQs

- 1** A patient is admitted following a motorcycle accident. He has fractured his left femur, tibia, fibula and pelvis. His blood pressure is 70/35 mmHg, pulse 140 beats/min, respiratory rate 35 breaths/min and Glasgow Coma Scale score 9/15. You wish to resuscitate the patient. Which one of the following procedures may be contraindicated in such a patient?

 - A Motorcycle helmet removal
 - B Urinary catheterization
 - C Neck line insertion
 - D Nasogastric tube insertion
 - E Intubation
- 2** A patient is admitted following an assault. On assessment, he has a stab wound to his chest. Clinically, he has a massive haemothorax and his Glasgow Coma Scale score is 4/15. Without further management, this patient will succumb to which cause of death first?

 - A Haemorrhagic shock
 - B Respiratory failure
 - C Airway compromise
 - D Intracranial haemorrhage
 - E Multiorgan failure
- 3** Which of the following techniques does not provide a definitive airway?

 - A Cricothyroidotomy
 - B Tracheostomy
 - C Nasotracheal tube
 - D Laryngeal mask airway
 - E Endotracheal tube
- 4** A patient is admitted following a road traffic accident. He has sustained significant blunt injury to his head, chest and abdomen and has a Glasgow Coma Scale score of 8/15. His saturations are poor at 89% on 15 L of oxygen via a non-rebreathing mask. You note bruising around both eyes and blood-stained fluid issuing from his left ear, which forms concentric rings when dripped on a white sheet. You wish to support his airway to improve oxygenation. The first choice of airway adjunct would be:

 - A Oropharyngeal airway
 - B Nasopharyngeal tube
 - C Laryngeal mask
 - D Intubation
 - E Positive pressure ventilation (continuous positive airway pressure)
- 5** A 35-year-old man is admitted after severing his arm on industrial machinery. His airway is patent and there is no identifiable hindrance to breathing. His pulse is 110 beats/min, blood pressure is 130/105 mmHg, and respiratory rate is 25 breaths/min. Which stage of shock is this patient in?

 - A Class I
 - B Class II
 - C Class III
 - D Class IV
 - E Impossible to say from given information
- 6** A 35-year-old butcher is admitted after inadvertently stabbing himself with a knife. His airway is patent and there is no identifiable hindrance to breathing. His pulse is 110 beats/min, BP 130/105 mmHg and respiratory rate 25 breaths/min. Assuming a body mass of 70 kg, what is the best estimate of the volume of blood lost?

 - A 400 mL
 - B 1000 mL
 - C 1800 mL
 - D 2500 mL
 - E Impossible to say from given information
- 7** A patient is admitted in haemorrhagic shock following a road traffic accident. A final year medical student has inserted a pink (20 G) cannula in the antecubital fossa. What rate of flow into the patient will this allow?

 - A 250 mL/min
 - B 170 mL/min
 - C 55 mL/min
 - D 25 mL/min
 - E 10 mL/min
- 8** A homeless man is admitted unresponsive after being found by police on a park bench. He has no external signs of injury. An oesophageal temperature probe records his core body temperature to be 34 °C. Which of the following management options is not routinely indicated in this case?

 - A Cardiac monitoring
 - B Warmed peritoneal lavage
 - C Warmed intravenous fluids
 - D Intravenous dextrose
 - E Blood alcohol and toxin screen

Answers 1-8

SBA MCQS

- 1 **B Urinary catheterization** — **UPDATED** A pelvic fracture may tear the urethra, so a urethral catheter is avoided until urethral injury is excluded (blood at meatus, retrograde urethrogram).
- 2 **C Airway compromise** — **UPDATED** A very low GCS indicates an inability to protect the airway, making airway compromise the most immediate threat.
- 3 **D Laryngeal mask airway** — **UPDATED** A definitive airway is a cuffed tube in the trachea (oro/nasotracheal, surgical); a laryngeal mask sits above the larynx and does not protect against aspiration.
- 4 **A Oropharyngeal airway** — **UPDATED** Raccoon eyes and CSF otorrhoea suggest a basal skull fracture, so a nasopharyngeal airway is avoided; an oropharyngeal airway is used pending a definitive airway.
- 5 **B Class II** — The vital signs indicate moderate blood loss corresponding to Class II shock.
- 6 **B 1000 mL** — **UPDATED** Tachycardia, narrowed pulse pressure and RR 25 with normal systolic BP is class II shock (15–30% loss, 750–1500 mL in a 70 kg adult), i.e. about 1000 mL.
- 7 **C 55 mL/min** — **UPDATED** A pink (20 G) cannula typically allows a flow rate of about 55 mL/min.
- 8 **B Warmed peritoneal lavage** — **UPDATED** At 34 °C (mild hypothermia) passive and active external rewarming with warmed IV fluids suffice; invasive core rewarming such as peritoneal lavage is for severe hypothermia.

3 QUESTIONS • AFTER PRETEST SURGERY

PreTest MCQs

- 1 A 33-year-old woman is seen in the ER with severe rectal bleeding. She has a history of ulcerative colitis. Her blood pressure is 95/56 mm Hg and her pulse is 110. Her base deficit is –8 mEq/L. What percent of her blood volume has been lost?
 - A <5%
 - B <15%
 - C 15% to 30%
 - D 30% to 40%
 - E >40%
- 2 Following a 2-hour firefighting episode, a 36-year-old fireman begins complaining of a throbbing headache, nausea, dizziness, and visual disturbances. He is taken to the ER, where his carboxyhemoglobin (COHb) level is found to be 31%. Which of the following is the most appropriate next step in his treatment?
 - A Begin an exchange transfusion.
 - B Treat with hyperbaric oxygen therapy
 - C Administer methylene blue
 - D Administer 100% oxygen by mask
 - E Perform flexible bronchoscopy
- 3 A 75-year-old man with a history of coronary artery disease, hypertension, and diabetes mellitus undergoes a right hemicolectomy for colon cancer. On the second postoperative day, he becomes hypotensive with depressed mental status and is immediately transferred to the intensive care unit. After intubation, an echocardiogram is performed which demonstrates a hypokinetic left ventricle with an ejection fraction of 15%. His central venous pressure is 18 mm Hg, and his hemoglobin is 10 g/dL. Which of the following is the most likely etiology for the patient's acute decompensation?
 - A Hemorrhagic shock
 - B Cardiogenic shock
 - C Septic shock
 - D Cardiac tamponade
 - E Pulmonary embolism of the main pulmonary arterial trunk

Answers 1-3

PRETEST MCQS

- 1 **D 30% to 40%** — **UPDATED** Hypotension with tachycardia and a base deficit of –6 to –10 indicates ATLS class III haemorrhage, a 30–40% loss of blood volume.
- 2 **D Administer 100% oxygen by mask** — **UPDATED** Give 100% oxygen via a tight-fitting mask at once (COHb half-life ~80 min vs ~5 h on air); hyperbaric oxygen is reserved for coma, seizures or severe cases.
- 3 **B Cardiogenic shock** — **UPDATED** Hypotension with a very low ejection fraction (15%) and high CVP in a patient with coronary disease is cardiogenic shock (pump failure, probably perioperative MI).

4 QUESTIONS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead MCQs

- 1 A 56-year-old woman is brought into the resuscitation room. Three days ago she suffered an insect bite to the abdomen which has now spread, causing redness across the whole of her abdomen. On examination, she is confused, with a temperature of 38.3°C, heart rate 106/min and blood pressure 100/56 mmHg. Which of the following is specific to the treatment of this type of shock?
 - A Antibiotics
 - B Antihistamines
 - C Atropine
 - D Fluids
 - E Inotropes
- 2 A 45-year-old man has walked into the emergency department following his involvement in a road traffic collision. On arrival at the hospital he is anxious, but otherwise fine. Later, he suddenly becomes faint. He is taken into the resuscitation room where he is found to have a heart rate of 46/min and a blood pressure of 80/48 mmHg. Primary examination is unremarkable and the patient has warm peripheries. Which of the following is the most likely cause of his symptoms?
 - A Cardiogenic shock
 - B Haemorrhagic shock
 - C Hypovolaemic shock
 - D Neurogenic shock
 - E Spinal shock
- 3 A 56-year-old man is one day post appendicectomy and becomes acutely short of breath. He has just been given his first dose of cyclizine. On arrival, he is breathless with heart rate 122/min, blood pressure 86/48 mmHg, saturations 85% in air. Which of the following would you administer first?
 - A Adrenaline
 - B Chlorphenamine
 - C Fluids
 - D Hydrocortisone
 - E Salbutamol
- 4 A 22-year-old man is brought in following a road traffic collision. He is suspected to have intra-abdominal injuries and there is bruising over his trunk. On arrival, he has a pulse rate of 140/min and his systolic blood pressure is 80 mmHg. Which of the following fluids would raise the patient's blood pressure most rapidly?
 - A 5% dextrose
 - B Fresh frozen plasma
 - C Packed red cells
 - D Mannitol
 - E Normal saline

Answers 1-4

GET AHEAD MCQS

- 1 **A Antibiotics** — **UPDATED** Septic shock requires prompt antibiotic therapy in addition to supportive measures.
- 2 **D Neurogenic shock** — Sudden loss of sympathetic tone leads to hypotension with bradycardia and warm peripheries.
- 3 **A Adrenaline** — Acute anaphylaxis is managed first with intramuscular adrenaline to relieve shock and bronchospasm.
- 4 **C Packed red cells** — **UPDATED** Red cells stay intravascular and restore oxygen-carrying capacity, whereas only about a quarter of infused saline remains intravascular; dextrose spreads through all body water.

1 QUESTIONS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course MCQs

- 1 A 54-year-old woman is 2 days post an open appendectomy for perforated appendicitis. She is tender in the lower abdomen, has a temperature of 38.9°C, blood pressure of 100/54 mmHg, pulse rate 95 beats/min, white cell count $21 \times 10^9/\text{L}$, and CRP 154 mg/L. What is the most likely diagnosis?

- A Cardiogenic shock
- B Hypovolaemic shock
- C Septic shock
- D Obstructive shock
- E Anaphylactic shock

Answer 1

CRASH COURSE MCQS

- 1 **C Septic shock** — Postoperative fever, high WCC/CRP, hypotension; consistent with sepsis.

Obesity & Bariatric Surgery

Indications for bariatric surgery, operations and their complications.

10

MCQs

0

EMQ SCENARIOS

3

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

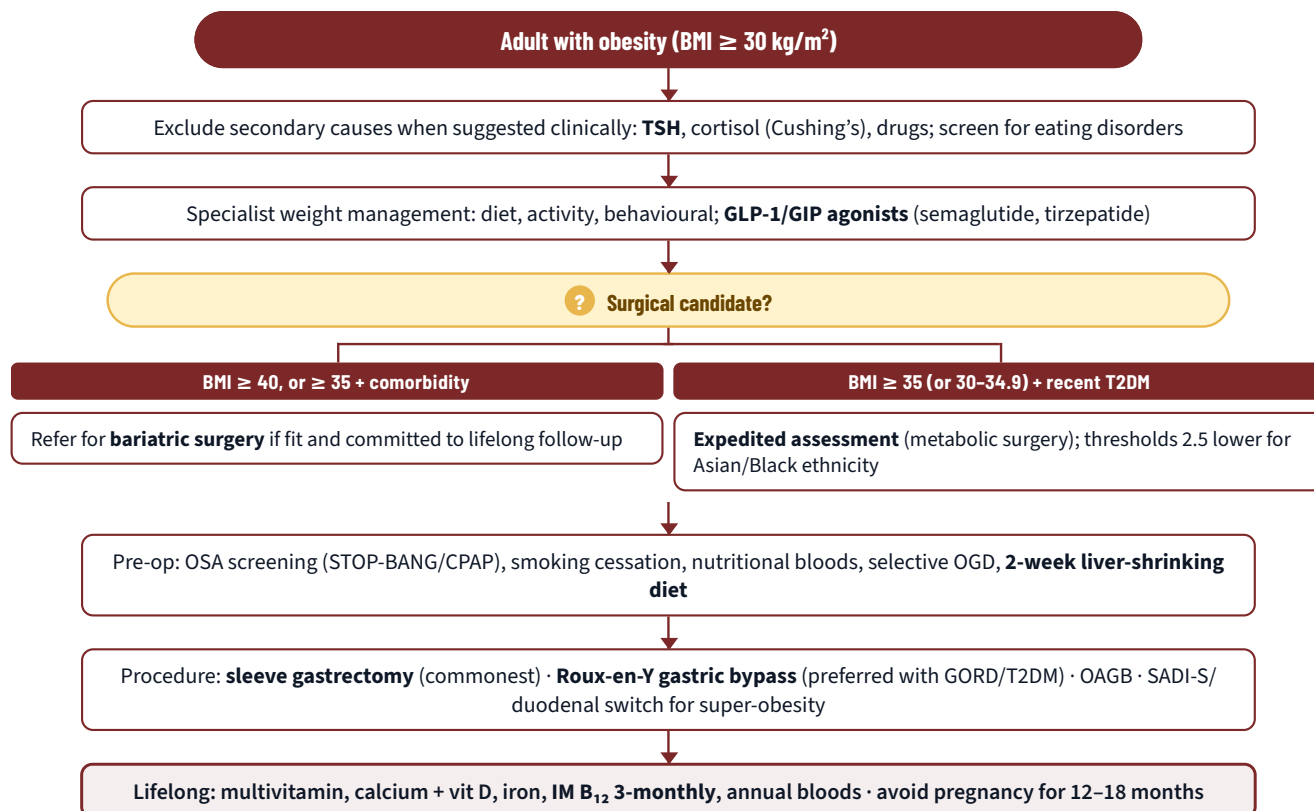
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Step by step

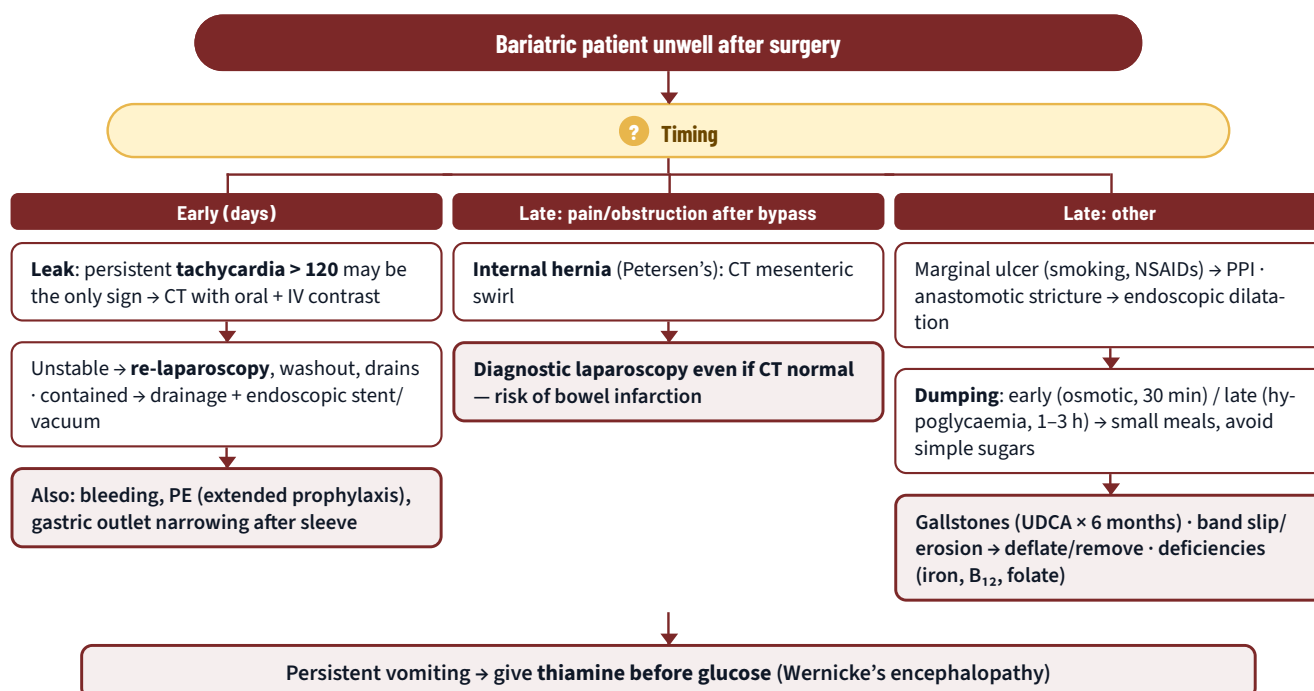
Selection and work-up for bariatric surgery

NICE NG246 (2025) • IFSO/ASMBS 2022 • BOMSS



Complications after bariatric surgery

BOMSS • IFSO 2022 • BAILEY & LOVE 28E



4 QUESTIONS • COLLEGE ESSAY QUESTIONS CONVERTED TO MCQS

Essay to MCQ MCQs

- 1 A 40-year-old woman with a BMI of 45 kg/m² has an HbA1c of 7% (53 mmol/mol) and a fasting blood glucose of 140 mg/dL (7.8 mmol/L). What metabolic disorder does she have?
 - A Hypoglycemia
 - B Hyperthyroidism
 - C Insulin resistance with impaired glucose control (type 2 diabetes)
 - D Metabolic alkalosis
 - E Dyslipidemia
- 2 A 40-year-old woman with a BMI of 45 kg/m² has type 2 diabetes (HbA1c 7%) despite lifestyle measures. What is the most appropriate surgical management of her metabolic disorder?
 - A Appendectomy
 - B Cholecystectomy
 - C Bariatric (metabolic) surgery, e.g. sleeve gastrectomy
 - D Colon resection
 - E Laparoscopic fundoplication
- 3 A 45-year-old man with a BMI of 37 kg/m² has a fasting blood glucose of 6.5 mmol/L (117 mg/dL) and an HbA1c of 6.2% (44 mmol/mol) on two occasions, despite 6 months of diet and exercise. Does he need any treatment for his blood sugar?
 - A No, because his fasting blood sugar is normal
 - B Yes, he should start insulin immediately
 - C No, because lifestyle modification is sufficient
 - D Yes, after lifestyle modifications, the patient should start Metformin
 - E Yes, he should start sulfonylureas
- 4 A 45-year-old obese man has fasting triglycerides of 600 mg/dL (6.8 mmol/L), LDL cholesterol 95 mg/dL (2.5 mmol/L) and HDL 38 mg/dL, despite 3 months of diet and exercise. Does he need any treatment for his lipid profile?
 - A No, his LDL is normal
 - B Yes, he should be started on statins
 - C No, lifestyle modification is enough
 - D Yes, he should be started on niacin
 - E Yes, due to hypertriglyceridemia, fibrates should be started if lifestyle modifications are not effective

Answers 1-4

ESSAY TO MCQ MCQS

- 1 **C Insulin resistance with impaired glucose control (type 2 diabetes)** — **UPDATED** HbA1c $\geq 6.5\%$ (48 mmol/mol) and fasting glucose ≥ 126 mg/dL (7.0 mmol/L) confirm type 2 diabetes, driven by obesity-related insulin resistance.
- 2 **C Bariatric (metabolic) surgery, e.g. sleeve gastrectomy** — **UPDATED** BMI ≥ 40 (or ≥ 35 with comorbidity such as diabetes) qualifies for bariatric/metabolic surgery, which often induces remission of type 2 diabetes.
- 3 **D Yes, after lifestyle modifications, the patient should start Metformin** — **UPDATED** He has prediabetes (non-diabetic hyperglycaemia); if intensive lifestyle change fails, metformin reduces progression to diabetes, especially when BMI ≥ 35 .
- 4 **E Yes, due to hypertriglyceridemia, fibrates should be started if lifestyle modifications are not effective** — **UPDATED** Severe hypertriglyceridaemia (>500 mg/dL) with normal LDL risks pancreatitis and is treated with a fibrate when lifestyle measures fail; statins mainly lower LDL.

5 QUESTIONS • AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love MCQs

- 1 Which of the following defines morbid obesity?
 - A BMI greater than 32 Kg/m²
 - B BMI greater than 35 Kg/m² with comorbidity
 - C BMI greater than 40 Kg/m²
 - D BMI greater than 45 Kg/m²
 - E BMI greater than 50 Kg/m²
- 2 Which of the following constitute the selection criteria for performing bariatric surgery as per international norms?
 - A Minimum of 5 years obesity
 - B Failure of conservative treatment
 - C No alcoholism or major untreated psychiatric illness
 - D Avoid if likely to get pregnant within 1 year
 - E Acceptable operative risk on preoperative assessment

- 3** Preoperative nutritional screening of morbidly obese patients includes which of the following?
- A** Vitamins A, D and E, parathyroid hormone
 - B** Glucocorticoids and mineralocorticoids
 - C** Magnesium, calcium and phosphate
 - D** Ferritin, vitamin B12, folate and coagulation screen
 - E** Zinc, selenium, copper and C-reactive protein
- 4** Roux-en-Y gastric bypass is which type of bariatric procedure?
- A** Restrictive
 - B** Malabsorptive
 - C** Combined restrictive and malabsorptive
 - D** Type of bowel transposition
 - E** Uses band prosthesis
- 5** Which of the following are specific complications of bariatric procedures?
- A** Internal herniation in gastric bypass and bilio-pancreatic diversion (BPD)
 - B** Staple line or anastomotic leak in gastric bypass, sleeve gastrectomy and BPD
 - C** Pouch dilatation in gastric band, gastric bypass and sleeve gastrectomy
 - D** Erosion in BPD and sleeve gastrectomy
 - E** Band slippage in gastric banding

Answers 1-5

BAILEY & LOVE MCQS

- 1** **BC** True: **B, C** — Morbid obesity is defined as a BMI ≥ 40 kg/m² or ≥ 35 kg/m² when significant comorbidities are present.
- 2** **ABCE** True: **A, B, C, E** — **UPDATED** D is a counselling point (avoid pregnancy for 12–18 months after surgery), not a selection criterion; the others are standard international selection criteria.
- 3** **ACDE** True: **A, C, D, E** — **UPDATED** B is false: corticosteroid hormones are not part of nutritional screening; fat-soluble vitamins, PTH, minerals, iron studies, B12/folate and trace elements are checked.
- 4** **C** **Combined restrictive and malabsorptive** — The procedure creates a small gastric pouch and bypasses part of the small intestine, providing both restrictive and malabsorptive effects.
- 5** **ABCE** True: **A, B, C, E** — **UPDATED** D is false: erosion complicates gastric banding, not BPD or sleeve gastrectomy; internal hernia, leaks, pouch/sleeve dilatation and band slippage are recognised.

1 QUESTIONS • AFTER LANGE REVIEW OF SURGERY

Lange MCQs

- 1** A 35-year-old morbidly obese woman with newly diagnosed type 2 diabetes is interested in gastric bypass to help her lose weight. Which one of these factors would preclude her from being considered a good candidate for surgery?
- A** body mass index (BMI) of 38
 - B** hypertension (HTN)
 - C** age
 - D** diabetes
 - E** inability to follow a diet

Answer 1

LANGE MCQS

- 1** **E** **inability to follow a diet** — Patient selection for bariatric surgery requires adherence to dietary guidelines.

Trauma & Perioperative EMQs

Extended matching questions on shock, trauma, burns, perioperative care and postoperative complications.

0

MCQS

105

EMQ SCENARIOS

5

SOURCES

0

REPEATED IDEAS

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Local Exams EMQs

SET 1 OF 2

39TH BATCH EXAMS

Block 2, Block 3, end-induction and final Papers 1 & 2 of the 39th batch (recalled)

THEME 1 Electrolyte disturbances

- A Hyponatraemia
- B Hypernatraemia
- C Hypokalaemia
- D Hyperkalaemia
- E Hypocalcaemia
- F Hypercalcaemia
- G Hypomagnesaemia
- H Hypermagnesaemia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An elderly woman becomes confused 2 days after surgery; plasma osmolality is low.
- 2 A man with extensive muscle destruction has tall peaked T waves on ECG.
- 3 A man has prominent U waves on ECG.
- 4 A malnourished alcoholic given fluids and nutrition has hypokalaemia that does not correct despite repeated potassium infusions.

THEME 2 Complications of fractures

- A Avascular necrosis
- B Fat embolism syndrome
- C Pulmonary embolism
- D Post-traumatic osteoarthritis
- E Compartment syndrome
- F Volkmann ischaemic contracture
- G Malunion
- H Delayed union
- I Nerve injury
- J Osteomyelitis
- K Gas gangrene
- L Sudeck atrophy (complex regional pain syndrome)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A soil-contaminated open fracture from a farm accident develops, days later, rapidly spreading swelling, severe pain, systemic toxicity and sepsis.
- 2 After multiple long-bone fractures a patient develops, within 24–72 hours, respiratory distress, confusion and a petechial rash.
- 3 A 19-year-old athlete with a tibial fracture has worsening pain hours later, worse on passive dorsiflexion; distal pulses are present.

THEME 3 Closure of skin defects

- A Split-thickness skin graft
- B Full-thickness skin graft
- C Axial (pedicled) flap
- D Regional muscle flap
- E Free muscle flap
- F Free fasciocutaneous flap
- G Local random-pattern flap
- H Healing by secondary intention
- I Delayed primary closure

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A superficial 14 × 10 cm burn wound of the back, after debridement, with a healthy vascular bed and no exposed structures.
- 2 A small nasal ala defect after skin cancer excision needs excellent contour and minimal contraction.
- 3 A fingertip wound < 1 cm with no exposed bone.
- 4 A cheek defect closed by rotating a flap around a known vascular pedicle.
- 5 A 24-year-old has an avulsion over the anterior tibia; after debridement bone without periosteum is exposed and the surrounding tissue is traumatised and previously irradiated.

THEME 4 Limb ischaemia

- A Chronic atherosclerotic ischaemia
- B Acute limb ischaemia
- C Reperfusion injury (compartment syndrome)
- D Leriche syndrome
- E Buerger disease
- F Aortic dissection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old with a mechanical mitral valve and infective endocarditis suddenly has a cold, pale, pulseless right leg with sensory loss below the knee; motor function is intact.
- 2 A patient has absent femoral pulses, buttock claudication and erectile dysfunction.
- 3 A patient develops pain and a tense swollen leg after emergency embolectomy.
- 4 A 30-year-old male smoker has ischaemia of the distal digits and previous superficial thrombophlebitis.

THEME 5 Choice of anaesthesia

- A GA with endotracheal intubation
- B GA with laryngeal mask airway
- C Spinal anaesthesia
- D Epidural anaesthesia
- E Peripheral nerve block
- F Local anaesthesia with sedation
- G Bier block (IV regional)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An asthmatic patient with wheeze needs ureteroscopic stone extraction.
- 2 A patient with COPD needs emergency laparotomy for a perforated viscus.
- 3 A patient with heart failure and CKD needs elective inguinal hernia repair.
- 4 Wide local excision of a basal cell carcinoma of the cheek.

THEME 6 Postoperative and surgical nutrition

- A IV fluids only
- B Total parenteral nutrition
- C Oral feeding
- D Feeding jejunostomy
- E Gastrostomy (PEG)
- F Nasogastric feeding

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An elderly woman has inoperable oesophageal cancer with severe dysphagia.
- 2 A man is stable after uncomplicated herniorrhaphy with normal bowel sounds.
- 3 A patient has undergone a Whipple procedure.
- 4 A patient with multiple fractures has prolonged ileus.
- 5 A patient with severe trauma and a liver injury.

Answers • Themes 1-6

LOCAL EXAMS EMQS

Theme 1 • Electrolyte disturbances

- 1 **A Hyponatraemia** — Postoperative confusion with low osmolality is hyponatraemia (ADH response plus hypotonic fluids).
- 2 **D Hyperkalaemia** — Rhabdomyolysis releases potassium; peaked T waves signal hyperkalaemia (give IV calcium first).
- 3 **C Hypokalaemia** — U waves, flat T waves and ST depression are signs of hypokalaemia.
- 4 **G Hypomagnesaemia** — Refractory hypokalaemia means hypomagnesaemia (renal K loss); replace magnesium first (and watch for refeeding).

Theme 2 • Complications of fractures

- 1 **K Gas gangrene** — Rapid spreading swelling and toxicity in a soil-contaminated wound is clostridial gas gangrene: radical debridement, penicillin, clindamycin.
- 2 **B Fat embolism syndrome** — Hypoxia, neurological signs and petechiae 1–3 days after long-bone fractures is fat embolism syndrome.
- 3 **E Compartment syndrome** — Pain out of proportion and on passive stretch is compartment syndrome; pulses persist until late. Urgent fasciotomy.

Theme 3 • Closure of skin defects

- 1 **A Split-thickness skin graft** — A large wound with a healthy vascular bed is covered by split-thickness skin graft.
- 2 **B Full-thickness skin graft** — Full-thickness grafts contract little and match facial skin, so suit small facial defects (local flaps are an alternative).
- 3 **H Healing by secondary intention** — Small fingertip wounds without exposed bone heal well by secondary intention.
- 4 **C Axial (pedicled) flap** — A flap based on a named vessel is an axial-pattern flap.

- 5 **E Free muscle flap** — Bare bone cannot take a graft and the local tissue is unusable, so a free muscle flap brings well-vascularised cover.

Theme 4 • Limb ischaemia

- 1 **B Acute limb ischaemia** — Sudden pain, pallor and pulselessness from an embolus is acute limb ischaemia; with intact motor function the limb is salvageable.
- 2 **D Leriche syndrome** — Aorto-iliac occlusion gives Leriche syndrome: buttock claudication, impotence and absent femoral pulses.
- 3 **C Reperfusion injury (compartment syndrome)** — Reperfusion after revascularisation causes oedema and compartment syndrome: fasciotomy.
- 4 **E Buerger disease** — Distal ischaemia with migratory thrombophlebitis in a young smoker is Buerger disease (thromboangiitis obliterans).

Theme 5 • Choice of anaesthesia

- 1 **C Spinal anaesthesia** — Spinal anaesthesia avoids airway instrumentation (bronchospasm) and suits ureteroscopy.
- 2 **A GA with endotracheal intubation** — Emergency laparotomy (full stomach) needs GA with rapid-sequence intubation to protect the airway.
- 3 **F Local anaesthesia with sedation** — Open inguinal hernia repair under local anaesthesia ($\pm$ sedation) avoids GA risks in high-risk patients.
- 4 **F Local anaesthesia with sedation** — Small facial skin excisions are done under local anaesthesia $\pm$ sedation.

Theme 6 • Postoperative and surgical nutrition

- 1 **E Gastrostomy (PEG)** — Inoperable oesophageal cancer needs long-term enteral access below the block: gastrostomy (or stent).

- 2 **C Oral feeding** — After uncomplicated surgery, early oral feeding is resumed.
- 3 **D Feeding jejunostomy** — A feeding jejunostomy placed at pancreaticoduodenectomy allows early enteral feeding beyond the anastomoses.

SET 2 OF 2

PREVIOUS YEARS

Previous-years college examinations

THEME 7 Postoperative complications

- A Anastomotic leak
- B Atelectasis
- C Hematoma formation
- D Pulmonary embolism
- E Wound dehiscence
- F Wound infection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 39-year-old male has a moderate-grade fever after one day from incisional hernia repair. Chest examination revealed reduced air entry over the right lower lobe. CXR shows right lower lobe collapse.

- 4 **B Total parenteral nutrition** — A non-functioning gut (prolonged ileus) needs parenteral nutrition.
- 5 **F Nasogastric feeding** — Critically ill trauma patients should start early enteral (nasogastric) feeding within 24–48 h if the gut works.

- 2 46-year-old female with a BMI of 51 underwent complicated laparoscopic cholecystectomy. 7 days post-op she developed SOB, tachycardia, and undetectable BP. ECG shows changes consistent with right ventricular strain.
- 3 A 37-year-old male underwent exploratory laparotomy for a bullet injury in the abdomen. The wound was initially healing well, but upon examination, it is observed that the surgical incision has separated and the underlying tissues are exposed.
- 4 A lady underwent sleeve gastrectomy 2 days earlier. She presents with tachycardia and tachypnoea. Chest examination is normal. Abdominal examination shows mild epigastric tenderness but normal-looking laparoscopic incisions. Gastrografin swallow shows contrast leaking from the staple line.

Answers · Theme 7

LOCAL EXAMS EMQS

Theme 7 · Postoperative complications

- 1 **B Atelectasis** — Fever on day 1 with reduced air entry and lobar collapse on CXR is atelectasis, the commonest cause of early postoperative pyrexia.
- 2 **D Pulmonary embolism** — Sudden breathlessness, tachycardia and circulatory collapse with right ventricular strain a week after surgery in a very obese patient is massive pulmonary embolism.

- 3 **E Wound dehiscence** — Separation of the healing incision exposing deeper tissues after laparotomy is wound dehiscence (burst abdomen).
- 4 **A Anastomotic leak** — Tachycardia and tachypnoea after sleeve gastrectomy suggest a staple-line leak even with minimal abdominal signs; contrast extravasation confirms it.

3 THEMES · 18 SCENARIOS · AFTER BAILEY & LOVE'S SHORT PRACTICE OF SURGERY MCQS & EMQS

Bailey & Love EMQs

THEME 1 Primary Survey and Resuscitation

- A Cardiac tamponade
- B Haemothorax
- C Hypoglycaemia
- D Major vessel injury
- E Myocardial infarction
- F Pneumonia
- G Pneumothorax
- H Severe head injury
- I Tension pneumothorax

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient presents following a fall from a third-story window and on primary survey is not maintaining adequate oxygen saturation on high-flow oxygen, is hypotensive, has a raised JVP with left tracheal deviation and the right hemi-thorax is hyper-resonant with no air entry.
- 2 A patient with a background of known alcohol excess presents with minor abrasions to the arms but no evidence of head injury. Chest is clear and heart sounds are normal. There is a reduced GCS 9/15 (E3, V2, M4) but no evidence of haemodynamic instability. Pupils are equal and reactive.

- 3** A patient presents with evidence of significant deep penetrating trauma at the level of the fifth/sixth intercostal space in left hemi-thorax. The patient has a raised JVP and muffled heart sounds, and is haemodynamically unstable with a tachycardia and hypotension.

THEME 2 Types of Shock and Blood Transfusion

- A** Anaphylactic shock
- B** Autologous blood transfusion
- C** Cardiogenic shock
- D** Hypovolaemic shock
- E** Mismatched blood transfusion
- F** Neurogenic shock
- G** Septic shock

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 50-year-old man is undergoing partial hepatectomy for secondary metastasis. The operation proceeded smoothly for the first couple of hours, after which the surgical team noticed unusual bleeding in the form of oozing from all the wound sites. The patient has two intravenous cannula sites through one of which he is on his first unit of blood.
- 2** A fit 30-year-old woman while gardening suddenly became very short of breath, had intense itching with rash and complained of a painful red spot on her arm. She has been brought to the A&E department and is hypotensive, hypoxic with warm peripheries.
- 3** A fit 25-year-old man fell from a height of 40 feet at a building site. He has been brought in unconscious with a GCS of 13 and flaccid limbs. He has been immobilised in a spinal frame and has an intravenous line.
- 4** A 70-year-old man of ASA anaesthetic category 3, underwent an emergency closure of a perforated duodenal ulcer. The anaesthetic and operation were uneventful. On the first postoperative day, he complained of feeling very unwell with a systolic blood pressure of 80 mm Hg with no unusual signs in his abdomen; there was impaired conscious level and peripheral vasoconstriction.
- 5** A 60-year-old man of ASA 1 anaesthetic risk underwent a total gastrectomy for stomach cancer. While in the ITU, 12 hours postoperatively, his BP has fallen to 80 mm Hg systolic, he has not put out any urine over the past 3 hours and he is hypoxic with oxygen saturation of 92%.
- 6** A 60-year-old woman has been admitted as an emergency with a 4-day history of severe right upper quadrant pain, vomiting, jaundice and intense pruritus and is very toxic - high temperature with rigors and hyperdynamic circulation.

- 7** A 35-year-old woman is due for a bilateral mastectomy. She has requested a bilateral breast reconstruction at the same time. However, she is a strict Jehovah's Witness and would under no circumstances have a heterologous blood transfusion.

THEME 3 Imaging Techniques

- A** Contrast enhanced computerised tomographic scan (CECT scan)
- B** Endoluminal (endoscopic) ultrasound (EUS)
- C** Intraoperative ultrasound (IOS)
- D** Magnetic resonance imaging scan (MRI scan)
- E** Plain x-ray
- F** Positron emission tomography scan/CT (PET/CT scan)
- G** Radionuclide imaging
- H** Ultrasound (US)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 68-year-old man complains of increasing dysphagia of 3 months' duration and has lost 25 kg in weight. Endoscopic biopsy confirms oesophageal carcinoma and CT shows no metastases. Which investigation best assesses the local (T and N) stage?
- 2** A 78-year-old man complains of gnawing backache associated with urinary frequency and poor stream for 4 months. Transrectal ultrasound (TRUS) biopsy showed a carcinoma of prostate.
- 3** A 32-year-old man while skiing sustained an injury to his right ankle following which it became very painful and grossly deformed.
- 4** A 45-year-old woman sustained a blunt injury to her lower chest and upper abdomen while horse riding. She was brought in with hypovolaemic shock. Having been resuscitated according to the ATLS protocol, she is now stable.
- 5** A 70-year-old man is undergoing a laparotomy for liver resection for metastases from colorectal carcinoma.
- 6** A 55-year-old woman presented with a lymph nodal lump over her right supraclavicular area. Excision biopsy showed a non-Hodgkin's lymphoma.
- 7** A 35-year-old woman has been admitted as an emergency with right upper quadrant pain, pyrexia and vomiting. On examination she is toxic, with a positive Murphy's sign.
- 8** A 62-year-old man presented with a large soft-tissue mass on the medial aspect of his left femoral triangle of 4 months duration. The mass is hard, immobile and tender.

Answers • Themes 1–3

Theme 1 • Primary Survey and Resuscitation

- I Tension pneumothorax** — Hypoxia, hypotension, raised JVP, trachea deviated to the left and a hyper-resonant silent right chest is a right tension pneumothorax: needle or finger decompression now.
- C Hypoglycaemia** — Reduced GCS in a drinker without head injury or shock suggests hypoglycaemia (alcohol impairs gluconeogenesis): check capillary glucose as part of 'D' in the primary survey.
- A Cardiac tamponade** — A penetrating left precordial wound with hypotension, raised JVP and muffled heart sounds (Beck's triad) is cardiac tamponade.

Theme 2 • Types of Shock and Blood Transfusion

- E Mismatched blood transfusion** — Sudden diffuse oozing from all wounds during the first unit of blood suggests an acute haemolytic (ABO-mismatch) reaction with DIC; stop the transfusion at once.
- A Anaphylactic shock** — Sudden breathlessness, itching, rash and hypotension with warm peripheries after a sting while gardening is anaphylactic shock.
- F Neurogenic shock** — A fall from height with flaccid limbs suggests a high spinal cord injury; loss of sympathetic tone causes neurogenic shock (hypotension, bradycardia, warm peripheries).
- C Cardiogenic shock** — Hypotension, confusion and vasoconstriction in a high-risk patient with a quiet abdomen after surgery suggests pump failure (perioperative MI): cardiogenic shock.
- D Hypovolaemic shock** — Hypotension, anuria and hypoxia 12 hours after a total gastrectomy suggest reactionary haemorrhage causing hypovolaemic shock.
- G Septic shock** — Pain, jaundice, fever with rigors and a hyperdynamic circulation is ascending cholangitis causing septic shock.

- B Autologous blood transfusion** — Jehovah's Witnesses refuse allogeneic blood; options include autologous techniques such as cell salvage (some accept it if kept in continuity), so check individual wishes.

Theme 3 • Imaging Techniques

- B Endoluminal (endoscopic) ultrasound (EUS)** — Endoscopic ultrasound is the most accurate test for depth of oesophageal wall invasion (T) and regional nodes (N) once CT has excluded metastases.
- G Radionuclide imaging** — Back pain in confirmed prostate cancer suggests bone metastases; a radionuclide (isotope) bone scan surveys the whole skeleton.
- E Plain x-ray** — A painful, deformed ankle after injury is a suspected fracture-dislocation: plain X-ray (after urgent reduction if the foot is threatened).
- A Contrast enhanced computerised tomographic scan (CECT scan)** — A haemodynamically stable patient after blunt thoraco-abdominal trauma is best assessed by contrast-enhanced CT for solid-organ and vascular injury.
- C Intraoperative ultrasound (IOS)** — Intraoperative ultrasound of the liver detects small occult metastases and defines their relation to vessels during resection of colorectal liver metastases.
- F Positron emission tomography scan/CT (PET/CT scan)** — PET-CT is the standard staging investigation for lymphoma, showing metabolically active nodal and extranodal disease.
- H Ultrasound (US)** — RUQ pain, fever and a positive Murphy's sign suggest acute cholecystitis; ultrasound is the first-line investigation.
- D Magnetic resonance imaging scan (MRI scan)** — A hard, fixed, growing deep soft-tissue mass suggests sarcoma; MRI is the best test for local staging before core biopsy.

5 THEMES • 25 SCENARIOS • AFTER GET AHEAD! SURGERY MCQS & EMQS

Get Ahead EMQs

THEME 1 Groin lumps

- A Epididymo-orchitis
- B False femoral aneurysm
- C Mumps orchitis
- D Psoas abscess
- E Saphena varix
- F Sebaceous cyst
- G Strangulated femoral hernia
- H Testicular torsion
- I Testicular tumour

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- A 22-year-old man presents with a swelling over his medial right thigh. On examination, the swelling is slightly tender and pulsatile but the patient otherwise feels well. The patient admits injecting heroin into the area earlier in the day.
- An 18-year-old man presents to the emergency department after developing mild pain in the left testicle while playing football. On examination, the left testicle is larger and more irregular compared with the right.

- 3** A 28-year-old man presents with a 3-day history of pain in his left testicle. On examination, there is a whitish discharge at the urethral meatus and the scrotum is swollen and tender to palpation.
- 4** A 37-year-old man presents with a painless swelling in his groin. On examination, the swelling is fluctuant, non-tender, and non-pulsatile. The patient does, however, admit to some tenderness in his back. He is currently taking rifampicin and isoniazid for a “chest infection.”
- 5** A 13-year-old boy presents with sudden-onset, severe pain in his left testicle, accompanied by lower abdominal pain and vomiting. On examination, the testicle is swollen, erythematous, and very tender to palpation. The left testicle is in its normal anatomical position.

THEME 2 DIAGNOSIS OF GROIN LUMPS

- A** Epididymo-orchitis
- B** False femoral aneurysm
- C** Hydrocele
- D** Mumps orchitis
- E** Reducible indirect inguinal hernia
- F** Saphena varix
- G** Sebaceous cyst
- H** Strangulated femoral hernia
- I** Strangulated inguinal hernia
- J** Testicular torsion
- K** Testicular tumour
- L** Varicocele

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 28-year-old man presents with a large left-sided scrotal swelling that is causing slight discomfort. On examination, the swelling is soft but does not transilluminate. You cannot get above it, and it disappears on lying flat; however, there is no cough impulse.
- 2** A 24-year-old man presents with a swelling in his right scrotum that has been increasing in size over the last year. On examination, the swelling is non-tender, fluctuant, and it transilluminates. It is possible to get above the swelling, but the testis is not palpable.
- 3** A 64-year-old man presents with a small lump on the skin of his left scrotum. On examination, the lump is tethered to the skin but not attached to any underlying structure, and it is not tender.
- 4** A 7-year-old boy who has been admitted to hospital with viral meningitis acutely develops a painful swelling in his testicle. On examination, the boy is pyrexial and the swelling is tender.
- 5** A 52-year-old woman presents with a swelling on her left thigh, below and lateral to the pubic tubercle. On examination, the swelling is bluish, non-tender, non-pulsatile, and compressible. The patient is otherwise well.

THEME 3 SHOCK

- A** Anaphylactic shock
- B** Cardiogenic shock
- C** Class I hypovolaemic shock
- D** Class II hypovolaemic shock
- E** Class III hypovolaemic shock
- F** Class IV hypovolaemic shock
- G** Neurogenic shock
- H** Septic shock
- I** Spinal shock

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 27-year-old man presents after haematemesis with four episodes of vomiting. He is drowsy, pulse 96/min, BP 130/70 mmHg, urine output 60 mL in the first hour.
- 2** A 54-year-old man with calf erythema and pain has HR 120/min, BP 100/65 mmHg, RR 30/min, T 38.5°C.
- 3** A 67-year-old diabetic presents feeling unwell. He is cool and clammy with elevated JVP, pulse 110/min, BP 90/50 mmHg.
- 4** A 24-year-old man after RTA with bilateral femoral fractures appears pale, pulse 130/min, BP 80/58 mmHg, RR 30/min, urine output 30 mL over 3 hours.
- 5** A 42-year-old woman after RTA suddenly becomes faint; peripheries are warm and dry, pulse 40/min, BP 82/48 mmHg.

THEME 4 TUMOUR MARKERS

- A** α -fetoprotein
- B** β -hCG
- C** CA 125
- D** CA 15-3
- E** CA 19-9
- F** Calcitonin
- G** Carcinoembryonic antigen
- H** Paraproteins
- I** Prostate-specific antigen
- J** S-100

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 47-year-old woman with a 3 cm firm breast lump fixed to skin.
- 2** A 76-year-old man with new-onset constipation, fresh blood in stools, and weight loss.
- 3** A 68-year-old man with difficulty passing urine and dribbling.
- 4** A 34-year-old woman with an irregular, pigmented 1 cm lesion on her leg that itches and bleeds.
- 5** A 62-year-old woman with painless progressive jaundice and a palpable gallbladder.

THEME 5 TUMOUR STAGING SYSTEMS

- A Ann Arbor
- B Breslow
- C Dukes A
- D Dukes B
- E Dukes C
- F Dukes D
- G Gleason

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 67-year-old man presents with difficulty passing urine. He has an elevated prostate-specific antigen, and a transrectal biopsy is performed.

- 2 A 72-year-old man has a sigmoid tumour resected. The pathologist notes that the tumour is confined to the bowel wall and that there is no lymph node involvement.
- 3 A 45-year-old man presents with a 4-week history of fever, night sweats and itching. He has lost 5 kg in weight over the last fortnight. On examination, enlarged lymph nodes are palpable in the neck.
- 4 A 65-year-old woman has a colon tumour resected successfully. The pathology report states that the tumour has invaded the bowel wall. The preoperative CT scan shows multiple metastases in the liver.
- 5 A 27-year-old woman presents with an irregularly pigmented lesion on her lower right leg that is subsequently excised.

Answers • Themes 1-5

GET AHEAD EMQs

Theme 1 • Groin lumps

- 1 **B False femoral aneurysm** — A pulsatile thigh swelling after injection suggests a false femoral aneurysm.
- 2 **I Testicular tumour** — Unilateral irregular testicular enlargement is suspicious for testicular tumour.
- 3 **A Epididymo-orchitis** — Urethral discharge with scrotal tenderness indicates epididymo-orchitis.
- 4 **D Psoas abscess** — Groin swelling with back tenderness in a TB patient suggests psoas abscess.
- 5 **H Testicular torsion** — Acute scrotal pain, swelling, and tenderness in a young boy confirm testicular torsion.

Theme 2 • DIAGNOSIS OF GROIN LUMPS

- 1 **L Varicocele** — Varicoceles are typically soft, reduce on lying down, and are most common on the left side. The inability to get above the swelling and absence of transillumination support this diagnosis.
- 2 **C Hydrocele** — A hydrocele is an accumulation of serous fluid in the scrotum, with transillumination and a fluctuant, non-tender mass where the testis may not be separately palpable.
- 3 **G Sebaceous cyst** — Sebaceous cysts are benign, superficial lesions tethered to the dermis and typically non-tender.
- 4 **D Mumps orchitis** — Mumps orchitis is a complication of mumps infection causing a painful, tender testicular swelling with systemic signs.
- 5 **F Saphena varix** — A saphena varix is a venous swelling near the saphenofemoral junction that is bluish, compressible, and located below and lateral to the pubic tubercle.

Theme 3 • SHOCK

- 1 **C Class I hypovolaemic shock** — Class I shock shows minimal signs despite minor blood loss.

- 2 **H Septic shock** — Septic shock presents with fever, tachycardia, hypotension in infection.
- 3 **B Cardiogenic shock** — Cool, clammy skin and raised JVP indicate cardiogenic shock.
- 4 **E Class III hypovolaemic shock** — Tachycardia, hypotension, and oliguria reflect Class III hypovolaemic shock.
- 5 **G Neurogenic shock** — Bradycardia with warm, dry peripheries indicates neurogenic shock.

Theme 4 • TUMOUR MARKERS

- 1 **D CA 15-3** — CA 15-3 is used in monitoring breast cancer.
- 2 **G Carcinoembryonic antigen** — CEA is a marker for colorectal cancer.
- 3 **I Prostate-specific antigen** — PSA is the marker for prostate pathology.
- 4 **J S-100** — S-100 is used in melanoma evaluation.
- 5 **E CA 19-9** — CA 19-9 is associated with pancreaticobiliary malignancies.

Theme 5 • TUMOUR STAGING SYSTEMS

- 1 **G Gleason** — Gleason score grades prostate cancer.
- 2 **C Dukes A** — Dukes A indicates tumour confined to bowel wall.
- 3 **A Ann Arbor** — Ann Arbor stages lymphoma with systemic B symptoms.
- 4 **F Dukes D** — Dukes D denotes distant metastases.
- 5 **B Breslow** — Breslow thickness stages melanoma.

3 THEMES • 13 SCENARIOS • AFTER CRASH COURSE SURGERY SBAS & EMQS

Crash Course EMQs

THEME 1 POSTOPERATIVE COMPLICATIONS

- A Atelectasis
- B Anastomotic dehiscence
- C Constipation
- D Deep vein thrombosis
- E Gravitational oedema
- F Haematoma
- G Pneumonia
- H Pseudo-obstruction
- I Urinary retention
- J Wound infection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 45-year-old man complains of increasing pain in the groin following a right inguinal hernia repair 5 days previously. On examination, the wound is red, swollen and tender.
- 2 A 70-year-old man who had a hip replacement 5 days previously complains of generalized abdominal discomfort. He has vomited four times, his bowels have not functioned for 5 days and he feels thirsty and has not passed urine for 8 hours. On examination, his abdomen is grossly distended and tympanitic, but there is no tenderness and he has decreased bowel sounds. The rectum is empty.
- 3 An 83-year-old woman had an emergency laparotomy for a bleeding duodenal ulcer 18 hours previously. She has become borderline pyrexial, and pulse oximetry shows saturation of 90% on 2 L of oxygen. Her blood pressure is normal, but her pulse rate is 110 beats/min. Her respiratory rate is 20 breaths/min and she has decreased breath sounds at the right base.
- 4 A 72-year-old woman who has had an anterior resection of the rectum suddenly develops severe abdominal pain 4 days postoperatively. On examination, she is unwell, with a pyrexia of 38°C, blood pressure of 100/60 mmHg, abdominal distension, a discharging wound and peritonism.
- 5 A 62-year-old woman develops discomfort of the right lower leg 7 days after a hysterectomy for carcinoma of the uterus. On examination, she has a pyrexia of 37.5°C and bilateral ankle oedema (right > left).

THEME 2 PREOPERATIVE INVESTIGATIONS

- A Arterial blood gases
- B Blood glucose
- C Carotid Doppler scan
- D Chest X-ray
- E Clotting screen
- F Coronary angiogram
- G Echocardiogram
- H ECG
- I Liver function tests
- J Spirometry
- K Troponin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 72-year-old woman has been admitted for a mastectomy. She gives a history of well-controlled hypertension for which she is on medication. On examination she looks well, and the examination of the cardiovascular system reveals an ejection systolic murmur in the aortic area.
- 2 A 50-year-old woman is admitted for a routine repair of a femoral hernia. She has chronic asthma for which she has been taking medication on a regular basis, but it is not always well controlled.
- 3 A 76-year-old patient is admitted as an emergency with symptoms of obstructive jaundice. An ultrasound shows the presence of a gallstone in the common bile duct. She is about to undergo an endoscopic retrograde cholangiopancreatography.
- 4 A 66-year-old man is admitted for a coronary artery bypass operation. During routine clerking, he describes an episode when he temporarily lost the vision in his right eye and some weakness of his arm, which has now recovered completely.

THEME 3 SCROTAL SWELLINGS

- A Direct inguinal hernia
- B Epididymal cyst
- C Hydrocele
- D Hydrocele of the cord
- E Indirect inguinal hernia
- F Lipoma
- G Lymph node
- H Spermatocele
- I Testicular tumour
- J Varicocele

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 40-year-old man presents with a tense lump in the scrotum which has appeared over the last 3 weeks and is mildly tender. It is transilluminable and the testis cannot be felt separately.
- 2 A 22-year-old man presents with a tender lump in his scrotum that his girlfriend noticed in the shower. He has lost a stone in weight over the last 2 months, and clinically, there is a lump attached.

- 3 A 30-year-old man presents with a large lump in the scrotum that came on a few days ago. It is large, tense and you cannot feel above it. It is nontender and non-transilluminable, and it has a cough impulse.
- 4 A 35-year-old man presents with a left-sided lump in the scrotum. He says that it disappears when he lies down. Upon palpation, the lump is inconsistent in nature and feels very mobile.

Answers • Themes 1–3

CRASH COURSE EMQs

Theme 1 • POSTOPERATIVE COMPLICATIONS

- 1 **J Wound infection** — A red, swollen, tender groin wound 5 days after hernia repair is a surgical site infection, which typically appears 5–7 days postoperatively.
- 2 **H Pseudo-obstruction** — Massive tympanitic colonic distension without tenderness, an empty rectum and reduced bowel sounds after orthopaedic surgery is acute colonic pseudo-obstruction (Ogilvie).
- 3 **A Atelectasis** — Low-grade pyrexia, tachycardia, hypoxia and reduced breath sounds at a base within 48 hours of laparotomy is basal atelectasis.
- 4 **B Anastomotic dehiscence** — Sudden pain, fever, hypotension, peritonism and wound discharge 4 days after anterior resection is anastomotic leak (dehiscence).
- 5 **D Deep vein thrombosis** — Calf discomfort, low-grade fever and asymmetrical (right > left) ankle oedema a week after pelvic cancer surgery is deep vein thrombosis.

Theme 2 • PREOPERATIVE INVESTIGATIONS

- 1 **G Echocardiogram** — An ejection systolic murmur in the aortic area suggests aortic stenosis, which must be assessed by echocardiography before anaesthesia.

- 2 **J Spirometry** — Poorly controlled asthma should be assessed with spirometry (peak flow/FEV₁) and optimised before elective surgery.
- 3 **E Clotting screen** — Obstructive jaundice impairs vitamin K absorption, so a clotting screen (INR) is essential before ERCP and sphincterotomy.
- 4 **C Carotid Doppler scan** — Transient monocular blindness and arm weakness are TIA symptoms; carotid Doppler is needed to detect significant stenosis before cardiac surgery.

Theme 3 • SCROTAL SWELLINGS

- 1 **C Hydrocele** — A tense, transilluminable swelling in which the testis cannot be felt separately is a hydrocele (scan the testis, as a tumour can cause a secondary hydrocele).
- 2 **I Testicular tumour** — A lump arising from the testis in a young man with weight loss is a testicular tumour until proven otherwise: ultrasound and tumour markers.
- 3 **E Indirect inguinal hernia** — A scrotal swelling one cannot get above, with a cough impulse and no transillumination, is an inguinoscrotal (indirect) hernia.
- 4 **J Varicocele** — A left-sided 'bag of worms' that empties on lying down is a varicocele (dilated pampiniform plexus).

5 THEMES • 20 SCENARIOS • Irfan SURGERY EMQ COLLECTION

Irfan EMQs

THEME 1 Feeding the surgical patient

- A Elemental diet
- B Normal oral intake
- C Total parenteral nutrition (TPN)
- D Nasojejunal feeding
- E Percutaneous endoscopic gastrostomy (PEG)
- F High-protein liquid supplements
- G 50% dextrose i.v.
- H Hartmann's solution i.v.
- I Percutaneous endoscopic jejunostomy
- J Nasogastric feeding

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 29-year-old man is admitted to the surgical ward following an acute exacerbation of Crohn's disease. He had a limited small bowel resection several months ago and has subsequently developed a high enterocutaneous fistula. On examination he appears malnourished.
- 2 A 40-year-old woman is involved in a road traffic accident and suffers significant head injuries with associated maxillofacial trauma. A prolonged recovery is expected.

- 3** A 57-year-old man with oesophageal cancer is severely malnourished due to significant dysphagia to solid food. He is able to tolerate small volumes of liquid feed with discomfort.
- 4** A 25-year-old homeless patient who has not eaten for 24 hours requires, later in the day, an incision and drainage procedure for a groin abscess.

THEME 2 The shocked surgical patient

- A** Hypovolaemia
- B** Sepsis
- C** Cardiac tamponade
- D** Tension pneumothorax
- E** Neurogenic shock
- F** Left ventricular failure
- G** Pulmonary embolism
- H** Anaphylaxis
- I** Acute adrenal failure
- J** Diabetic ketoacidosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 65-year-old patient is brought to A&E following a trauma call. On examination he has right-sided chest wall tenderness; X-ray shows a proximal displaced femoral fracture. The airway is patent, chest expansion is reduced, breath sounds are vesicular and JVP is not visible, respiratory rate 20/min, pulse 110/min, BP 90/60 mmHg.
- 2** A 35-year-old patient complains of shortness of breath and palpitations shortly after admission for acute cholecystitis. Blood pressure drops suddenly to 90/40 mmHg with tachycardia; there is marked erythema around the IV cannula.
- 3** A 68-year-old patient who had an abdominal aortic aneurysm repair was in significant pain postoperatively and was seen by the acute pain team about 15 minutes ago. His blood pressure has now dropped to 90/50 mmHg with a pulse of 60/min; he is slightly dizzy, though urine output is adequate and CVP is +12.
- 4** A 74-year-old patient is brought to A&E after being found collapsed at home; on examination his abdomen is rigid, pulse is 120/min, BP is 100/60 mmHg and temperature is 38°C.

THEME 3 Complications of blood transfusion

- A** Hyperkalaemia
- B** Iron overload
- C** Transfusion-related acute lung injury
- D** Hypothermia
- E** Non-haemolytic febrile reaction
- F** Haemolytic transfusion reaction
- G** Thrombophlebitis
- H** Air embolus
- I** Delayed haemolytic transfusion reaction
- J** Hypocalcaemia
- K** Viral infection
- L** Circulatory overload

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 75-year-old patient is found to be acutely short of breath after receiving her third unit of packed red cells. On examination, fine end-inspiratory crackles are heard up to mid zones in both lung fields.
- 2** A 37-year-old patient complains of headache and abdominal pain a few minutes after red cell transfusion; temperature is 38.5°C, pulse 120/min, BP 100/60 mmHg.
- 3** A 50-year-old patient presents with jaundice 5 days after receiving a red cell transfusion.
- 4** A 24-year-old patient is noted to be flushed a few minutes after starting a red cell transfusion; temperature is 37.8°C and observations are otherwise stable.

THEME 4 Head injury

- A** Scalp haematoma
- B** Concussion syndrome
- C** Diffuse axonal injury
- D** Basal skull fracture
- E** Subarachnoid haemorrhage
- F** Subdural haematoma
- G** Post-concussive syndrome
- H** Cerebral contusion
- I** Extradural haemorrhage

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 24-year-old student is brought to A&E after being assaulted outside a club. CT reveals small areas of haemorrhage in the frontal region and his GCS drops from 14 to 12 over 24 hours.
- 2** A 75-year-old woman, following a fall, presents with fluctuating consciousness; CT shows a hyperdense crescentic lesion against the inner left frontal skull with moderate midline shift.
- 3** A 42-year-old man is found unconscious on the street with bruising around both eyes and a clear nasal discharge.

- 4 A 25-year-old was involved in a high-speed road traffic accident 48 hours ago. He has been comatose since the injury. There are no major abnormalities on CT.

THEME 5 Chest trauma

- A Isolated rib fracture
- B Aortic transection
- C Simple pneumothorax
- D Flail chest
- E Pulmonary contusion
- F Cardiac tamponade
- G Haemothorax
- H Diaphragmatic rupture
- I Diaphragmatic contusion
- J Myocardial contusion
- K Pleural effusion
- L Tension pneumothorax

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 45-year-old male is brought to A&E after suffering multiple stab wounds to the chest. He is in respiratory distress with poor expansion on the right side, tracheal deviation to the left and distended neck veins.
- 2 A 45-year-old patient involved in a road traffic accident presents with bruising all over the chest and exhibits shallow tachypnoea with indrawing of the chest wall.
- 3 A 45-year-old man is stabbed in the right side of the chest. Chest X-ray shows a whiteout of the right lung field.
- 4 A 26-year-old man is brought to A&E in extremis after a single stab wound to the left side of the chest. He is tachypnoeic, heart sounds are muffled and blood pressure is 90/55 mmHg despite IV fluid resuscitation.

Answers · Themes 1-5

IRFAN EMQS

Theme 1 · Feeding the surgical patient

- 1 **C Total parenteral nutrition (TPN)** — The patient with an enterocutaneous fistula and inflamed bowel secondary to Crohn's disease is unlikely to absorb enteral feeds; hence, TPN is indicated.
- 2 **E Percutaneous endoscopic gastrostomy (PEG)** — In the presence of maxillofacial trauma, it is unsafe to insert a nasogastric/nasojejunal tube so PEG feeding is the best alternative for a prolonged recovery.
- 3 **J Nasogastric feeding** — Severe dysphagia due to oesophageal cancer precludes oral solid intake; bypassing the obstruction via a nasogastric tube (enteral feeding) is optimal.
- 4 **H Hartmann's solution i.v.** — For a patient scheduled for a minor procedure with short-term fasting, avoidance of enteral feeding with IV fluid supplementation is appropriate.

Theme 2 · The shocked surgical patient

- 1 **A Hypovolaemia** — A femoral fracture with tachycardia, hypotension and flat neck veins after trauma means blood loss: hypovolaemic shock.
- 2 **H Anaphylaxis** — Sudden breathlessness, hypotension and erythema at the cannula site soon after IV drugs (e.g. antibiotics) suggests anaphylaxis.
- 3 **E Neurogenic shock** — Hypotension with bradycardia and a normal CVP just after an epidural top-up reflects sympathetic blockade (neurogenic-type shock), not hypovolaemia.
- 4 **B Sepsis** — Collapse with a rigid abdomen, fever and tachycardia suggests perforation with peritonitis causing septic shock.

Theme 3 · Complications of blood transfusion

- 1 **L Circulatory overload** — Breathlessness with bibasal crackles in an elderly patient after several units is transfusion-associated circulatory overload: stop, sit up and give a diuretic.
- 2 **F Haemolytic transfusion reaction** — Fever, tachycardia, hypotension, headache and loin/abdominal pain within minutes

of starting red cells is an acute haemolytic (usually ABO-incompatible) reaction.

- 3 **I Delayed haemolytic transfusion reaction** — Jaundice 5–10 days after transfusion is a delayed haemolytic reaction from an anamnestic antibody response to a minor red-cell antigen.
- 4 **E Non-haemolytic febrile reaction** — Isolated mild fever and flushing with stable observations is a non-haemolytic febrile reaction (cytokines/leucocyte antibodies): slow the transfusion and give paracetamol.

Theme 4 · Head injury

- 1 **H Cerebral contusion** — Small areas of frontal haemorrhage on CT after an assault with a slowly falling GCS (oedema around the bruise) is a cerebral contusion.
- 2 **F Subdural haematoma** — A crescentic (concave) hyperdense collection with fluctuating consciousness after a fall in an elderly patient is an acute subdural haematoma.
- 3 **D Basal skull fracture** — Bilateral periorbital (racoon) bruising with CSF rhinorrhoea indicates an anterior basal skull fracture.
- 4 **C Diffuse axonal injury** — Coma from the moment of a high-speed (acceleration–deceleration) injury with a near-normal CT is diffuse axonal injury; MRI shows the shearing lesions.

Theme 5 · Chest trauma

- 1 **L Tension pneumothorax** — Respiratory distress, poor right-sided expansion, tracheal deviation to the left and distended neck veins is a right tension pneumothorax.
- 2 **D Flail chest** — Paradoxical indrawing of a chest-wall segment after blunt trauma is a flail chest (two or more ribs each broken in two places).
- 3 **G Haemothorax** — A penetrating chest wound with whiteout of the hemithorax on an erect film is a haemothorax: chest drain, thoracotomy if ≥ 1500 mL.

- 4 **F Cardiac tamponade** — A left chest stab wound with muffled heart sounds and hypotension unresponsive to fluid is cardiac tamponade.

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